MECHANICAL ENGINEERING DRAWING

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PREFACE

From the experience while teaching Mechanical Engineering Drawing to the undergraduate students, the authors have observed the problems faced by the students in some specific areas. Of them the most important one is that most of the books, which are currently available on Mechanical Engineering Drawing, do not emphasize much on the solution of problems. As a result, it really becomes difficult for the beginners to grasp the clear concept of the drawing. From the interaction with the students, the authors have the feelings that they have the keen interest to learn but become helpless for want of a book of their choice incorporating a reasonable number of problem solutions in the systematic way. They often express their feelings considering Mechanical Engineering Drawing as a difficult subject.

Another experience of the authors from the interaction with the students is that they always want to avoid the lengthy description of the various items of drawing. It has been observed in many books that the sizes of the drawing of many parts have been made small due to space problem. Additionally, after providing the dimensions, the drawings really appear as clumsy ones, which the students do not prefer.

Based on the fact as mentioned above, effort has been given to write this book. The solutions of some interesting and essential problems have been incorporated in this book. In each chapter, every item relevant to the practical problems has been considered with examples side by side. The authors think that this will help the students enormously in solving many other problems with much confidence without taking help from the others. The authors believe that this book will guide the students to understand the basic idea of Mechanical Engineering Drawing in the simpler way and the subject would not appear to them as the difficult one any more.

Each necessary item of drawing has been described in the concise form highlighting the important points only and avoiding the elaborate description. Simple example of drawing has been taken into consideration in the systematic way so that the students do not have any tendency to avoid rather find interest to learn. A compromise has been followed about the size of the drawing so that the students do not consider them as clumsy.

The most common elements of Mechanical Engineering Drawing have been considered in this book. It covers the undergraduate mechanical engineering drawing courses, which is taught in the different technical universities, institutions, polytechnic and AMIE in Bangladesh. The professionals of the various organizations will also be benefited from this book to a great extent in making their working drawings. Since the primary objective was to give the basic idea about Mechanical Engineering Drawing, as such, the complicated drawings have been avoided. However, if a student goes through the whole book sincerely, it will definitely help him in solving the complicated problems as well.

A separate chapter on AutoCAD consisting of the basic idea has been incorporated in this book. The authors think that this will guide the students in building their background to perform the drawing using AutoCAD. Every important element of AutoCAD has been presented with example side by side in the systematic way, so that the students can use AutoCAD without taking assistance from the others.

This book contains Mechanical Engineering Drawing only, as such if in any place only drawing is mentioned; it will indicate Mechanical Engineering Drawing (Mechanical Drawing). Throughout the books the SI Units have been used. Though the authors have tried their utmost to write the book in accomplishing the objectives as mentioned, there may be some deficiency and error in the book. In this regard, the authors wholeheartedly welcome any suggestion for its further improvement.

The authors would like to thank Mr. Zunaid Shams, Drafting Instructor of the Department of Mechanical Engineering, Bangladesh University of Engineering and Technology (BUET), Dhaka for his assistance during preparing the AutoCAD chapter.

June, 2006 The Authors

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CHAPTER 1

BASIC ELEMENTS OF MECHANICAL DRAWING

1.1 *Introduction*

From the ancient time people are communicating their thoughts and views by means of pictures, which is called the graphic language. A drawing is a graphic representation of a real thing or an idea. On the other hand drafting is the graphic language because it uses pictures to communicate the thoughts and ideas. People have been using the drawings for many centuries to express their ideas and concepts. Engineering drawing is a kind of drawing used by the engineers and technologists. It is the communication medium between the various persons involved in the design and manufacture of machines, building, bridges etc. For that matter it is often called graphic language. Mechanical engineering drawing (mechanical drawing) is one class of engineering drawing that is used to manufacture the various machine elements. In this book only the mechanical drawing will be taken into consideration for discussion.

When a drawing is performed without using instruments i.e. done by the pencils and erasers only, it is known as the freehand drawing. It is very important for preliminary design work. The drawing can be done using different drafting instruments. The instruments which are used to perform the drawing are called drafting instruments such as, drawing board, T-square, triangles, compasses, pencils etc. It may also be done with the help of a computer. For the complicated design the drawings performed by a computer will be convenient. Drawings done by a computer become faster and any modification can be incorporated in that easily.

1.2 *Drafting Instruments*

The most common types of drafting instruments are described in brief in the following subsections.

1.2.1 *Drawing Board*

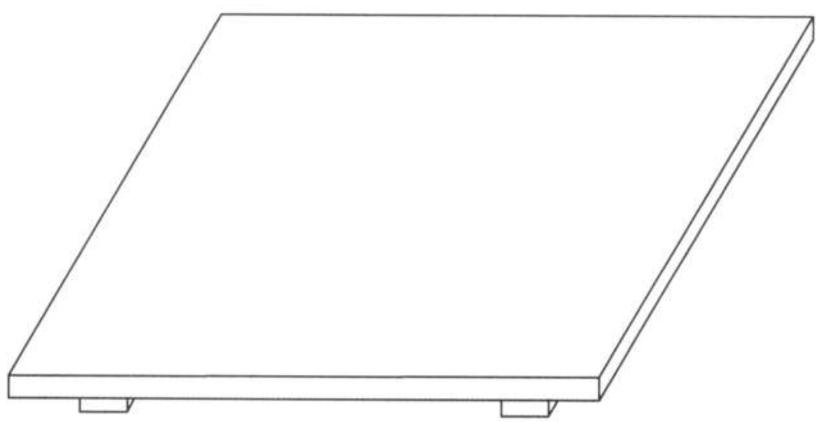


Figure 1.1: Drawing Board

Drawing board is used for placing the drawing paper on it with the help of either cellophane tape or board pin. One may use a drafting table in place of a drawing board. A drafting table is a kind of worktable with the adjustable top used for drawing. The stand of the drafting table is made in such a way that the top can be adjusted at any desired angle according to the convenience of the user. But the students usually use drawing board of the type as shown in Figure 1.1, which is made of wood. Sometimes a steel edge is attached with the board for better edge and wear resistant. Another type of drawing board may be with a parallel straightedge shown in Figure 1.2; as such it does not require a separate T-square.

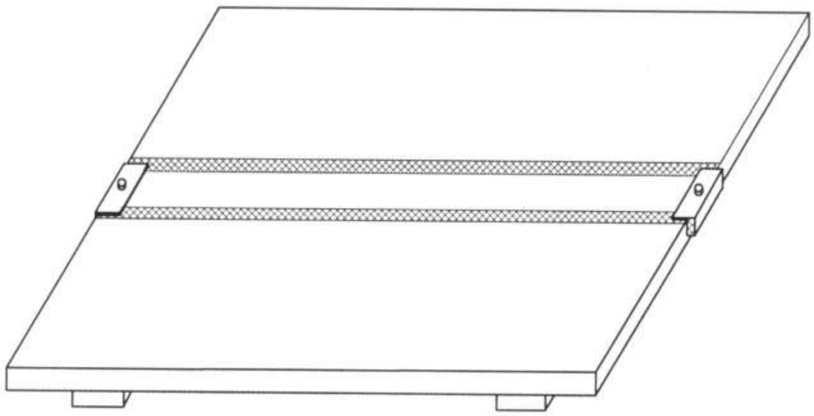


Figure 1.2: Drawing Board with a Parallel Straightedge

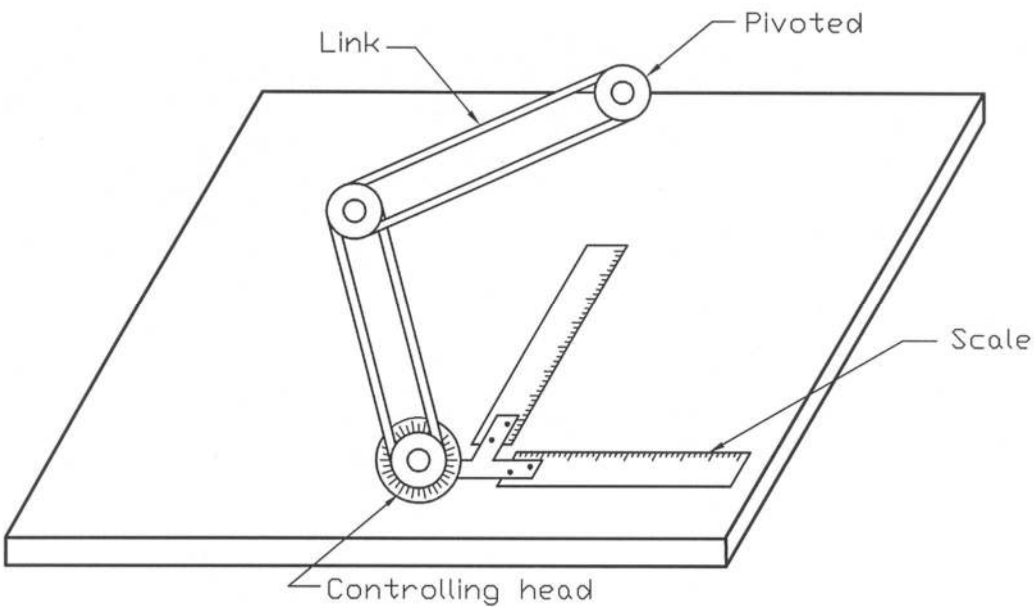


Figure 1.3: A Drafting Machine

A drafting machine may be attached with the drafting table (Figure 1.3), which replaces T-square, triangles, scales and protractor. The machine consists of two arms or links pivoted at one end from the top of the table. At the other end, it consists of a controlling head. The links are arranged in such a way that the controlling head can be moved to any location on the table. The controlling head contains a circular graduated scale in degrees and attached with two scales set at right angle with each other. The pair of scales can be set at any desired angle needed for the drawing. Using drafting machine one can perform drawing conveniently and speedily. A parallel straightedge similar to that shown in Figure 1.2 may also be used on the drafting table in place of a drafting machine. The ends of the straightedge are controlled by a chord and pulley system, which allows the straightedge to move up and down.

1.2.2 ***T-Square***

T-square is used to draw horizontal line. A T-square is shown in Figure 1.4. Usually a student uses a T-square on a drawing board. The T-square is commonly made of wood with its side made of transparent plastic. The head of the T-square is placed on the left edge of the board for the right-handed person and vice versa. To draw the parallel lines by the T-square the head of it has to be held firmly against the edge of the board.

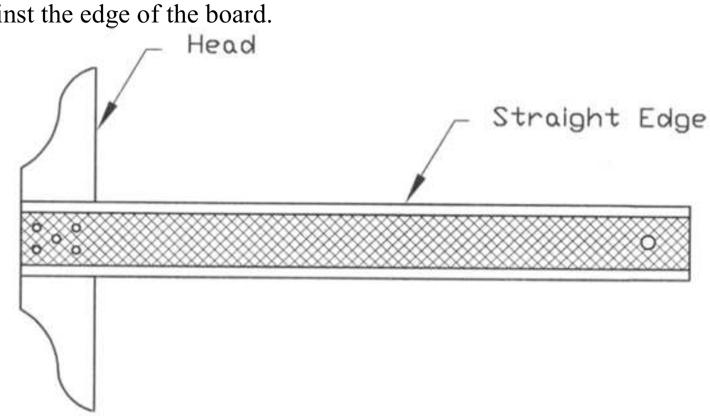


Figure 1.4: T-Square

1.2.3 ***Triangles***

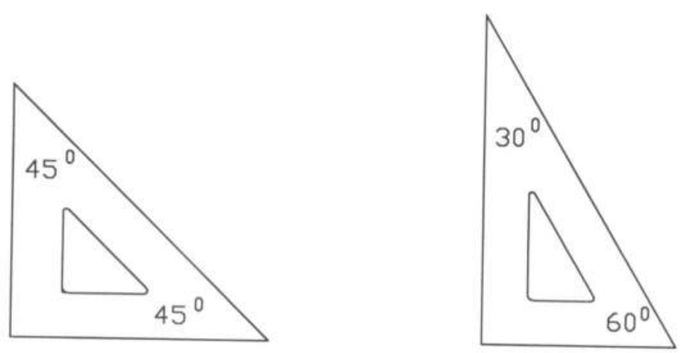


Figure 1.5: 45^0 and 30^0- 60^0 Triangles

In Figure 1.5, 45^0 and 30^0-60^0 triangles have been shown. They are usually made of clear plastic. Sometimes they may be colored, but the uncolored ones are usually preferable. They are used together with the straightedge of the T-square to draw the vertical and the inclined lines.

1.2.4 ***Scales***

The features of the cross-sections of the triangular scale and the flat scale are shown in Figure 1.6. Different types of scales have been shown in Figure 1.7. Scales are used for measurements only, not for drawing lines. When the actual size of the object is drawn, it is called full scale or scale 1:1. Often it becomes necessary to reduce the scale or enlarge the scale to represent an object. For an object of big size, reduced scale is necessary; on the other hand for a tiny object enlarged scale is required. An enlarged scale is 2:1 i.e. double the exact size of the object and the reduced scale is 1:2 i.e. half the exact size of the object. The scale to which the drawing is to be performed is indicated on the scale, so it has to be chosen accordingly. They are classified as metric scale, engineers scale, mechanical engineers scale, decimal scale and architects scale as shown in Figure 1.7.

A triangular scale consists of six scales. Mainly two types of scales exist. One type includes both the inch and the metric (millimeter, centimeter, meter) subtypes in a decimal base. On the other hand the other type is based on a foot-inch system. Scales may be made of wood, plastic or metal of which plastic scale is the best because of its low cost, accuracy and good visibility.

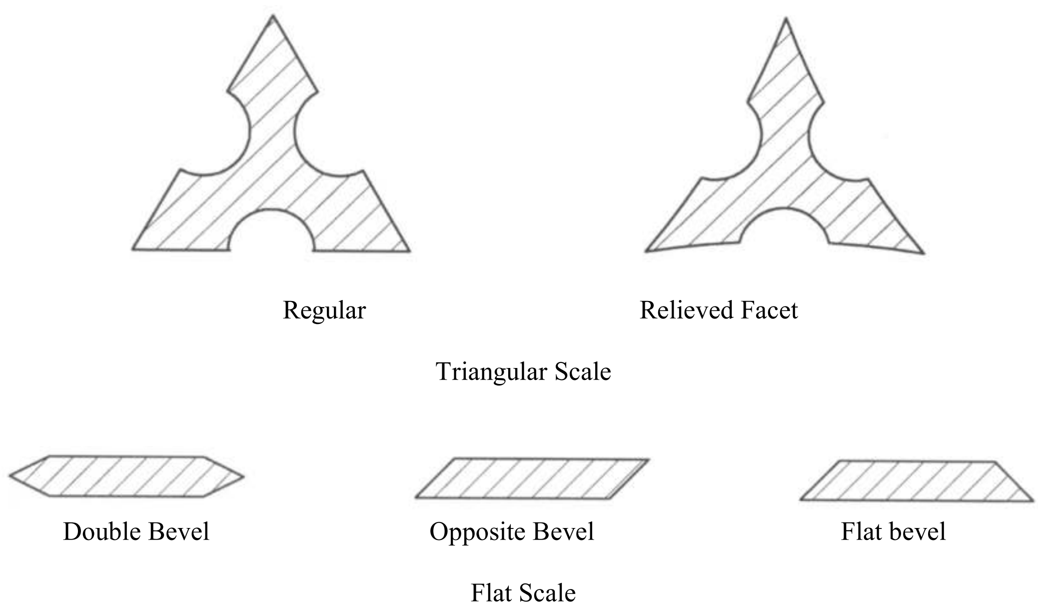
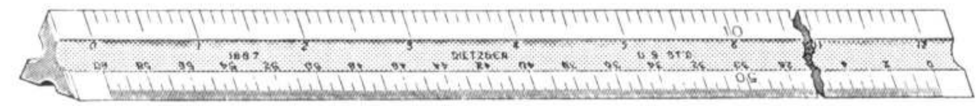
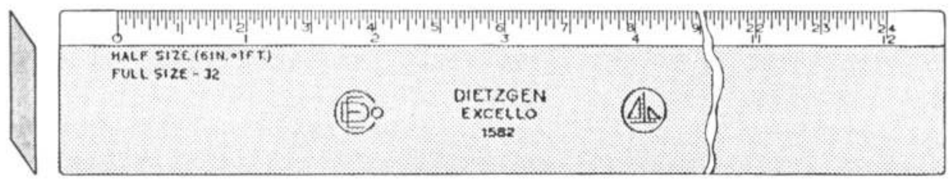


Figure 1.6: Cross Sections of Various Scales

(a) Metric Scale

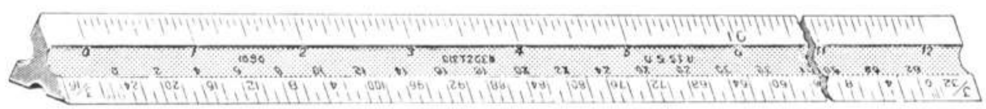


(b) Engineers Scale



(c) Mechanical Engineers Scale

(d) Decimal Scale



(e) Architects Scale

Figure 1.7: Different Types of Scales

1.2.5 ***Compasses***

A compass is a common drafting instrument. It is used to draw a circle or an arc. Compasses may be of different types. The most common types of compasses such as, friction head compass, drop bow compass, combination bow compass, compass with extender beam and beam compass are shown in Figures 1.8 to 1.12 respectively.

Figure 1.8: Friction Head Compass

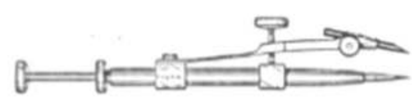
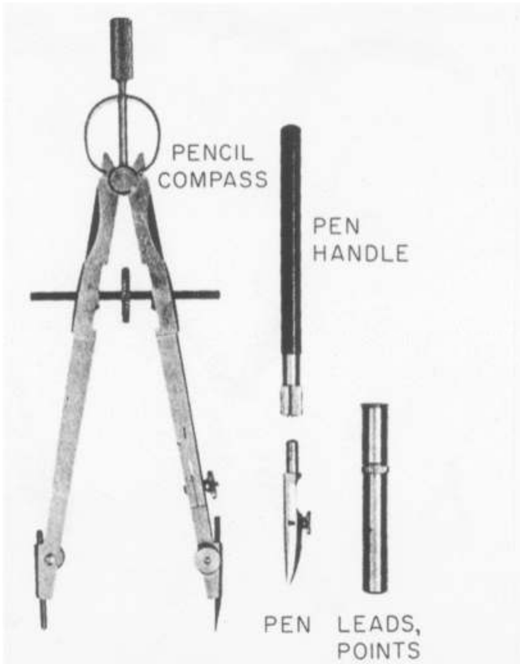


Figure 1.9: Drop Bow Compass



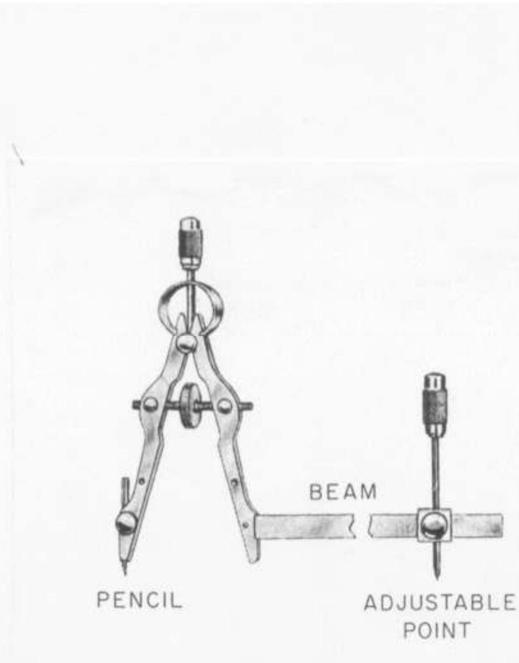


Figure 1.10: Combination Bow Compass Figure 1.11: Compass with Extender Beam

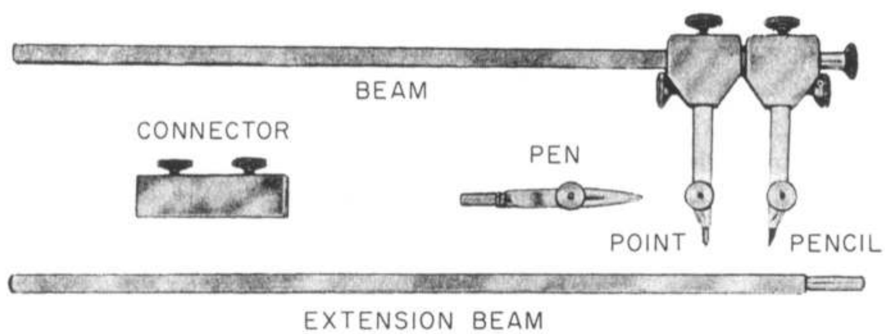


Figure 1.12: Beam Compass

Friction head compass is normally used in many drawings. Bow compass is preferred to friction head compass by many users because of its adjustment facility by a screw. Drop bow compass is used when a small circle is to be drawn. To draw a large circle the compass with extender beam or beam compass is used.

1.2.6 ***Dividers***

In order to transfer the measurement quickly in the drawing, a divider is used. A divider is similar to a compass. The compass may be used as a divider replacing the lead point with a steel pin. They are available in different sizes and designs. A most common divider is shown in Figure 1.13.

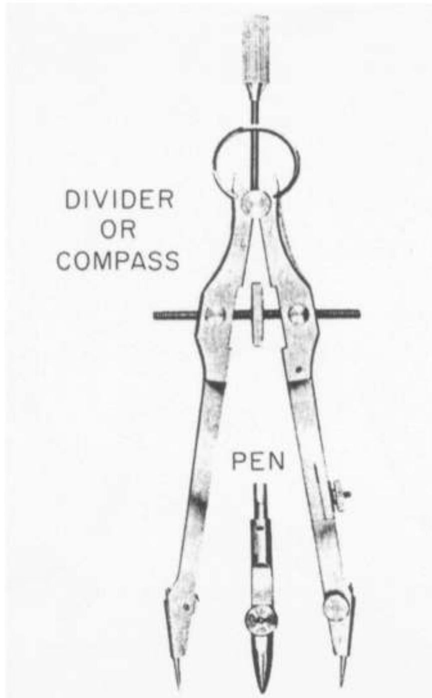


Figure 1.13: Divider

1.2.7 ***Paper***

Two types of paper are commonly used. They are detail paper and white drawing paper. Detail paper is used mainly for pencil work while the white drawing paper is used for finished drawings. Another type is the tracing paper, which is translucent in nature. It can be used both for pencil and ink work.

1.2.8 ***Pencils and Leads***

Different types of pencils are available in the market. One type is ordinary pencil with lead set into the wood. Another type is the mechanical pencil. Since it becomes time-consuming to sharpen a pencil, a thin lead pencil of mechanical type with 0.5 mm lead is preferred. It is not necessary to sharpen. Often 0.3 mm lead is available which is used to draw extra fine line.

Lead made of graphite is available in various grades of hardness. They may be classified as hard grade, medium grade and soft grade. Hard grade includes 9H, 8H, 7H, 6H, 5H and 4H; of them 9H is the hardest and 4H is the softest one. Medium grade includes 3H, 2H, H, F, HB and B. Of them 3H is the hardest and B is the softest one. Soft grade includes 2B, 3B, 4B, 5B, 6B and 7B. Of them 2B is the hardest and 7B is the softest one. The selection of proper grade of lead is important for quality drawing. One has to be careful in selecting a lead because very hard lead might penetrate the drawing, on the other hand soft lead may smear. The beginners may start with the 2H, H, F, HB and B leads.

1.2.9 ***Protractors***

To measure or lay out a particular angle, which is not a multiple of 15^0, the protractors may be necessary. It must be chosen of larger size in order to make the measurement accurately. The most common type of protractor is shown in Figure 1.14. The protractor as shown in Figure 1.15, includes a vernier in order to measure the angle with higher accuracy.

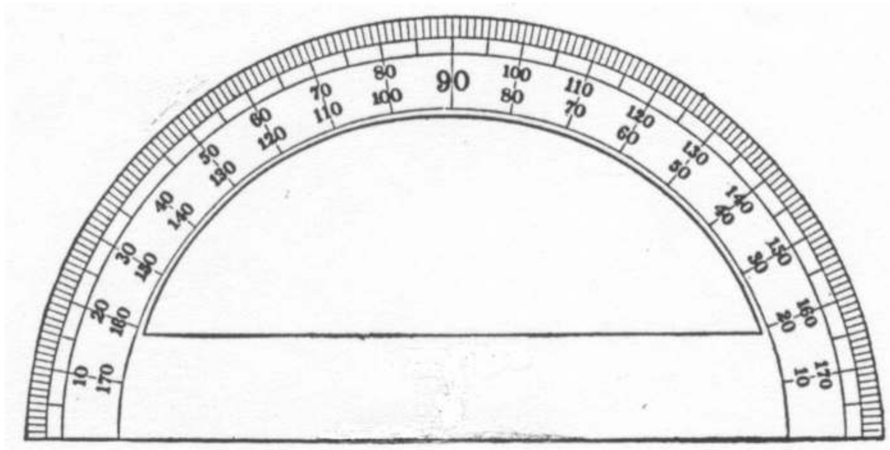


Figure 1.14: Protractor

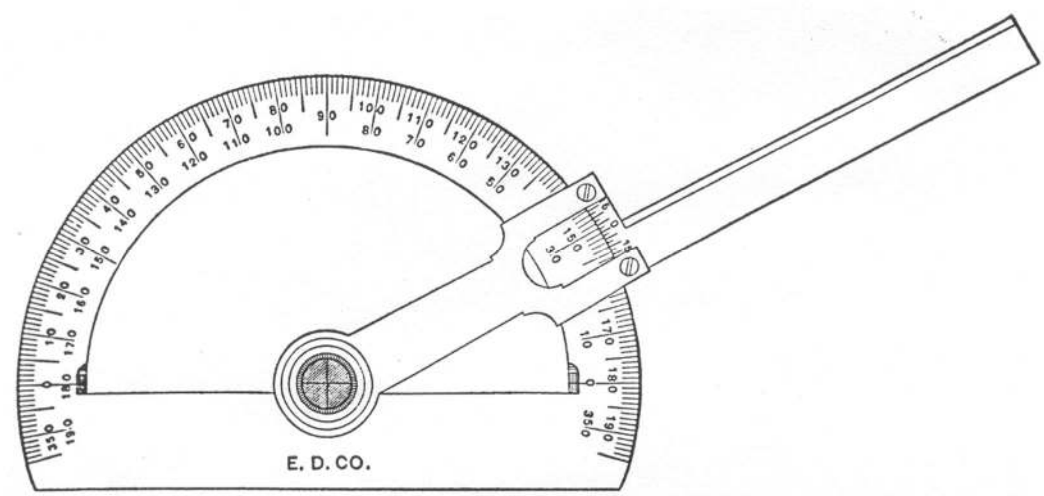
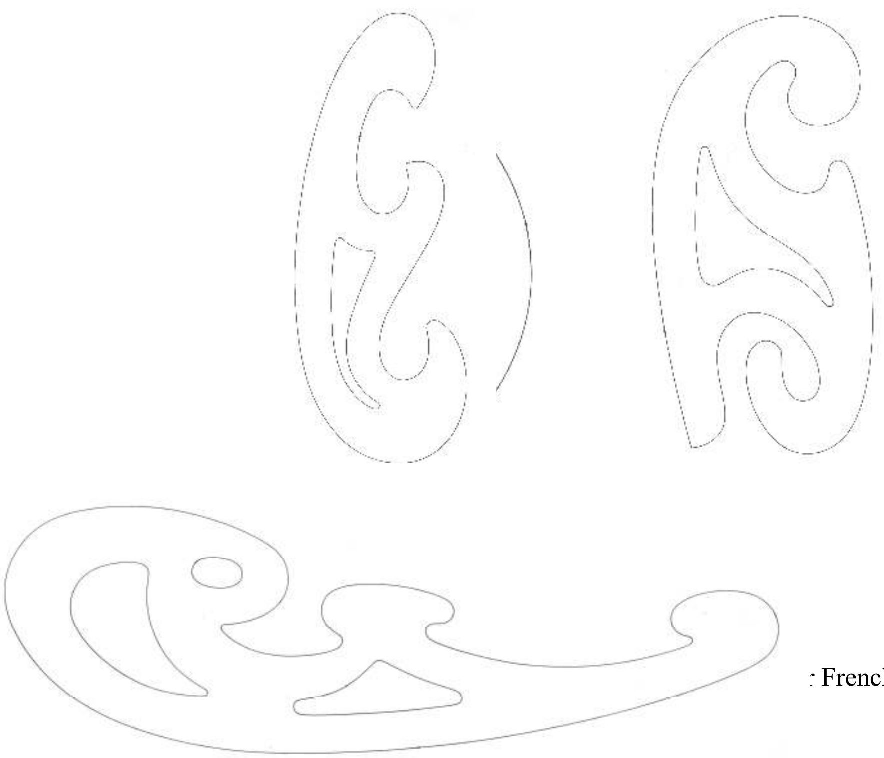


Figure 1.15: Protractor With Vernier

1.2.10 ***Irregular Curves***

To draw curved lines other than circles or circular arcs, an irregular or French curve is generally used. Many different sizes and forms of irregular curves are available. Of them some common ones are shown in Figure 1.16. The patterns for these curves are made on the basis of various combinations of ellipses, parabola, hyperbola and other mathematical curves.



French Curves

Figure 1.16: Irregular or French Curves

Besides these many other instruments such as, adjustable triangle, template, cardboard scale, pencil eraser, cellophane tape/board pin, pencil sharpener etc. may be necessary. A piece of cloth or tissue may be required to keep the paper clean removing dirt particle.

1.3 ***Sizes of Drawing Paper***

Drawing paper sizes of the metric standard from A0 to A4 have been shown in Table 1.1. These sizes are according to ISO 5457: 1999 (ISO stands for International Standard Organization).

Table 1.1: Standard Metric Paper Sizes

Paper Size	Border Size (b mm × l mm)	Overall Paper Size (b_0 mm × l_0 mm)
A0	821 × 1159	841 × 1189
A1	574 × 811	594 × 841
A2	400 × 564	420 × 594
A3	277 × 390	297 × 420
A4	190 × 267	210 × 297

The length and breadth of the paper sizes are shown in Figure 1.17 while in Figure 1.18 the way, how to obtain the different paper sizes has been provided. Paper of A0 size has an area of 1 m^2 with length to breadth ratio 1: √2. Each other size can be obtained by simply halving the preceding sheet on the longer side. However, the length to breadth ratio of each sheet remains constant at 1: √2.

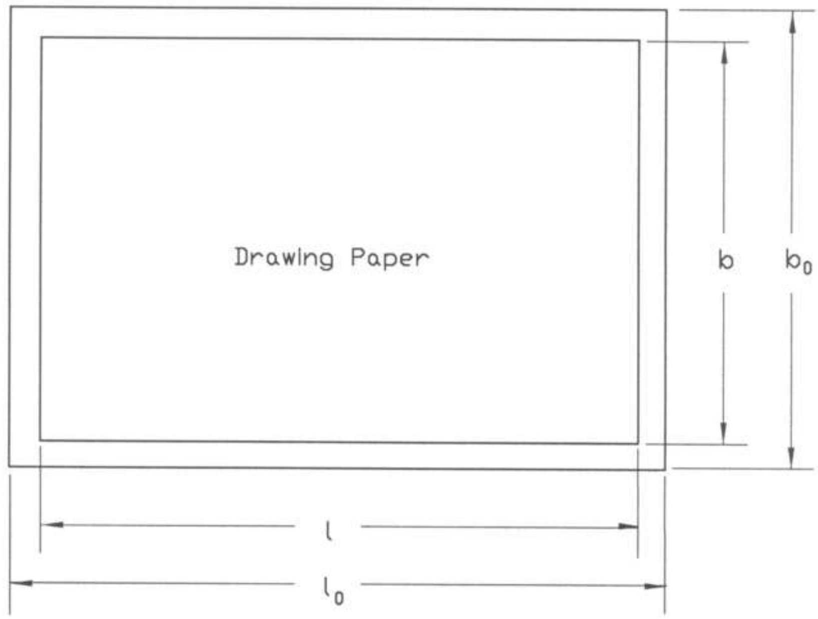


Figure 1.17: Length and Breadth of Paper

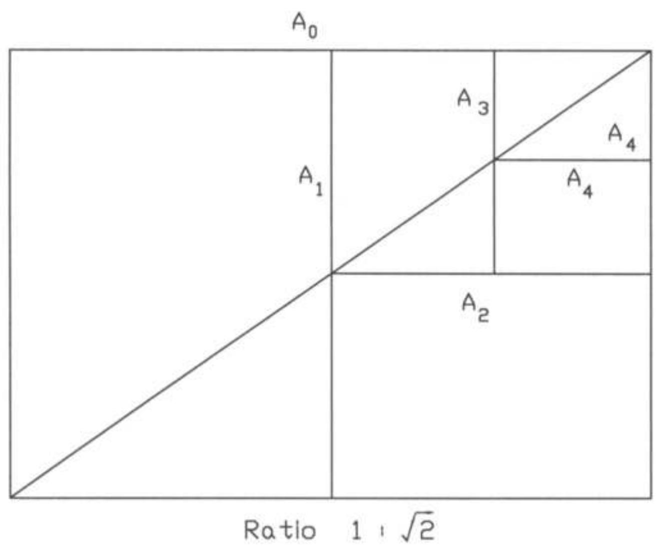


Figure 1.18: Metric Drawing paper

1.4 ***Uses of Common Instruments***

In regard to the uses of the most common drawing instruments, a brief description is provided below.

1.4.1 *Use of Pencils*

When an ordinary pencil is used, it has to be sharpened by the help of a pencil sharpener. The lead of the pencil has to be made pointed using a sand paper pad or a file. The lead of a mechanical pencil may be sharpened similarly. While drawing a line uniform pressure has to be put on the pencil and it has to be rotated while drawing the line. This will make the line uniform and keep the pencil sharp. In case of using the thin lead pencils i.e. either 0.5 mm or 0.3 mm lead, no sharpening is required.

1.4.2 *Drawing Lines With T-Square*

The T-square has straight edges. It is used to draw the horizontal lines. Holding the head of the T-square against the left edge of the drawing board firmly, the desired horizontal line may be drawn (Figure 1.19). Horizontal line should always be drawn from the left to the right. Care should be taken so that the pencil remains close to the edge of the T-square and the angle must not change during drawing the line.

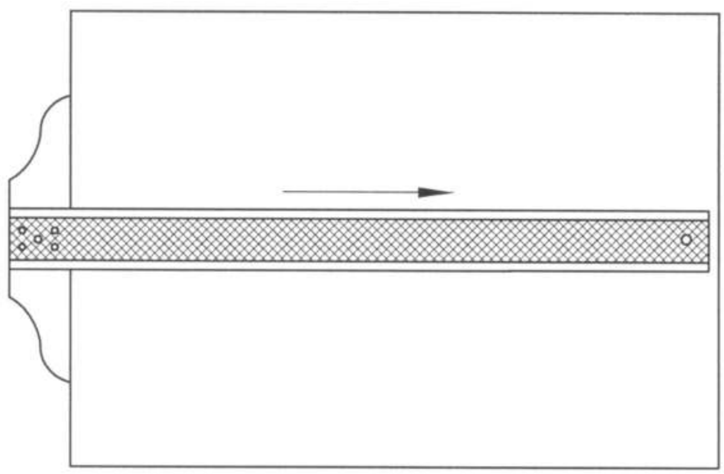


Figure 1.19: Drawing a Horizontal Line

1.4.3 *Drawing Lines With Triangles*

A vertical line can be drawn with the triangle and T-square. Holding the T-square firmly against the left edge of the board by the thumb and the triangle against the edge of the T-square, a vertical line (Figure 1.20) may be drawn. Line has to be drawn from the bottom to the top. In order to ensure accuracy in the line, the extreme corner of the triangle has to be avoided, which can be done by placing the T-square below the lower end of the line.

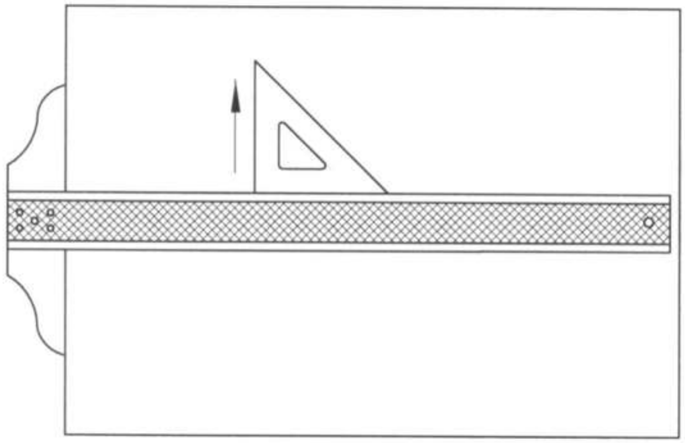
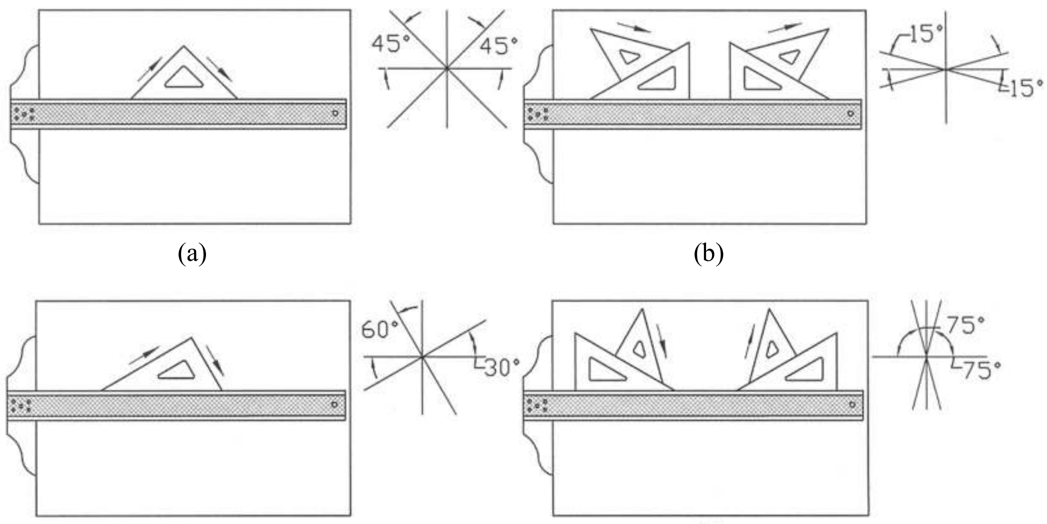


Figure 1.20: Drawing a Vertical Line



(a) (b)

(c) (d)

Figure 1.21: Drawing Inclined Lines

Any inclined line at 15^0 to 75^0 with a step of 15^0 can be drawn by the triangles. During drawing the lines it has to be remembered that for the left slope, the line has to be drawn from the bottom while for the right slope, it has to be drawn from the top to the bottom (Figure 1.21).

1.4.4 ***Drawing Circle/Arc With Compass***

A compass is used to draw a circle or an arc. Always the compass lead will be softer because same impression while drawing line cannot be imparted by a compass. To obtain the identical lining, this technique should be followed. A circle or an arc may be drawn rotating the compass lead in the clockwise direction as shown in Figure 1.22.

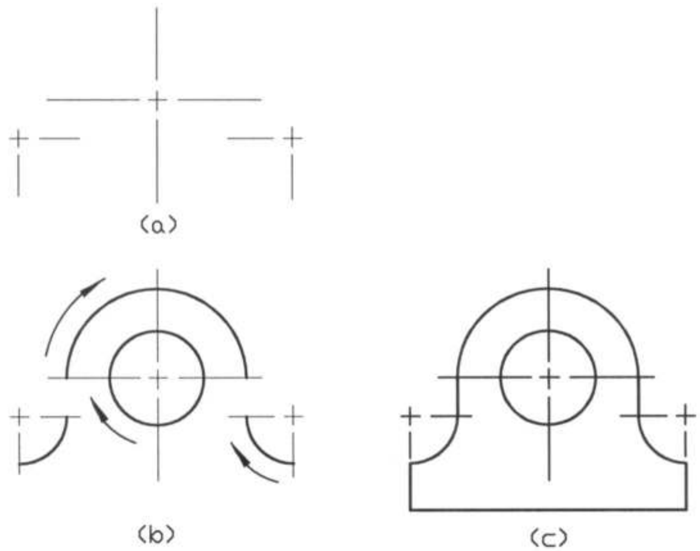


Figure 1.22: Drawing Circles and Arcs

1.4.5 ***Use of Irregular curves***

The use of an irregular or French curve has been shown in Figure 1.23.

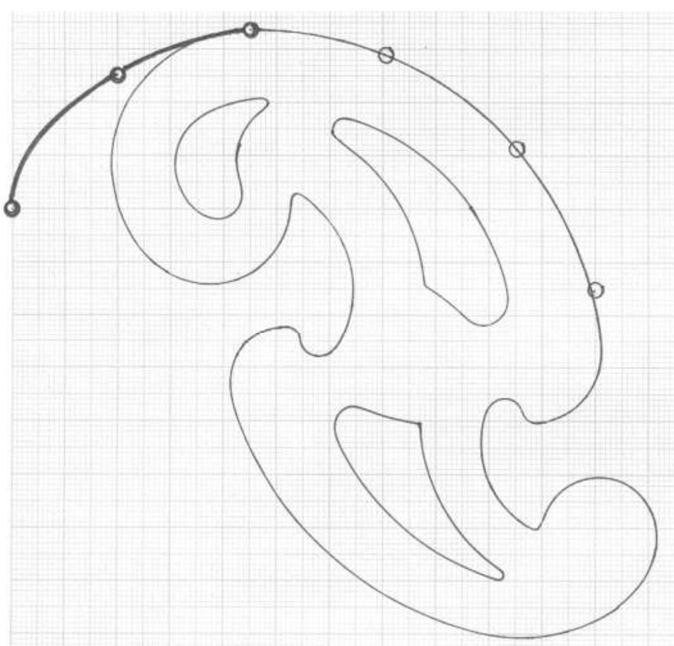


Figure 1.23: Drawing Curve Line

When a curved line is to be drawn with many points, an irregular or French curve may be used. Before drawing the curved line by the irregular curve, a light pencil line may be drawn freehand through the points. Then the various segments of the irregular curve should be matched with the freehand curve until the best matching is found. Next the curved line should be drawn by a pencil or ink as desired along the edge of the irregular curve. If all the points can not be included at a time to draw the curved line, some of the points are considered first; next the other points are considered and this process is continued until the curved line is completed including all the points. While drawing the curved line care should be taken so that there occurs no abrupt change in the curvature of the line.

1.5 *Types of Lines*

Different types of lines are used in the drawing. In Table 1.2, types of lines with their uses have been presented. Usually two types of widths are used for the lines; they are thick and thin. Thick lines are between 0.5 mm to 0.8 mm wide while thin lines are between 0.3 mm to 0.5 mm wide. According to ANSI Y14.2 - 1995 (ANSI stands for American National Standards Institute) the thick line is about 0.7 mm wide while the thin line is about 0.35 mm wide. Visible, cutting plane and short break lines are thick lines, on the other hand hidden, center, extension, leader, dimension, section, phantom and long break lines are thin. However, the actual width of each line is governed by the size and style of the drawing. If the size of the drawing is larger, the width of the line becomes higher. According to ISO 128: 1982(E), the cutting plane line is also shown in Table 1.2.

Table 1.2: Types of Lines with Their Usage (contd.)

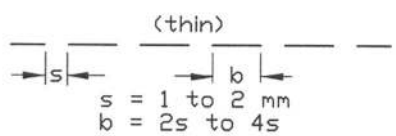
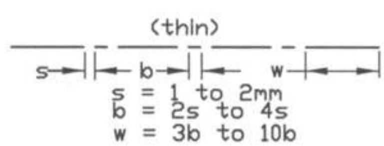
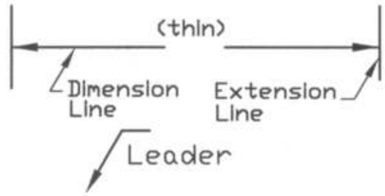
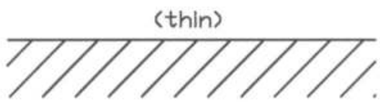
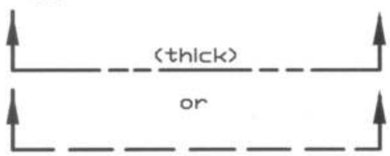
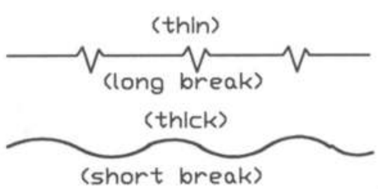
Sl No.	Types of Line	Usage
1	Visible line / object line 	to indicate all visible outlines of an object. It shows the shape of an object.
2	Hidden line / dashed line 	to represent the hidden edge of an object. It must begin and end with a dash touching the visible lines. Dashes that show hidden lines usually touch each other at intersection.
3	Center line 	to show the center line of holes, pitch line.

Table 1.2: Types of Lines with Their Usage (contd.)

Sl No.	Types of Line	Usage
4	Extension line, dimension line and Leaders 	to show dimension of an object extension line, dimension line and leaders are used.
5	Section line 	to indicate the cut portion of an object.
6	Cutting plane line 	to show the imaginary cutting of an object
7	ISO Cutting Plane Line 	
8	Break line 	to show a break on the object. It shortens the view of a long part.
9	Phantom line/repeat line 	to show the alternate position of an object or the position of an adjacent part.

When a pencil is used to perform the drawing manually, thin lines are often drawn with H or 2H lead grades while thick lines are drawn with softer leads, such as F or H. Relatively softer lead is used for the thick line and harder lead is used for the thin line. The line widths for the ink drawing are virtually identical to the pencil drawing with some exception that the widths for visible and hidden lines are slightly thicker. First very light construction lines should be drawn to give the main shape of the object in the different views. As such, it becomes convenient to modify and erase them easily. Then the final lines are drawn following the thickness as mentioned.

Line gage has been shown in Figure 1.24. It gives the idea about the various widths of lines. One can practice drawing lines of various widths and determine the widths of them from the line gage. Thus the control on drawing lines of different widths can be achieved. However, when a drawing is done with the help of a computer using AutoCAD, the lines of different widths can be used easily in accordance with the necessity.

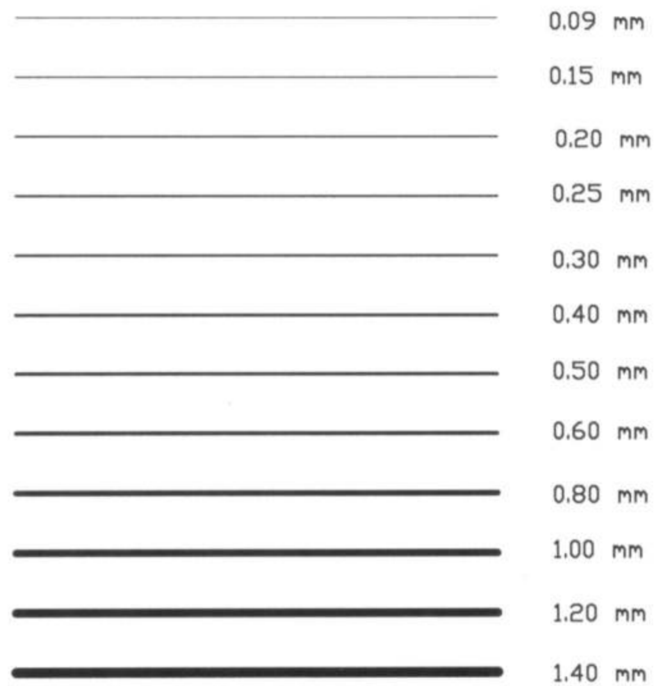
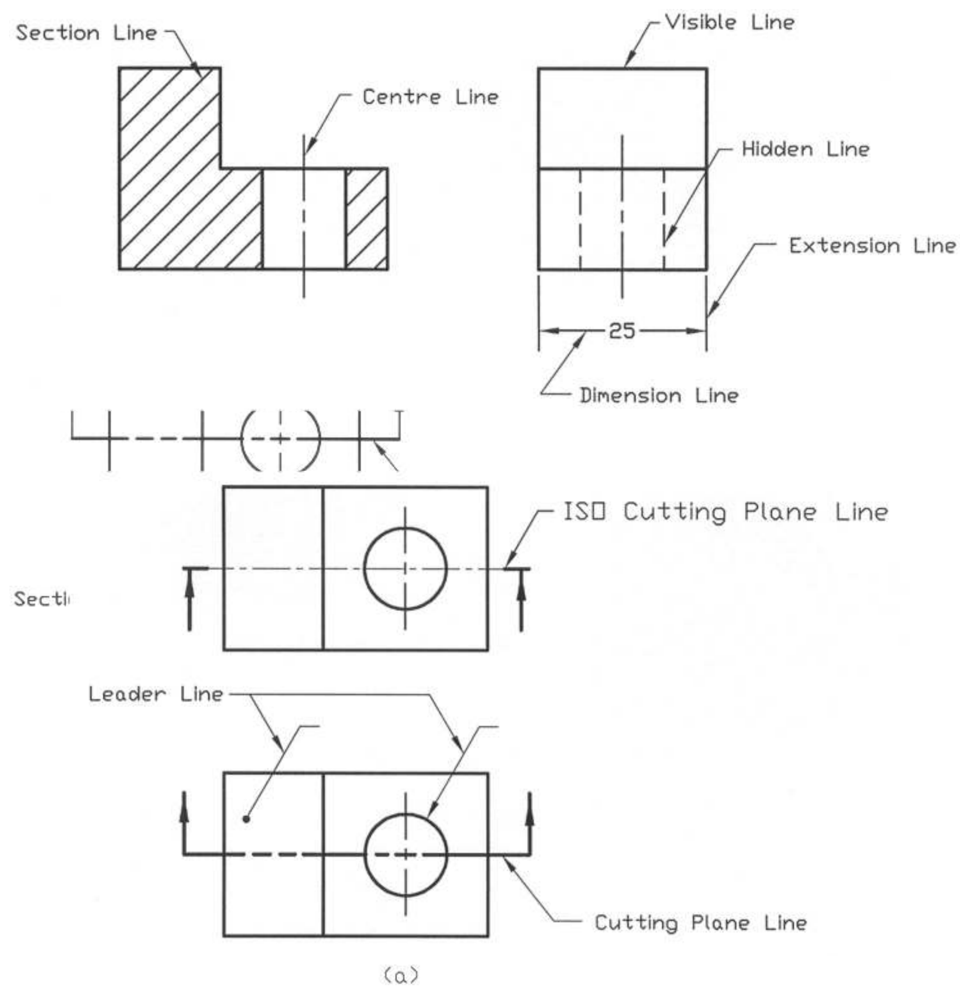
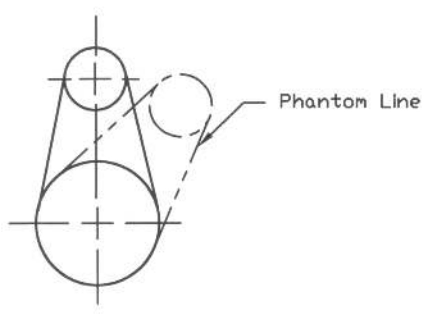


Figure 1.24: Line Gage

In Figures 1.25 to 1.28, the applications of the various lines have been shown. The uses of the visible, hidden, center, extension, dimension, and section lines have been shown in Figure 1.25 while the uses of leader and cutting plane lines have been shown in Figure 1.26. The use of ISO cutting plane line is also shown in Figure 1.26. The applications of phantom and break lines have been shown in Figures 1.27 and 1.28 respectively.





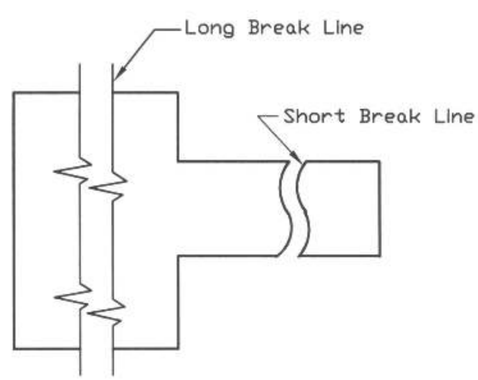


Figure 1.27: Use of Phantom Line

Figure 1.28: Usage of Long and Short Break Lines

1.6 *Sheet Layout and Title*

Layout of a drawing sheet showing their margin and title block is given in Figure 1.29. The title block is placed in the bottom right corner of the sheet. The margin/border line may be chosen in accordance with Table 1.1. The title block is shown in Figure 1.30.

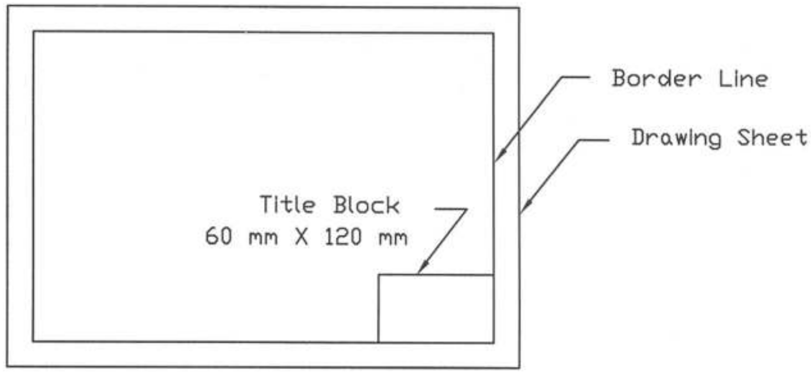


Figure 1.29: Layout of Drawing Sheet

BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
TITLE:		
SCALE:	MAT:	
NAME:		
DEPT:	ROLL:	DATE:

Figure 1.30: Title Block

The layouts of the drawing sheet and the title block are provided for use by the students. There is no rigid rule in regard to the size of the title block. However, there may be slight difference in sizes attributed by the different institutions. The manufacturing company uses title block of different sizes to satisfy their specific requirements. The title block of a manufacturing company may include the name of the company, part name, part number, material, clearance, tolerance, surface finish, operation, drawn, checked, approved, scale, date, sheet number, order or contract number etc. as is required by the company. (In Chapter 10 uses of some of them are provided). When the list of the parts will be required, then just above the title block it is given in the form as shown in Table 1.3 (an example). The location of the table containing the list of parts is shown in Figure 1.31.

When the drawing is performed in the tracing paper directly, the listing of the parts may be inserted from the bottom to the top of the table in order to accommodate further parts at the top, if necessary. However, in case of drawing done by AutoCAD, it may be inserted from the top to the bottom as in Table 1.3, because here at any location new accommodation of parts can be done.

Table 1.3: List of Parts

PT. No.	Part Name	Mat.	Qty.	Notes
1	Shaft	AISI C1045	1	
2	Pulley	ASTM 35	2	Enamel Paint
3	Key	AISI C1020	2	

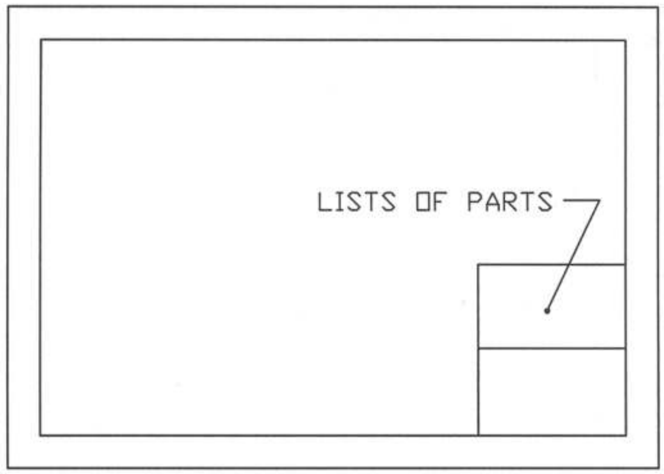


Figure 1.31: Layout of a Drawing Sheet Showing Location of List of Parts

1.7 *Scale of Drawings*

Scale is the ratio of the linear dimension of an element of a machine part as represented in the original drawing to the real linear dimension of the same element of the machine part itself. The scales of drawings are of three categories.

(a) Full Size: This consists of the ratio 1:1. When the part is drawn based on its actual size, the full size i.e. scale 1:1 is attained.

(b) Enlargement Scale: This is a scale whose ratio is larger than 1:1. To draw a part of smaller size this scale is necessary. The part is drawn in the bigger size than the actual one. When the drawing is done making the size of the part double to its actual one, the scale becomes 2:1. If the drawing is done with 50 times the actual size of the part, the scale is 50:1.

(c) Reduction Scale: This is a scale whose ratio is smaller than 1:1. This scale becomes necessary in order to draw the part of bigger size. The part is drawn in the smaller size than the actual one. When the drawing is made with half the actual size of the part, the scale is 1:2. If the part is drawn with one-tenth of its actual size, the scale becomes 1:10.

Whatever may be the scale for the drawing of the part, the dimensions of the part in the drawing always remain the same as those of the actual size of the part. According to ISO 5455:1979 (E), the recommended scales for use in technical drawings are provided in Table 1.4. If it is necessary for special application to enlarge or reduce the scale it may be done by multiplying with a factor of 10. However, in the exceptional cases, intermediate scales may be used other than the recommended scales.

Table 1.4: Recommended Scales of International Standard

Category	Recommended Scales		
Enlargement Scale	50:1	20:1	10:1
	5:1	2:1	
Full Size		1:1	
Reduction Scale	1:2	1:5	1:10
	1:20	1:50	1:100
	1:200	1: 500	1: 1000
	1:2000	1: 5000	1:10000

1.8 *Lettering*

Lettering is an important part of drawing. For the description, figured dimension, notes on material, finish, title etc. lettering is essential. Most of the lettering is done in single stroke either in vertical or in inclined manner. However, only one style of lettering should be used throughout the drawing. Lettering may be done either freehand or by templates. In Figure 1.32, vertical gothic alphabet has been shown. It is commonly used for all types of mechanical drawings. This type of letter is easier to make and read. Each letter has been shown in a square to show the relative proportion of the height and width of the letter. It can be observed that the heights and widths of the letters **A, O, Q, T, V, X, Y** and **Z** are same. On the other hand the widths of the letters **M** and **W** are more than their heights. The heights of the rest letters are more than their widths. The heights of all the numerals are more than their widths.

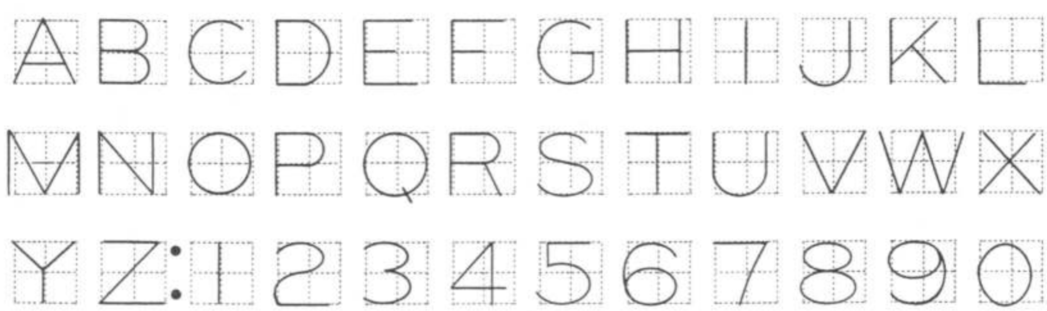


Figure 1.32: Vertical Gothic Alphabet

For application on technical drawings and associated documents, the characteristics of lettering specified by ISO 3098 – 2: 2000 are provided in Appendix 1.

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.***

Prob. 1.1: Draw a tangent at a point on a circle.

Prob. 1.2: Draw an arc tangent of radius 7 mm between a straight line and a circle of diameter 40 mm, the minimum distance between the center of the circle and the straight line being 30 mm.

Prob. 1.3: Draw a reverse curve tangent having 15 mm radius of curvature to three lines, the angles between the consecutive lines are (a) 90^0 (b) 60^0.

Prob. 1.4: Draw an arc tangent of radius 60 mm to the two circles each of 30 mm diameters; the center distance between the circles is 50 mm.

Prob. 1.5: Draw a hexagon inscribing a circle of diameter 60 mm.

Prob. 1.6: Draw a regular pentagon inscribing a circle of radius 30 mm.

Prob. 1.7: Draw an octagon outscribing a circle of radius 30 mm.

Prob. 1.8: Draw an ellipse with major and minor diameters of 60 mm and 40 mm respectively by the **Concentric Circle method**.

Prob. 1.9: Draw an ellipse with major and minor diameters of 60 mm and 40 mm respectively by the **Parallelogram method**.

Prob. 1.10: Draw an ellipse with major and minor diameters of 60 mm and 40 mm respectively by the **Four-Centered Approximate method**.

Prob. 1.11: Draw a tangent to an ellipse from a given outside point, which must not lie on a line parallel to either the major or minor axis and passing through the intersecting point on the curve with either the major or minor axis. The major and minor diameters of the ellipse are 60 mm and 40 mm respectively.

Prob. 1.12: Draw a tangent to an ellipse at a given point A on the curve, which must not lie on either the major or minor axis. The major and minor diameters of the ellipse are 60 mm and 40 mm respectively.

Prob. 1.13: Draw a parabola by the **Parallelogram method**.

Prob. 1.14: Draw a parabola by the **Offset method**.

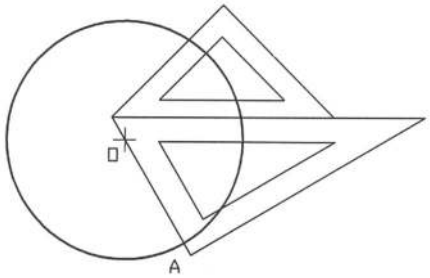
Prob. 1.15: Draw an involute profile on a base circle of diameter 65 mm.

Solutions for Example Problems

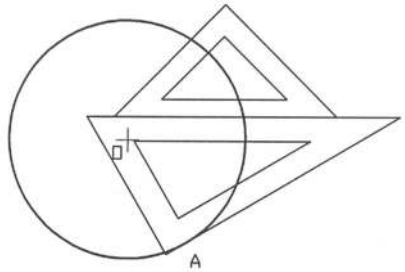
Solution of P1.1

Procedure:
The steps as mentioned below are followed in order to draw a tangent at a point on a circle.

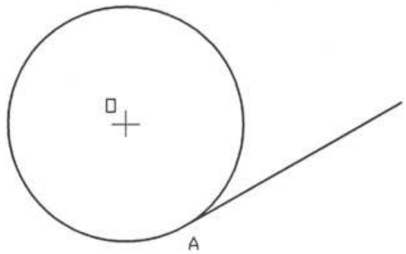
- First a circle is to be drawn.
- Next the point A on the circumference of the circle has to be selected where the tangent will be drawn (Fig. S1.1).
- Then a Triangle is to be placed such that its one edge joins the center of the circle and the point A.
- Then another Triangle has to be placed as shown in the figure so that their edges coincide with each other.
- Next the first Triangle has to be moved along the edge of the Second Triangle whose position is kept fixed as in the figure until the edge of the First Triangle just touches the point A.
- Now a line is to be drawn through the point A, which is the required tangent.



(a)



(b)



(c)

Fig. S1.1

Solution of P1.2

Procedure:

The steps in drawing an arc tangent between a straight line and a circle are given below:

- A circle of 40 mm diameter with center O and a straight line AB at a distance of 30 mm from the center of the circle are drawn (Fig. S1.2).
- Then an arc EF of radius equal to the radius of the circle plus 7 mm is drawn. A line CD at a distance of 7 mm from AB and parallel to AB is drawn. The line CD and the arc EF intersect at the point G.
- With the center G and a radius of 7 mm, the required arc tangent is drawn which touches the circle and the straight line AB at the points P and Q respectively.

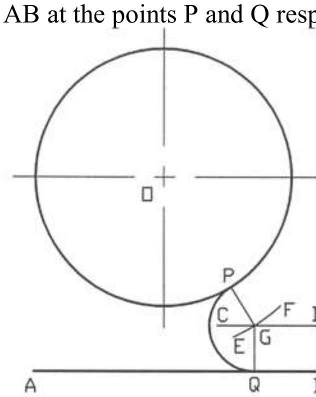


Fig. S1.2

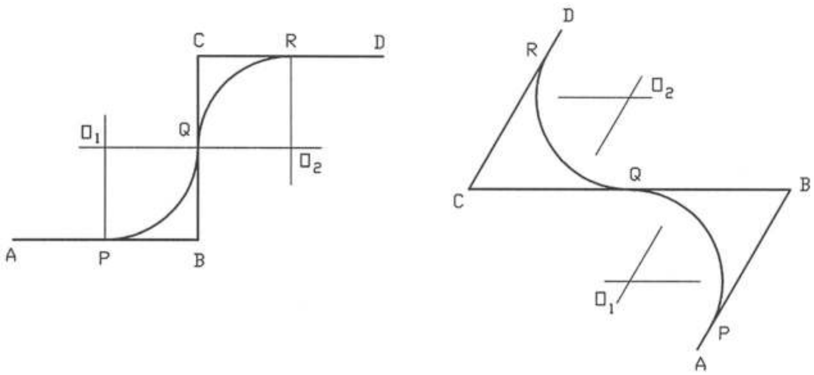
Solution of P1.3

Procedure:

The steps to draw a reverse curve tangent having 15 mm radius of curvature, are given below:

- Three lines AB, BC and CD are drawn with angle of 90^0 between AB and BC, BC and CD (Fig. S1.3a).
- Then lines parallel to AB, BC and CD are drawn at a distance of 15 mm from each of them as shown in the figure. The parallel lines intersect at the points $O_1$and O_2.
- Now with O_1 and O_2 as the centers and 15 mm radius, the required reverse curve tangent is drawn which touches the lines AB, BC and CD at the points P, Q and R respectively.

The same procedure is followed to draw a reverse curve tangent with angle of 60^0 between the consecutive lines (Fig. S1.3b).



(a) (b)

Fig. S1.3

Solution of P1.4

Procedure:

The procedure of drawing an arc tangent on the two circles are as follows:

- Two circles each of 30 mm diameters with centers A and B are drawn side by side such that the center distance between the circles is 50 mm (Fig. S1.4).
- Then with centers A and B, the two arcs each with radius 45 mm, which is equal to the difference of the arc tangent radius 60 mm and the circle radius 15 mm are drawn that intersect at O.
- Now taking O as the center the required arc tangent PQ is drawn with radius of 60 mm, which touches the two circles at the points P and Q.

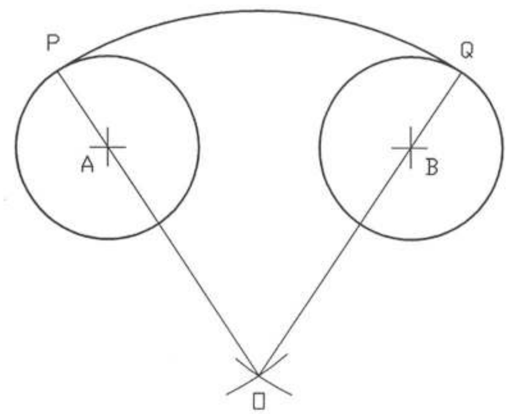


Fig. S1.4

Solution of P1.5

Procedure:
In order to draw a hexagon the steps as mentioned below have to be followed:

- A circle of diameter 60 mm is to be drawn first.
- Then equal segment each of 30 mm has to be drawn on the circumference of the circle, which are marked by the points 1, 2, 3, 4, 5 and 6 (Fig. S1.5).
- Next each two consecutive points as shown in the figure have to be added.

In this way the required hexagon is obtained.

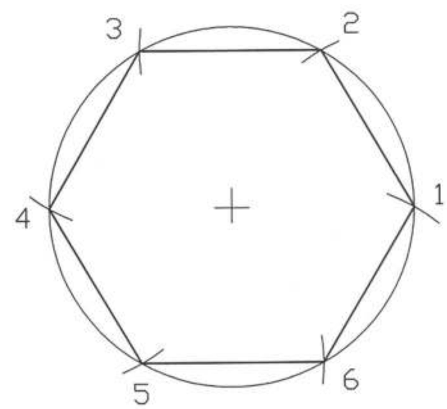


Fig. S1.5

Solution of P1.6

Procedure:
The following steps are to be followed to draw a regular pentagon inscribing a circle.

- A circle of 30 mm radius is drawn first and then a diameter MN is drawn (Fig. S1.6).
- From the center of the circle O, a perpendicular OP is drawn.
- Now on the line ON a mid point U is obtained.
- Taking U as the center and PU as the radius an arc PV is drawn.

- Now considering P as the center and PV as the radius, another arc QV is drawn.
- PQ is the side of the regular pentagon. The other sides are drawn taking equal segment as PQ on the circumference of the circle. Now the required pentagon P-Q-R-S-T is completed.

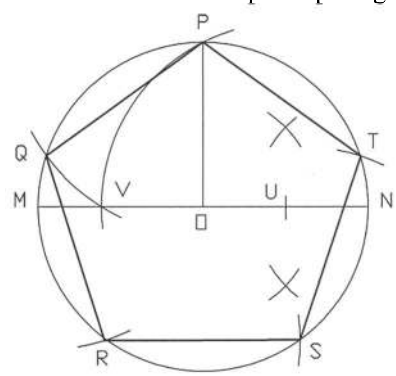


Fig. S1.6

Solution of P1.7

Procedure:

The procedure of drawing an octagon outscribing a circle is as follows:

- First a circle of radius 30 mm is drawn.
- Then a square is drawn such that its sides are tangent to the circle (Fig. S1.7).
- Next the diagonals of the square are drawn.
- Then the tangents are drawn at the points of intersection of the diagonals and the circumference of the circle and the required octagon 1-2-3-4-5-6-8 is obtained.

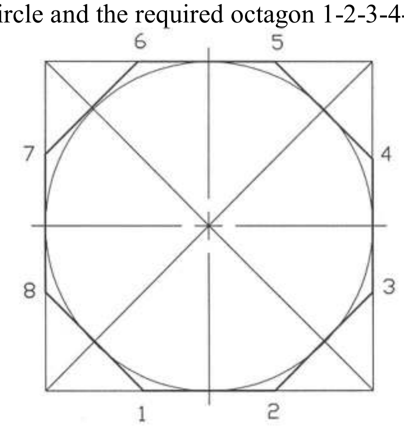


Fig. S1.7

Solution of P1.8

Procedure:

The procedure for drawing an ellipse by the Concentric Circle method is described below:

- Two circles are drawn with the major and minor diameters of 60 mm and 40 mm respectively (Fig. S1.8).
- Then the circle is divided into a number of equal radial divisions. They intersect both the inner and the outer circles of minor and major diameters respectively.

- Now from the intersecting points of the outer circle and the inner circle vertical and horizontal lines are drawn respectively. The intersecting points of the vertical and the horizontal lines give the locus of the ellipse. They are joined to obtain the required ellipse.

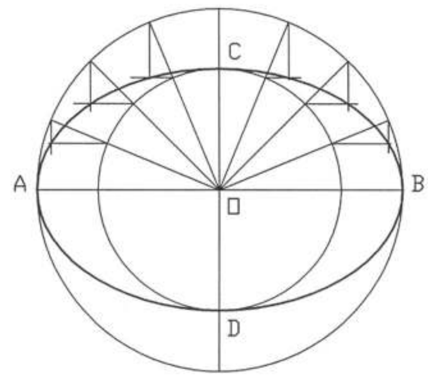


Fig. S1.8

Solution of P1.9

Procedure:

The procedure of drawing an ellipse by the Parallelogram method is given below:

- AB and CD are the major and minor diameters of the ellipse. A parallelogram PQRS is drawn with the major and minor diameters (Fig. S1.9).
- Now AB is divided into a number of equal parts and then PQ and RS are also divided into the same number of equal parts. They are marked with 1, 2 and 3.
- Then through the points 1, 2 and 3, lines are drawn from C and D. The intersecting points of the lines are the locus of the ellipse. Now the required ellipse is completed.

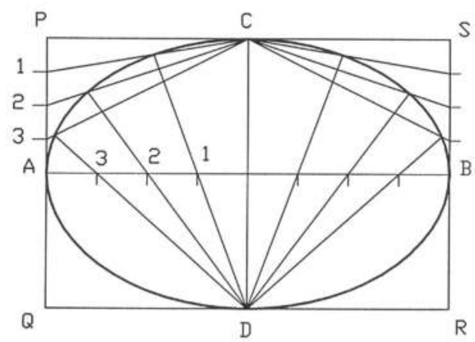


Fig. S1.9

Solution of P1.10

Procedure:

In order to draw an ellipse by the Four-Centered Approximate method, the steps as mentioned below are to be followed:

- PQ and RS are the major and the minor diameters of the ellipse. They intersect at the point O (Fig. S1.10).
- An arc is drawn with center O and radius OP. It intersects the extended minor axis at the point A.

- Now PR is joined and an arc of radius RA is drawn taking R as the center. It intersects PR at the point B.
- Then PB is bisected and a perpendicular is drawn through the bisecting point C. The perpendicular intersects the extended minor axis at the point D. CD intersects PO at the point E.
- $OE^{/}$ is drawn equal to OE and $OD^{/}$ is drawn equal to OD. $DE^{/}$, $D^{/}E$ and $D^{/}E^{/}$ are added and extended.
- Taking D and $D^{/}$ as the centers and DR or $D^{/}S$ as the radius two arcs MRN and KSL are drawn respectively. Also taking E and $E^{/}$ as the centers and PE or $QE^{/}$ as the radius another two arcs KPM and NQL are drawn respectively. Thus the required ellipse is completed.

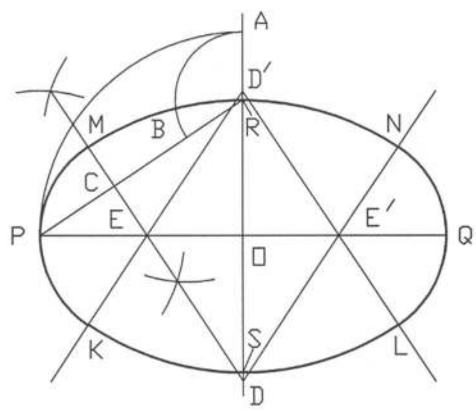


Fig. S1.10

Solution of P1.11

Procedure:

To draw a tangent to an ellipse from an outside point A, the steps as mentioned below are to be followed:

- The focuses of the ellipse are obtained first. Taking radius equal to half of the major axis and with R as the center two arcs are drawn. They intersect the major axis PQ at the points F_1 and F_2, which are the focuses of the ellipse (Fig. S1.11).
- Now with the given outside point A as the center and AF_2 as the radius an arc MF_2N is drawn.
- Then with F_1 as the center and PQ as the radius two arcs are drawn which intersect the arc MF_2N at the points M and N.
- F_1N and F_1M intersect the ellipse at the points T_1 and T_2 respectively. AT_1 and AT_2 are joined and extended to obtain the required tangents.

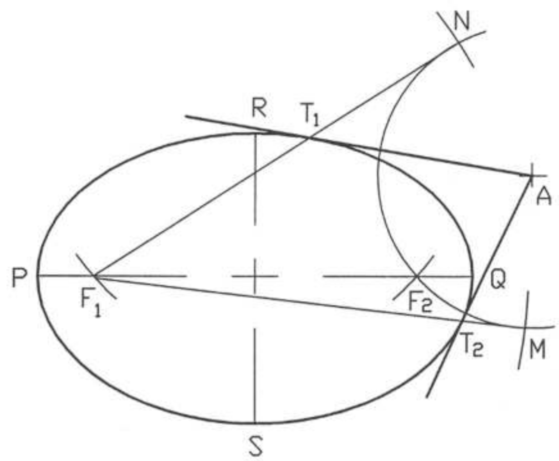


Fig. S1.11

Solution of P1.12

Procedure:
The procedure of drawing a tangent at a given point A on the curve is given below:

- First the focuses F_1 and F_2 of the ellipse as in Prob. 1.11 are obtained (Fig. S1.12).
- Then from the given point A on the curve, two lines AF_1 and AF_2 are drawn and F_2A is extended up to C.
- Now the angle F_1AC is bisected. The bisecting line AB is the required tangent.

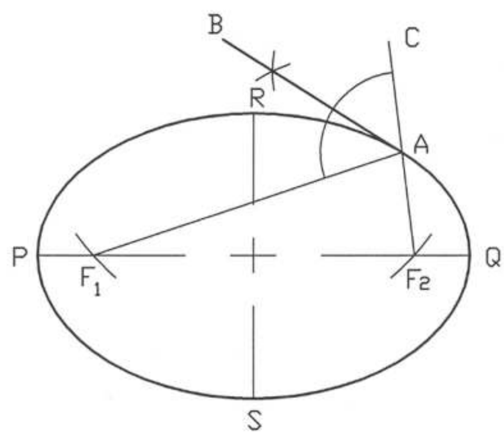


Fig. S1.12

Solution of P1.13

Procedure:

The procedure to draw a parabola by the parallelogram method is given below:

- The rectangle PQRS with sides PQ and QR is drawn first. The axis of the parabola OA is drawn parallel to QR passing through the mid-point of PQ (Fig. S1.13).
- Now QR is divided into a number of equal divisions, which are marked with 1, 2, 3 and 4. QO is divided into the same number of equal divisions and they are also marked with 1, 2, 3 and 4.
- Then the points 1, 2, 3 and 4 on QR are joined with the point O. From the points 1, 2, 3 and 4 on OQ lines are drawn parallel to OA. They intersect with the respective lines from QR.
- Now the curve is drawn through the intersecting points as in the figure to obtain the required parabola.

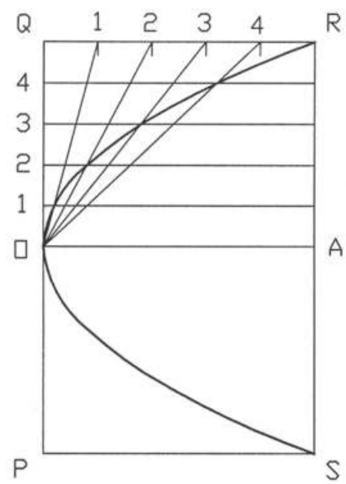


Fig. S1.13

Solution of P1.14

Procedure:

The steps to draw a parabola by the offset method are given as follows:

- First of all a parallelogram PQRS is constructed with sides PQ and QR. The axis OA is drawn parallel to the side QR and passing through the mid-point of PQ (Fig. S1.14).
- Then OQ is divided into a number of five equal parts and they are marked with 1, 2, 3, 4 and 5.
- Now QR is to be divided into $5^2 = 25$ equal parts. The divisions on QR are indicated by 1^2, 2^2, 3^2, 4^2 and 5^2.
- The parallel lines are drawn through the points 1, 2, 3 and 4 from both OQ and QR. The intersecting points of them are the locus of the parabola.
- Finally the intersecting points are joined to obtain the required parabola.

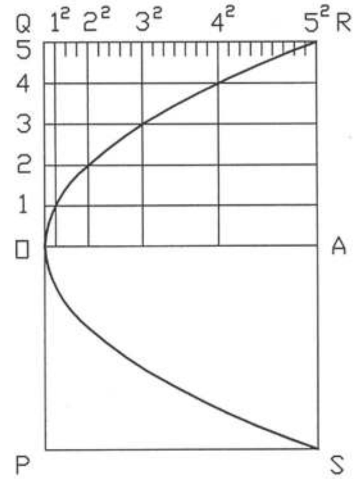


Fig. S1.14

Solution of P1.15

Procedure:

The procedure of drawing an involute profile is given below:

- The base circle of diameter 65 mm is drawn and a point 'a' is taken on its circumference, then a tangent am is drawn through the point 'a' on the base circle (Fig. S1.15).
- Now from the point 'a' the line am and the circumference are divided equally into a number of points b, c, d, e, f and b_1, c_1, d_1, e_1, f_1 respectively.
- Next to find the locus of the involute profile a tangent is drawn through the point f_1 on the base circle and the circle is drawn through the point f with center o. They intersect at the point f_2.
- Similarly the other points b_2, c_2, d_2, e_2 are obtained and a smooth curve is drawn through the points to find the required involute profile.

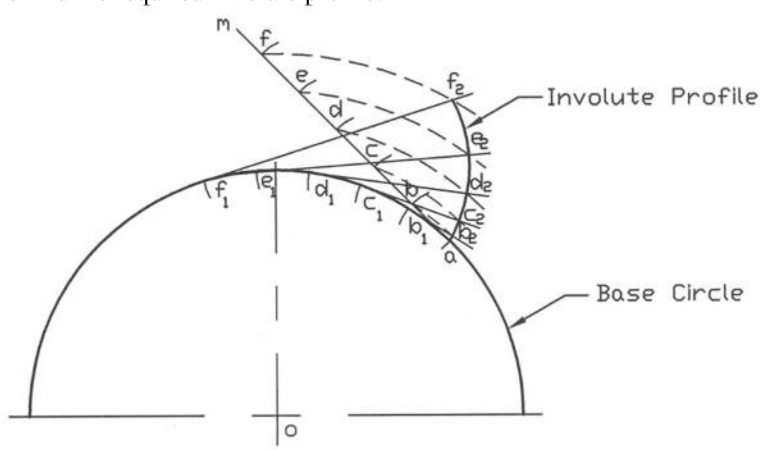


Fig. S1.15

Problems

Prob. 1.16: Describe different types of compasses and mention their applications.

Prob. 1.17: Discuss about the most common types of protractors. Mention when a protractor is used for drawing.

Prob. 1.18: With the help of T-square and Triangles divide 360^0 with a step of 15^0.

Prob. 1.19: Discuss different types of scales and mention their applications.

Prob. 1.20: What is the function of a French curve? Plot a curve with the help of French curve using the following data (values are in mm).

X = 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100
Y = 0.0, 9.0, 14.5, 17.0, 18.8, 21.2, 21.5, 22.2, 22.0, 20.7, 18.3

Prob. 1.21: Draw the following geometric features:

a) A circle of 50 mm diameter showing centerline.
b) Two concentric circles one of 50 mm and the other of 70 mm showing a cutting plane line at the center.
c) A bar of length 250 mm and thickness 10 mm showing a breaking line across its length.
d) A square of 50 mm side with a circle of 30 mm diameter at the middle with hidden line.

Prob. 1.22: Make a title block of your own choice using (a) vertical gothic alphabet (b) ISO lettering A vertical (c) ISO lettering A inclined.

Prob. 1.23: Draw a square and then divide it into eight equal triangular area using T-square and Triangles.

Prob. 1.24: Draw a 75 mm x 45 mm rectangle and divide it into 15 squares each of 15 mm x 15 mm sides; then draw 15 mm diameter circle in each square so that the sides of the square become tangent to the circle.

Prob. 1.25: Draw 10 mm rounds on the four corners of a square of side 30 mm.

Prob. 1.26: Draw an arc of radius 10 mm between two intersecting straight lines with an (a) acute angle of 30^0 and (b) obtuse angle of 120^0.

Prob. 1.27: Draw an arc tangent of radius 10 mm at right angle corner.

Prob. 1.28: Draw an arc tangent of 7 mm radius to two circles each of 30 mm diameters having their center distance at 40 mm.

Prob. 1.29: Draw a triangle inscribing a circle of 60 mm diameter.

Prob. 1.30: Draw a hexagon outscribing a circle of radius 25 mm using Triangles.

Prob. 1.31: Draw two concentric circles of 50 mm and 80 mm respectively. On the circumference of the inner circle draw eight radial slots at equal inter-space each of 5 mm width in the circumferential direction and 7.5 mm length in the radial direction.

CHAPTER 2

DIMENSIONING

2.1 *Introduction*

To manufacture a part, dimensioning plays a significant role. Engineering drawing without dimensioning is meaningless. If a drawing of a part is done and the scale is mentioned, it does not become sufficient for manufacturing. By direct measurement from the drawing the part cannot be produced accurately for many reasons. Whatever may be the scale of the part, the actual size dimensions have to be always mentioned on the part. Dimensions are indicated on the drawings by arrowheads, extension lines, dimension lines, leaders, figures, notes, symbols etc. in order to define the geometric characteristics such as, lengths, diameters, angles, locations etc. The lines used in the dimensions are thin compared to the visible lines. The dimensions must be clear, concise and always allow the single interpretation. Standard rules of dimensioning have to be followed unless it becomes essential.

2.2 *Arrowheads*

The important part of the dimensioning is the arrowhead. The arrowhead may be drawn in accordance with Figure 2.1. Arrowheads are usually drawn freehand. However, all arrowheads have to be identical in shape and size throughout the drawing unless it becomes essential. Sometimes it becomes necessary to shorten them due to space limitation. The length of the arrowhead may vary depending on the size of the drawing. The approximate length of the arrowhead may be 3 mm. However, for the larger drawing it may be a little bit larger in size. The approximate ratio of the length to width of the arrowhead is 3:1 as shown in the figure. The arrowhead must touch the line. It must not be either away from the line or cross the line.

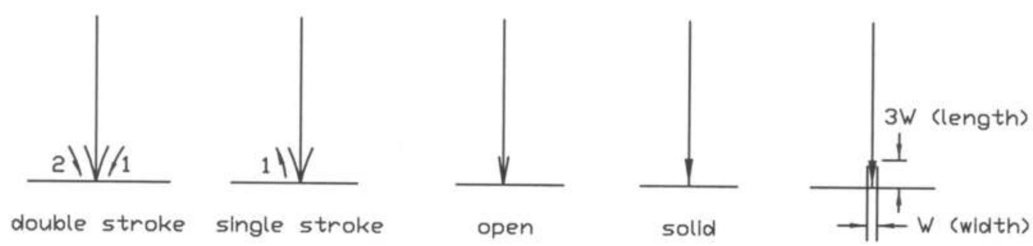


Figure 2.1: Arrowheads

2.3 *Extension Line, Dimension Line and Leaders*

Extension and dimension lines are used to show dimension of a part. Extension lines indicate the point or line on the drawing to which the dimension is applied while dimension lines show the dimensions. On the other hand, leaders are used to present note, dimension, symbol, item number or part number on the drawings. They are thin lines. Extension and dimension lines are introduced in Figure 2.2.

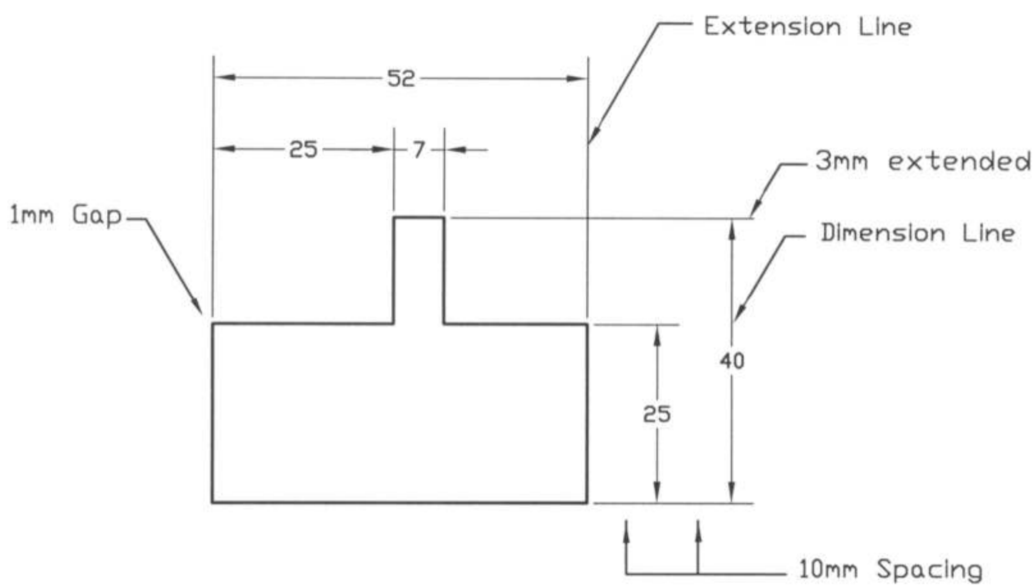


Figure 2.2: Extension and Dimension Lines

The following rules are to be remembered while inserting dimensions on a part.

2.3.1 ***Extension Line***

(1) A gap of 1 mm has to be kept in between the extension line and the visible line.
(2) An extension line should extend about 3 mm from the outermost dimension line.
(3) Extension lines may cross each other without a break. They may also cross the visible lines without a break. If the extension lines cross arrowheads or dimension lines close to arrowheads, a break may be permissible.
(4) Centerlines may be used as extension lines. However, it should not be broken while passing the circle.
(5) Extension lines are usually drawn perpendicular to dimension lines, where there is overcrowding extension lines may be drawn at an oblique angle.

2.3.2 ***Dimension Line***

(1) Dimension line should be approximately 10 mm away from the visible line. The spacing between the consecutive parallel dimension lines may also be considered as 10 mm.
(2) Dimension lines are broken near the middle to allow space for dimension.
(3) Dimension lines should usually be placed outside the view unless it becomes necessary.
(4) When the space in between the extension lines is too small to insert dimension line completed with arrowhead, it may be provided outside the extension line.
(5) To accomplish it the shortest dimension line should be placed nearest to the outline of the part.
(6) Centerlines should never be used as dimension lines.
(7) In case of extra long dimension line or crowding of dimensions, dimension lines inside the view may be permissible.
(8) When several dimension lines are required side by side along a line, it is recommended to stagger them.

(9) The dimension line for a part, which cannot be presented completely on the drawings due to its large distance, the free end is terminated in a double arrowhead pointing in the direction in which it could be completed.

2.3.3 ***Leaders***

(1) A leader should always be inclined at an angle of 60^0 preferably and 45^0 occasionally (not horizontal or vertical) with a 3 mm horizontal bar.
(2) A leader is either terminated by an arrowhead on a line or a small dot of about 1.5 mm diameter within the outline of the part.
(3) Leaders should not be drawn bent except special circumstances.
(4) Leaders should not cross each other, however, they may be drawn parallel to each other.
(5) To direct a circle or an arc the leader should be so drawn, if it is imagined to extend it must pass through the center of the circle or the arc.
(6) All notes and dimensions in a leader have to be provided in the horizontal direction.

In Figures 2.3 to 2.8, some uses of extension lines have been presented.

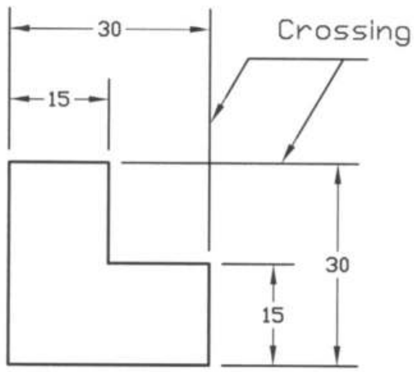


Figure 2.3: Crossing of Extension Lines

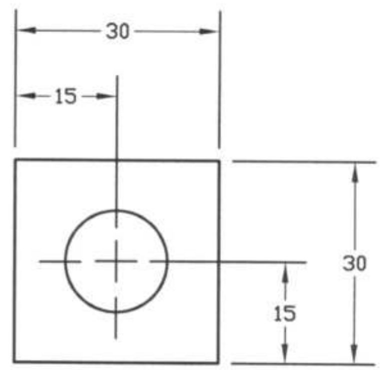


Figure 2.4: Center Line as Extension Line

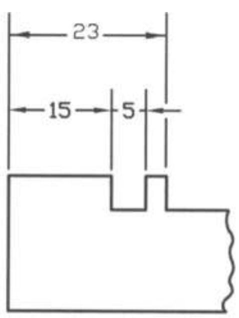


Figure 2.5: Break in Extension Line

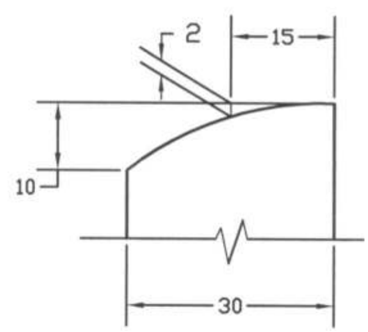


Figure 2.6: Oblique Extension Line

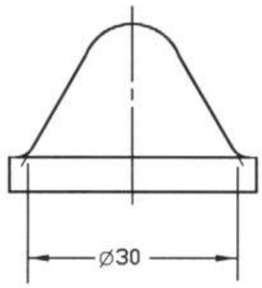


Figure 2.7: Extension Line From Point

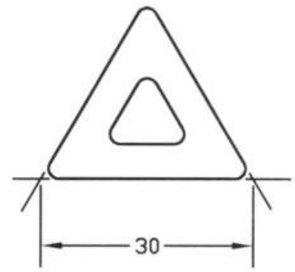


Figure 2.8: Extension Line From Point

The crossing of extension lines is shown in Figure 2.3 while in Figure 2.4 it is shown that the centerline is used as the extension line for dimensioning. In Figure 2.5, the break in extension line is made. The use of oblique extension line is presented in Figure 2.6. Extension lines have been shown from the points in Figures 2.7 and 2.8.

Some uses of dimension lines have been shown in Figures 2.9 to 2.13.

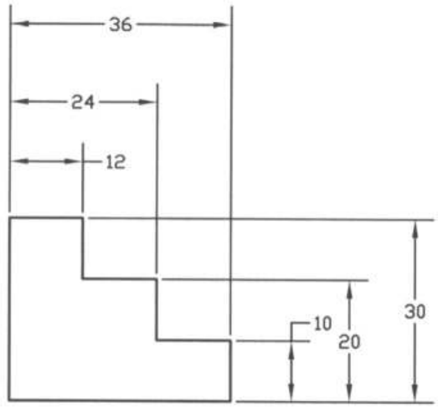


Figure 2.9: Staggered Dimensioning

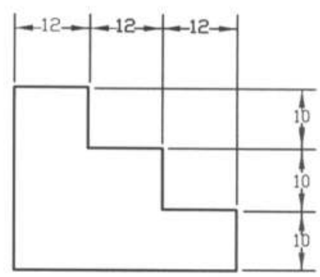


Figure 2.10: In-Line Dimensioning

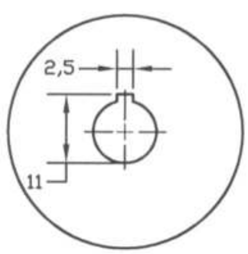


Figure 2.11: Dimensioning Inside View
Avoiding Long Extension Line

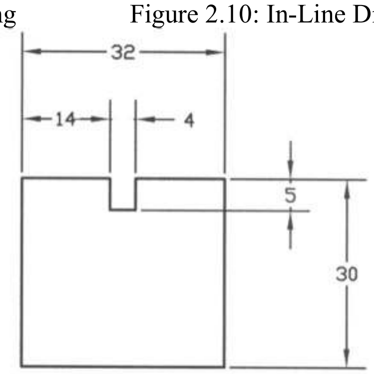


Figure 2.12: Dimensioning Outside
With Extension Line
Crossing Visible Line

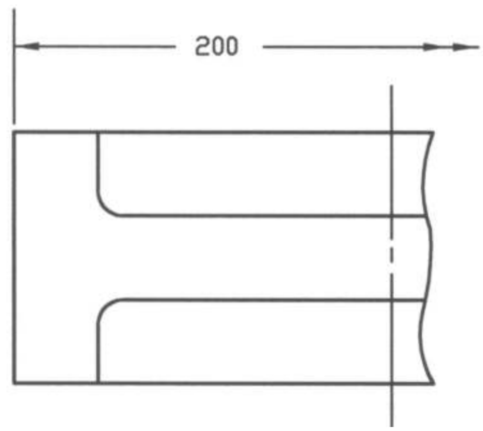


Figure 2.13: Dimensioning For Part With Large Distance

The uses of staggered dimensioning and in-line dimensioning are presented in Figures 2.9 and 2.10 respectively. Staggered dimensioning is considered as good and in-line dimensioning is considered as weak. To avoid long extension line often the dimensioning is done inside the view, which is shown in Figure 2.11. While in Figure 2.12 the dimensioning is shown outside the view where the extension line crosses the visible line. Dimensioning for part with large distance is shown in Figure 2.13.

Some usage of leaders has been given in Figure 2.14.

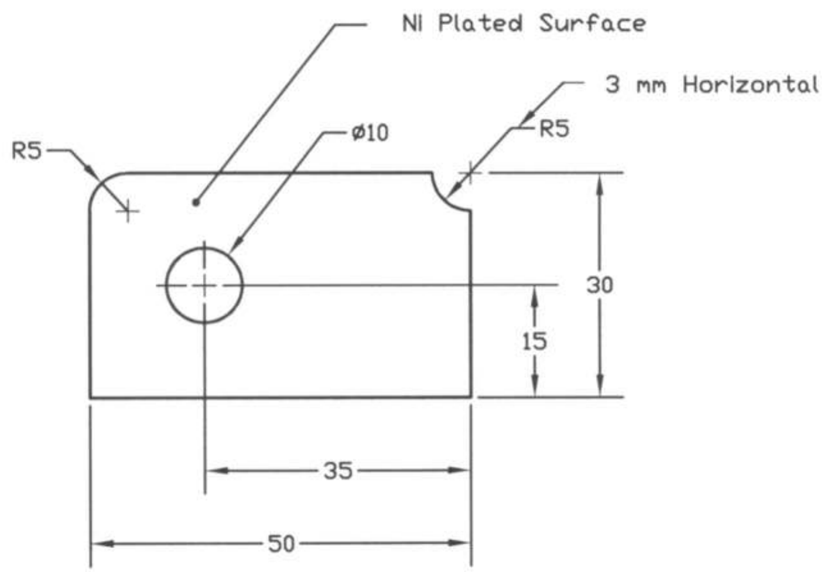


Figure 2.14: Usage of Leaders

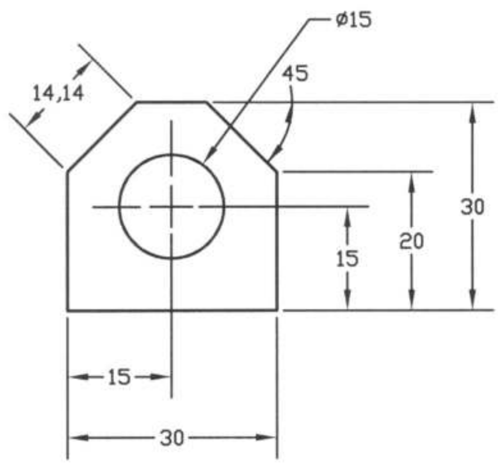
Some other rules of dimensioning are provided below, which have to be remembered while dimensioning.

(1) Dimensioning should be given between the views in general.

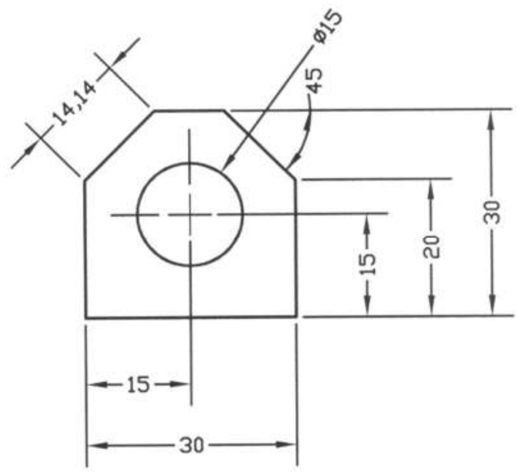
(2) Dimensioning should not be duplicated in other views.
(3) Dimensioning should be made in such a way so that no subtraction or addition is required to define or locate a feature.
(4) Dimensioning may be inserted on relatively larger view in order to make it clear.
(5) One system of dimensions either unidirectional or aligned has to be used throughout the drawing.
(6) Dimensioning to hidden lines should be avoided in general. To accomplish that a sectional view or broken-out section may be used to place dimensions.
(7) Dimensioning should be made on the view, which represents the shape of the part best.

2.4 *Direction of Dimensions*

In Figure 2.15 the direction of dimensions has been shown. Direction of dimensions is done in either of two systems: unidirectional or aligned. The unidirectional system is often called 'Horizontal System'. In the unidirectional system all the dimensions (figures) are oriented to be read from the bottom of the drawing while in the aligned system the dimensions (figures) are oriented to be read from the bottom or right side of the drawing. The unidirectional system is preferred to aligned system.



(a) Unidirectional

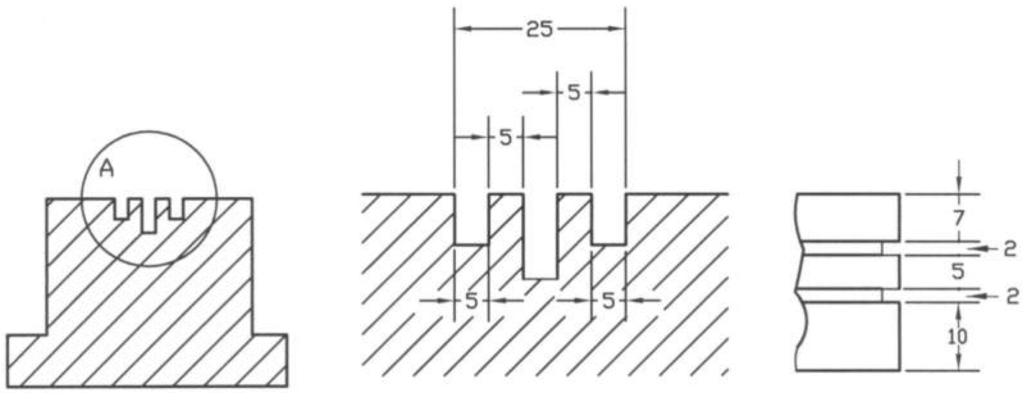


(b) Aligned

Figure 2.15: Direction of Dimensions

2.5 *Dimensioning in Limited Space*

To show dimensions in a limited space, sometimes it becomes necessary to enlarge the portion for clear dimensioning. Such dimensioning is shown in Figures 2.16a and 2.16b. In Figure 2.16c, the use of smaller circular dot in place of arrowhead has been shown due to space limitation.



(a) Portion to be Enlarged (b) Enlarged View at 'A' (c) Use of Small Circular Dot

Figure 2.16: Dimensioning in Limited Space

2.6 *Dimensioning of Angles*

Dimensioning of angles has been presented in Figure 2.17. Here the dimension line is the arc whose center is at the intersecting point of the two sides of the angle. The angle is read horizontally. But in the aligned system for the large arc, it is made aligned with respect to the dimension arc. Dimensioning of angle as represented in Figure 2.17e is preferable to that in Figure 2.17f.

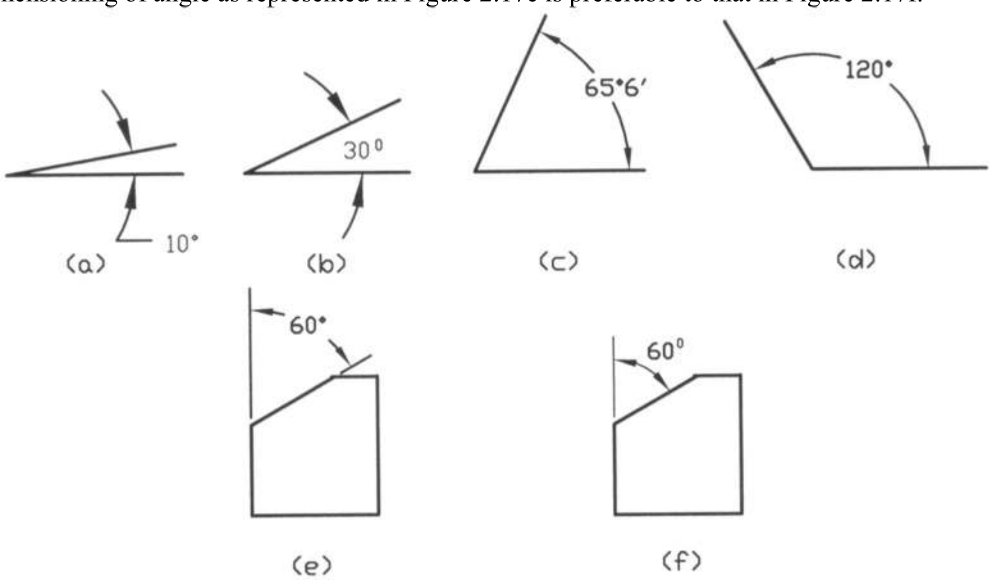
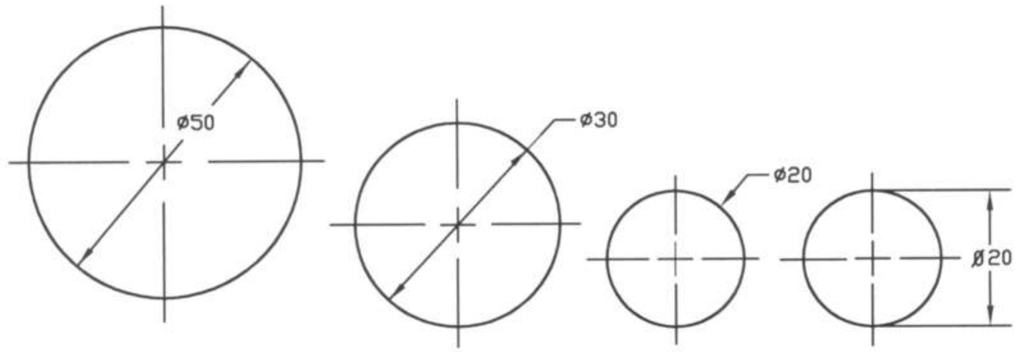


Figure 2.17: Dimensioning of Angles

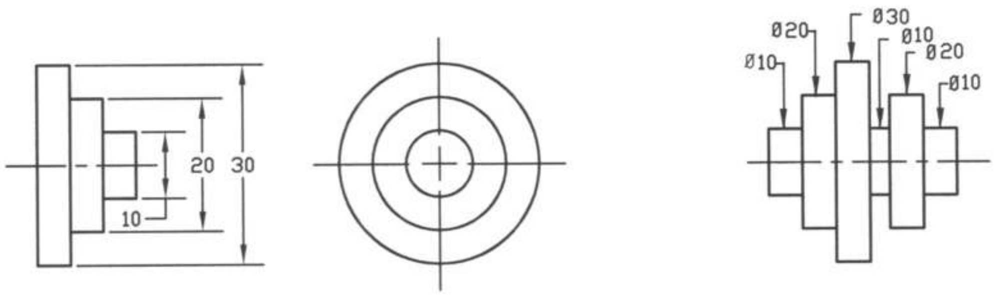
2.7 *Dimensioning in Circular Features*

Dimensioning in the circular features for the diameters has been shown in Figures 2.18 and 2.19 and that for the radii is shown in Figure 2.20. In the two views drawing, it is recommended that dimensioning should be made on the longitudinal view (Figure 2.19). Each center has been located by a cross mark as shown in the figure.



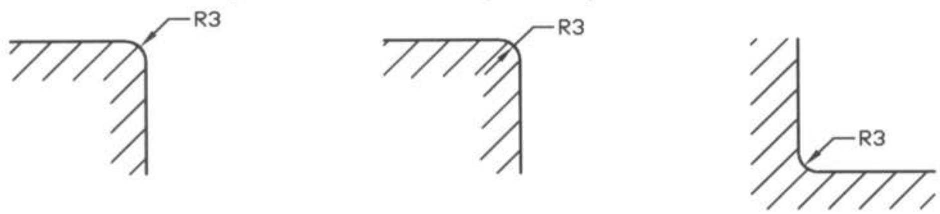
(a) (b) (c) (d)

Figure 2.18: Dimensioning in Single Diameter



(e) (f)

Figure 2.19: Dimensioning in Multiple Diameters



(a) (b) (c)

(d) (e)

Figure 2.20: Dimensioning in Radii

2.8 ***Dimensioning in Cylindrical Holes***

Dimensioning on various types of cylindrical holes has been shown in Figures 2.21 to 2.24.

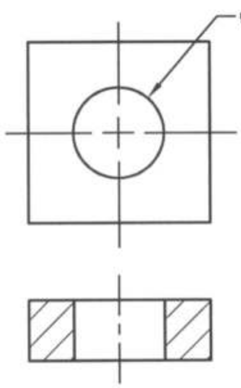


Figure 2.21: Through Hole

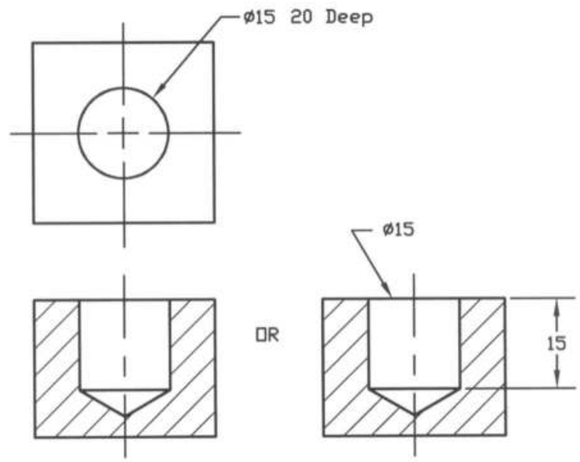


Figure 2.22: Blind Hole

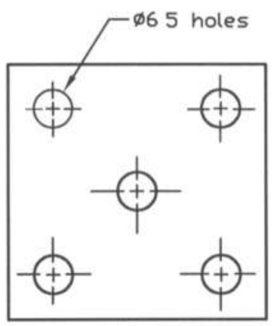


Figure 2.23: Group of Holes

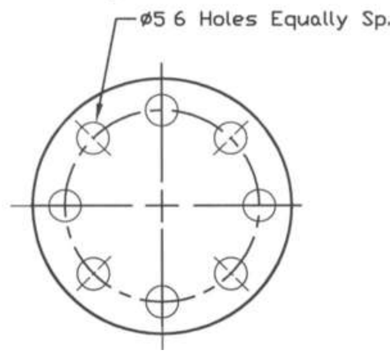


Figure 2.24: Group of Holes

2.9 ***Dimensioning in Slotted Holes***

In Figure 2.25, dimensioning of a slotted hole has been shown.

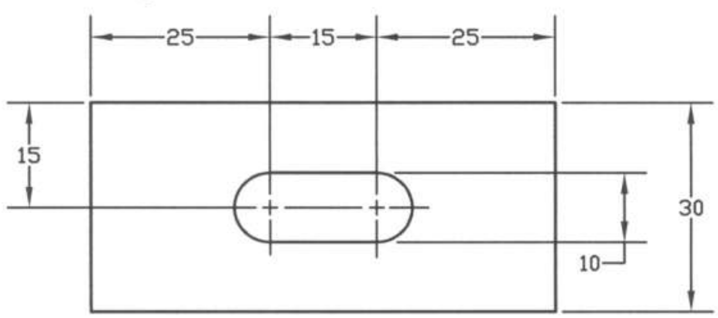


Figure 2.25: Dimensioning in Slotted Hole

2.10 ***Oblique Dimensioning***

In Figure 2.26, dimensioning in oblique feature has been given.

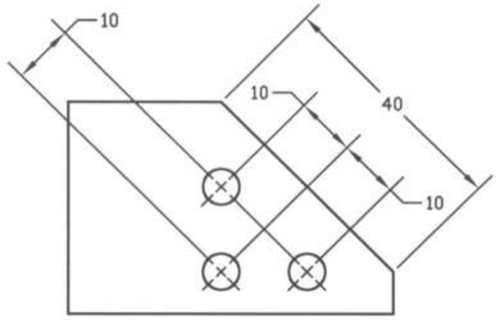


Figure 2.26: Dimensioning in Oblique Feature

2.11 ***Dimensioning in Countersink, Counter bore and Spot face***

Dimensioning for the countersink, counter bore and spot face have been shown in Figures 2.27, 2.28 and 2.29 respectively.

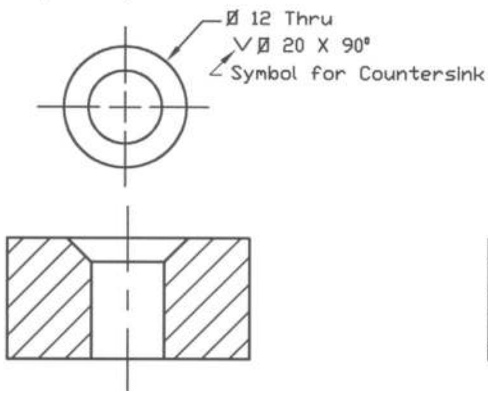


Figure 2.27: Dimensioning in Countersink

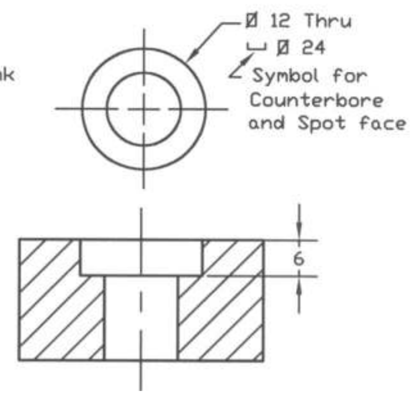


Figure 2.28: Dimensioning in Counter bore

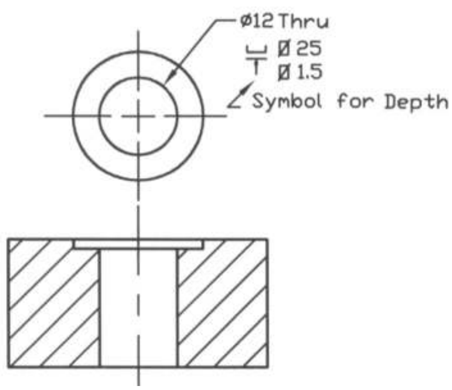


Figure 2.29: Dimensioning in Spot face

2.12 ***Dimensioning in Chamfers and Undercut***

In Figures 2.30 and 2.31, the methods of dimensioning in external and internal chamfers are shown respectively while in Figure 2.32 dimensioning in undercut is shown.

Figure 2.30: Dimensioning in External Chamfers

Figure 2.31: Dimensioning in Internal Chamfers

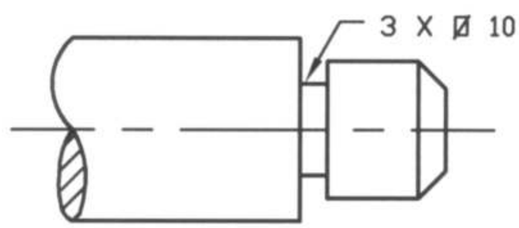
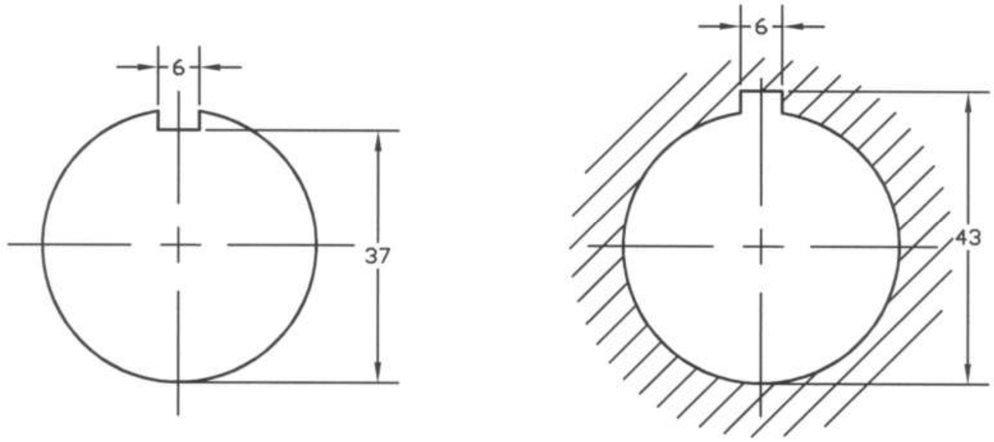


Figure 2.32: Dimensioning in Undercut

2.13 ***Dimensioning in Keyways***

In Figure 2.33, dimensioning for the keyways is shown.

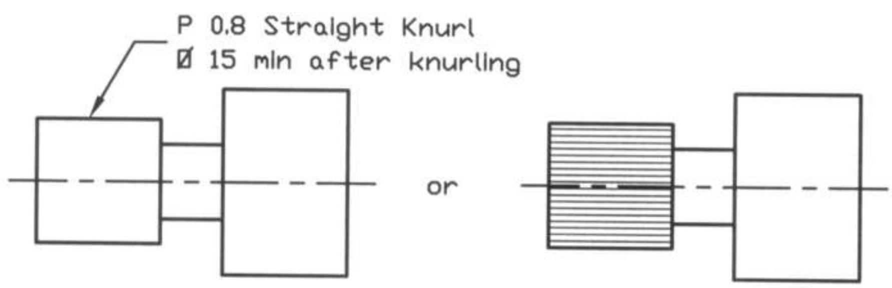


(a) Shaft (b) Hub

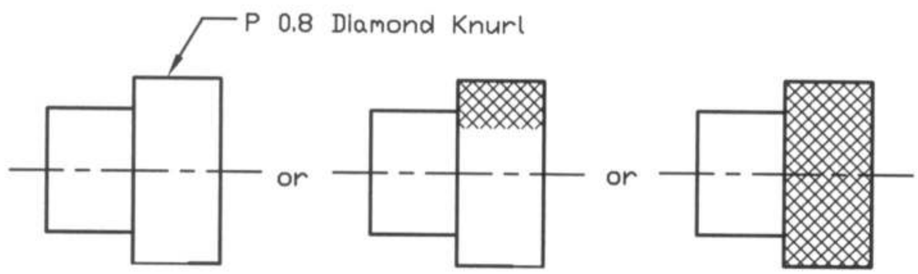
Figure 2.33: Dimensioning in Keyways

2.14 ***Dimensioning in Knurls***

It has been shown in Figure 2.34, how to give dimension in knurls. Knurling is specified in terms of type, pitch and diameter before and after knurling. The letter P indicates the pitch diameter. In Figure 2.34a the diameter after knurling has been mentioned as an example.



(a) Straight Knurl



(b) Diamond Knurl

Figure 2.34: Dimensioning in Knurls

2.15 ***Machining Finish***

It often requires machining the surface of a part, which is manufactured by casting, molding or forging. In the drawing it has to be shown by the symbols where machine finish is necessary.

2.15.1 ***Machining Symbol***

To indicate the surface finish by machining, a machining symbol (Finish Marks) is used on the surface of the machine part drawing. In Figures 2.35a and 2.35b machining symbols with the standard sizes are shown while in Figures 2.35c and 2.35d, the uses of machining symbols are shown. Either of the machining symbols as shown in the figure may be used. However, the machining symbol as shown in Figure 2.35c is used mostly now a day. It can be noted from here that the degrees of roughness of the surface is not mentioned here. The machining symbol always appears on the side view on the surface or it may be used on the extension line where the space is limited. When all the surfaces of a machine part is required to be machine finished, then instead of using machining symbol, a note is provided as, "Finish all over".

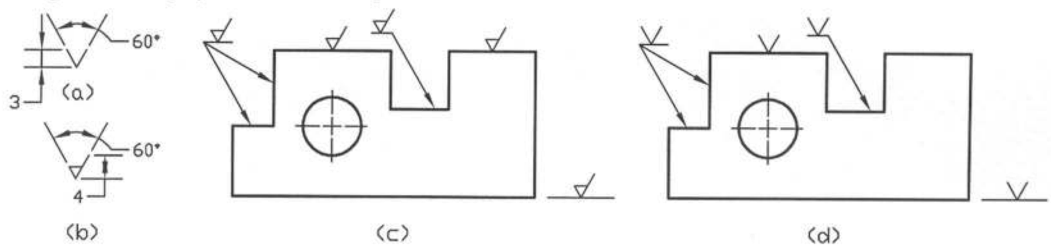


Figure 2.35: Use of Finish Marks

When notes are provided for dimensions in case of drilling, boring and reaming etc., no machining symbol is given. For part manufactured from the cold work, no machining symbol is provided.

2.15.2 ***Surface Texture Symbol***

In Figure 2.36a the surface texture characteristics have been illustrated. **Surface roughness** consists of finer irregularities in the surface texture (in the form of small peaks and valleys), which results from the inherent action in the production process (produced by tool cutting edges and feed), though the surface apparently appears to be smooth in the naked eye. In the production process a regular pattern of tool marks is produced on the surface, which is called the **lay direction** of the surface as shown in Figure 2.36a. **Roughness height** (R_t) is the peak to valley distance while **roughness width** is the maximum permissible spacing between the repetitive units of the surface pattern (distance between the two adjoining waves), which is measured in millimeters. **Waviness** indicates the larger undulation of the surface texture as shown in Figure 2.36a and is measured in millimeters.

Roughness average value (R_a) is defined as the average value of the deviation (both above and below) of the surface measured in a normal direction from the mean line in a profile over a sampling length called **roughness width cutoff** (l). Roughness average value is measured in micrometer (μm) while roughness width cutoff is measured in millimeters. The higher will be the roughness average value (R_a) the higher will be the roughness of the surface or vice versa. The mean line is so chosen that the total area above and that below will be equal i.e. ($a_1 + a_2 + a_3$) equals ($b_1 + b_2 + b_3$) as shown in Figure 2.36b. Mathematically R_a is determined from,

$$R_a = \frac{(a_1 + a_2 + a_3) + (b_1 + b_2 + b_3)}{l}$$

For a pure triangular profile $R_a = R_t/4$. In Figure 2.36c the size of surface texture symbol as provided by ANSI Y14.36 is shown. This standard also provides the preferred roughness average values (R_a) as shown in Table 2.1. The recommended values are usually used. The roughness grade numbers corresponding to the recommended values are also provided in the table. The symbols in Figures 2.36d, 2.36e and 2.36f respectively indicate that the metal removal is optional, obligatory and prohibited.

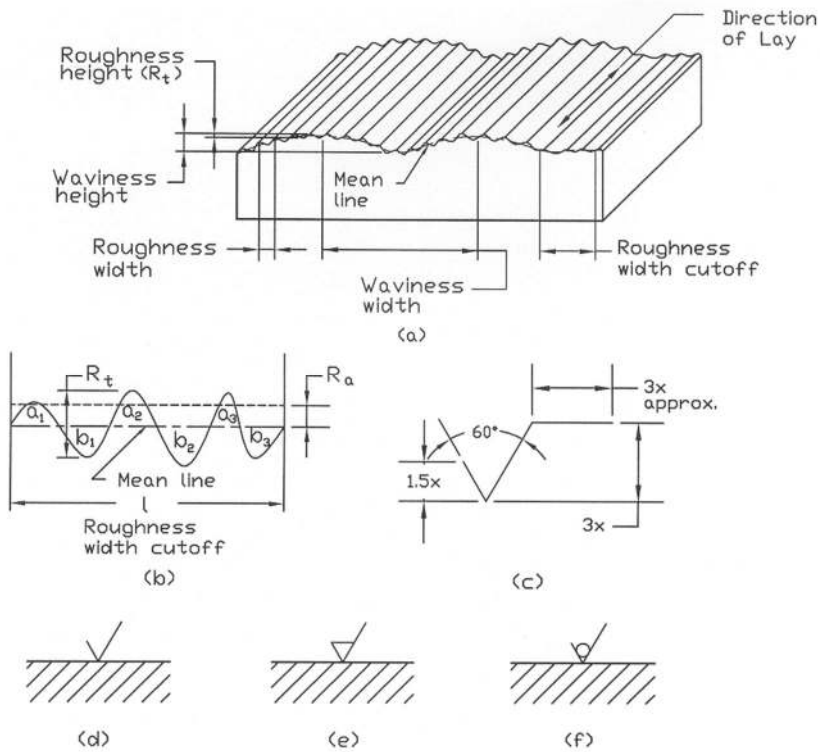


Figure 2.36: Surface Texture Characteristics and Symbol

Table 2.1: Preferred Roughness Average Values, R_a (μm)

0.012	0.125	**0.40** (N5)	1.25	4.0	**12.5** (N10)
0.025 (N1)	0.15	0.50	**1.60** (N7)	5.0	15
0.050 (N2)	**0.20** (N4)	0.63	2.0	**6.3** (N9)	20
0.075	0.25	**0.80** (N6)	2.5	8.0	**25** (N11)
0.10 (N3)	0.32	1.00	**3.2** (N8)	10.0	**50** (N12)

Bold face numbers indicate recommended values.
The numbers within the parenthesis () indicate roughness grade numbers.

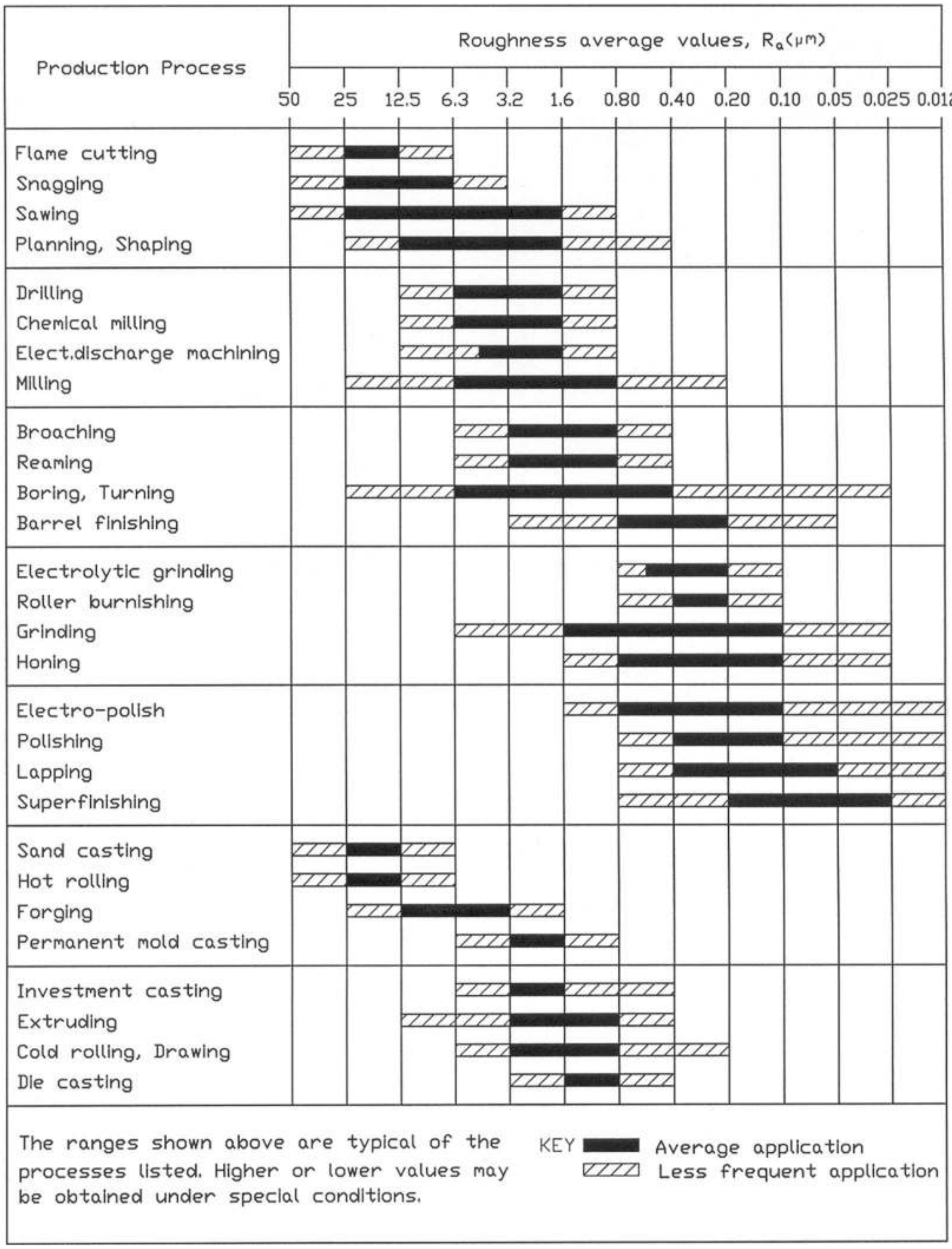


Figure 2.37: Range of Roughness Average Values (R_a) in μm

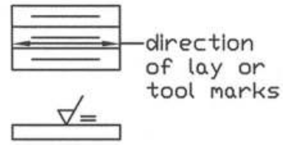
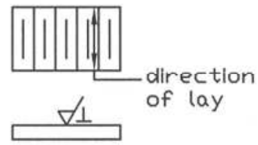
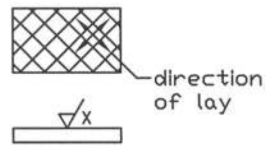
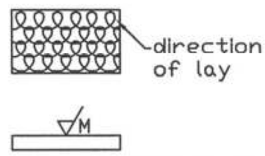
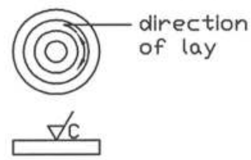
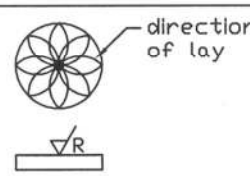
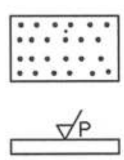
Symbol	Description	Example
=	Lay is parallel to the plane of projection of the view in which the symbol is used.	 direction of lay or tool marks
⊥	Lay is perpendicular to the plane of projection of the view in which the symbol is used.	 direction of lay
X	Lay is crossed in two oblique directions relative to the plane of projection of the view in which the symbol is used.	 direction of lay
M	Lay is multi-directional	 direction of lay
C	Lay is approximately circular relative to the center of the surface to which the symbol is applied.	 direction of lay
R	Lay is approximately radial relative to the center of the surface to which the symbol is applied.	 direction of lay
P	Lay is particulate, non-directional or protuberant	

Figure 2.38: Lay Symbols

The range of roughness average values (R_a) for various production processes are shown in Figure 2.37 while the lay symbols for the common surface patterns are specified in Figure 2.38. These lay symbols are according to ISO 1302: 1992 or ANSI Y14.36 – 1978. The symbol as shown in Figure 2.39a indicates that maximum and minimum roughness average values are respectively 1.6μm and 0.8μm. While the symbol in Figure 2.39b represents that the roughness average value is 3.2μm and any value lower than this is acceptable. In Figure 2.39c, the roughness height is 1.6μm, roughness width is 0.13mm, waviness height is 0.05mm and waviness width is 50mm etc. are shown as an example to specify the roughness characteristics in detail. When necessary the detail roughness characteristics may be used. In Figure 2.39d the use of the surface texture symbol is shown. When the required surface texture is to be produced by a particular method, that method can be indicated in words over a line connected to the longer arm of the symbols as shown in Figures 2.39e and 2.39f.

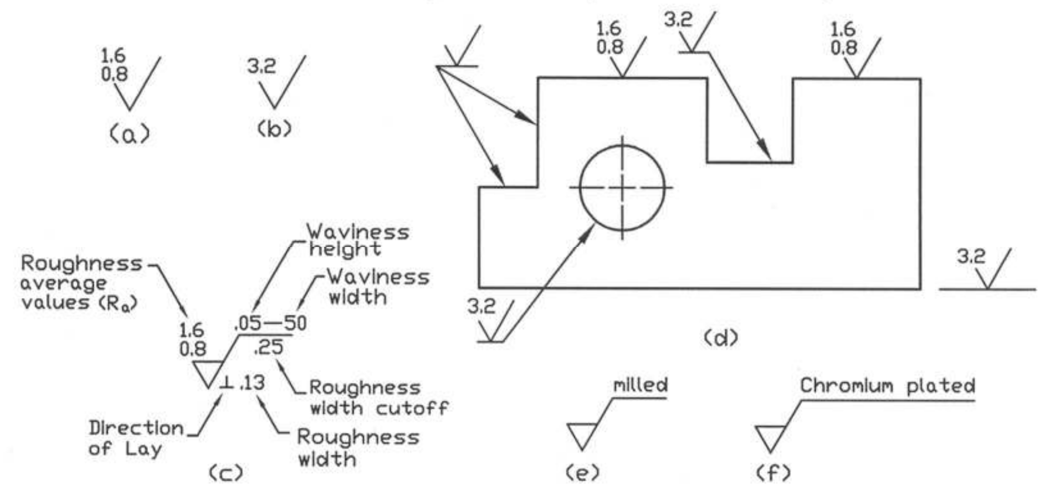


Figure 2.39: Use of Surface Texture Symbol

2.16 ***Limits and Fits***

The terms of limits and fits for the mating parts e.g. hole and shaft, are illustrated in Figure 2.40. The hole and shaft can be applied for any part having two parallel faces such as, slot and key respectively. The various terms for the limits and fits are defined as follows:

Basic size: It is the size of a part to which all the limits of variation are applied to obtain the final dimension of the mating parts.

Actual size: It is the actual measured dimension of the part.

Tolerance: It is the difference between the maximum and the minimum limits of size for a hole or shaft.

Deviation: Deviation is the difference between the basic size and the actual size.

Upper deviation: It is the difference between the maximum limit and the basic size.

Lower deviation: It is the difference between the minimum limit and the basic size.

Fundamental deviation: It is either the upper or lower deviation, which is closest to the basic size.

Fits: The degree of tightness or looseness between the two mating parts is called the fit of the parts. They are of three types such as, clearance fit, interference fit and transition fit. Different types of fits are shown in Figure 2.41.

Clearance: It is the amount by which the size of the actual shaft is less than the size of the mating hole in the assembly.
Interference: It is the amount by which the size of the actual shaft is larger than the size of the mating hole in the assembly.
Clearance fit: In clearance fit the size limits of the mating parts are so selected that clearance between them always occurs.
Interference fit: In interference fit the size limits of the mating parts are so selected that interference between them always occurs.
Transition fit: In the transition fit the size limits of the two mating parts are so selected that either clearance or interference between them may occur.
Basis of limits systems are of two types: Hole basis system and shaft basis system. Basis of Limit systems are shown in Figure 2.42.
Hole basis system: In the hole basis system, the system of fits corresponds to the basic hole size i.e. the hole is kept as the constant member.
Shaft basis system: In the shaft basis system, the system of fits corresponds to the basic shaft size i.e. the shaft is kept as the constant member.

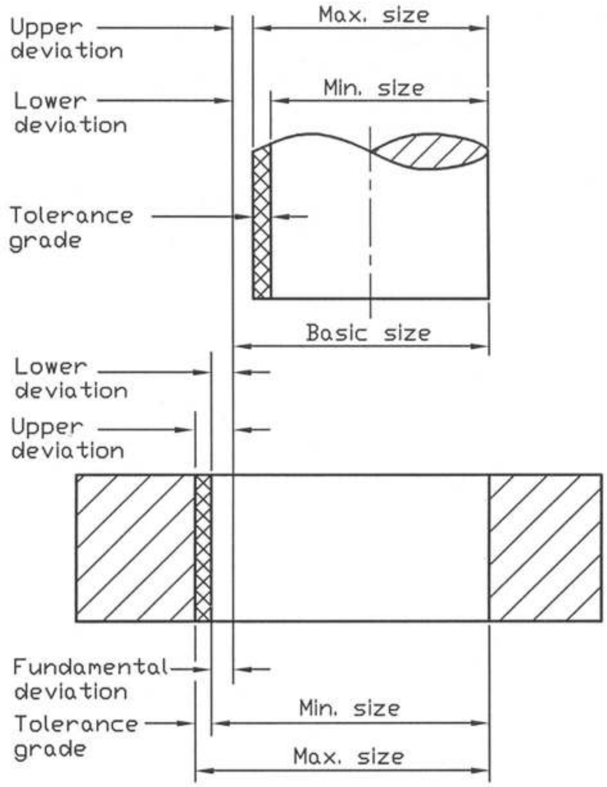
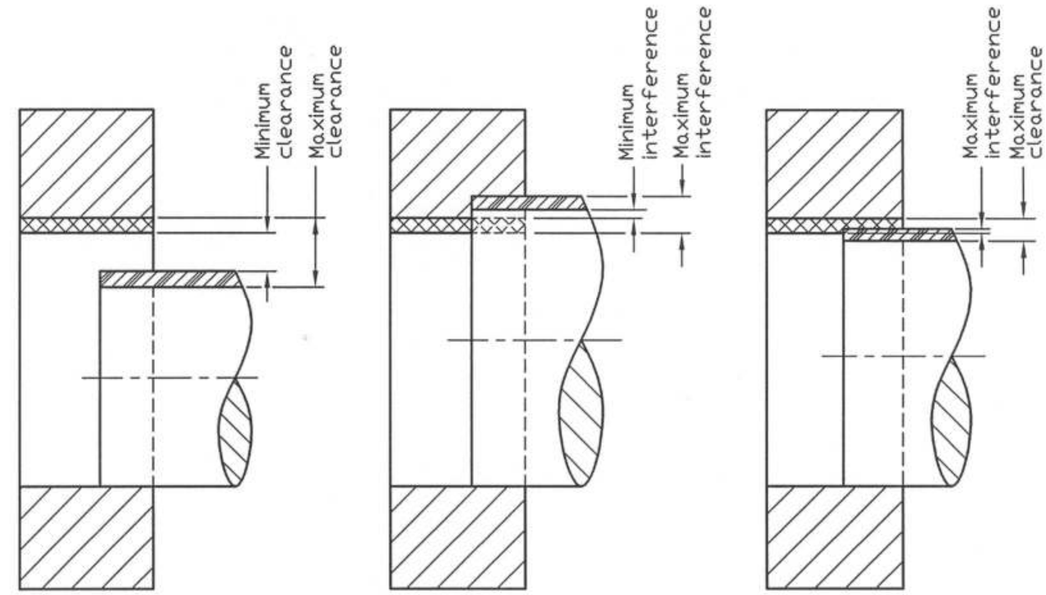
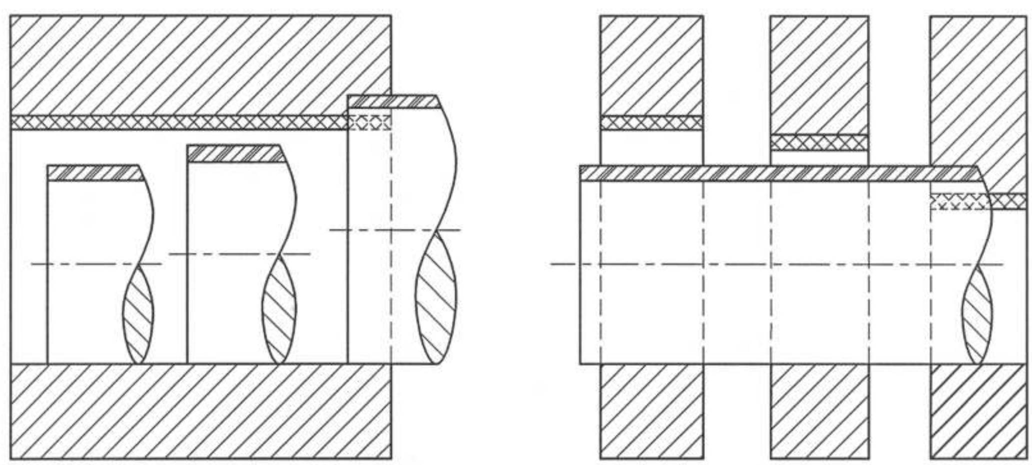


Figure 2.40: Terms of Limits and Fits



Clearance Fits | Interference Fits | Transition Fits

Figure 2.41: Types of Fits



Hole Basis System | Shaft Basis System

Figure 7.42: Basis of Limit Systems

The variation in the size of the part (hole or shaft) is expressed by **IT** numbers, where **IT** indicates the international tolerance grade, which varies with the basic size and the type of fits. In the hole basis system when a fit is designated by a symbol **H8/f7**, the capital letter **H** represents the fundamental deviation and the numeral **8** indicates the tolerance grade of **IT8** for the hole. On the other hand the lower case letter **f** represents the fundamental deviation and the numeral **7** indicates the tolerance grade of **IT7** for the shaft. In the hole basis system, only the letter **H** is used for the hole and the letters **c, d, f, g, h, k, n, p, s** and **u** are usually used for the shaft to represent fundamental deviations. In the shaft basis system the capital letters **C, D, F, G, H, K, N, P, S** and **U** are usually used for the hole and the lower case letter **h** is used for the shaft to represent fundamental deviations. Preferred hole basis fits and shaft basis fits are provided in **Appendices 23** and **24** respectively in the tabulated form according to ANSI B4.2 – 1978 (R 1984) while identical fits for the mating parts are also introduced by ISO 286: 1988. Thus one can use the values directly from the table.

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.***

Prob. 2.1 –2.12: Show dimensions on the views given below in Fig. P2.1 – P2.12 (The scale of each view is half size).

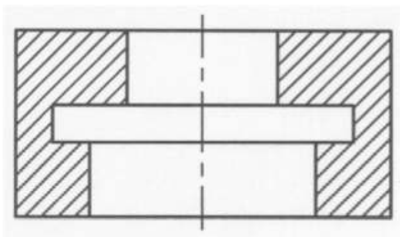


Fig. P2.1

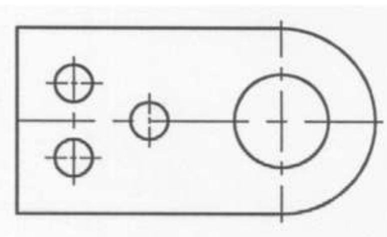


Fig. P2.2

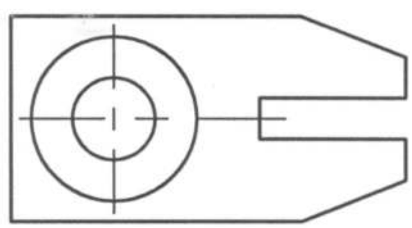


Fig. P2.3

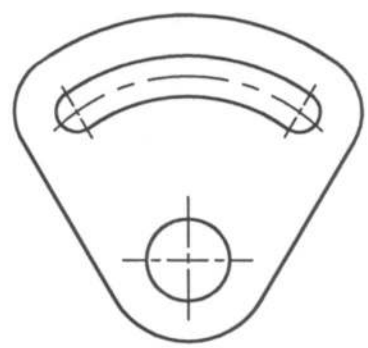


Fig. P2.4

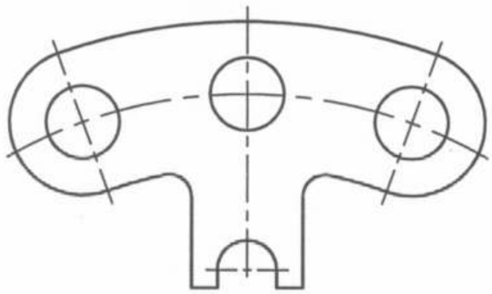


Fig. P2.5

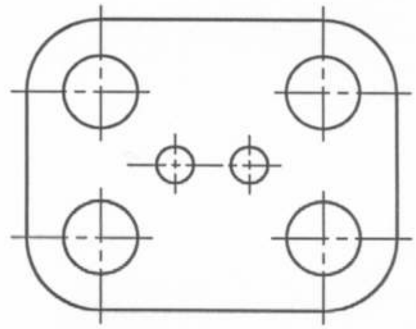


Fig. P2.6

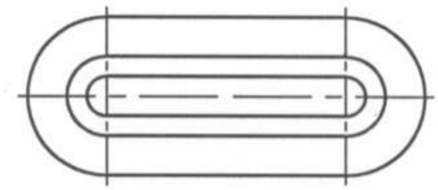


Fig. P2.7

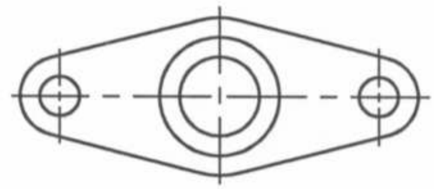


Fig. P2.8

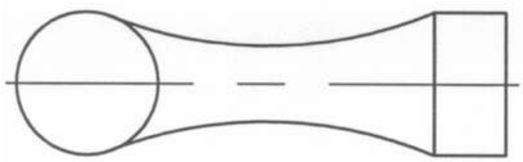


Fig. P2.9

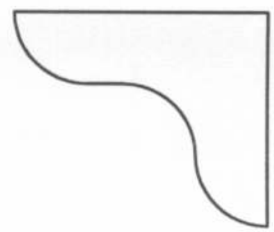


Fig. P2.10

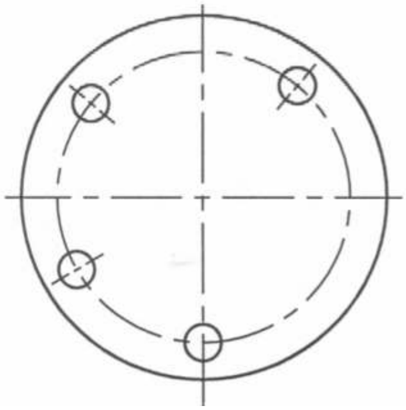


Fig. P2.11

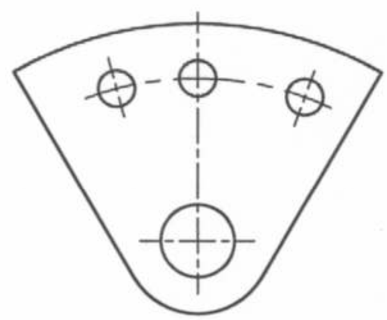
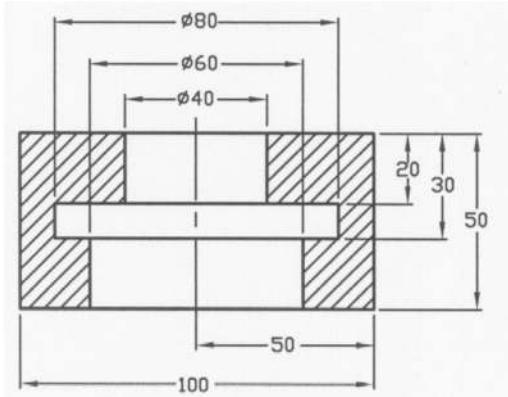
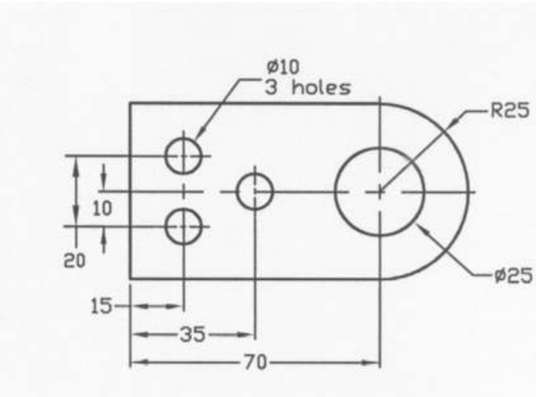


Fig. P2.12

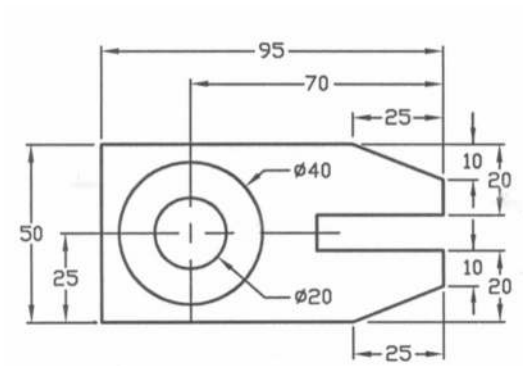
Solutions for Example Problems



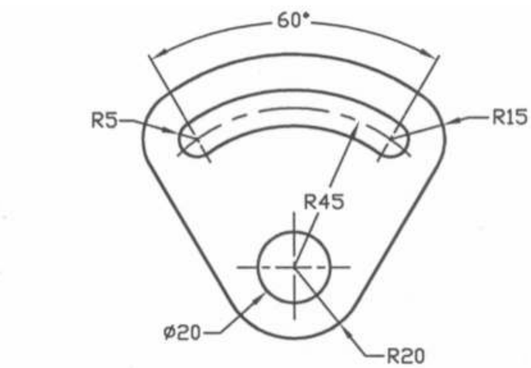
Solution of P2.1



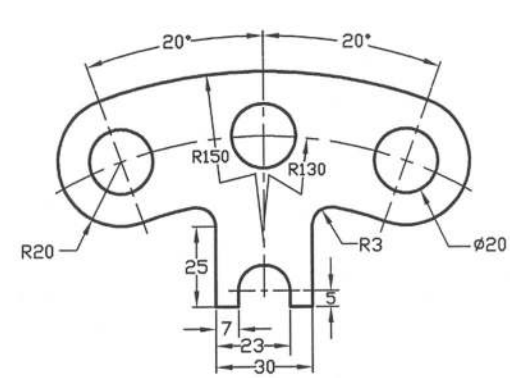
Solution of P2.2



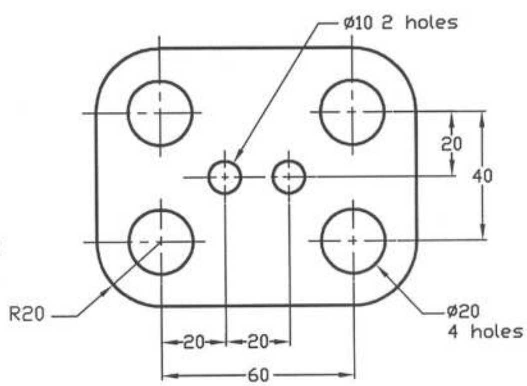
Solution of P2.3



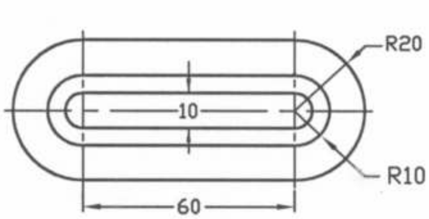
Solution of P2.4



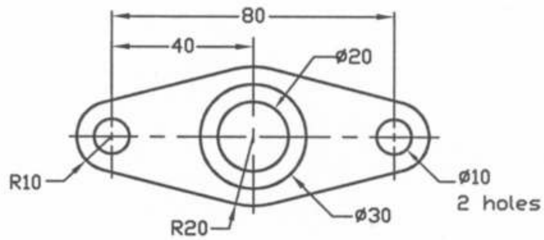
Solution of P2.5



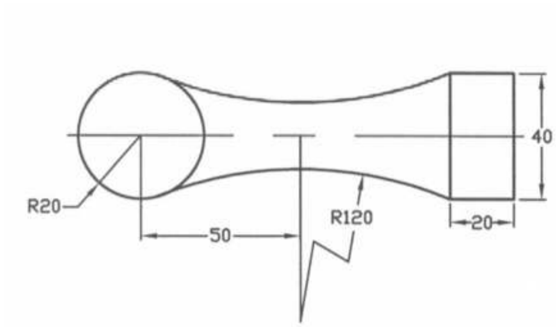
Solution of P2.6



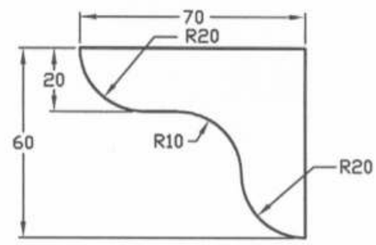
Solution of P2.7



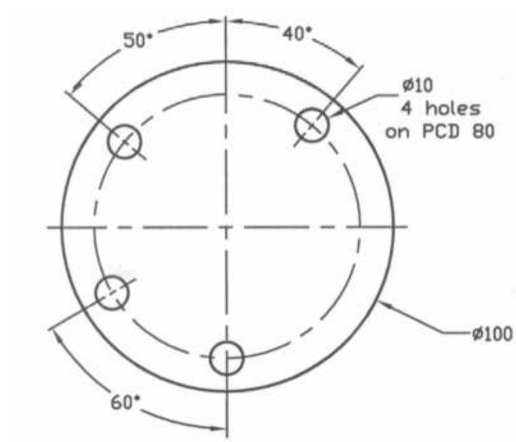
Solution of P2.8



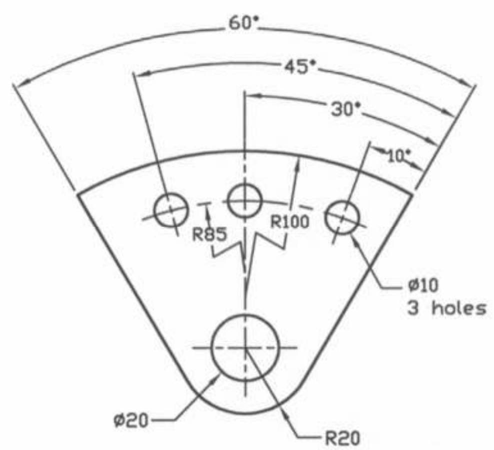
Solution of P2.9



Solution of P2.10



Solution of P2.11



Solution of P2.12

Problems

Prob. 2.13 – 2.20: Show appropriate dimensions required for the views given below in Fig. P2.13 – P2.20 (The scale of each view is half size).

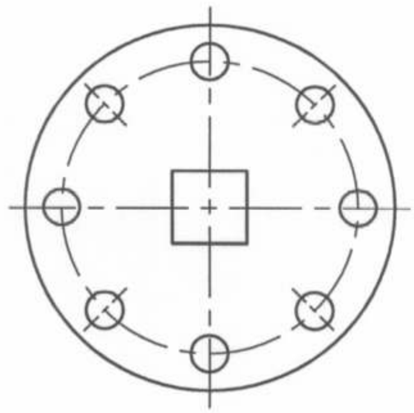


Fig. P2.13

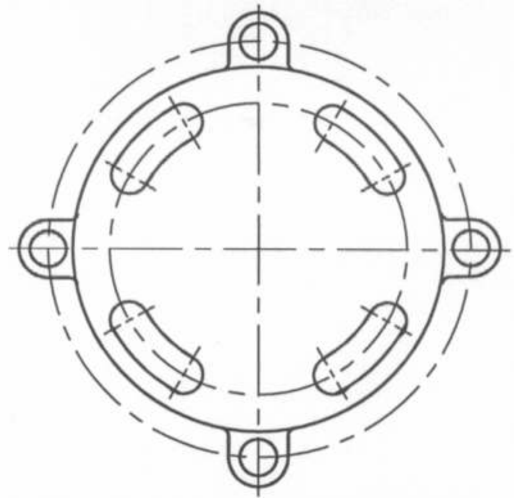


Fig. P2.14

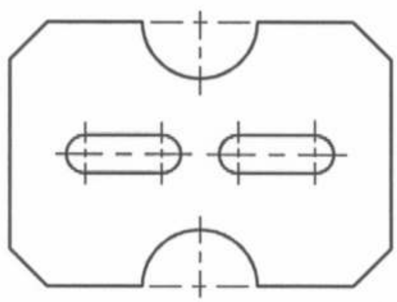


Fig. P2.15

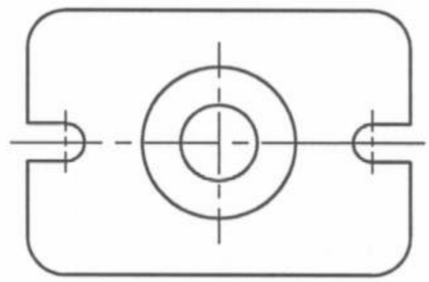


Fig. P2.16

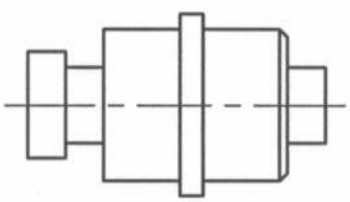


Fig. P2.17

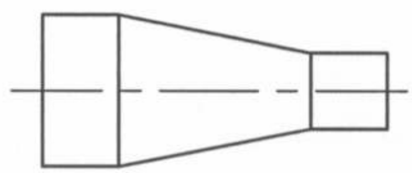


Fig. P2.18

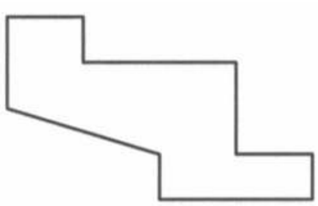


Fig. P2.19

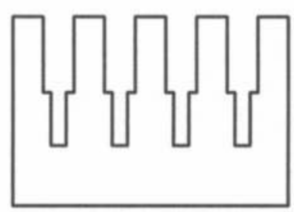


Fig. P2.20

CHAPTER 3

ORTHOGRAPHIC PROJECTION

3.1 *Introduction*

The purpose of mechanical engineering drawing is to indicate the shape and size of an object or a machine part. All objects have three dimensions such as length, breadth and height. The exact shape of an object may be produced with the help of projection. **Projection** is the process in which the rays of sight taken in a particular direction from an object to form an image on a plane called **plane of projection** or picture plane. The image on the plane is called the view of the object. There are various types of projections, such as orthographic, oblique and perspective depending on the direction of the rays of sight.

3.2 *Orthographic Projection*

When the rays of sight are made in a perpendicular direction to the plane of projection, it is called **orthographic projection**. The word orthographic is obtained from the Greek words: orthos, meaning straight, correct, at right angles to; and graphikus, meaning to write or describe by drawing lines. In the orthographic projection the observer stands in the infinite distance so that the rays of the sight appear to be parallel with each other theoretically (Figure 3.1). The perpendiculars, which are used to draw the view in the orthographic projection, are called the **projectors**. When a view is generated using perpendiculars from all the points of the object to the picture plane, it is called **orthographic view**.

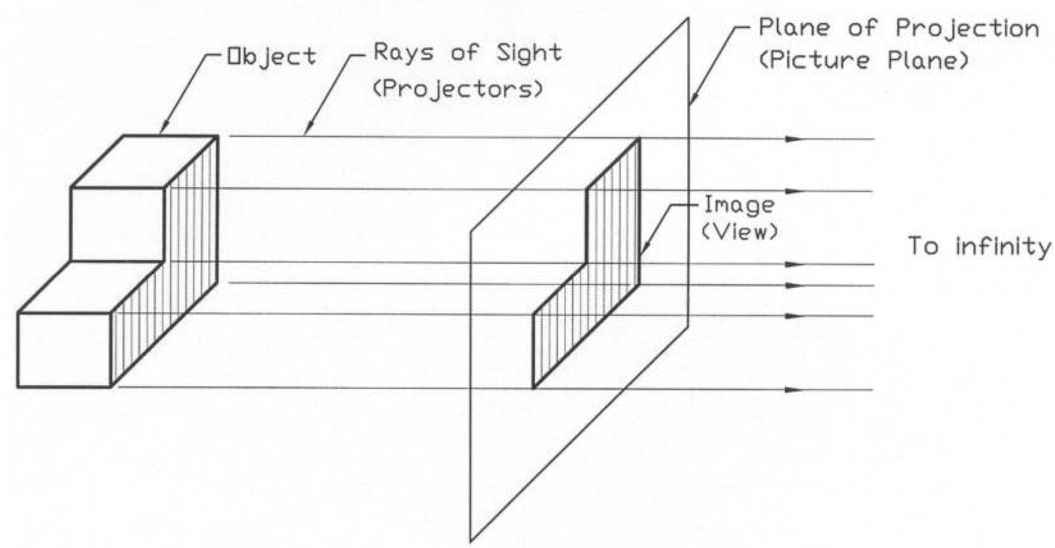


Figure 3.1: Orthographic Projection

In the oblique projection the rays of sight are parallel with each other but they are at an angle (not perpendicular) to the plane of projection. While in the perspective projection the rays of sight occur at an angle to the plane of projection but they are not parallel with each other, rather they converge to a point (Figure 3.2); as if an observer sees the image of the object on the plane of projection from his eye located at that point.

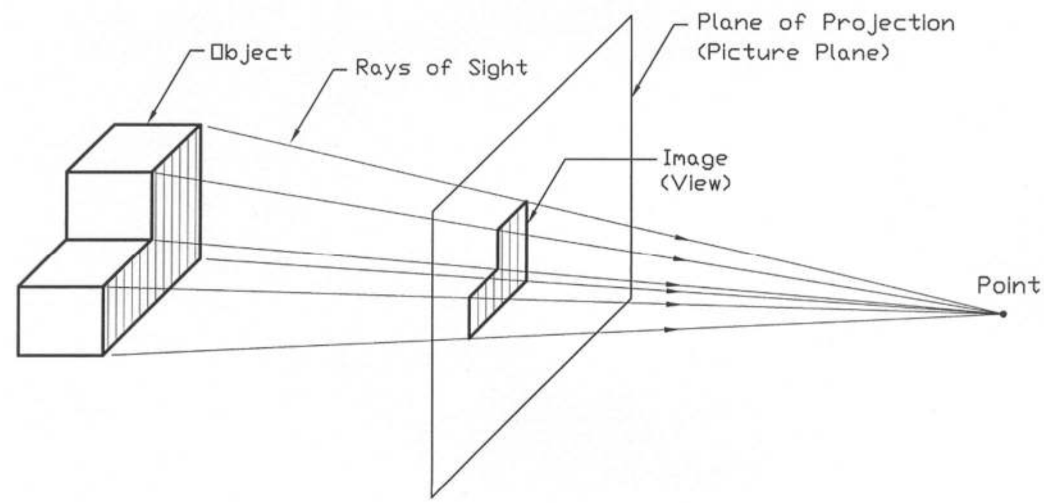


Figure 3.2: Perspective Projection

More than one plane may be required to represent the object completely. In that case the planes are positioned horizontally and vertically at right angles to each other. Total six possible views may be obtained such as, front, back, top, bottom, left and right sides. However, six views are rarely required. The number of views should be just sufficient to represent the shape of the object completely. For most of the objects, those adjacent views are necessary of which front view is the common one. Two views and sometimes one view may be good enough to represent an object completely.

3.3 ***Types of Projections***

Orthographic projection can be produced in four different angles (known as dihedral angles), which are formed by horizontal, and vertical planes. The angles are shown in Figure 3.3. The first and the third angles are used only for the projection. First angle is used for the First angle projection while third angle is used for the Third angle projection. In Figure 3.4 the difference of first and third angle projections has been illustrated showing the relative positions of the observer, object and the picture plane. The object is positioned in the first quadrant for the first angle projection and in the third quadrant for the third angle projection. In order to eliminate the clumsiness, only one image for each projection has been taken into consideration in this figure.

The observer is located at infinity so that the rays of sight become parallel and fall in perpendicular direction to the picture plane. In the Third angle projection the picture plane is placed in between the object and the observer. On the other hand in the First angle projection the object is placed in between the observer and the picture plane. In Bangladesh usually the Third angle projection is used. They are used in U.S.A., Canada and many other countries. On the other hand First angle projection is used in many European and other countries. However, only the Third angle projection will be taken into consideration throughout this book.

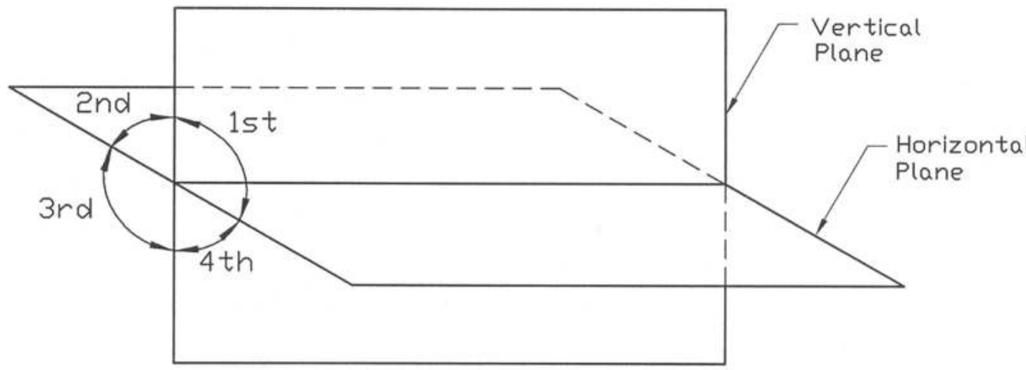


Figure 3.3: Four Different Angles

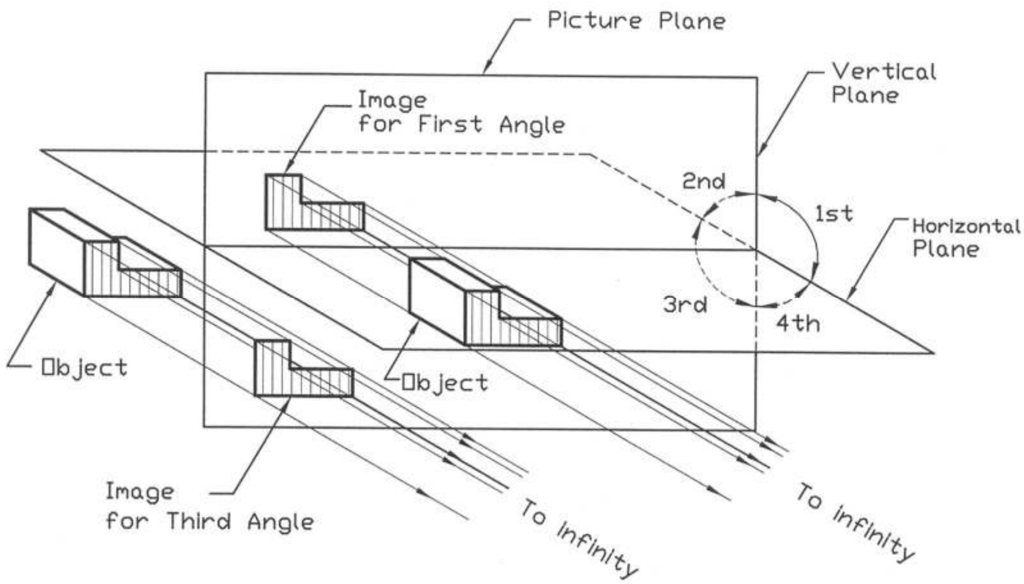


Figure 3.4: Difference Between First and Third Angle Projections

3.4 ***Third Angle Projection***

The development of the Third angle projection has been illustrated in Figures 3.5 to 3.8. In Figure 3.5, a six-sided transparent box in the third angle projection has been shown. This transparent box with an object showing views on the various planes has been presented in Figure 3.6. There are six sides but here only four sides are considered to represent the object. The two other sides rear and bottom, have been omitted. The projection plane upon which the front view is projected is called the frontal plane (a-b-c-d is the frontal plane) and that upon which the top view is projected is called the horizontal plane (d-c-g-h is the horizontal plane) as shown in the figure. While the projection plane upon which the side view is projected is called the profile plane (a-d-h-e is the profile plane).

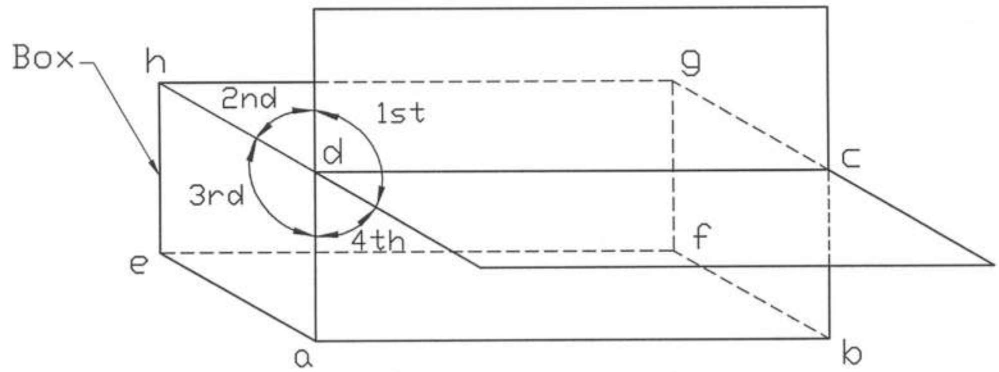


Figure 3.5: Six-Sided Transparent Box in Third Angle Projection

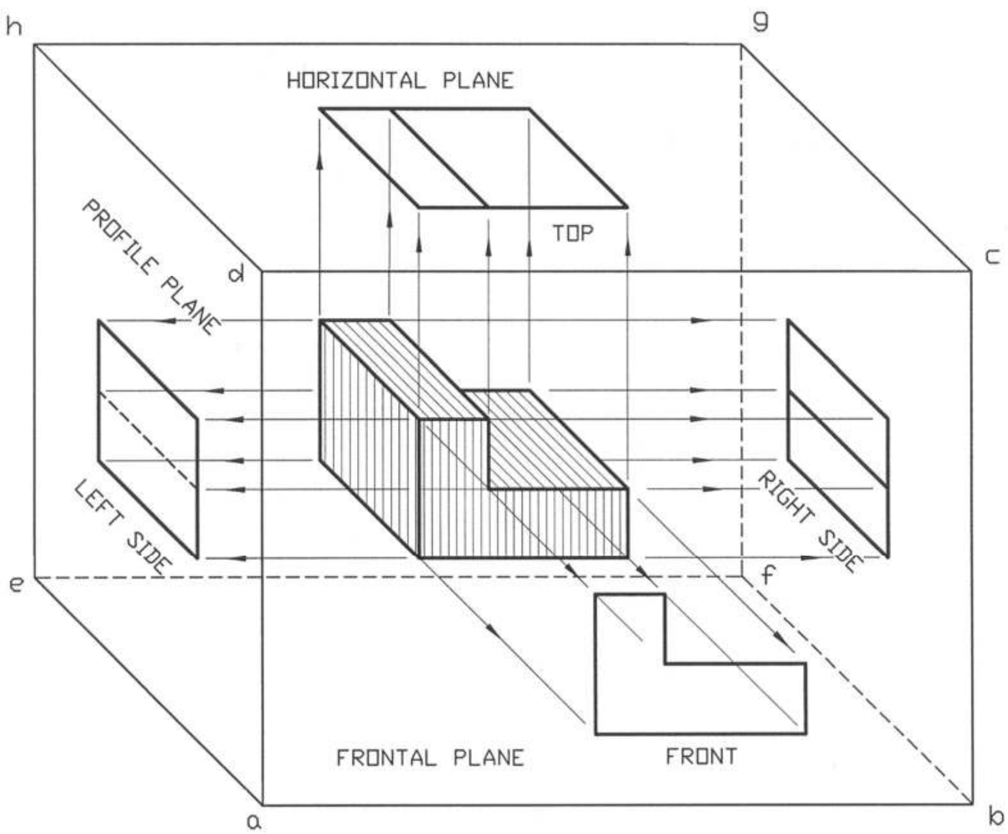


Figure 3.6: Views in Third Angle Projection

The process of unfolding the box has been shown in Figure 3.7. The views on the sides are also given here. In Figure 3.8 the relative positions of the views on the sides of the unfolding box have been provided.

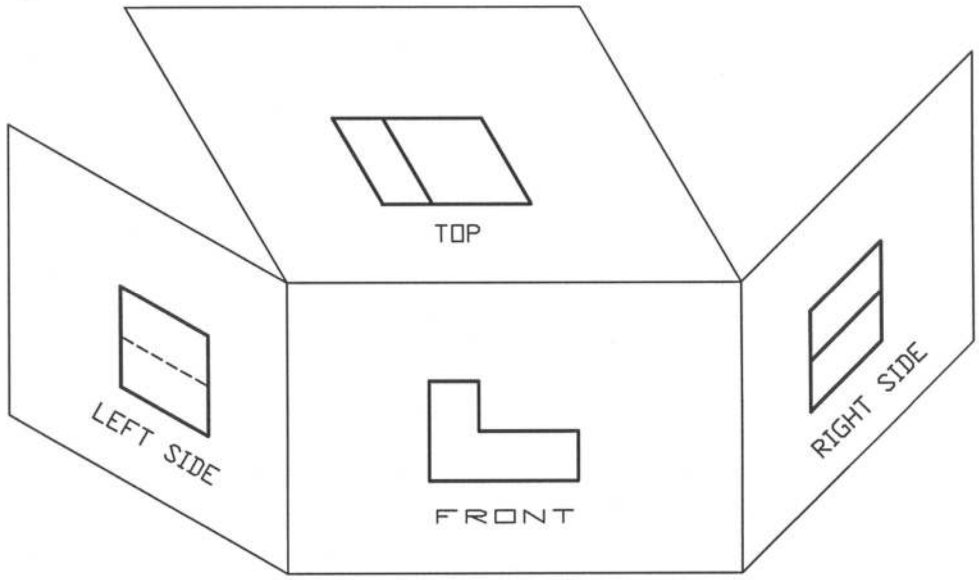


Figure 3.7: Views on Sides of Unfolding Box in Third Angle Projection

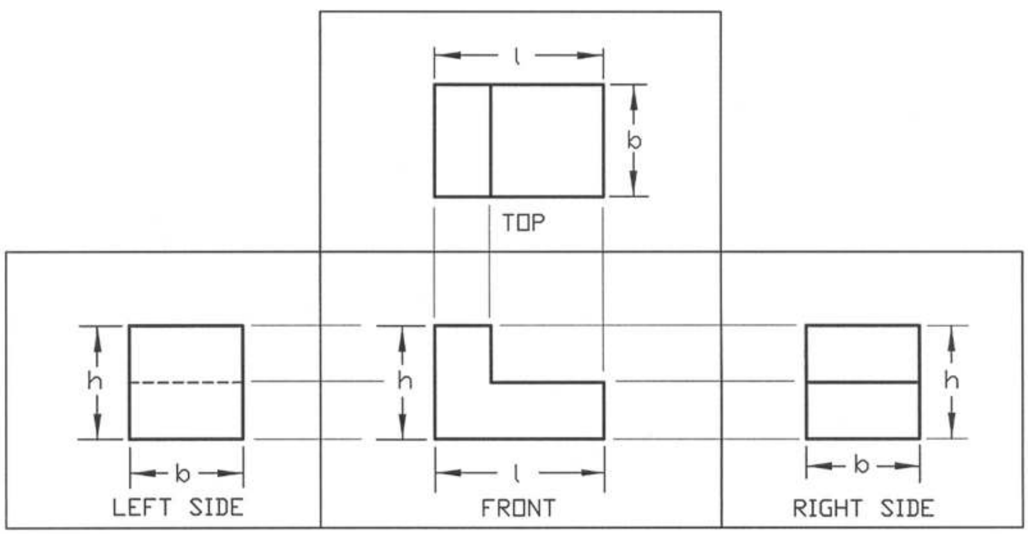


Figure 3.8: Relative Positions of Views on Sides of Unfolding Box in Third Angle Projection

3.5 ***First Angle Projection***

The development of the First angle projection has been illustrated in Figures 3.9 to 3.12. The six-sided transparent box for the First angle projection has been shown in Figure 3.9. While an object showing the four views on the sides of the transparent box has been presented in Figure 3.10. In this figure front view is shown in the frontal plane (e-f-g-h is the frontal plane), top view in the horizontal plane (a-b-f-e is the horizontal plane) and side view on the profile plane (a-d-h-e is the profile plane). Views on the sides of the unfolding box in the First angle projection are given in Figure 3.11. On the other hand the relative positions of the views on the sides of the unfolding box have been shown in Figure 3.12.

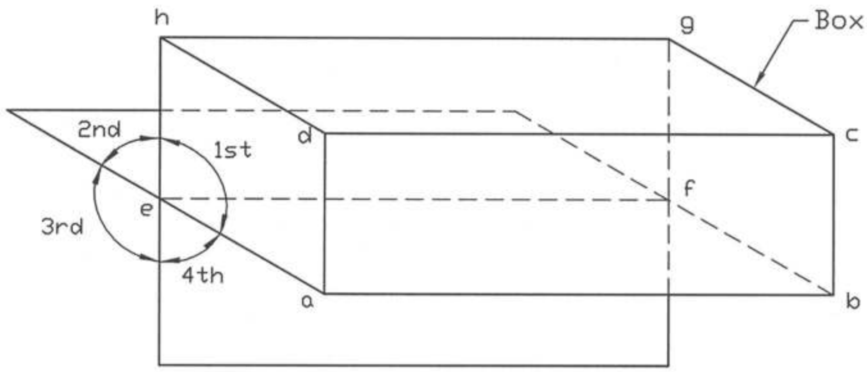


Figure 3.9: Six-Sided Transparent Box in First Angle Projection

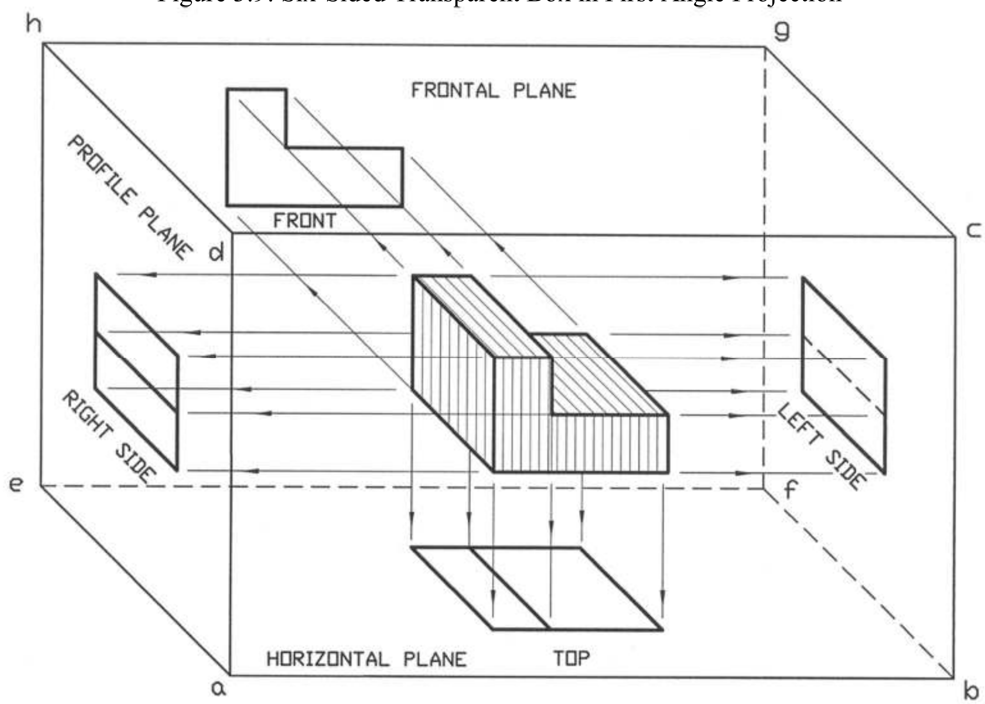


Figure 3.10: Views in First Angle Projection

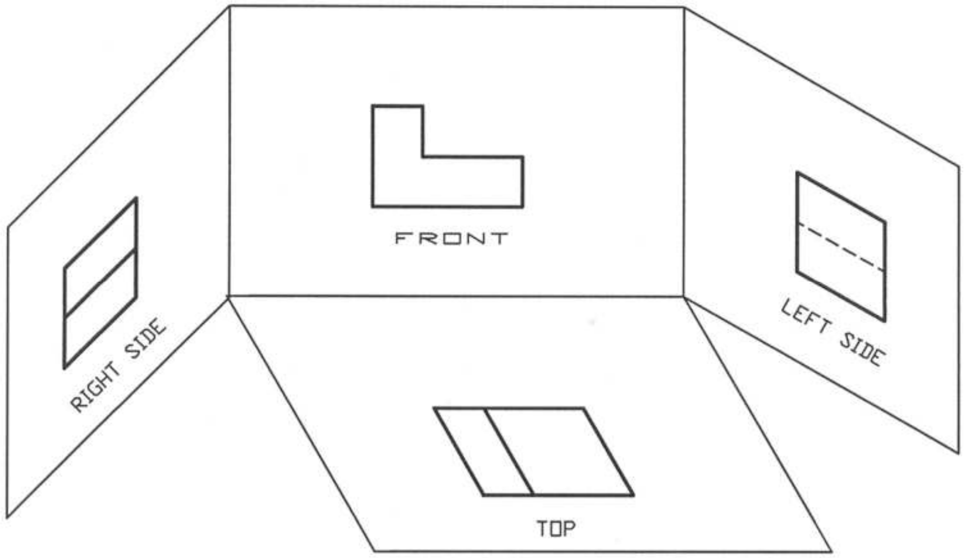


Figure 3.11: Views on Sides of Unfolding Box in First Angle Projection

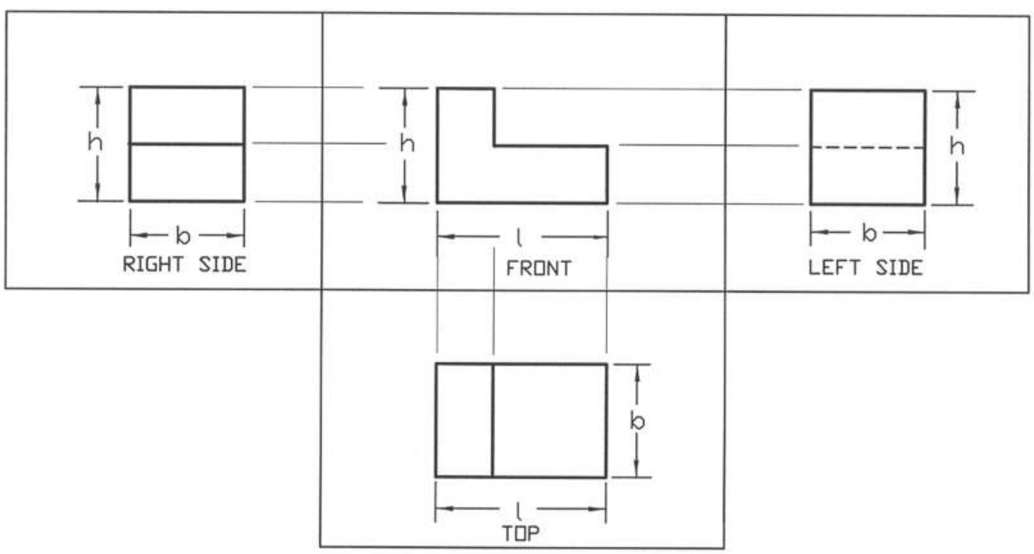


Figure 3.12: Relative Positions of Views on Sides of Unfolding Box in First Angle Projection

3.6 ***Method of Projecting Views***

There are several methods of projecting views in the orthographic projection, which are shown in Figures 3.13a to 3.13d. In the projection there is a relationship of the different views. It is usual practice to draw the front view first, then the top and side views are drawn in reference to the front view with the help of the vertical and horizontal projection lines. This can be done using T-square, triangles and compasses when drawing is performed manually. In making projection lines as shown in Figure 3.13b it requires to use the compass in addition to T-square and triangles. The projection lines between the different views have been shown in Figure 3.13.

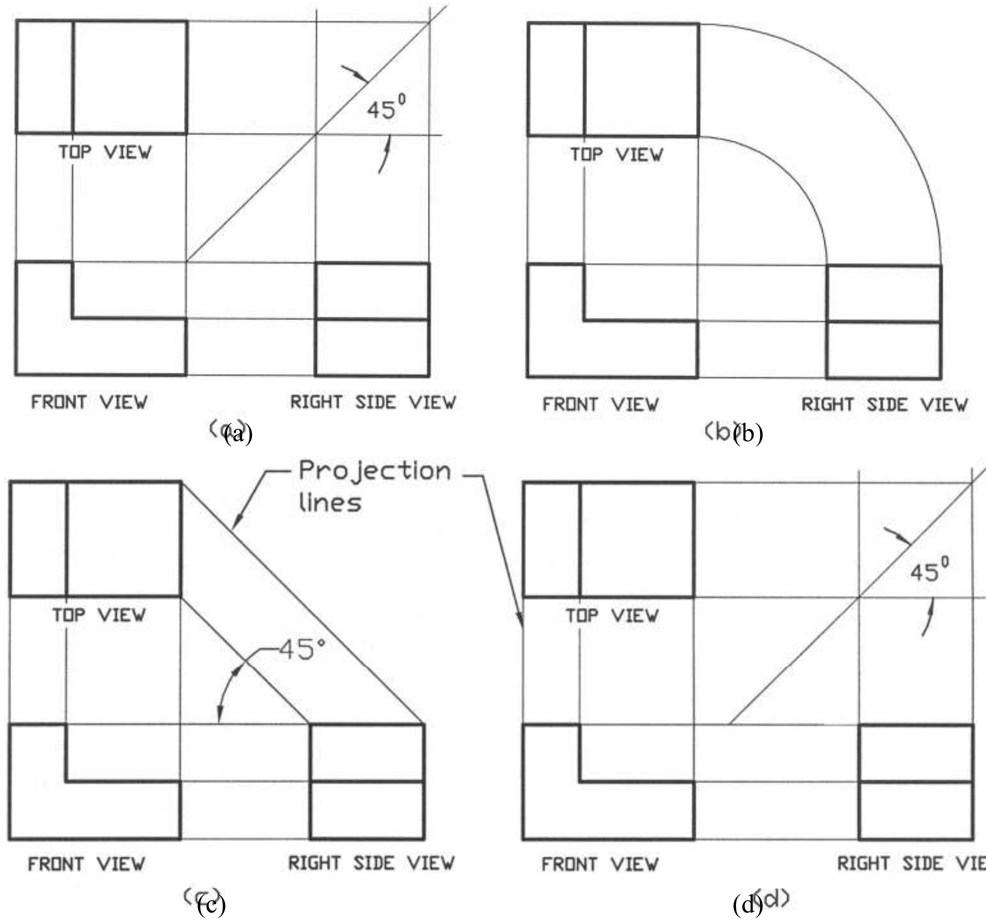


Figure 3.13: Different Methods of Projecting Views

The spacing between the views has to be determined beforehand. It will depend on many factors such as, space required for the dimensions, space required for writing views and space required for writing essential notes. The space should be sufficient in order to give the dimensions avoiding any crowding. However, the excessive space should be avoided. The beginners may choose the space in between the views as 30 to 40 mm. In Figures 3.13a to 3.13c the interspacing between the views has been made as equal. However, if it is necessary different spacing may be used as well; an example of that is shown in Figure 3.13d. Any method of projecting views as described in Figure 3.13 may be used. However, the projection as described in Figure 3.13a is mostly used. In Figure 3.14, the orthographic projection of an object has been provided as a further example.

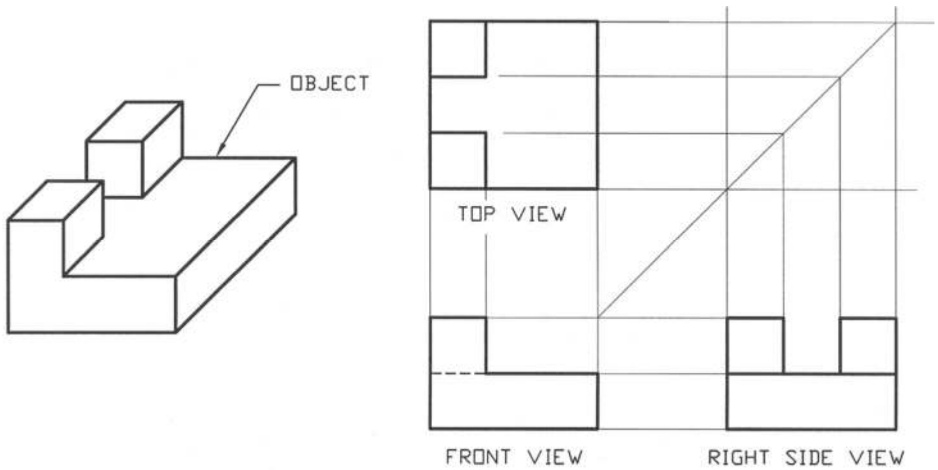


Figure 3.14: Orthographic Projection

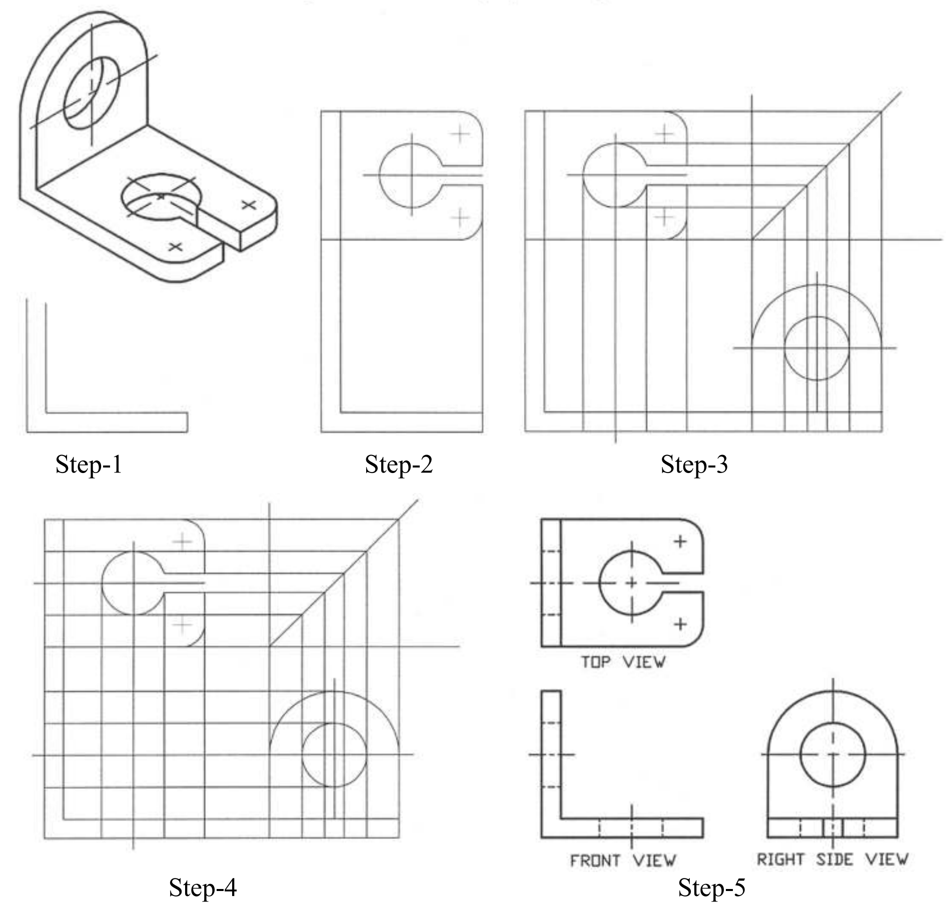


Figure 3.15: Typical Steps in Drawing a Part

In Figure 3.15 typical steps in drawing a part have been illustrated. The projection has been started from the front view but before completion of the front view, other views have been projected in parallel. It is observed that the front view is also projected in reference to the top and sides views. It can be noted here that the thin lines have been used in projecting the preliminary views. When the views are projected completely, the projection lines are erased out and the views are completed using the lines of appropriate thickness as shown in the figure. In Figure 3.16 an example of the first angle orthographic projection is shown.

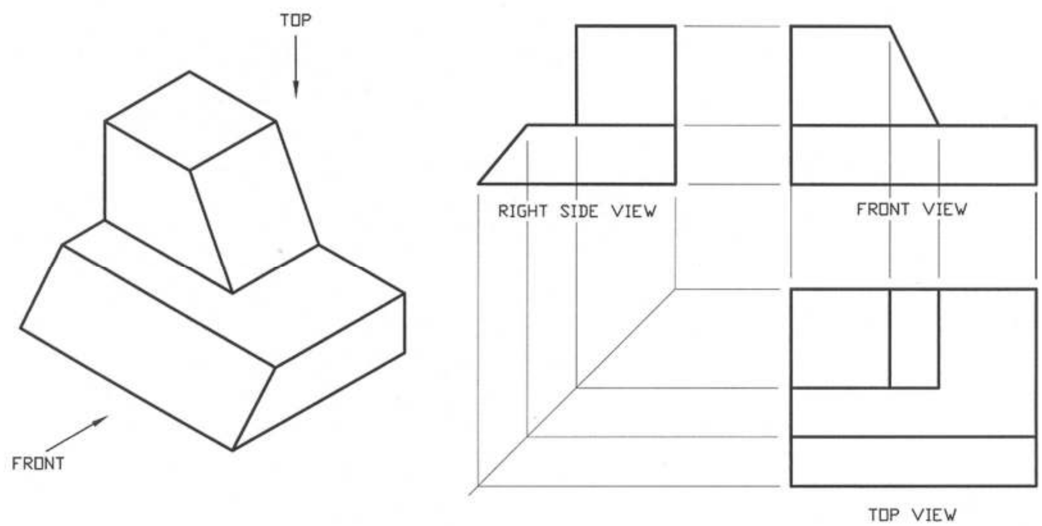
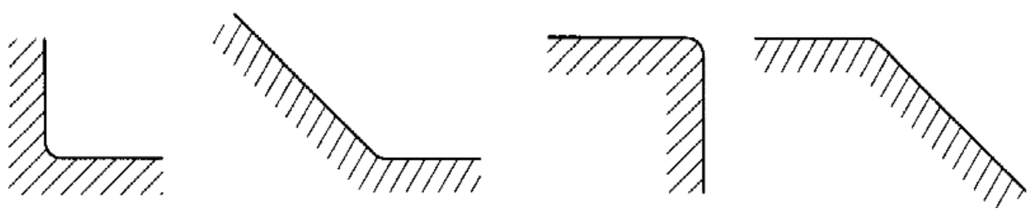


Figure 3.16: First Angle Projection

3.7 ***Fillets and Rounds***

The sharp internal corners are made rounded in a casting in order to avoid the possibility of stress crack. These rounded internal corners are called fillets. On the other hand the external sharp corners are made rounded for safety and appearance. These rounded external corners are called rounds. In Figure 3.17 fillets and rounds are shown.



Fillets Rounds

Figure 3.17: Fillets and Rounds

3.8 ***Projection Symbol***

Projection symbols according to ISO 128 – 1982(E) for Third Angle and First Angle are shown respectively in Figure 3.18a and Figure 3.18b. It is usual practice to locate the ISO projection symbol in the lower right hand corner of the drawing adjacent to the title block as shown in Figure 3.18c.

From the symbol one can easily identify the type of projection (First Angle or Third Angle) used to perform the drawing.

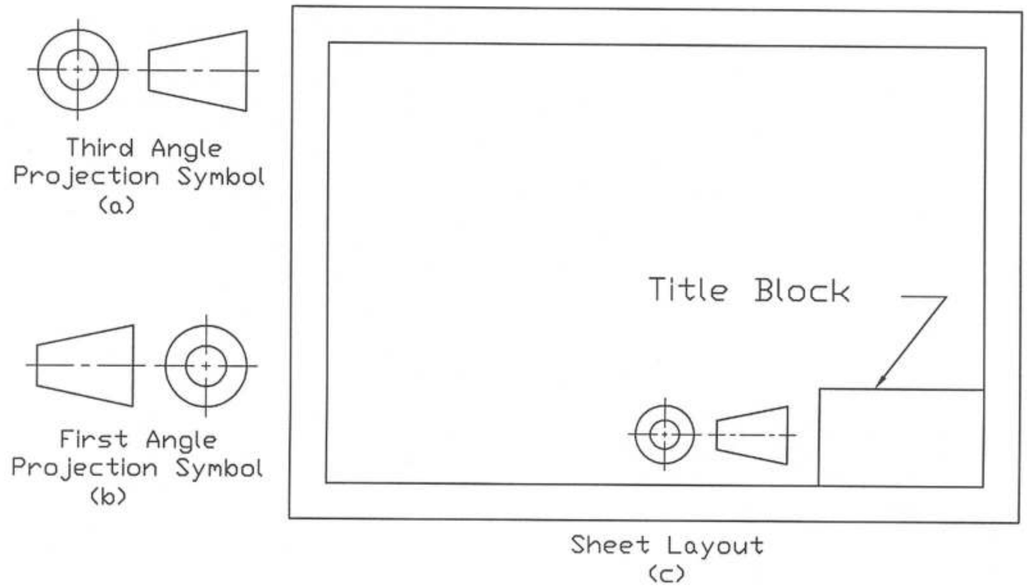


Figure 3.18: Projection Symbol

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.*** The scaled drawings are provided on the squared paper.

Prob. P3.1 - P3.24: Draw the necessary orthographic views of each of the objects (Fig. P3.1 – P3.24) to best represent them on the squared or plain papers. Free hand drawings may be done.

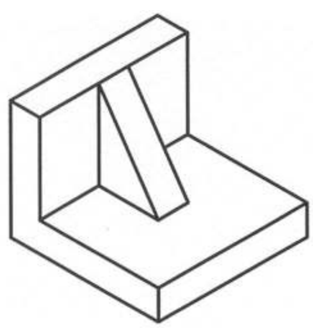


Fig. P3.1

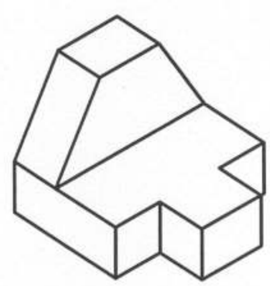


Fig. P3.2

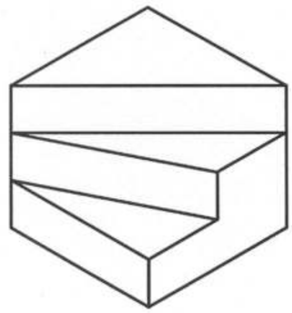


Fig. P3.3

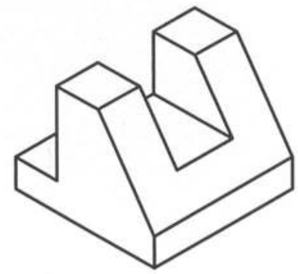


Fig. P3.4

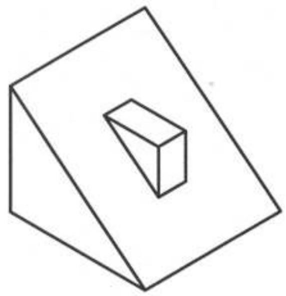


Fig. P3.5

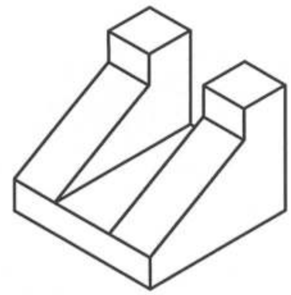


Fig. P3.6

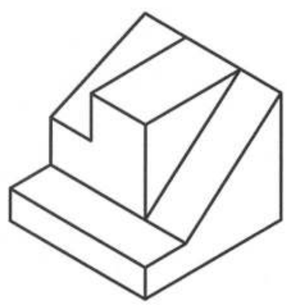


Fig. P3.7

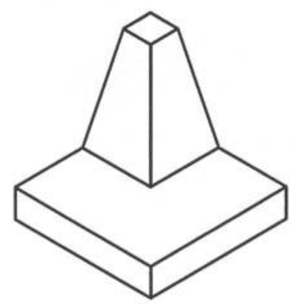


Fig. P3.8

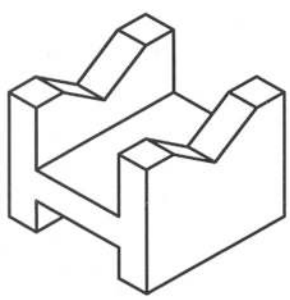


Fig. P3.9

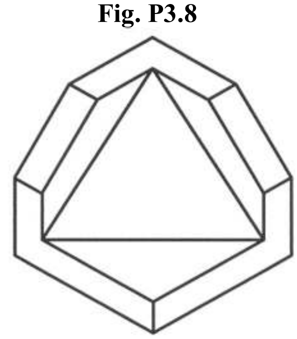


Fig. P3.10

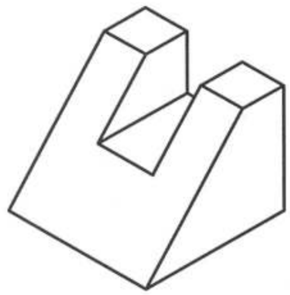


Fig. P3.11

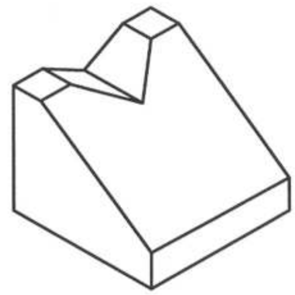


Fig. P3.12

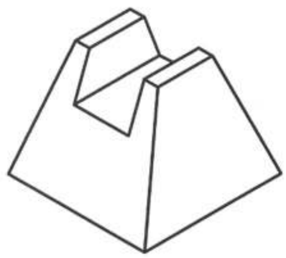


Fig. P3.13

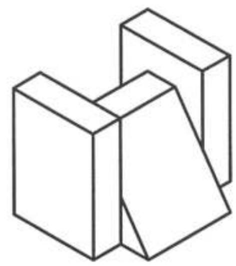


Fig. P3.14

Fig. P3.15

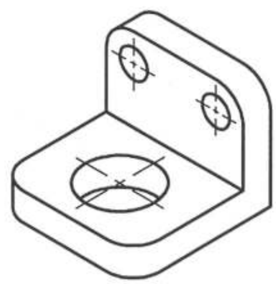


Fig. P3.16

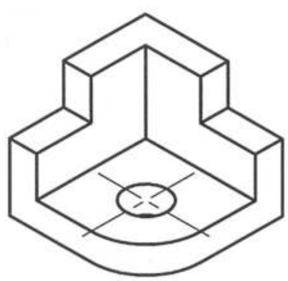


Fig. P3.17

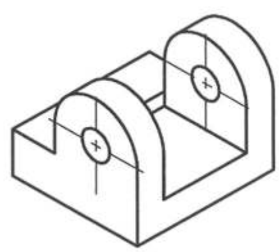


Fig. P3.18

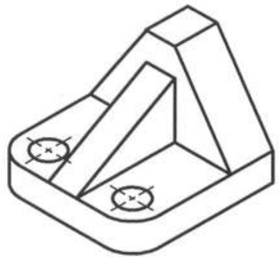


Fig. P3.19

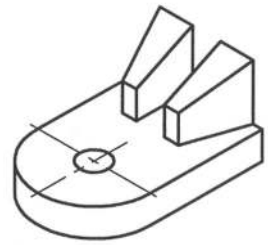


Fig. P3.20

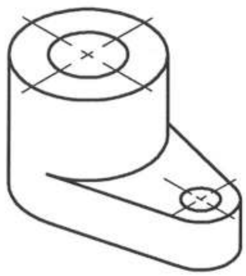


Fig. P3.21

Fig. P3.22

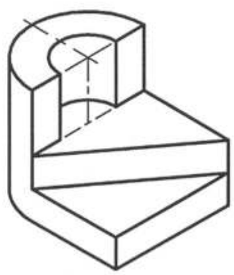


Fig. P3.23

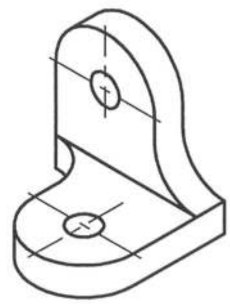


Fig. P3.24

Prob. P3.25 – P3.40: Complete the missing views from the given views as shown in Fig. P3.25 – P3.40.

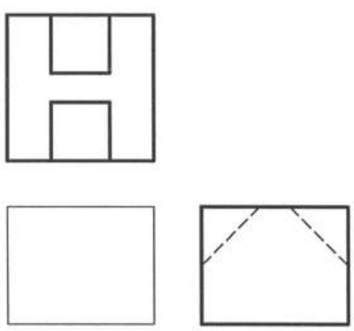


Fig. P3.25

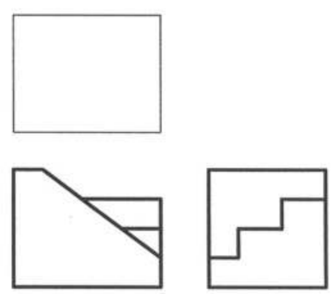


Fig. P3.26

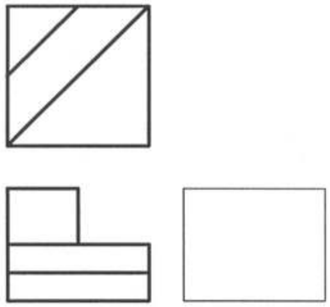


Fig. P3.27

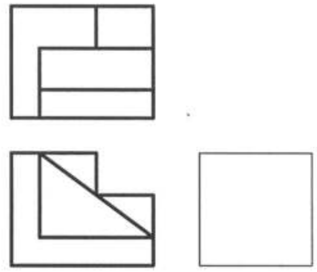


Fig. P3.28

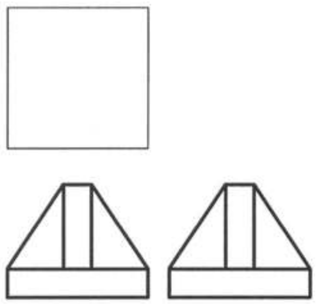


Fig. P3.29

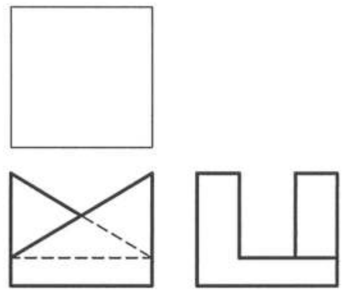


Fig. P3.30

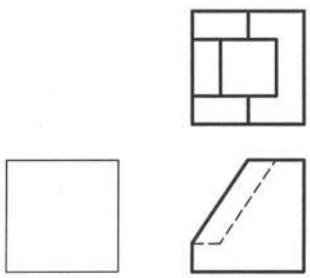


Fig. P3.31

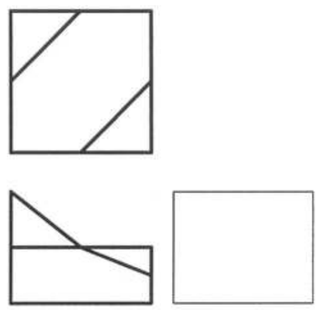


Fig. P3.32

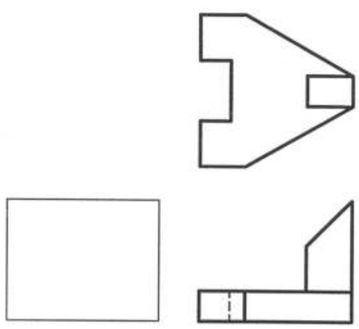


Fig. P3.33

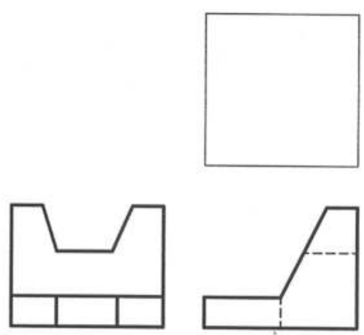


Fig. P3.34

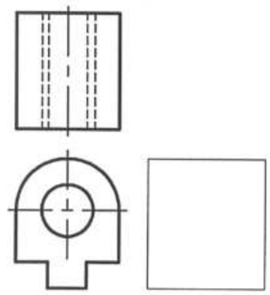


Fig. P3.35

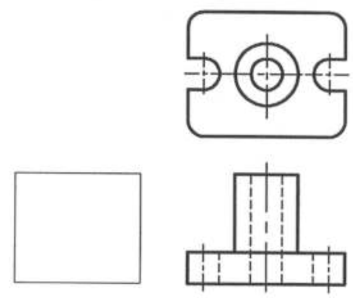


Fig. P3.36

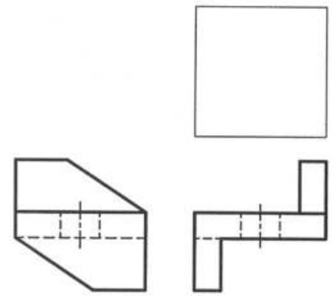


Fig. P3.37

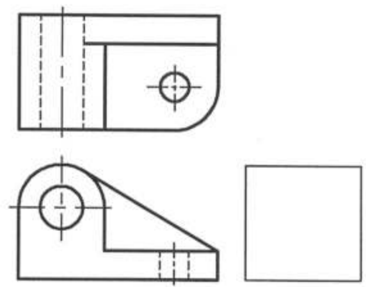


Fig. P3.38

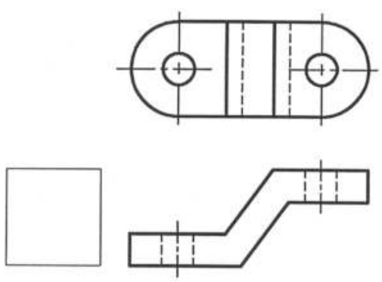


Fig. P3.39

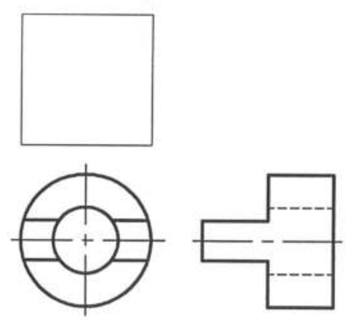


Fig. P3.40

Prob. P3.41 – P3.57: Draw the missing line/lines in the views as shown in Fig. P3.41 – P3.57.

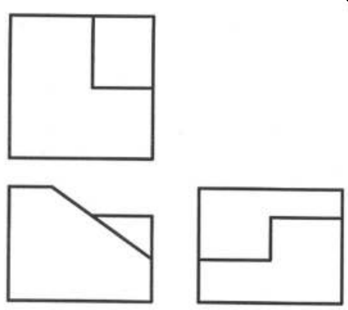


Fig. P3.41

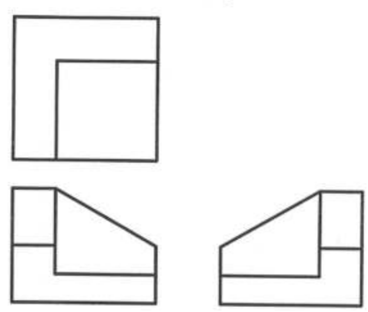


Fig. P3.42

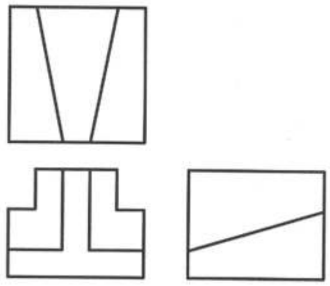


Fig. P3.43

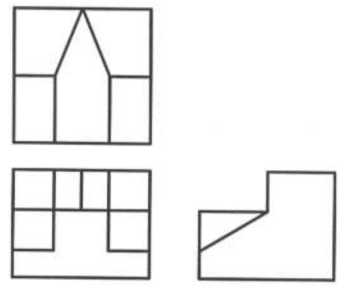


Fig. P3.44

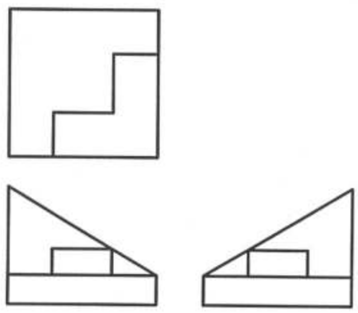


Fig. P3.45

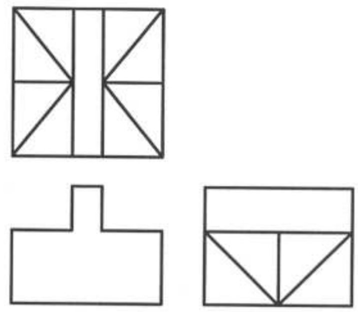


Fig. P3.46

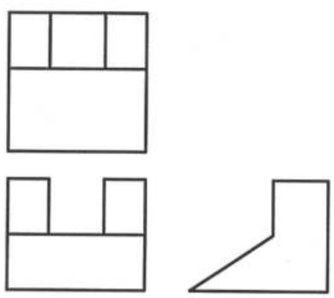


Fig. P3.47

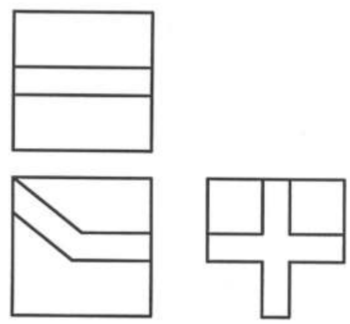


Fig. P3.48

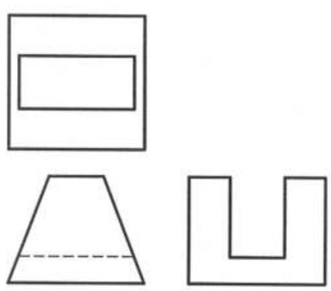


Fig. P3.49

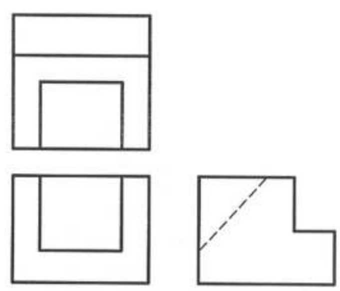


Fig. P3.50

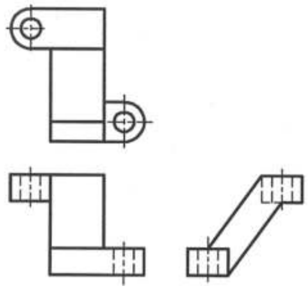


Fig. P3.51

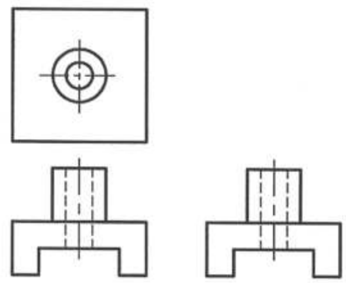


Fig. P3.52

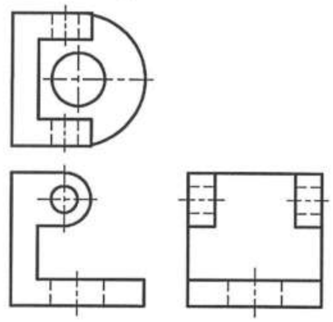


Fig. P3.53

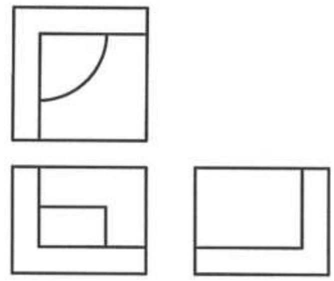


Fig. P3.54

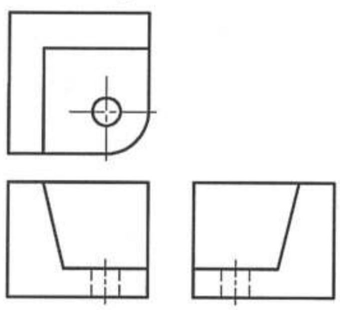


Fig. P3.55

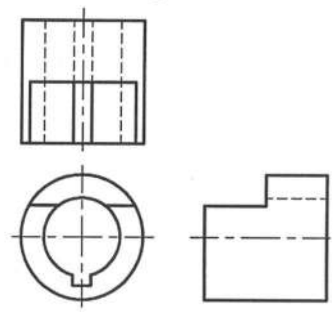


Fig. P3.56

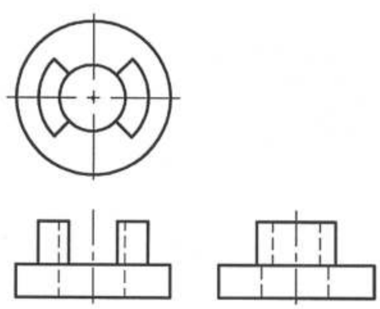


Fig. P3.57

Prob. P3.58: Draw top, front and right side views of the holder as shown in Fig. P3.58.

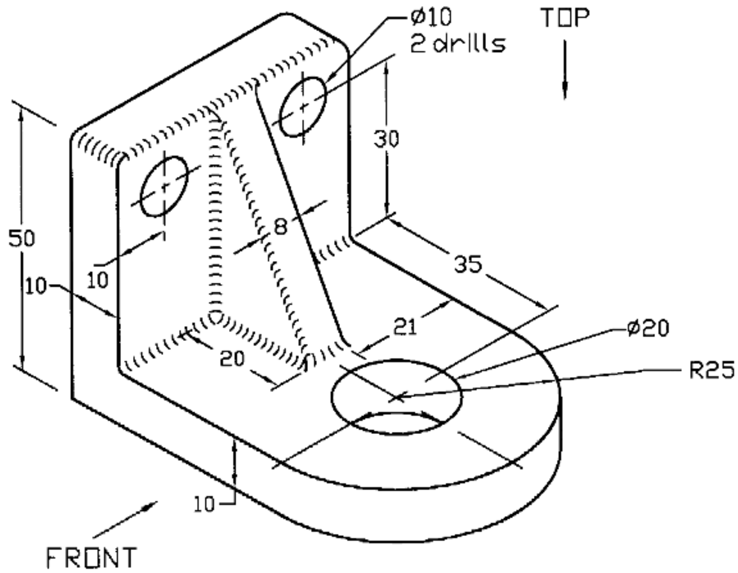


Fig. P3.58

Prob. P3.59: Draw the top, front and right side views of the support as shown in Fig. P3.59.

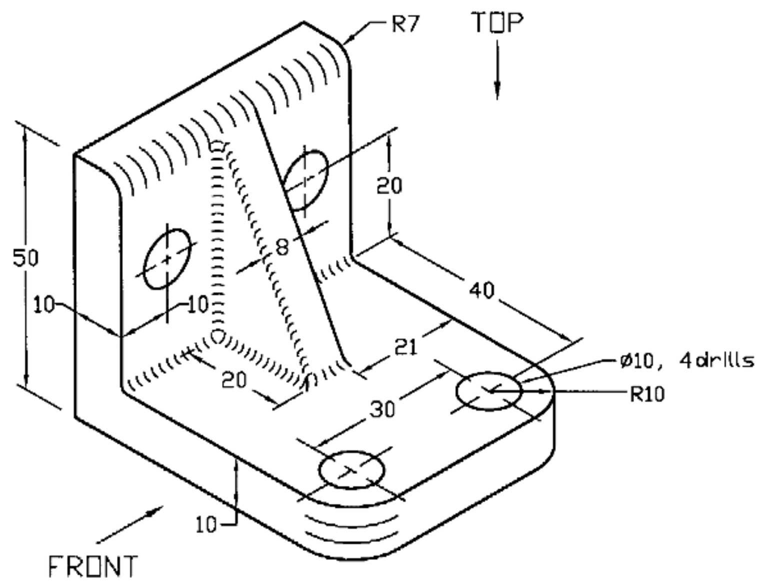


Fig. P3.59

Prob. P3.60: Draw top, front and right side views of the bracket as shown in Fig. P3.60.

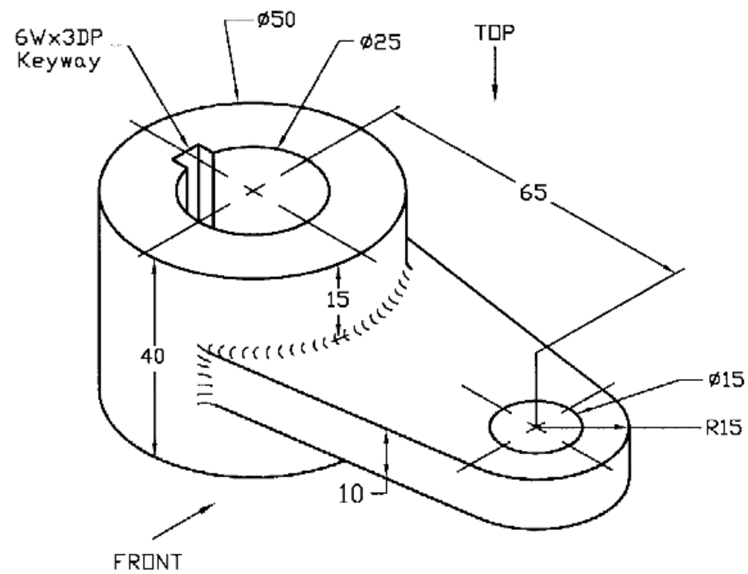


Fig. P3.60

Prob. P3.61: Draw the top, front and right side views of the bracket as shown in Fig. P3.61.

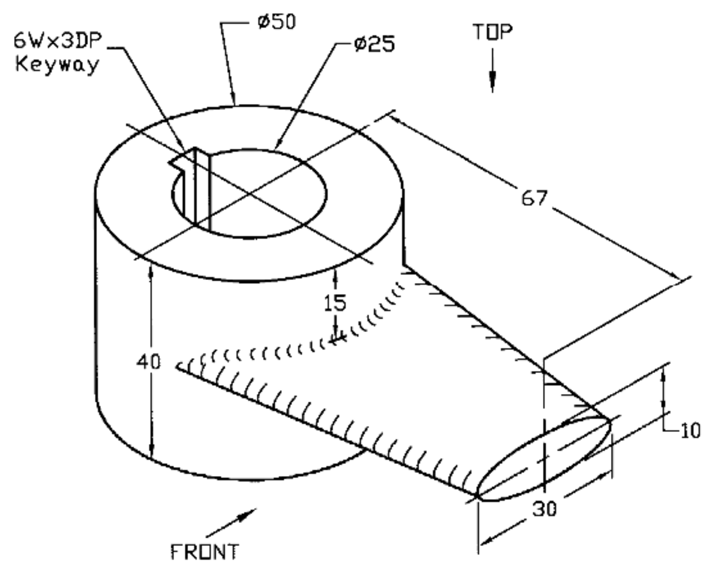


Fig. P3.61

Prob. P3.62: Draw the top, front and right side views of the fixer as shown in Fig. P3.62.

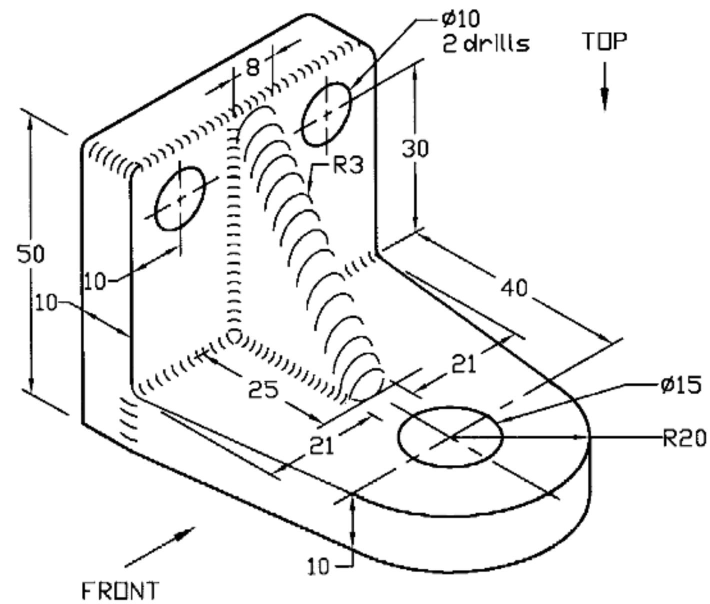


Fig. P3.62

Prob. P3.63: Draw the top, front and right side views of the guide block as shown in Fig. P3.63.

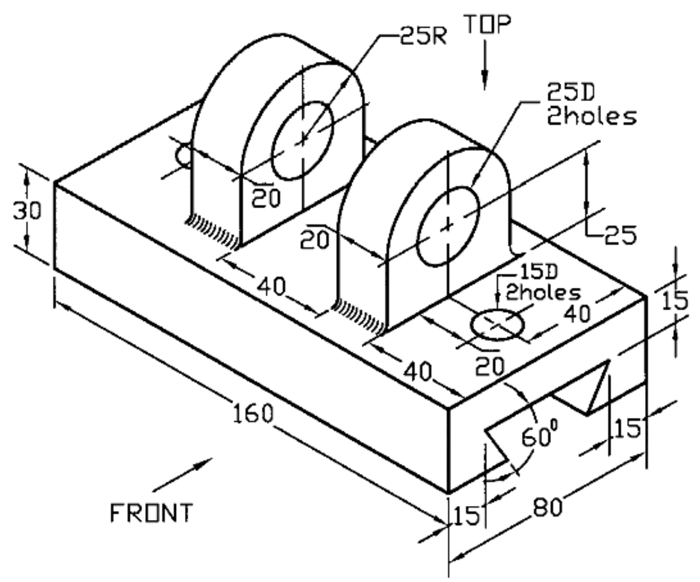


Fig. P3.63

Prob. P3.64: Draw the top, front and left side views of the bearing as shown in Fig. P3.64.

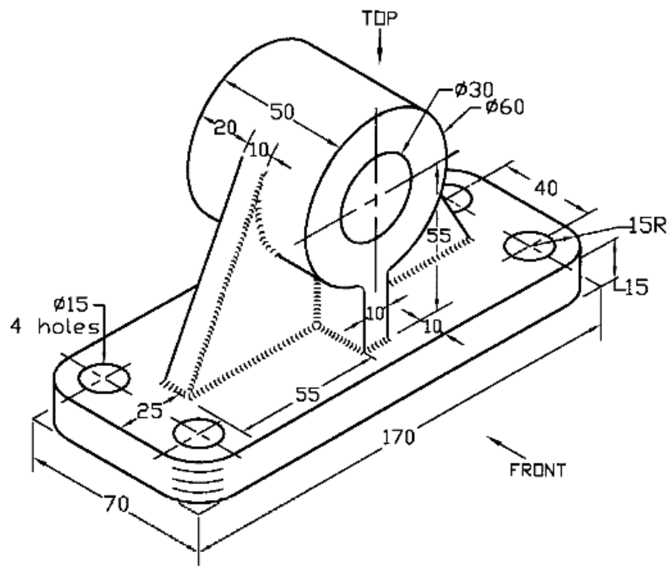


Fig. P3.64

Prob. P3.65: Draw the top, front and left side views of the end bracket as shown in Fig. P3.65.

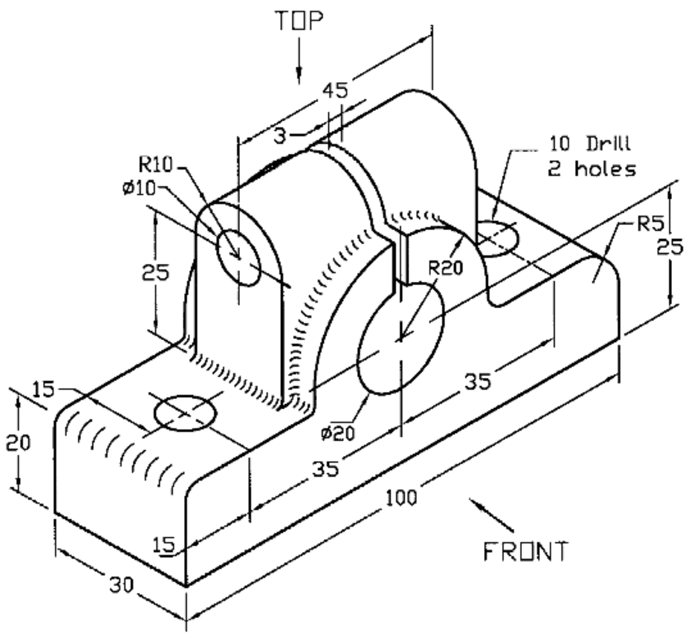


Fig. P3.65

Prob. P3.66: Draw the top, front and left side views of the bearing bracket as shown in Fig. P3.66.

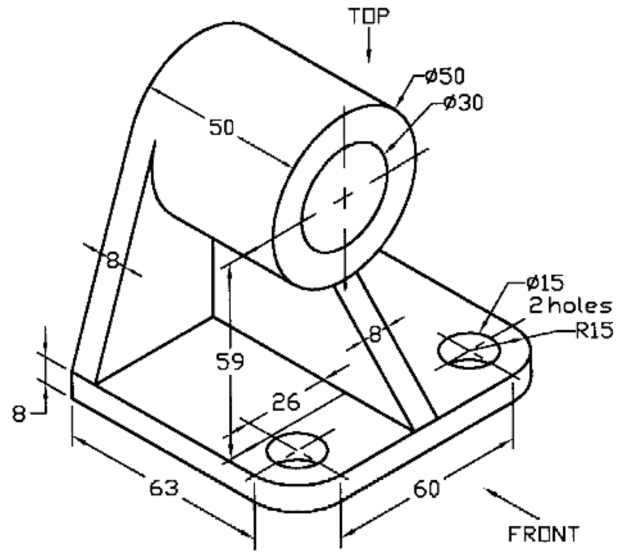


Fig. P3.66

Prob. P3.67: Draw the top, front and right side views of the bearing bracket as shown in Fig. P3.67.

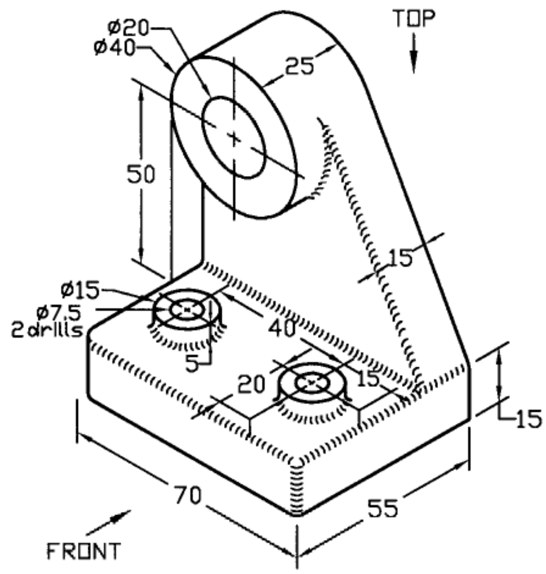
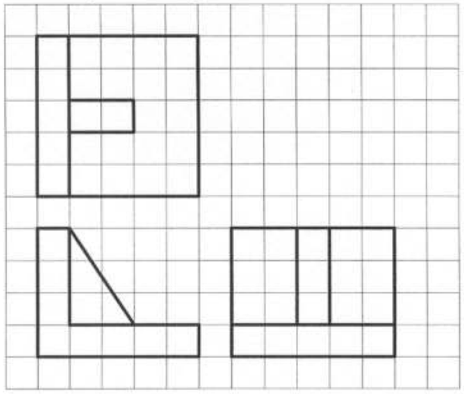
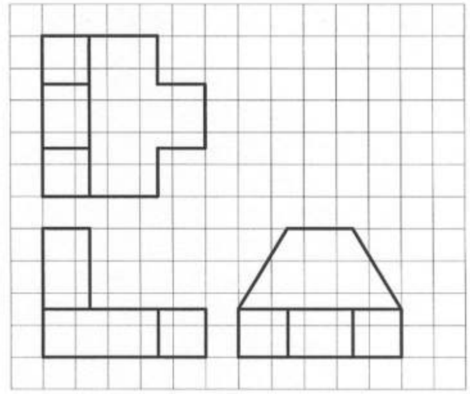


Fig. P3.67

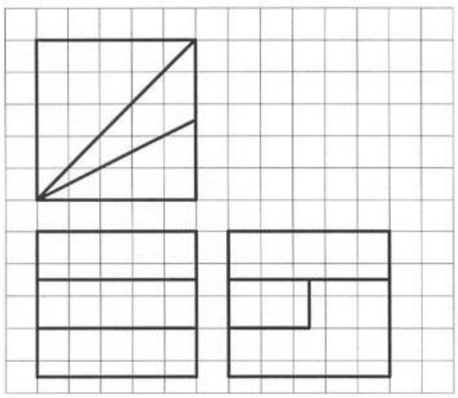
Solutions for Example Problems



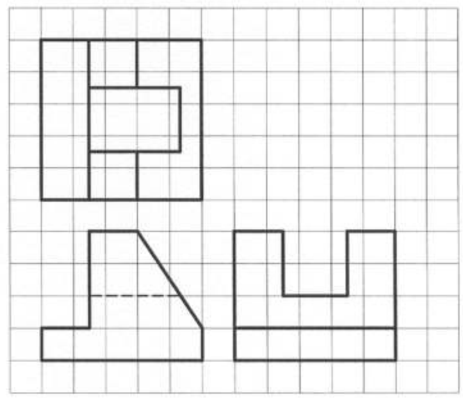
Solution of P3.1



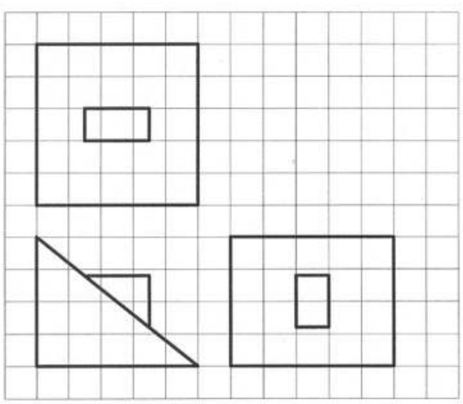
Solution of P3.2



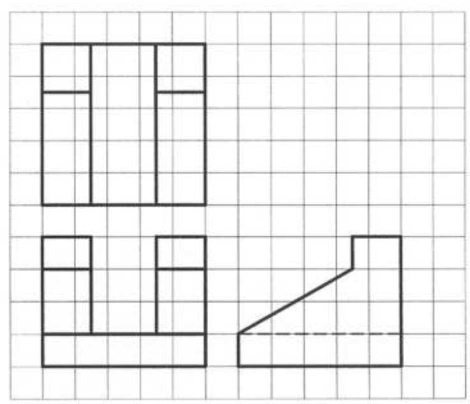
Solution of P3.3



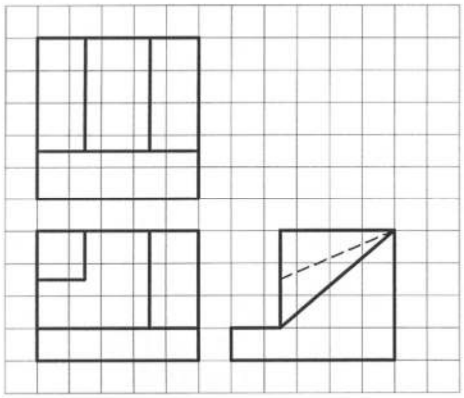
Solution of P3.4



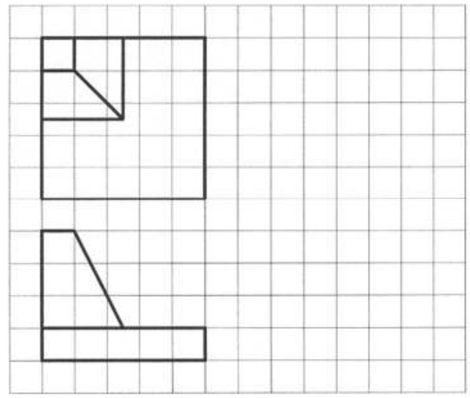
Solution of P3.5



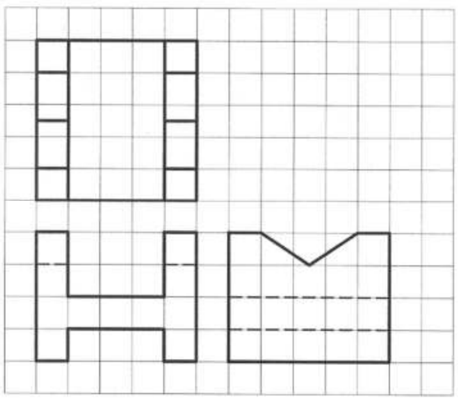
Solution of P3.6



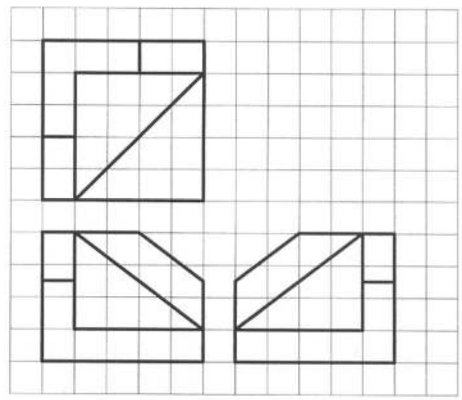
Solution of P3.7



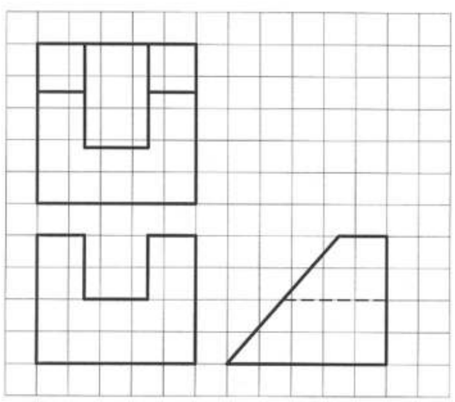
Solution of P3.8



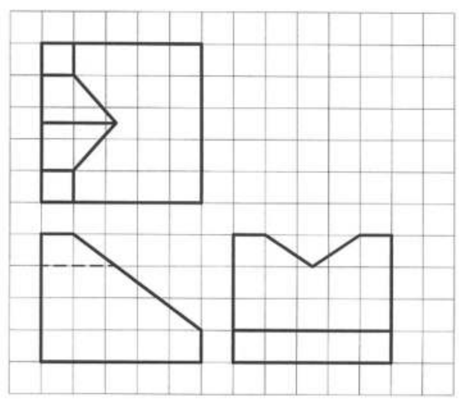
Solution of P3.9



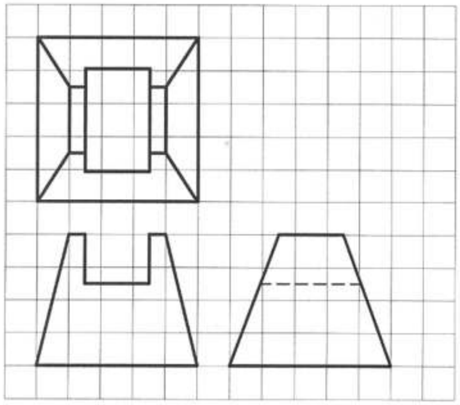
Solution of P3.10



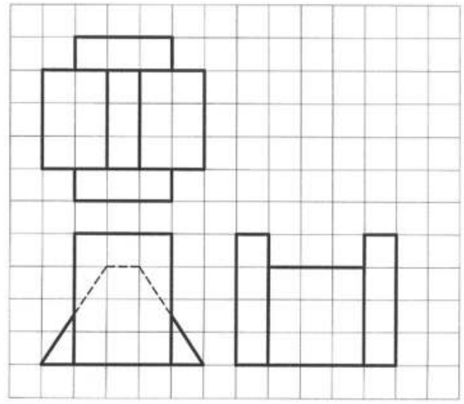
Solution of P3.11



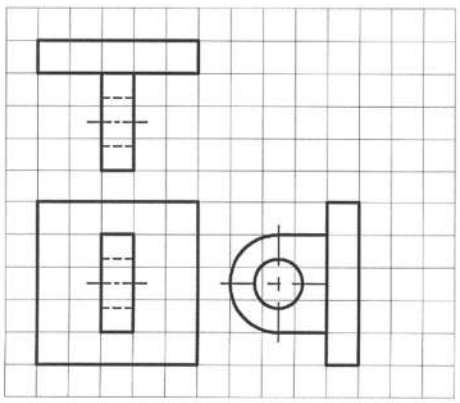
Solution of P3.12



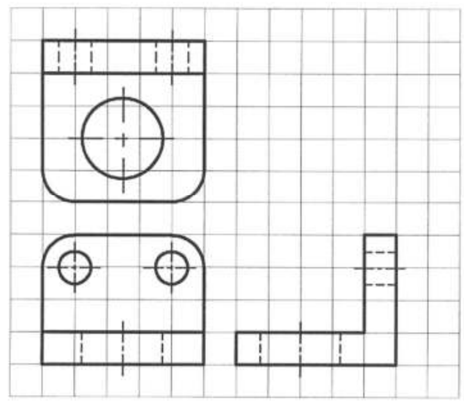
Solution of P3.13



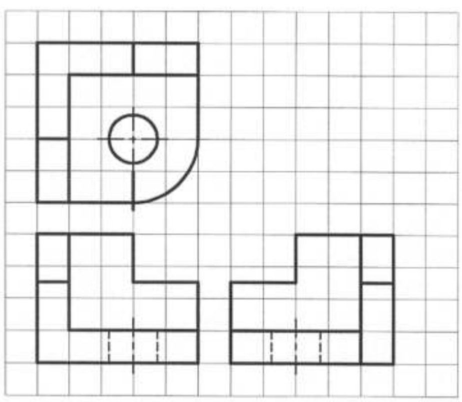
Solution of P3.14



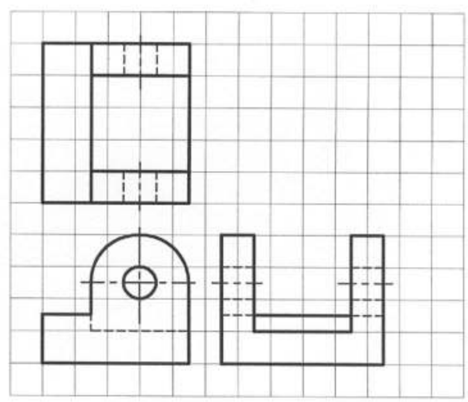
Solution of P3.15



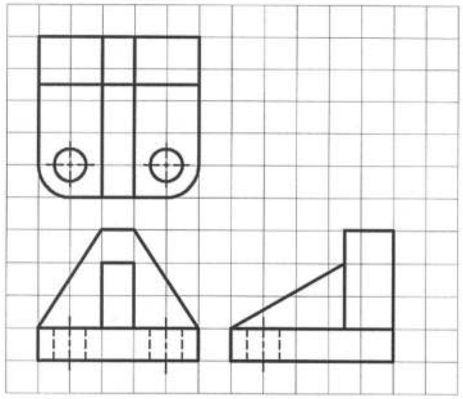
Solution of P3.16



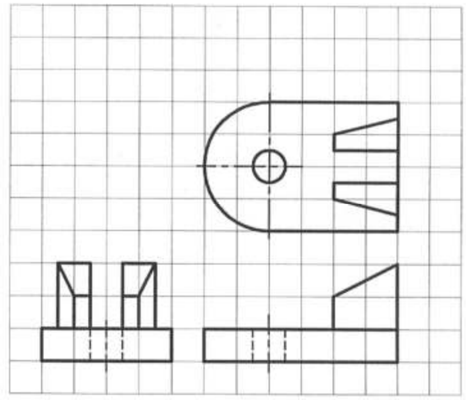
Solution of P3.17



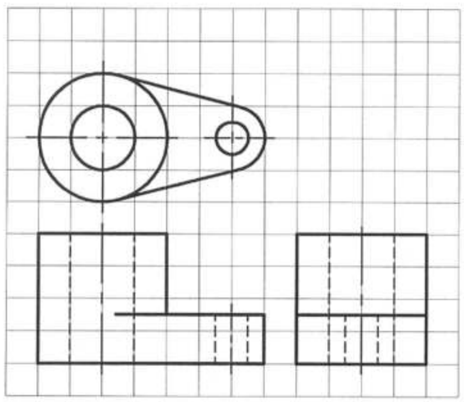
Solution of P3.18



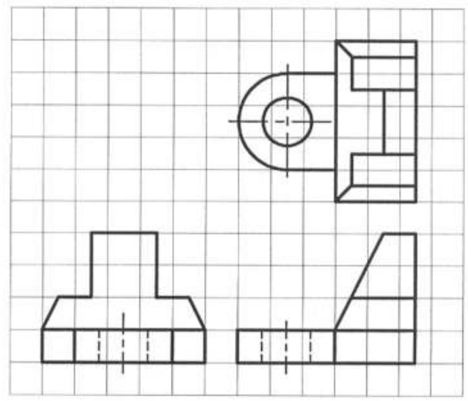
Solution of P3.19



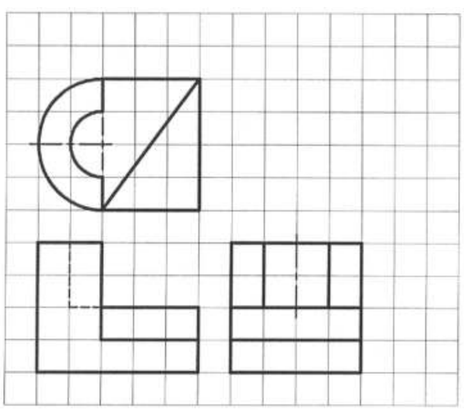
Solution of P3.20



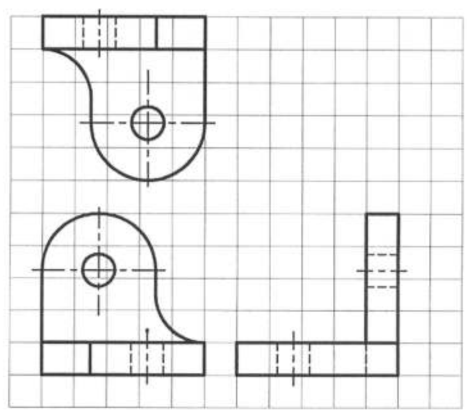
Solution of P3.21



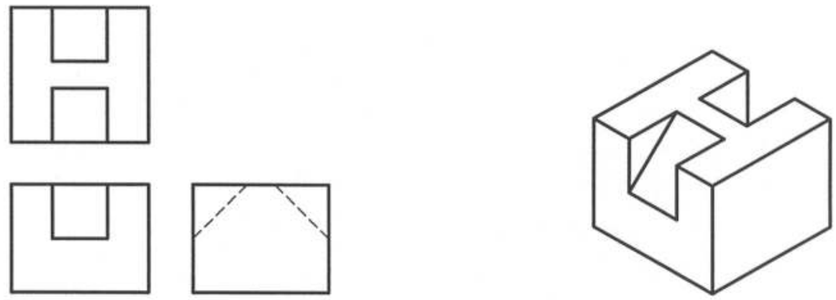
Solution of P3.22



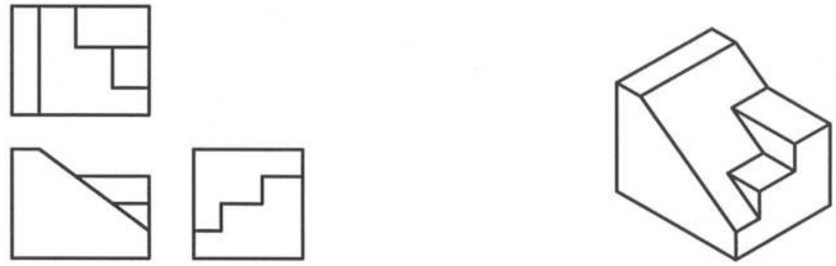
Solution of P3.23



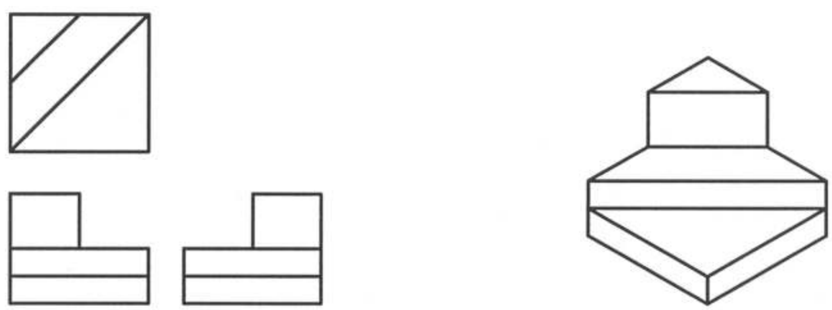
Solution of P3.24



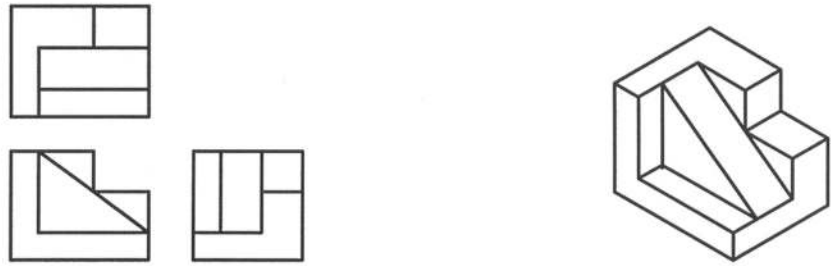
Solution of P3.25



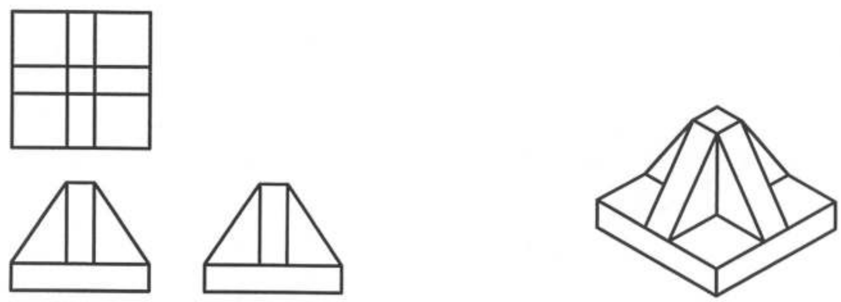
Solution of P3.26



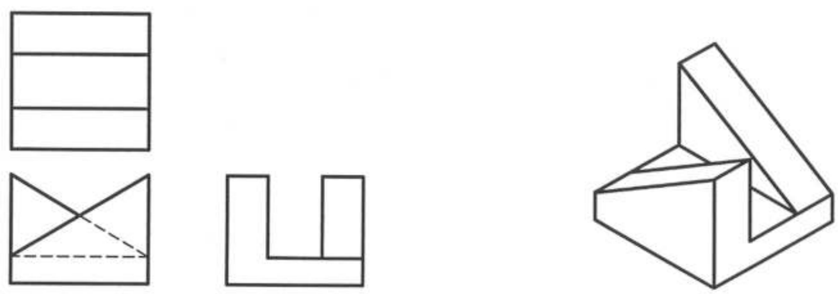
Solution of P3.27



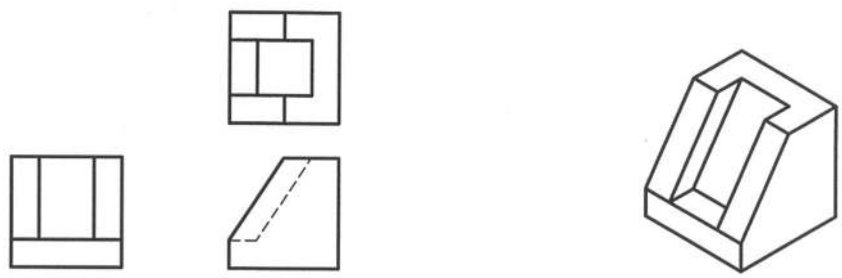
Solution of P3.28



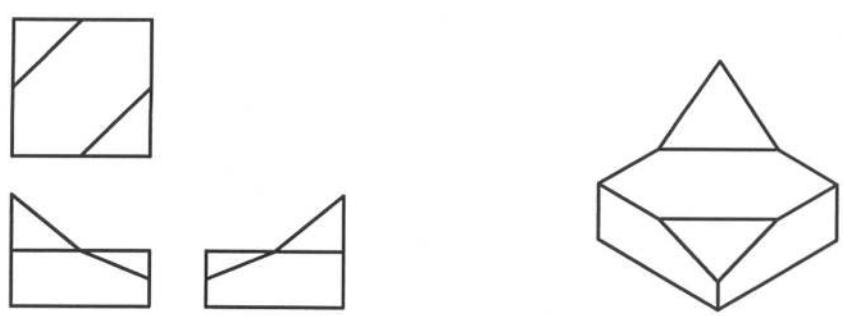
Solution of P3.29



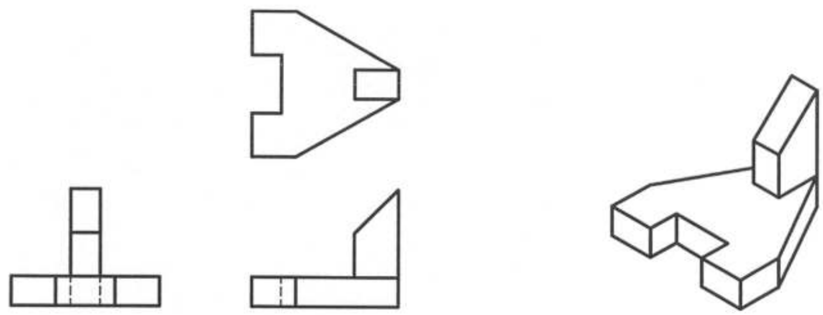
Solution of P3.30



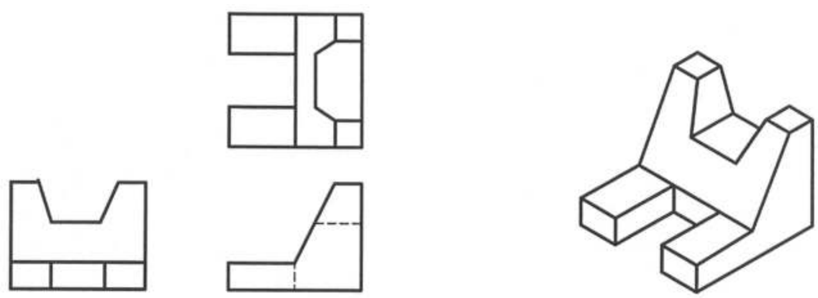
Solution of P3.31



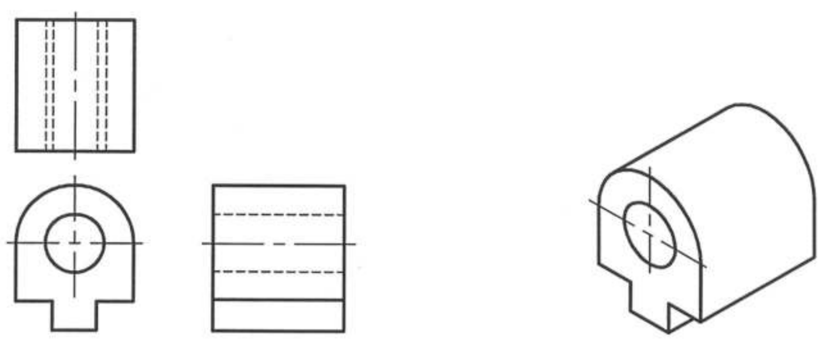
Solution of P3.32



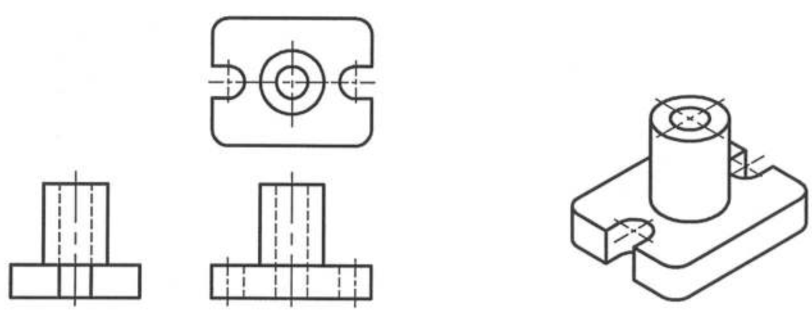
Solution of P3.33



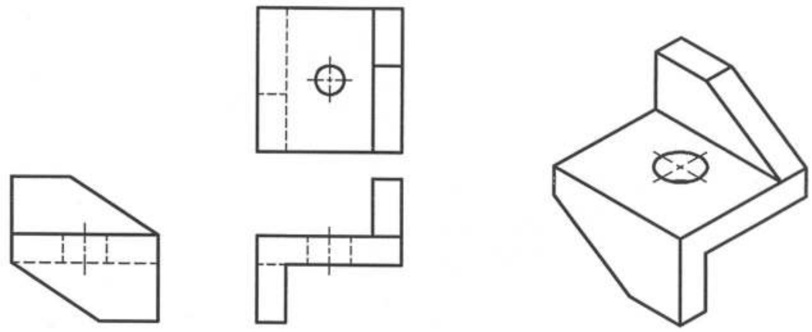
Solution of P3.34



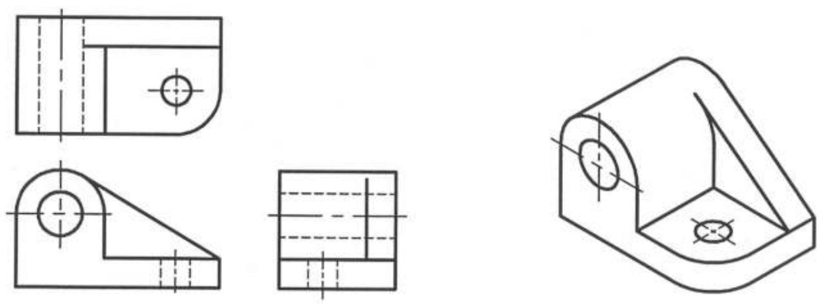
Solution of P3.35



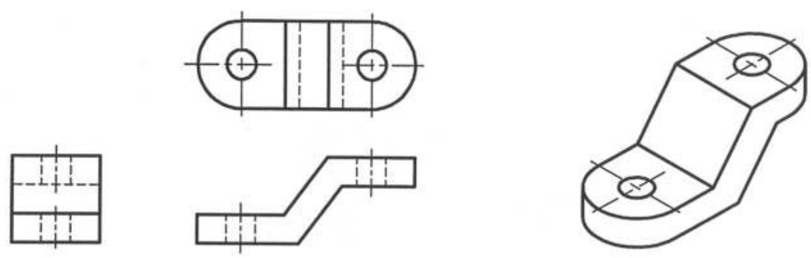
Solution of P3.36



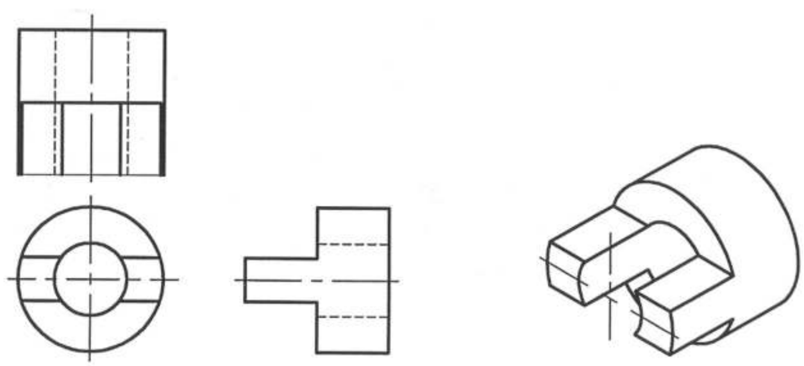
Solution of P3.37



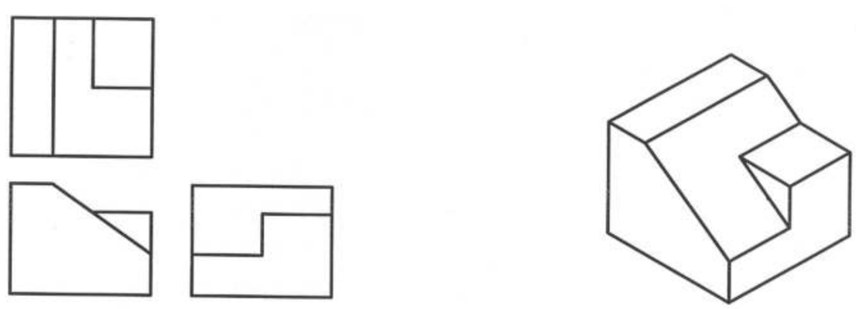
Solution of P3.38



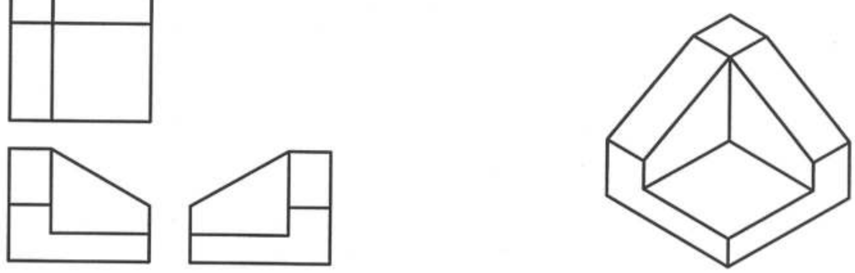
Solution of P3.39



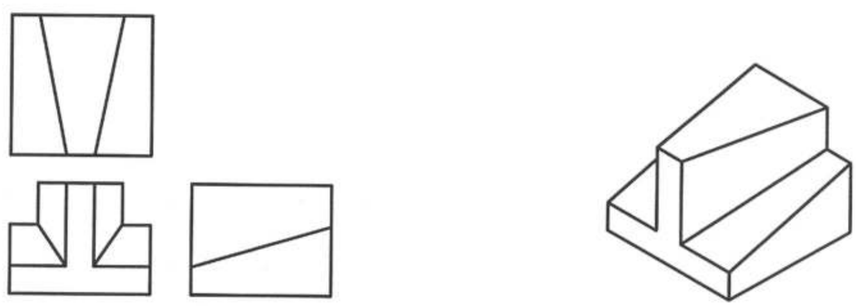
Solution of P3.40



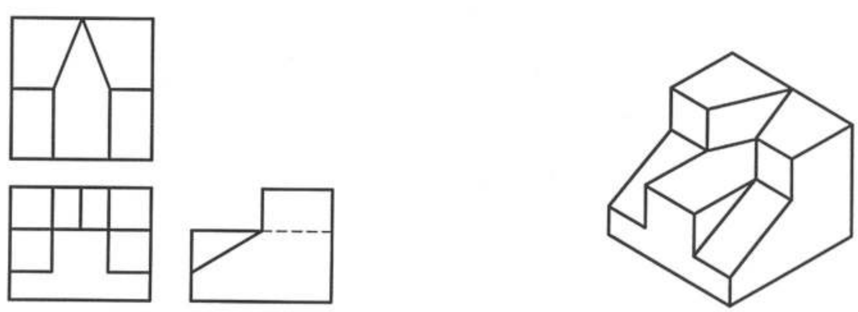
Solution of P3.41



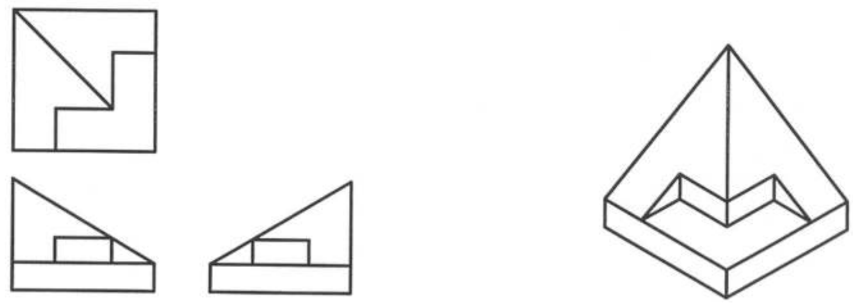
Solution of P3.42



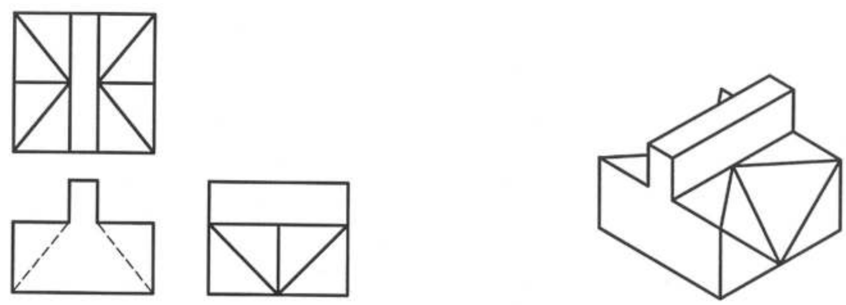
Solution of P3.43



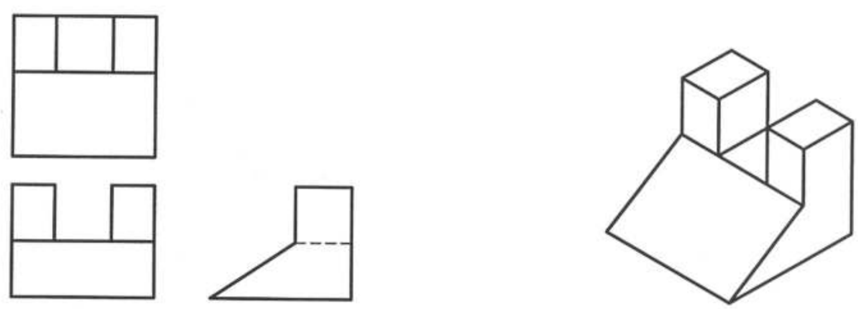
Solution of P3.44



Solution of P3.45



Solution of P3.46



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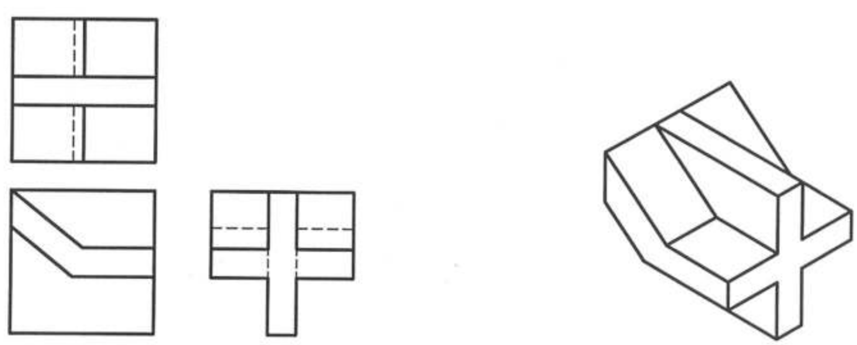

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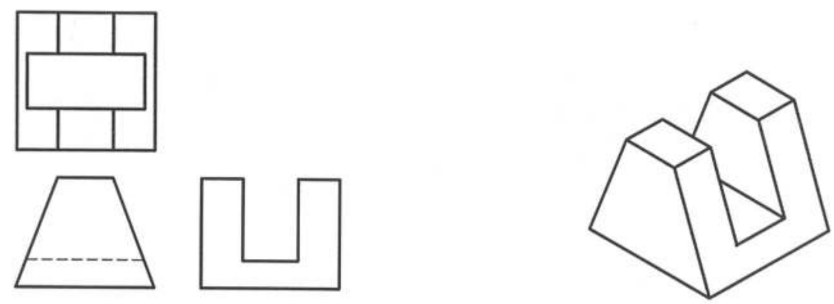
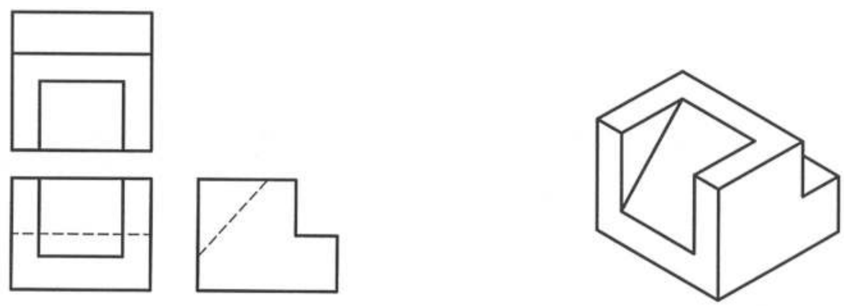
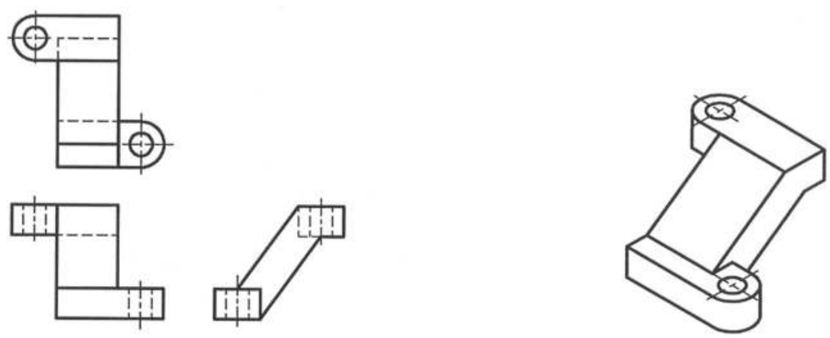


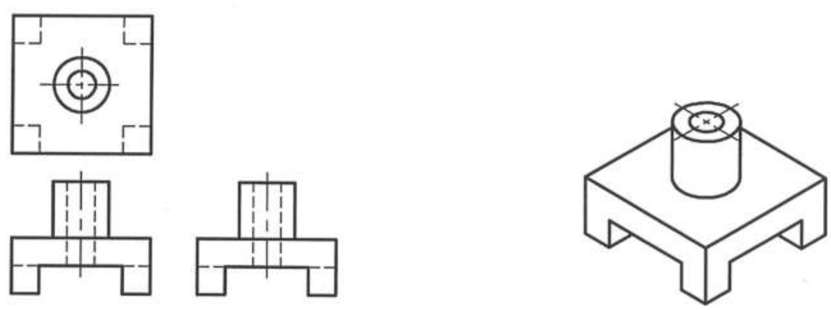
Solution of P3.49



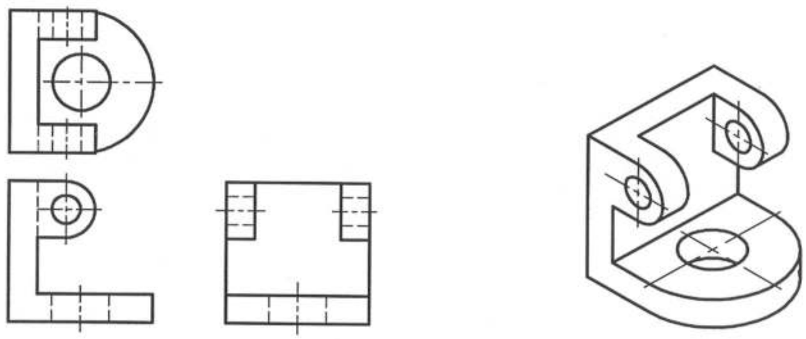
Solution of P3.50



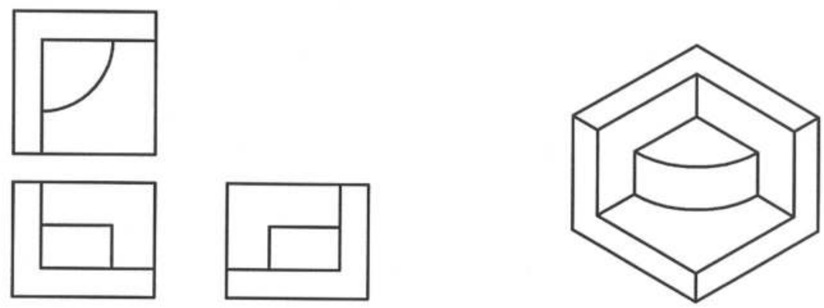
Solution of P3.51



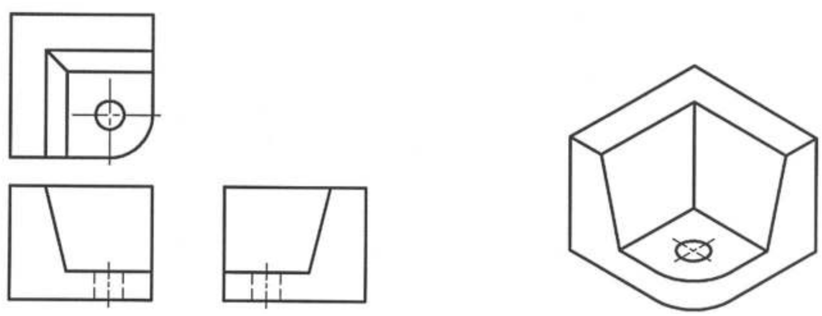
Solution of P3.52



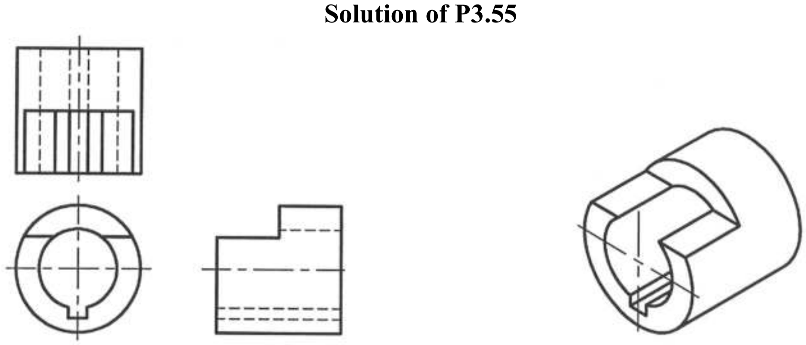
Solution of P3.53



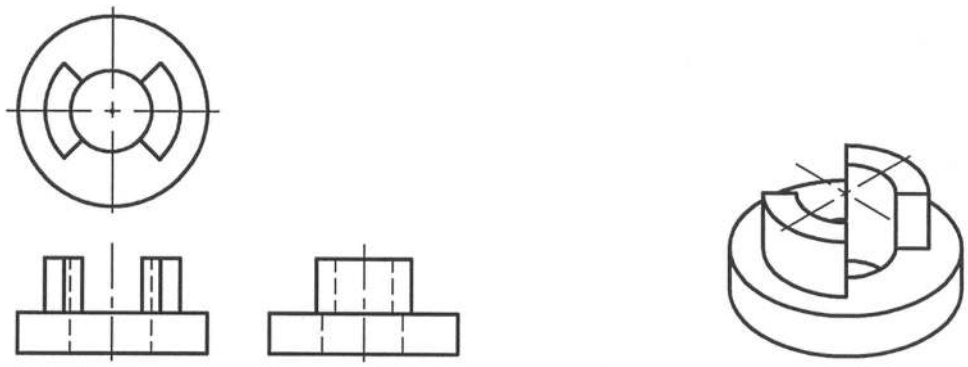
Solution of P3.54



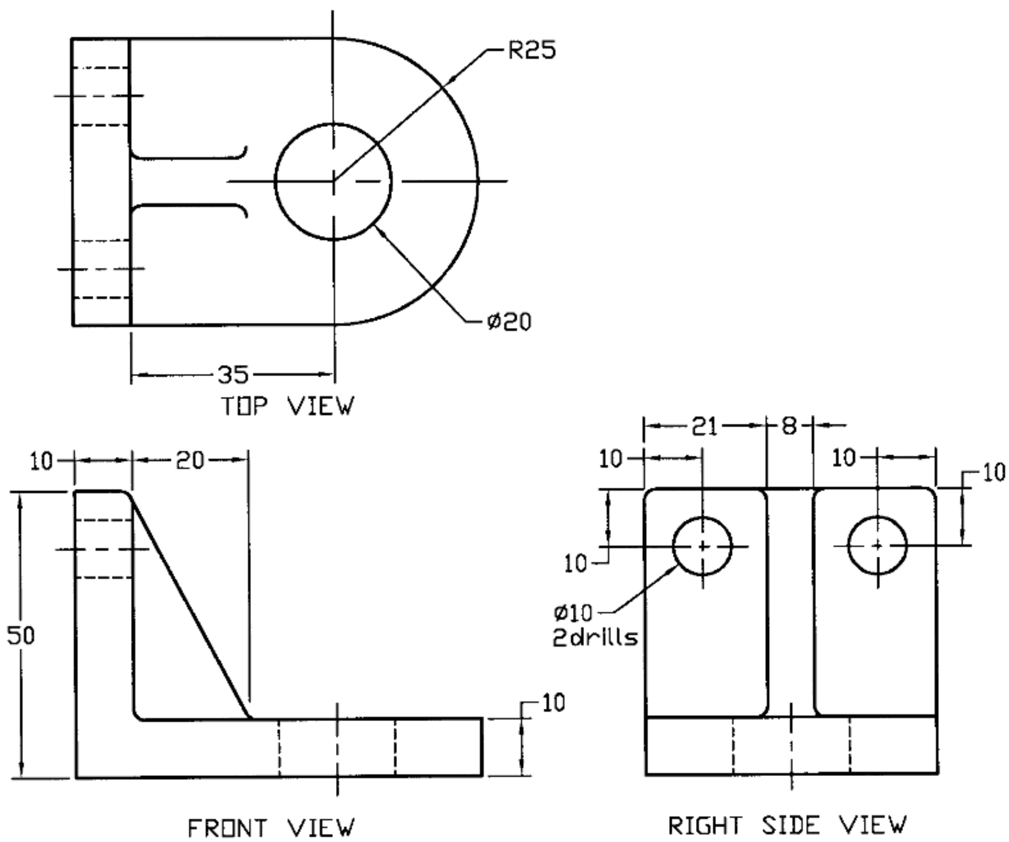
Solution of P3.55



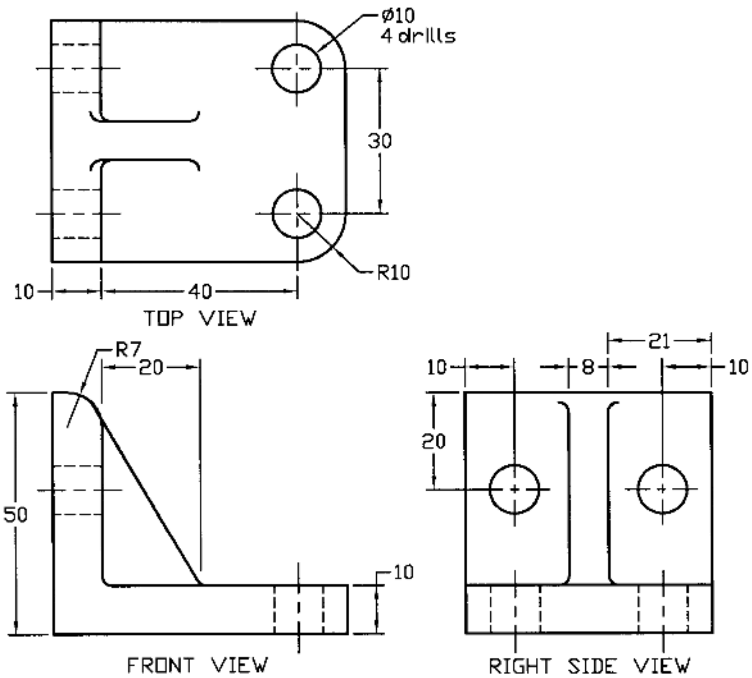
Solution of P3.56



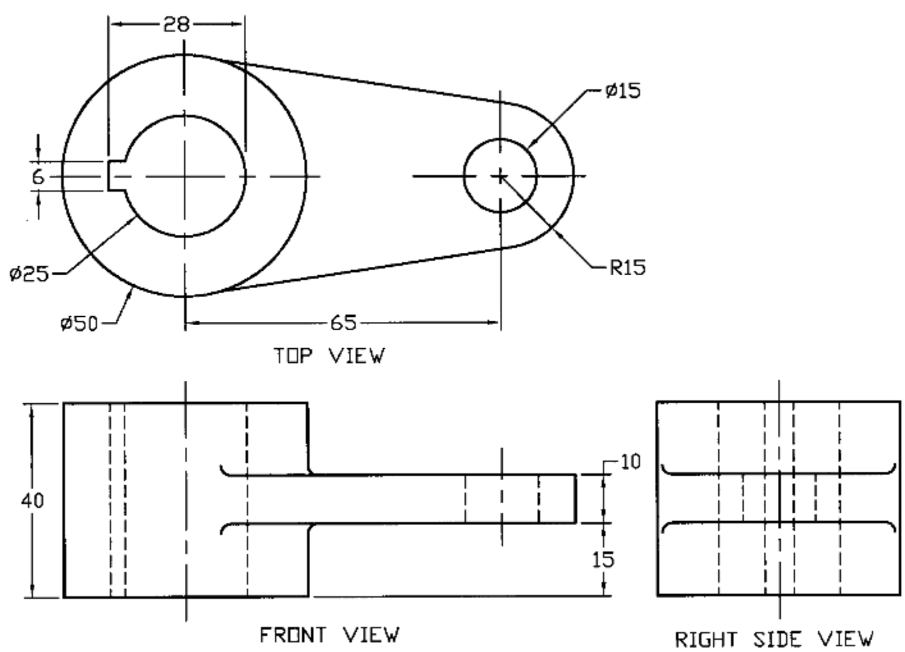
Solution of P3.57



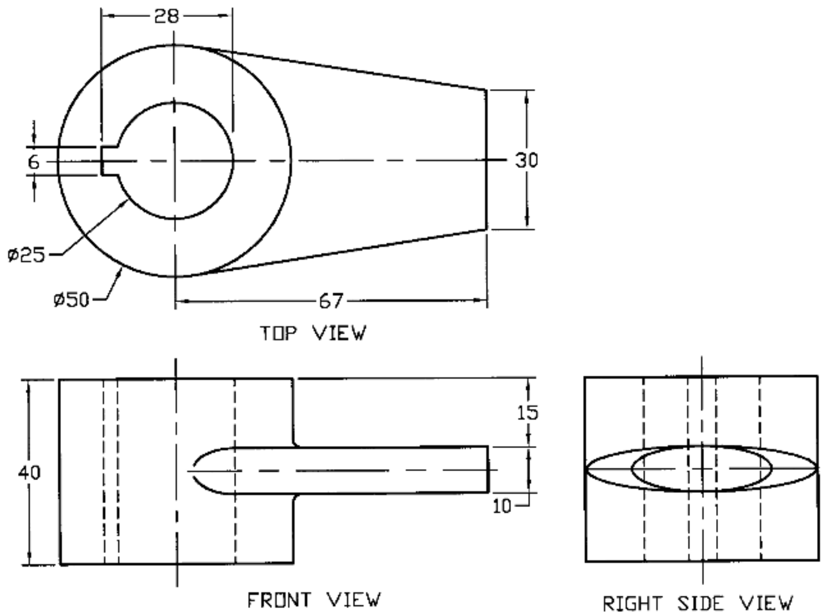
Solution of P3.58



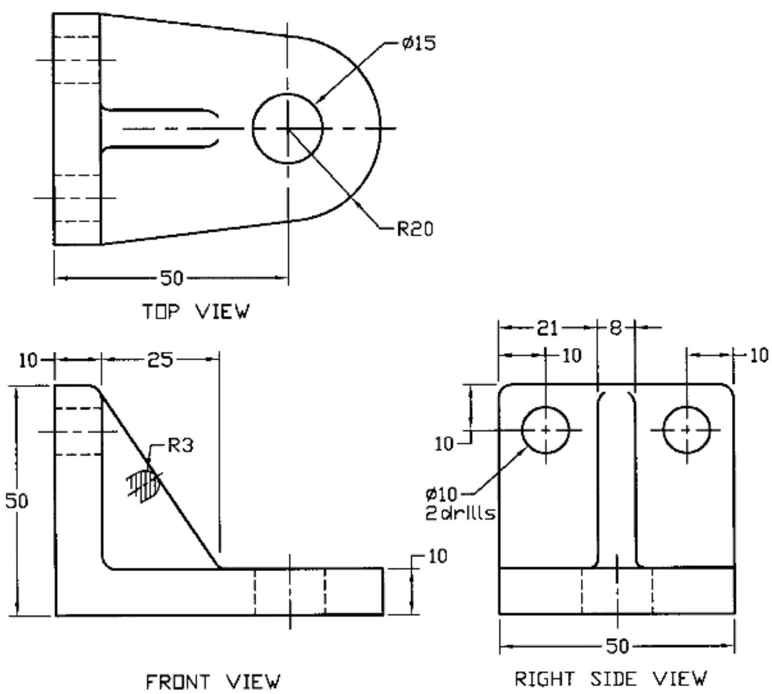
Solution of P3.59



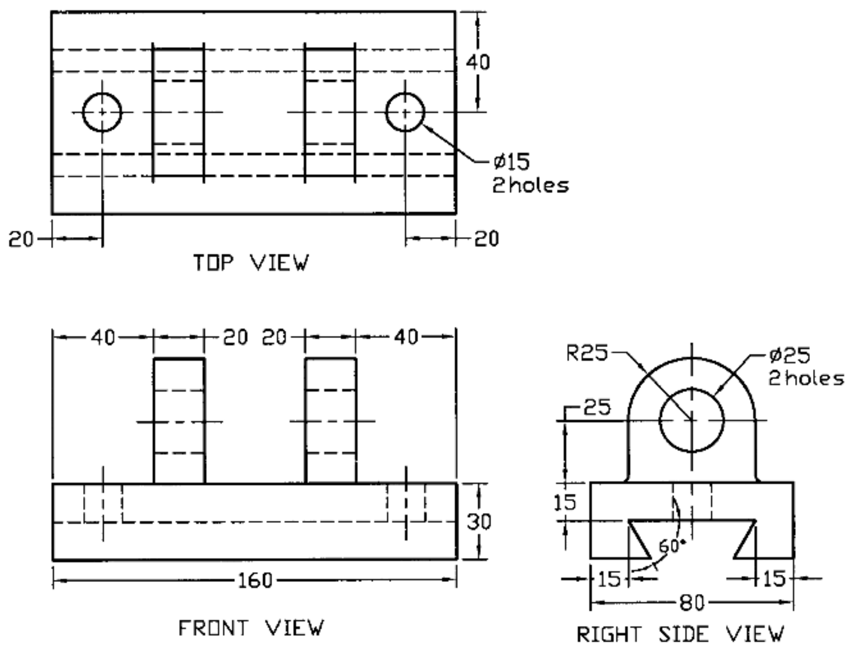
Solution of P3.60



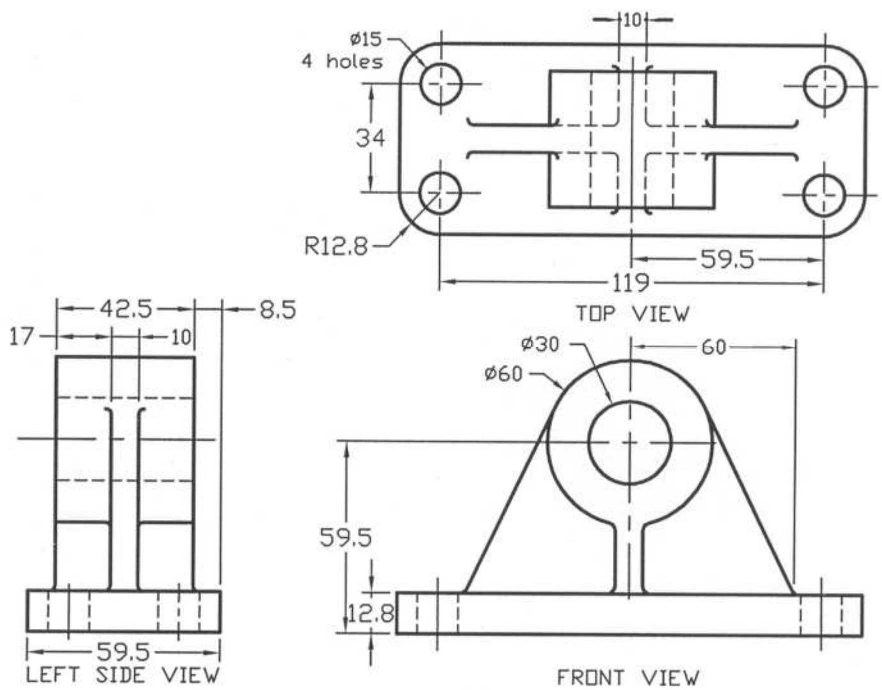
Solution of P3.61



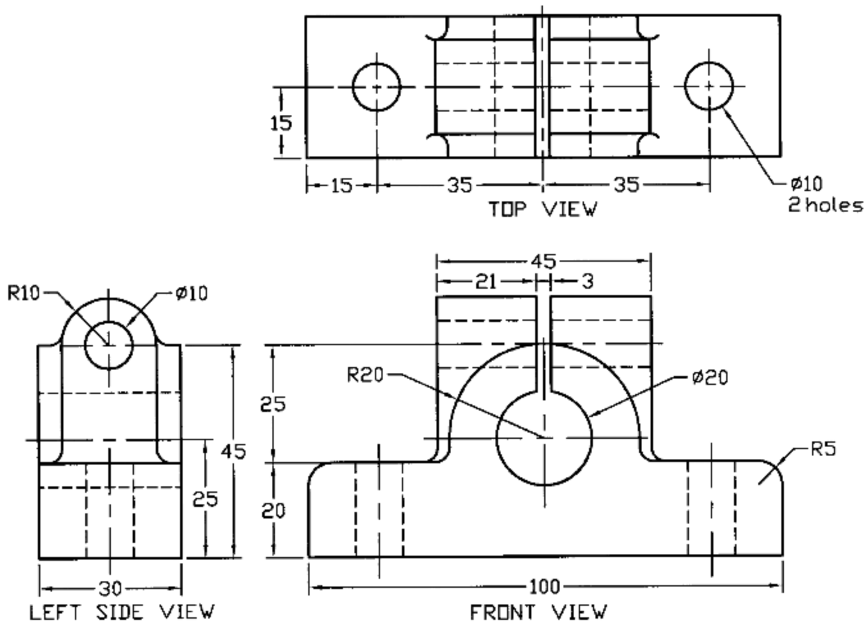
Solution of P3.62



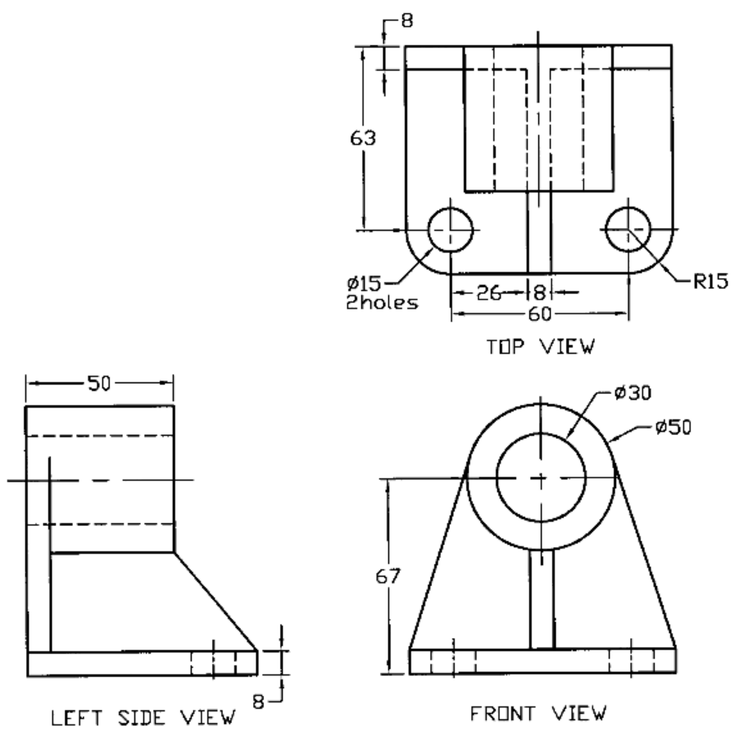
Solution of P3.63



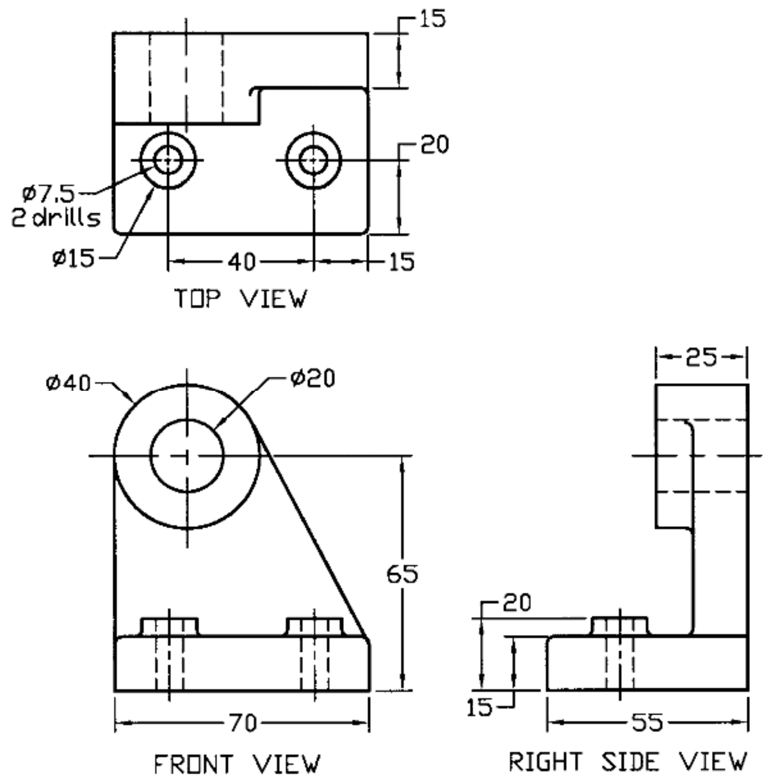
Solution of P3.64



Solution of P3.65



Solution of P3.66



Solution of P3.67

Problems

Prob. P3.68 – P3.91: Draw necessary free hand orthographic projections of the objects as shown in Fig. P3.68 – P3.91. Free hand drawings may be done.

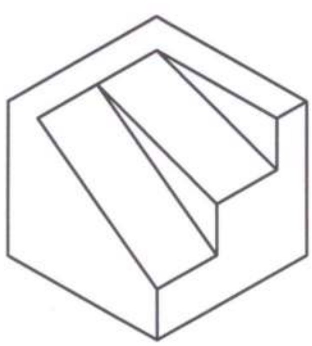


Fig. P3.68

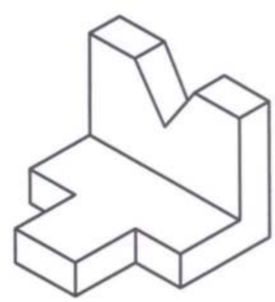


Fig. P3.69

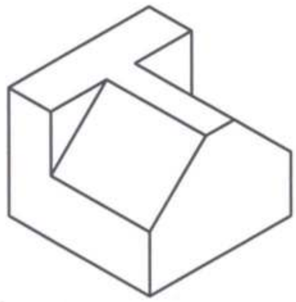


Fig. P3.70

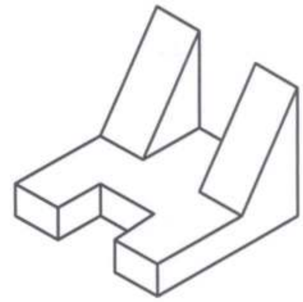


Fig. P3.71

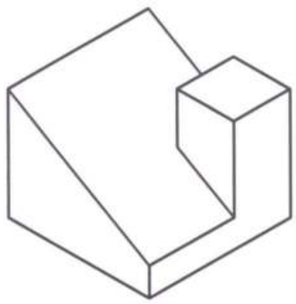


Fig. P3.72

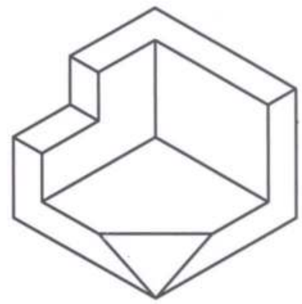


Fig. P3.73

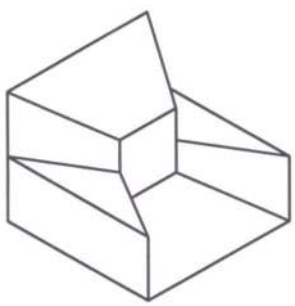


Fig. P3.74

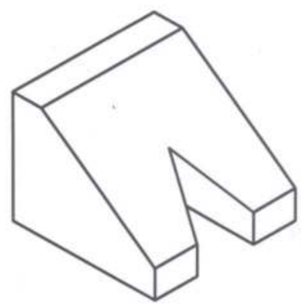


Fig. P3.75

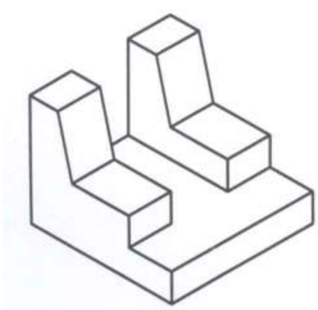


Fig. P3.76

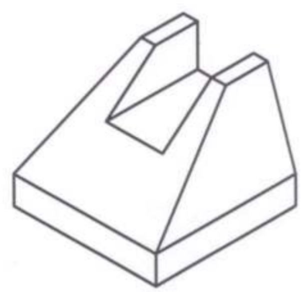


Fig. P3.77

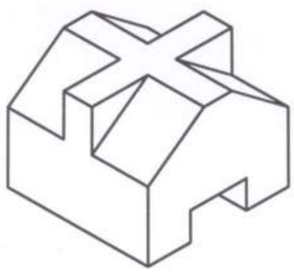


Fig. P3.78

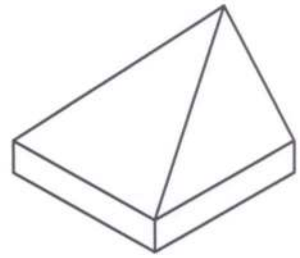


Fig. P3.79

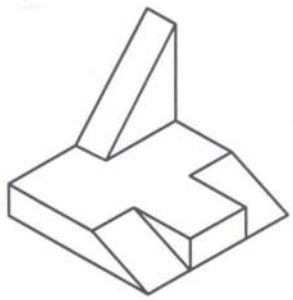


Fig. P3.80

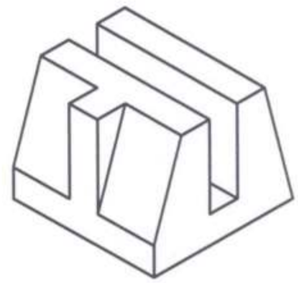


Fig. P3.81

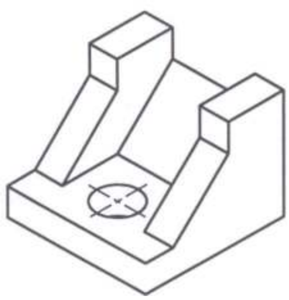


Fig. P3.82

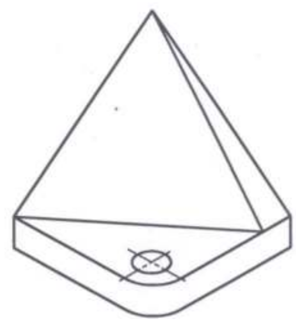


Fig. P3.83

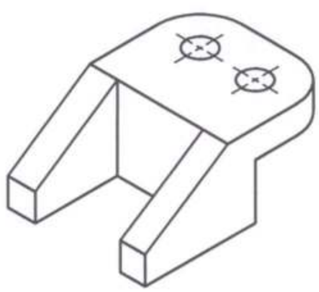


Fig. P3.84

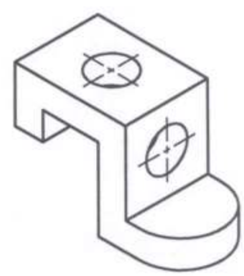


Fig. P3.85

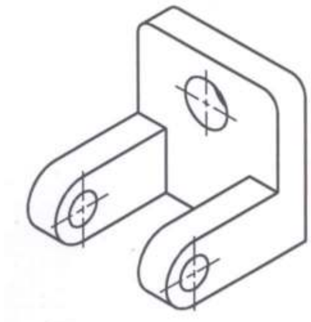


Fig. P3.86

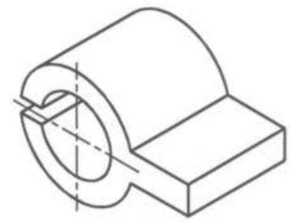


Fig. P3.87

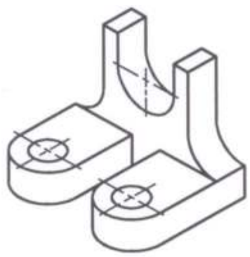


Fig. P3.88

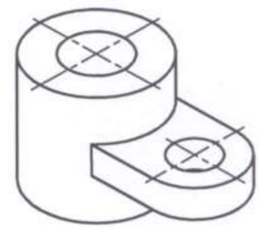


Fig. P3.89

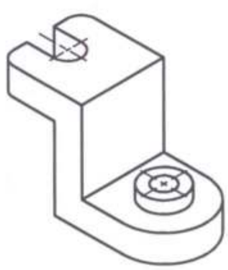


Fig. P3.90

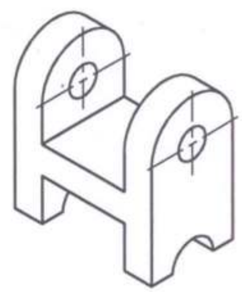


Fig. P3.91

P3.92 – P3.107: Complete the missing view from the views as shown in Fig. P3.92 – P3.107.

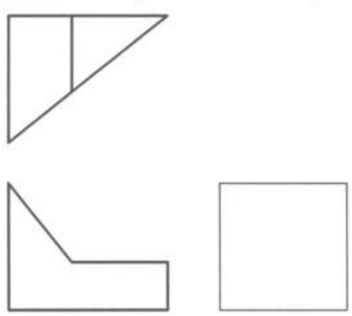


Fig. P3.92

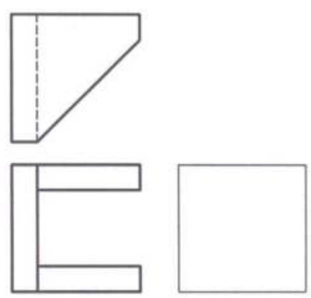


Fig. P3.93

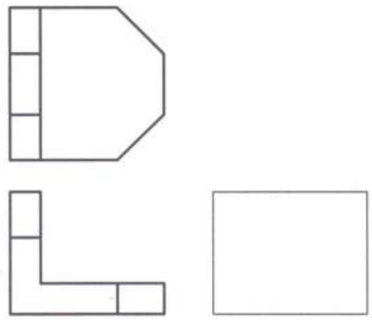


Fig. P3.94

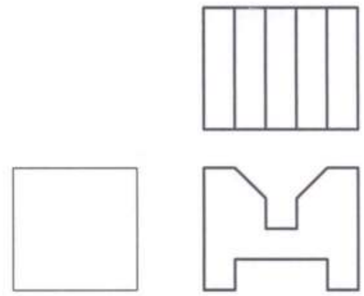


Fig. P3.95

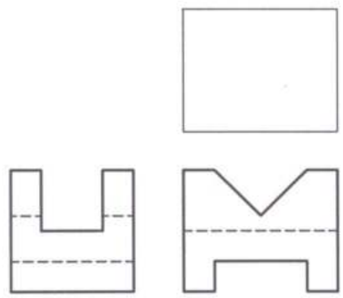


Fig. P3.96

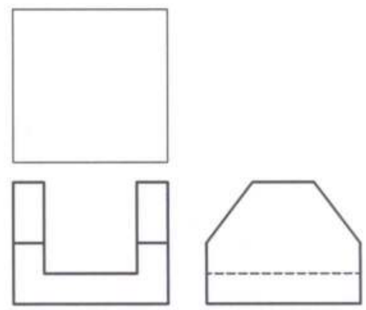


Fig. P3.97

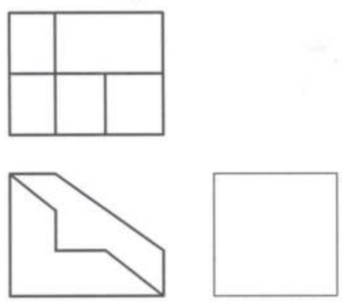


Fig. P3.98

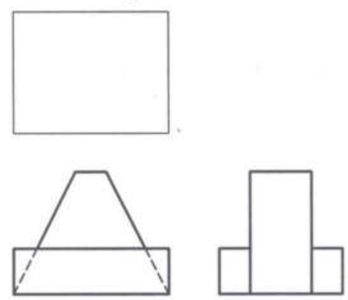


Fig. P3.99

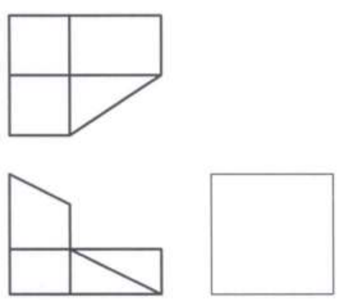


Fig. P3.100

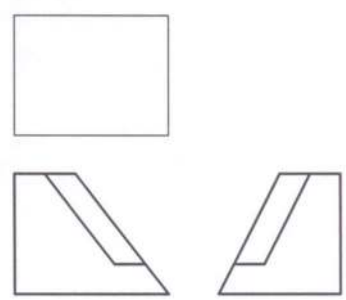


Fig. P3.101

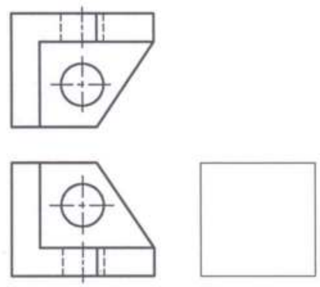


Fig. P3.102

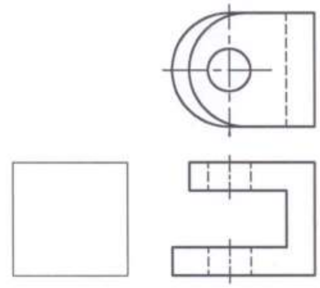


Fig. P3.103

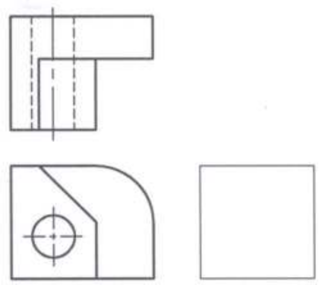


Fig. P3.104

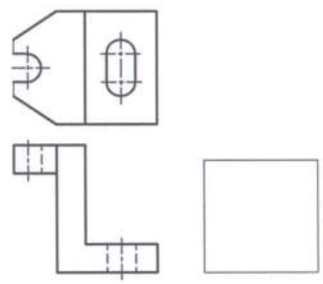


Fig. P3.105

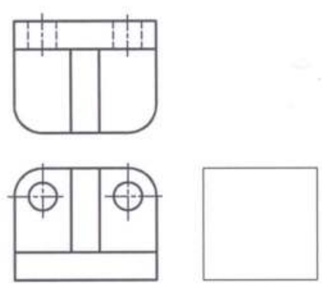


Fig. P3.106

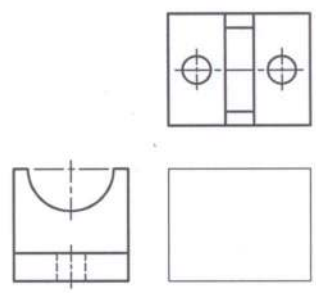


Fig. P3.107

Prob. P3.108 – P3.123: Draw the missing line/lines in the views as shown in Fig. P3.108 – P3.123.

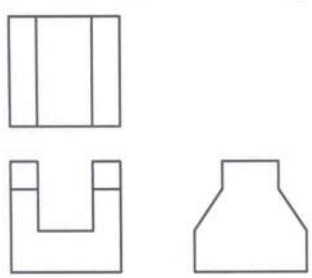


Fig. P3.108

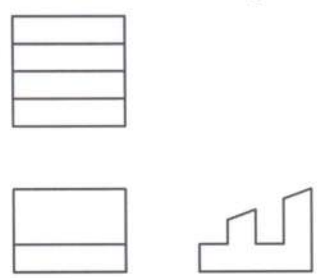


Fig. P3.109

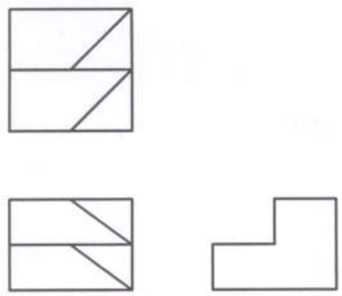


Fig. P3.110

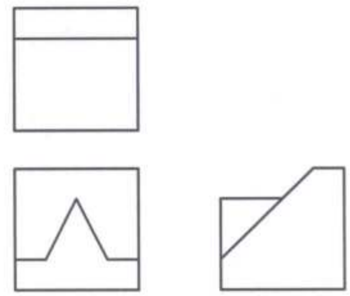


Fig. P3.111

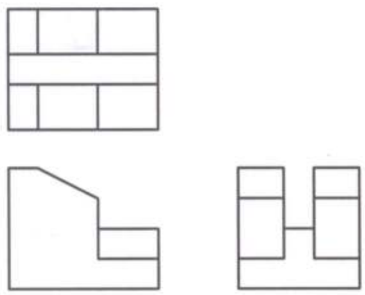


Fig. P3.112

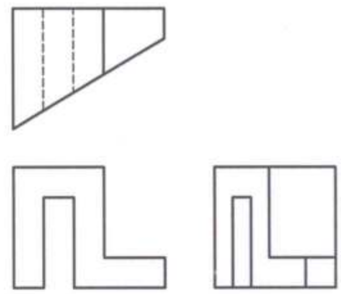


Fig. P3.113

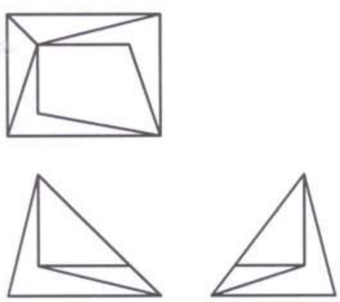


Fig. P3.114

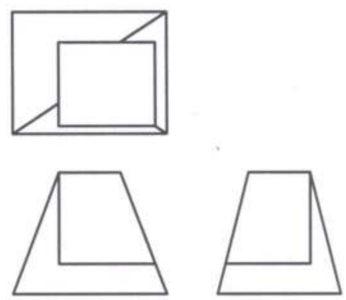


Fig. P3.115

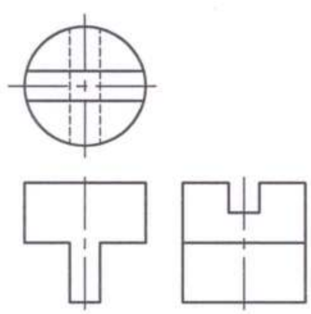


Fig. P3.116

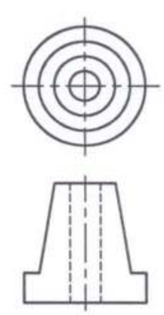


Fig. P3.117

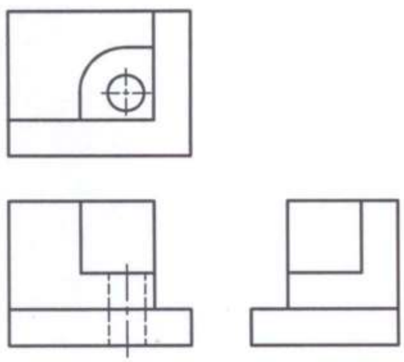


Fig. P3.118

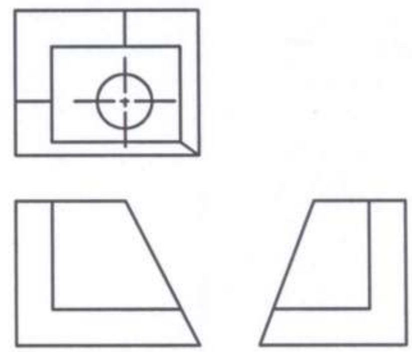


Fig. P3.119

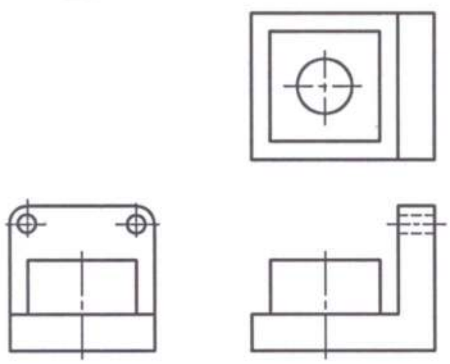


Fig. P3.120

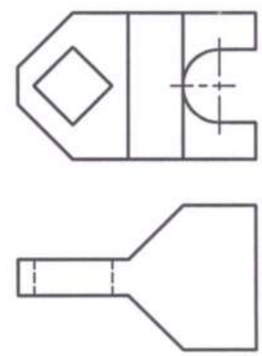


Fig. P3.121

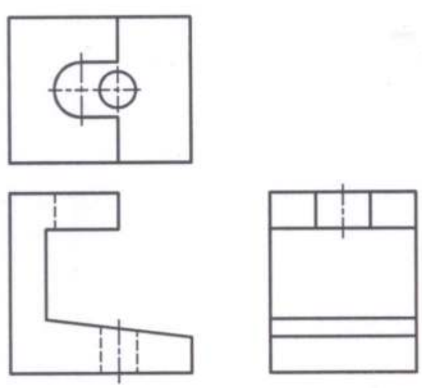


Fig. P3.122

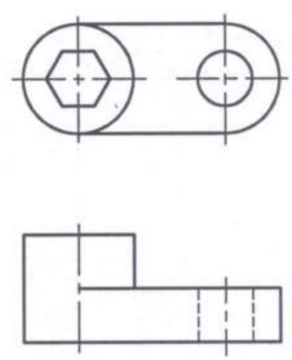


Fig. P3.123

Prob. P3.124: Draw the top, front and left side views of the V-slide as shown in Fig. P3.124.

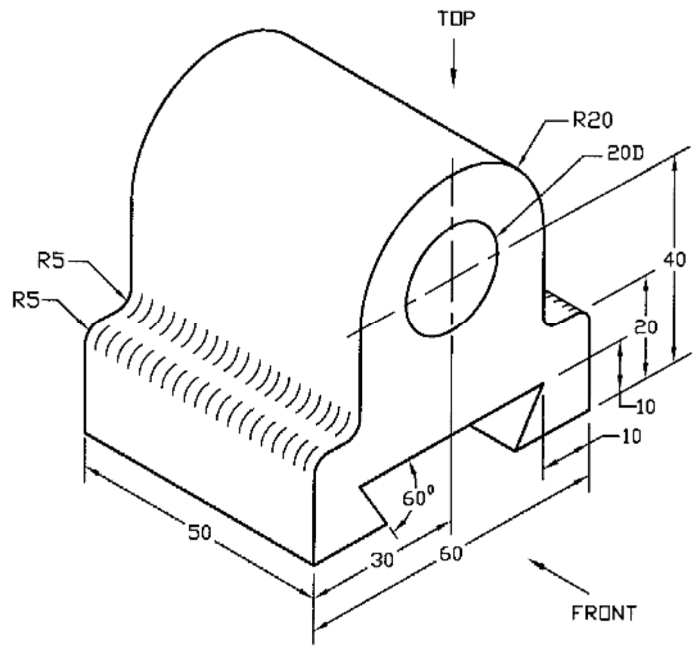


Fig. P3.124

Prob. P3.125: Draw the top, front and right side views of the hanger support as shown in Fig. P3.125.

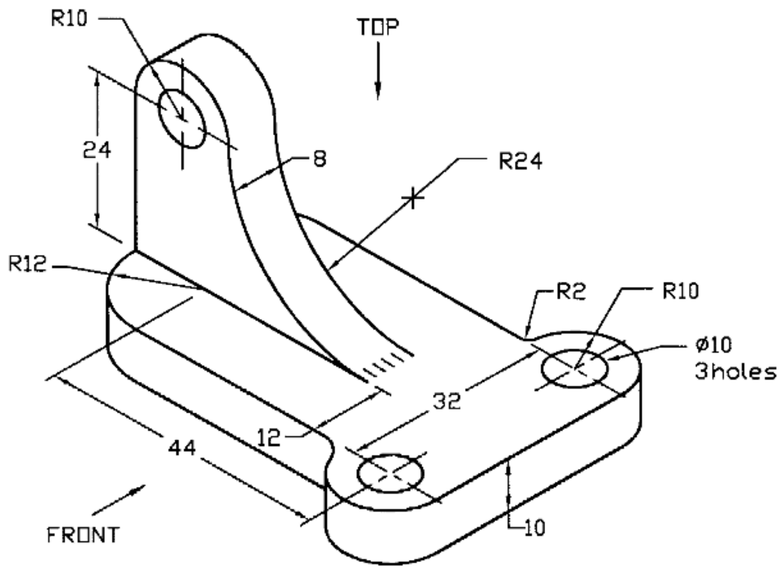


Fig. P3.125

Prob. P3.126: Draw the top, front and left side views of the wedge block as shown in Fig. P3.126.

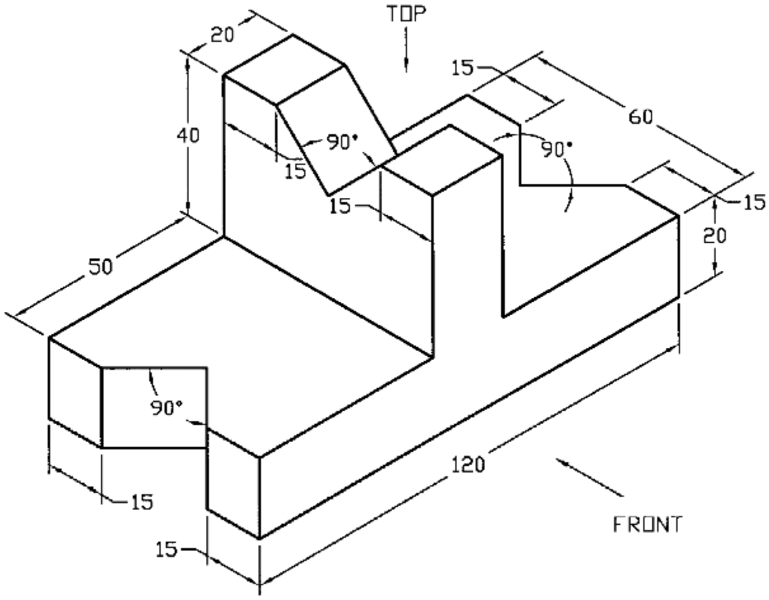


Fig. P3.126

Prob. P3.127: Draw the top and front views of the cylinder as shown in Fig. P3.127.

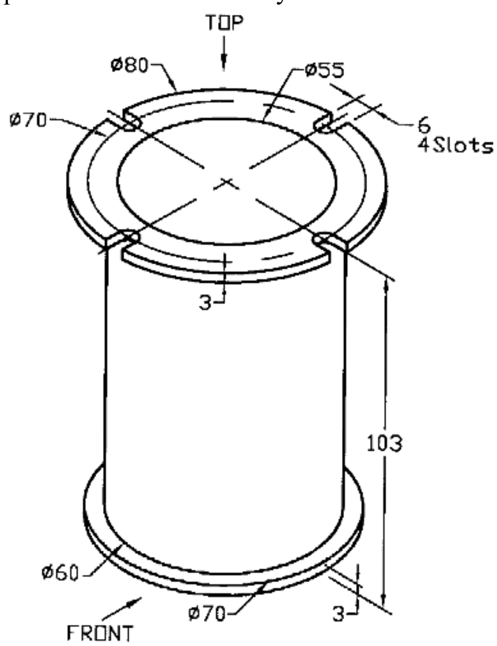


Fig. P3.127

Prob. P3.128: Draw the top, front and left side views of the bearing bracket as shown in Fig. P3.128.

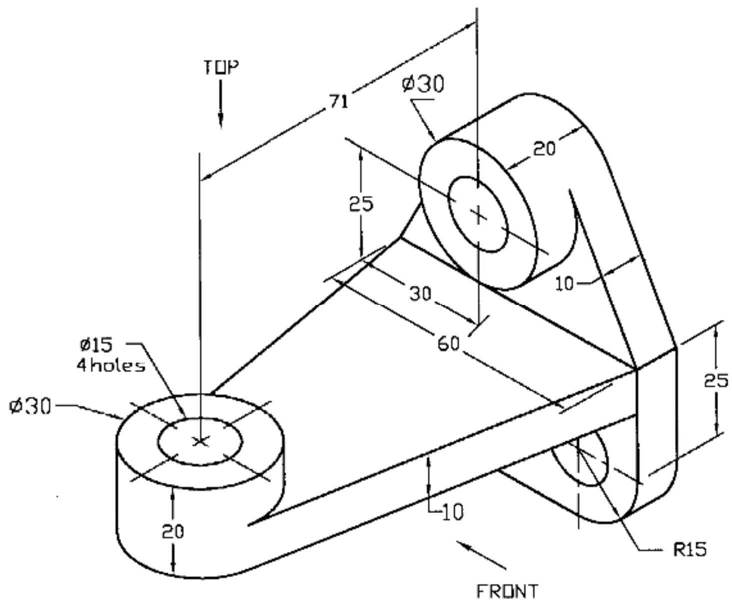


Fig. P3.128

Prob. P3.129: Draw the top, front and right side views of the support block as shown in Fig. P3.129.

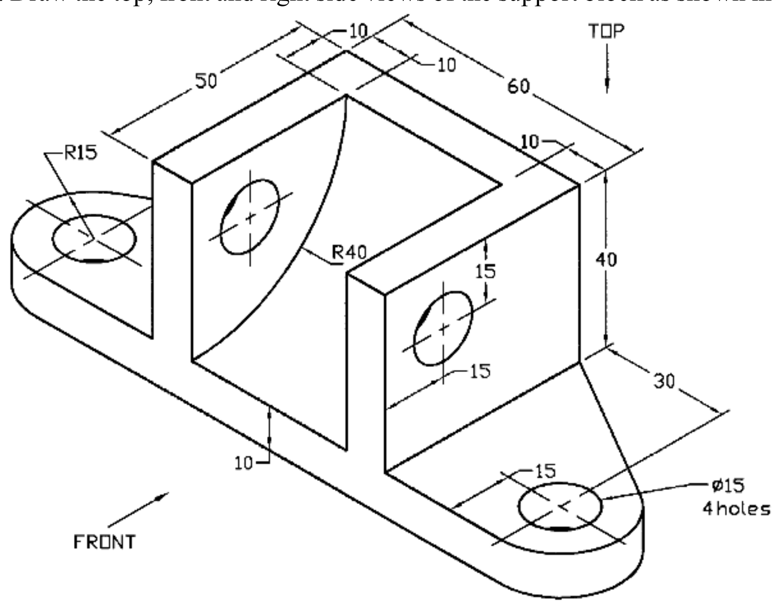


Fig. P3.129

Prob. P3.130: Draw the top, front and right side views of the bracket as shown in Fig. P3.130 (Fillets are 3mm).

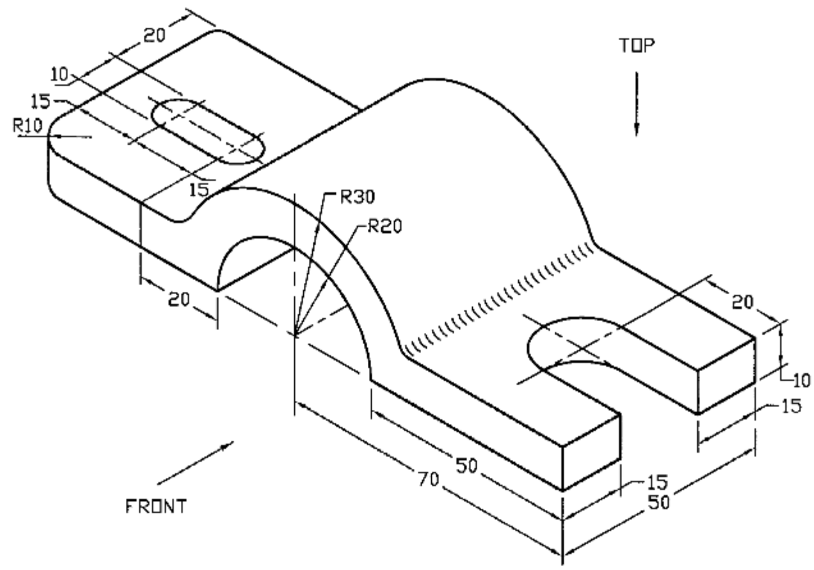


Fig. P3.130

Prob. P3.131: Draw the top, front and right side views of the bracket as shown in Fig. P3.131.

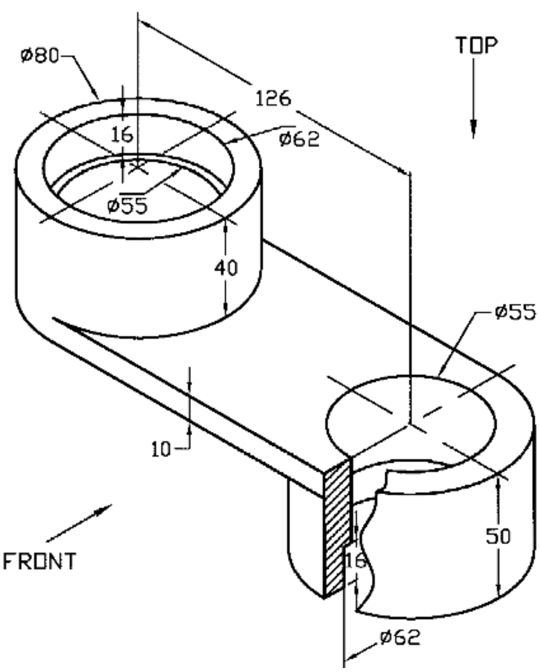


Fig. P3.131

CHAPTER 4

SECTIONAL VIEWS AND CONVENTIONS

4.1 *Introduction*

The main objective of a drawing is to represent the size and shape of an object clearly. Sometimes it so happens that lots of hidden lines are required to show the interior detail of an object. As such it becomes very difficult to visualize the object thereby jeopardizing the primary objective of the drawing. In order to overcome this problem, a sectional view is often necessary. A sectional view is that view, which is seen beyond the imaginary cutting plane through an object at right angle to the direction of sight. It represents the interior construction or details of hidden features clearly to the users.

The exposed or cut surface is identified with the help of section lining or cross-hatching. Hidden lines and details behind the cutting-plane line are usually omitted unless they are essential for clarity or dimensioning. A sectional view frequently replaces one of the regular views. A regular front view may be replaced by a front sectional view.

4.2 *Generating Sectional View*

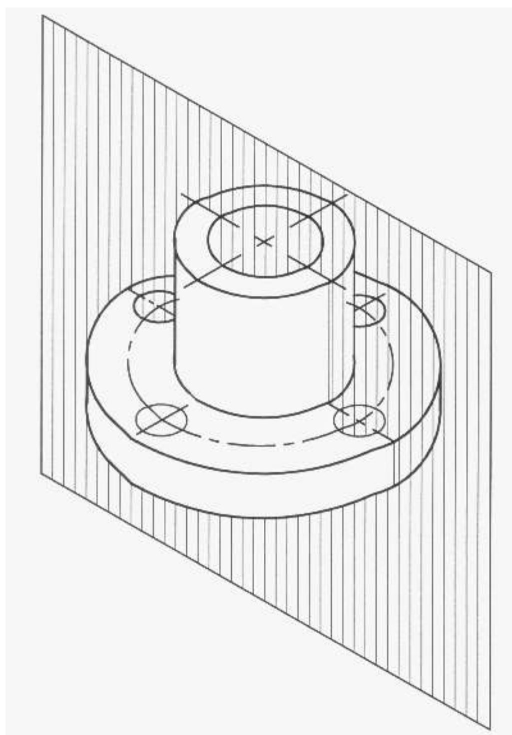


Figure 4.1: Object with Cutting Plane Through Mid-Section

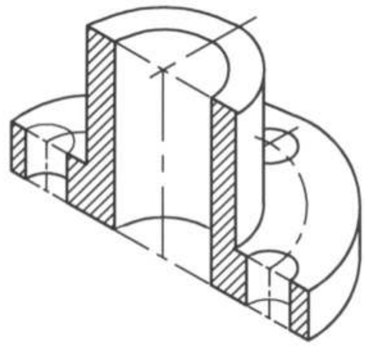
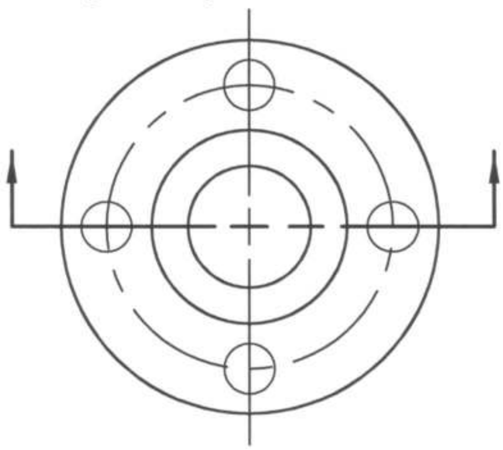
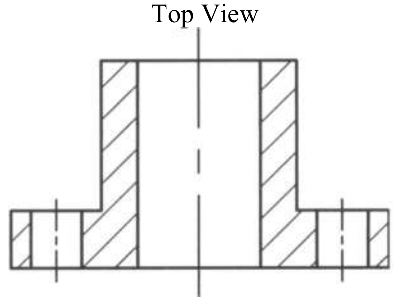


Figure 4.2: Section After Cutting and Removal of Front Portion

It has been represented in Figures 4.1 and 4.2 how to generate sectional view. Figure 4.1 shows a cutting-plane passing through the mid-section of the object while Figure 4.2 shows the section of the object after removal of the front portion. The interior of the object has become very clear after cutting. A cutting-plane line represents the plane, along which the object is cut. In Figure 4.3, the cutting plane line has been shown on the view of the object, along which section has been made. The sectional view has also been given in Figure 4.3.



Top View



Sectional View

Figure 4.3: Views Showing Cutting Plane and Section

4.3 ***Section Lining***

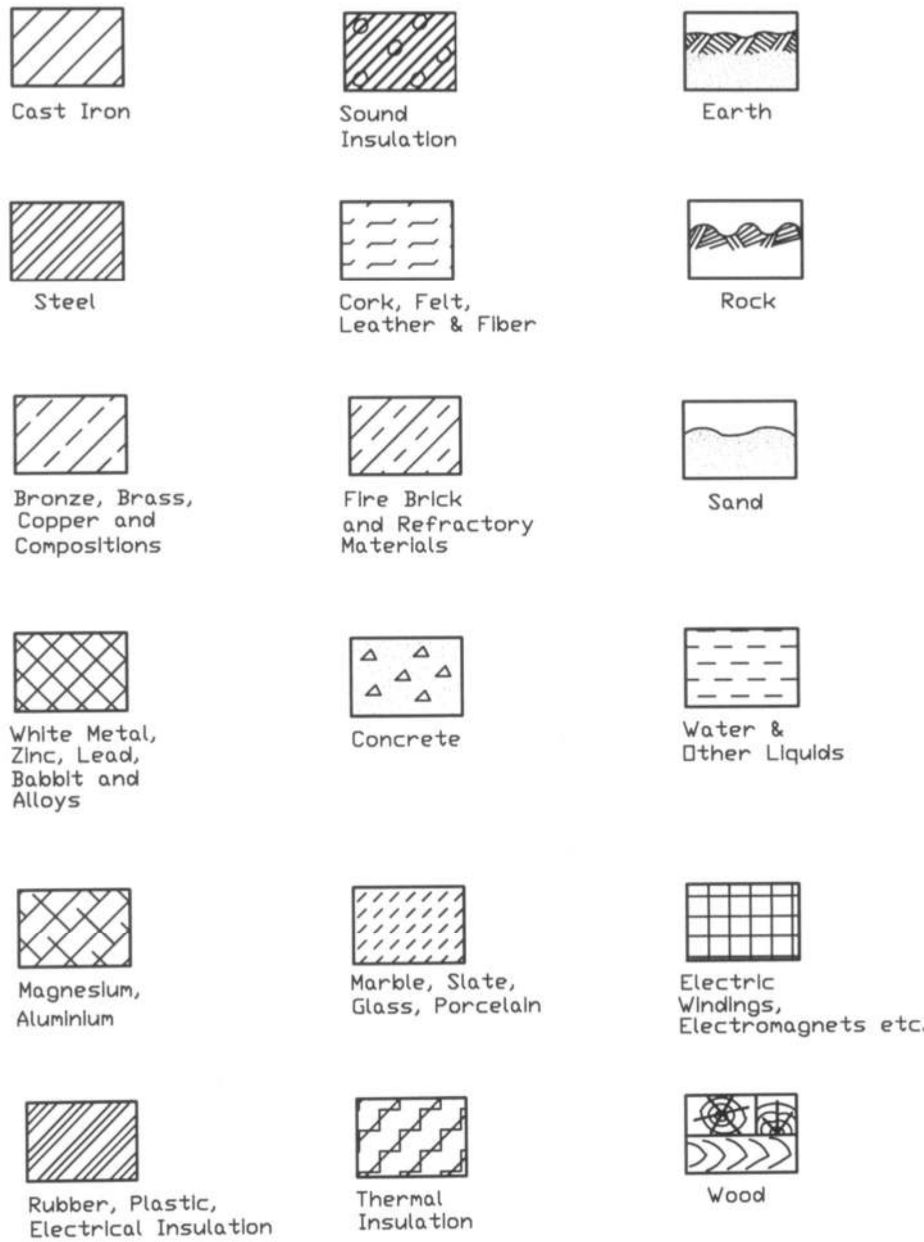


Figure 4.4: Symbols for Section Lining

Section lining is often called cross-hatching. The purpose of section line is to indicate the surface that has been cut hypothetically thereby clarifying the internal shape of the object and the material from which the object is made of. The symbols of section lining for various materials have been given in Figure 4.4.

The lines used to indicate section are thin and they are usually drawn at an angle of 45^0 to the major outline of the object. The spacing of the lines has to be reasonably uniform for good appearance. The pitch or in other words the perpendicular distance between the consecutive lines may vary between 1 to 3 mm.

When two adjacent pieces are to be crosshatched in an assembly drawing, they may be done in opposite directions. However, for more than two pieces, section angle other than 45^0 such as 30^0 or 60^0 may be used. Alternatively, all pieces may be sectioned at the same angle of 45^0 but with different pitch. If a part is so shaped that section lining at 45^0 runs parallel to its principal outlines, another direction may be chosen (Figure 4.5a). For the large area, section lines may be provided only on the edges of the area (Figure 4.5b) depending on the size of the surface to be sectioned.

When the thickness is small such as for packing, gaskets, sheet metal, plastic sheet etc., the section lining may be omitted and the area may be filled in completely (4.5c). Dimensions or other lettering should not be placed in sectional areas in general. When it becomes essential omit the section lining for them where they have to be inserted (Figure 4.5d).

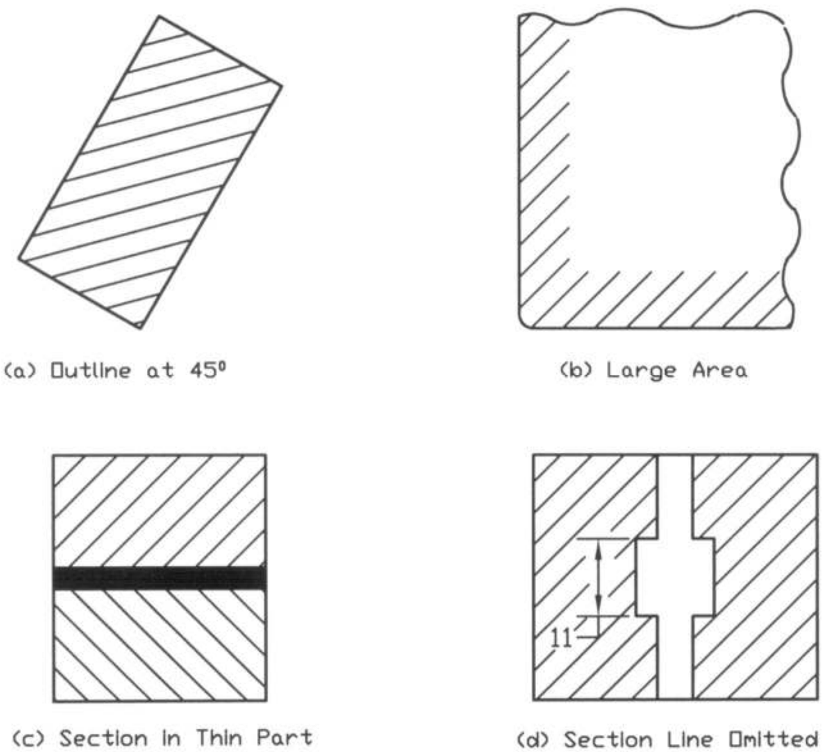


Figure 4.5: Sectioning in Special Features

4.4 ***Types of Sections***

There are many types of sections, which are used in engineering drawings. These are explained in concise form in the following subsections.

4.4.1 *Full Section*

When a cutting plane passes through the entire object, the section obtained is called the full section of the object. The cutting plane may pass through the object either straight or change direction while passing through. Figure 4.2 shows the full section of the object and Figure 4.3 provides the full section view of the same object.

4.4.2 *Half Section*

Sometimes for the symmetric object, one half is drawn in section and the other half is drawn as a regular exterior view. Cutting plane line is extended half way across, then in forward direction being perpendicular. Figure 4.6 shows the removal of the quarter portion of the object. In Figure 4.7 the cutting plane line has been shown. On the section view either a center line (Figure 4.8) or a visible line (Figure 4.9) is used to divide the sectional half from the unsectioned half of the drawing.

A half section has the advantage of showing both the interior and the exterior of the object on one view without using hidden lines. However, when dimensioning is to be given, hidden line may be added to accomplish that (Figure 4.10). In the assembly drawing half section may be used showing both internal and external construction where only overall and center-to-center dimensions are necessary.

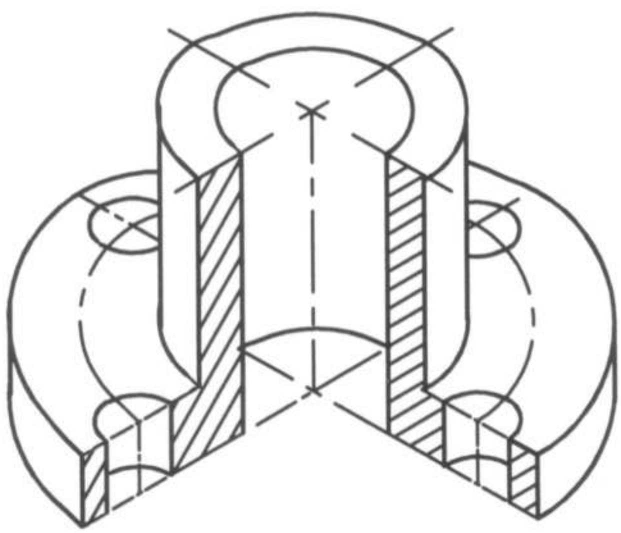


Figure 4.6: Object With Half Section

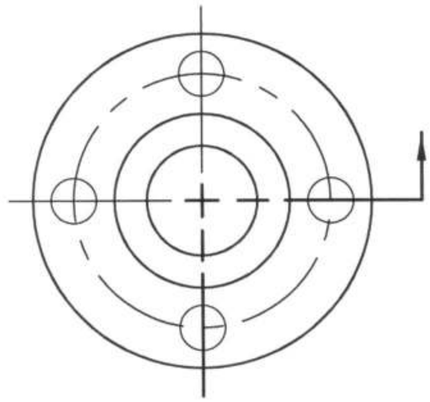


Figure 4.7: View With Cutting Plane Line

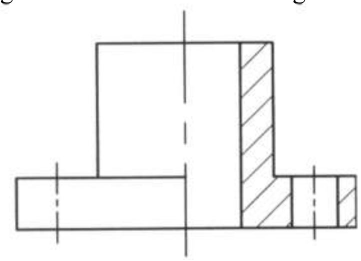


Figure 4.8: View With Half Section
(Divided With Center Line)

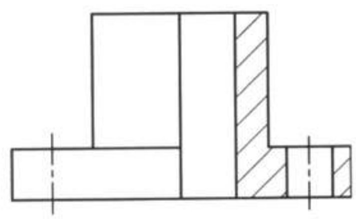


Figure 4.9: View With Half Section
(Divided With Visible Line)

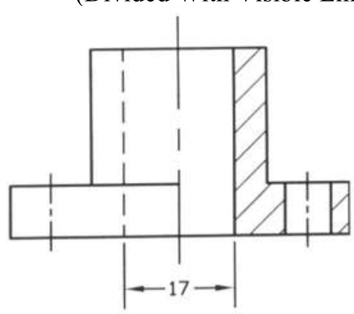


Figure 4.10: View With Half Section
(For Dimensioning)

4.4.3 ***Broken-Out Section***

When an interior portion of an object is needed to visualize by a section, but full or half section cannot be made because that removes some essential feature of the object. The section is made directly on the exterior view. An irregular freehand line is drawn to limit the break. An object with broken-out section and its view are shown in Figures 4.11 and 4.12 respectively.

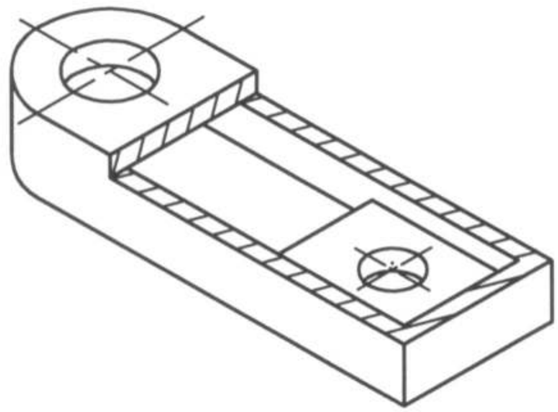


Figure 4.11: Object with Broken-Out Section

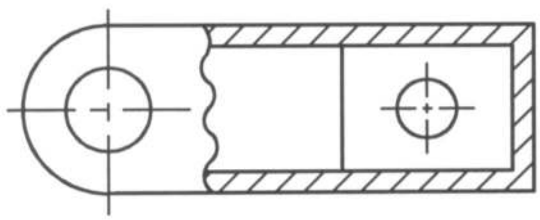


Figure 4.12: View with Broken-Out Section

4.4.4 ***Revolved Section***

Sometimes a revolved or rotated section becomes necessary to illustrate the shape of a bar, arm, spoke, rib etc. An imaginary cutting plane is assumed to pass across them to obtain the section, then the section is rotated by 90^0. In Figure 4.13, revolved T-section with break and elliptical section have been shown. It can be observed that when the top and bottom lines are not parallel, revolved section with break becomes suitable.

When the revolved sections are removed to other locations of the sheet, they are known as removed section (Figure 4.13). When space limitation appears for revolved section or dimensioning cannot be provided, removed section is required. Sometimes several sections are necessary for the part with gradual change in shape and size, and then removed sections become useful. Often the sections are drawn at a larger scale to show the dimensions; removed section becomes suitable for that.

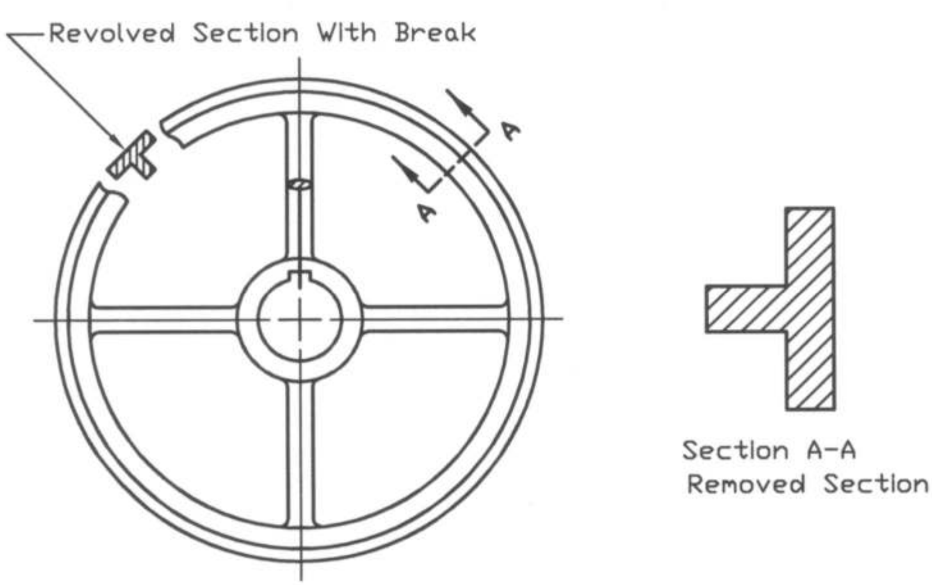
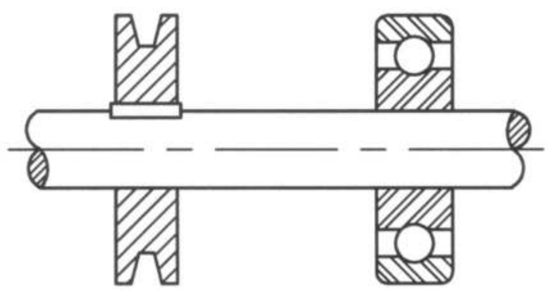


Figure 4.13: Revolved Section

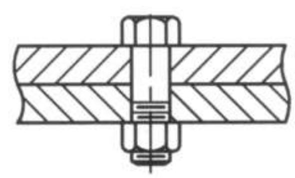
4.5 ***Parts Not Sectioned***

Shafts, bolts, nuts, rivets, balls, pins and keys are not sectioned when they are in the line of cutting plane. They are always left in full view. They have been shown in Figure 4.14.



(a) Shaft, Key, Ball

(b) Rivet



(c) Bolt

Figure 4.14: Parts not to be Sectioned

4.6 ***Ribs in Section***

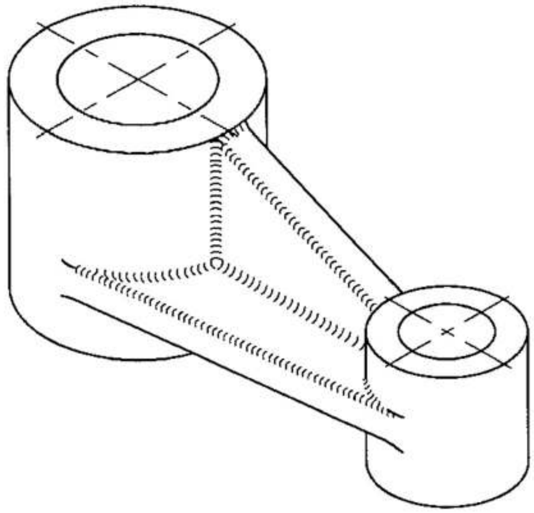
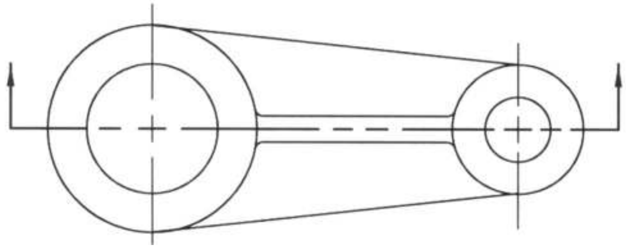
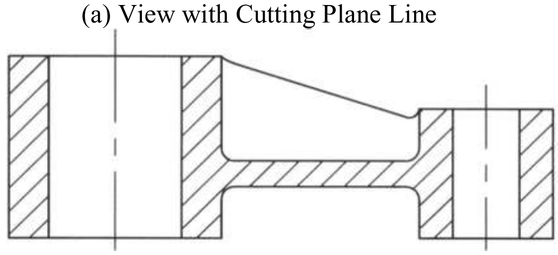


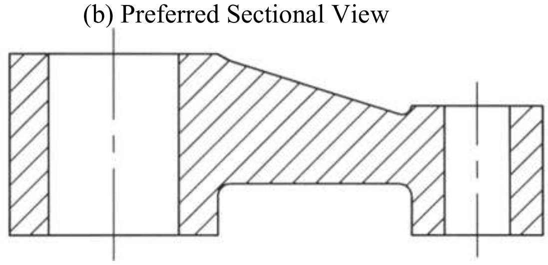
Figure 4.15: Object with Rib



(a) View with Cutting Plane Line



(b) Preferred Sectional View



(c) True Sectional View

Figure 4.16: Sectional View with Ribs

An object with ribs and its sectional view are shown in Figure 4.15 and 4.16 respectively. When the cutting plane passes longitudinally through the center of a rib or web, section lining is omitted there

(Figure 4.16b). Providing section lining throughout including the ribs the view cannot describe the object clearly (Figure 4.16c) rather it makes the drawing misleading. On the other hand, when the cutting plane cuts the ribs transversely, that is, at right angle to its length, it is always sectioned.

An object with odd number of ribs and its sectional view are shown in Figures 4.17 and 4.18 respectively. When the ribs are odd in number, the true section cannot illustrate the shape of the object properly (Figure 4.18b). In that case if the rib as well as the hole are aligned or rotated as such, the view becomes symmetric and makes a good relationship (Figure 4.18c). This can describe the object clearly.

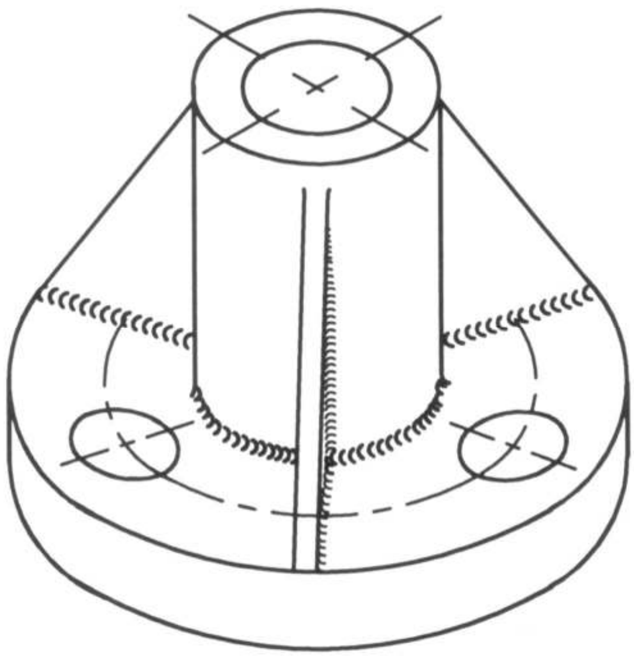
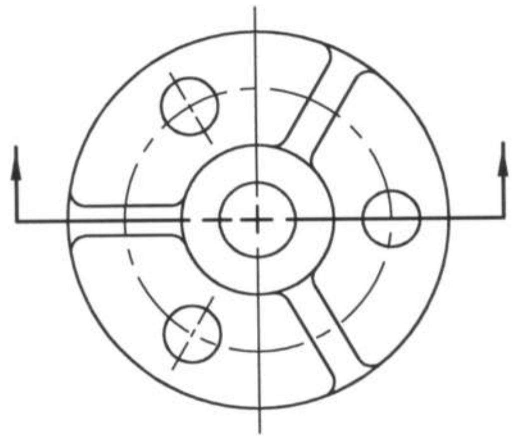
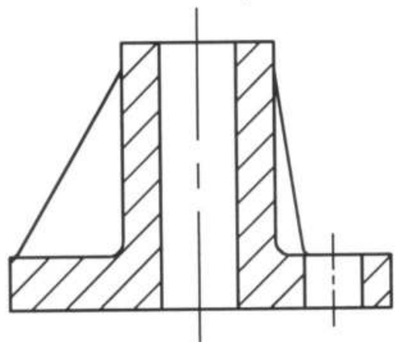


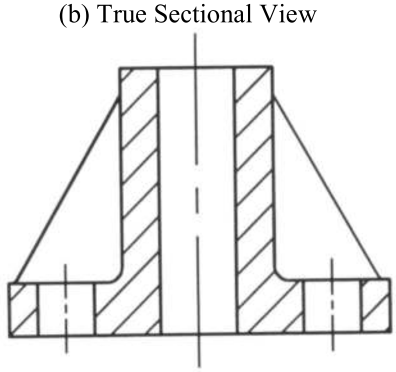
Figure 4.17: Object with Odd Number of Ribs



(a) View with Cutting Plane Line



(b) True Sectional View



(c) Preferred Sectional View

Figure 4.18: Sectional View with Odd Number of Ribs

4.7 ***Spokes in Section***

In Figures 4.19 and 4.20, a wheel with odd number of spokes and its sectional view are shown respectively. Figure 4.20a represents a view with cutting plane line while Figures 4.20b and 4.20c represent the true sectional view and preferred sectional view respectively. The true sectional view does not illustrate the object clearly while the preferred sectional view represents the object clearly. In the preferred sectional view the spoke has been made aligned to present a true relationship.

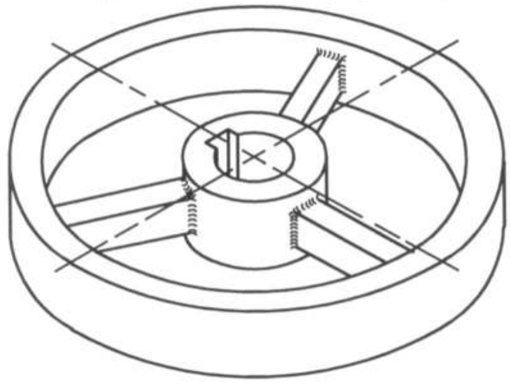
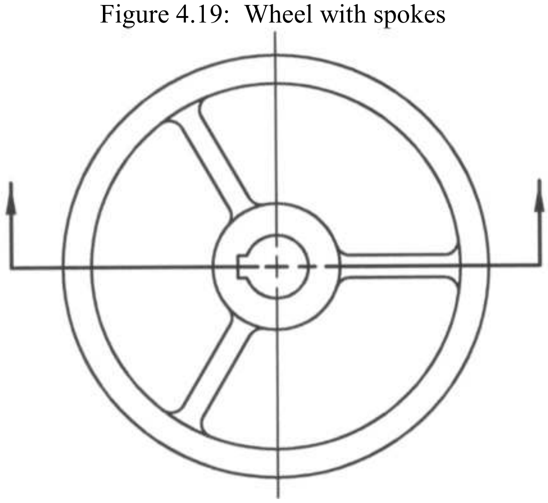
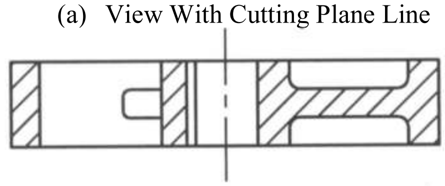


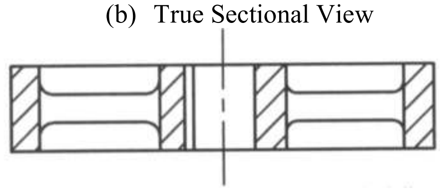
Figure 4.19: Wheel with spokes



(a) View With Cutting Plane Line



(b) True Sectional View



(c) Preferred Sectional View

Figure 4.20: Sectional View with Odd Number of Spokes

4.8 ***Lugs in Section***

In Figure 4.21, an object with a lug has been presented. A view with cutting plane line of this object has been given in Figure 4.22a. The sectional view has been shown in Figure 4.22b, where the lug has not been sectioned.

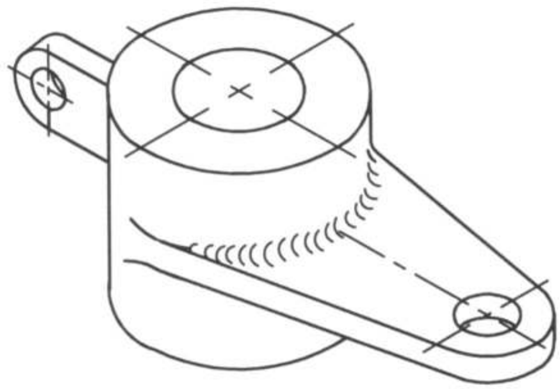
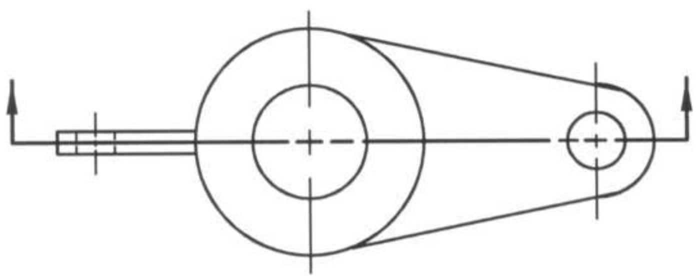
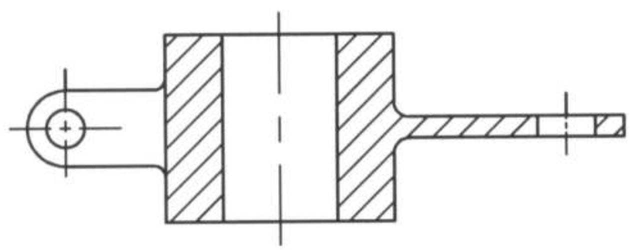


Figure 4.21: An Object With a Lug



(a) View With Cutting Plane Line



(b) Preferred Sectional View

Figure 4.22: Sectional View with Lug

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.***

Prob. 4.1: Draw the top view and a suitable sectional view of the bearing shown in Fig. P4.1 (Fillets and Radii are 3 mm).

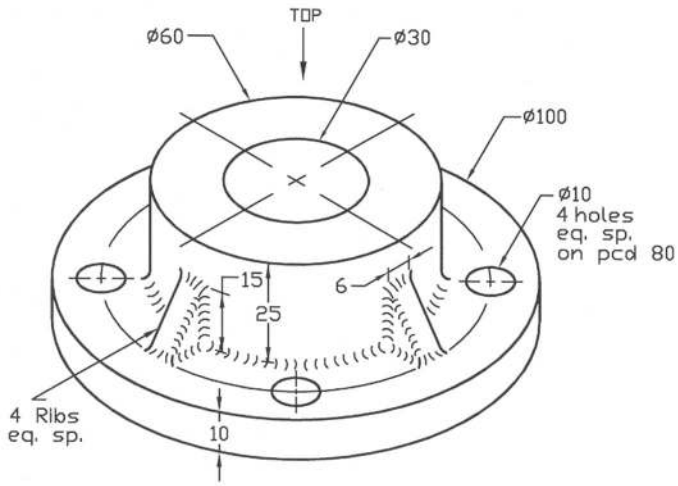


Fig. P4.1

Prob. 4.2: Draw the top and front sectional views of the base plate as shown in Fig. P4.2 (Fillets and Radii are 3 mm).

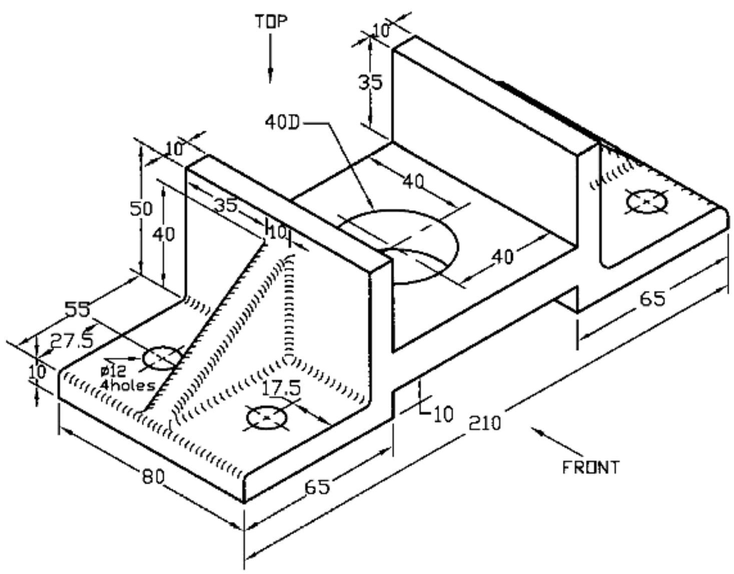


Fig. P4.2

Prob. 4.3: Draw the top view and a suitable sectional view of the bearing retainer shown in Fig. P4.3.

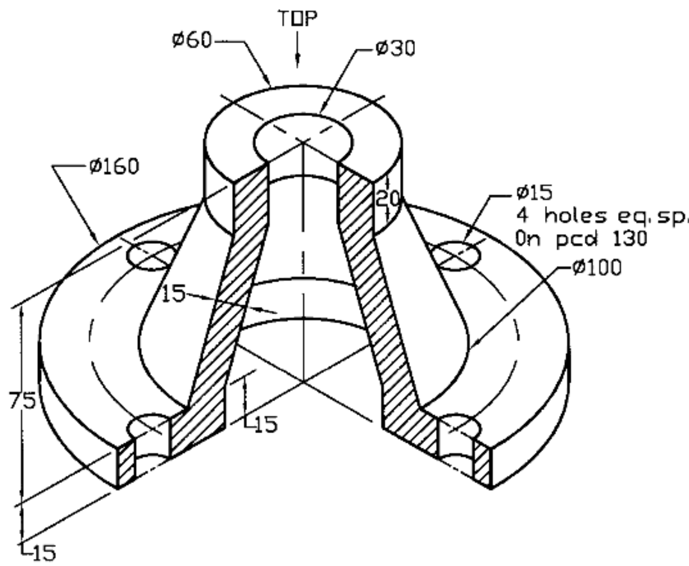


Fig. P4.3

Prob. 4.4: Draw the top, front sectional and right side views of the slider block as shown in Fig. P4.4 (Fillets and Radii are 3 mm).

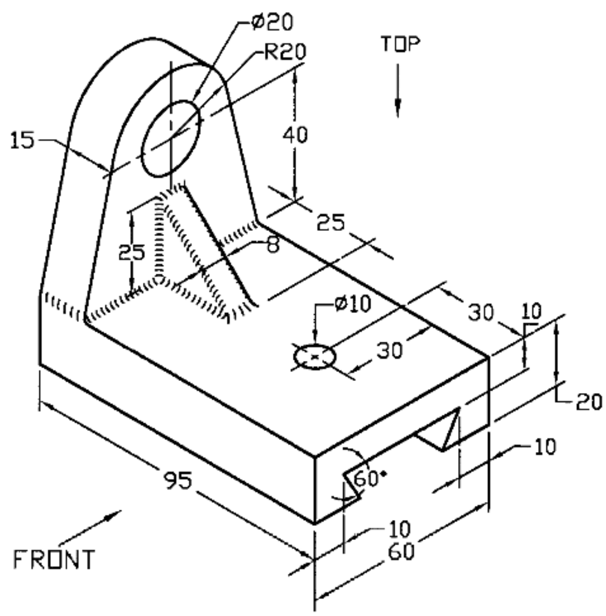


Fig. P4.4

Prob. 4.5: Draw the top view and a suitable sectional view of the rotor shown in Fig. P4.5 (Fillets are 3 mm).

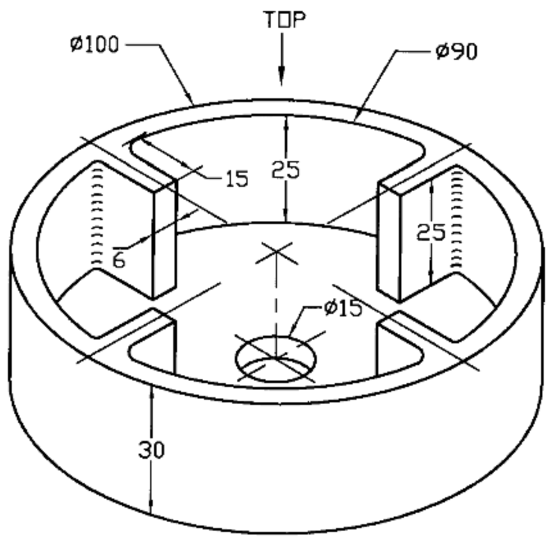


Fig. P4.5

Prob. 4.6: Draw the top, front sectional and left side views of the tool post as shown in Fig. P4.6 (Fillets are 3 mm).

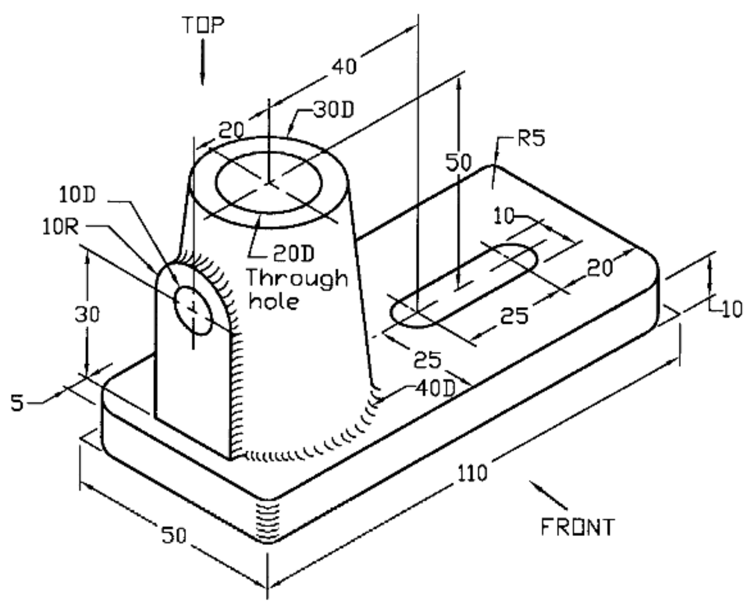


Fig. P4.6

Prob. 4.7: Draw the front view and a suitable sectional view of the flat pulley shown in Fig. P4.7 (Fillets are 3 mm).

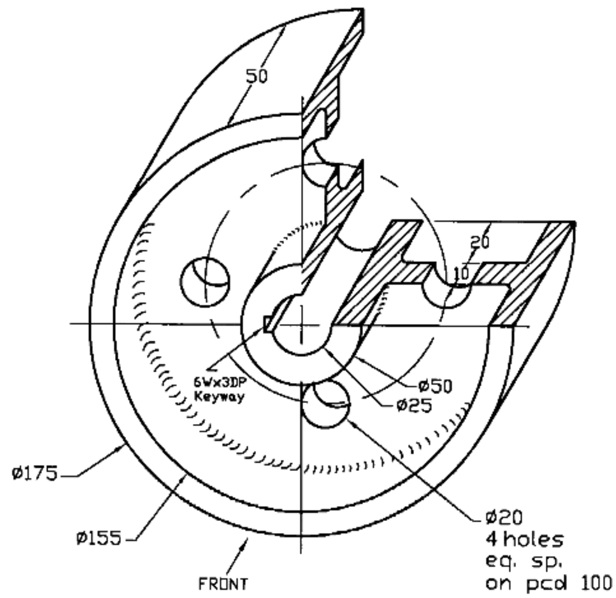


Fig. P4.7

Prob. 4.8: Draw the top and front sectional views of the offset bearing shown in Fig. P4.8.

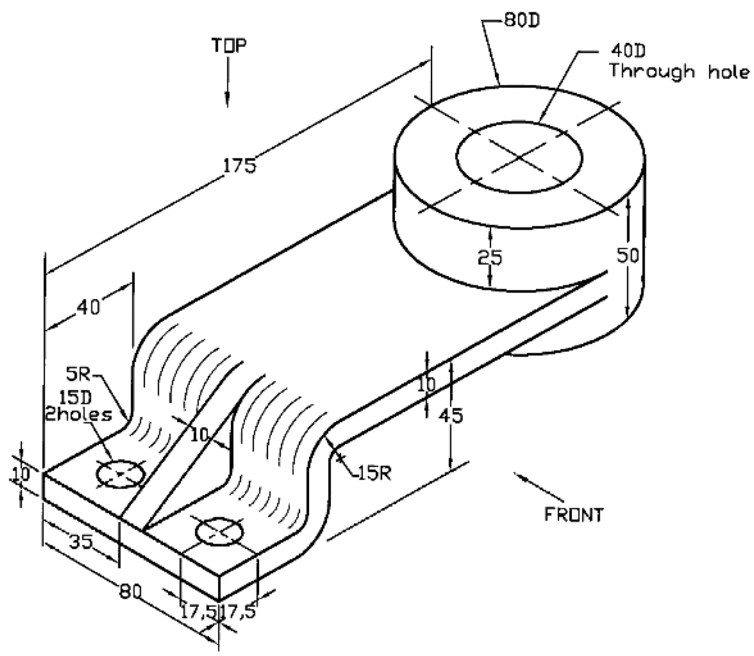


Fig. P4.8

Prob. 4.9: Draw the top, front sectional and right side views of the rod support as shown in Fig. P4.9 (Fillets and Radii are 3 mm).

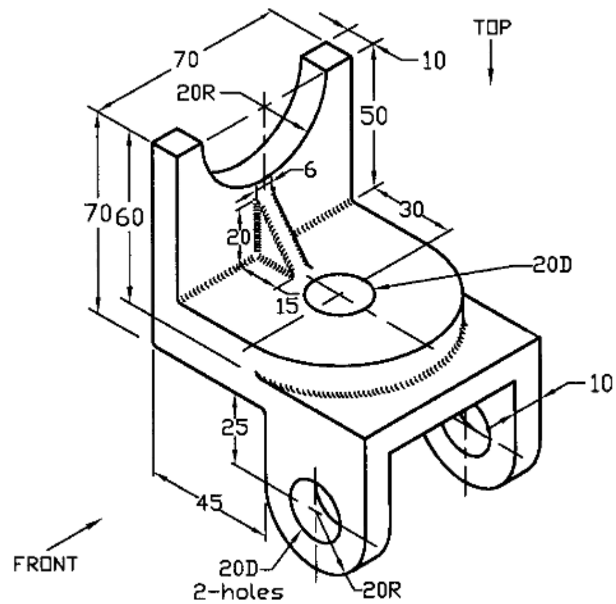


Fig. P4.9

Prob. 4.10: Draw the top view and a suitable sectional view of the rotary yoke shown in Fig. P4.10 (Fillets and Radii are 3 mm).

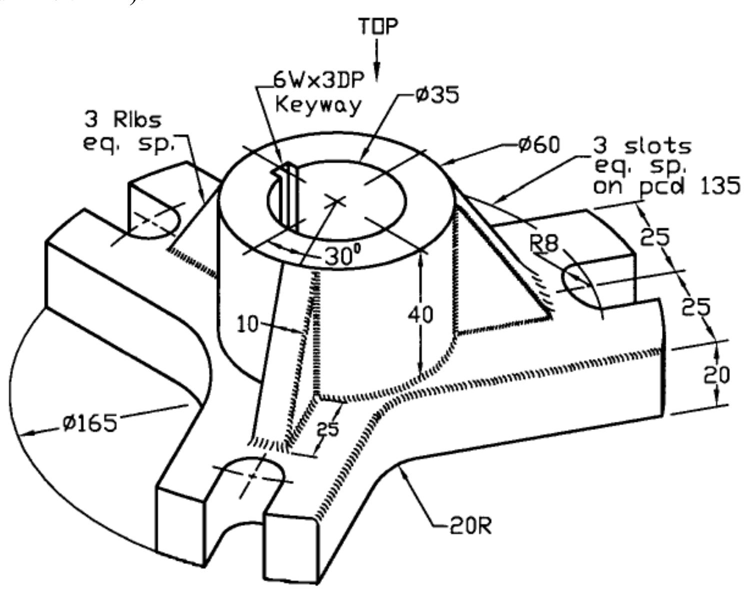


Fig. P4.10

Prob. 4.11: Draw the top, front sectional and left side views of the guide plate as shown in Fig. P4.11 (Fillets and Radii are 3 mm).

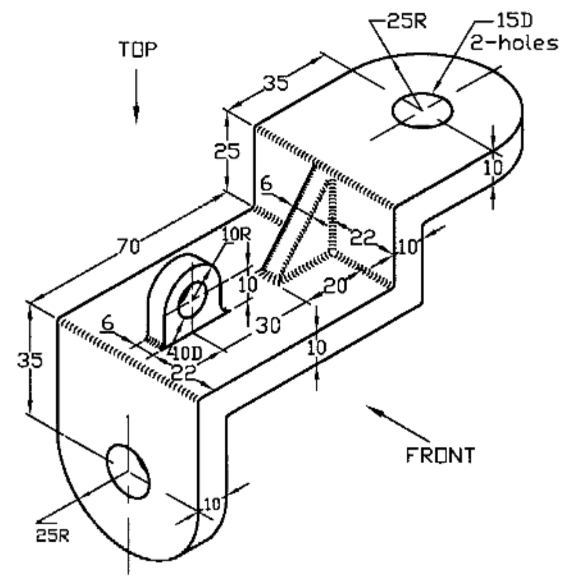


Fig. P4.11

Prob. 4.12: Draw the top and front sectional views of the clamping bracket shown in Fig. P4.12 (Fillets and Radii are 3 mm).

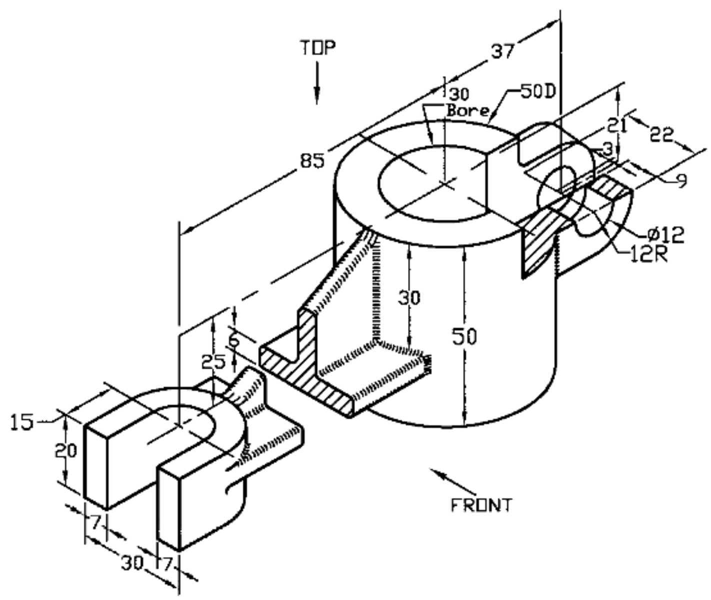


Fig. P4.12

Prob. 4.13: Draw the top, front sectional and left side views of the mounting bracket shown in Fig. P4.13 (Fillets and Radii are 3 mm).

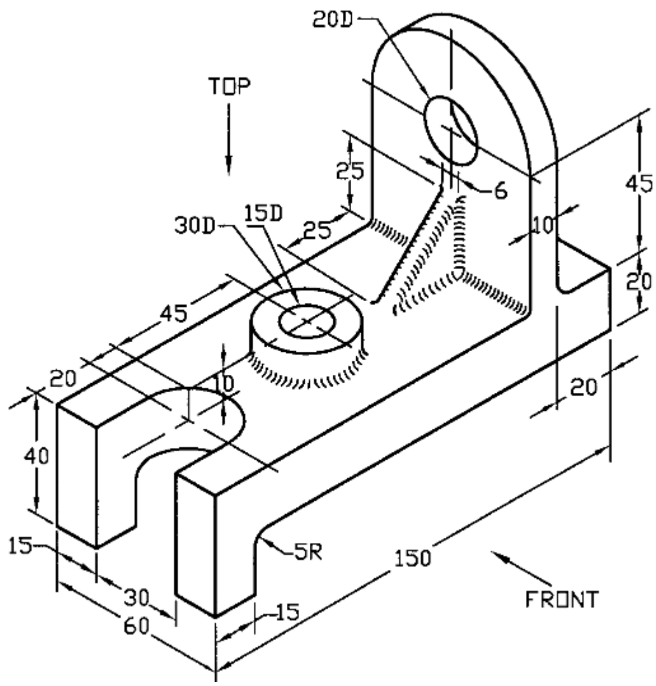


Fig. P4.13

Prob. 4.14: Draw the top, front sectional and right side views of the column collar as shown in Fig. P4.14 (Fillets are 3 mm).

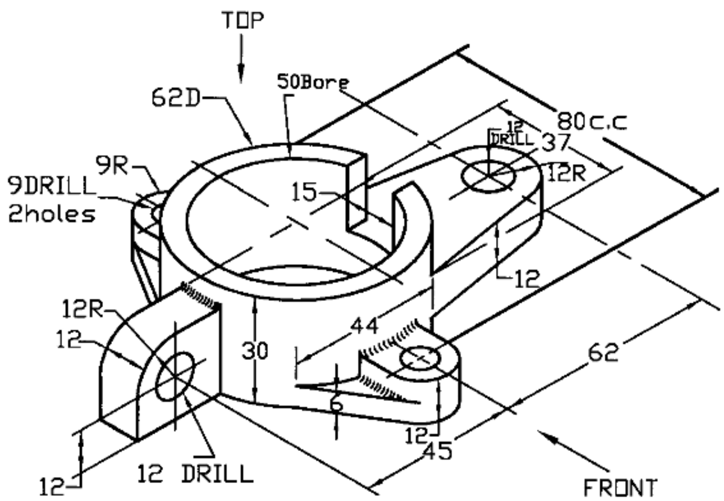
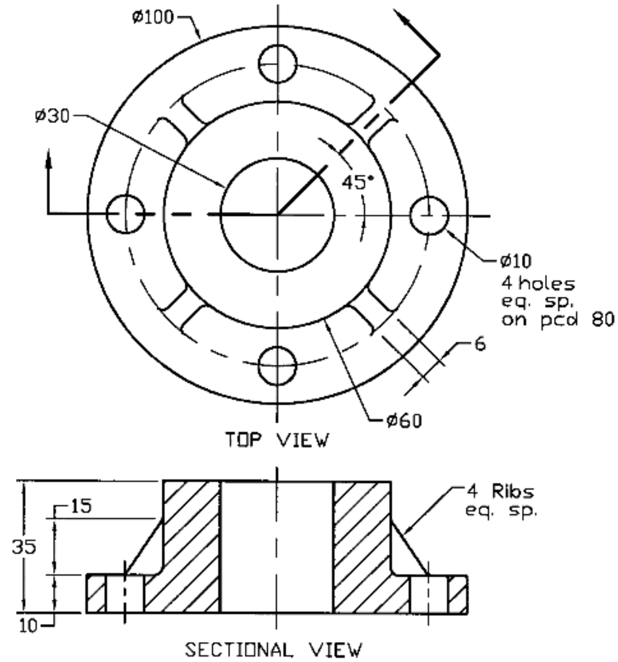
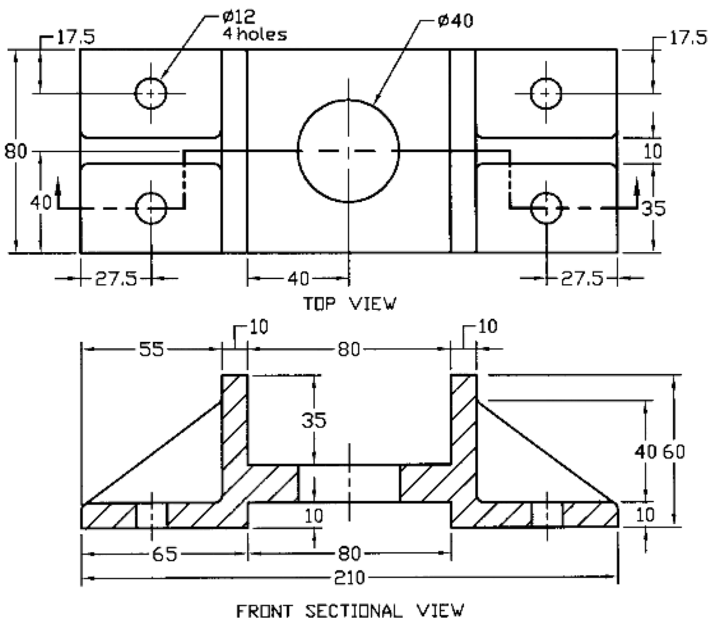


Fig. P4.14

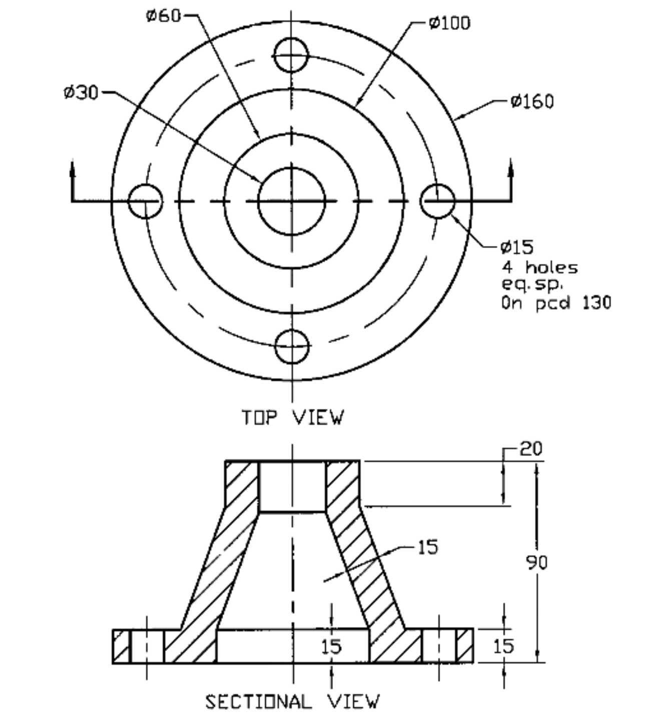
Solutions for Example Problems



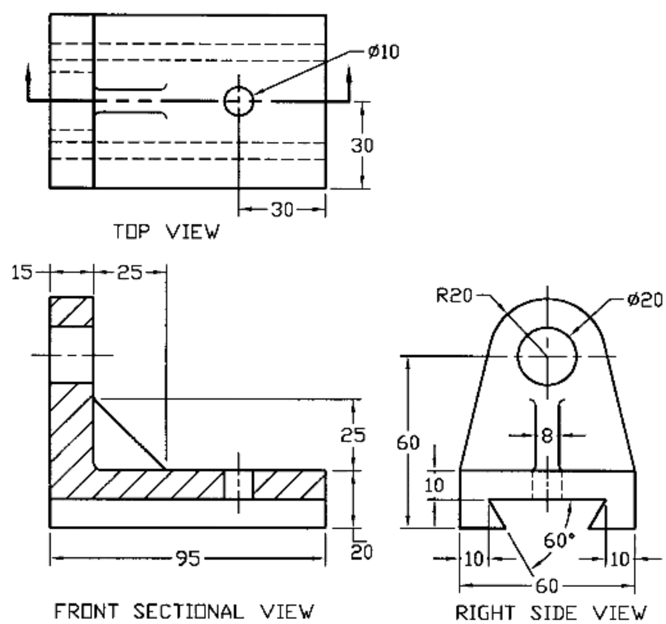
Solution of P4.1



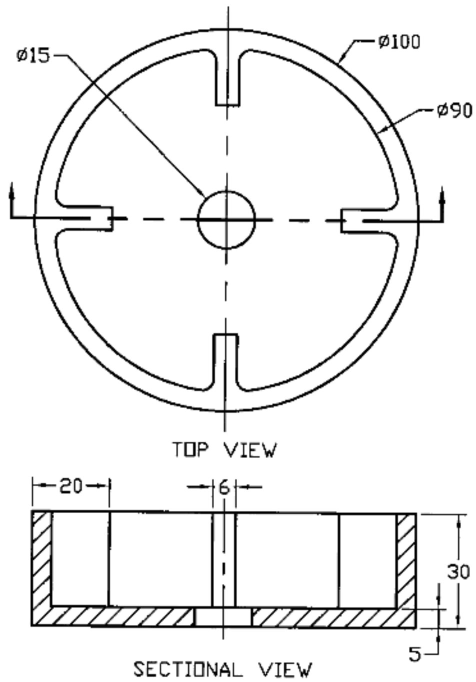
Solution of P4.2



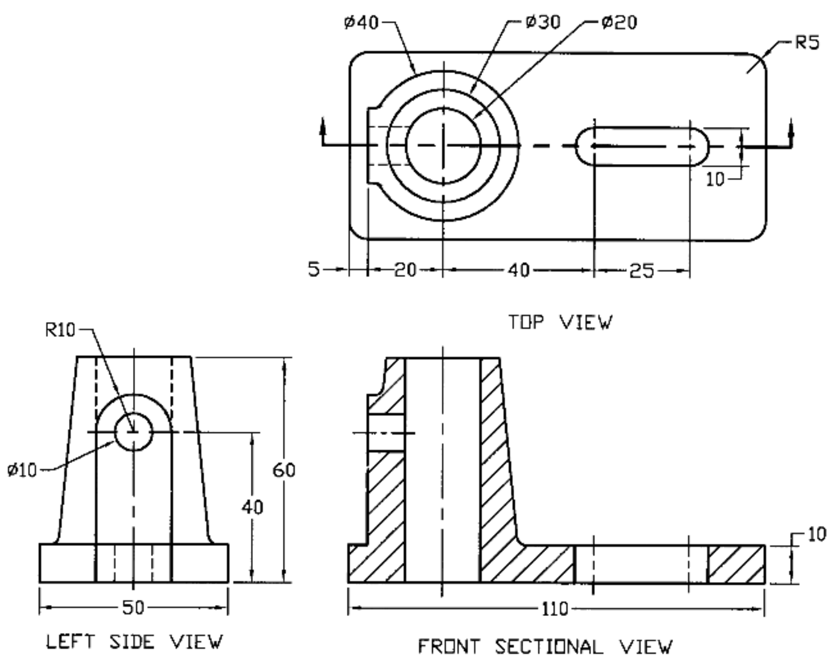
Solution of P4.3



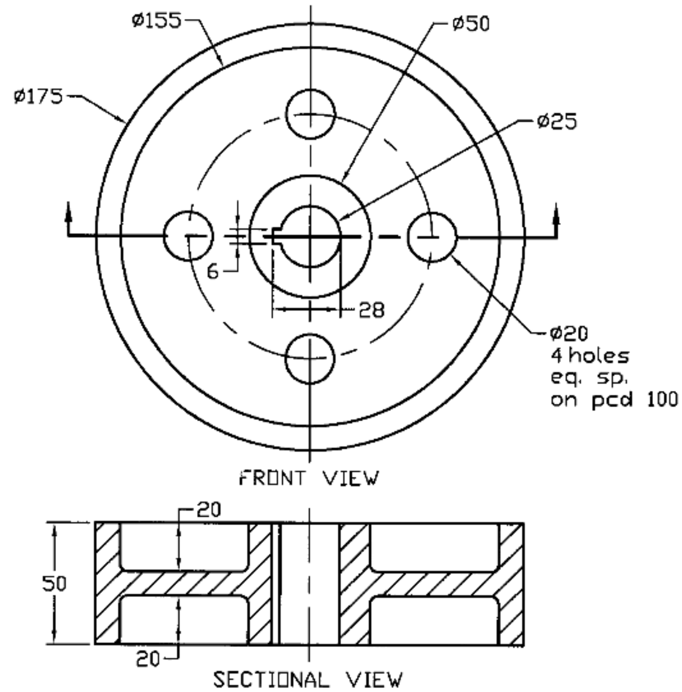
Solution of P4.4



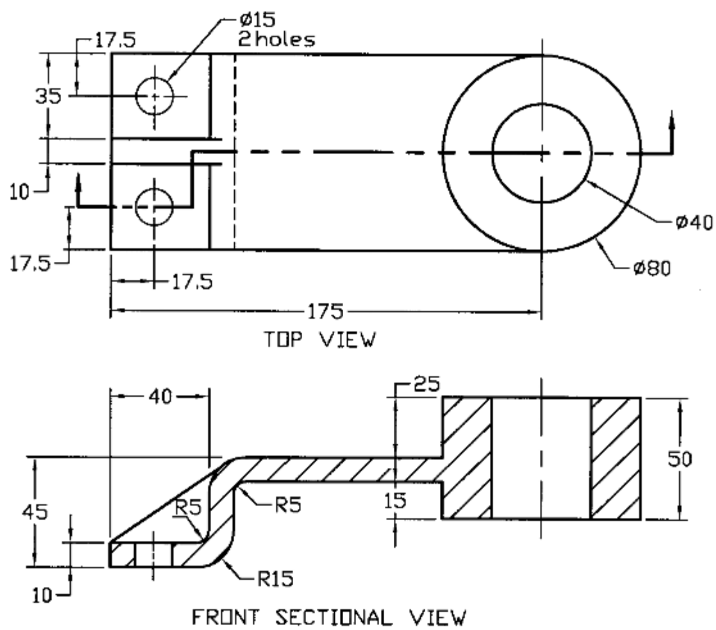
Solution of P4.5



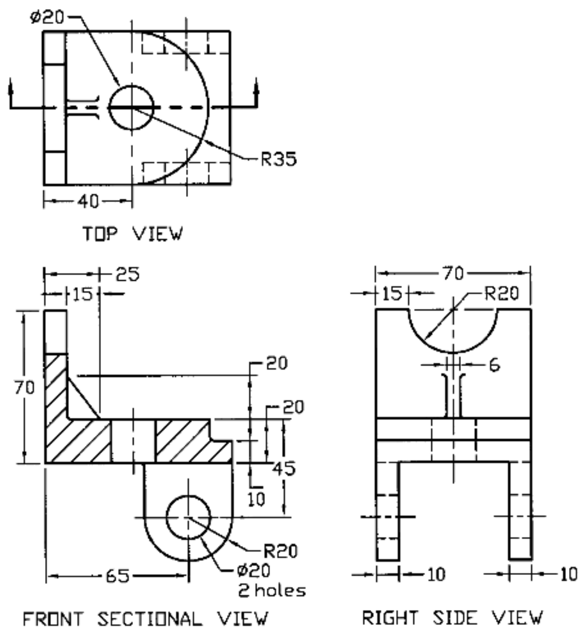
Solution of P4.6



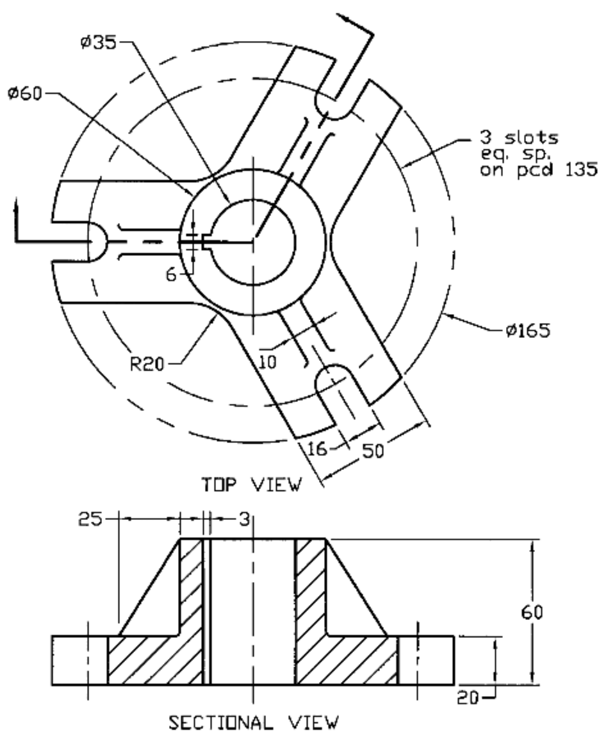
Solution of P4.7



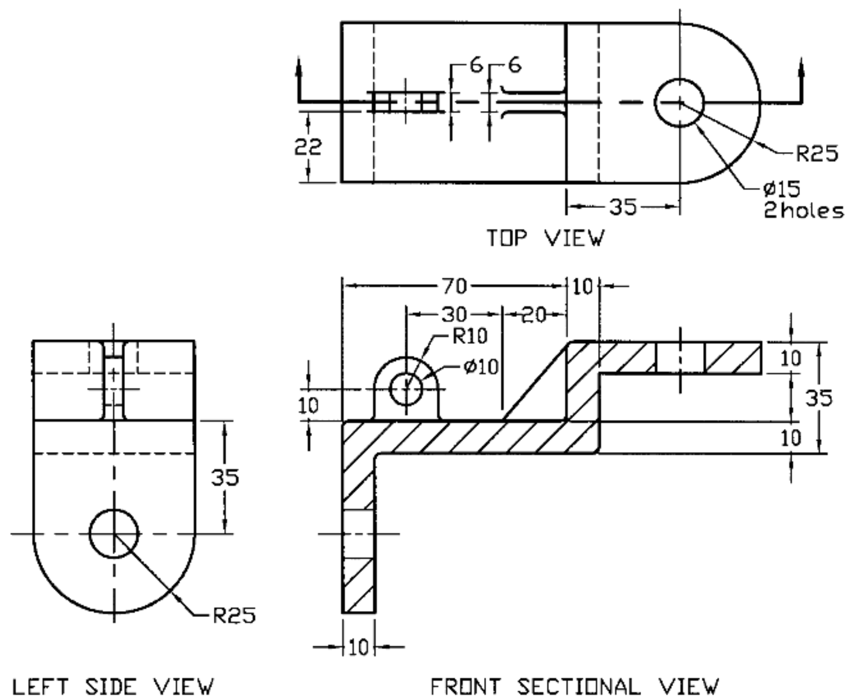
Solution of P4.8



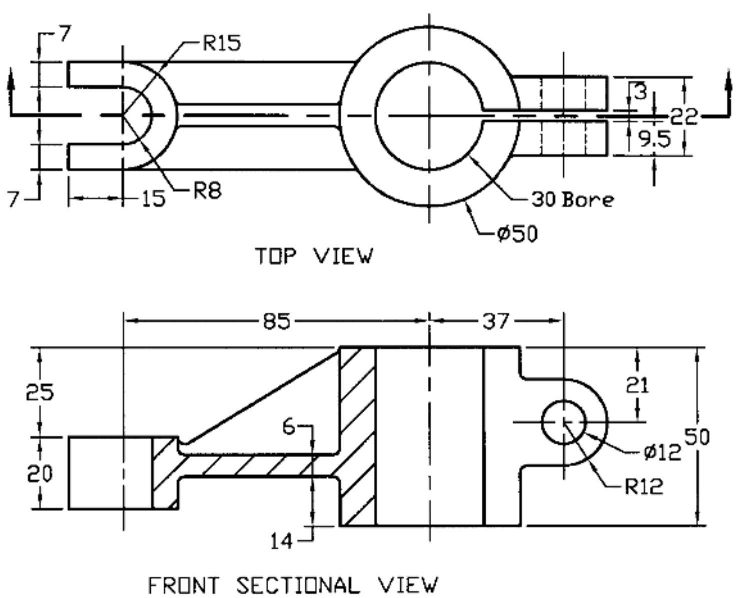
Solution of P4.9



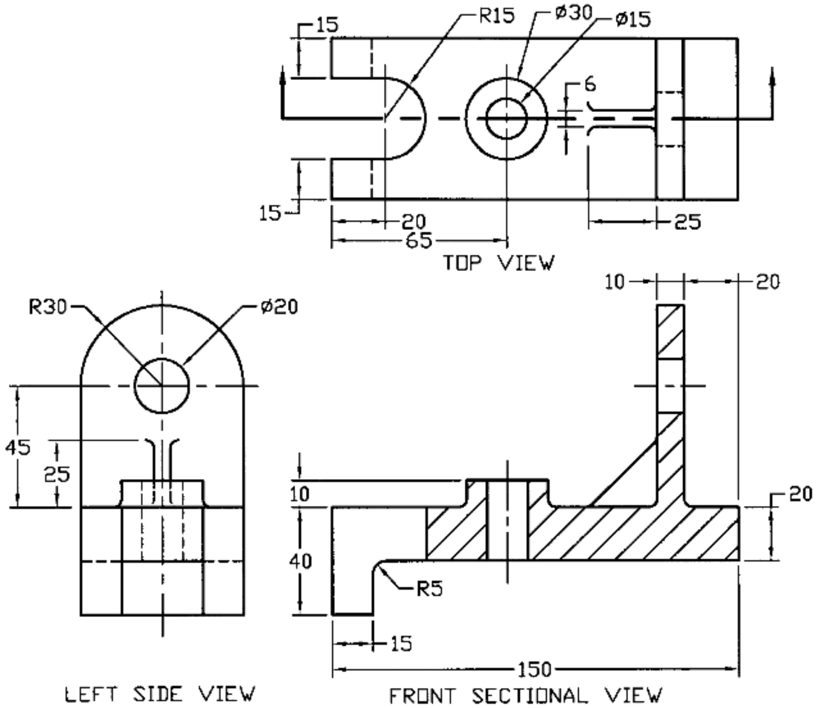
Solution of P4.10



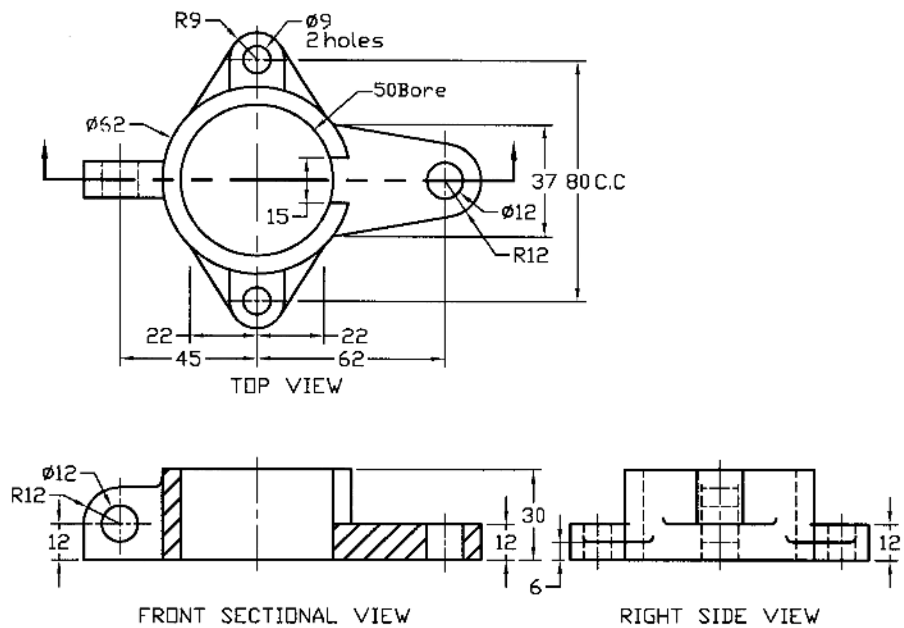
Solution of P4.11



Solution of P4.12



Solution of P4.13



Solution of P4.14

Problems

Prob. 4.15: Draw the top, front sectional and right side views of the support block shown in Fig. P4.15 (Fillets and radii are 3mm).

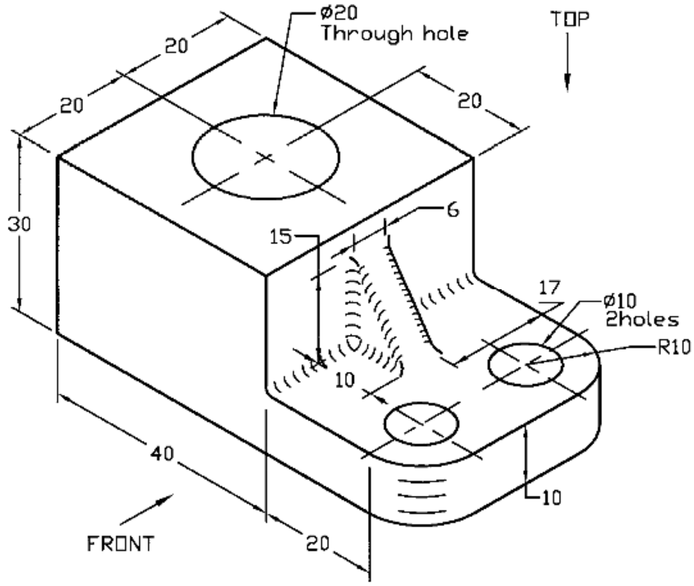


Fig. P4.15

Prob. 4.16: Draw the top, front sectional and right side views of the yoke shown in Fig. P4.16.

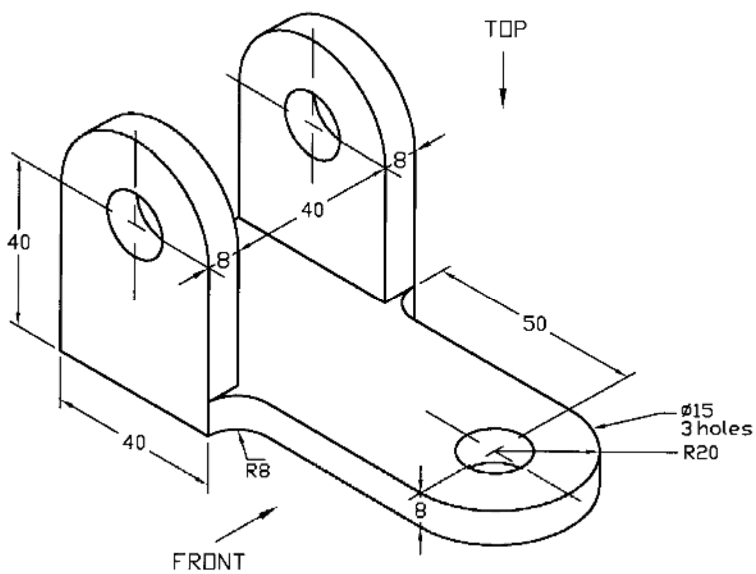


Fig. P4.16

Prob. 4.17: Draw the top and front sectional views of the channel frame shown in Fig. P4.17.

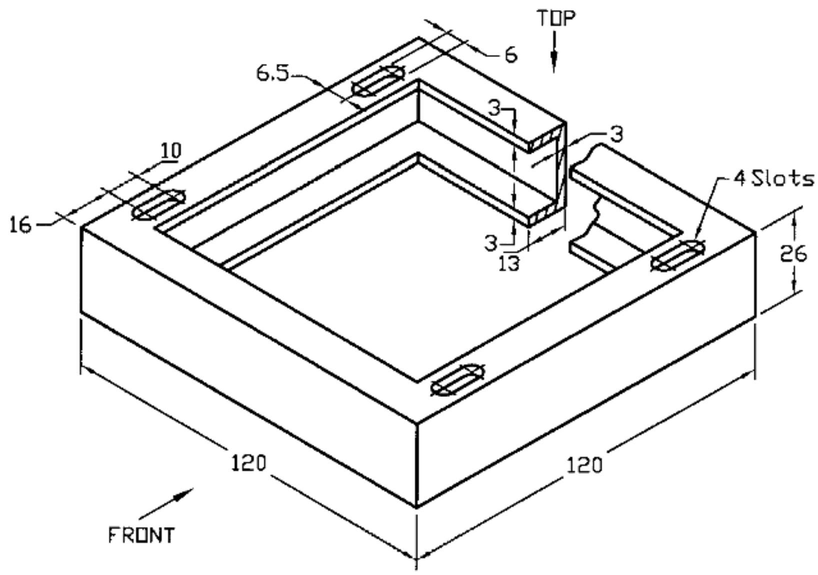


Fig. P4.17

Prob. 4.18: Draw the top, front sectional and left side views of the column base shown in Fig. P4.18 (Fillets are 3mm).

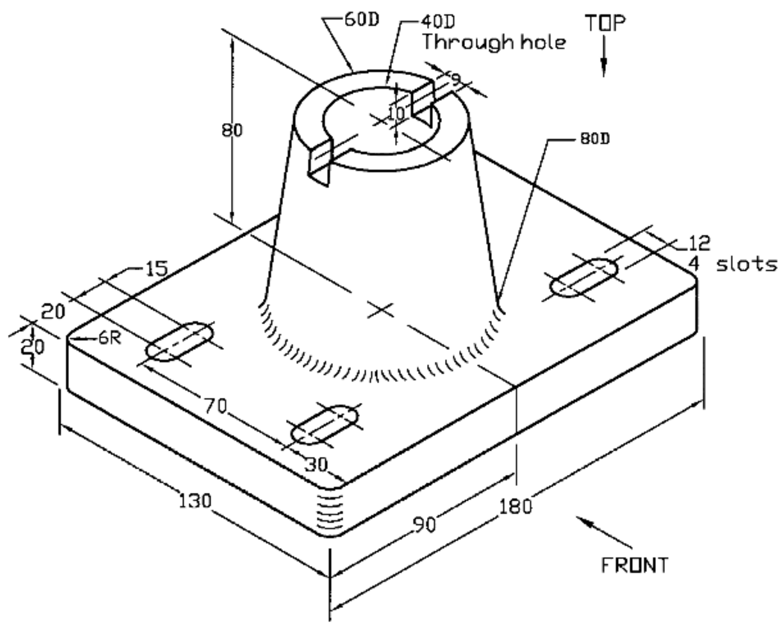


Fig. P4.18

Prob. 4.19: Draw the top and front sectional views of the support plate shown in Fig. P4.19.

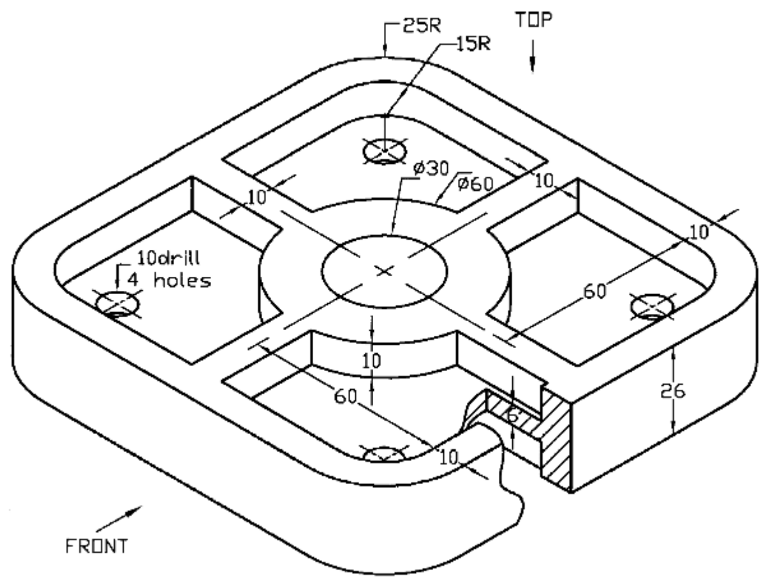


Fig. P4.19

Prob. 4.20: Draw the front view and a suitable sectional view of the bearing as shown in Fig. P4.20.

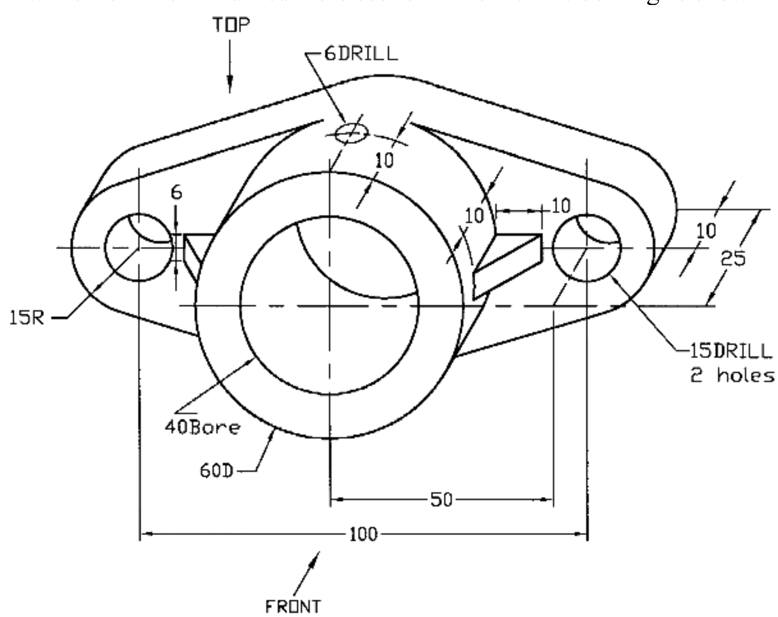


Fig. P4.20

Prob. 4.21: Draw the top, front sectional and left side views of the special bearing as shown in Fig. P4.21 (Fillets and radii are 3mm).

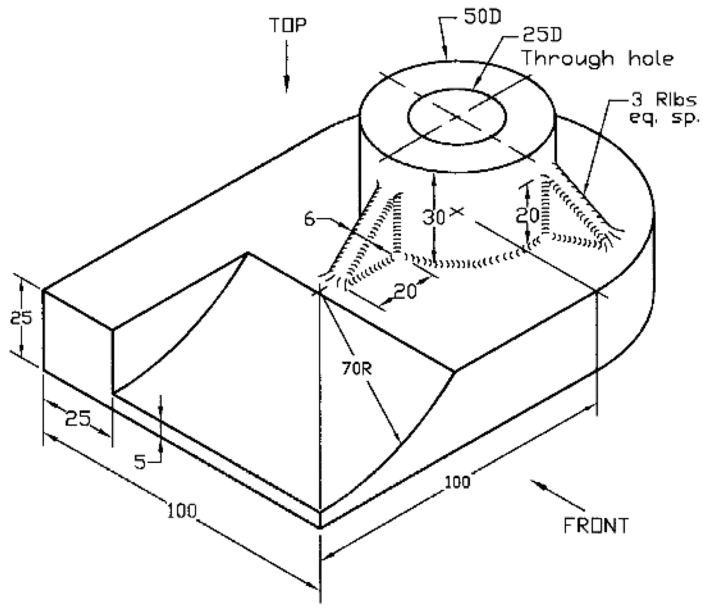


Fig. P4.21

Prob. 4.22: Draw the front view and a suitable sectional view of the column support as shown in Fig. P4.22 (Fillets and radii are 3mm).

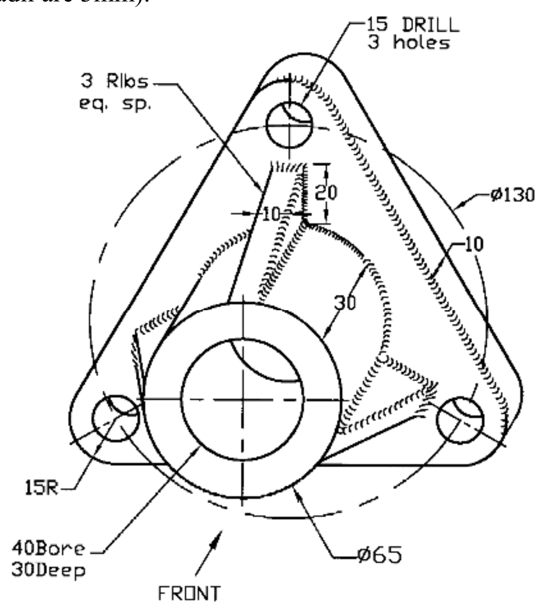


Fig. P4.22

Prob. 4.23: Draw the front view and a suitable sectional view of the bearing shown in Fig. P4.23 (Fillets and radii are 3mm).

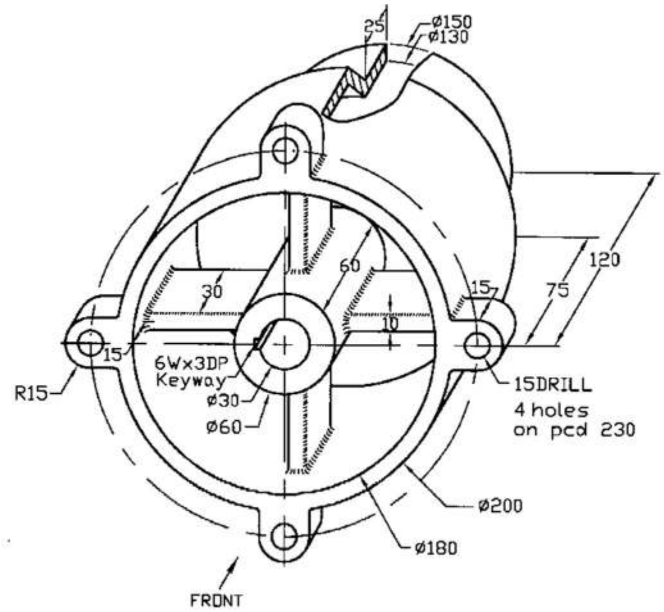


Fig. P4.23

Prob. 4.24: Draw the top, front sectional and left side views of the end plate shown in Fig. P4.24 (Fillets and radii are 3mm).

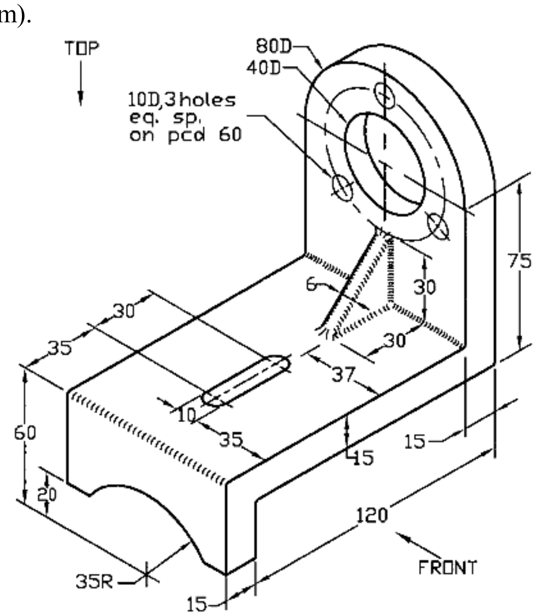


Fig. P4.24

Prob. 4.25: Draw the top view and a suitable sectional view of the V-pulley as shown in Fig. P4.25 (Fillets are 3mm).

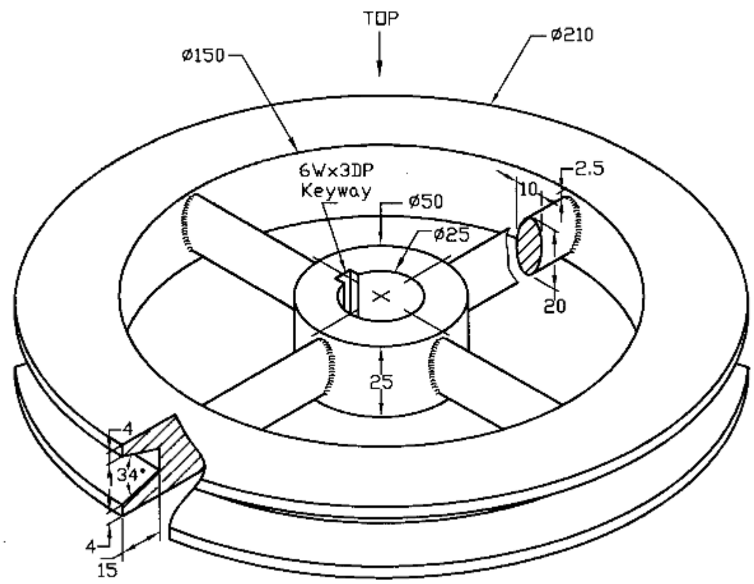


Fig. P4.25

Prob. 4.26: Draw the top, front sectional and left side views of the bracket shown in Fig. P4.26 (Fillets are 3mm).

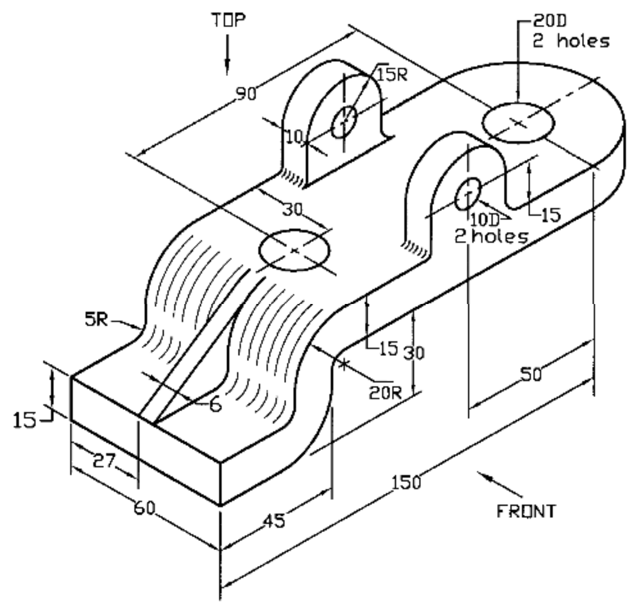


Fig. P4.26

CHAPTER 5

AUXILIARY VIEW

5.1: ***Introduction***

In orthographic projection sometimes it happens that some surfaces of the machine parts are not parallel to the plane of projection i.e. they are at an angle. These surfaces are called inclined surfaces. The regular orthographic view of the inclined surface does not give the exact shape; it gives the distorted shape of the surface. Thus the clear feature of the inclined surface of the object cannot be obtained from the regular view. As a result the main objective of the drawing is lost. In order to overcome this problem an auxiliary view is necessary.

A picture plane is assumed parallel to the inclined surface called auxiliary projection plane and orthographic projection on that plane is made. It gives the true shape of the inclined surface and preserves the objective of the drawing. Sometimes an auxiliary view replaces one of the regular views. For some machine part it may happen that some portion of the surface is inclined and the rest portion is not. In that case to represent the total surface of the object partial views are drawn. The partial view is drawn in order to avoid distortion and complication. In auxiliary view, hidden lines are avoided unless they become necessary to represent the view clearly.

5.2: ***Generating Auxiliary View***

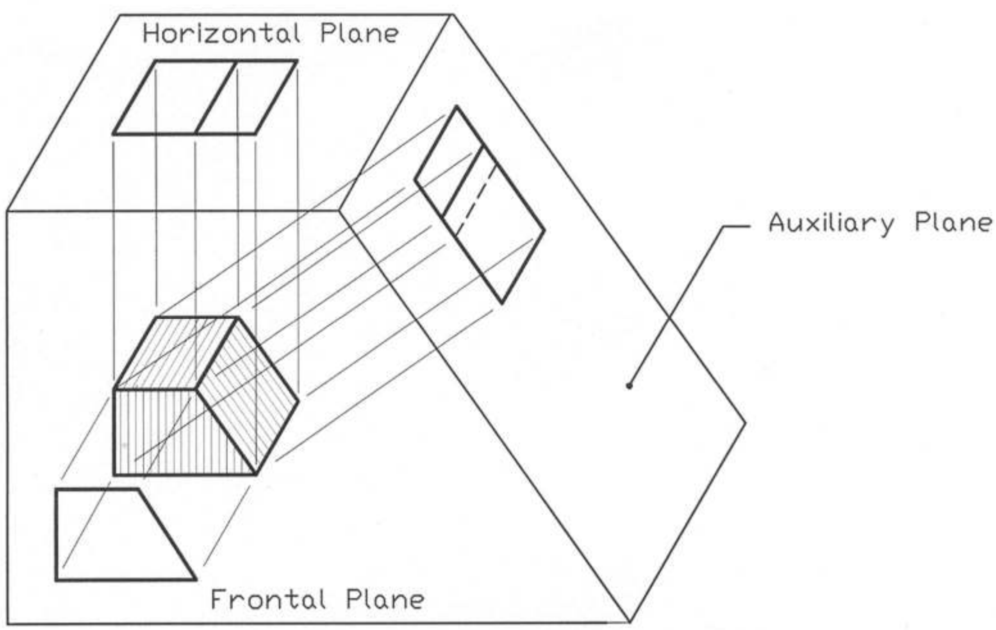


Figure 5.1: Auxiliary View on Inclined Plane

It is described in this section how to generate an auxiliary view. A simple object is placed inside a transparent box (Figure 5.1) each side of which is called the picture plane. The inclined plane which, is parallel to the inclined surface of the object, is called the auxiliary plane as shown in the figure. Other planes are horizontal and frontal planes. In Figure 5.2 the views of the object with the box in the unfolding condition have been shown while in Figure 5.3 the relative positions of the views have been provided.

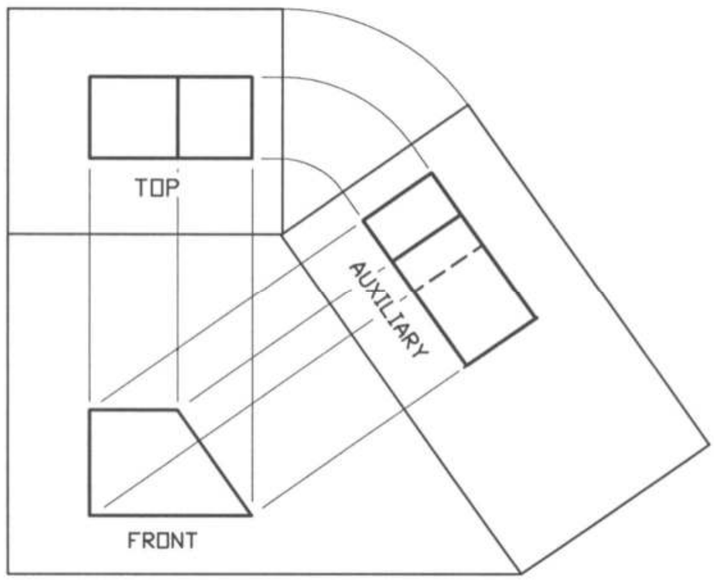


Figure 5.2: Views with Box in Unfolding Condition

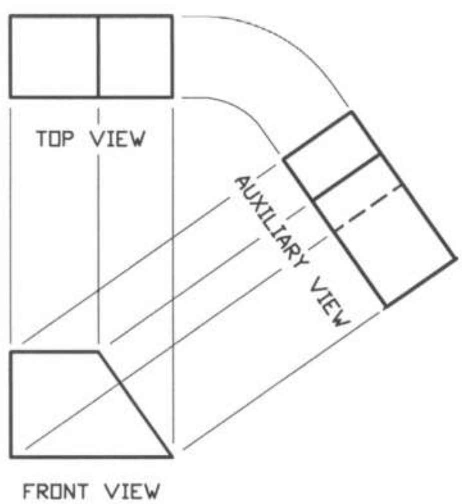


Figure 5.3: Relative Positions of Views

In order to give further example, another object with inclined surface having some complication has been chosen which can be seen in Figure 5.4.The regular orthographic views of the object with inclined surface (Figure 5.4) are given in Figure 5.5. While the necessary views of the same object including an auxiliary view have been presented in Figure 5.6.

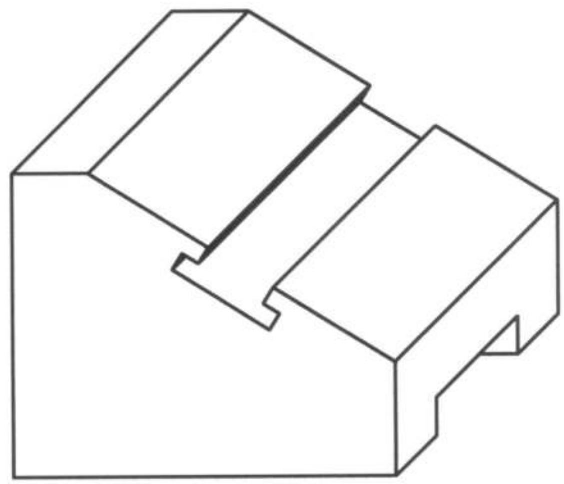


Figure 5.4: Object with Inclined Surface

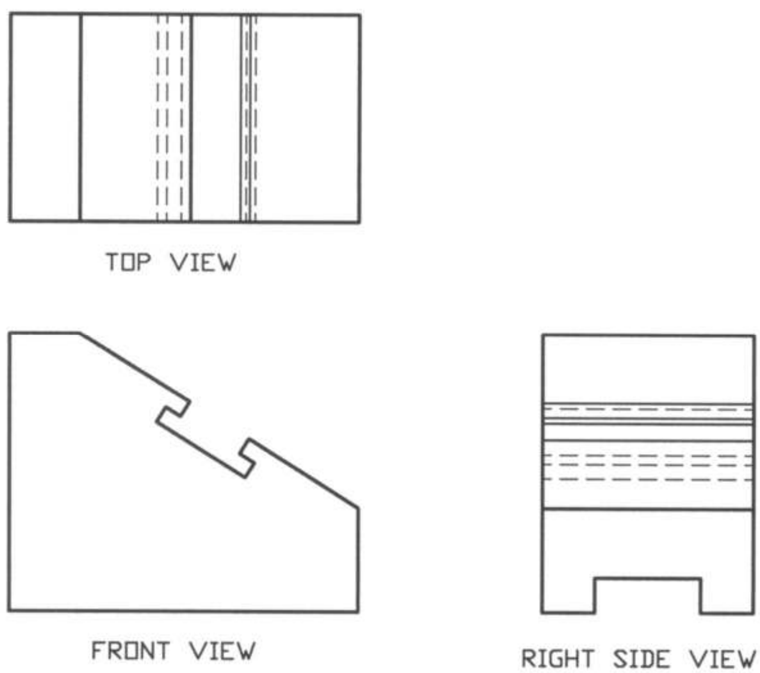


Figure 5.5: Regular Orthographic Views

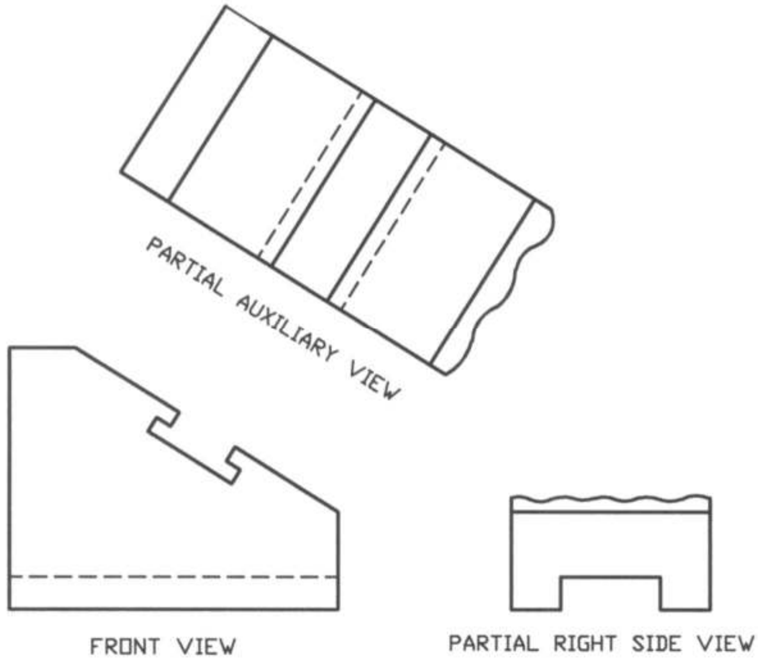


Figure 5.6: Auxiliary View

It is obvious from Figure 5.5 that the regular orthographic views cannot produce the true shape of the object for the inclined surface. As such it becomes very difficult to use them as the detail drawing to manufacture. Because they do not give the clear picture about the inclined surface and in addition the size of the inclined surface is shortened. In order to overcome this problem, auxiliary view becomes essential as shown in Figure 5.6.

It can be observed from Figure 5.6 that, the top view has been dropped and it has been replaced by auxiliary view. Top view is no more necessary for this particular case. It is also seen from this figure that the right side view has been drawn partially. As such, it is called partial right side view.

The auxiliary view gives the true shape of the inclined surface of the object and it becomes clear for understanding. If the auxiliary view includes the feature of the inclined portion of the object only omitting the view of the non-inclined portion (if there is any), it is called the partial auxiliary view. To eliminate distortion a partial auxiliary view is used. On the other hand, if the entire surface is taken under consideration the view becomes a complete auxiliary view. Sometimes a complete auxiliary view is drawn where no distortion appears. Usually a break line is used to indicate the imaginary break in the partial views.

5.3 *Auxiliary View with Circular Feature*

In Figure 5.7, an object with inclined surface containing a circular feature is shown. The regular orthographic views of the object (Figure 5.7) are shown in Figure 5.8. While the essential views including an auxiliary view are shown in Figure 5.9.

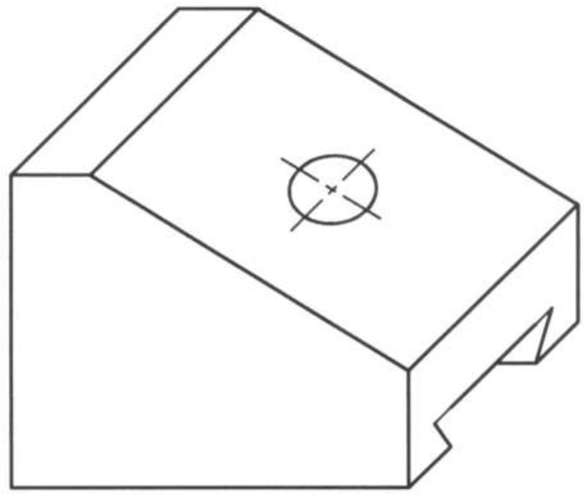


Figure 5.7: Object with Inclined Surface Containing a Hole

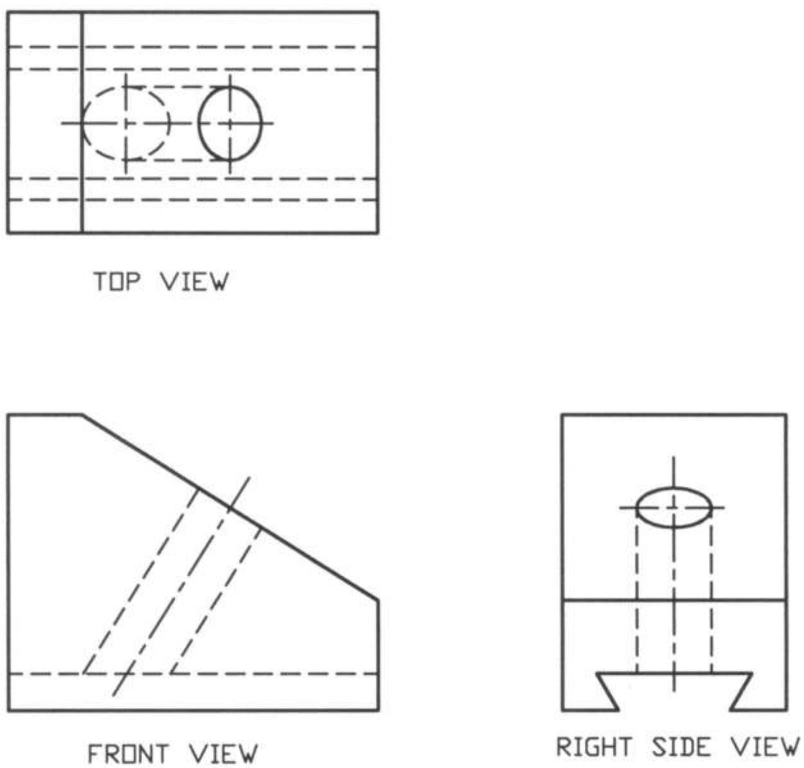


Figure 5.8: Regular Orthographic Views.

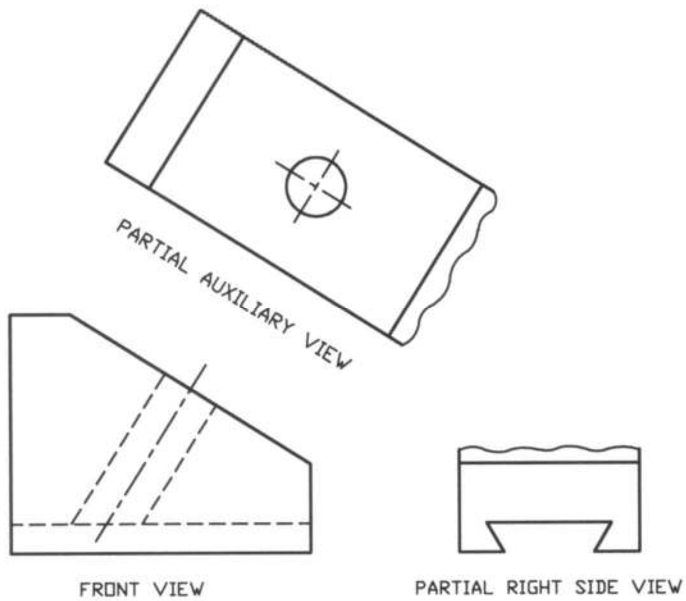


Figure 5.9: Auxiliary View

It is seen from Figure 5.8 that the shape of the hole in the inclined surface has turned into the shape of an ellipse i. e. the true shape of the circle has been distorted. In addition the views have become complicated to understand. To avoid this problem an auxiliary view is required as shown in Figure 5.9. Here a partial auxiliary view and a partial right side view have been drawn. The top view has been omitted because it is not required to represent the object.

5.4: ***Multiple Auxiliary Views***

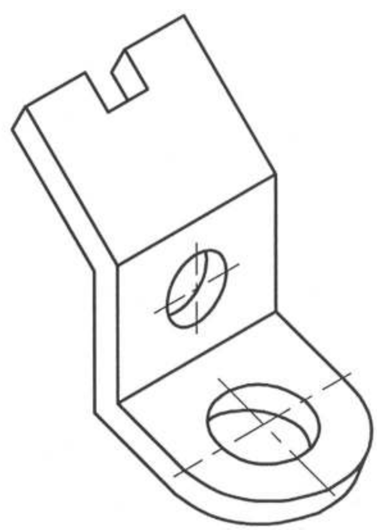


Figure 5.10: Object with two inclined surfaces

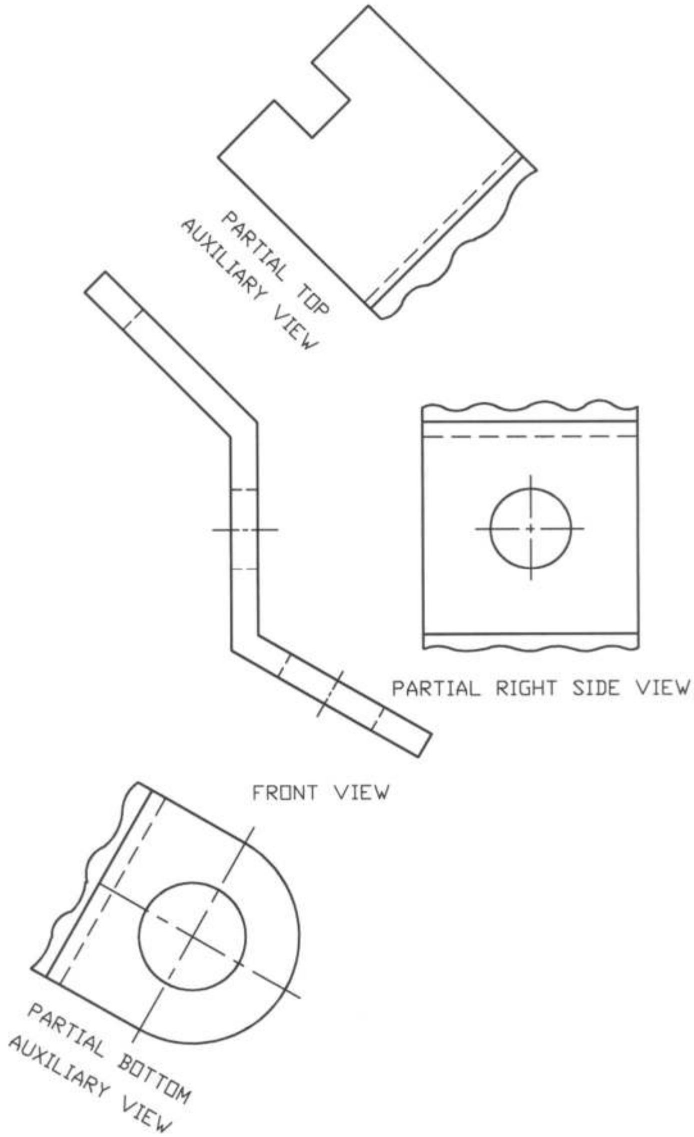


Figure 5.11: Multiple Auxiliary Views

There are some objects, which have several inclined surfaces. They require multiple auxiliary views to represent them clearly. In Figure 5.10 an object with two inclined surfaces are given. In Figure 5.11 the necessary views of the object as shown in Figure 5.10 have been provided. In this figure two auxiliary views have become necessary to represent the two inclined surfaces of the object. Partial auxiliary views have been drawn and a partial right side view has been provided to avoid distortion.

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.***

Prob. 5.1: Draw front, partial bottom, left and right auxiliary views of the connector plate as shown in Fig. P5.1.

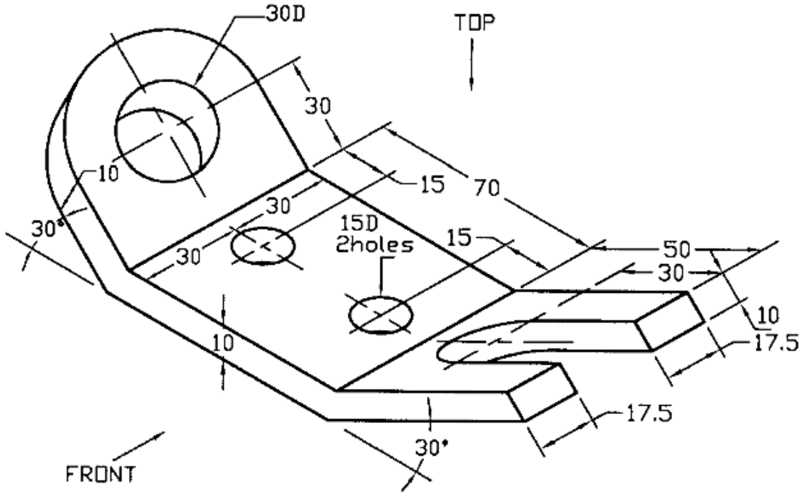


Fig. P5.1

Prob. 5.2: Draw front, partial left, partial top and auxiliary views of the supporting plate as shown in Fig. P5.2.

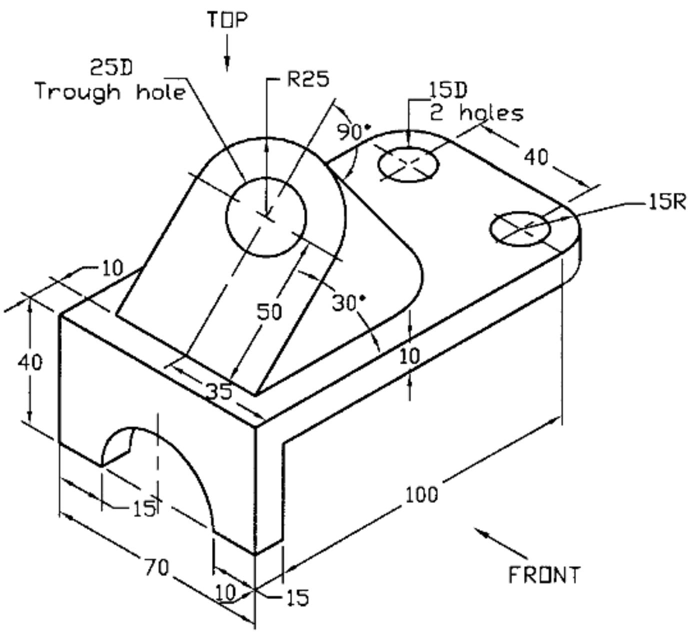


Fig. P5.2

Prob. 5.3: Draw front, partial left side, partial top and auxiliary views of the supporting bracket as shown in Fig. P5.3.

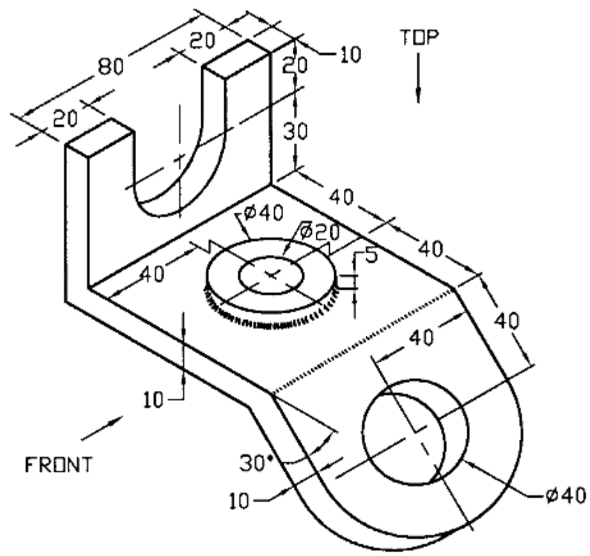


Fig. P5.3

Prob. 5.4: Draw front, partial top and auxiliary views of the holder as shown in Fig. P5.4.

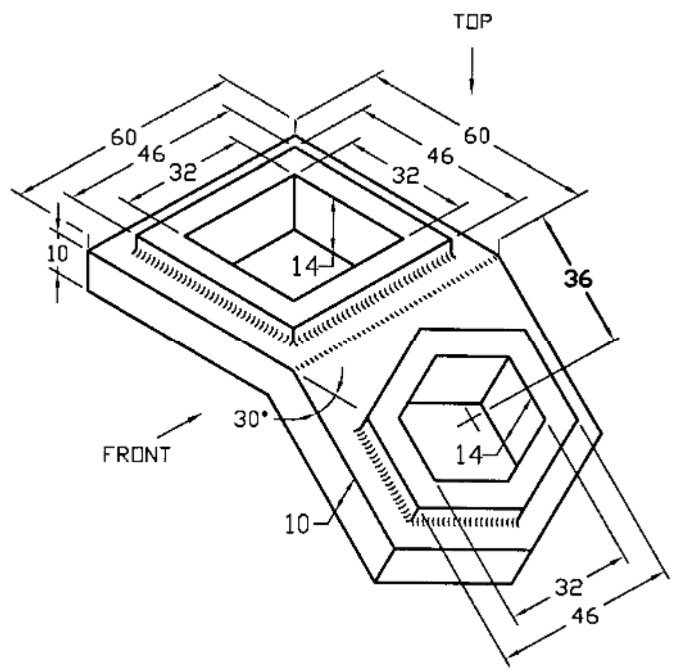


Fig. P5.4

Prob. 5.5: Draw front, partial bottom and auxiliary views of the fixer plate as shown in Fig. P5.5.

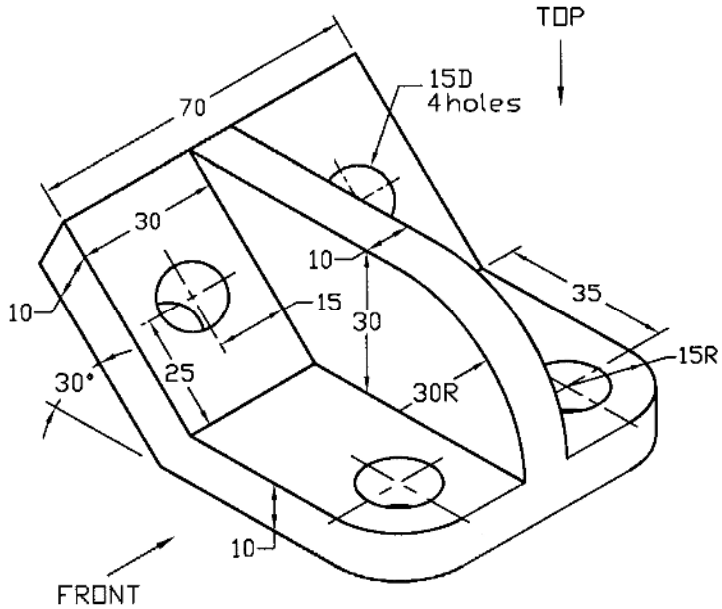


Fig. P5.5

Prob. 5.6: Draw front, partial bottom, partial left side and auxiliary views of the clip as shown in Fig. P5.6.

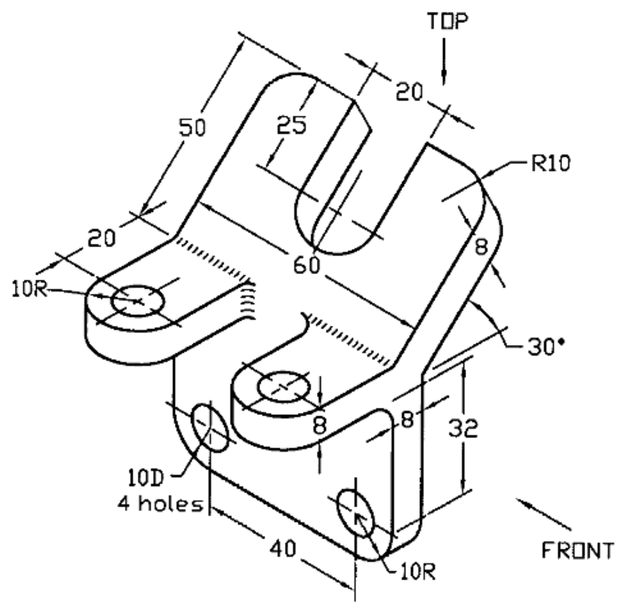


Fig. P5.6

Prob. 5.7: Draw front, partial bottom and auxiliary views of the support bracket as shown in Fig. P5.7.

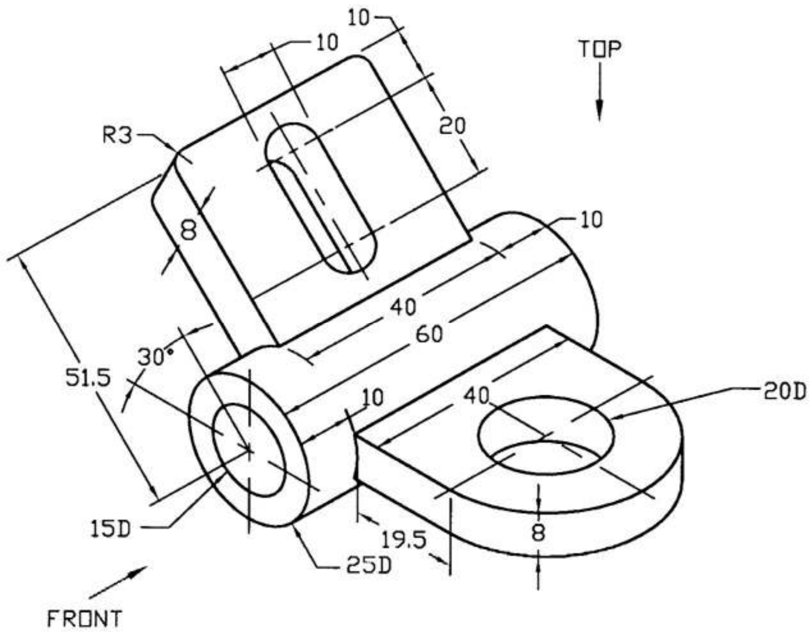


Fig. P5.7

Prob. 5.8: Draw front, partial top and auxiliary views of the angle bracket as shown in Fig. P5.8.

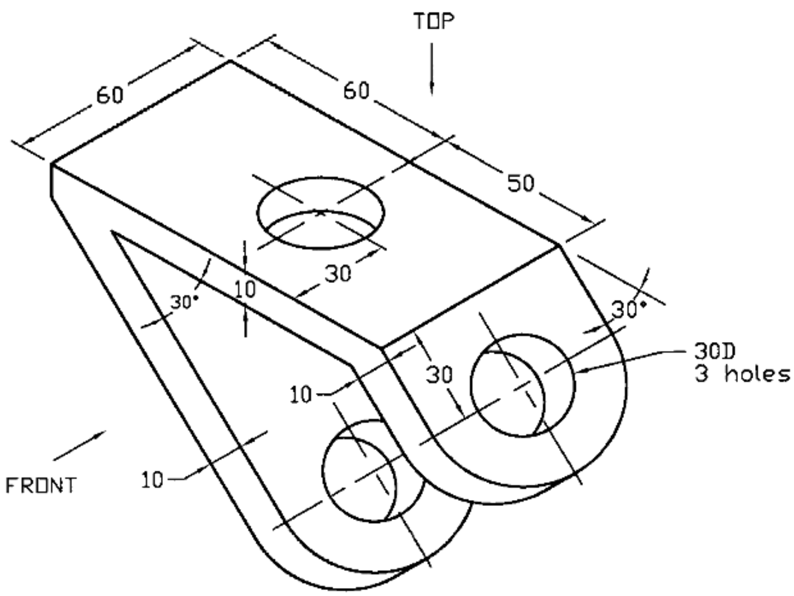


Fig. P5.8

Prob. 5.9: Draw top, partial front and auxiliary views of the wedge block as shown in Fig. P5.9.

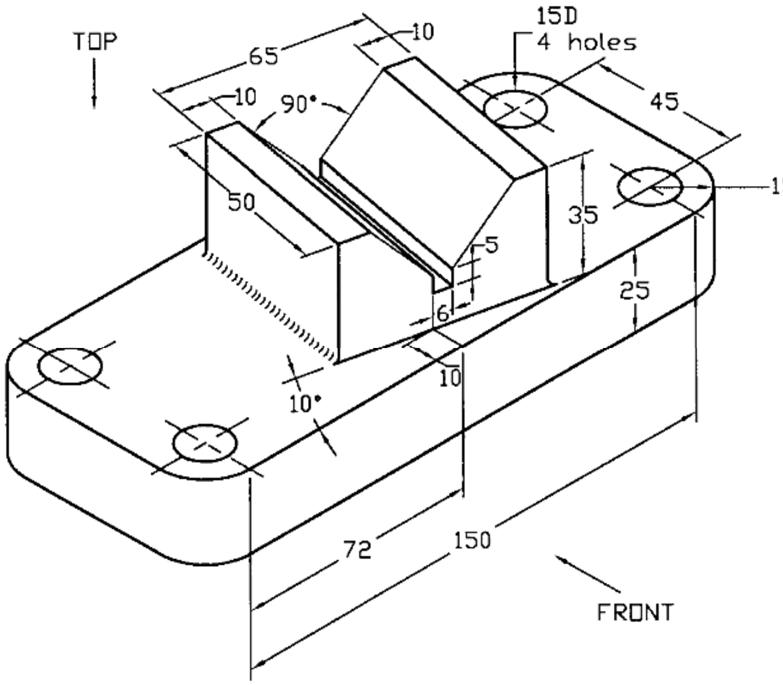


Fig. P5.9

Prob. 5.10: Draw top, partial front and auxiliary views of the connector as shown in Fig. P5.10.

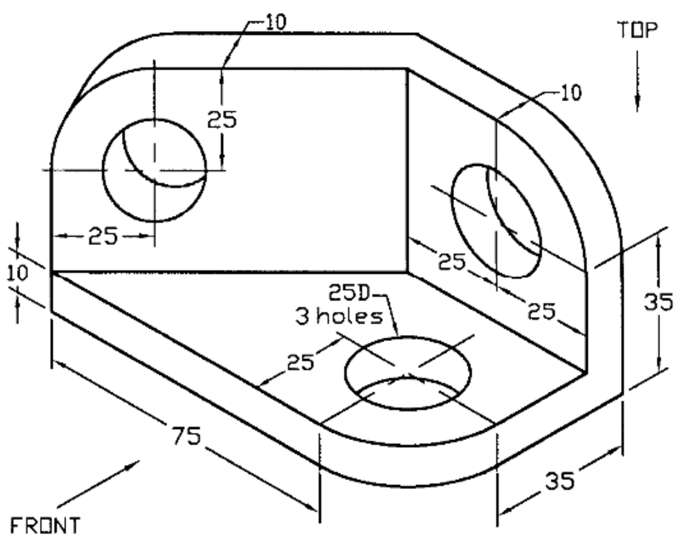


Fig. P5.10

Prob. 5.11: Draw front, partial bottom, left and right auxiliary views of the guide bracket as shown in Fig. P5.11.

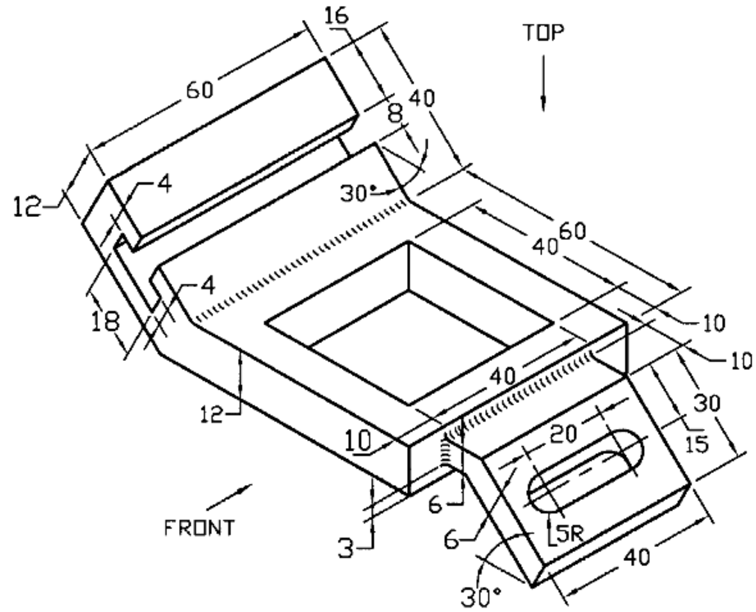


Fig. P5.11

Prob. 5.12: Draw front, partial bottom, partial right side and auxiliary views of the bracket as shown in Fig. P5.12.

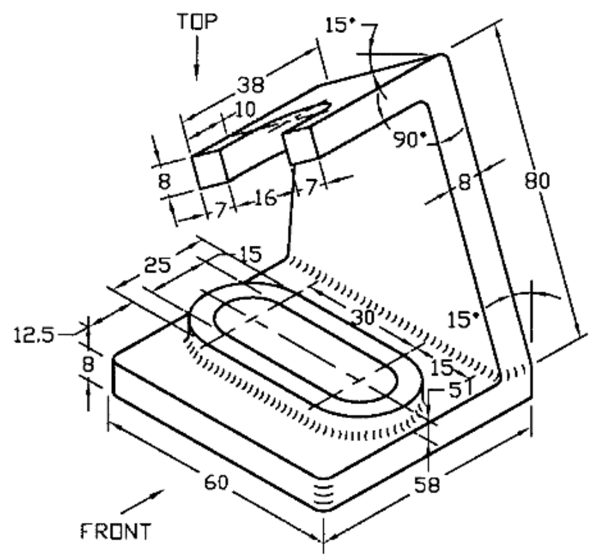
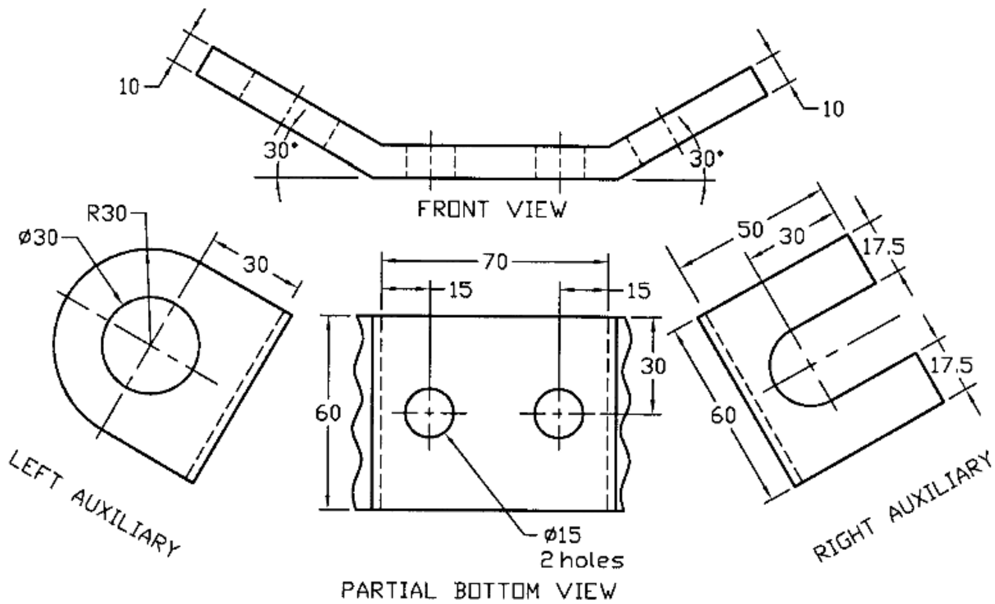
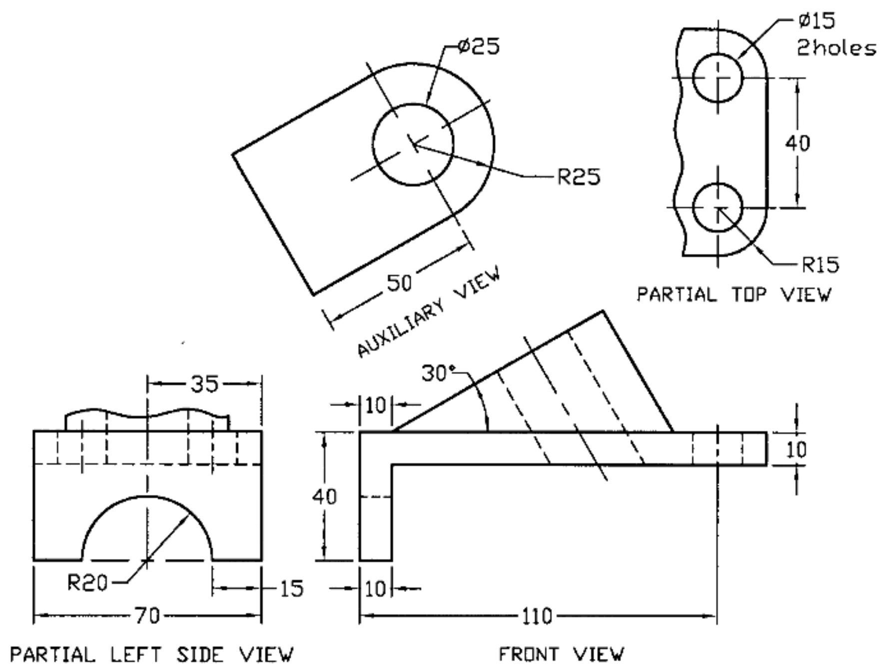


Fig. P5.12

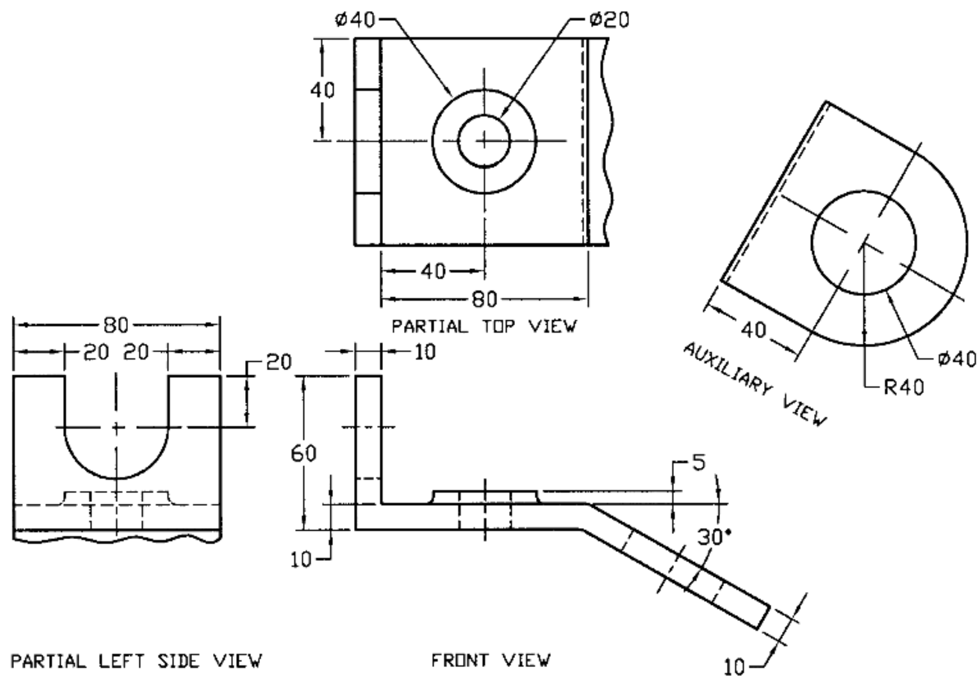
Solutions for Example problems



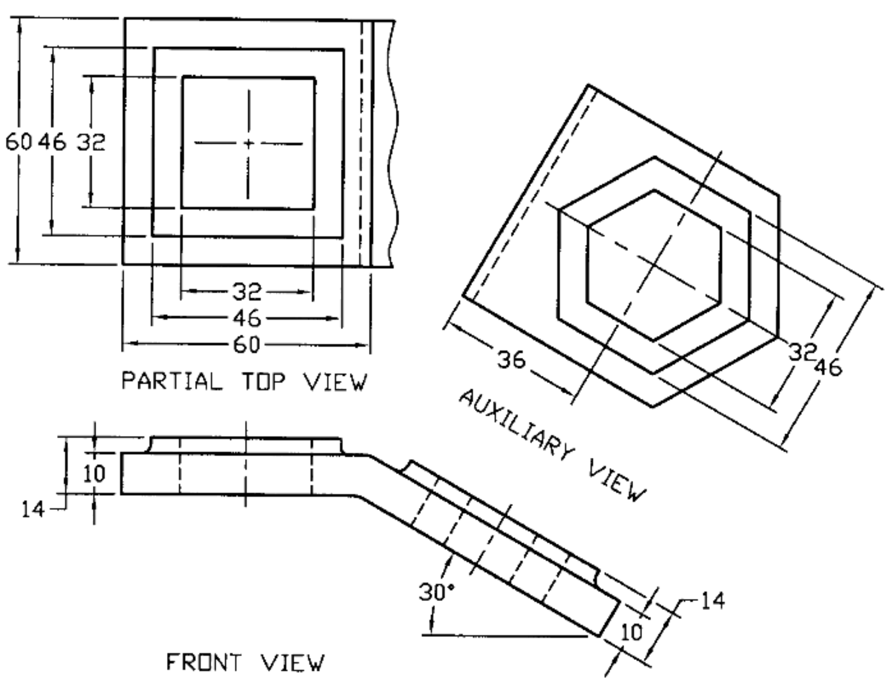
Solution of P5.1



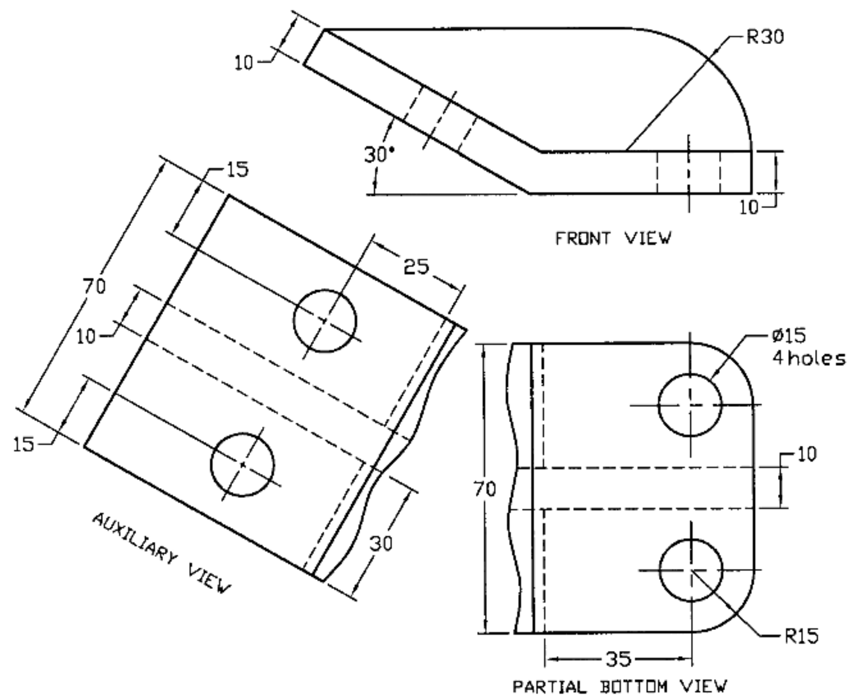
Solution of P5.2



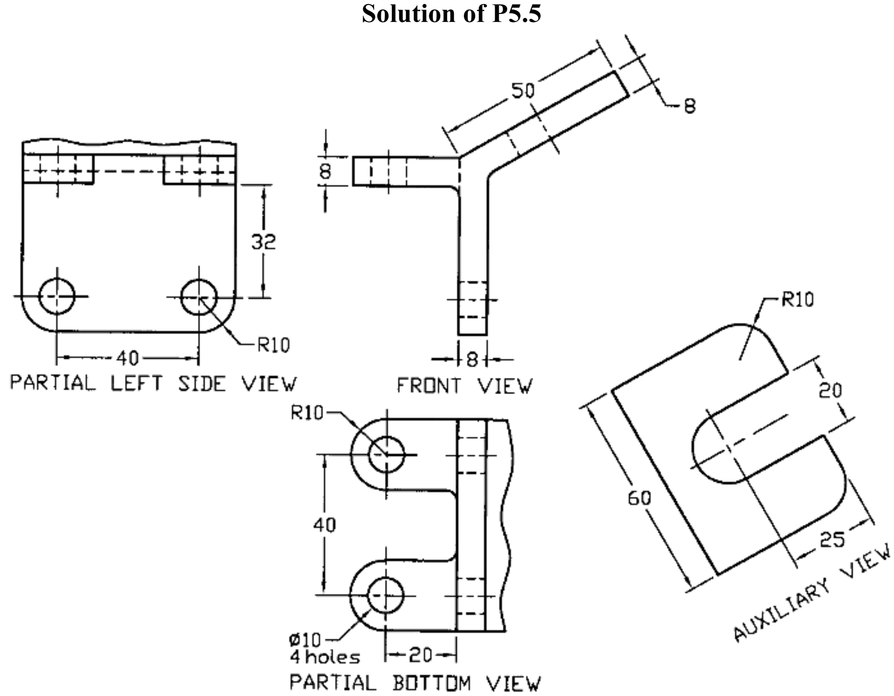
Solution of P5.3



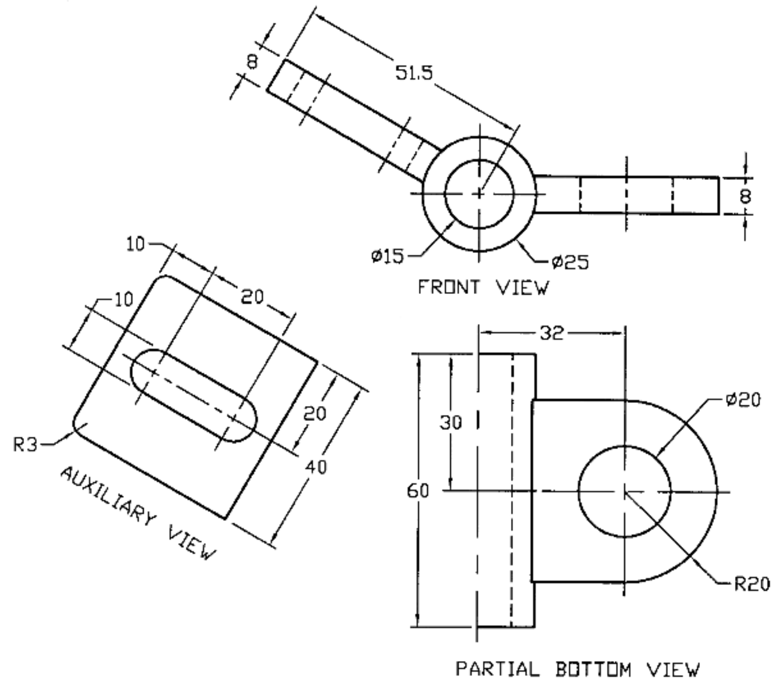
Solution of P5.4



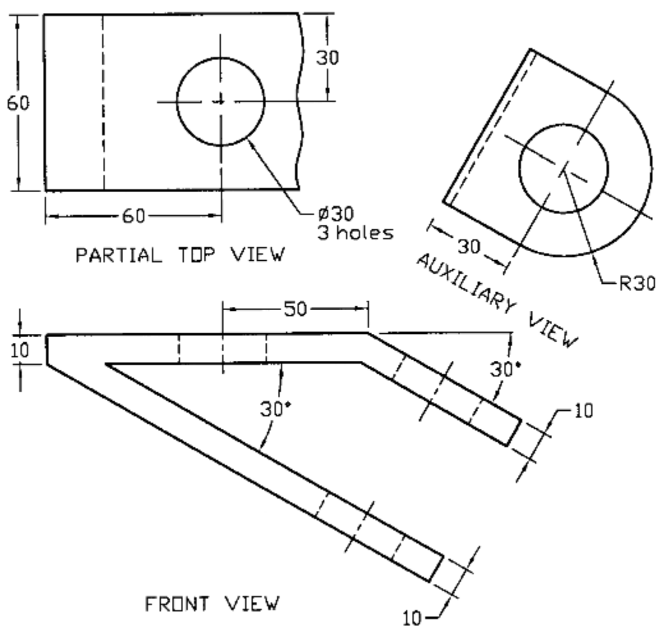
Solution of P5.5



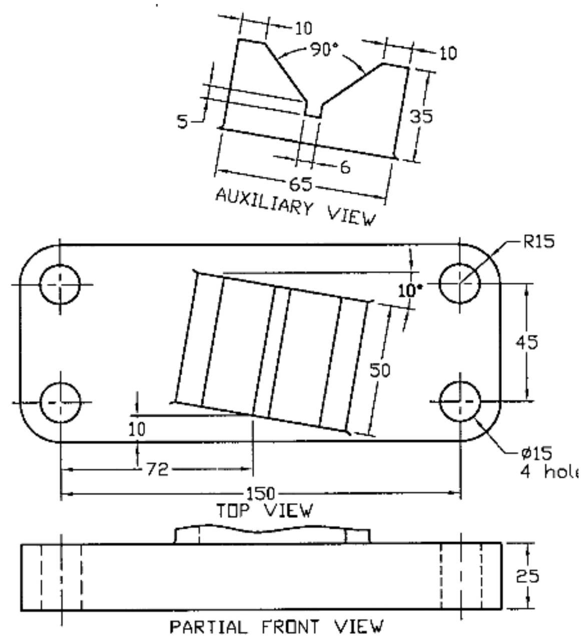
Solution of P5.6



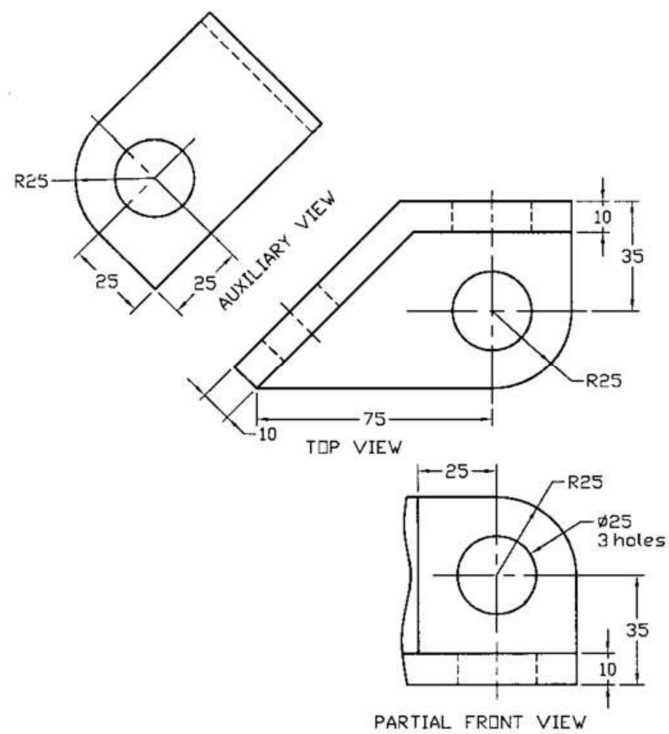
Solution of P5.7



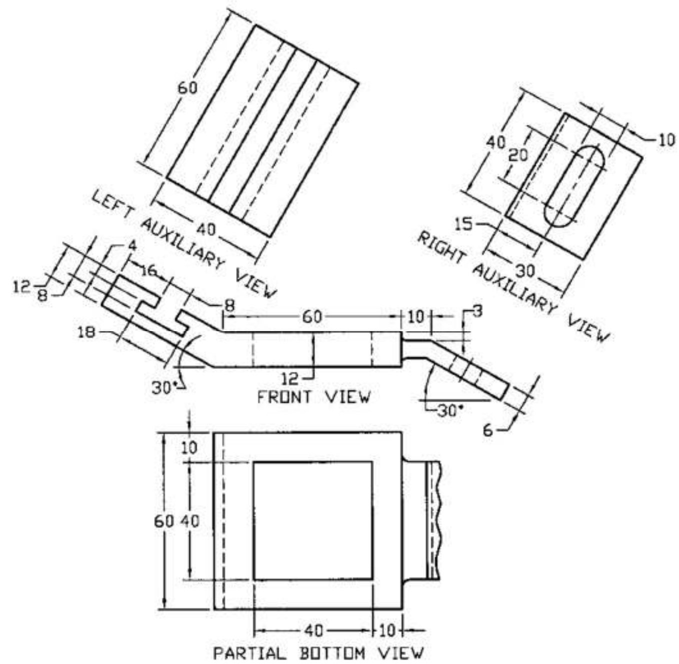
Solution of P5.8



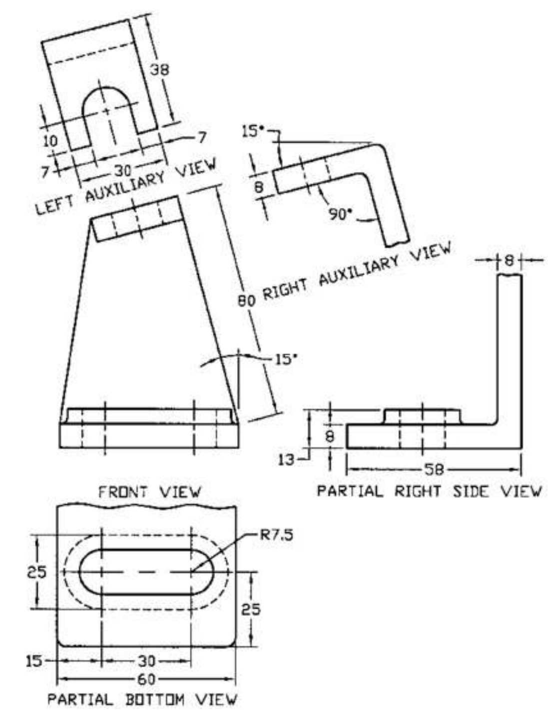
Solution of P5.9



Solution of P5.10



Solution of P5.11



Solution of P5.12

Problems

Prob. 5.13: Draw front, partial top and auxiliary views of the guide block shown in Fig. P5.13.

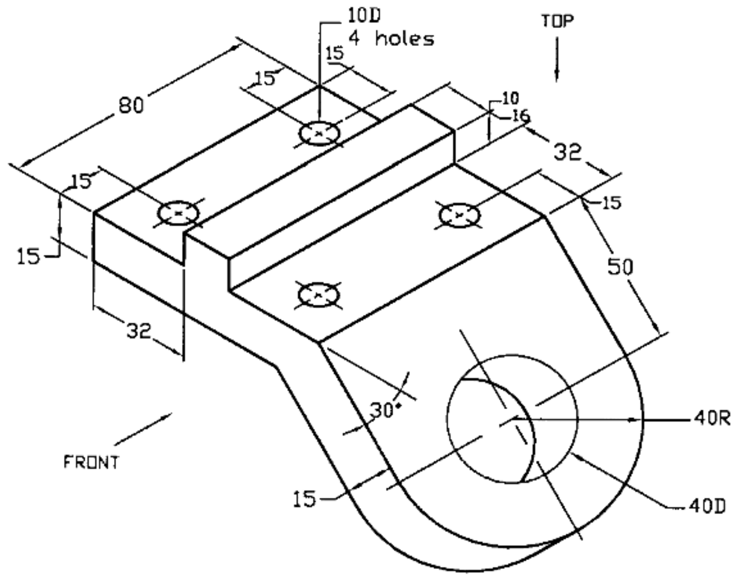


Fig. P5.13

Prob. 5.14: Draw front, partial top, partial right side and auxiliary views of the mounting bracket shown in Fig. P5.14.

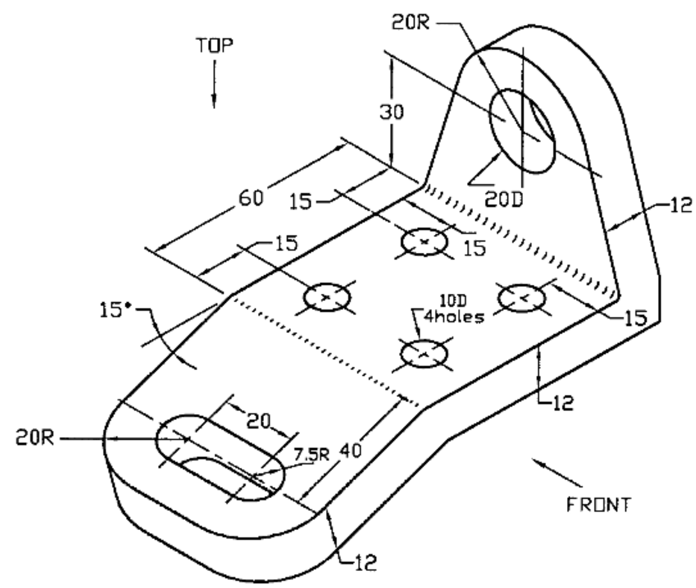


Fig. P5.14

Prob. 5.15: Draw front, partial top and auxiliary views of the clip shown in Fig. P5.15.

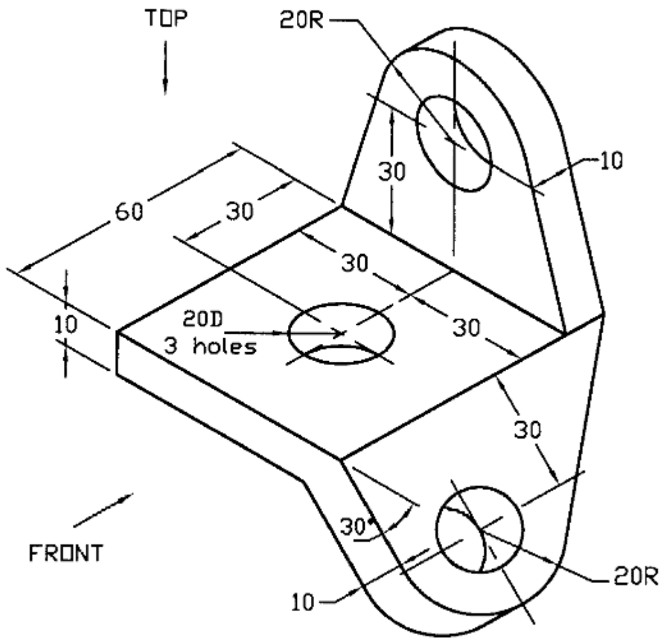


Fig. P5.15

Prob. 5.16: Draw the necessary views of the support base shown in Fig. P5.16.

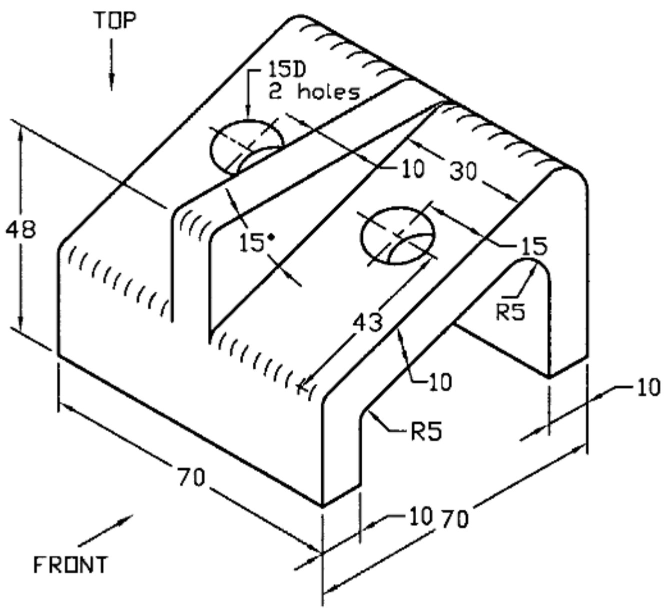


Fig. P5.16

Prob. 5.17: Draw the front, partial top, left and right auxiliary views of the support bracket shown in Fig. P5.17.

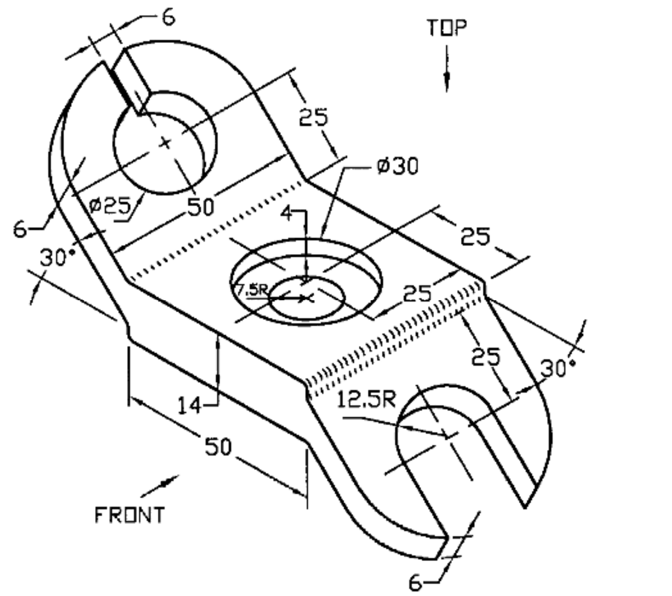


Fig. P5.17

Prob. 5.18: Draw the necessary views of the connector block shown in Fig. P5.18.

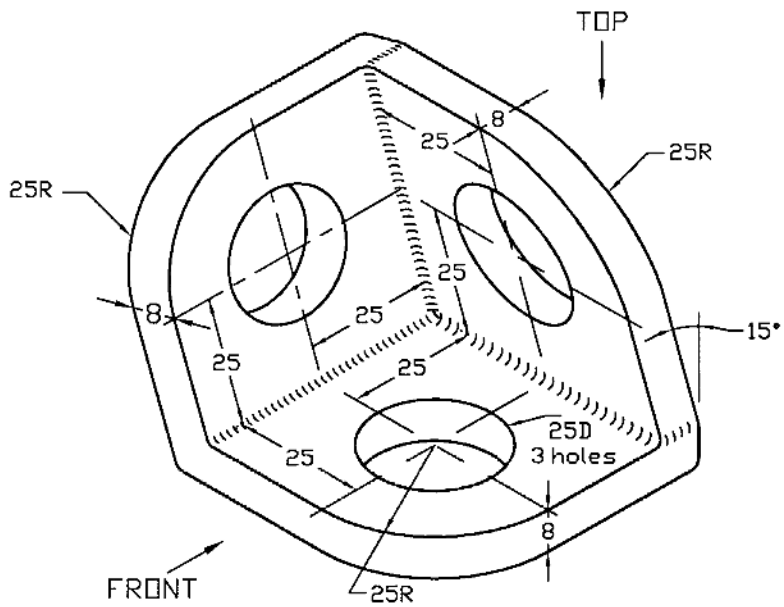


Fig. P5.18

Prob. 5.19: Draw the necessary views of the clip shown in Fig. P5.19.

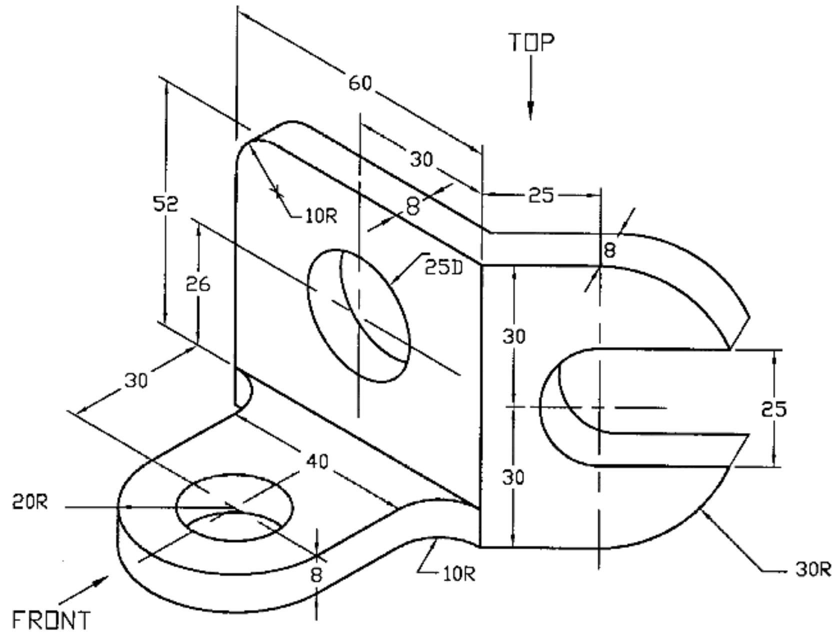


Fig. P5.19

Prob. 5.20: Draw the necessary views of the angle bracket shown in Fig. P5.20.

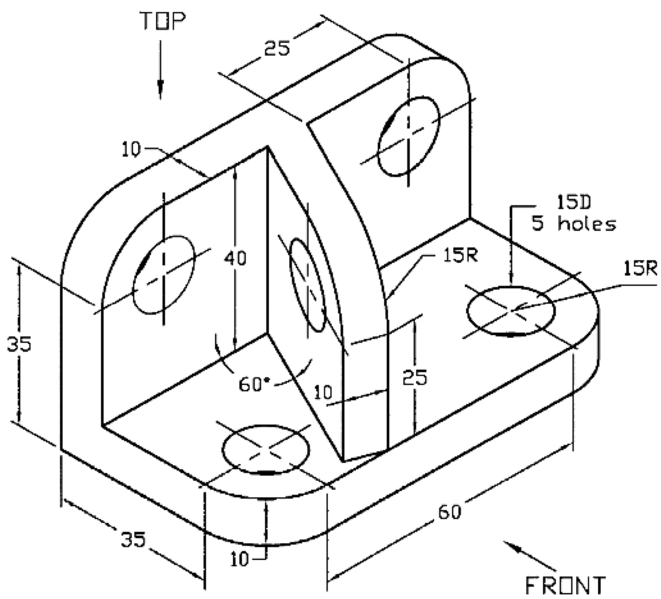


Fig. P5.20

Prob. 5.21: Draw the top, partial front and auxiliary views of the pin support base shown in Fig. P5.21.

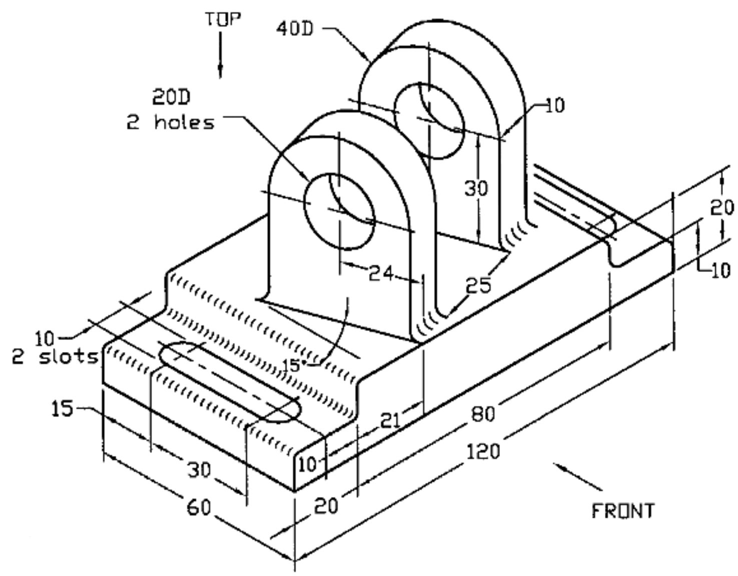


Fig. P5.21

Prob. 5.22: Draw the necessary views of the elbow shown in Fig. P5.22.

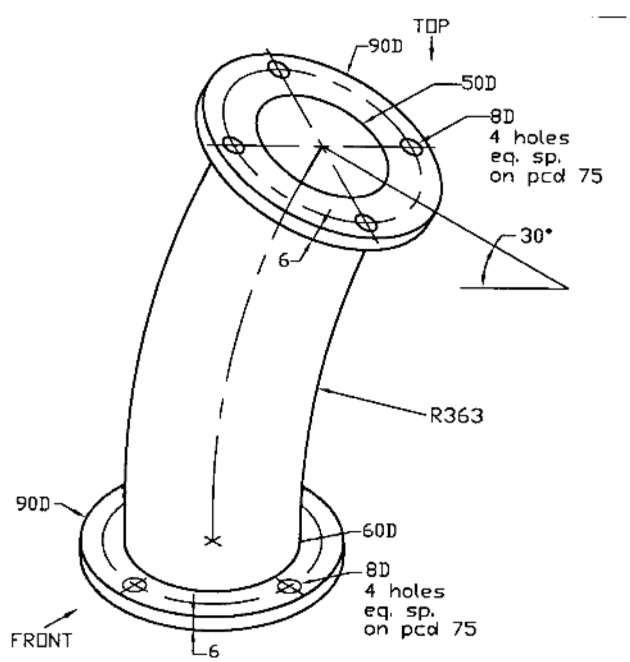


Fig. P5.22

Prob. 5.23: Draw the necessary views of the fork bracket shown in Fig. P5.23.

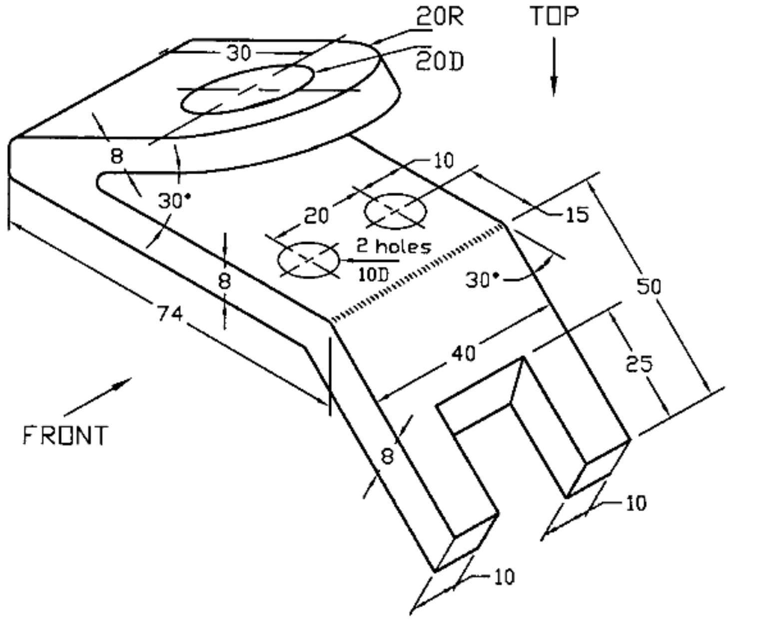


Fig. P5.23

Prob. 5.24: Draw the necessary views of the duct shown in Fig. P5.24.

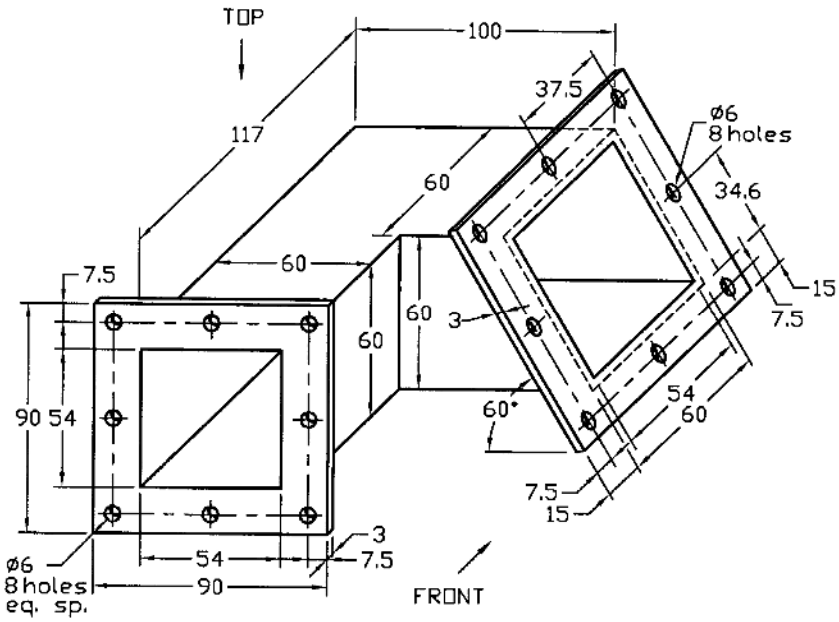


Fig. P5.24

Prob. 5.25: Draw the necessary views of the angle frame as shown in Fig. P5.25.

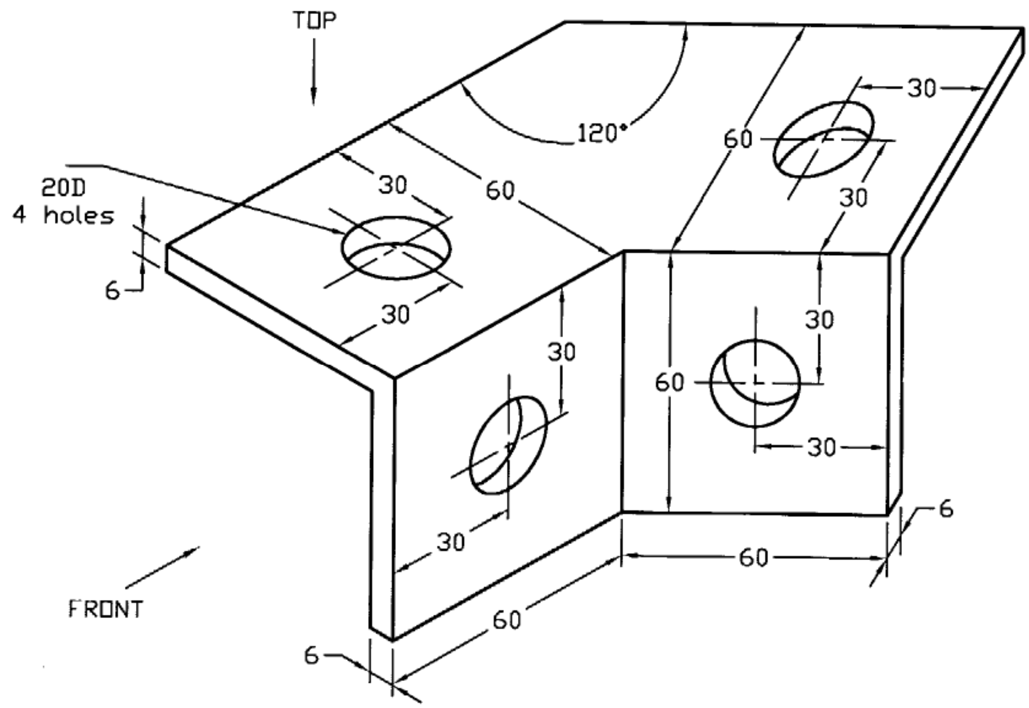

TOP
120°
30
60
20D
4 holes
30
60
6
30
30
60
30
30
60
30
30
FRONT
6
60
6
60

Fig. P5.25

CHAPTER 6

PICTORIAL DRAWING

6.1 *Introduction*

Pictorial view is a three dimensional one which is used to visualize an object in one view. On the other hand, the orthographic views such as, top view, front view, side view etc. which are separated from each other provide necessary information about dimensions, material, surface finish etc. to manufacture the object. With the help of pictorial view complicated engineering drawings can easily be communicated to the people who do not have sufficient training in understanding the orthographic views. A pictorial view provides the main dimensions of the object only. It cannot be used as a working drawing. It is used only to visualize the object. Pictorial drawing is classified as:

(1) Axonometric
(2) Oblique
(3) Perspective

6.2 *Perspective View*

The perspective drawing is relatively complicated to produce and is used mainly by the architects. A perspective projection is not suitable for working drawing. An example of a perspective drawing has been shown in Figure 6.1. Perspective drawing is the view drawn looking an object with normal vision. When an object goes away from the observer, it appears smaller and ultimately vanishes at a point. However, the engineers use only the oblique and axonometric drawing.

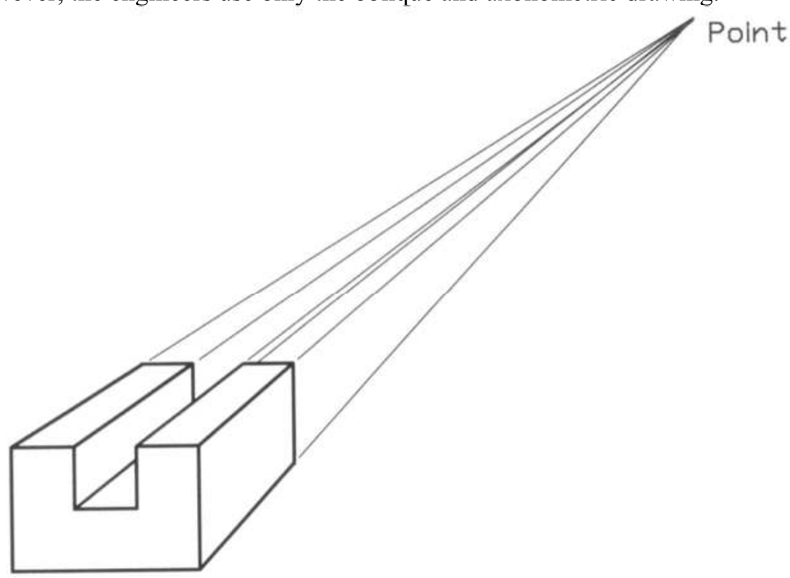


Figure 6.1: One Point Perspective View

6.3 *Axonometric Projection*

In the axonometric projection the three faces of an object are seen on the plane of projection (picture plane) and the projection lines from the object are perpendicular to the plane of projection. To understand an axonometric projection, a cube is considered. If the cube is placed behind a vertical transparent plane in such a way that one of the faces of the cube is parallel to the plane, the view on the plane based on orthographic projection will be of square shape that is represented by the shaded view as shown in Figure 6.2. Top and right side views have also been shown in the figure.

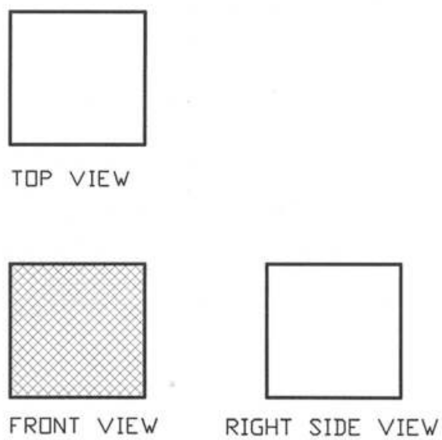


Figure 6.2: Views with One Face Parallel To the Projection Plane

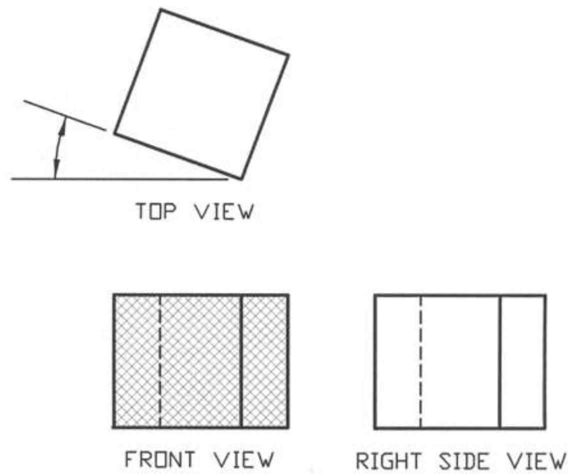


Figure 6.3: Views after Rotation About Vertical Axis

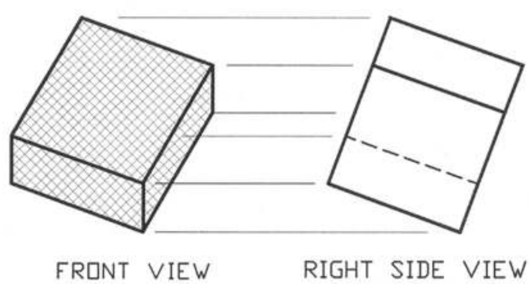


Figure 6.4: Views after Rotation about Horizontal Axis (Tilting)

If the cube is now rotated by any angle less than 90^0 about the vertical axis, the view on the plane will be similar to that represented by the shaded view as shown in Figure 6.3. Top and side views have also been shown. Now if the cube just from this position is rotated by any angle less than 90^0 about the horizontal axis that is parallel to the projection plane, the view on the plane will appear similar to that shown by the shaded view in Figure 6.4. Right side view has also been presented in this figure. The angle at which the cube is rotated may be of infinite number; as a result the number of axonometric positions may be infinite. It is usual practice to use three types of positions. Accordingly axonometric projections are of three types:

(1) Isometric

(2) Dimetric
(3) Trimetric

6.4 ***Isometric Projection***

To obtain the isometric projection, the cube is rotated by 45^0 about the vertical axis (Figure 6.5) from the position as shown in Figure 6.2, and then rotated about the horizontal axis, which is parallel to the projection plane in such a way that the top face of it makes a slope of $35^0 16^/$ approximately (Figure 6.6). In this position the diagonal OE passing through O, is perpendicular to the projection plane and the three mutually perpendicular edges OA, OB and OC make angles of 120^0 with each other. The projection of this particular position as represented by the front view in Figure 6.6 is called the isometric projection. (iso means same and metric means measurement, thus isometric means same measurement).

According to the geometrical principle when the top surface of the cube makes a slope of $35^0 16^/$, then the angles between the edges OA, OB and OC of the cube become 120^0 on the projection plane. The length OA as represented by the right side view of Figure 6.6, is the actual length (100%) and its projected length OA as shown in the front view in the same figure is shortened to 81% approximately. The isometric axes OA, OB and OC are foreshortened equally because they are at the same angle of 120^0 to the projection plane.

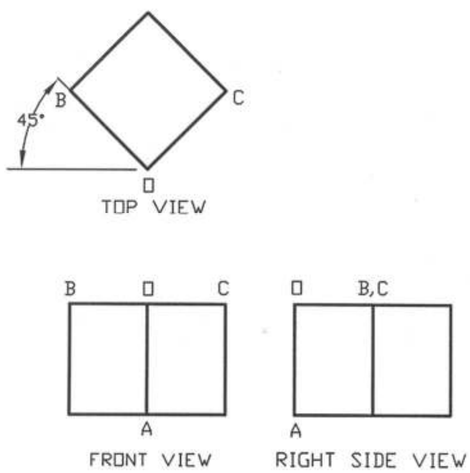


Figure 6.5: Views after Rotation About Vertical Axis by 45^0

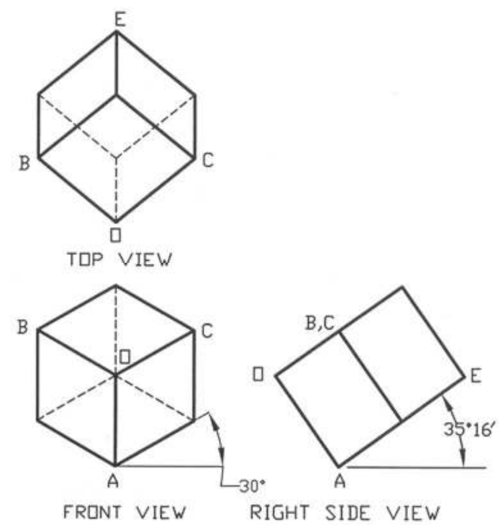


Figure 6.6: Views after Rotation About Horizontal Axis by $35^0 16^/$

In Figures 6.7 and 6.8 an isometric projection and isometric drawing are shown. An isometric drawing differs from an isometric projection in that the isometric drawing is done in actual size while the isometric projection is drawn foreshortened. If the actual size is 100%, the size in the isometric projection is $100\cos(35^0 16^/)$ i.e. 81% approximately. As a matter of fact it is usually convenient to make the isometric drawing rather than the isometric projection.

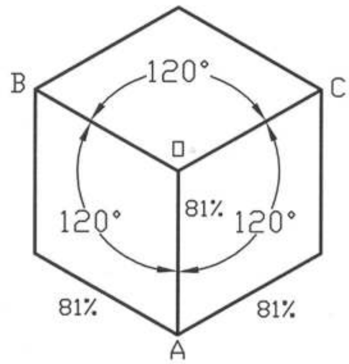


Figure 6.7: Isometric Projection

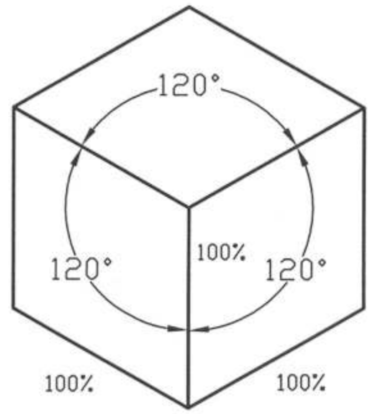


Figure 6.8: Isometric Drawing

Here the lines OA, OB and OC are called isometric axes. Any line parallel to the isometric axis is called isometric line. The line which, is not parallel to the isometric axis, is called non-isometric line. The planes of the faces of the isometric cube and all other planes, which are parallel to them, are called isometric planes.

In the isometric projection, the isometric lines can be drawn using isometric scale (Figure 6.9). The isometric line is shortened to 81% approximately, which is obtained from the isometric scale directly.

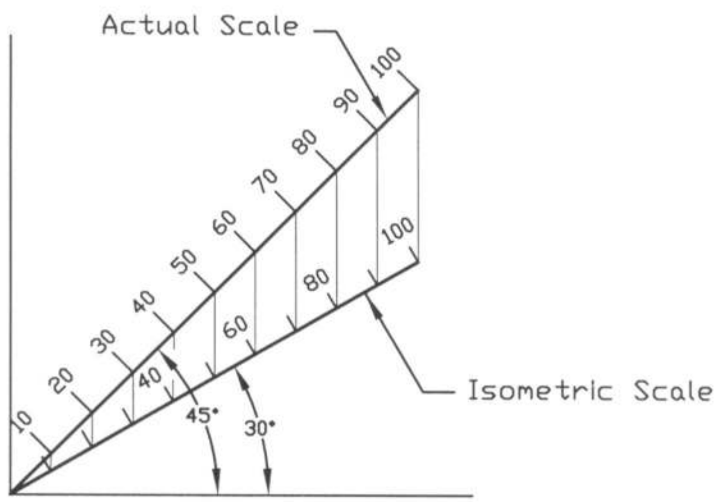
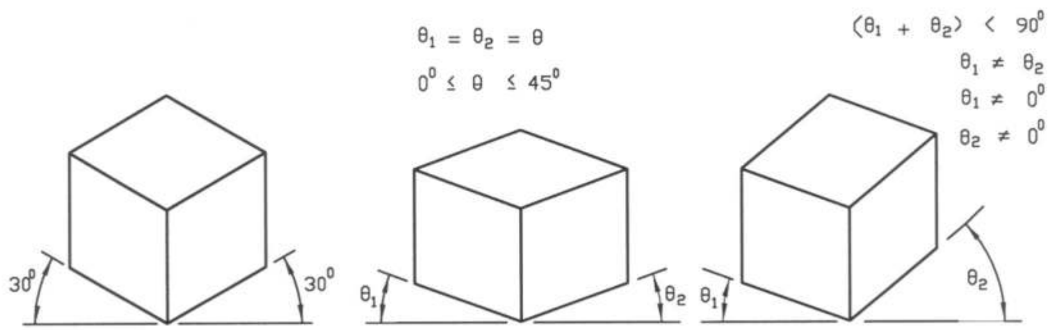


Figure 6.9: Isometric Scale

In dimetric projection, the two angles (θ_1, θ_2) as shown in Figure 6.10 are kept equal and may be any angle between 0^0 and 45^0 except 30^0. While in the trimetric projection the angles (θ_1, θ_2) are not equal; sum of them is less than 90^0 but neither angle is 0^0. The comparisons of the isometric, dimetric and trimetric projections have been shown in Figure 6.10.



(a) Isometric Projection (b) Dimetric Projection (c) Trimetric Projection

Figure 6.10: Types of Axonometric Projection

6.5 *Making Isometric View*

A simplified procedure is provided below to draw an isometric view from the orthographic projection. In Figure 6.11, the orthographic projection of an object is given from where an isometric view will be drawn. The steps as mentioned below are to be followed in order to make the isometric view.

(1) An isometric parallelepiped is drawn (Figure 6.12) in such a way that the length (l), breadth (b) and height (h) of the parallelepiped are respectively equal to those of the orthographic projection (Figure 6.11). The lines will be very thin so that they can be erased easily when required.

(2) Now the views are drawn on the respective side of the isometric parallelepiped keeping appropriate relationship with each other (Figure 6.13). In drawing views the unnecessary lines may be avoided otherwise, the drawing will be clumsy. The horizontal lines of the view will be parallel to the isometric axes OB and OC while the vertical lines will be parallel to the isometric axis OA. Each line will be thin.

(3) Next shifting of the necessary surfaces is made keeping conformity with the views. Surface a-b-c-d has been shifted to the new position a'-b'-c'-d' and surface p-q-r-s has been shifted to the new position p'-q'-r'-s' (Figure 6.14).

(4) The unnecessary lines are now erased and the isometric view is completed making the lines thick (Figure 6.15).

(5) Finally from the generated isometric view the orthographic projection may be verified to confirm the drawing whether it has been done properly.

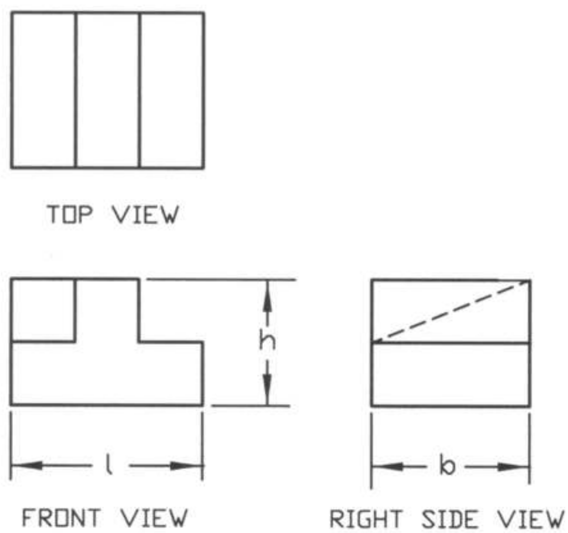


Figure 6.11: Orthogonal Projection

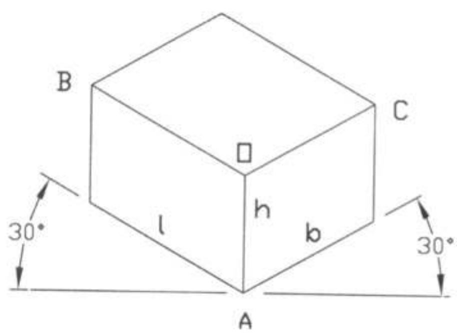


Figure 6.12: Isometric Parallelepiped

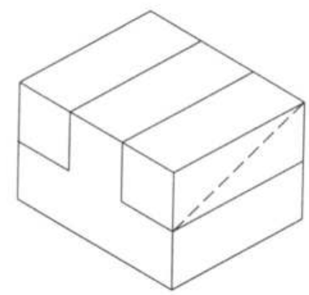


Figure 6.13: Views on Each Side

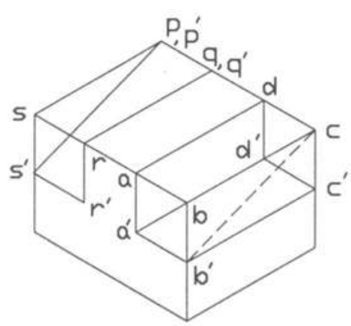


Figure 6.14: Shifting of Surfaces

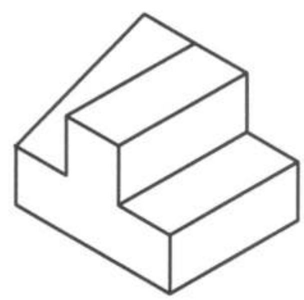


Figure 6.15: Isometric View

6.6 ***Making Isometric View with Circular Features***

A circle in the orthographic projection is transformed into the shape of an ellipse. In Figure 6.16 the orthographic projection of an object with circular feature is provided. To make the isometric view the steps as mentioned below are followed.

(1) A square touching the circle is drawn with sides equal to the diameter of the circle on the orthographic projection (Figure 6.17).

(2) Now the isometric parallelepiped (Figure 6.18) is drawn as done earlier.

(3) Next the views on the surfaces of the parallelepiped (Figure 6.19) are drawn with the square of the circle omitting the circle itself. The square is turned into the shape of a rhombus. The unnecessary lines may be avoided.

(4) Now the lines on the rhombus are drawn as shown in Figure 6.20. Then the four centers c_1, c_2, c_3, and c_4 are located.

(5) The two arcs are drawn with radii c_1b and c_2a with respect to the centers c_1 and c_2 respectively. Next two other arcs are drawn with radii c_3a and c_4b with respect to the centers c_3 and c_4 respectively (Figure 6.21).

(6) Now the unnecessary lines are erased and the isometric view is completed making the lines thick (Figure 6.22).

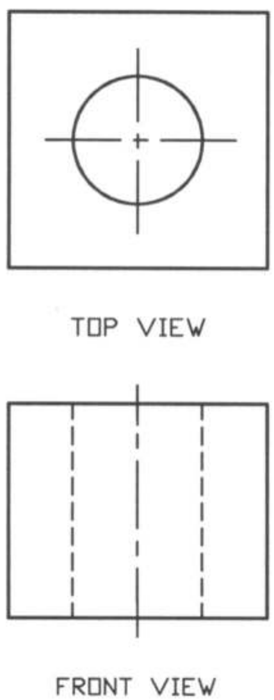


Figure 6.16: Orthographic Projection

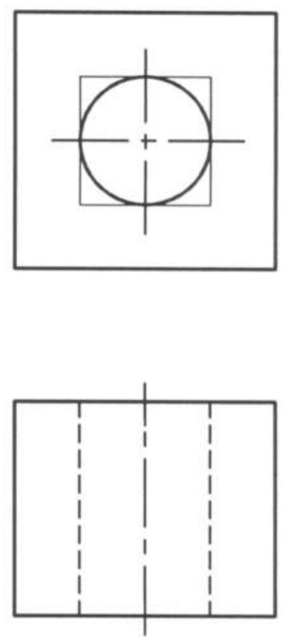


Figure 6.17: Square Touching Circle

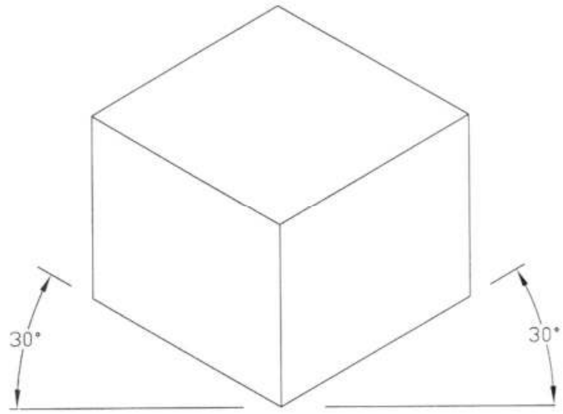


Figure 6.18: Isometric Parallelepiped

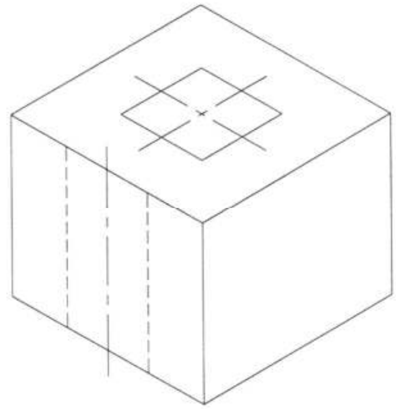


Figure 6.19: Views on Faces of Parallelepiped

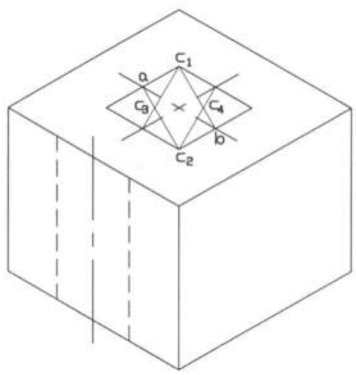


Figure 6.20: Location of Centers

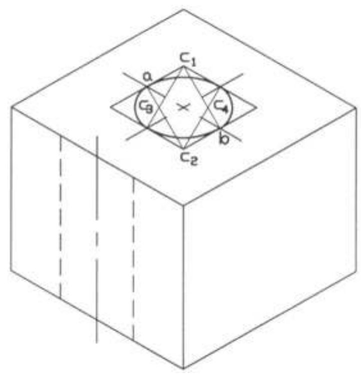


Figure 6.21: Drawing Arcs

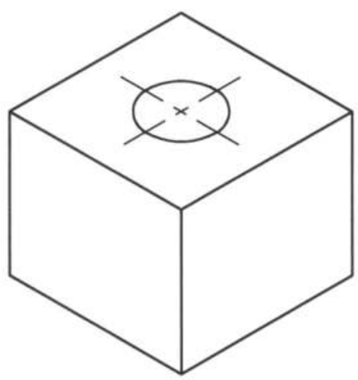


Figure 6.22: Isometric Circle

The orthographic projection of an object incorporating the circular feature is shown in Figure 6.23. The various steps are represented graphically in Figures 6.24 to 6.31 in order to generate an isometric view. This is provided as a further example.

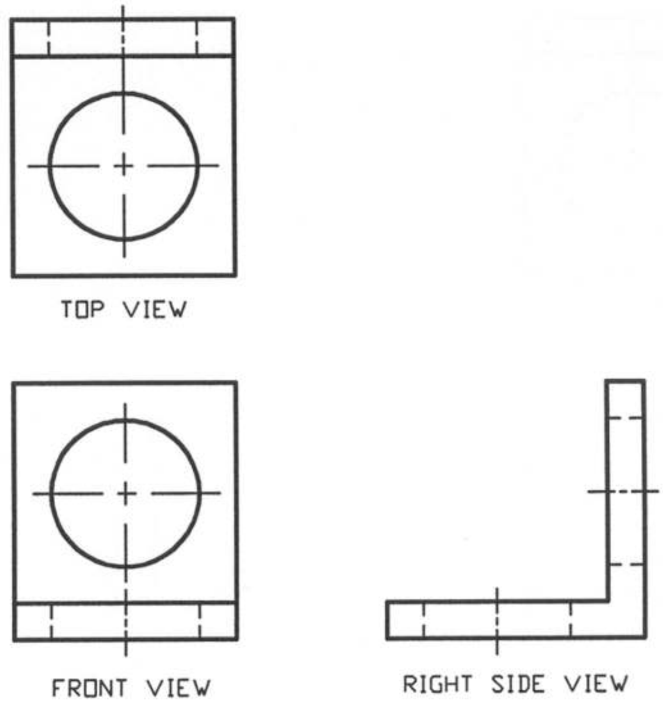


Figure 6.23: Orthographic Projection

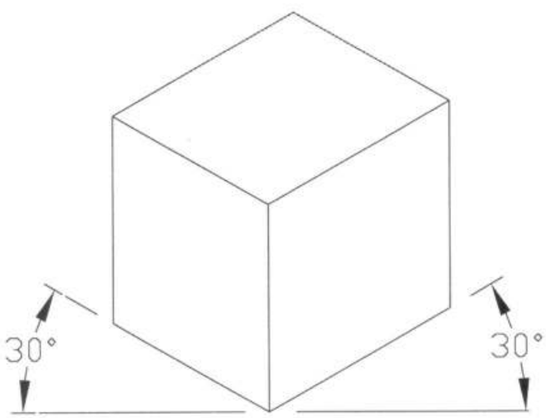


Figure 6.24: Making Parallelepiped

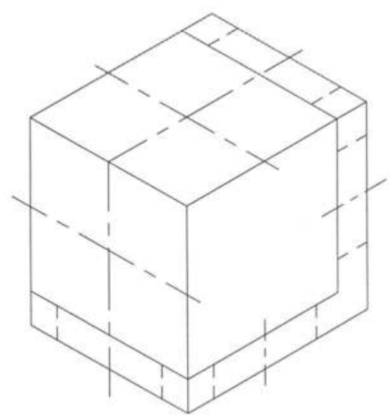


Figure 6.25: Views on Faces

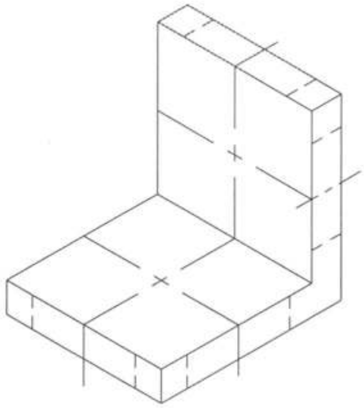


Figure 6.26: Shifting Faces

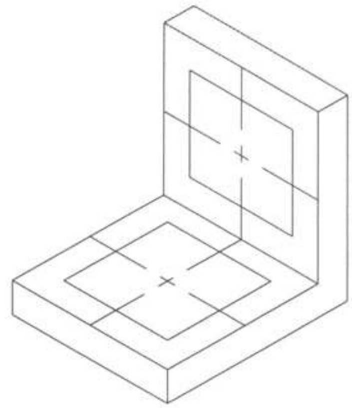


Figure 6.27: Rhombuses for Circles

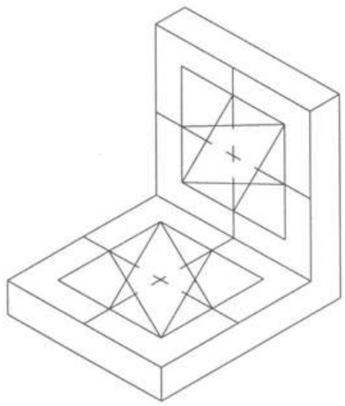


Figure 6.28: Locating Centers

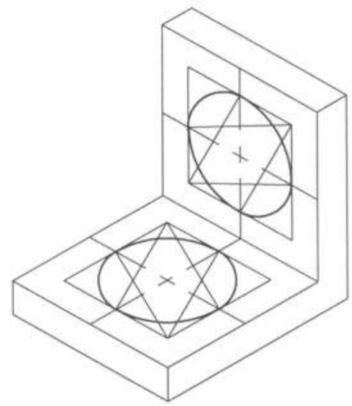


Figure 6.29: Making Ellipses

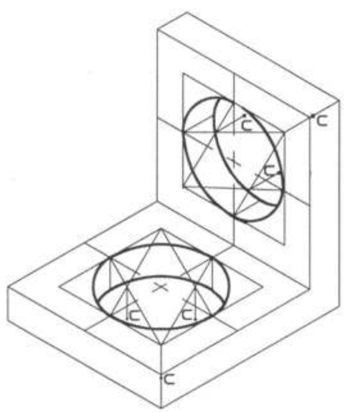


Figure 6.30: Ellipses on Other Faces

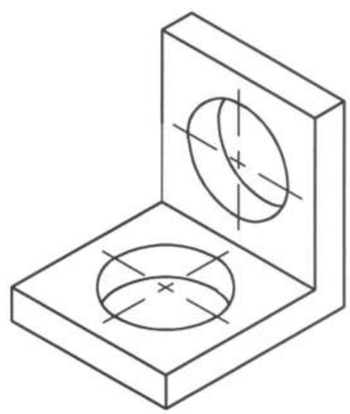


Figure 6.31: Complete Isometric View

6.7 ***Oblique Projection***

In generating an isometric projection, one looks in a perpendicular direction through the projection plane at an object, which is positioned in rotating condition with respect to the projection plane (rotated about vertical axis and tilted). But in producing oblique projection opposite phenomenon occurs. The object is positioned with its major face parallel to the projection plane. Now an observer looks at the object through the projection plane at an oblique angle unlike in the case of isometric projection. In Figure 6.32 an oblique projection has been presented.

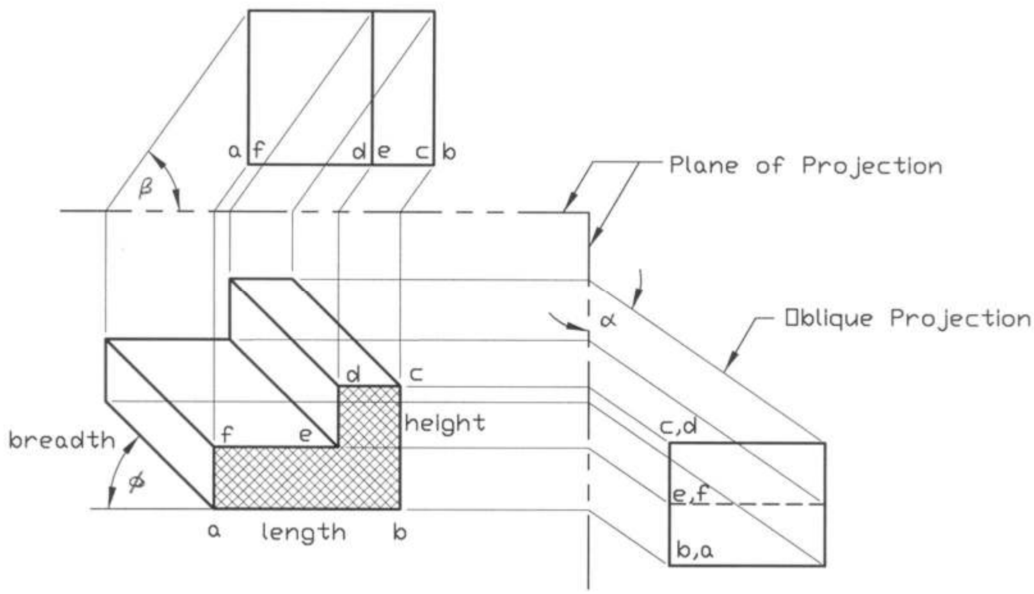


Figure 6.32: Oblique Projection

The axes of length and height are perpendicular to each other as shown in the figure. The axis of breadth can vary in angle ϕ and in length. Angle ϕ and the breadth of the object may be any value but usual practice is to keep ϕ at 30^0 or 45^0 for convenience. The surface of the object, which is parallel to the frontal plane of projection, represents the true shape as shown by the shaded area in the figure. This is the basic character of the oblique projection. As a result any surface of the object containing circle, curve or irregular shape, which is parallel to the frontal plane represents the exact shape. That is why the oblique projection is used mainly for the objects containing circle, curve or irregular features on one face or on parallel faces. For objects of this type oblique projection is easier to draw and give dimension.

In Figure 6.33 some oblique views are provided. When the breadth equals the full size (L), it is called cavalier and when the breadth equals the half size (.5L), it is called cabinet. Cabinet drawing looks pleasant compared to cavalier one. In case of the breadth in between the half and the full size (.5L<l<L), it is known as general.

In making an oblique view three mutually perpendicular axes passing through O (Figure 6.35) are chosen, one is horizontal, one is vertical and the other one is at an angle. On the three axes the height (h), width (w) and depth (d) are chosen in accordance with the orthographic projection (Figure 6.34). Now the centers are located and the view is completed. The interesting point can be noted that, as the frontal face contains the circle, no transformation into ellipse is necessary. All the circles in this feature remain as they are. It makes the drawing easier.

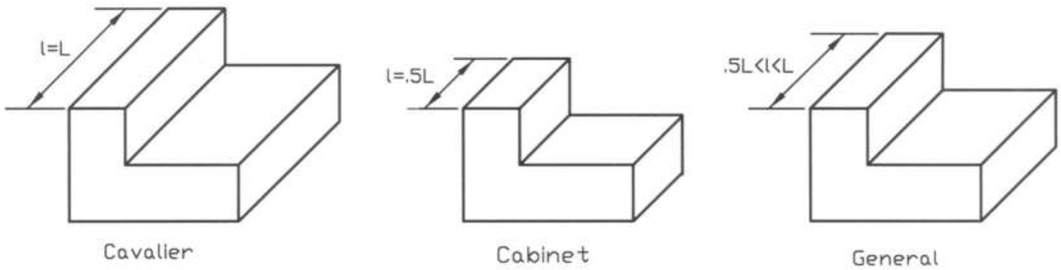


Figure 6.33: Oblique Views

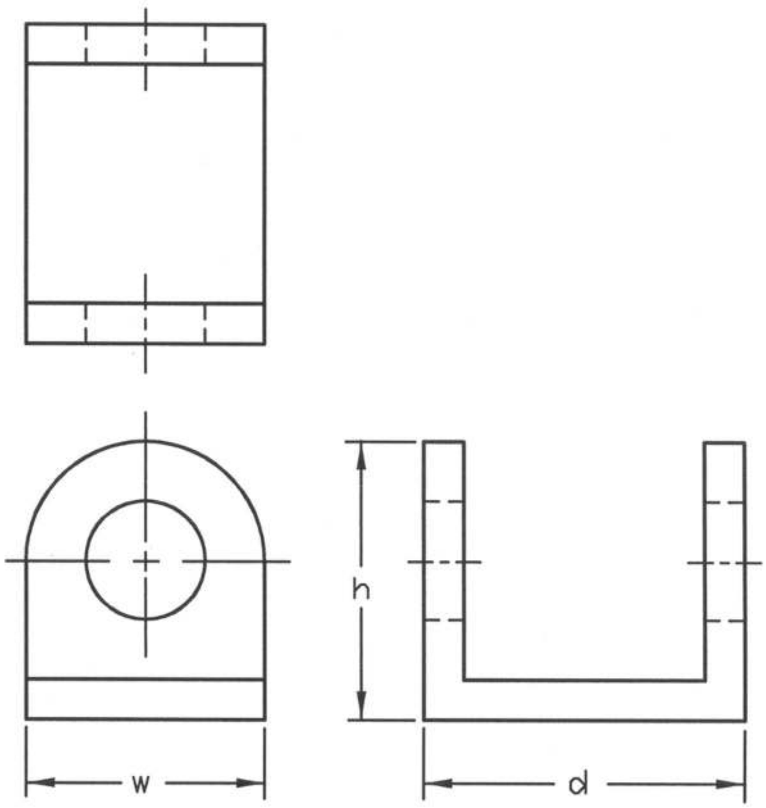


Figure 6.34: Orthographic Projection

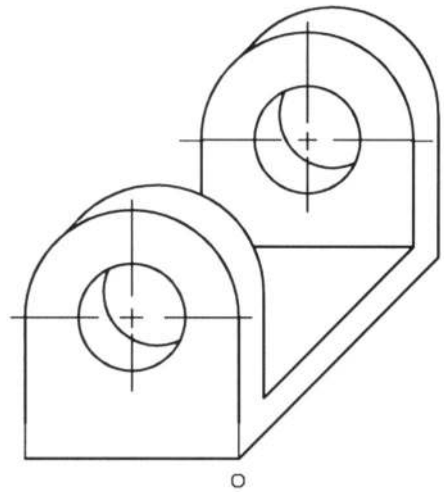


Figure 6.35: Oblique View with circular Feature

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.***

Prob. 6.1- 6.24: Draw the isometric sketches of the objects from the following views as shown in Fig. P6.1 to P6.24. The free hand Sketches may be done. Here the scaled drawings are provided.

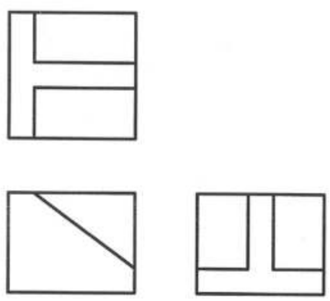


Fig. P6.1

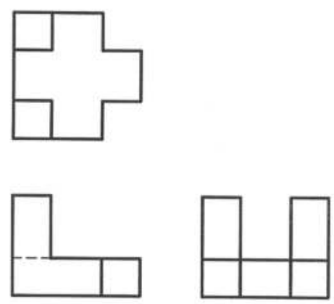


Fig. P6.2

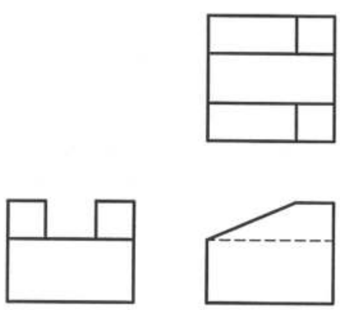


Fig. P6.3

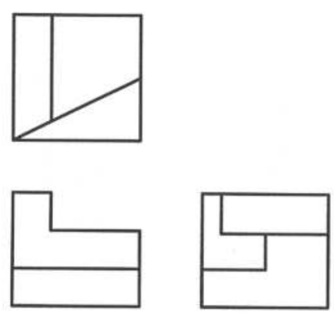


Fig. P6.4

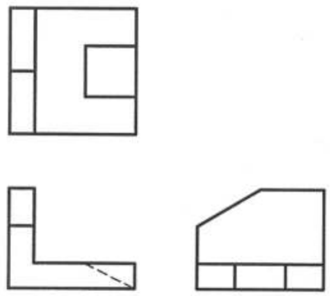


Fig. P6.5

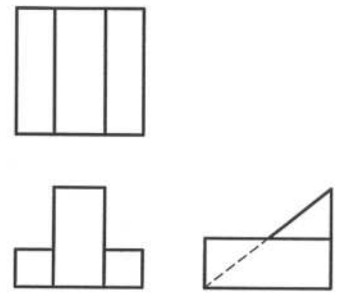


Fig. P6.6

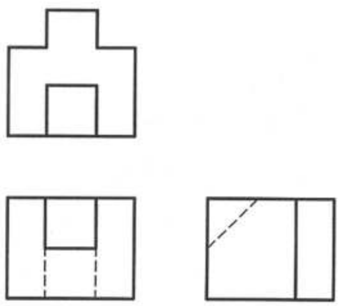


Fig. P6.7

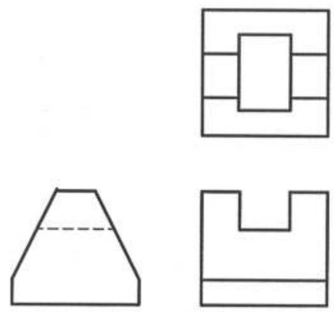


Fig. P6.8

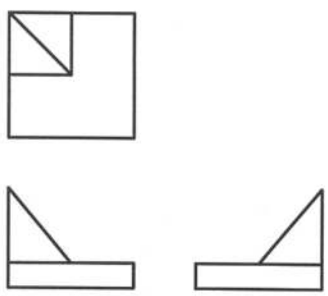


Fig. P6.9

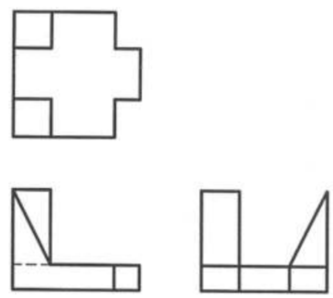


Fig. P6.10

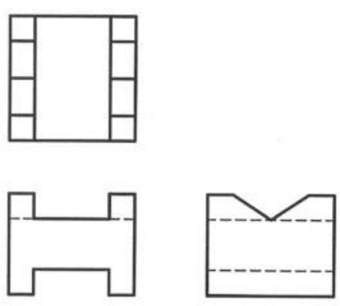


Fig. P6.11

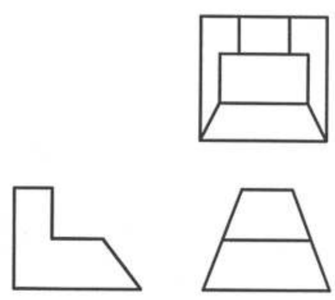


Fig. P6.12

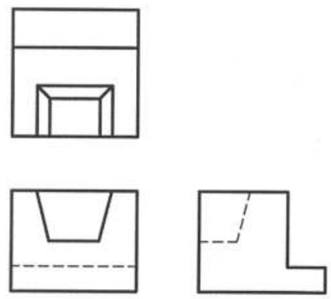


Fig. P6.13

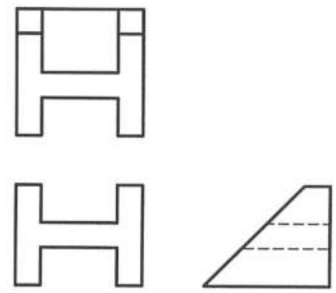


Fig. P6.14

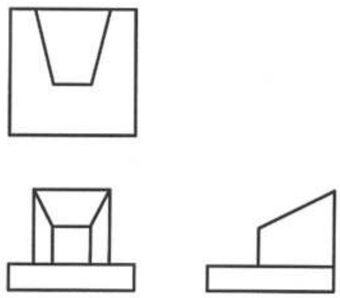


Fig. P6.15

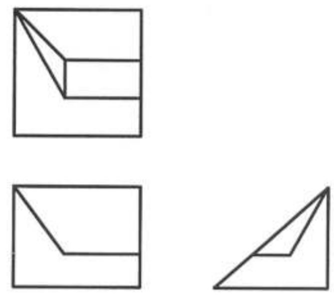


Fig. P6.16

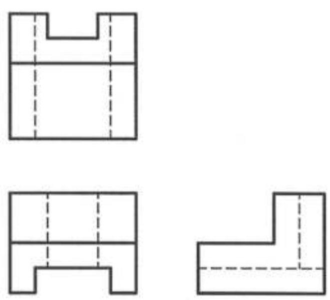


Fig. P6.17

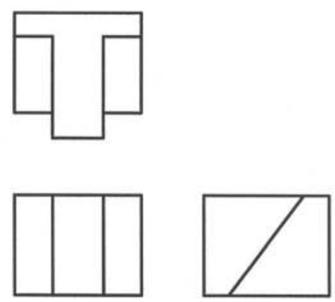


Fig. P6.18

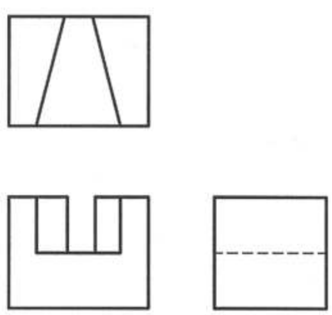


Fig. P6.19

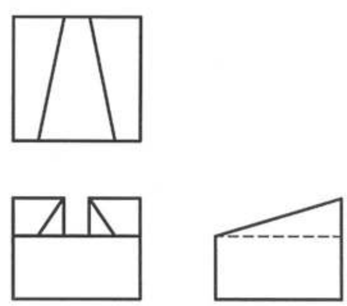


Fig. P6.20

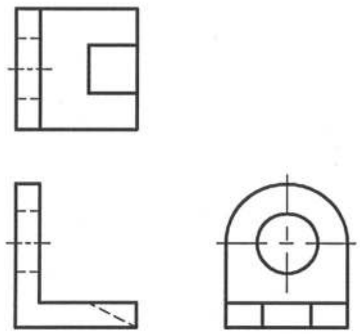


Fig. P6.21

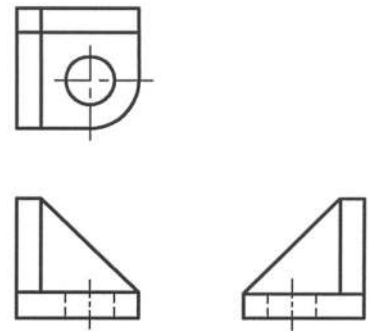


Fig. P6.22

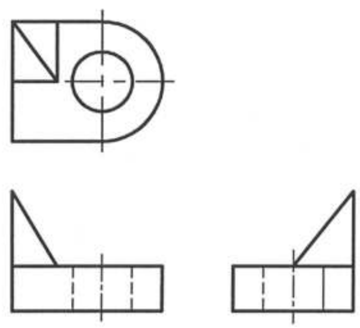


Fig. P6.23

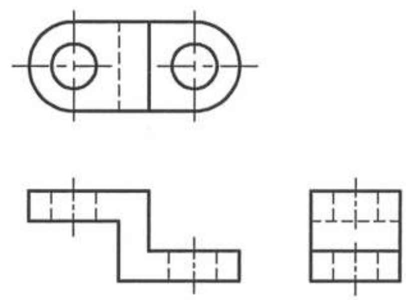


Fig. P6.24

Prob. 6.25- 6.45: Draw the isometric drawing of the objects from the following views as shown in Fig. P6.25 to P6.45.

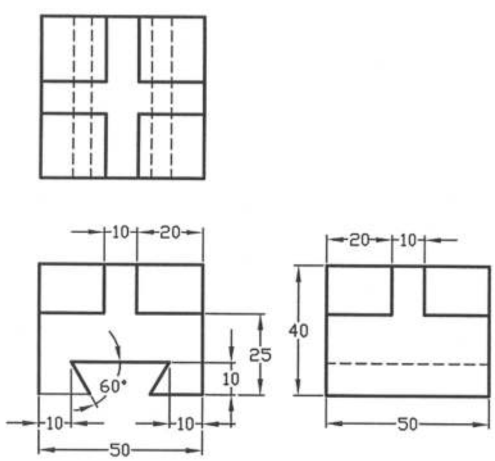


Fig. P6.25

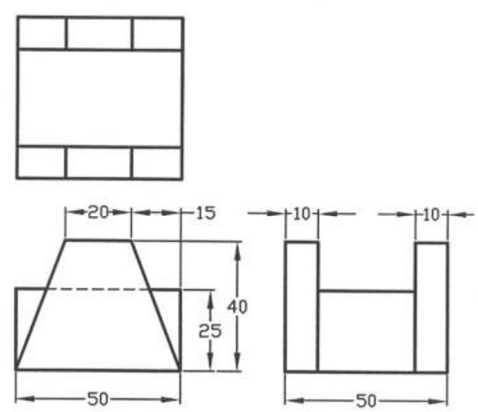


Fig. P6.26

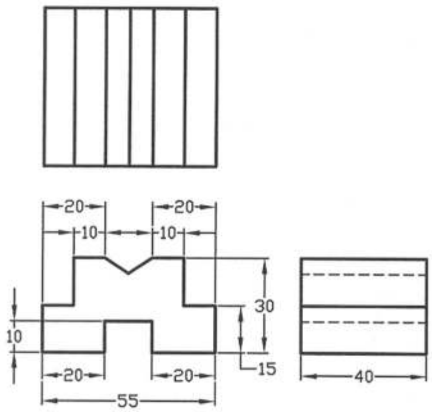


Fig. P6.27

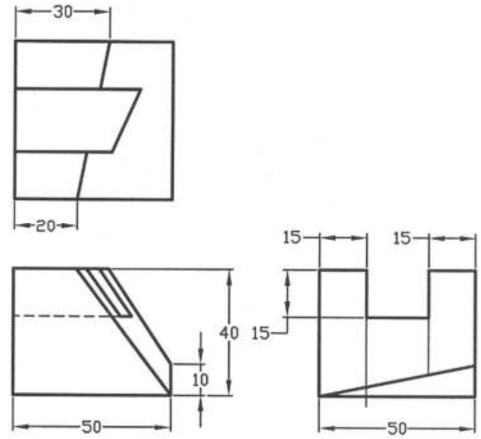


Fig. P6.28

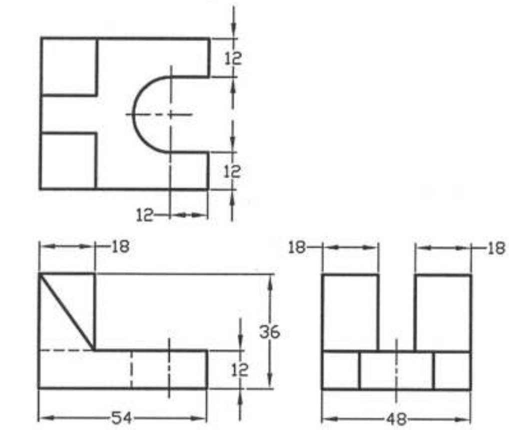


Fig. P6.29

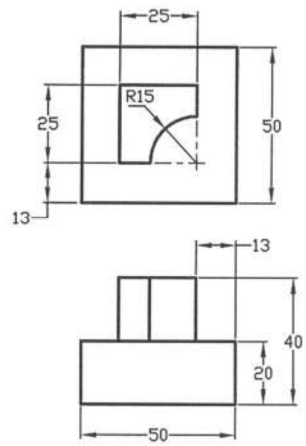


Fig. P6.30

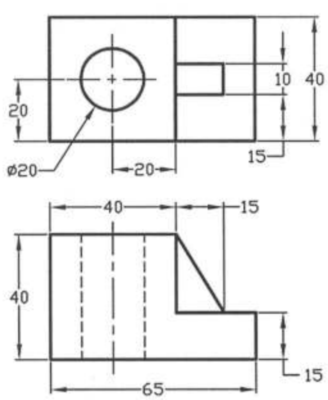


Fig. P6.31

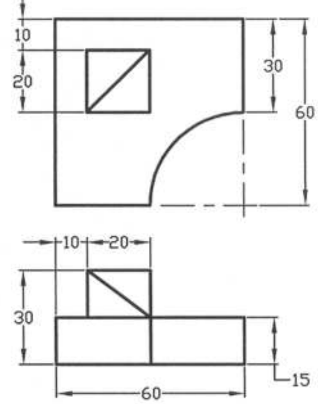


Fig. P6.32

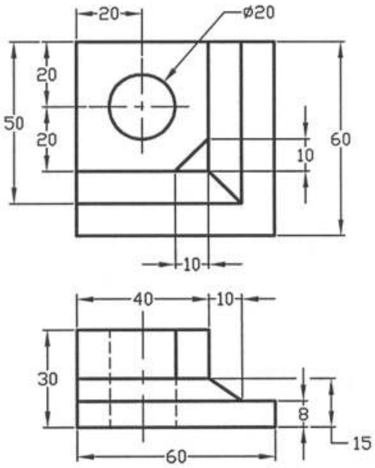


Fig. P6.33

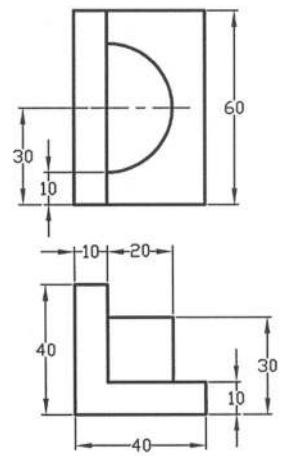


Fig. P6.34

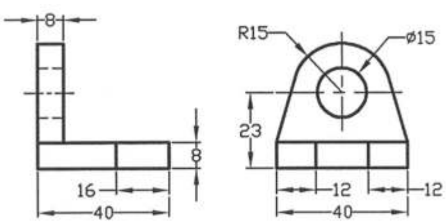


Fig. P6.35

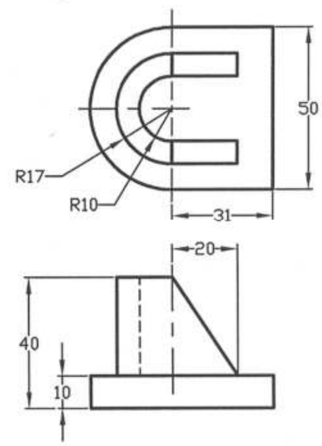


Fig. P6.36

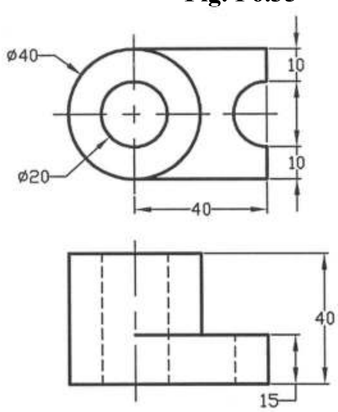


Fig. P6.37

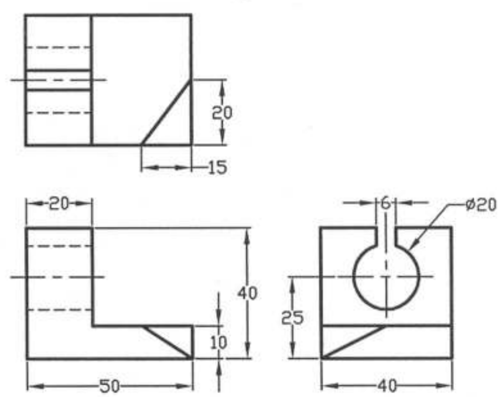


Fig. P6.38

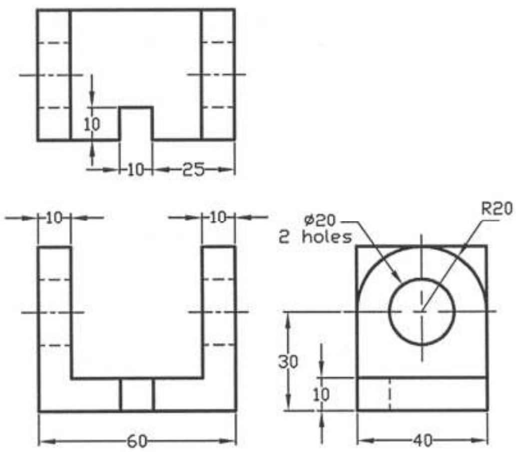


Fig. P6.39

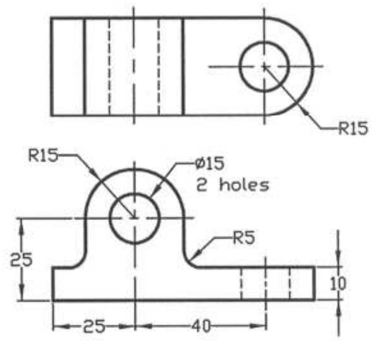


Fig. P6.40

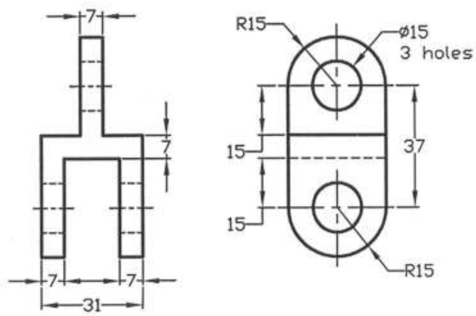


Fig. P6.41

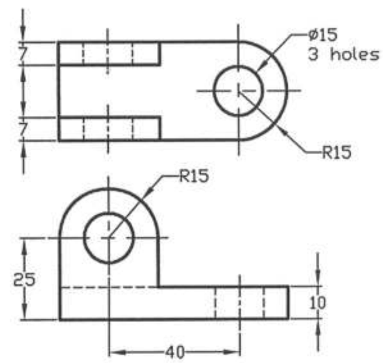


Fig. P6.42

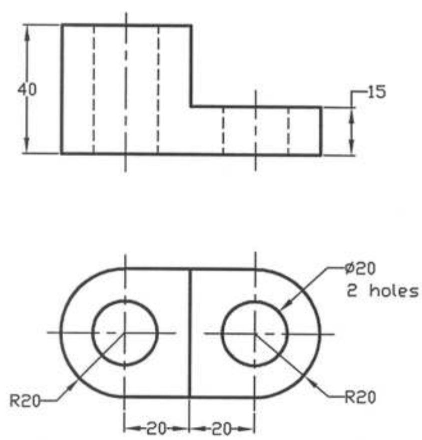


Fig. P6.43

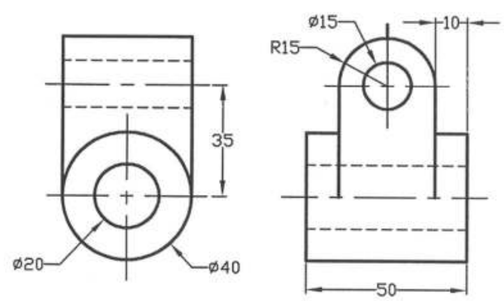


Fig. P6.44

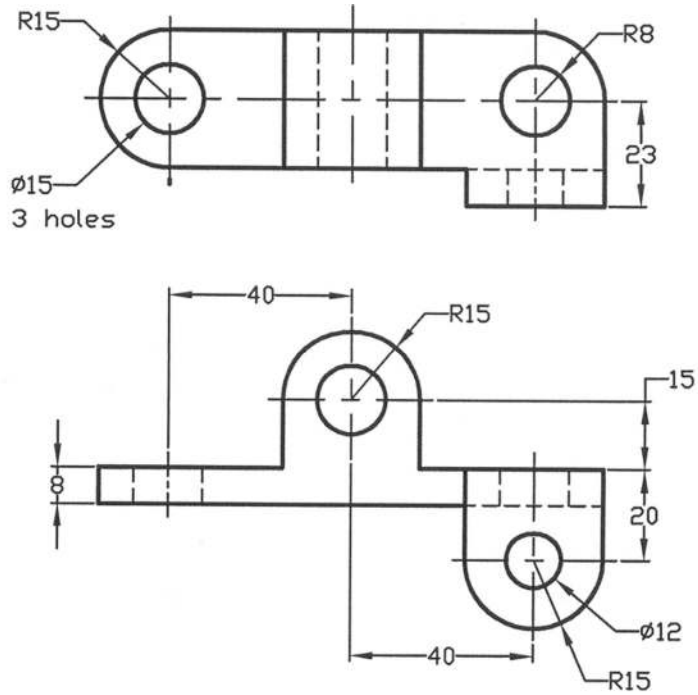


Fig. P6.45

Prob. 6.46- 6.51: Draw the oblique views (cavalier) of the objects from the following views as shown in Figures P6.46 to P6.51.

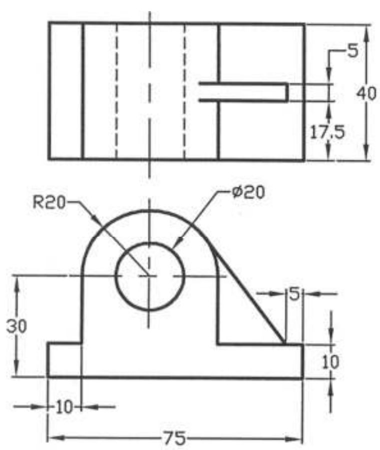


Fig. P6.46

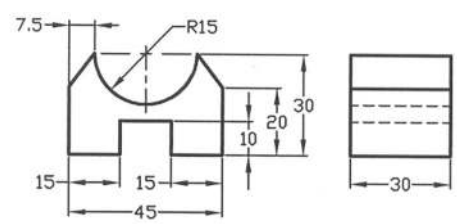


Fig. P6.47

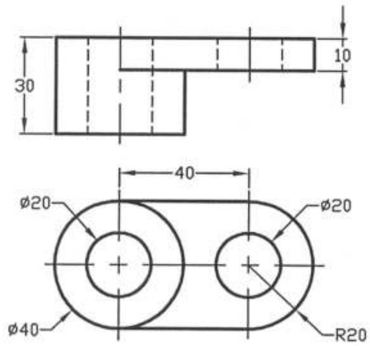


Fig. P6.48

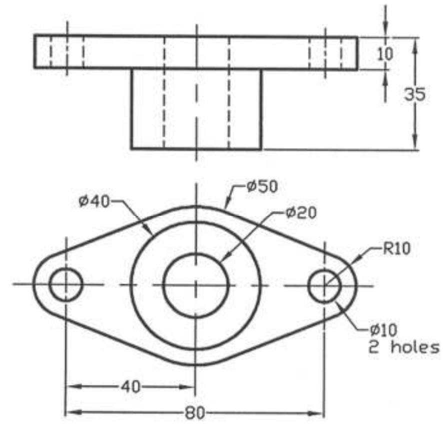


Fig. P6.49

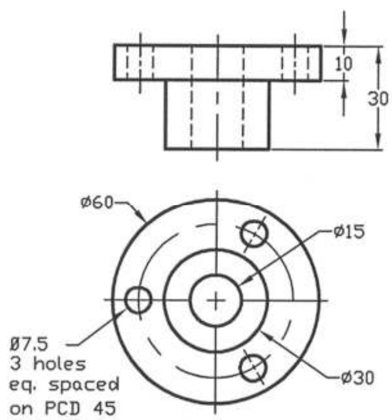


Fig. P6.50

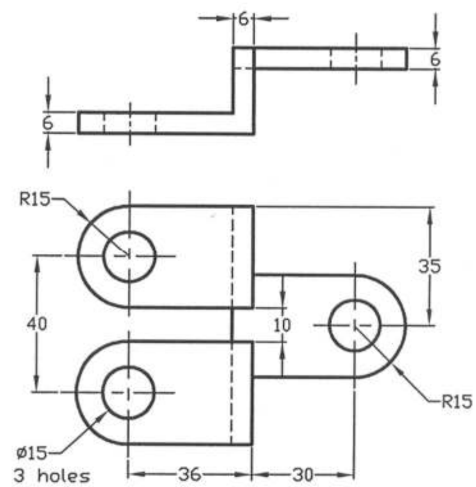
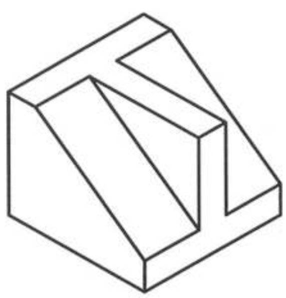
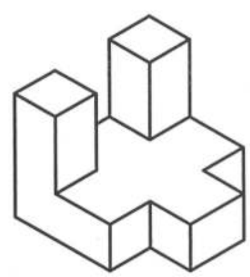


Fig. P6.51

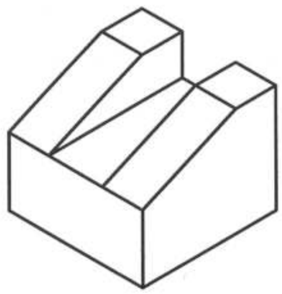
Solutions for Example Problems



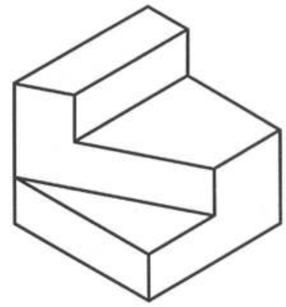
Solution of P6.1



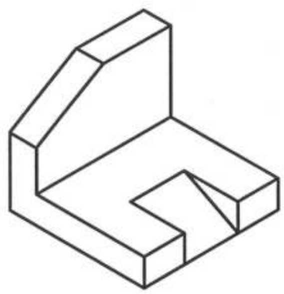
Solution of P6.2



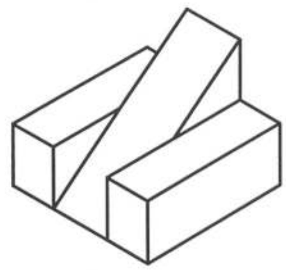
Solution of P6.3



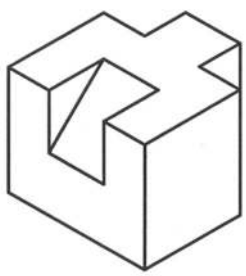
Solution of P6.4



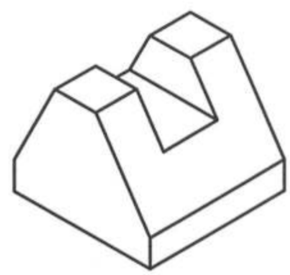
Solution of P6.5



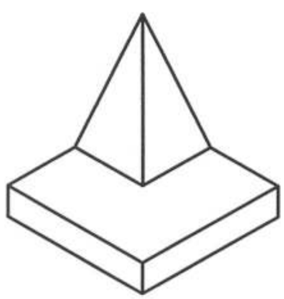
Solution of P6.6



Solution of P6.7

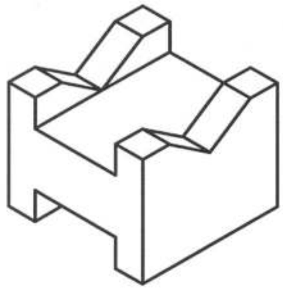


Solution of P6.8

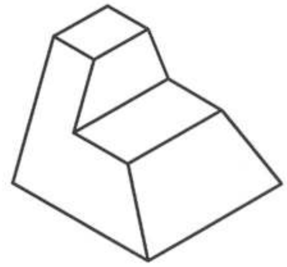


Solution of P6.9

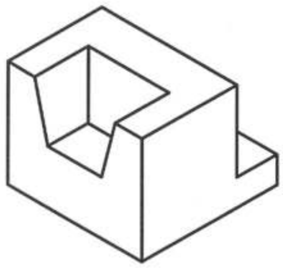
Solution of P6.10



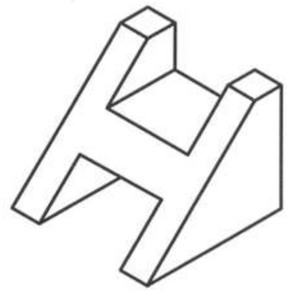
Solution of P6.11



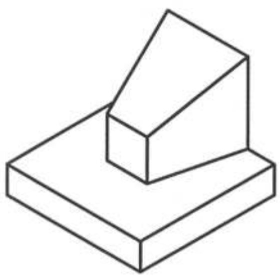
Solution of P6.12



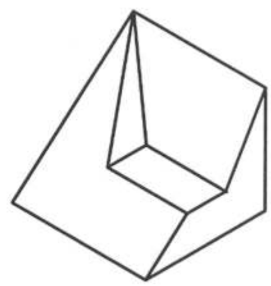
Solution of P6.13



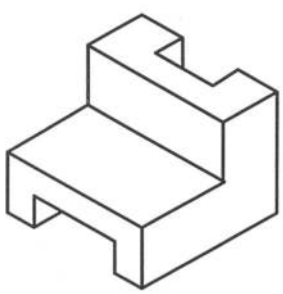
Solution of P6.14



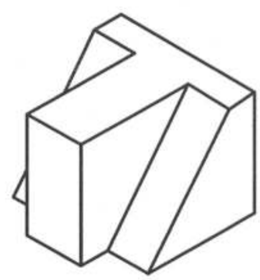
Solution of P6.15



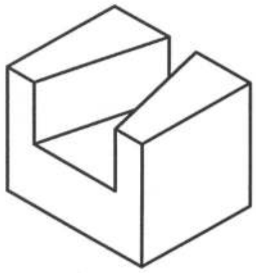
Solution of P6.16



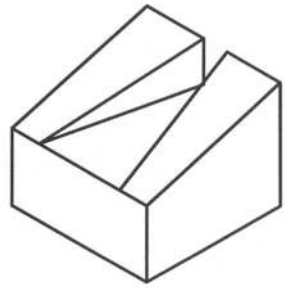
Solution of P6.17



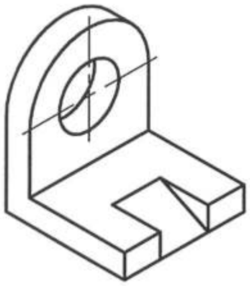
Solution of P6.18



Solution of P6.19

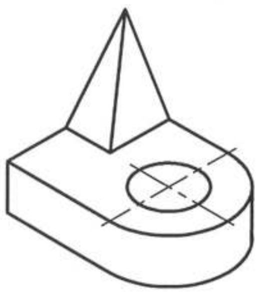


Solution of P6.20

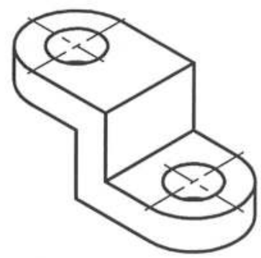


Solution of P6.21

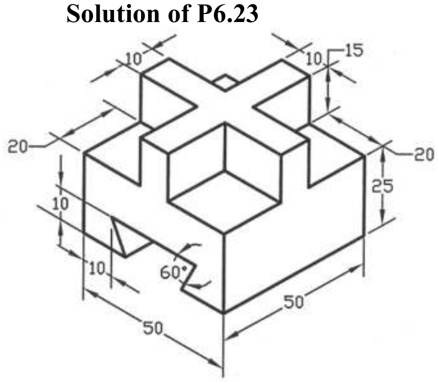
Solution of P6.22



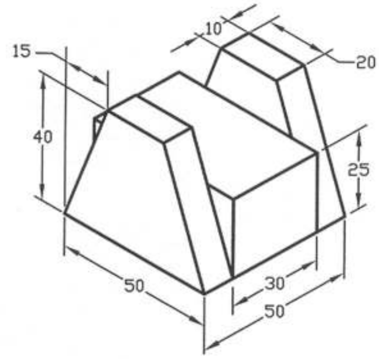
Solution of P6.23



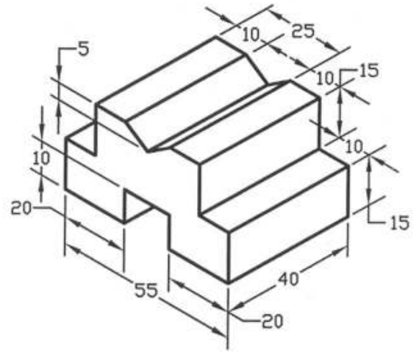
Solution of P6.24



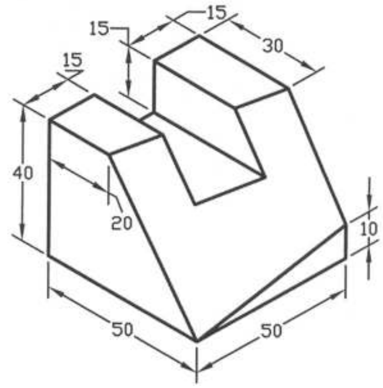
Solution of P6.25



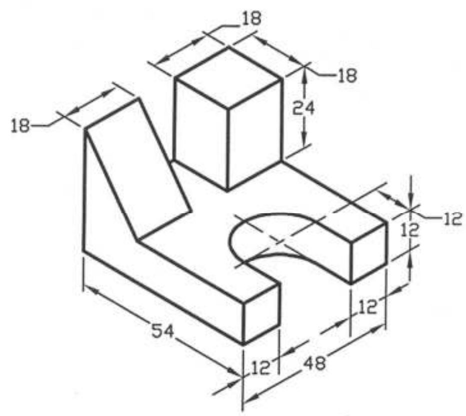
Solution of P6.26



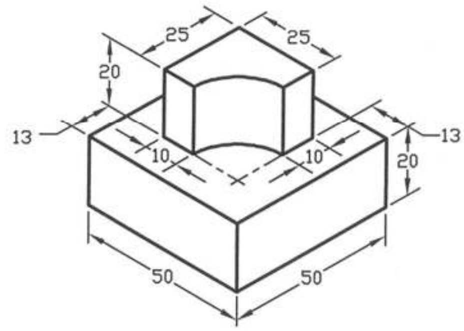
Solution of P6.27



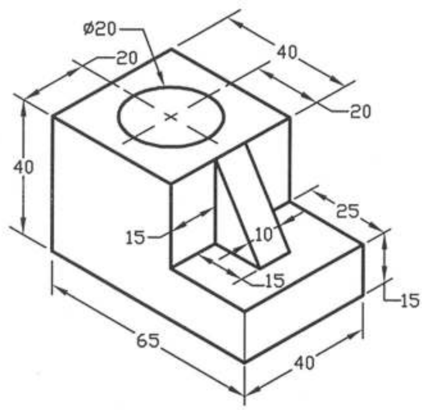
Solution of P6.28



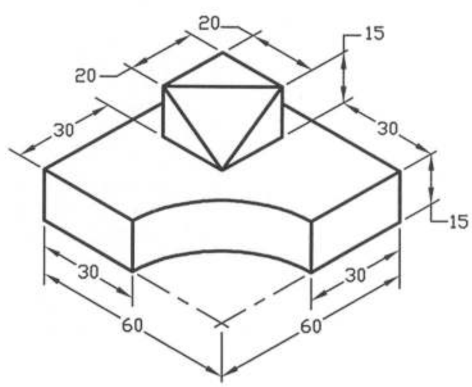
Solution of P6.29



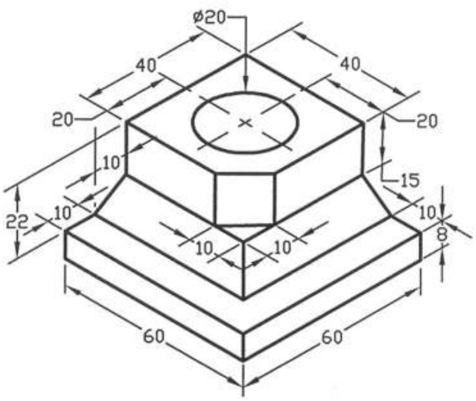
Solution of P6.30



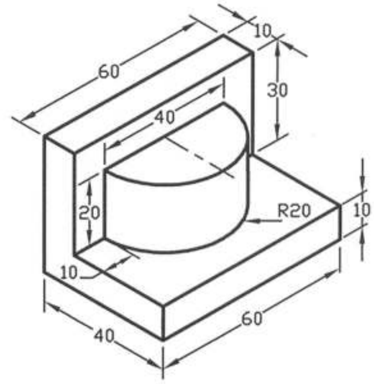
Solution of P6.31



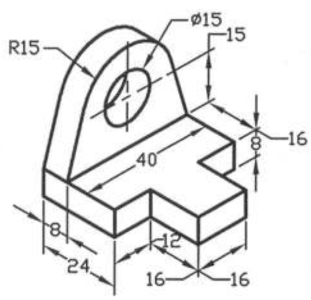
Solution of P6.32



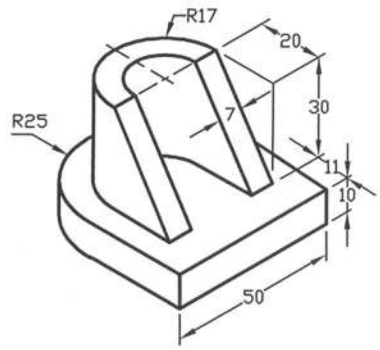
Solution of P6.33



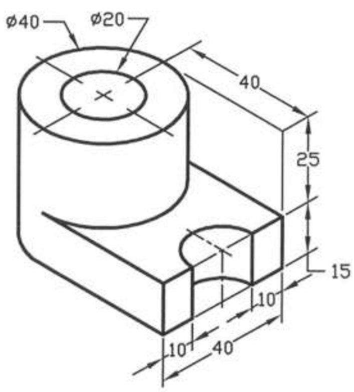
Solution of P6.34



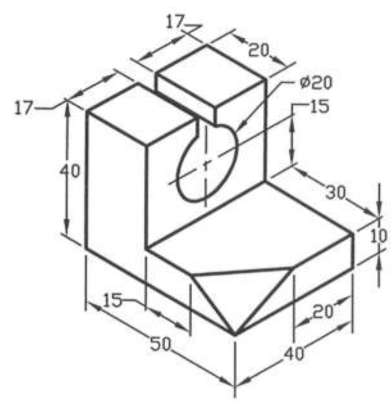
Solution of P6.35



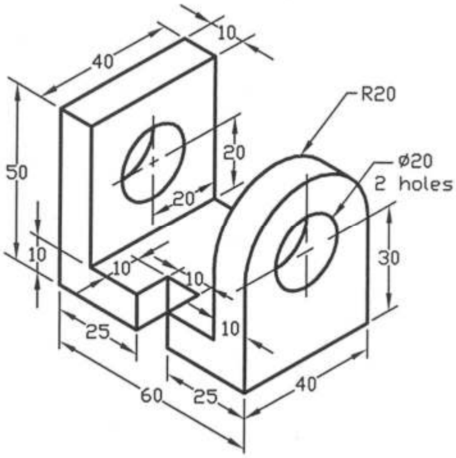
Solution of P6.36



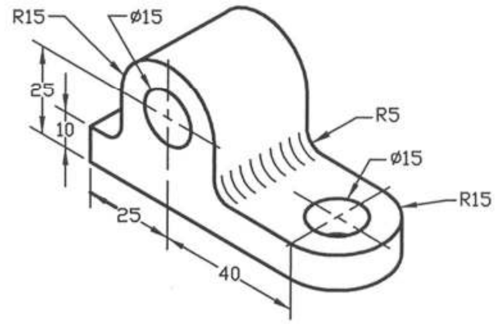
Solution of P6.37



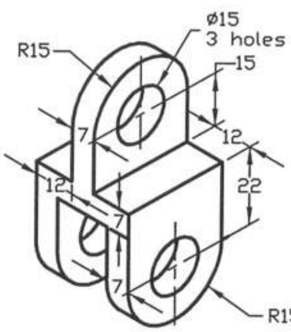
Solution of P6.38



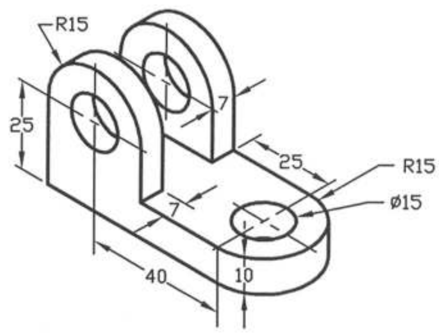
Solution of P6.39



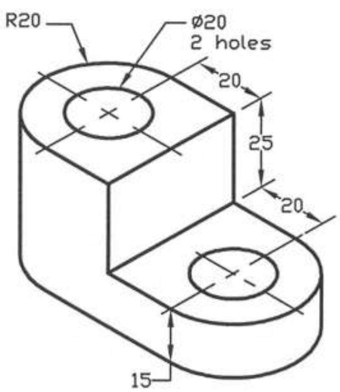
Solution of P6.40



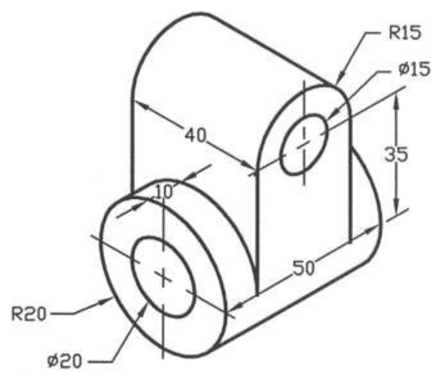
Solution of P6.41



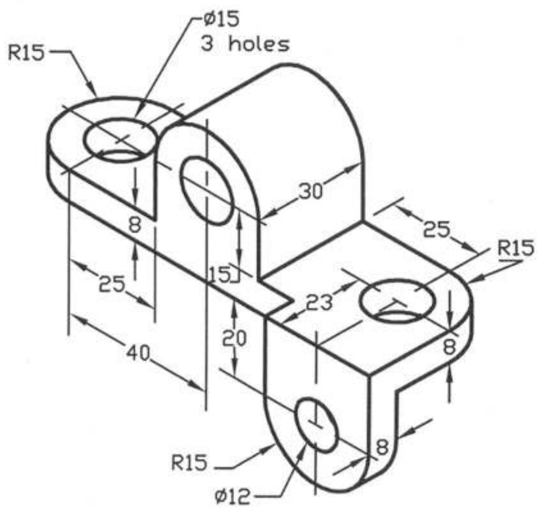
Solution of P6.42



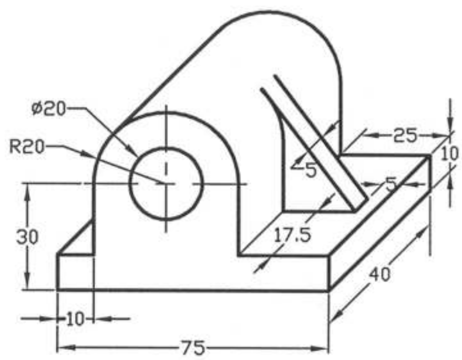
Solution of P6.43



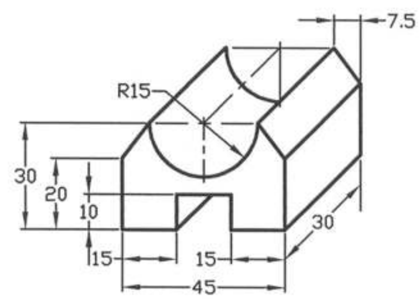
Solution of P6.44



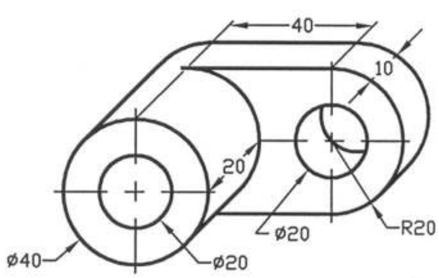
Solution of P6.45



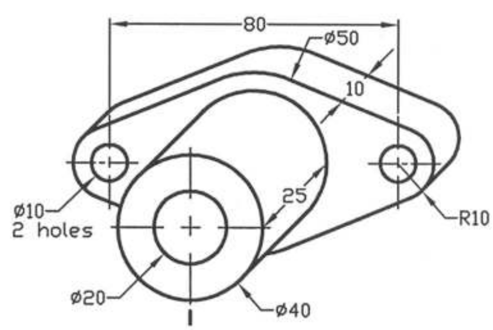
Solution of P6.46



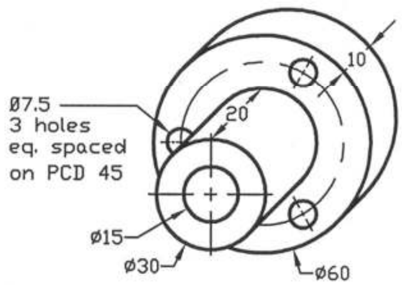
Solution of P6.47



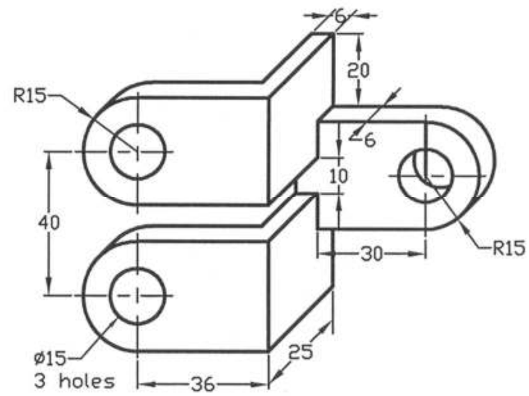
Solution of P6.48



Solution of P6.49



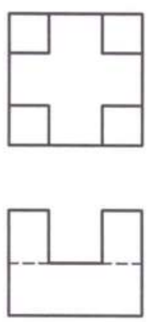
Solution of P6.50



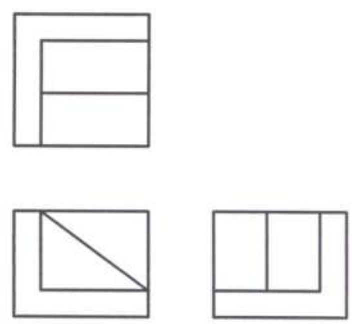
Solution of P6.51

Problems

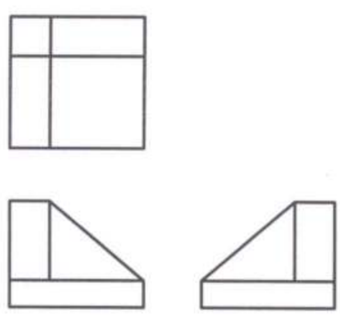
Prob. 6.52- 6.67: Draw the free hand isometric sketches of the objects from the following views as shown in Fig. P6.52 to P6.67.



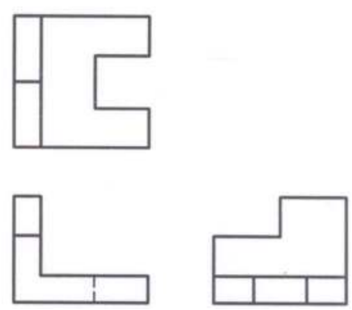
Prob. P6.52



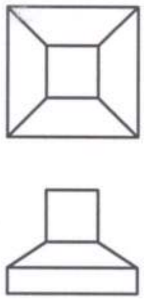
Prob. P6.53



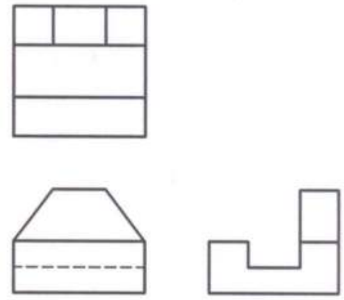
Prob. P6.54



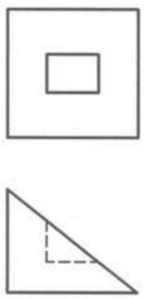
Prob. P6.55



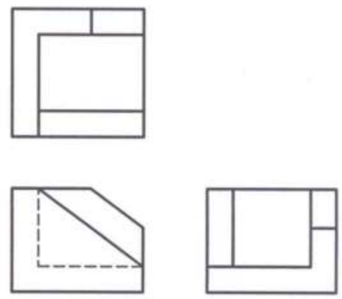
Prob. P6.56



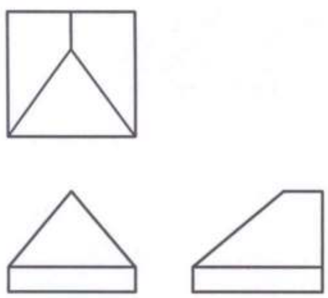
Prob. P6.57



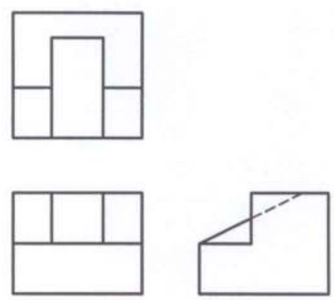
Prob. P6.58



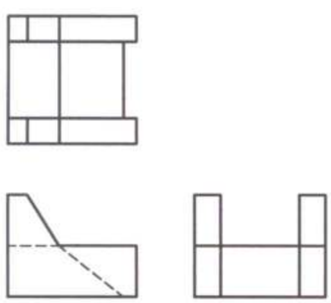
Prob. P6.59



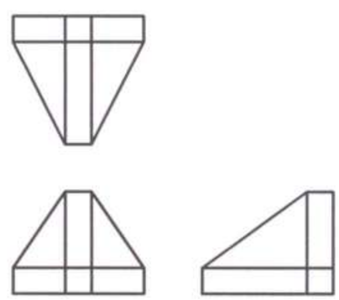
Prob. P6.60



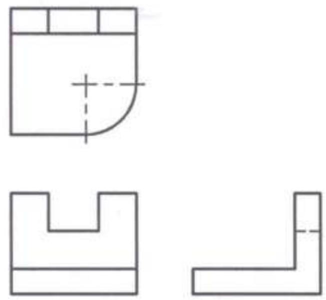
Prob. P6.61



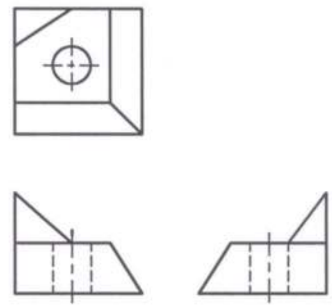
Prob. P6.62



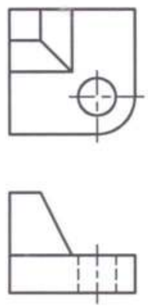
Prob. P6.63



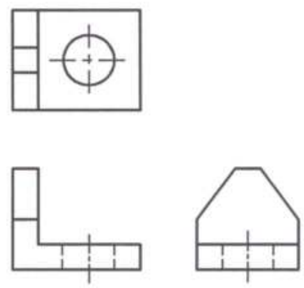
Prob. P6.64



Prob. P6.65

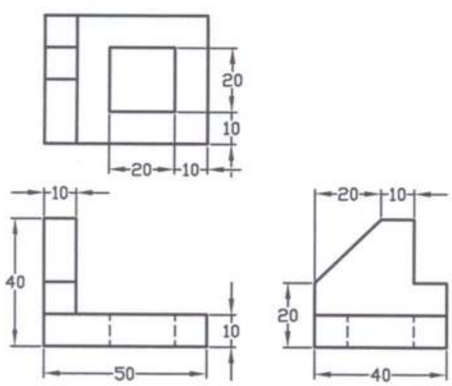


Prob. P6.66

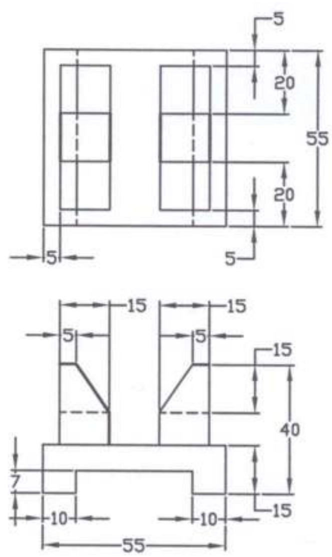


Prob. P6.67

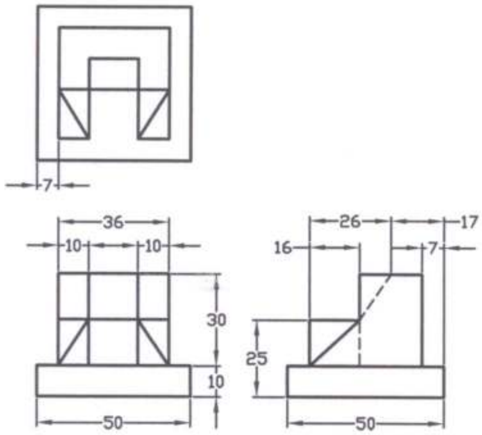
Prob. 6.68 - 6.85: Draw the isometric drawing of the objects from the following views as shown in Fig. P6.68 to P6.85.



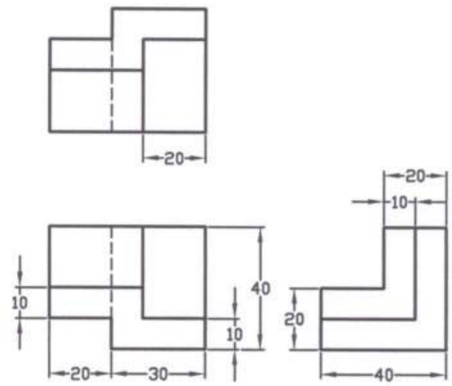
Prob. P6.68



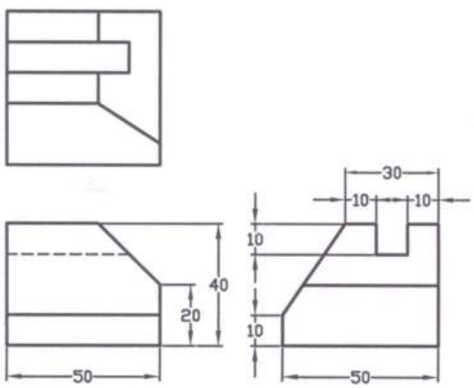
Prob. P6.69



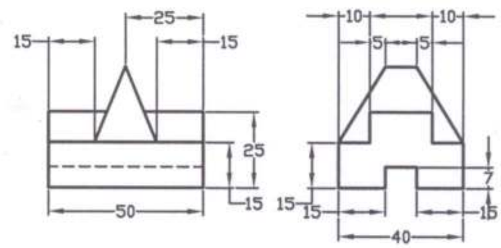
Prob. P6.70



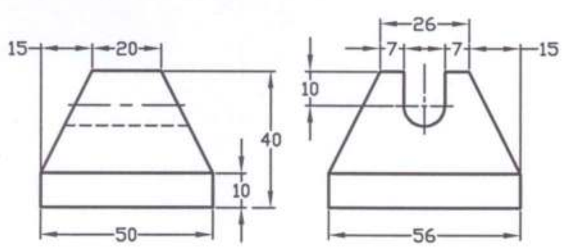
Prob. P6.71



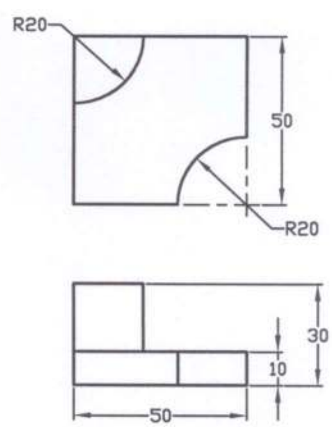
Prob. P6.72



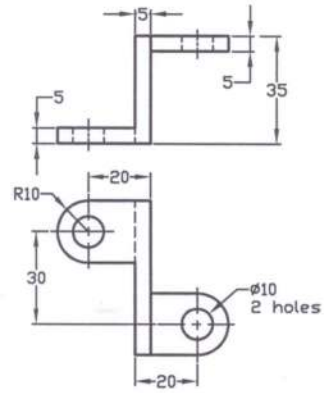
Prob. P6.73



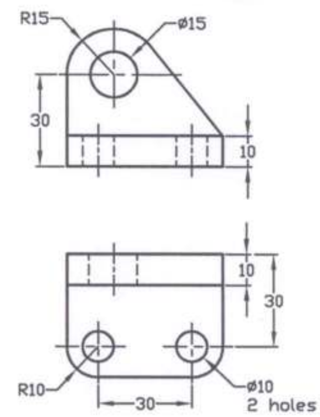
Prob. P6.74



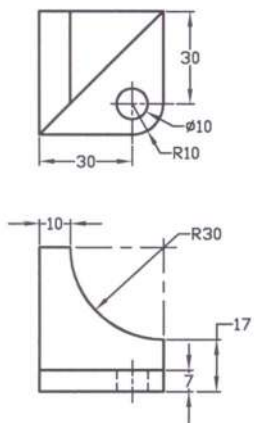
Prob. P6.75



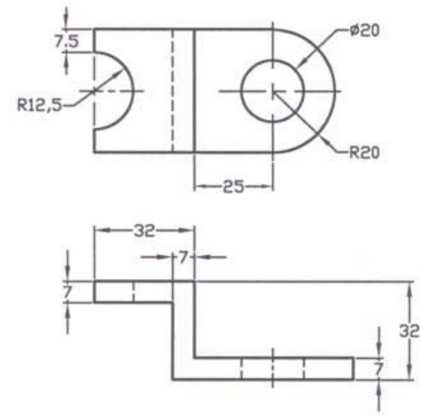
Prob. P6.76



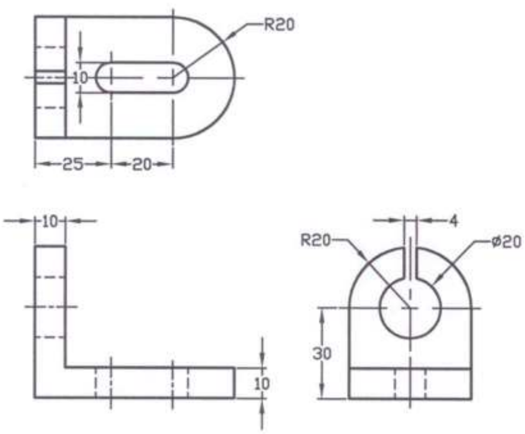
Prob. P6.77



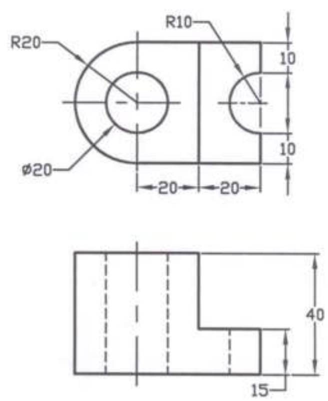
Prob. P6.78



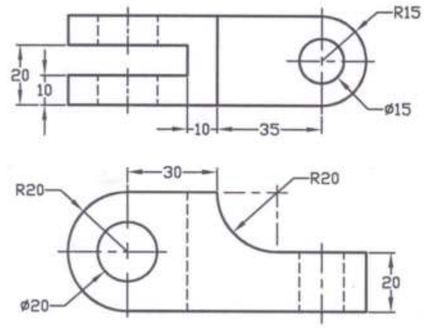
Prob. P6.79



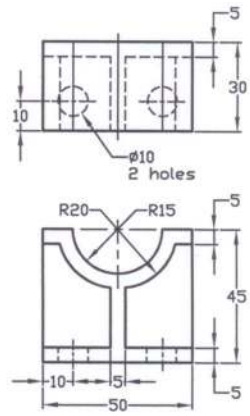
Prob. P6.80



Prob. P6.81



Prob. P6.82



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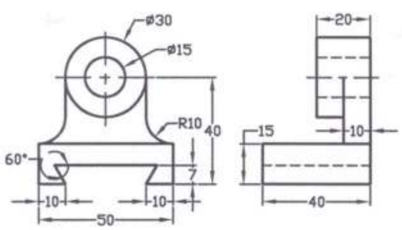
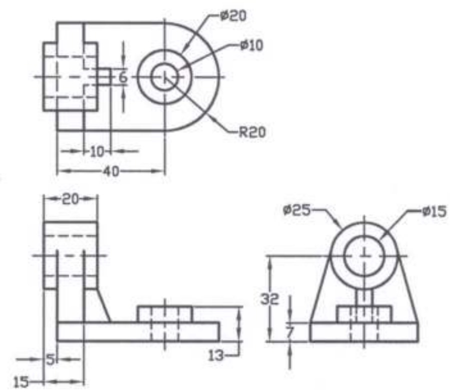
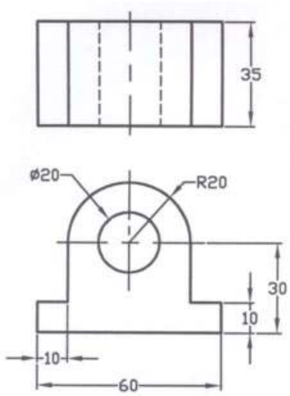




Prob. P6.84 **Prob. P6.85**

Prob. 6.86- 6.91: Draw the oblique views (cavalier) of the objects from the following views as shown in Fig. P6.86 to P6.91.



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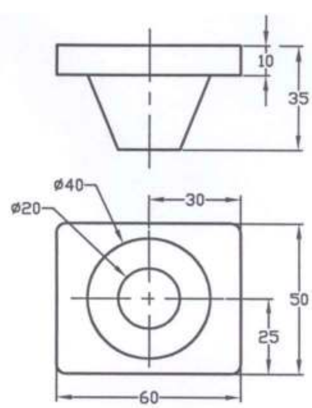

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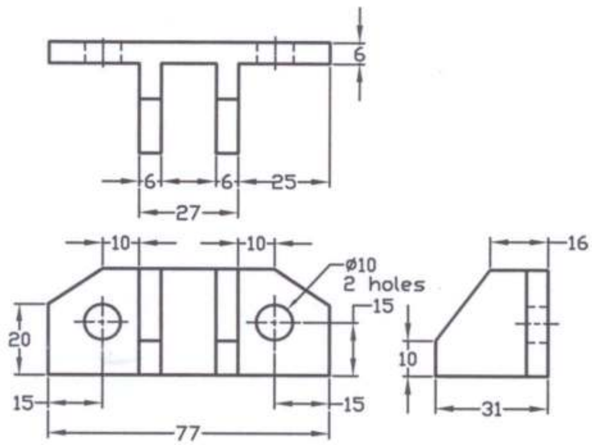
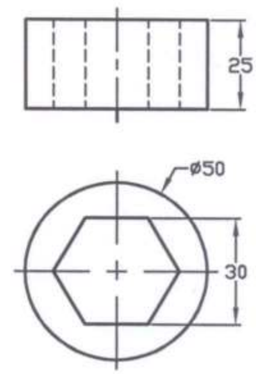
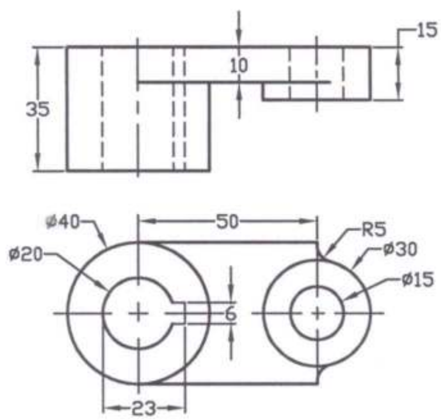


Prob. P6.88



Prob. P6.89



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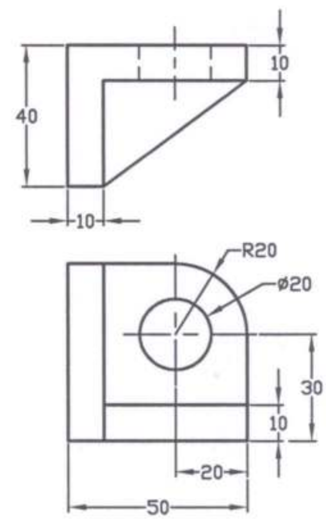


Prob. P6.91

CHAPTER 7

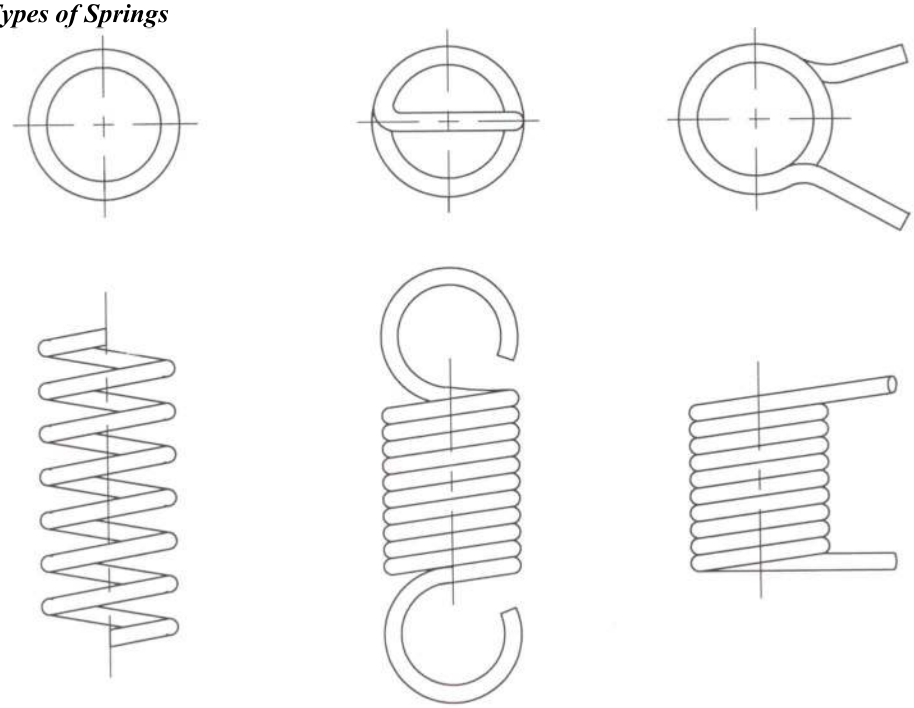
SPRINGS, FASTENERS, PULLEYS AND GEARS

7.1 ***Introduction***

In this chapter the most common types of springs, fasteners, pulleys and gears have been taken into consideration for their drawings. These components are very important for the machines. Fasteners are of two types: one is permanent and the other is removable. Rivets and welds are permanent fasteners. On the other hand bolts and nuts, keys and pins are removable fasteners. The thorough knowledge of the graphic representation of most of the common types of springs and fasteners, pulleys and gears are important for drawings. This will lead to the further development of knowledge in the practical field.

According to ISO 724: 1973, the standard dimensions of some metric screw threads are presented in Appendix 8 and the tap drill sizes for specific metric screw threads are presented in Appendix 9. The dimensions of some common cap screws are provided in Appendix 10. The standard hexagon-head bolts and hexagon-head nuts are given in Appendices 11 and 12 while some common washer sizes are presented in Appendix 13.

7.2 ***Types of Springs***

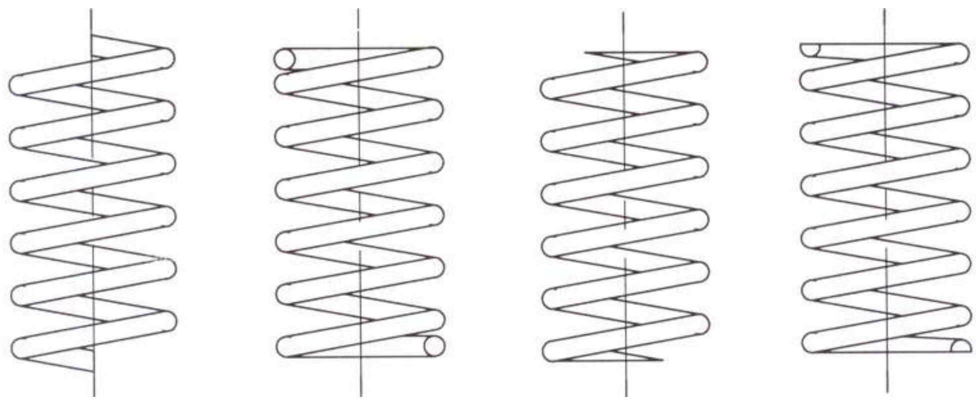


(a) Compression (b) Extension (c) Torsion

Figure 7.1: Helical Springs

A spring is a machine element, which stores energy when deflected. Springs may be classified as helical spring and flat spring mainly. Again helical springs may be classified as compression spring, extension spring and torsion spring. A compression spring offers resistance to the compressive force; an extension spring offers resistance to the tensile force while the torsion spring offers resistance to the twisting force. In Figure 7.1 compression, extension and torsion helical springs have been shown.

The compression helical springs with different end conditions have been presented in Figure 7.2. They are plain end, squared or closed end, plain and ground end and squared and ground end. The helical compression spring with squared and ground end is mostly used.



(a) Plain End (b) Squared End (c) Plain and Ground End (d) Squared and Ground End

Figure 7.2: Compression Helical Springs with Different End Conditions

7.3 ***Drawings of Springs***

On working drawings, the true projections of the helical springs are avoided in order to save unnecessary wastage of time. The spring may be represented using straight lines in place of helical curves. This is known as the schematic drawing. In Figures 7.3, 7.4 and 7.5 the schematic drawings of the helical compression spring, extension spring and torsion spring have been shown respectively. When the drawing of the spring is too small, schematic representation becomes suitable. The helical compression spring, which has been shown here, consists of squared end. The total coils of the compression spring shown at the left of Figure 7.3 are 4½ while that of the compression spring shown at the right are 4. The total coils of the extension spring as shown in Figure 7.4 are 5.

Another way of graphic representation of the springs has been shown in Figure 7.6. Here the sections of the compression helical springs with plain and ground end have been shown. When the area in section is large, then the section lines may be given. If the area of the section is small due to the smaller wire diameter, the area is made solid black as shown in the figure. This presentation is usually recommended for the assembly drawing where section may be necessary. In the schematic drawings of the springs, the detailed information of the springs such as the length, outside diameter, wire diameter, pitch, number of coils etc. are necessary for manufacture.

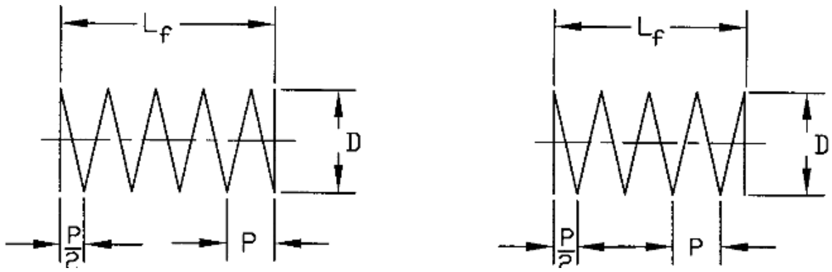


Figure 7.3: Schematic Drawings of Helical Compression Spring

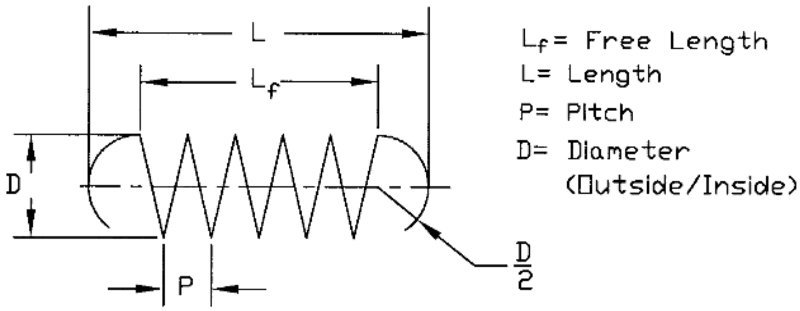


Figure 7.4: Schematic Drawings of Helical Extension Spring

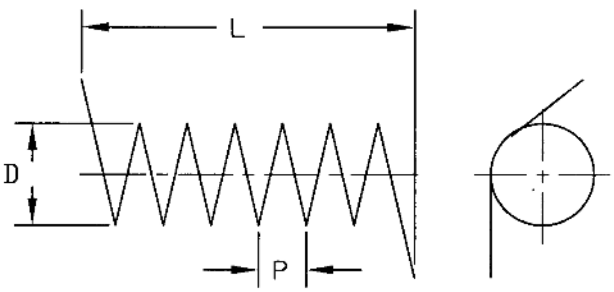
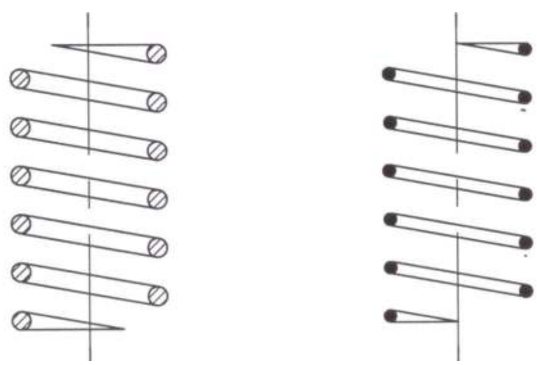


Figure 7.5: Schematic Drawings of Helical Torsion Spring



(a) (b)

Figure 7.6: Detailed Drawings of Helical Springs

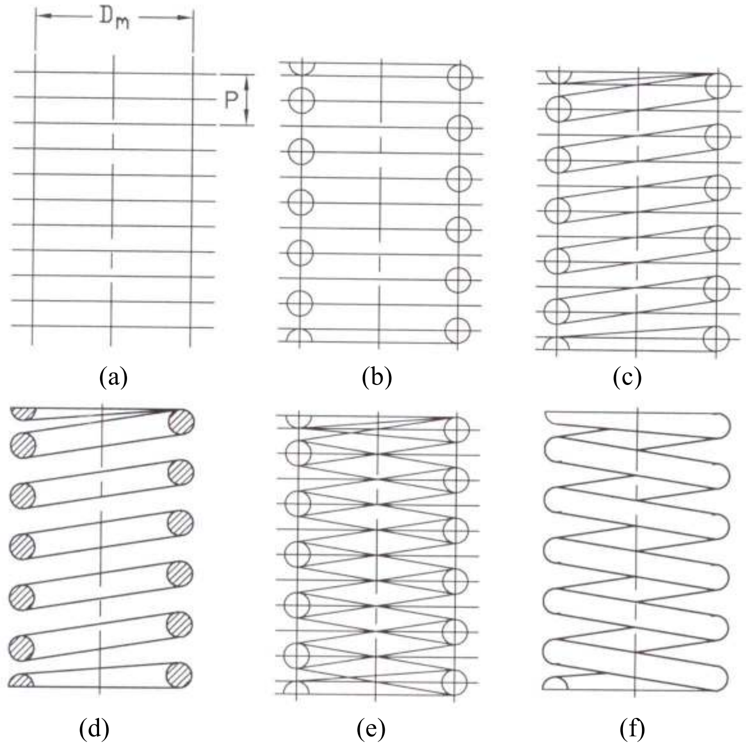


Figure 7.7: Steps of Drawing Helical Compression Springs

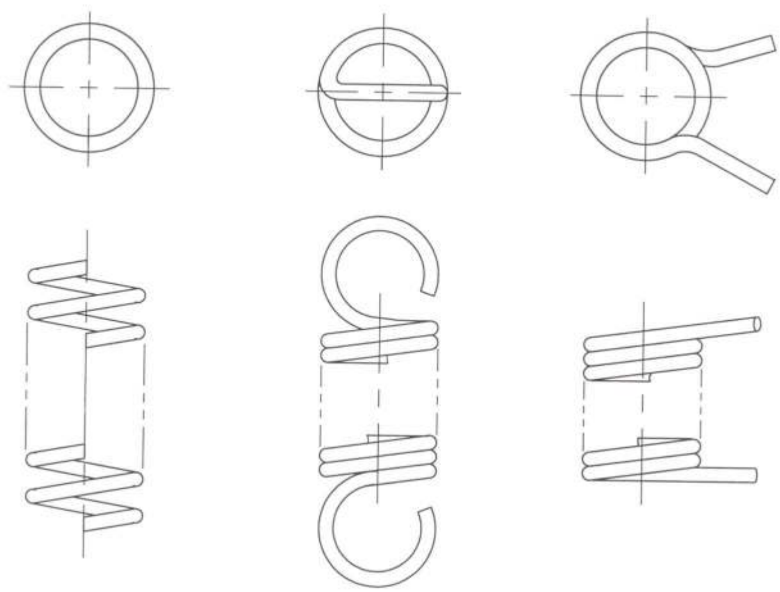
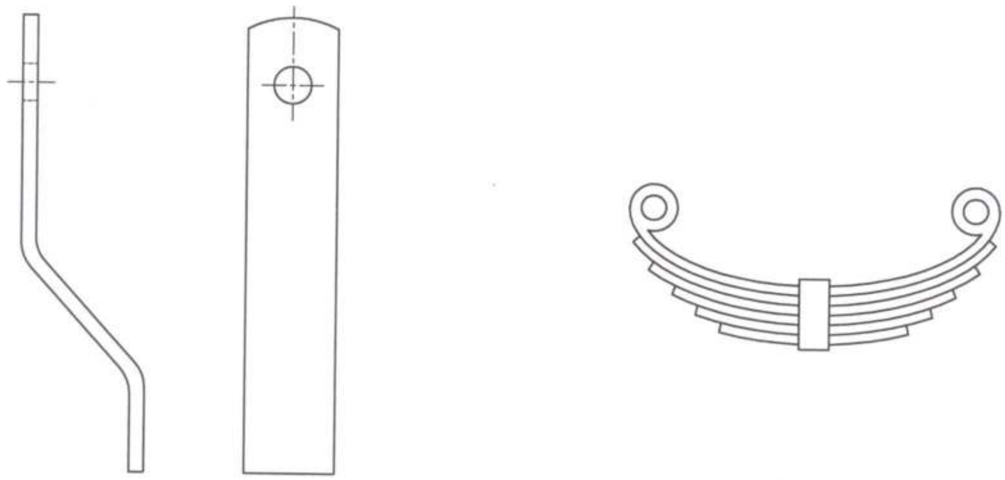


Figure 7.8: Drawings of Springs Using Phantom Lines

In Figure 7.7 the steps of drawing the helical compression springs have been presented. The ends of the springs are considered as squared and ground. In this figure the symbols D_m and P indicate the mean diameter and the pitch of the spring. Sometimes to save time the long springs may be drawn using the phantom lines as shown in Figure 7.8. This is done where a complete picture of the spring is not required.

In Figure 7.9 the drawings of flat springs have been shown. They are made from the flat or the strip material. In Figure 7.9a, a typical simple flat spring has been shown while in Figure 7.9b; a leaf spring has been shown. It consists of several strips.



(a) Simple Flat Spring (b) Leaf Spring

Figure 7.9: Flat Springs

7.4 *Screw-Thread Terminology*

In Figure 7.10, the different elements of a screw thread have been shown. A ridge of uniform section, which forms the helix, is called the **screw-thread or thread**. The thread on the external surface and that on the internal surface of a cylinder or cone are called the **external and internal thread** respectively. The largest and the smallest diameters of a screw-thread are called the **major and the minor diameters** of a screw-thread. The distance from a point on a screw-thread on a corresponding point on a next screw-thread is called the **pitch**. The diameter of an imaginary cylinder passing through the thread such that the width of the thread and that of the groove are equal is called the **pitch diameter**.

The thread on a cylinder is called a straight thread while that on a cone is called the taper thread. The profile of the thread is called the **thread form**. The top surface joining the two surfaces of a thread is called the **crest** while the bottom surface joining the sides of two adjacent threads is called the **root**. The distance between the crest and the root measured to the normal to the axis is called the **depth** of the thread. A thread which when viewed axially winds in clockwise and receding direction is called the **right-hand** (RH) thread. Threads are always right-hand unless otherwise specified. The cross-section of a thread cut by a plane passing through the axis of the thread is called the **form of the thread**.

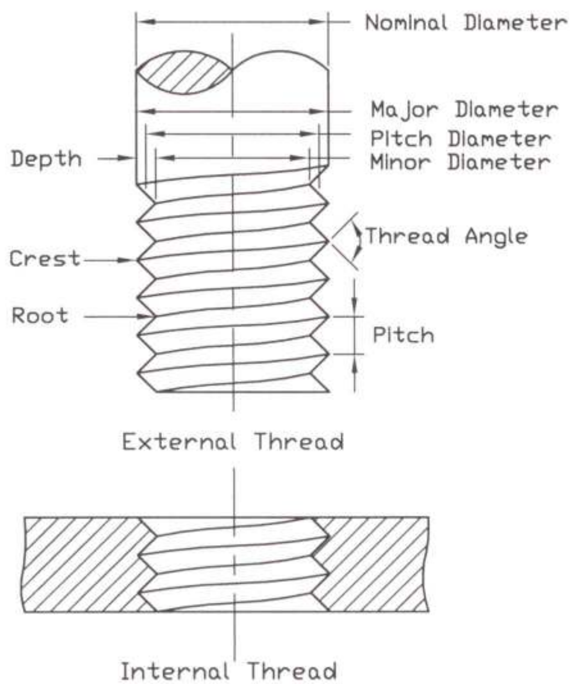


Figure 7.10: Different Elements of a Screw-Thread

7.5 *Forms of Screw Threads*

In Figure 7.11, the different forms of screw-threads have been presented. The dimensions shown in the figure do not incorporate the clearance. In practical field there will be clearance between the external and the internal threads.

At present the **sharp-V** thread is of theoretical interest rather than the practical one. This thread is of simple type and it is used rarely. To maintain the sharp edge is difficult. It was originally called the United States Standard thread, or the Sellers thread.

The **American National** thread is obtained making the crests and roots of the sharp-V thread flat. It is stronger thread. The United States, Canada and Great Britain agreed to make the standard thread modifying the American National thread. The crest of the external thread is made either flat or rounded and the root is made rounded. This thread is called the **Unified** thread.

The International Screw Thread Fasteners agreed to develop a new screw thread known as the **Metric** thread. The crest and root of this thread are flat. Often they are made rounded. This thread has the similarity with the American National and the Unified threads but the depth of the thread is less.

Another form of the thread is the **Square** thread having its thread of the square type. The face of the thread is at right angle to the axis of the screw thread. It is used for transmission of power. Because of some disadvantages of the square thread, another type of thread known as **ACME** has been developed. It is stronger than the square thread and easier to cut. It permits the use of a

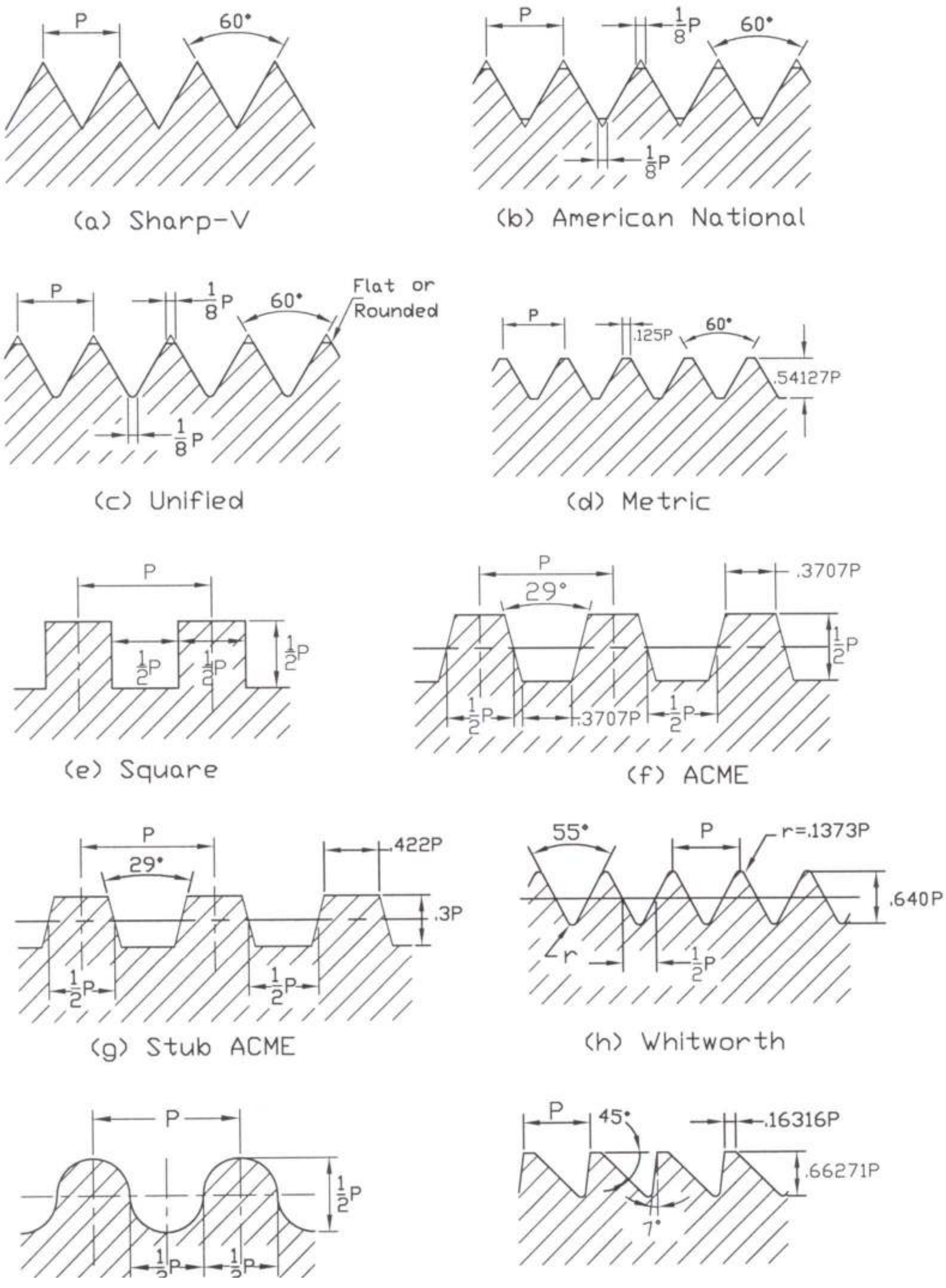


Figure 7.11: Different Forms of Screw-Threads

disengaging or split nut. **Stub ACME** is a modified version of ACME thread. It is also used for transmission of power. It is a strong thread and suitable for limited space.

Another type of thread is the **Whitworth** thread of British standard. It is being replaced by the Unified thread.

When the threads are to be molded or rolled in sheet metal the **Knuckle** thread is used. It is used in modified form in electric bulbs and sockets, bottle tops etc.

When the power transmission is required in one direction, **buttress** thread is used. It is used to withhold the pressure breechblocks of large guns; hence it is often called "**Breechblock**" thread.

7.6 *Thread Symbols*

Threads are represented by either of the three conventions – (a) simplified (b) schematic/regular and (c) detailed. For threads under approximately 25 mm diameter, simplified or schematic symbols are used. The simplified symbols are usually used for the detail drawings and the schematic symbols are used for the assembly drawings. When the diameter of the thread on the drawing is over 25 mm approximately a detailed representation is usually used.

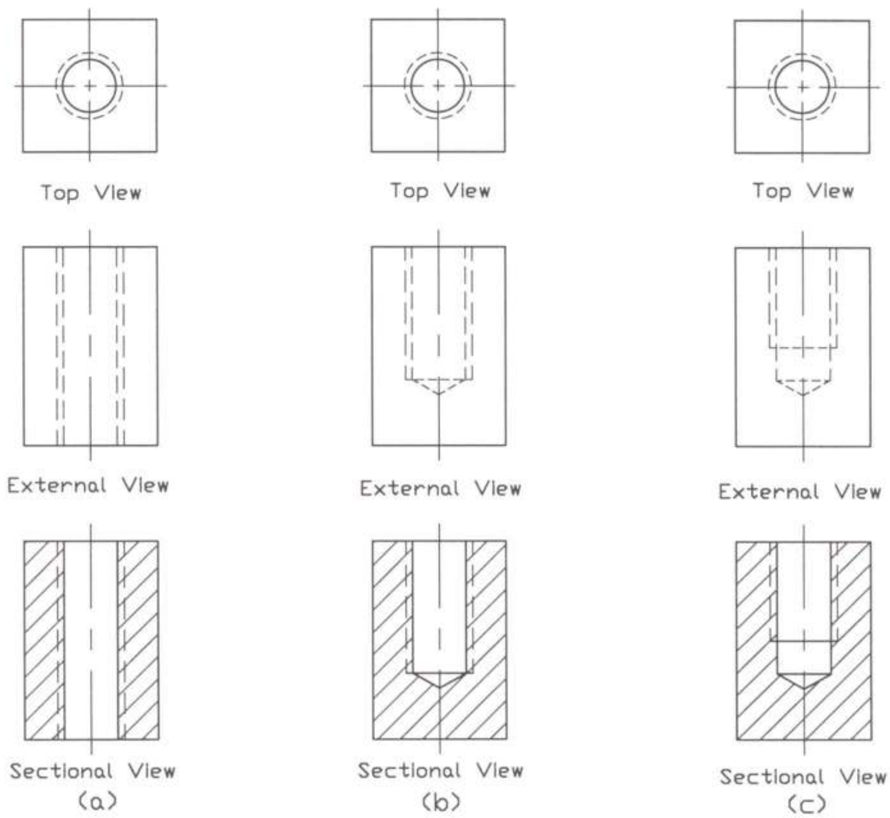


Figure 7.12: Simplified Symbols for Internal Threads

The symbols are same for all forms of threads, such as unified, metric, square, ACME. In order to avoid unnecessary labor in drawing the true projection of the thread, simplified or schematic drawing is done. The detailed representation is a close approximation to the true projection of the thread.

In Figures 7.12 and 7.13 respectively simplified and schematic/regular symbols for the internal threads have been presented. It can be seen from these figures that there are differences between the simplified and the schematic representations in case of thread in section only. For the external views in elevation for both the simplified and schematic representation, they are identical.

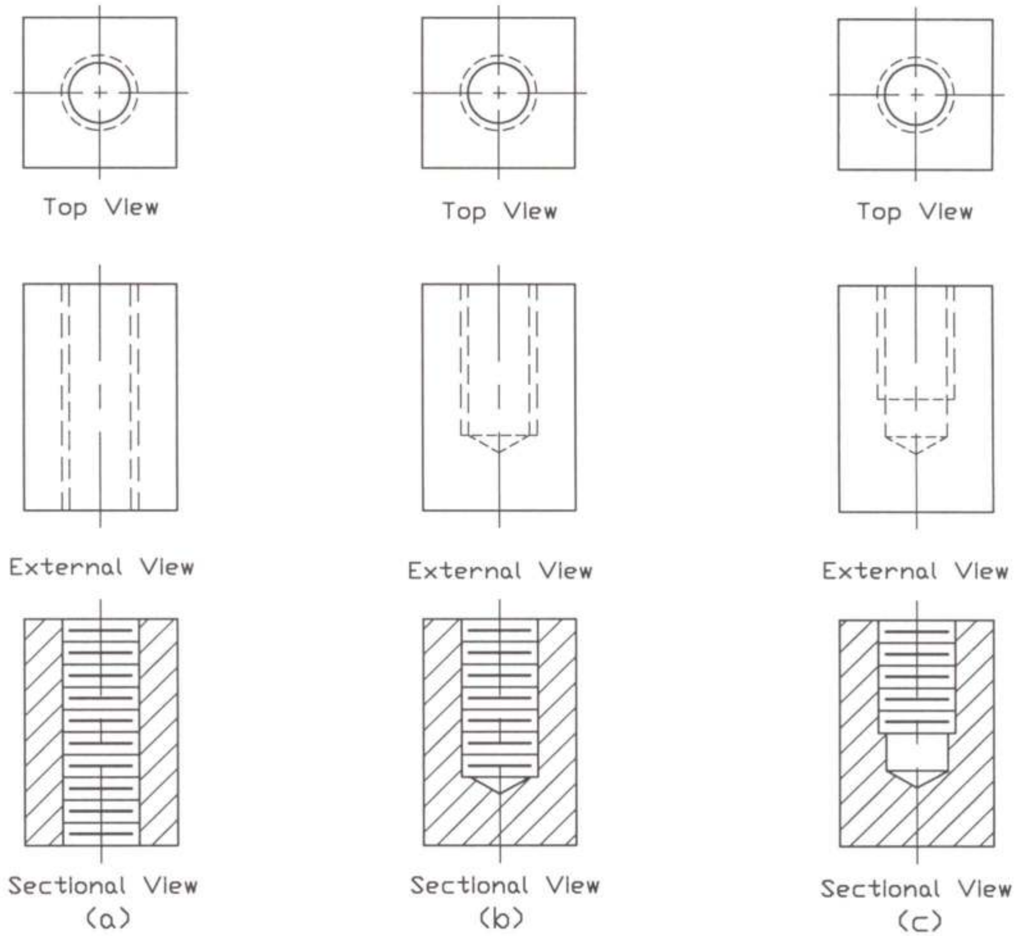
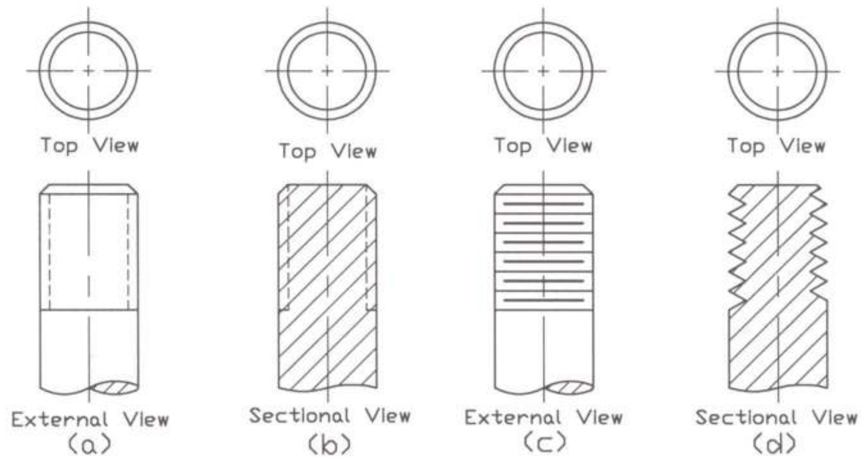


Figure 7.13: Schematic Symbols for Internal Threads

In Figures 7.12 (a) and 7.13 (a), hole and tap have been made throughout. For the bottoming tap as shown in Figures 7.12 (b) and 7.13 (b), the thread length equals the depth of the drill. For the blind tapped hole, it is usual convention to draw the drill depth equal to at least three schematic pitches beyond the thread length, which have been presented in Figures 7.12 (c) and 7.13 (c). To show the hidden thread by two parallel dashed lines, they should be made staggered as shown in the figures.



Simplified Schematic

Figure 7.14: Simplified and Schematic Symbols for External Threads

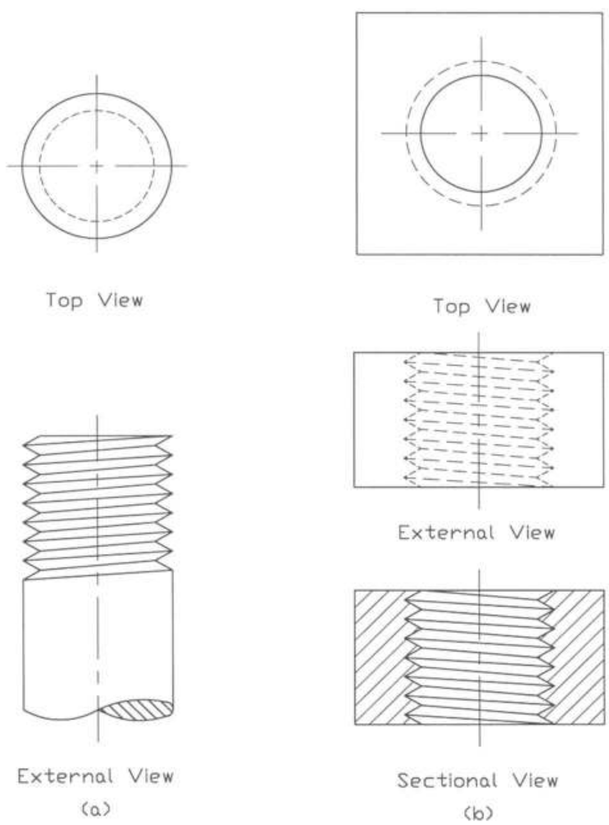


Figure 7.15: Detailed Representation of External and Internal Threads.

It may be mentioned that the pitch of the thread need not be drawn to the scale of the pitch; rather it should be drawn in such a way so that it does not look clumsy. The root should be drawn in the middle of the pitch and it must be thicker.

In Figure 7.14 both simplified and schematic thread symbols have been shown. Simplified symbols are shown in Figures 7.14 (a) and (b) while schematic symbols are shown in Figures 7.14 (c) and (d). It is observed that for the external thread in section hidden lines have been used for the simplified representation in Figure 7.14 (b) while V's have been used for the schematic representation in Figure 7.14 (d). Without V's thread section in schematic representation cannot be made clear. However, the V's are drawn at 60^0, which in turn control the pitch of the thread. Exact pitch of the thread is not required. In Figure 7.14 (a) and (c), the external views of the thread have been shown.

Detailed representations of external and internal threads have been shown in Figure 7.15. External view of external thread in elevation is shown in Figure 7.15 (a) while both external and sectional views of internal thread are shown in Figure 7.15 (b). The true helical shape of the thread has been avoided here. The lines have been made straight to avoid unnecessary labour. Despite, it represents almost the true feature of the thread.

7.7 ***Steps of Drawing Threads***

Steps in drawing the internal thread have been presented in Figure 7.16. In this figure steps (a) to (d) are for the simplified representation while steps (a) to (b) and (e) to (h) are for the schematic representation. Complete external and sectional views in elevation of both the simplified and the schematic representations can be seen in this figure. It can be noted here that the external view in elevation for both the simplified and the schematic representations are identical. There is difference in case of the sectional view only. It can be further noted here that the actual pitch and depth are not considered for drawing the thread symbols but for only the larger sizes of the threads, the actual pitch and true depth of the thread may be considered.

In Figure 7.17, steps in drawing the external thread symbol have been shown. In this figure steps from (a) to (e) are for the simplified representation while steps (a) to (c) and (f) to (j) are for the schematic representation. Complete external and sectional views in elevation of the simplified and schematic representations can be seen from this figure.

Steps in drawing the detailed representation of the thread have been shown in Figure 7.18. Major diameter, thread length and the pitch values are needed to draw the detailed representation of the thread. In step (e) the complete view of the thread has been shown. It can be seen from this figure that the detailed representation of the thread does not present the actual helical shape of the thread. Here it has been made straight in order to simplify the drawing. Still the view shows almost the true shape of the thread.

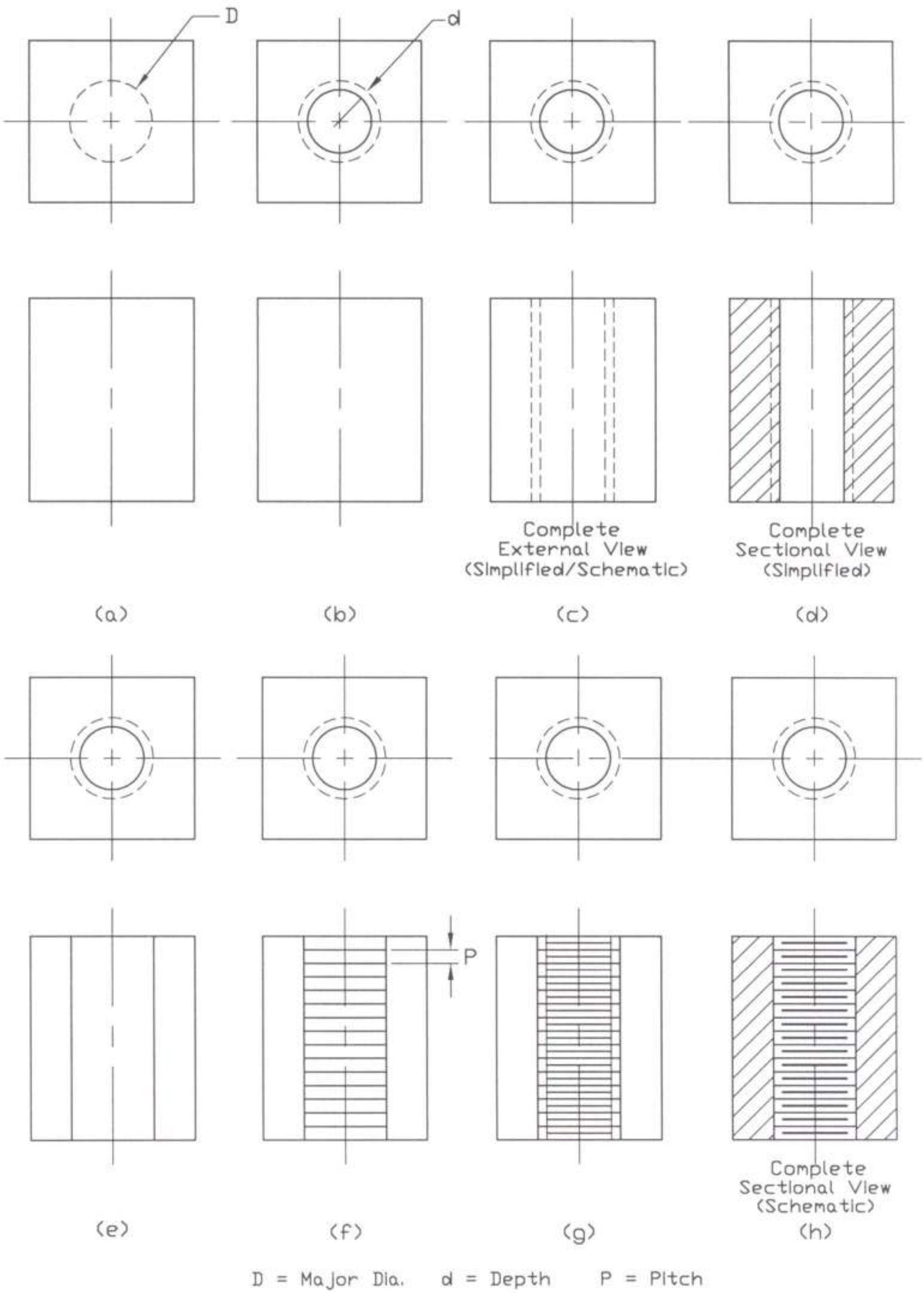


Figure 7.16: Steps in Drawing Internal Thread

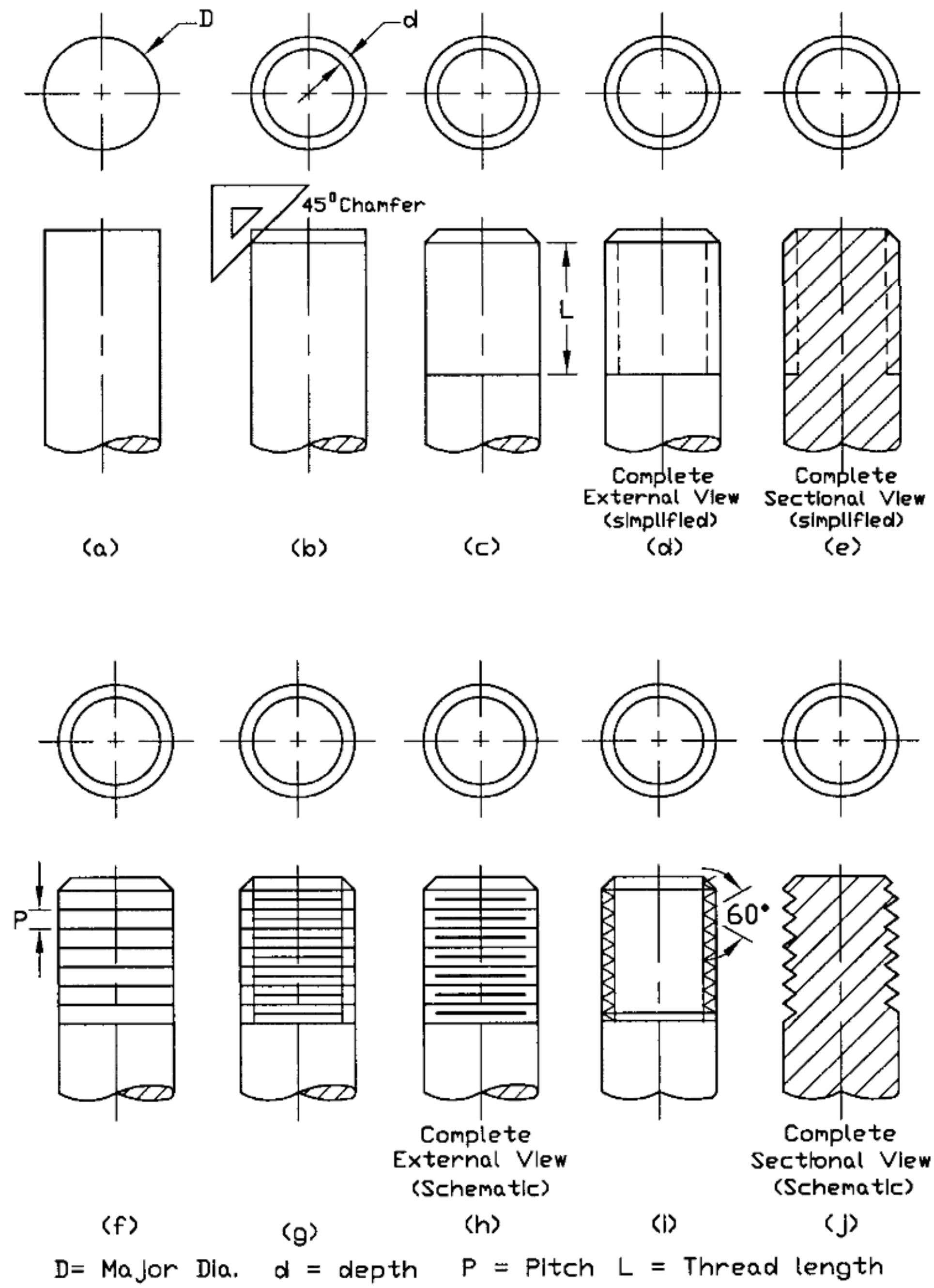


Figure 7.17: Steps in Drawing External Thread

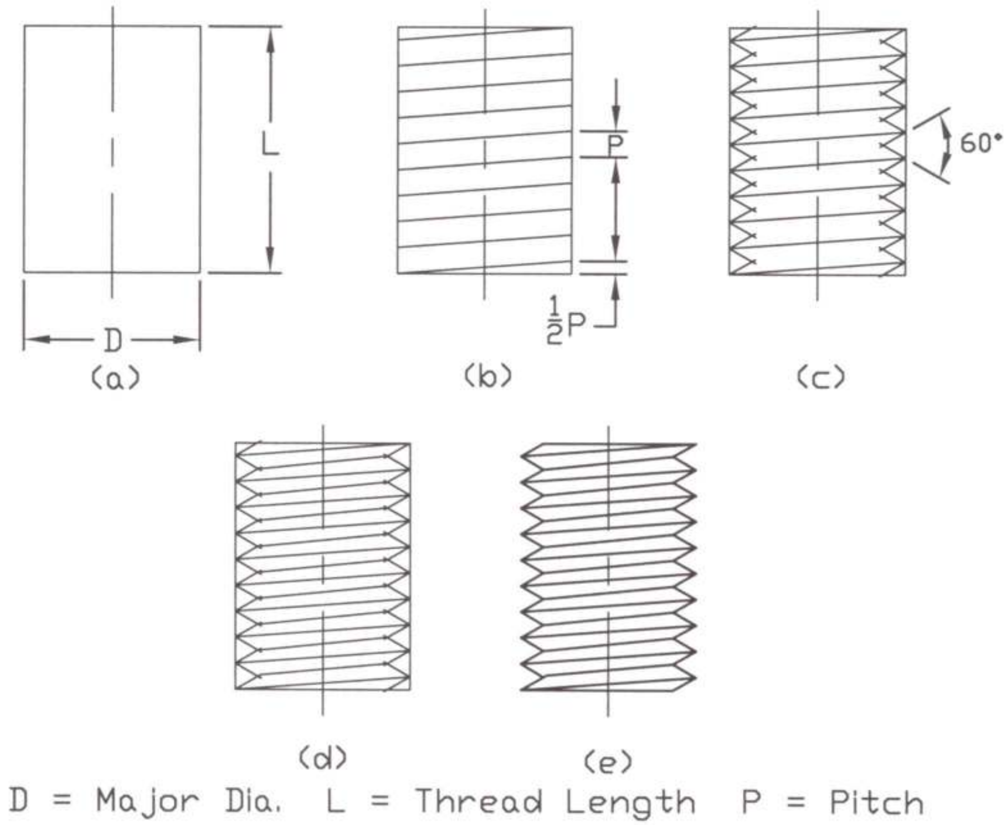


Figure 7.18: Steps in Drawing Detailed Thread

7.8 ***Steps of Drawing Bolt Head***

The steps as mentioned below are followed in drawing the bolt head.

- First of all the top view of the bolt head is drawn as shown in Figures 7.19 and 7.20 for hexagonal and square bolt heads respectively.
- Then the diameter of the bolt (d), the height of the bolt head (H) and the thickness of the washer face (1mm) if it is required, are drawn. And the vertical lines to represent the sides of the bolt are drawn. (D = 1.5d and H =2/3d for regular series).
- Next the centers (c) of the chamfer arcs are obtained.
- Then the chamfer arcs are drawn with respect to the centres (c).
- Finally the bolt head is completed.

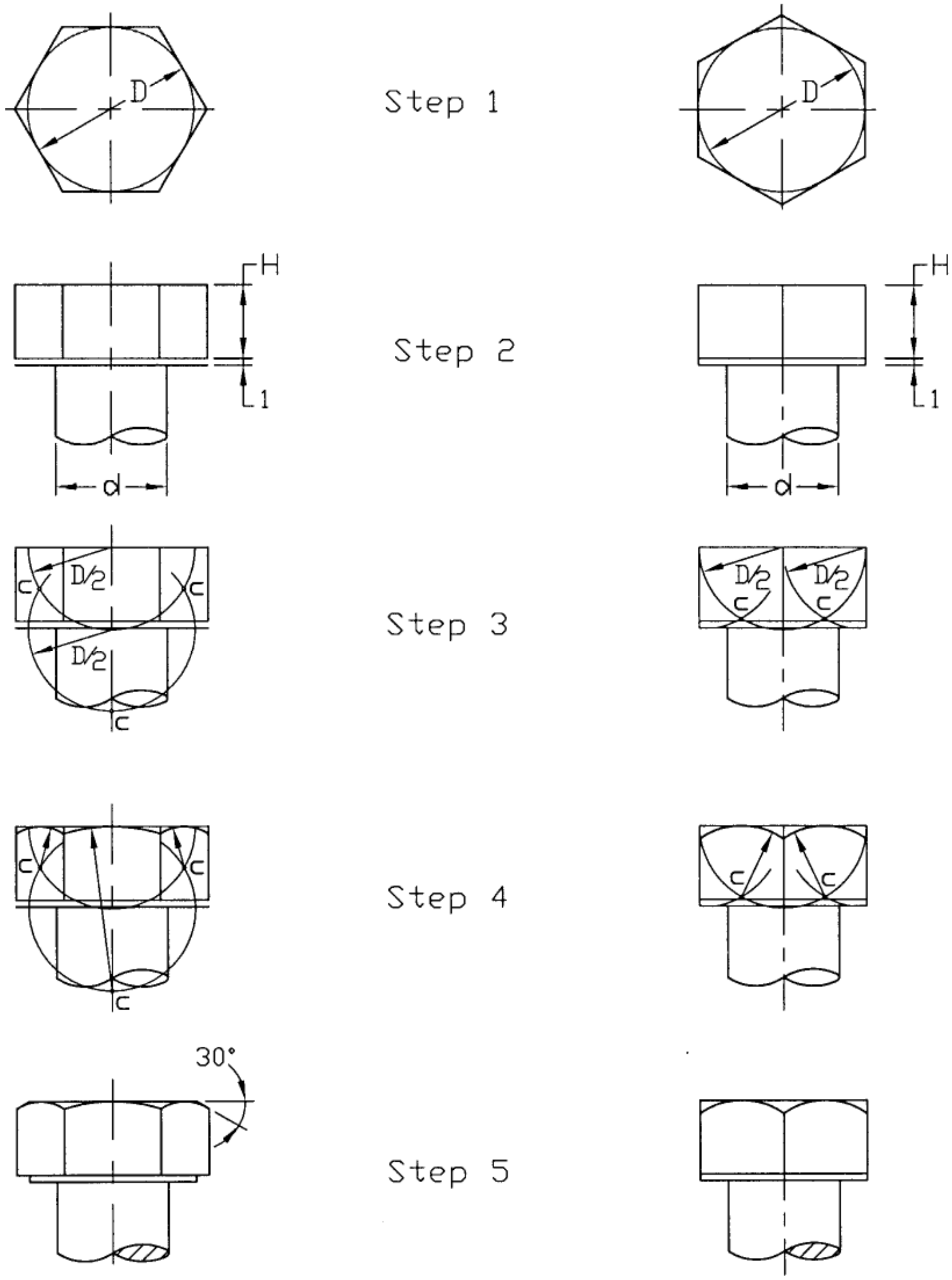


Figure 7.19: Steps in Drawing Hexagonal Bolt head

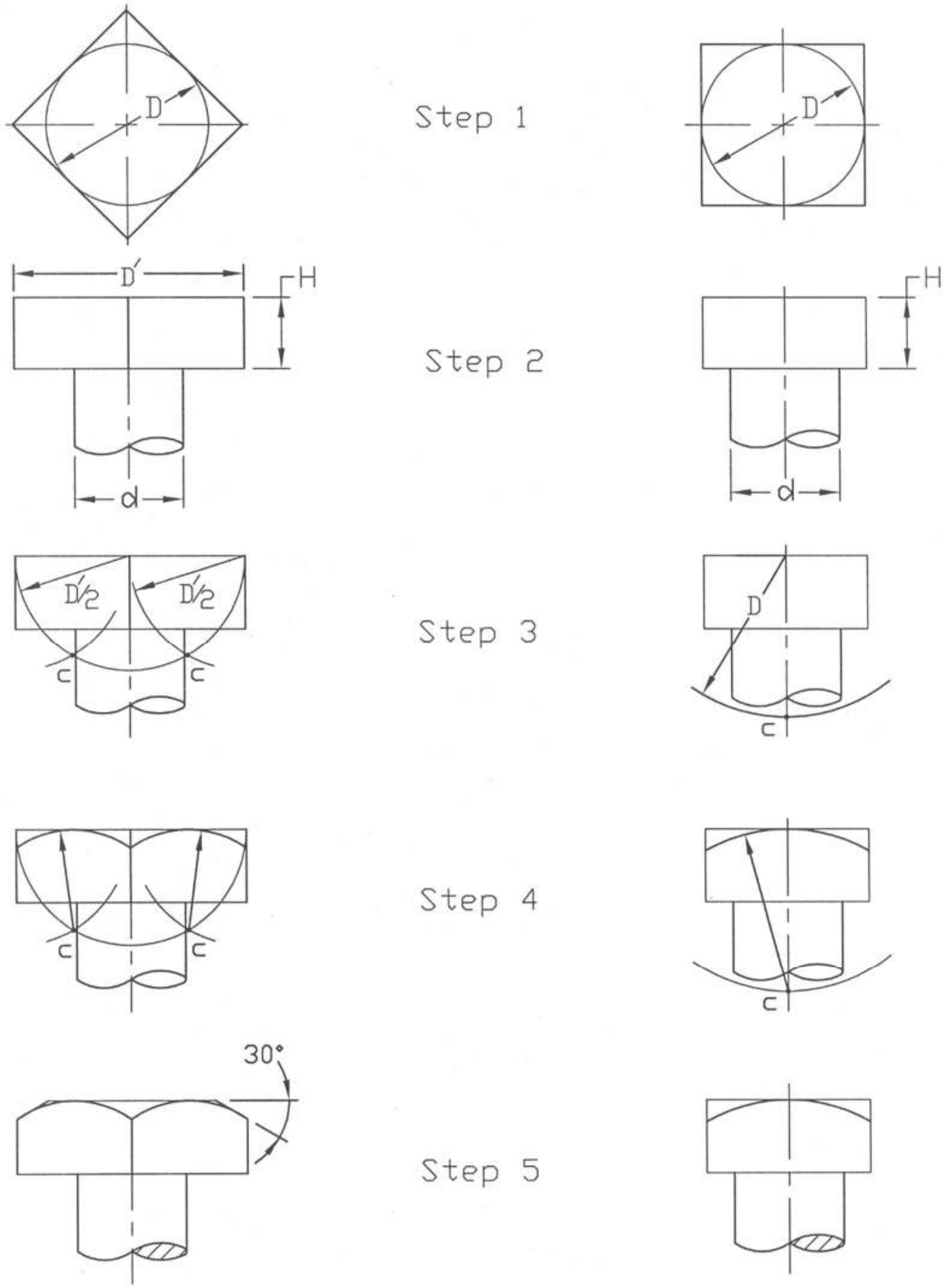
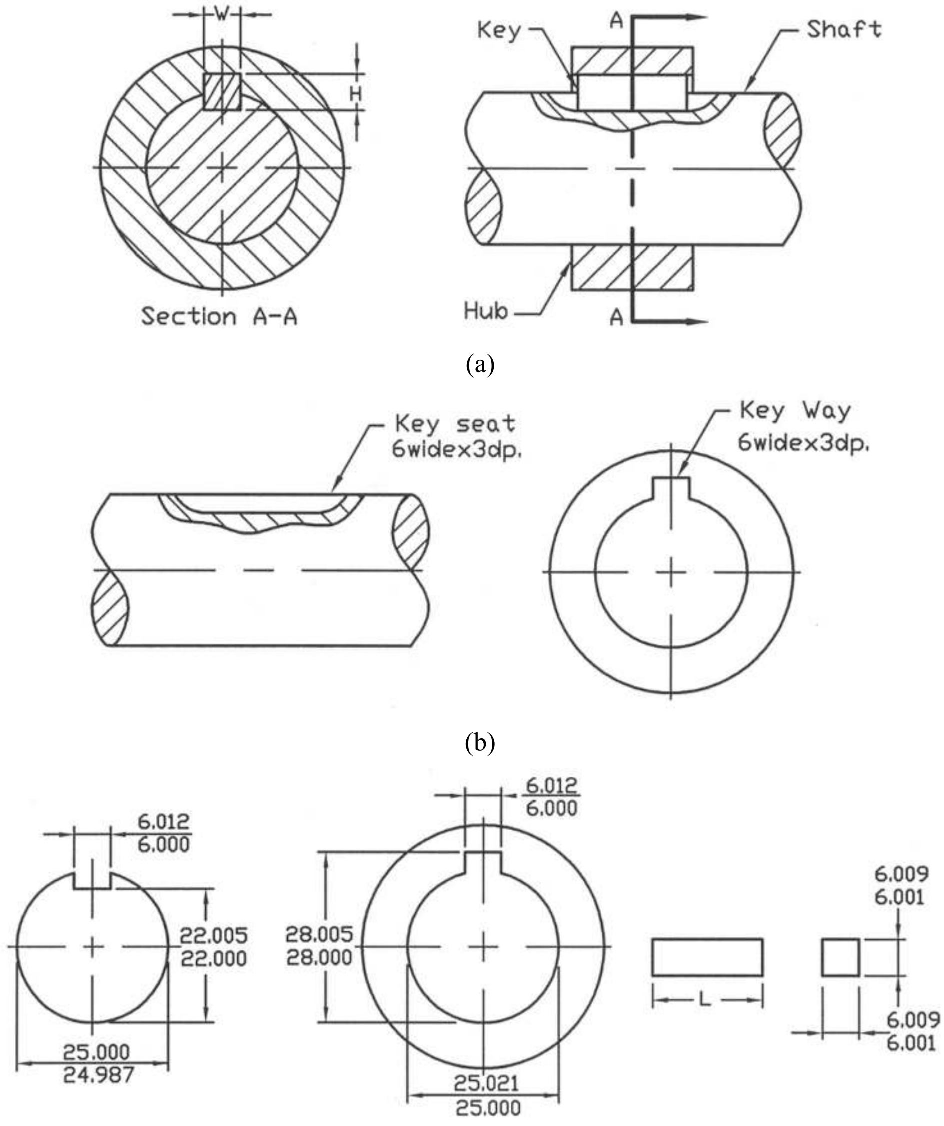


Figure 7.20: Steps in Drawing Square Bolt Head

7.9 ***Key***

In order to fasten a wheel, gear, pulley, sprocket etc. with a shaft, a key may be used. To insert the key a groove is made in the shaft, called key seat while a groove is made in the hub of the wheel, gear, pulley, sprocket etc., called the key way. A portion of the key lies in the key seat of the shaft and the rest portion fits into the key way in the hub, thus fastening them together so that there occurs no relative motion between the shaft and the hub.



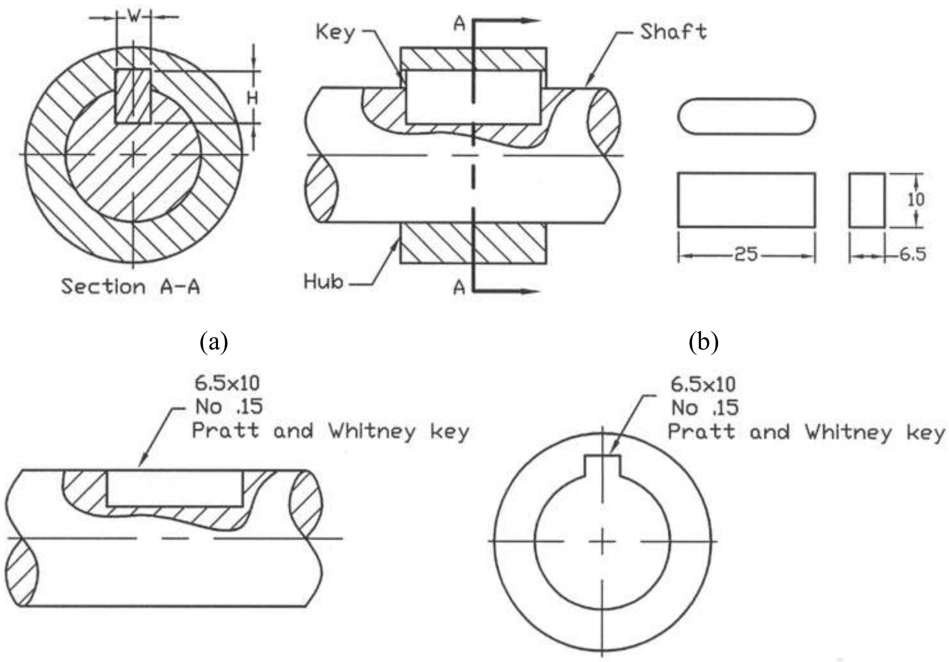
(a) (b) (c) (d)

Figure 7.21: Square Key

Keys are of various types such as, square key, flat key, Gib-head key, Pratt and Whitney key, Woodruff key etc. Of them the simplest one is the square key or flat key. Square and flat keys are widely used in the industry. The width of the square or flat key is approximately one-quarter of the shaft diameter. The standard dimensions of the square and flat keys are provided in Appendix 14. The square and flat keys are also available with 1:100 taper on their top surfaces, which are called square taper and flat taper keys. The key way in the hub is made taper to accommodate the taper key. The Gib-head key is same as the square or flat taper key having an additional head to give it an easy removal.

A shaft and hub fastened together with a square key is shown in Figure 7.21a. A flat key is also used in the similar way. For unit production where machinist is expected to fit the key, nominal dimensions are given only mentioning the width and depth as shown in Figure 7.21b. For the interchangeable assembly and for mass production the limit dimensions for the key way and key seat are shown in Figure 7.21c, while the limit dimensions of a square key are shown in Figure 7.21d.

The Pratt and Whitney key has the rectangular cross-section with the rounded ends. Two-third of this key sits in the shaft and one-third sits in the hub. A shaft and hub fastened with a Pratt and Whitney key is shown in Figure 7.22a. The standard dimensions of the Pratt and Whitney key are provided in Appendix 15. It is the standard practice to make the depth of the key seat and that of the key way as equal and half the width of the key respectively. The nominal dimensions of a Pratt and Whitney key are shown in Figure 7.22b while the nominal dimensions of key seat and key way, are shown in Figure 7.22c.

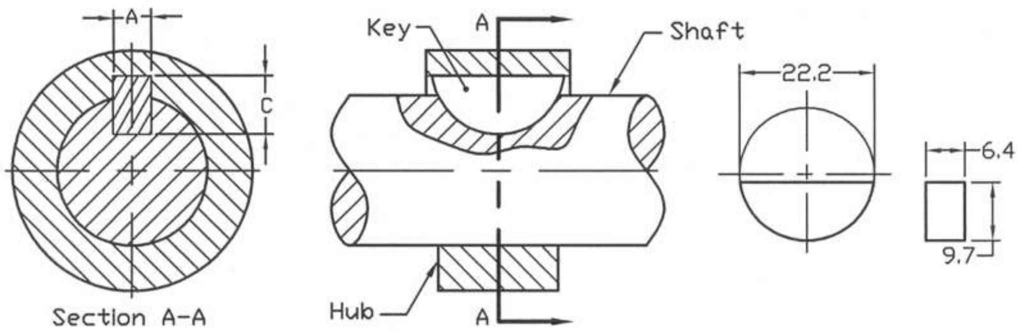


(c)

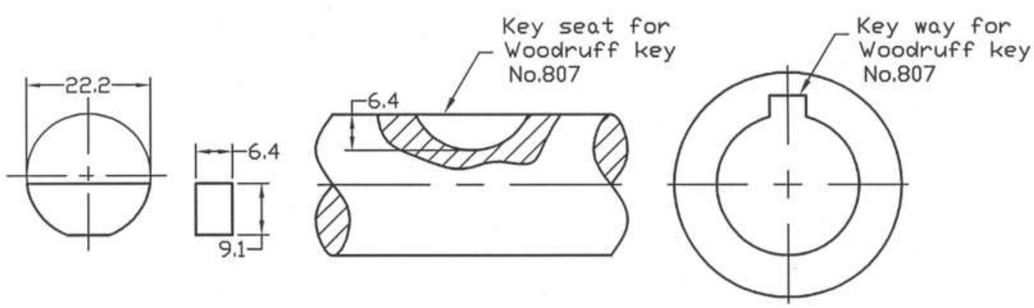
Figure 7.22: Pratt and Whitney key

A shaft and hub fastened with a Woodruff key is shown in Figure 7.23a. The key is of the shape of a semi-cylindrical disc, often at the bottom it is flattened as shown in Figure 7.23b and Figure 7.23c.

The nominal dimensions of them are also provided in these figures. The key seat in the shaft is of the semi-cylindrical shape and the depth is so made that half the width of the key extends above the shaft and into the hub where there is a rectangular key way. In Figure 7.23d the key seat and key way are shown. The standard dimensions of the Woodruff key are given in Appendix 16.



(a) (b)



(c) (d)

Figure 7.23: Woodruff Key

7.10 ***Rivet Joints***

To make permanent fastenings generally between pieces of sheet and rolled metal, rivets are used. The rivets may be made of wrought iron, soft steel, copper or occasionally other metals. Rivets are inserted through the holes made on the metal pieces to be fastened and then the ends are formed over to securely hold them. Holes may be made by punching, drilling or punching and then reaming. The holes are slightly larger than the shank diameter of the rivet. The riveted parts cannot be disassembled for maintenance or replacement without driving out the rivet.

Rivet joints are of two types such as, lap and butt. In Figure 7.24 the types of rivet joints are shown. In Figure 7.25, heads of large rivets are shown. Large rivets are used in structural work of buildings and bridges. They are also used in ship, boiler and tank construction. The button head and flat top countersunk head are commonly used in structural work while button head and cone head are commonly used in tank and boiler construction. For light work small rivets are used. Some small rivets are illustrated in Figure 7.26.

The rivets, which are inserted in the structure at the shop, are called the shop rivets while that are inserted in the site, are called the field rivets. The conventional rivet symbols for the shop rivets and field rivets are shown in Figure 7.27 in accordance with the American and Canadian Institutes of Steel Construction. For the shop rivets the diameter of the head of the rivet is shown and for the field rivets the diameter of the shank is shown on the drawing. Rivet diameters are usually chosen as $D = 1.2\sqrt{t}$ to $1.4\sqrt{t}$, where t is the metal thickness. Large rivets are available in diameters from 12mm to 45mm by increment of 3mm.

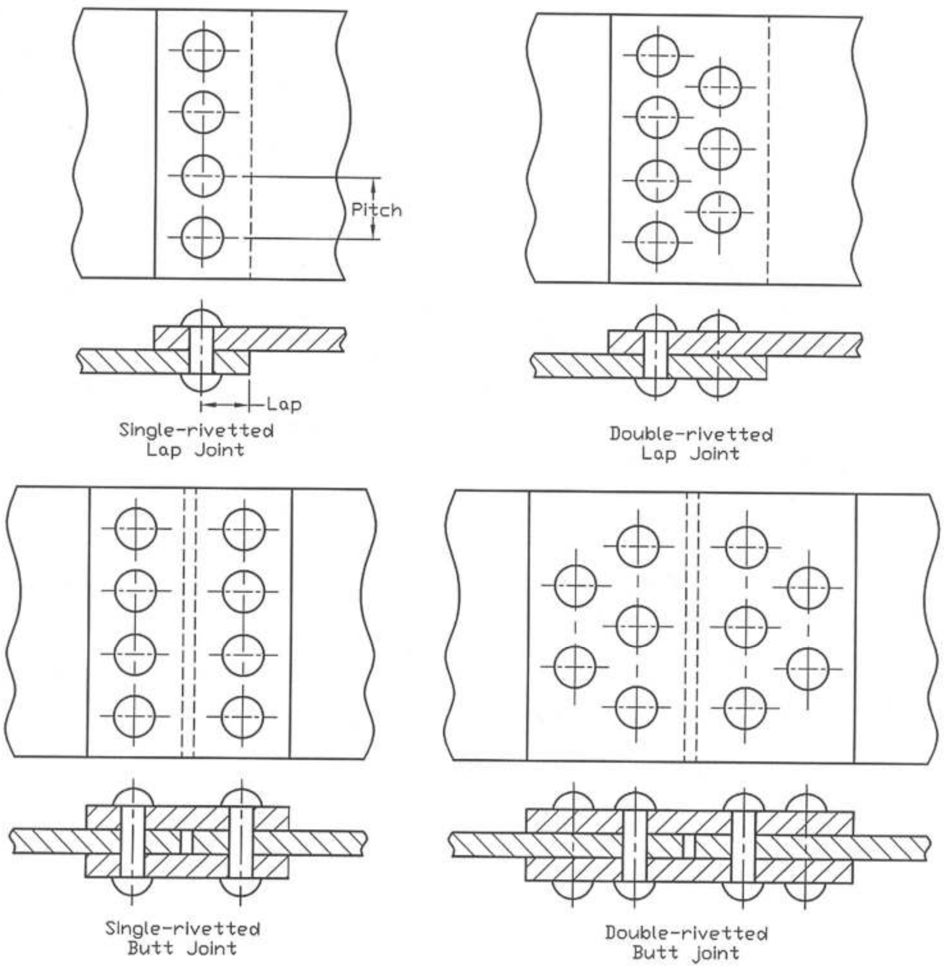


Figure 7.24: Types of Rivet Joints

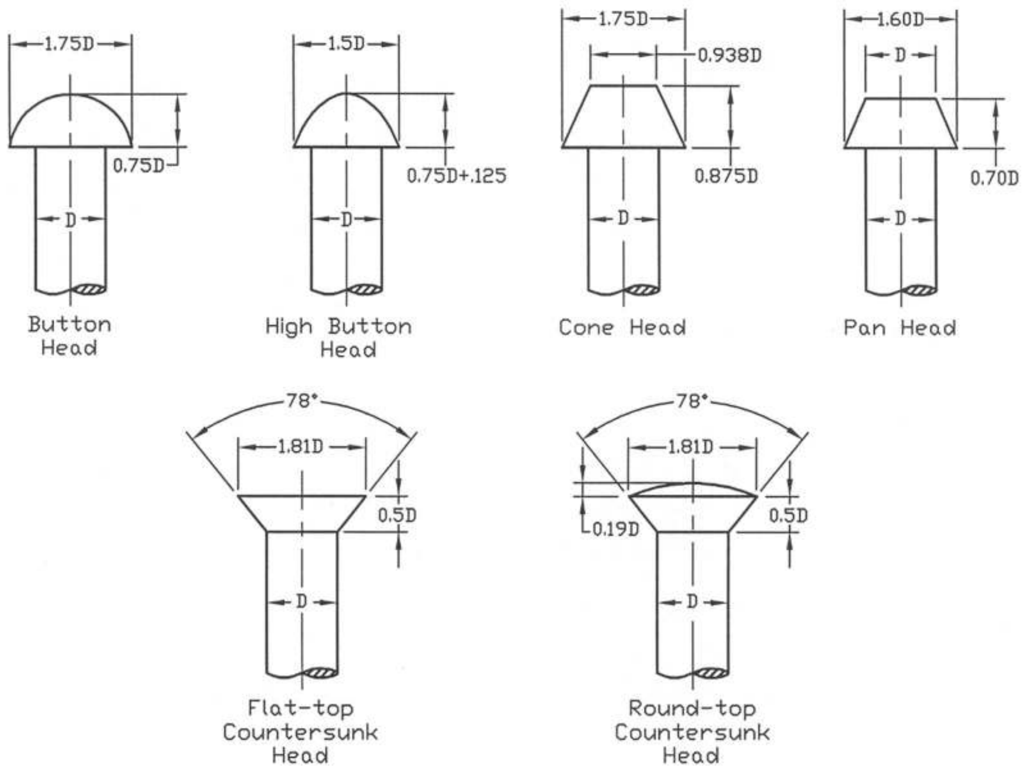


Figure 7.25: Heads of Large Rivets

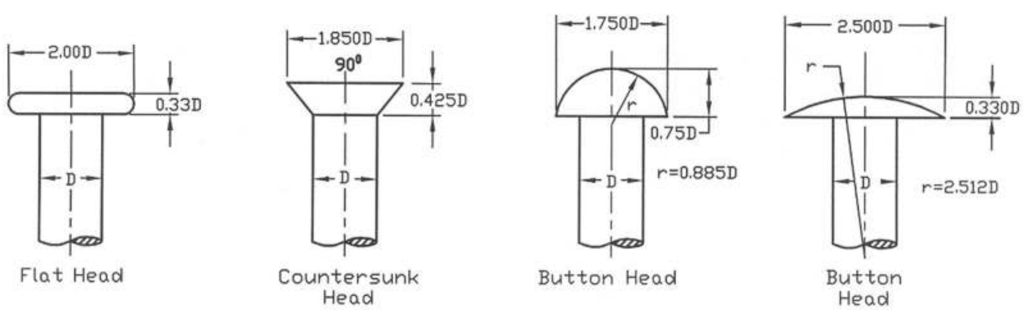


Figure 7.26: Heads of Small Rivets

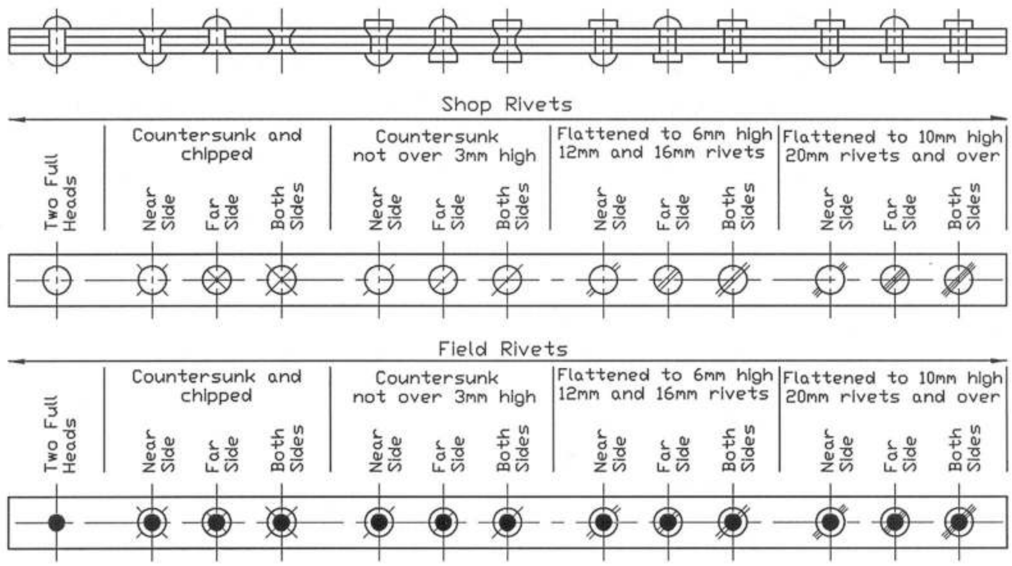


Figure 7.27: Conventional Symbols for Rivets

7.11 ***Weld Joints***

For permanent fastening between machine parts weld joints are used. In Figure 7.28 different types of welded joints are shown.

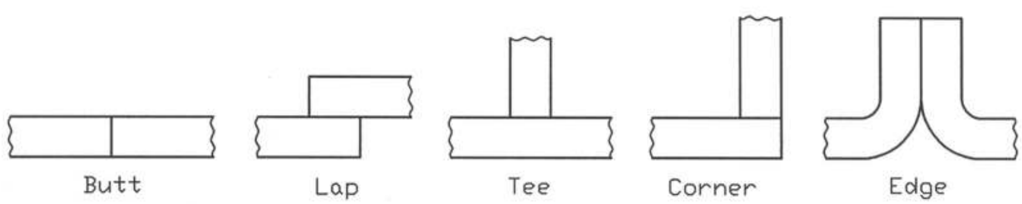


Figure 7.28: Types of Welded Joints

The names of the types of joints are based on the geometric relationship between the two parts. However, for each type of the joints different types of weld can be used. In Figure 7.29, some common types of welds with their symbols according to ANSI/AWS A2.4 – 86 are shown, where AWS stands for American Welding Society. Since the welding symbol simplifies the drawing, both time and money can be saved using it. Some supplementary welding symbols are shown in Figure 7.30. The applications of some welding symbols are presented in Figure 7.31.To show the weld on the arrow side and the other side the weld symbols are respectively located on and below the leader as shown in the figure. For welds on both sides (arrow and other sides), the weld symbols are placed both on and below the leader. The necessary dimensions of welds are also shown with the symbols.

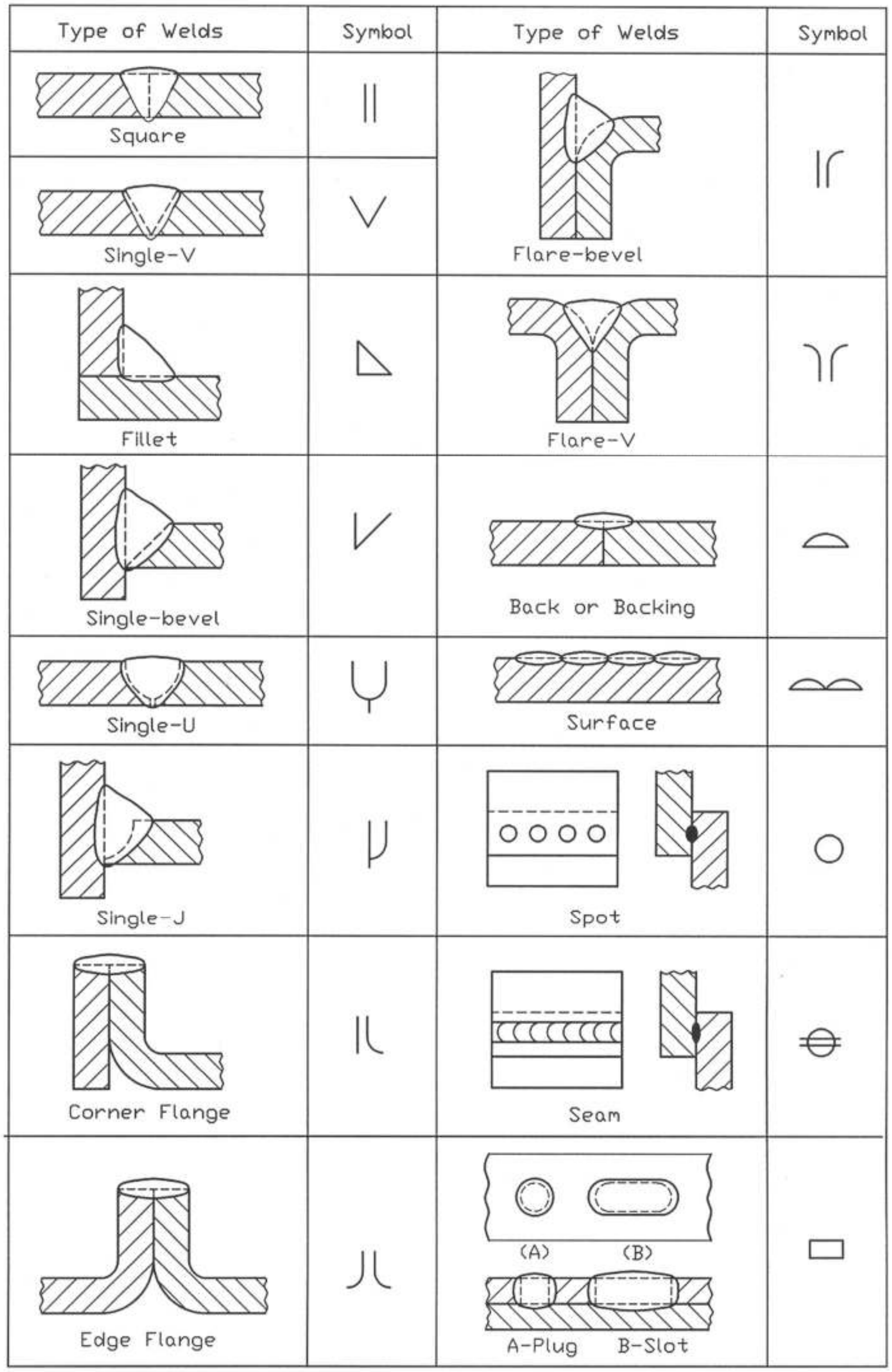


Figure 7.29: Welding Symbols

In Figure 7.32, the applications of some supplementary welding symbols are provided. On the opposite side of a single V-groove weld and a single bevel-groove weld the application of back welds are shown in Figure 7.33. The applications of some resistance welds such as, spot and seam welds are shown in Figure 7.34. The spot and seam welds have been given in the center and the weld symbols have been shown accordingly. These welds may be done on the arrow side and the other side as well.

Back or backing is a bead type weld may be done on the opposite side of a single weld. **Melt thru** indicates that the joint is to have 100 per cent penetration in the weld made from only one side. **Flush** indicates that a melt-thru weld is to be made flush with the surface by grinding, chipping or other mechanical means (say, machining). **Field weld** indicates that the weld is to be done completely at the site or field. In **plug or slot** weld a hole or slot is made in one of the parts to be welded and welding is done inside the slot. **Weld all around** indicates that the welding is done around the particular joint.

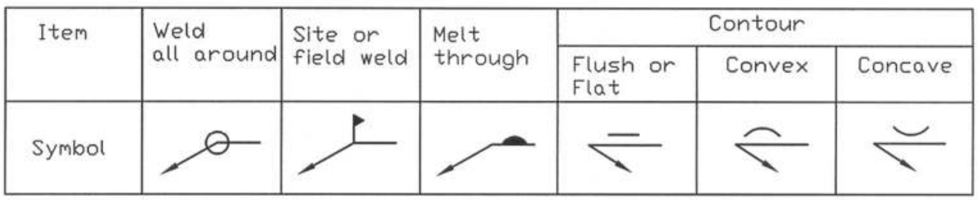


Figure 7.30: Supplementary Welding Symbols

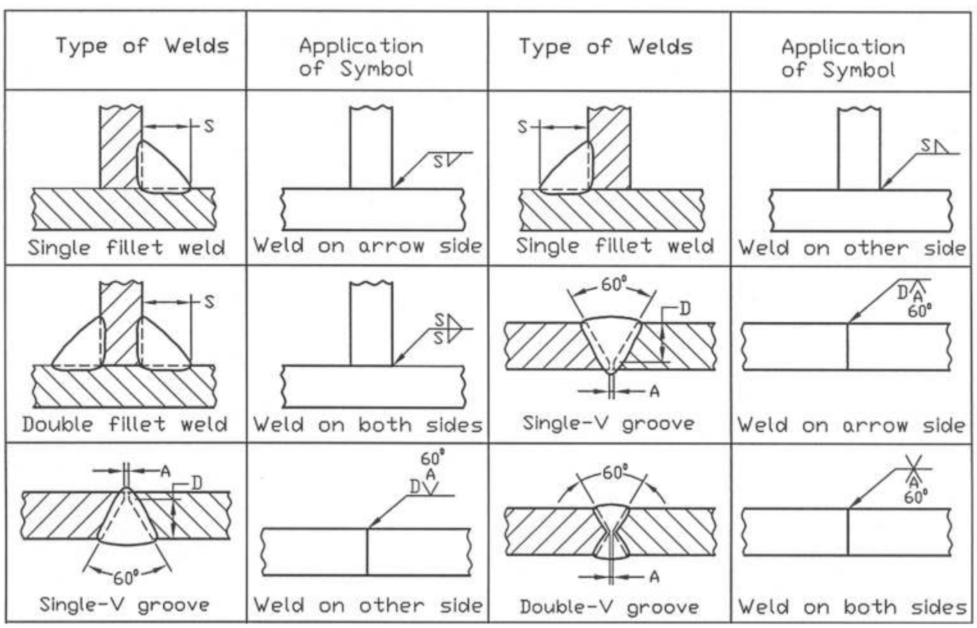


Figure 7.31: Applications of Welding Symbols (Contd.)

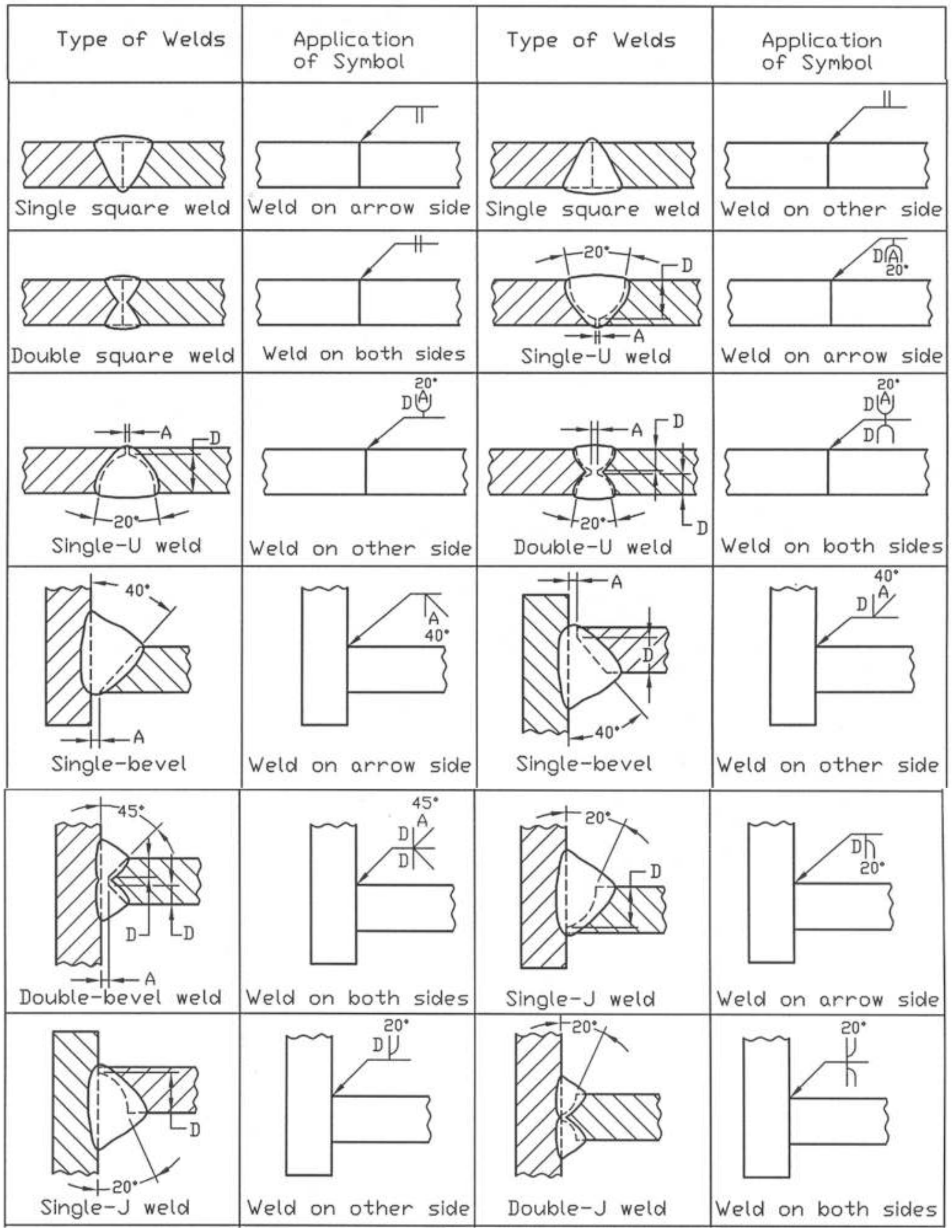


Figure 7.31: Applications of Welding Symbols (Contd.)

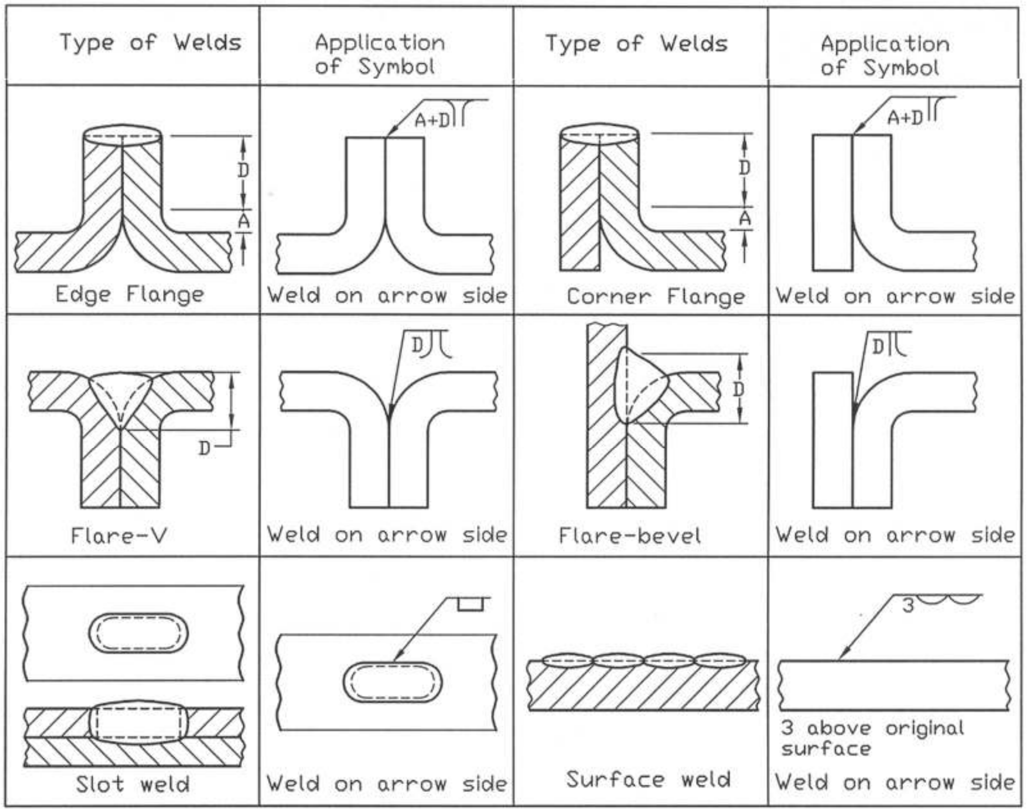


Figure 7.31: Applications of Welding Symbols (contd.)

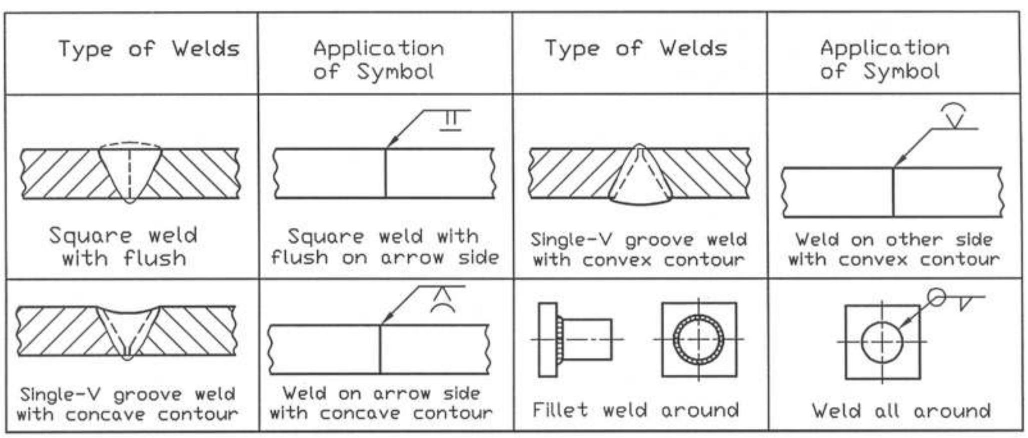


Figure 7.32: Application of Supplementary Welding Symbols

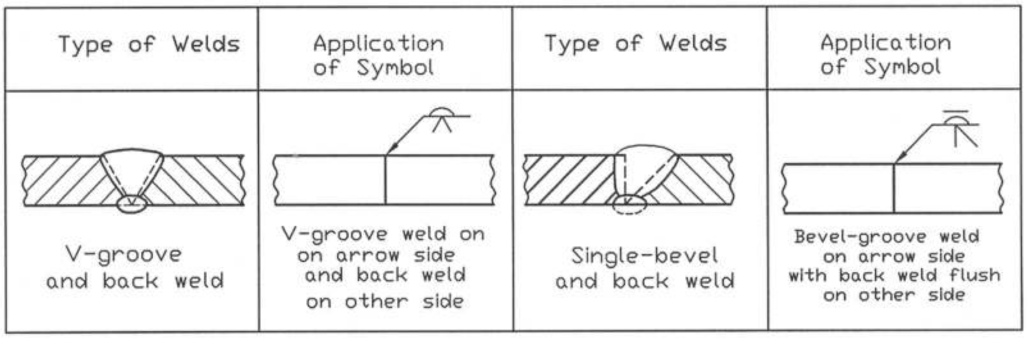


Figure 7.33: Application of Back or Backing Welding Symbols

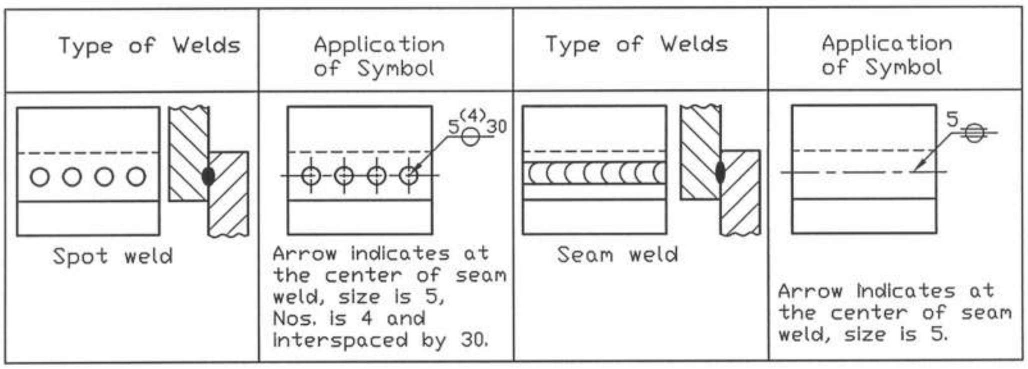


Figure 7.34: Application of Resistance Welding Symbols

7.12 ***Pulley***

Pulleys are used to transmit power from one shaft to another with the help of belts such as, flat belt, V-belt, round belt etc. Pulleys may be of different types such as, crowned pulleys and grooved pulleys. Crowned pulleys are used to transmit power by the flat belt while the grooved pulleys are used to transmit power by the V-belt or round belt. When the surface of the rim of pulley is made of a little convex shape, it is called crowned pulley. On the other hand to transmit power by the V-belt the pulley contains V-groove. The nomenclature of a crowned pulley is shown in Figure 7.35.

In accordance with the Indian standard IS: 2122 (Part 1) – 1973 (R1986), the recommended widths of the pulley for flat belts are provided in the Table 7.1.

Table 7.1: Recommended Width of Pulley for Flat Belts

Belt width (mm)	up to 125	125 - 250	250 -375	375 – 500
Width of pulley higher than belt width by (mm)	13	25	38	50

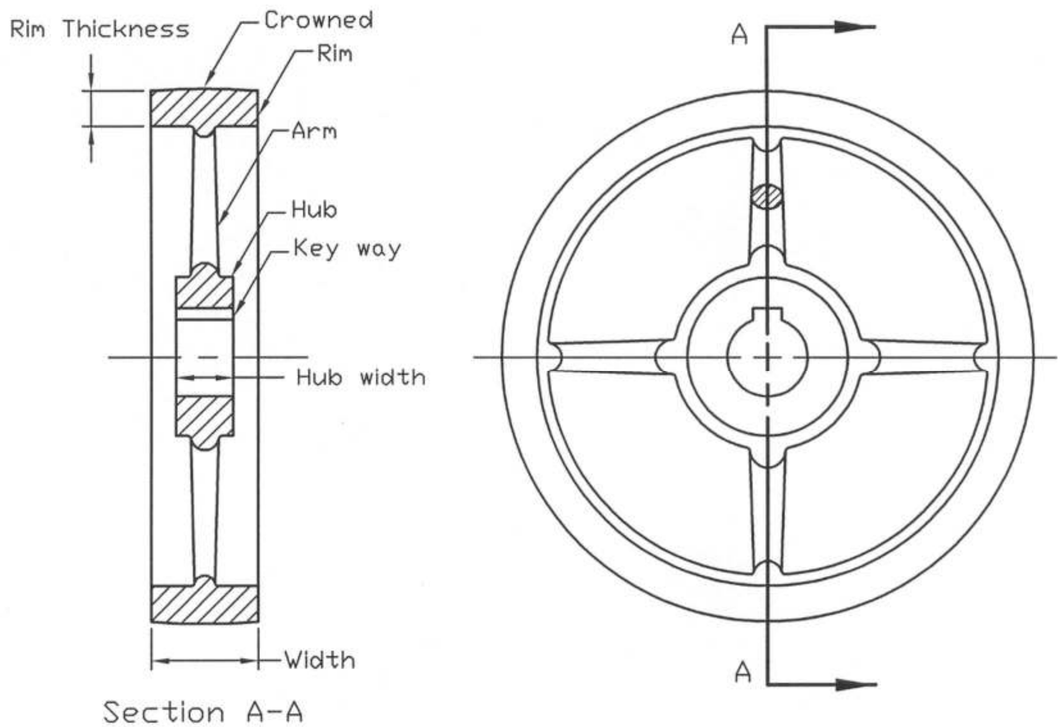


Figure 7.35: Nomenclature of a Crowned Pulley

According to the American Leather belting Association (ALBA), the minimum diameters of the pulleys for the flat leather belts are given in Table 7.2.

Table 7.2: Recommended Minimum Pulley Diameters for Flat Leather Belts

Belt Speed (m/s)	Single Ply		Double Ply			Triple Ply	
	Medium t = 4.5	Heavy t = 5.0	Light t = 7.0	Medium t = 8.0	Heavy t = 9.0	Medium t = 12.0	Heavy t = 13.5
Up to 13	63	75	100	125	200	400	500
13 – 20	75	90	115	150	230	450	550
20 - 30	90	100	125	175	250	500	600

The recommended values of crown heights against pulley diameters for flat belt drive are shown in Table 7.3 in accordance with the standard ISO.

Table 7.3: Recommended Values of Crown height

Pulley Diameter (mm)	Crown Height (mm)	Pulley Diameter (mm)	Crown Height (mm)	
			w ≤ 250mm	w > 250mm
40, 50, 63.5	0.3	320, 355	0.8	0.8
70, 80	0.3	320, 355	1.0	1.0
90, 100, 115	0.3	570, 635, 710	1.25	1.25
125, 140	0.4	800, 900	1.25	1.50
160, 180	0.5	1010	1.25	1.50
200, 225	0.6	1140, 1270, 1420	1.50	2.00
250, 285	0.8	1600, 1800, 2030	1.78	2.50

Face width, B = (n-1) e +2f
Note: Crown should be rounded, not angled.

The recommended values of the minimum pitch diameter of V-pulley for the different belt sections are shown in Table 7.4.

Table 7.4: Recommended Minimum Pitch Diameter for V- Pulley

Belt Section	A	B	C	D	E
Minimum Pitch Diameter. (mm)	75	135	225	325	540

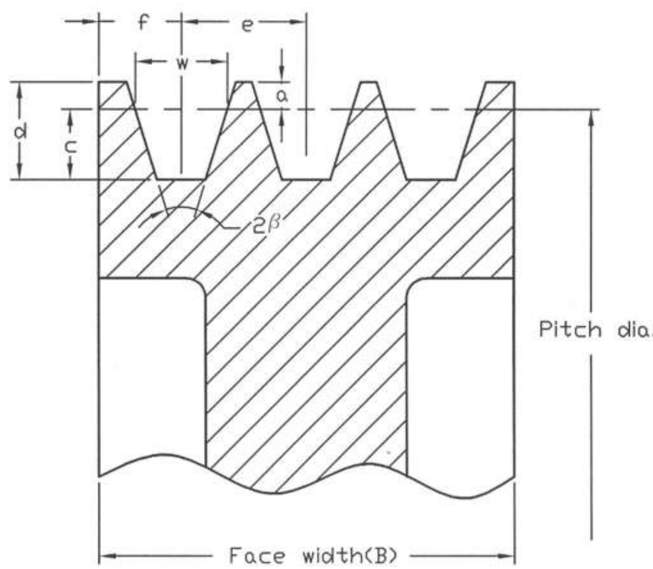


Figure 7.36: V-Pulley

A V-pulley is shown in Figure 7.36. V-belts have various sections such as, A, B, C, D and E. For the different belt sections the recommended dimensions of the standard V-grooved pulley according to IS: 2494 – 1974 are given in Table 7.5.

Table 7.5: Dimensions of Standard V-grooved Pulleys

Type of belts	w	d	a	c	f	e	No. of grooves (n)	Groove angle (2β) in deg
A	11	12	3.3	8.7	10	15	6	32, 34, 38
B	14	15	4.2	10.8	12.5	19	9	32, 34, 38
C	19	20	5.7	14.3	17	25.5	14	34, 36, 38
D	27	28	8.1	19.9	24	37	14	34, 36, 38
E	32	33	9.6	23.4	29	44.5	20	_____

7.13 *Gears*

Most of the machines cannot be imagined without gears. Gears are of different types. **Spur gear** and **helical gear** are used to transmit power between two parallel shafts. **Bevel gear** is used to transmit power between intersecting shafts. They may be either at right angle or at any other angle. **Worm gear** is used to transmit power between non-intersecting shafts, which are usually at right angle with each other. Besides these a **spur gear** and a **rack** may be used to convert the rotary motion to the linear motion.

According to ISO 2203: 1973(E), the conventional representations of various gears in first angle projection are shown in Figures 7.37 to 7.47. However, the same principle can be used to the third angle projection. In order to eliminate the unnecessary drawing time, conventional representation is preferred. In this representation the drawing of the actual gear tooth profile is avoided as shown in the figures.

In Figures 7.37, 7.38 and 7.39, a spur gear, bevel gear and worm gear with section are shown respectively. A spur gear is represented in the unsectioned condition in Figure 7.40. In Figure 7.41 a gear is represented with one tooth while in Figure 7.42, representations of both helical and double helical gears are provided using three thin continuous lines. In Figures 7.43 to 7.47, the conventional representations of engagement of gears are shown. The engagement of spur gears and that of bevel gears with section are respectively shown in Figures 7.43 and 7.44. The engagement of two spur gears, helical gears and double helical gears are provided in Figure 7.45 in the unsectioned condition. While in Figures 7.46 and 7.47, the engagement of a rack and pinion and that of worm and worm gears are respectively presented.

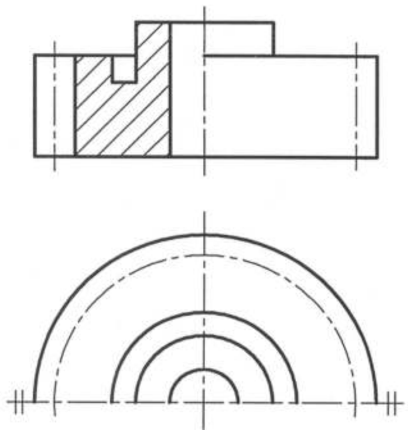


Figure 7.37: Spur Gear With Section

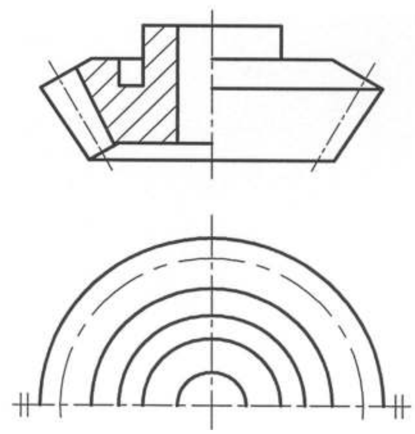


Figure 7.38: Bevel Gear With Section

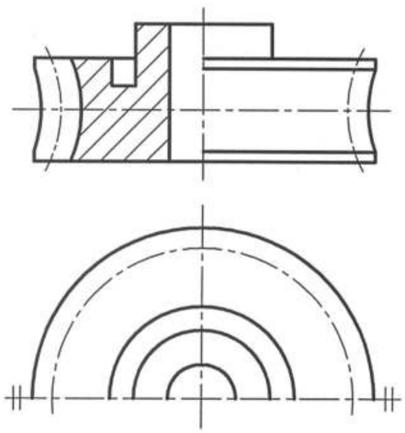


Figure 7.39: Worm Gear With Section

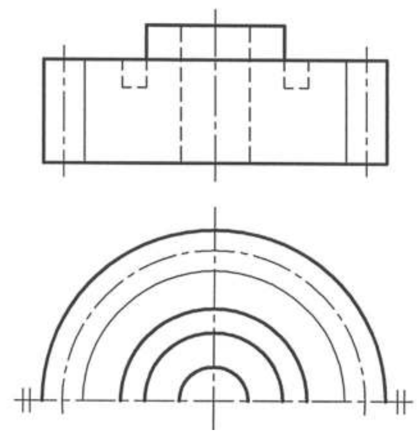


Figure 7.40: Spur Gear Without Section

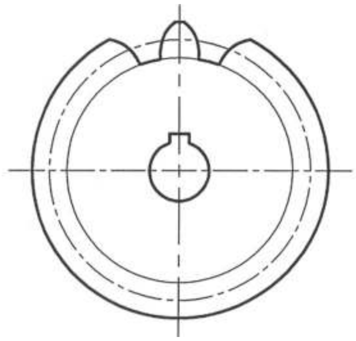
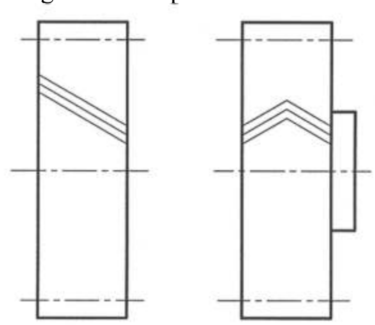


Figure 7.41: Spur Gear



Helical Gear Double Helical

Figure 7.42: Helical Gear

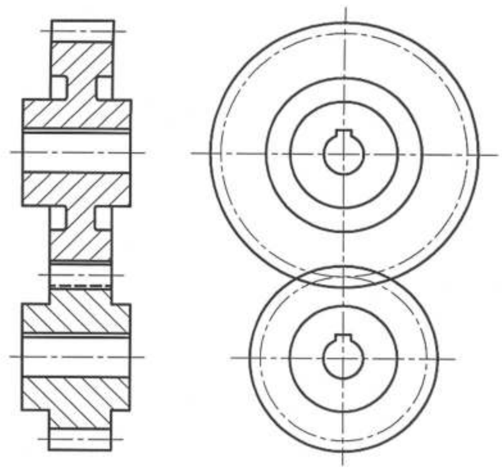


Figure 7.43: Engagement of Spur Gear

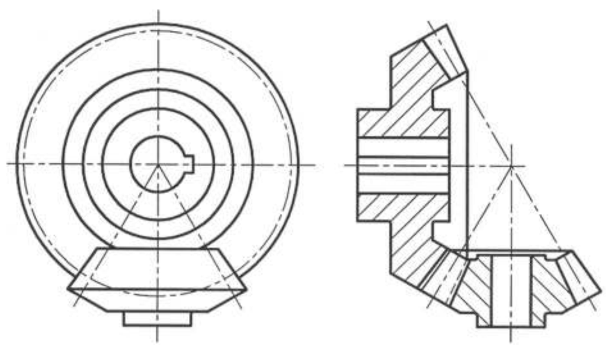
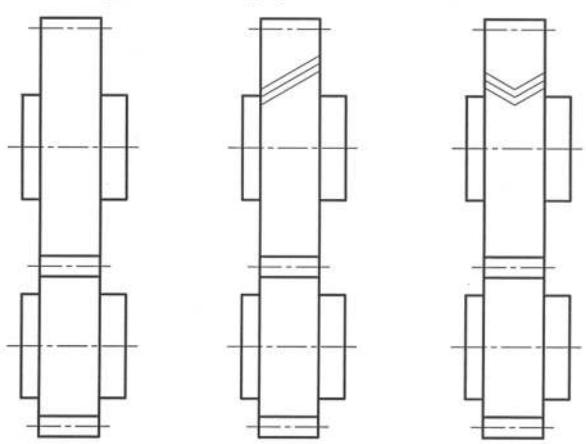


Figure 7.44: Engagement of Bevel Gears



Spur Gears Helical Gears Double Helical Gears

Figure 7.45: Engagement of Spur, Helical and Double helical Gears

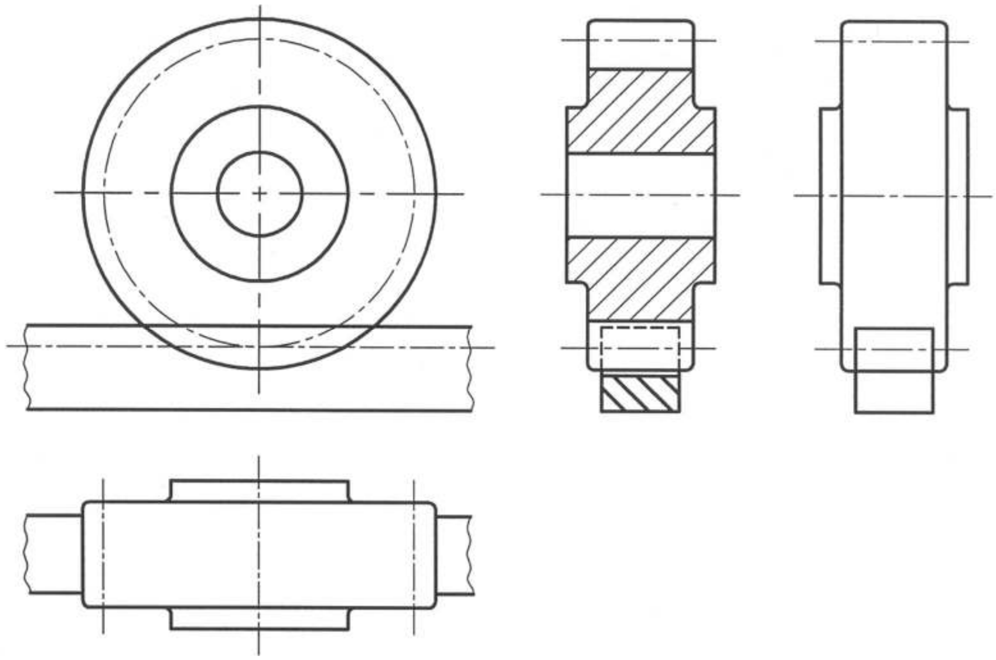


Figure 7.46: Engagement of Rack and Pinion

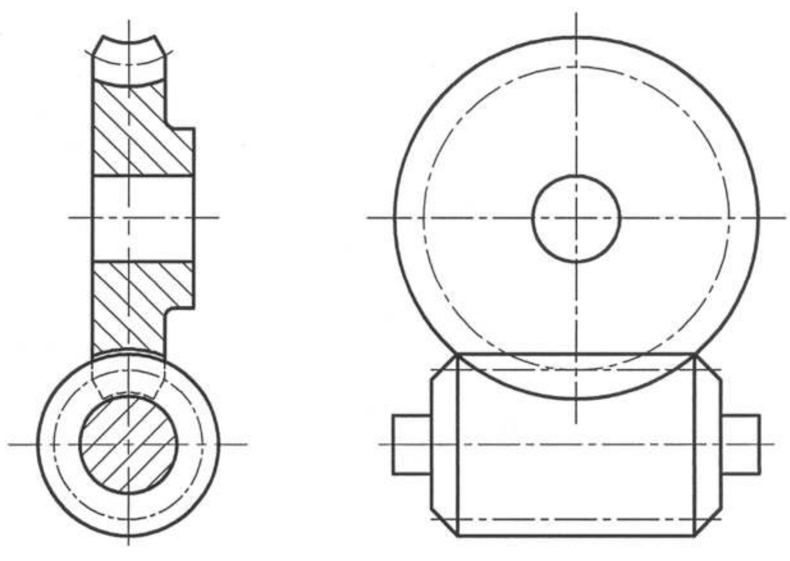


Figure 7.47: Engagement of Worm and Worm Gear

In Figure 7.48, the terminology of a spur gear is illustrated. According to the SI unit, module is the index of the tooth size of a gear. Module (m) is defined as the ratio of the pitch diameter (d) to the number of tooth of the gear (N) i.e. $m = d / N$. The circular pitch (p) is defined as the distance measured on the pitch circle from a point on one tooth to a corresponding point on the adjacent tooth i.e. $p = \pi d / N = \pi m$. The most common form of the tooth is of the involute profile. The recommended range of the face width (F) of gear is $3p \le F \le 5p$. The standard pressure angles (ϕ) are $14\frac{1}{2}^0$, 20^0 and 25^0 of which 20^0 is widely used. Standard values of some terms of a spur gear for $\phi = 20^0$ and 25^0 are provided in Table 7.6.

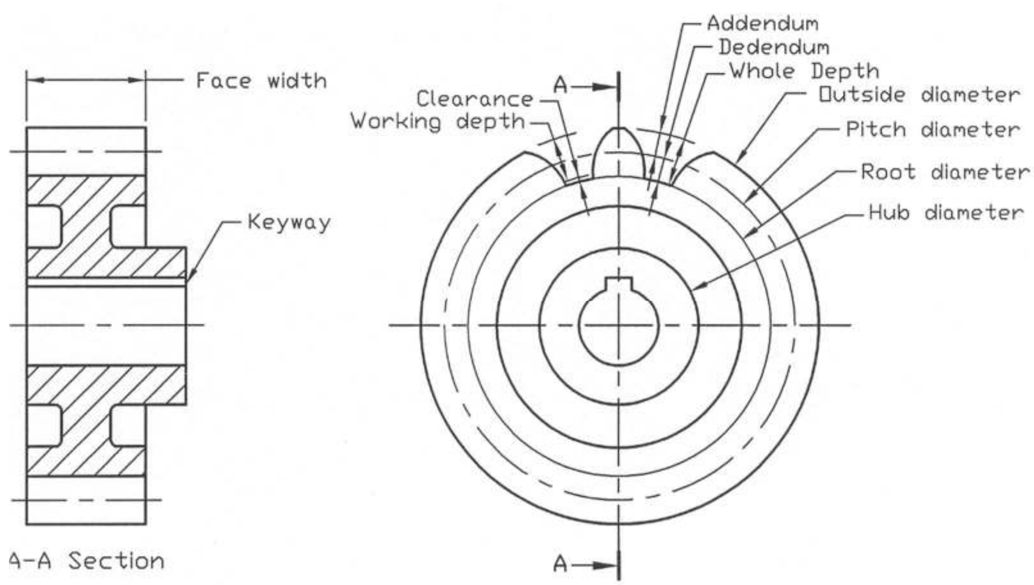


Figure 7.48: Spur Gear Nomenclature

Table 7.6: Standard Values of Some Terms of a Spur Gear

Term	Addendum	Dedendum	Working Depth	Whole Depth	Clearance (min.)	Min. No. of Teeth	
						$\phi = 20^0$	$\phi = 25^0$
Value	1m	1.25m	2m	2.25m	0.25m	18	12

The preferred module for general uses is shown in Table 7.7.The terminology of a bevel gear is shown in Figure 7.49. The standard tooth proportions for 20^0 straight-bevel gear, are provided in Table7.8. Here N_P and N_G indicate the number of teeth of the pinion and gear respectively. While d_P and d_G indicate the pitch diameter of pinion and gear respectively.

Table 7.7: Preferred Module (m) in General Uses

1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 16, 20, 25, 32, 40, 50

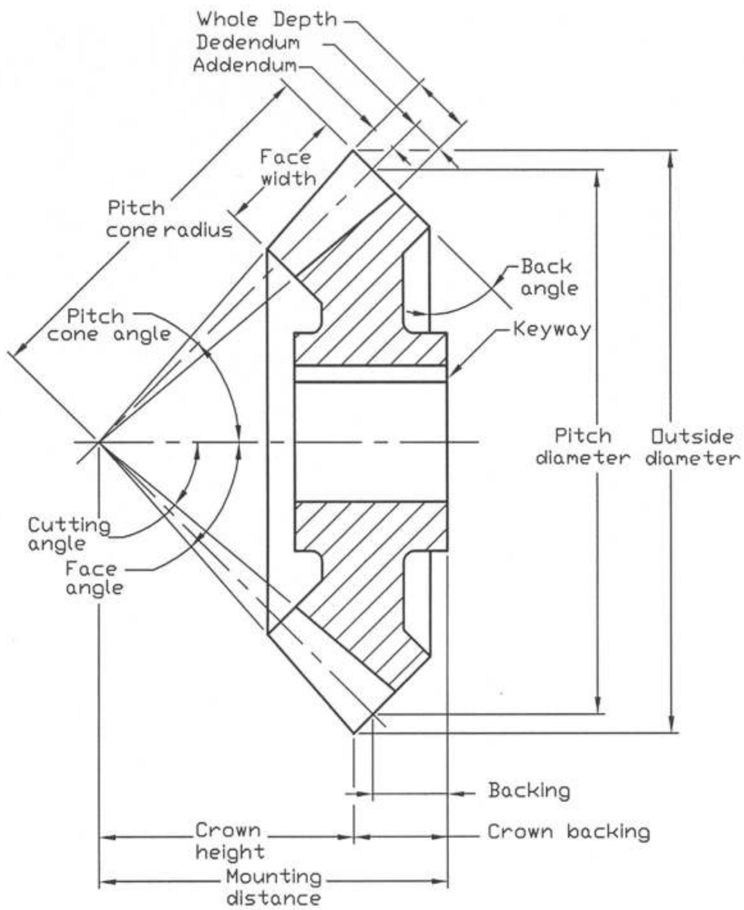


Figure 7.49: Bevel-Gear Nomenclature

Table 7.8: Standard Tooth Proportions for 20^0 Straight Bevel-Gear (contd.)

Term	Formula
Pitch cone angle	$\gamma = \tan^{-1}(N_P / N_G)$ for pinion $\Gamma = \tan^{-1}(N_G / N_P)$ for gear
Pitch cone radius	$A_O = \dfrac{d_P / 2}{\sin\gamma}$ for pinion $A_O = \dfrac{d_G / 2}{\sin\Gamma}$ for gear

Table 7.8: Standard Tooth Proportions for 20^0 Straight Bevel-Gear (contd.)

Term	Formula					
Back angle	Pitch cone angle (γ or Γ)					
Working depth	$h_k = 2.0m$					
Clearance	c = (0.188m) + 0.050					
Addendum of gear	$a_G = 0.54m + \frac{0.460m}{(m_{90})^2}$					
Gear ratio	$m_G = N_G/N_P$					
Equivalent 90^0 ratio	$m_{90} = m_G$, when $\Sigma = 90^0$ $m_{90} = \sqrt{(m_G \cos\gamma/\cos\Gamma)}$, when $\Sigma \neq 90^0$					
Face width	$F = A_o/3$ or $F = 10m$, whichever is smaller					
Minimum Number of pinion	pinion	16	15	14	13	
	Gear	16	17	20	30	

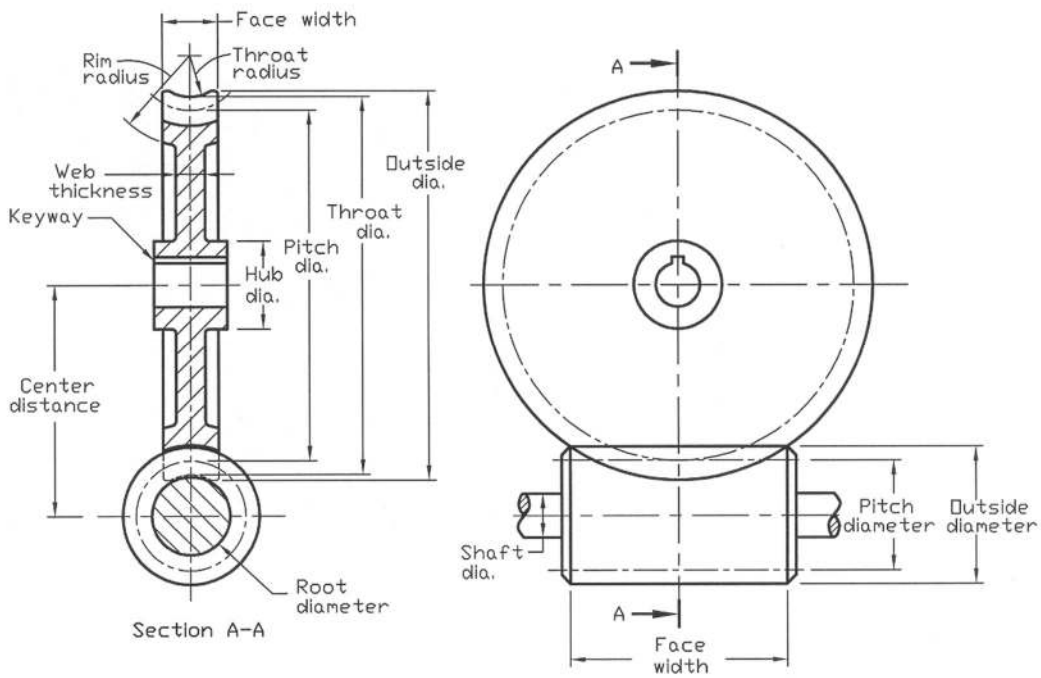


Figure 7.50: Nomenclature of Worm and Gear

The terminology of a worm and gear is shown in Figure 7.50. The standard values of some essential terms of a worm and gear are given in Table 7.9. The module of the gear is indicated by m whereas N_w and N_g are respectively the number of teeth of the worm and the worm gear and ϕ_n is the normal pressure angle of the gear. Usually the face width of the worm is made equal to the length between two points on the circumference of the outside diameter of the gear as shown in Figure 7.50.

Table 7.9: Standard Values of Some Terms of Worm and Gear (contd.)

Term	Formula
Circular pitch of gear	$P_t = \pi m$
Axial pitch of worm	$P_x = P_t$
Lead	$L = P_x N_w$
Pitch diameter of worm	$d_w \approx 3P_t$ according to AGMA*
Pitch diameter of gear	$d_g = \dfrac{N_g P_t}{\pi}$
Face width of gear	$b_{\max} \leq 0.5d_o$ $b = 0.5d_w$ may be chosen
Center distance	$C = (d_w + d_g)/2$
Lead angle	$\lambda_w = \tan^{-1}(L/\pi d_w)$
Addendum	$a = 0.3683P_x$ for $\phi_n = 20^0$ $a = 0.2865P_x$ for $\phi_n = 25^0$
Dedendum	$a_G = 0.3683P_x$ for $\phi_n = 20^0$ $a_G = 0.3314P_x$ for $\phi_n = 25^0$
Throat radius	$R_t = \dfrac{d_w}{2} - a$
Rim radius	$R_r = \dfrac{d_w}{2} + P_x$

Table 7.9: Standard Values of Some Terms of Worm and Gear (contd.)

Term	Formula
Outside diameter of worm	$d_o = d_w + 2a$

Throat diameter of gear	$d_t = d_g + 2a$
Outside diameter of gear	$d_{g_o} = d_t + 0.4775P_t$ for 1 or 2 threads $d_{g_o} = d_t + 0.3183P_t$ for 3 or 4 threads

* AGMA stands for American Gear Manufacturers Association

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.***

Prob. 7.1: Draw top view and side view with both simplified and schematic thread symbols of the connector as shown in Fig. P7.1.

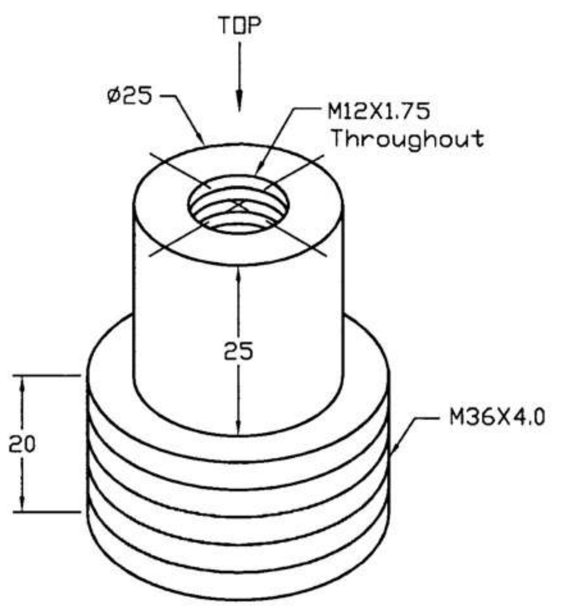


Fig. P7.1

Prob. 7.2: Draw top view and front sectional view with simplified and schematic thread symbols of the lock nut as shown in Fig. P7.2.

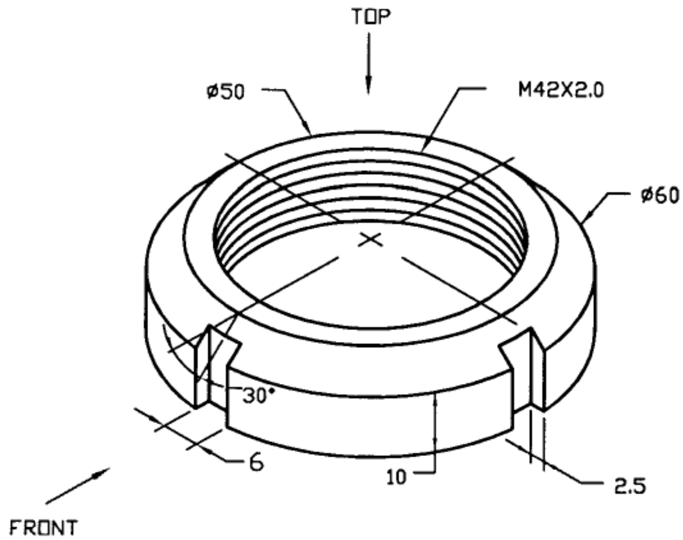


Fig. P7.2

Prob. 7.3: Draw two necessary views of the bolt as shown in Fig. P7.3. Provide both the simplified and schematic thread symbols.

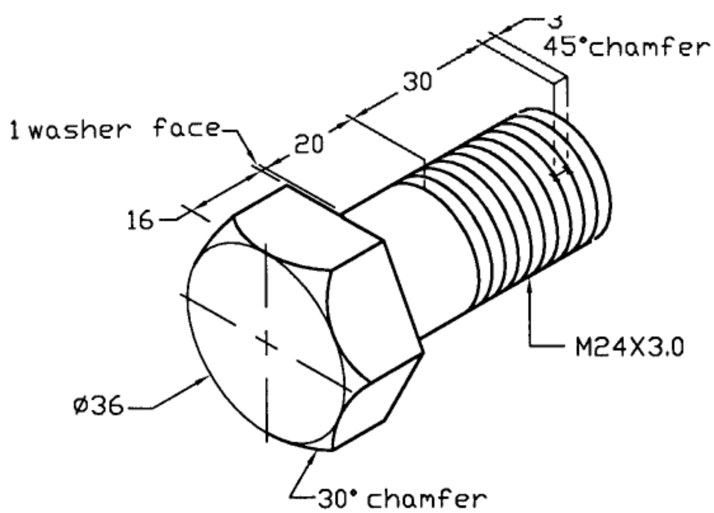


Fig. P7.3

Prob. 7.4: Draw top view, front sectional view with schematic thread symbol and right hand side view of the distributor as shown in Fig. P7.4.

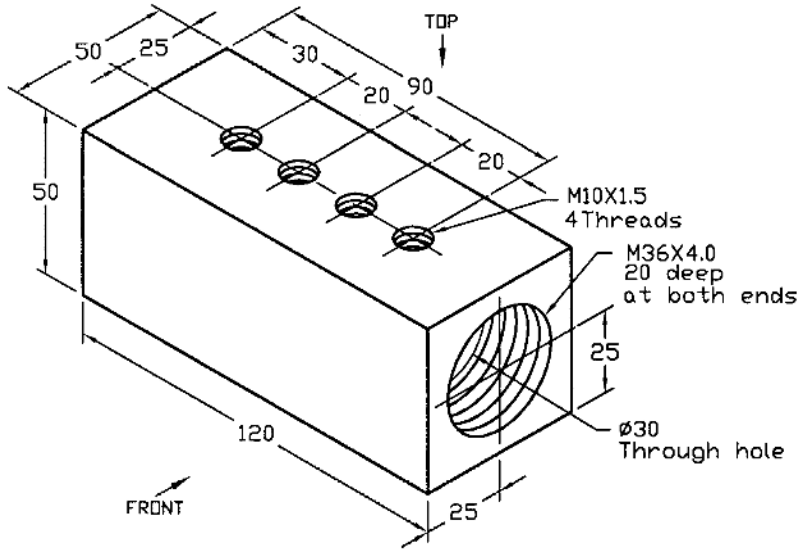


Fig. P7.4

Prob. 7.5: Draw top view and front sectional view with schematic thread symbol of the collar seal as shown in Fig. P7.5.

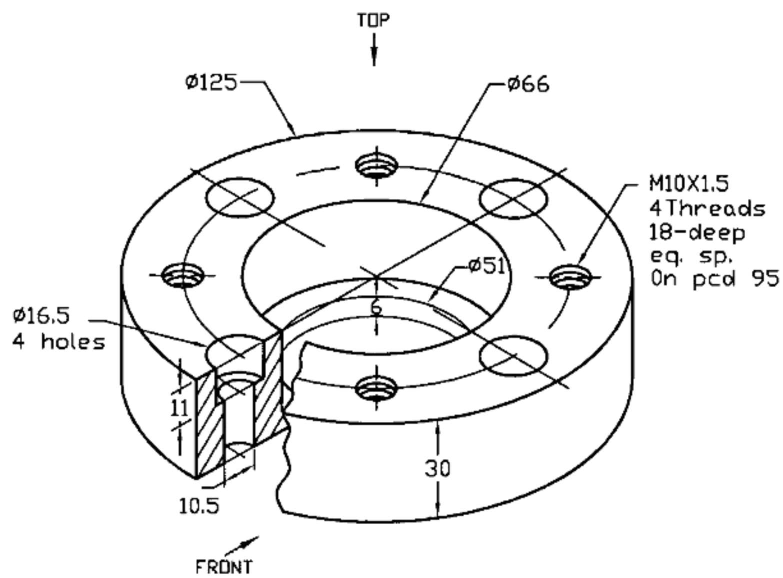


Fig. P7.5

Prob. 7.6 Draw top view and any suitable sectional view with simplified thread symbol of the bearing support as shown in Fig. P7.6.

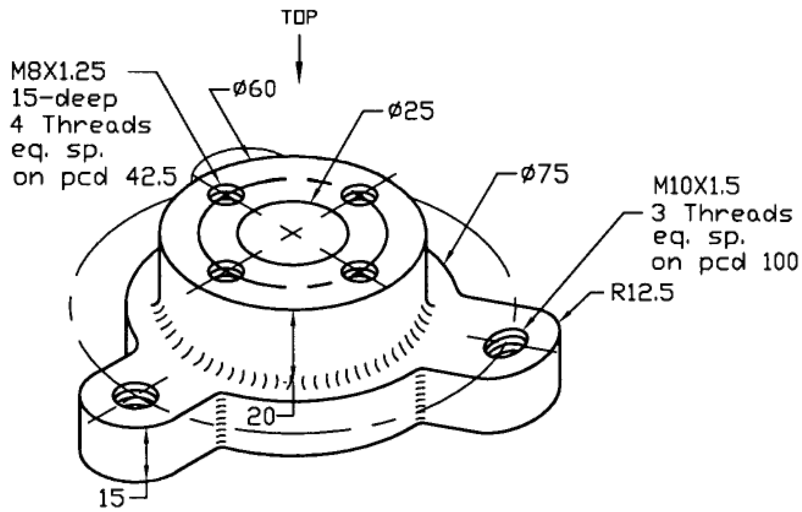


Fig. P7.6

Prob. 7.7: Draw front view, partial top view and right auxiliary view with schematic thread symbol of the angle bracket as shown in Fig. P7.7.

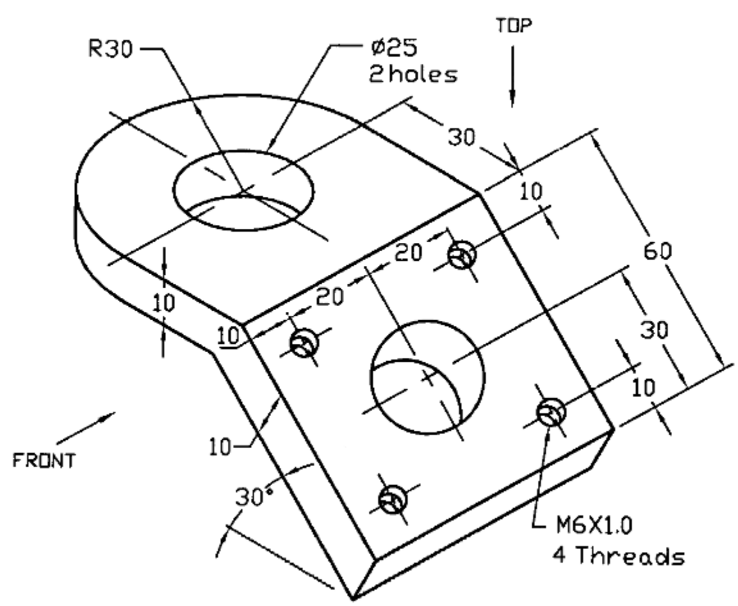


Fig. P7.7

Prob. 7.8: Draw top and front views with schematic and simplified thread symbols of the end plate as shown in Fig. P7.8.

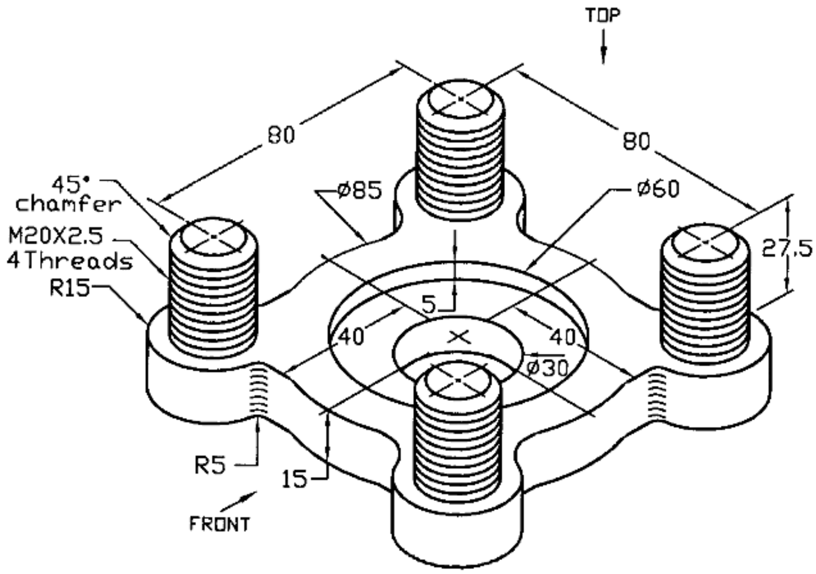


Fig. P7.8

Prob. 7.9: Draw two views of a spur gear of which one is sectional to best represent it with the following specifications:

- Pitch circle diameter = 200
- Number of teeth = 40
- Module = 5
- Pressure Angle = 20^0
- Addendum = 5
- Whole depth = 11.25
- Face/rim width = 50
- Inside diameter of hub = 30
- Outside diameter of hub = 60
- Hub width = 60
- Inside diameter of rim = 150
- Web thickness = 15

There is web in between the rim and hub and it is placed symmetrically with respect to the rim and hub widths. Fillets and rounds are 3. There is a keyway of width 8 and depth 4 to fix a shaft. Material is AISI 1020 steel, as rolled.

Prob.7.10: Draw two necessary views of a straight bevel -gear of which one is sectional with the following specifications:

- Number of teeth = 20
- Pitch diameter = 100
- Module = 5
- Addendum = 5
- Clearance = 0.99

- Working depth = 10
- Face width = 24
- Shaft diameter = 22
- Web thickness = 15
- Hub diameter = 40
- Hub length = 29
- Keyway = 6 wide x 3 dp.
- Pitch cone angle = 45^0
- Pressure angle = 20^0

Choose any other necessary dimensions required for your drawing. The material is gray cast iron (ASTM 30).

Prob. 7.11: Draw the necessary views of a worm and gear in a worm gear–set with the specifications as provided below.

- Number of teeth of worm = 3
- Number of teeth of gear = 30
- Pitch diameter of worm = 75
- Pitch diameter of gear = 240
- Face width of gear = 38
- Normal pressure angle = 20^0
- Lead angle = 17.8^0
- Module = 8
- Axial pitch/Transverse circular pitch = 25.13
- Addendum = 9.26
- Dedendum = 9.26
- Shaft diameter of gear = 30
- Hub diameter of gear = 60
- Length of hub = 50
- Keyway = 8 wide x 4 dp.
- Shaft diameter of worm = 30
- Web thickness = 20
- Throat radius = 28.24
- Rim radius = 62.63

The shafts of worm and gear are set at right angle to each other. Worm material is hardened steel and gear material is bronze.

Prob. 7.12: An object, which is fabricated by welding, is shown in Fig. P7.12. Draw the top, front sectional and right side views of the object showing the required welding symbols.

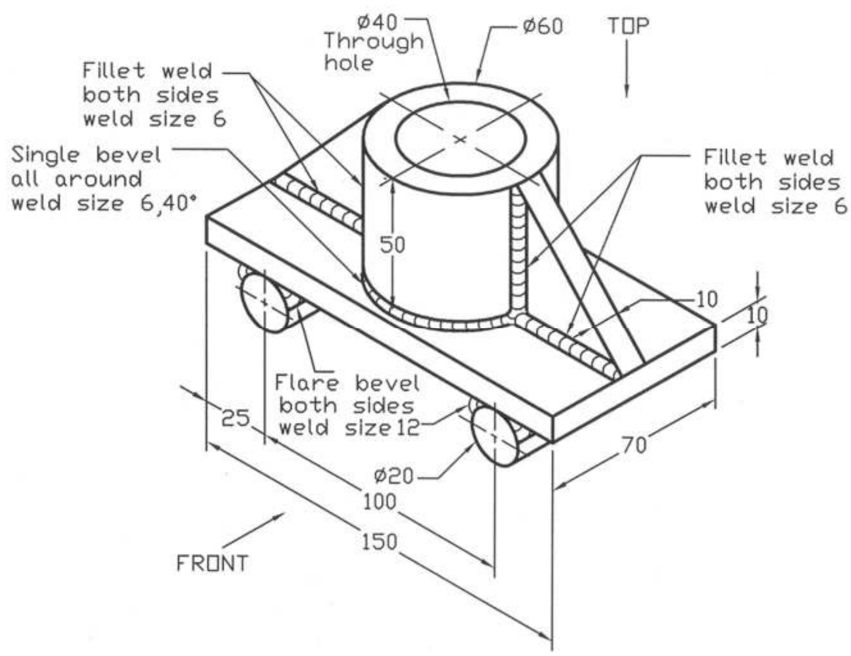
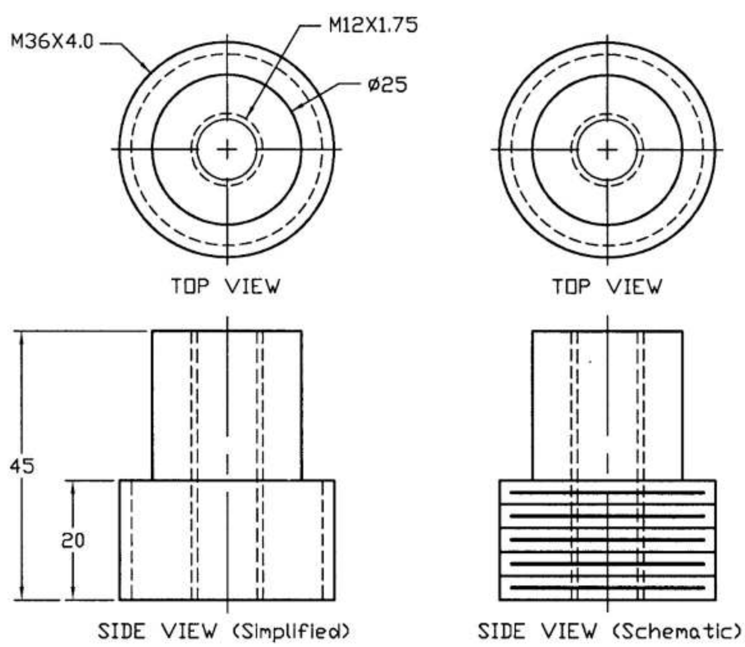
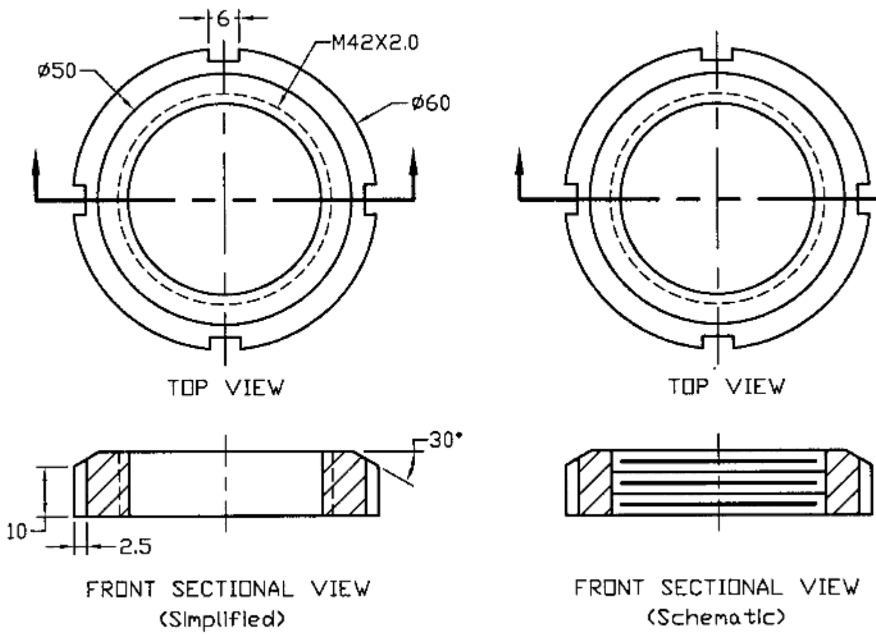


Fig. P7.12

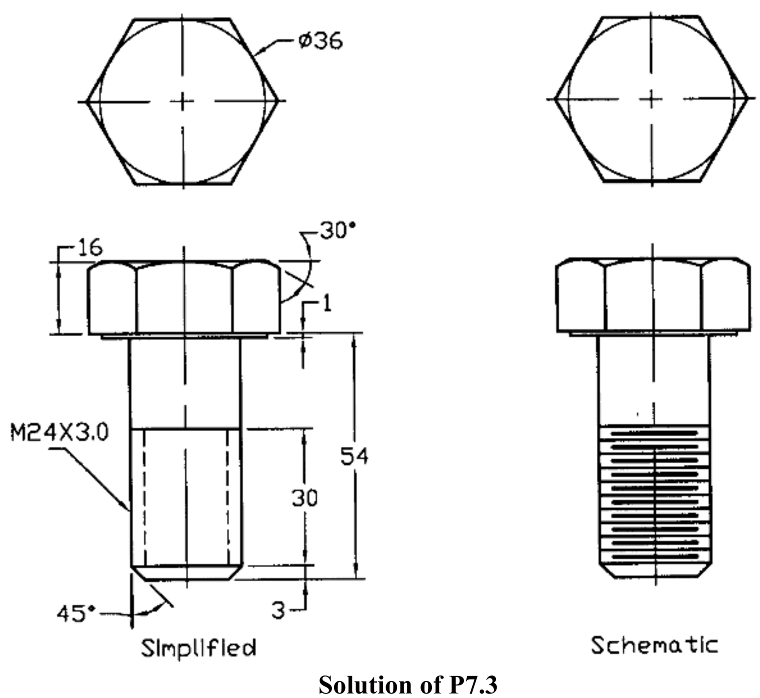
Solutions for Example problems



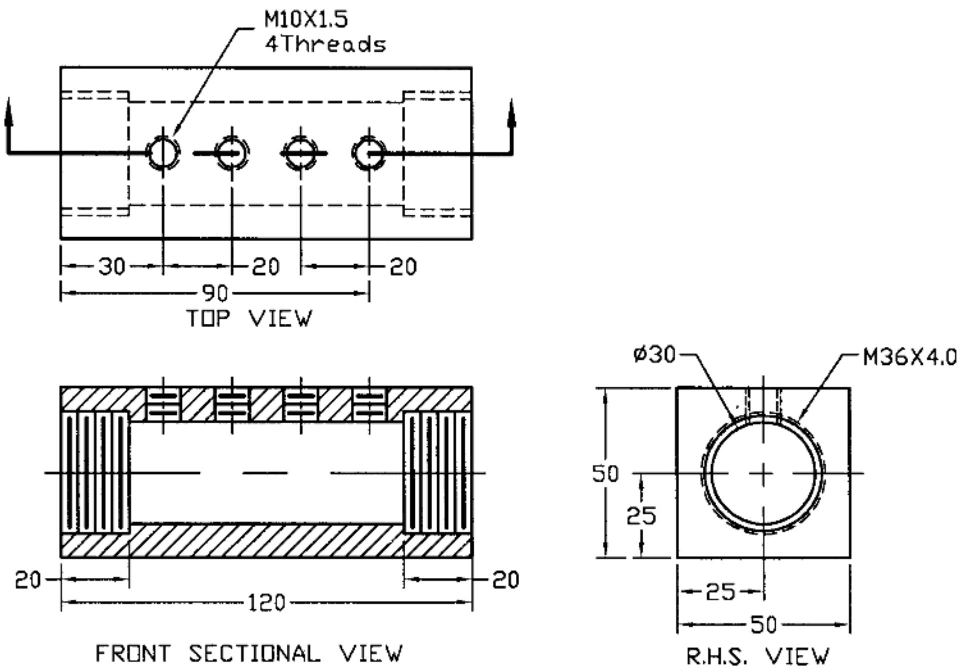
Solution of P7.1



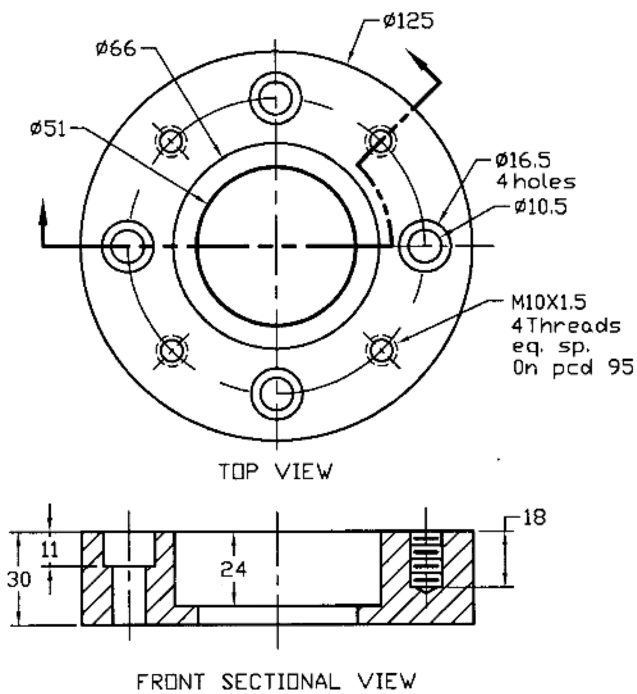
Solution of P7.2



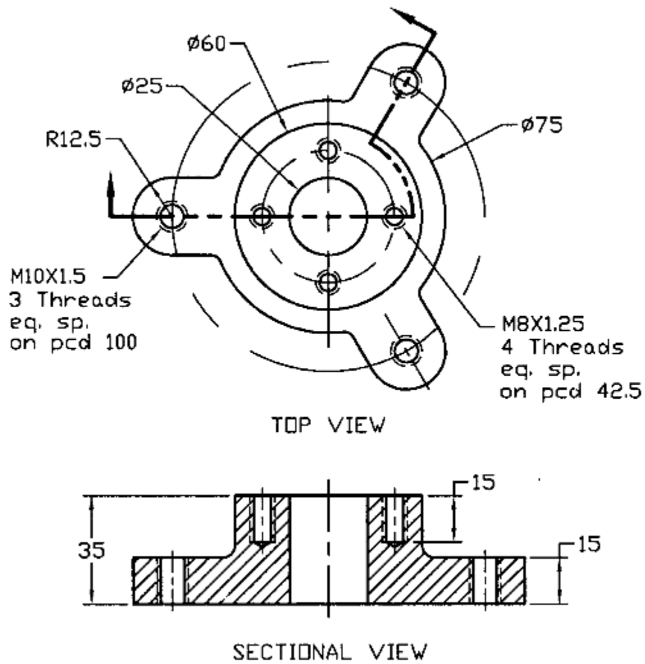
Solution of P7.3



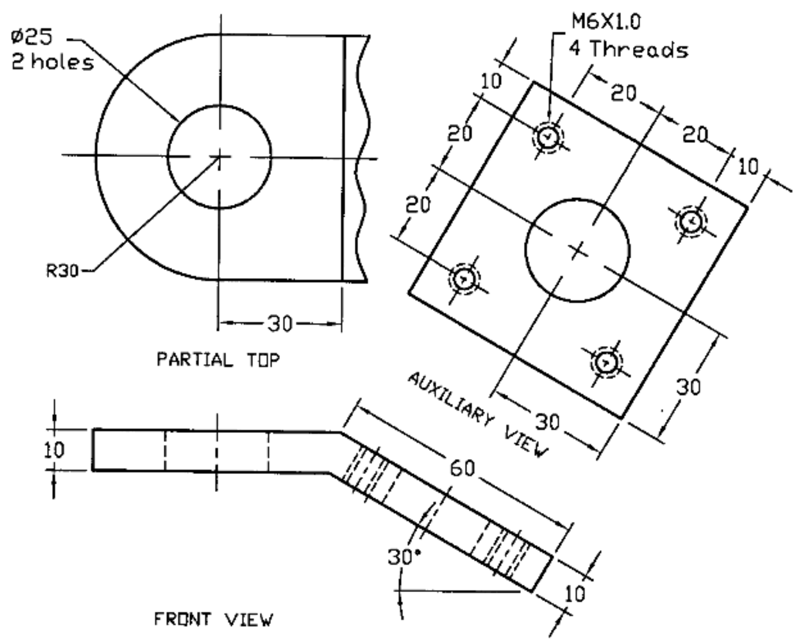
Solution of P7.4



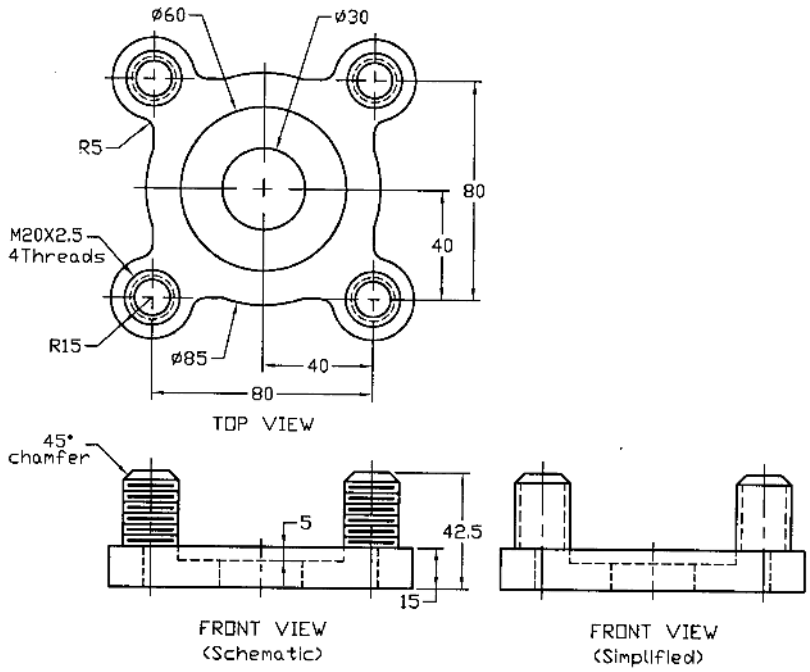
Solution of P7.5



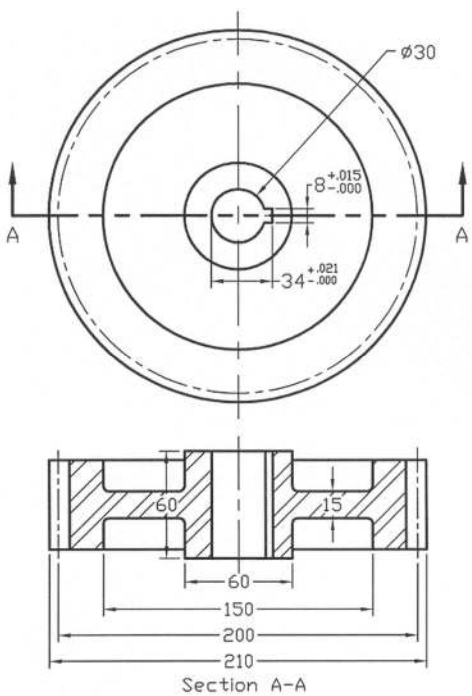
Solution of P7.6



Solution of P7.7

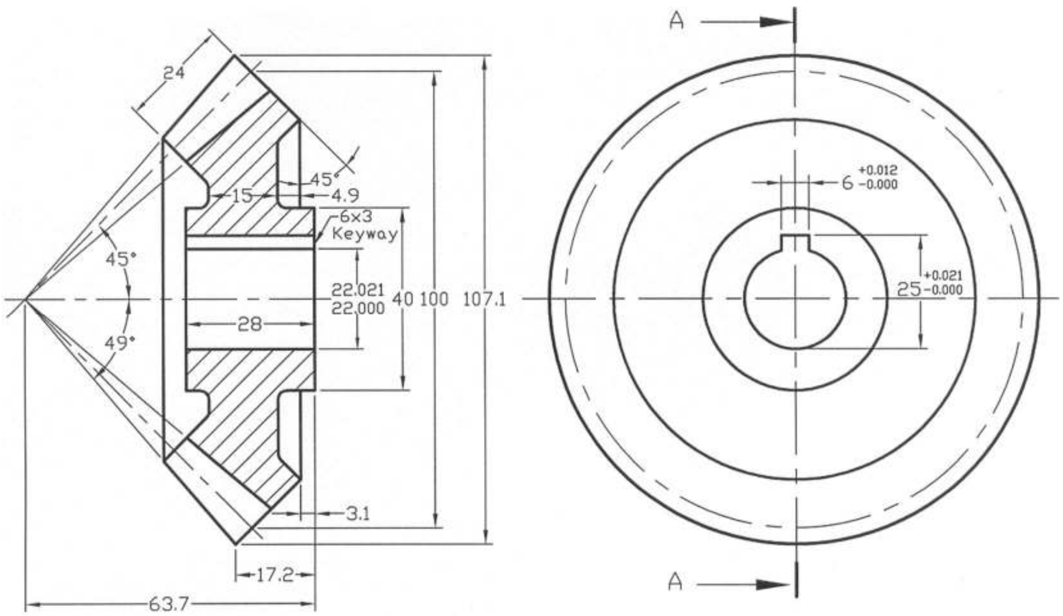


Solution of P7.8



Data of Spur Gear (Involute Profile)	
Pitch diameter	200
Module	5
No. of teeth	40
Pressure angle	20^0
Addendum	5
Whole depth	11.25

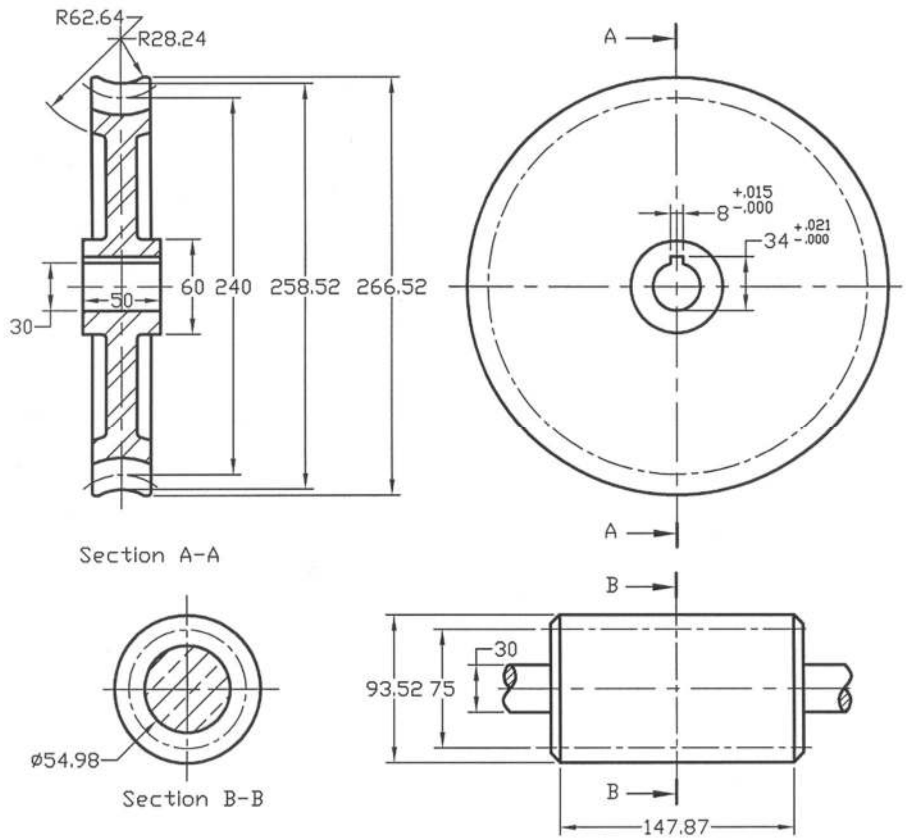
Solution of P 7.9



Section A-A

Data of Strait Bevel-Gear (Involute Profile)	
Pitch diameter	100
Module	5
No. of teeth	20
Pressure angle	20°
Addendum	5
Whole depth	10.99

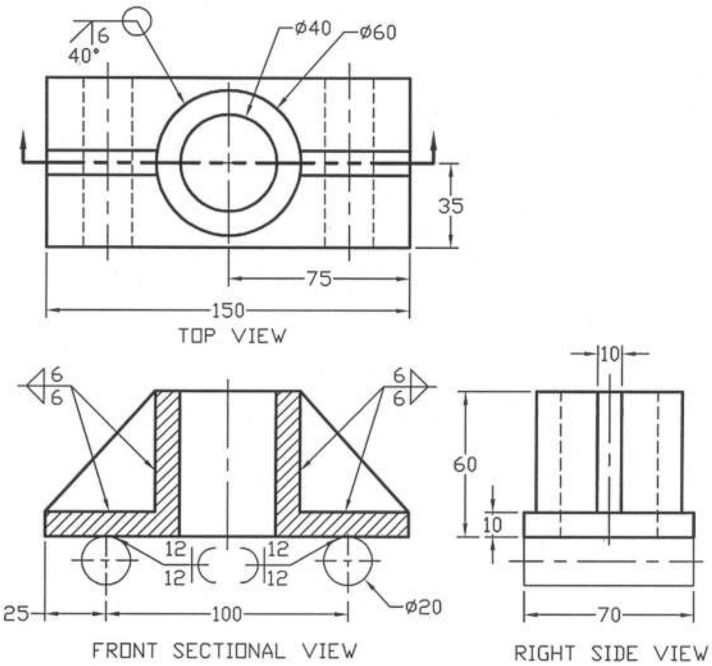
Solution of P 7.10



Data of Gear	
Pitch diameter	240
Module	8
No. of teeth	30
Pressure angle	20⁰
Addendum	9.26
Dedendum	9.26
Circular pitch	25.13
Helix angle	17.8⁰

Data of Worm	
Pitch diameter	75
Module	8
No. of teeth	3
Pressure angle	20⁰
Addendum	9.26
Axial pitch	25.13
Lead angle	17.8⁰

Solution of P 7.11



Solution of P 7.12

Problems

Prob. 7.13: Draw top view and front view with simplified and schematic thread symbols of the plug as shown in Fig. P7.13.

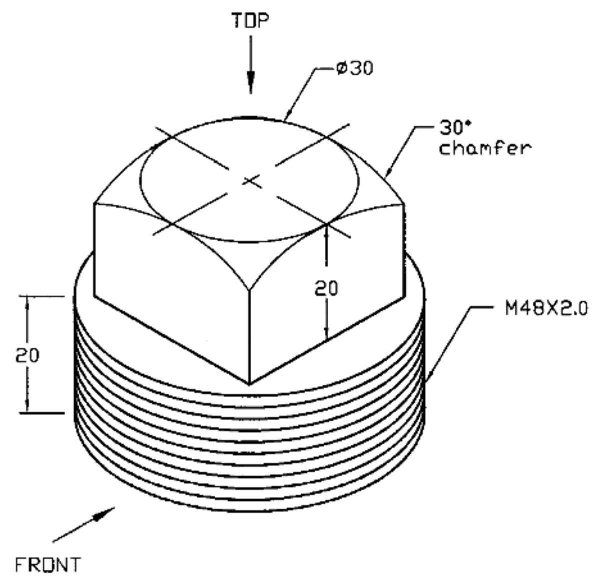


Fig. P7.13

Prob. 7.14: Draw top view and front sectional view with simplified and schematic thread symbols of the nut as shown in Fig. P7.14.

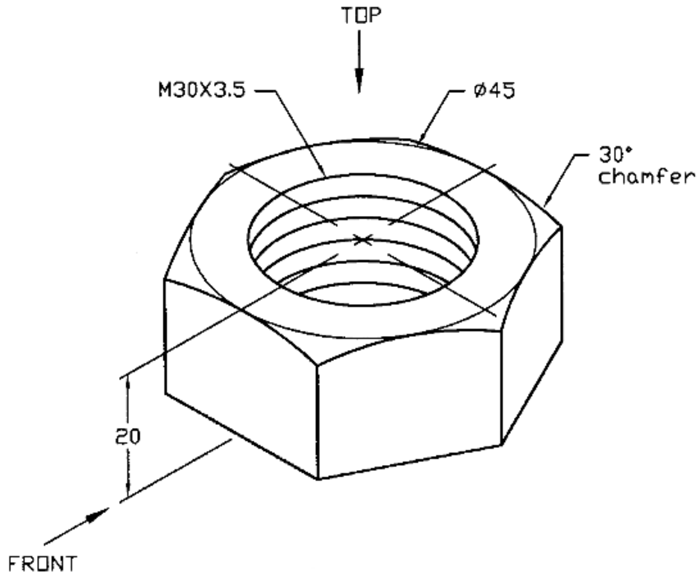


Fig. P7.14

Prob. 7.15: Draw top view and any suitable side view with simplified and schematic thread symbols of the nipple as shown in Fig. P7.15.

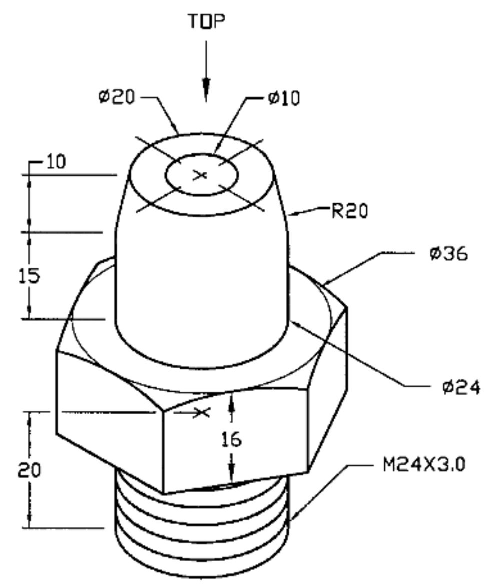


Fig. P7.15

Prob. 7.16: Draw top view and front sectional view with simplified and schematic thread symbols of the abutment collar as shown in Fig. P7.16.

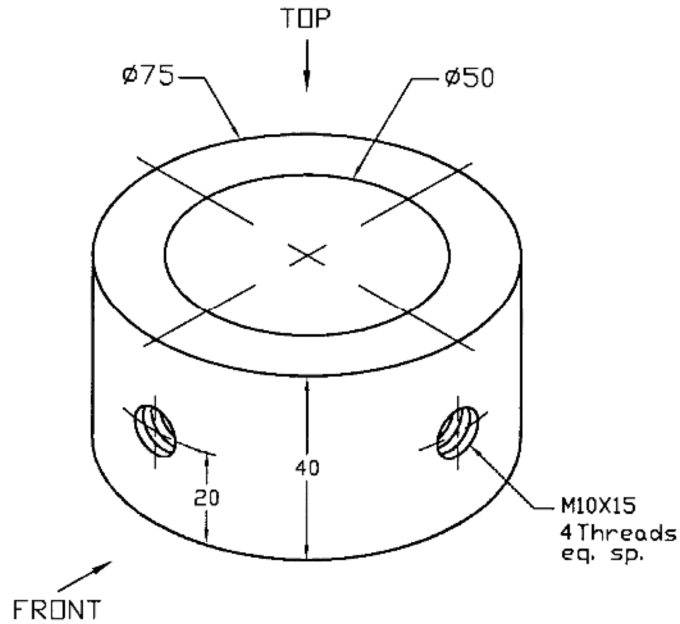


Fig. P7.16

Prob. 7.17: Draw suitable views of the 90^0 elbow as shown in Fig. P7.17 using both simplified and schematic thread symbols.

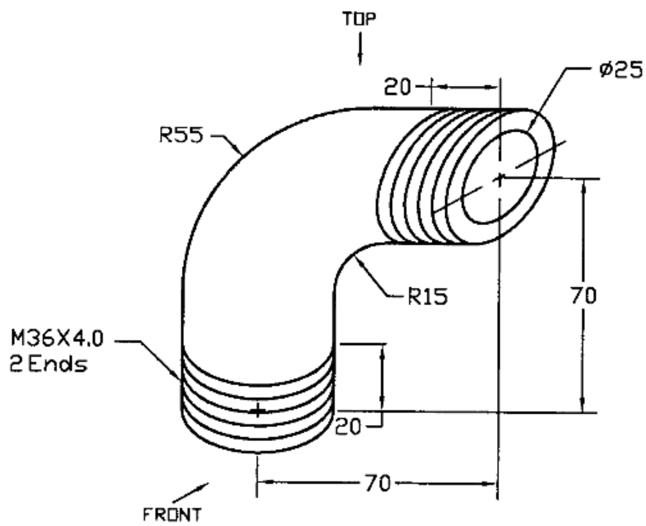


Fig. P7.17

Prob. 7.18: Draw top view and front sectional view with simplified and schematic thread symbols of the bearing housing as shown in Fig. P7.18.

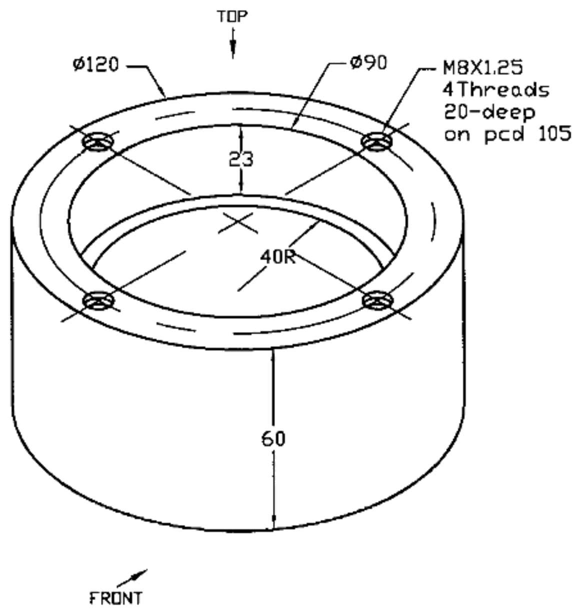


Fig. P7.18

Prob. 7.19: Draw suitable views of the holder block as shown in Fig. P7.19.

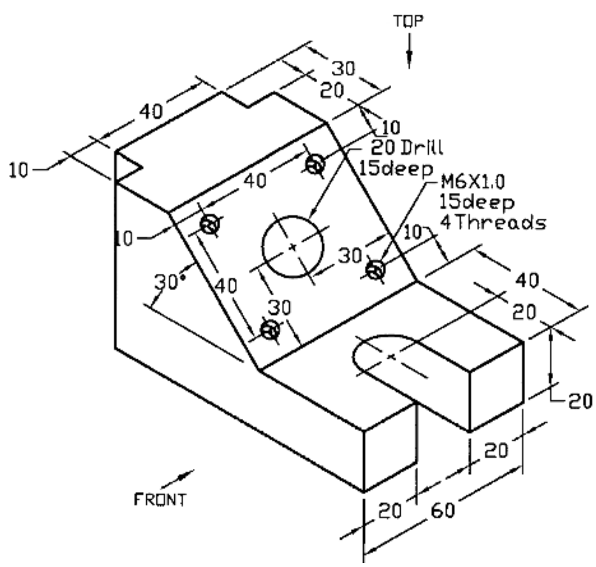


Fig. P7.19

Prob. 7.20: Draw suitable views of the holder block as shown in Fig. P7.20.

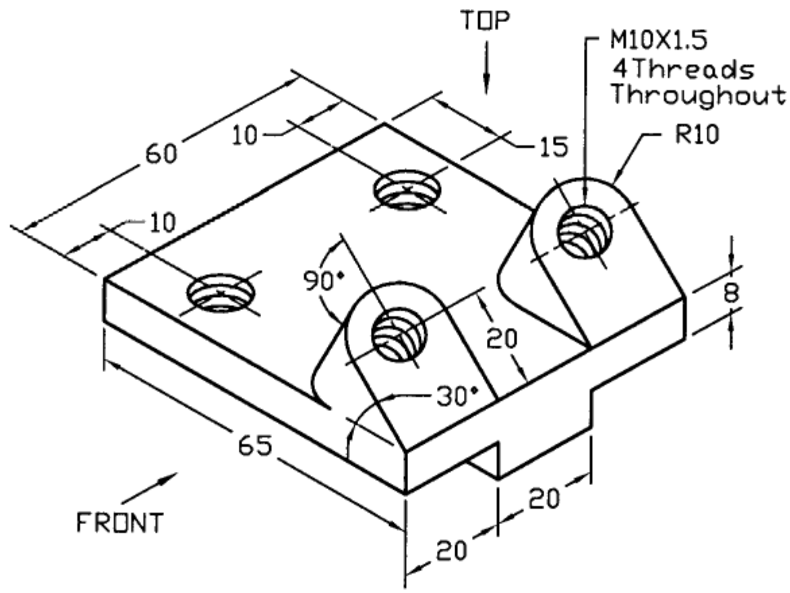


Fig. P7.20

Prob. 7.21: For B-type belt, draw two views of a V-grooved pulley of which one is sectional to best represent it in accordance with the following specifications.

- Pitch diameter of pulley = 125
- Shaft diameter = 25
- Hub diameter = 45
- Hub length = 35
- Keyway = 6 wide x 3 dp.
- Web thickness = 8
- Inside diameter of rim = 85

The V-groove dimensions will be according to the Indian Standard (IS). The material is AISI 1030 steel, as rolled. The hub will be placed symmetrically with respect to the rim.

Prob. 7.22: Draw two views of a spur gear of which one is sectional, to best represent it with the following specifications:

- Pitch circle diameter = 160
- Number of teeth = 40
- Module = 4
- Pressure Angle = 20^{0}
- Addendum = 4
- Whole depth = 9
- Face/rim width = 40
- Inside diameter of hub = 25
- Outside diameter of hub = 50
- Hub length = 40
- Inside diameter of rim = 118

- Web thickness = 12

There is web in between the rim and hub and it is placed symmetrically with respect to the rim and hub widths. Fillets and rounds are 3. There is a keyway of width 6 and depth 3 to fix a shaft. Provide necessary tolerances according to ISO. Material is gray cast iron (ASTM 30).

Prob.7.23: Draw two views of a straight bevel gear of which one is sectional with the following specifications:

- Number of teeth = 24
- Pitch diameter = 120
- Module = 5
- Addendum = 5
- Clearance = 0.99
- Working depth = 10
- Face width = 30
- Shaft diameter = 28
- Web thickness = 16
- Hub diameter = 50
- Hub length = 40
- Keyway = 6 wide x 3 dp.
- Pitch cone angle = 42^0
- Pressure angle = 20^0

Choose any other dimensions required for your drawing. The material is AISI 1020 steel, as rolled.

Prob.7.24: Draw the necessary views of a worm and gear in the worm gear–set with the specifications as given below:

- Number of teeth of worm = 3
- Number of teeth of gear = 30
- Pitch diameter of worm = 57
- Pitch diameter of gear = 180
- Face width of gear = 28
- Normal pressure angle = 20^0
- Lead angle = 23^0
- Module = 6
- Axial pitch/Transverse circular pitch = 18.85
- Addendum = 6.943
- Dedendum = 6.943
- Throat radius = 21.56
- Rim radius = 47.35

The shafts of worm and gear are set at right angle to each other. Worm material is hardened steel and gear material is bronze. Choose the dimensions of the worm shaft, gear hub etc. that are necessary for your drawing. Keep provision for a woodruff keyway in the hub of the gear.

CHAPTER 8

INTERSECTIONS

8.1 *Introduction*

The concept of intersection plays an important role in many engineering applications. In the orthographic drawings lines of intersections between the various surfaces of different objects are common. Intersection of objects is called the locus of points of contact of the objects. The intersection of two planes is given by a straight line. The intersection between a plane and a cylinder gives either a circle or an ellipse. Some examples of intersections including simple geometric forms will be given in this chapter.

8.2 *Some Common Objects*

Before going to discuss about the intersection, it will be relevant to introduce some very common types of objects, which are often used in the intersection. In Figures 8.1, 8.2, 8.3 and 8.4 respectively, some common types of prisms, some common types of pyramids, some common types of cylinders and some common types of cones have been shown. In Figure 8.1, it can be seen that a right prism has faces and lateral surfaces perpendicular to the bases while an oblique prism has faces and lateral edges oblique to the bases.

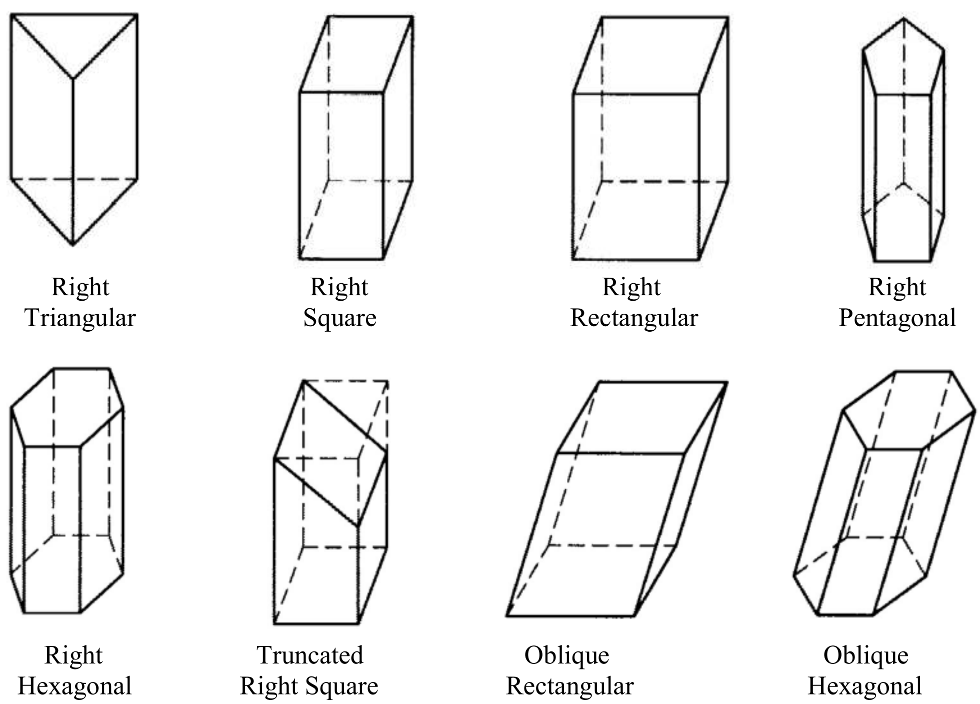


Figure 8.1: Some Common Types of Prisms

In case of right pyramids, cylinders and cones the axis is perpendicular to the base of them. However, when the pyramids, cylinders and the cones are oblique the axis is not perpendicular to the base. When the object is cut with a plane not parallel to the base, it is called truncated. If a cone or a pyramid is cut with a plane parallel to its base, it is called a frustum. In Figure 8.4 an oblique circular frustum is shown.

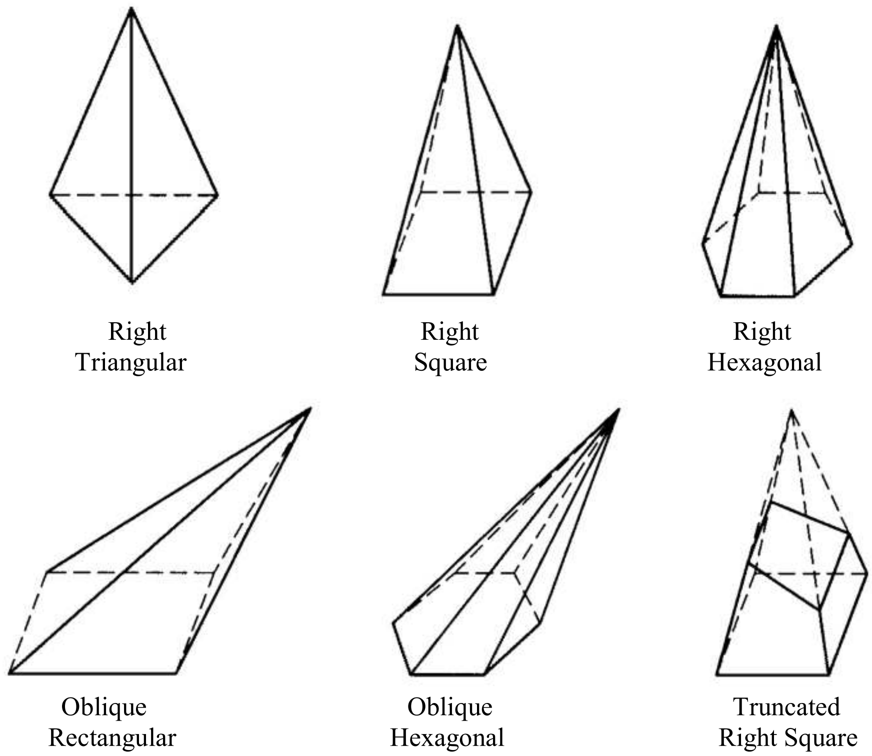


Figure 8.2: Some Common Types of Pyramids

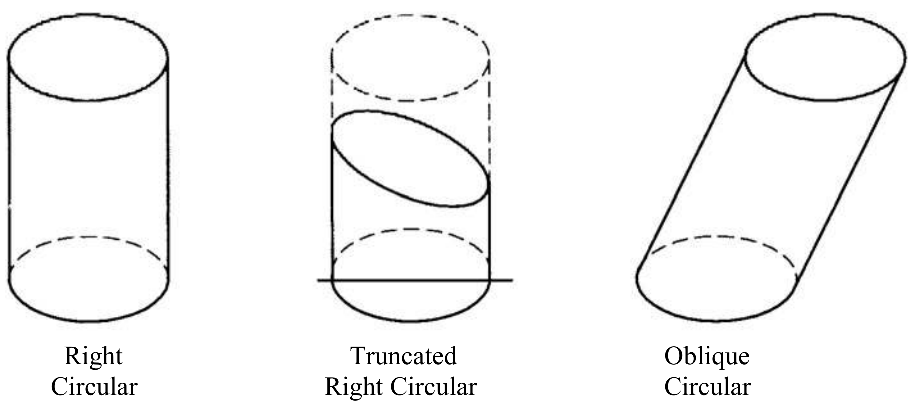


Figure 8.3: Some Common Types of Cylinders

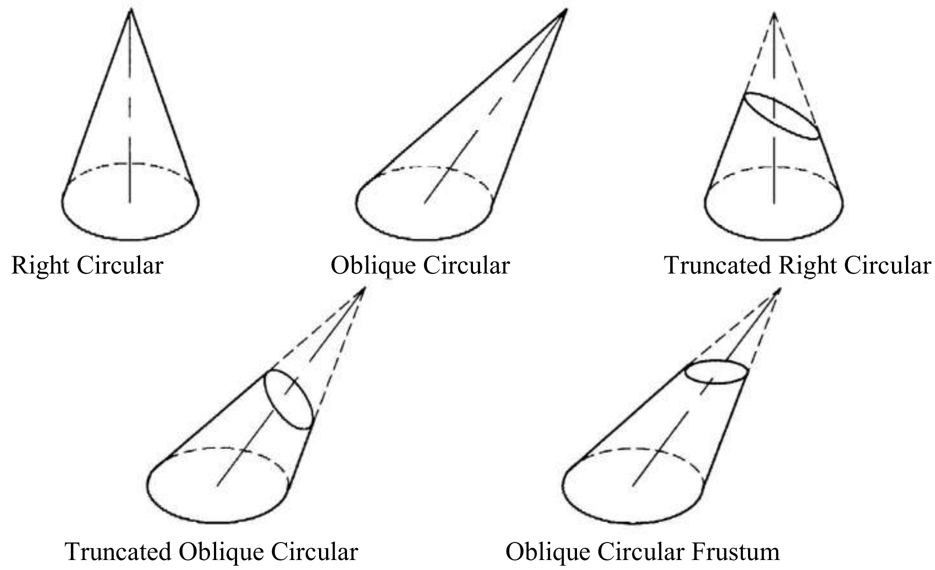


Figure 8.4: Some Common Types of Cones

8.3 ***Intersection of a Plane and a Cone***

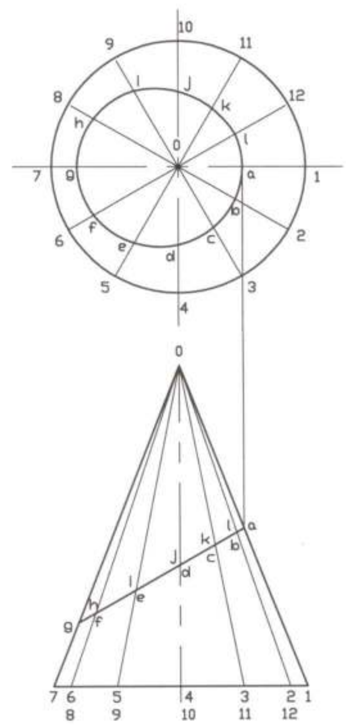


Figure 8.5: Intersection of a Plane and Right Circular Cone

In Figure 8.5 intersection of a plane and a right circular cone has been shown. The steps to determine the line of intersection are as follows:

- First the base circle of the cone in the top view is divided into a number of equal divisions represented by the lines 01, 02, 03, 04 etc.
- Then these division lines are drawn in the front view taking projection from the top view.
- Next in the front view from the intersecting points of the lines 01, 02, 03, 04 etc. with the plane at a, b, c, d etc. respectively, the projections are drawn in the top view.
- Now the intersecting points of the projection lines and the lines 01, 02, 03, 04 etc. in the top view are joined with the help of smooth curve.

8.4 ***Intersection of Two Prisms at Right Angles***

In Figure 8.6 intersection of two prisms at right angles has been presented. The steps of intersection are as follows:

- First the end view of the horizontal prism is marked with 1,2,3 and 4 and the corresponding points in the top view are marked.
- Next the projections are taken from the top and the end view to the front view to find the required lines of intersection.

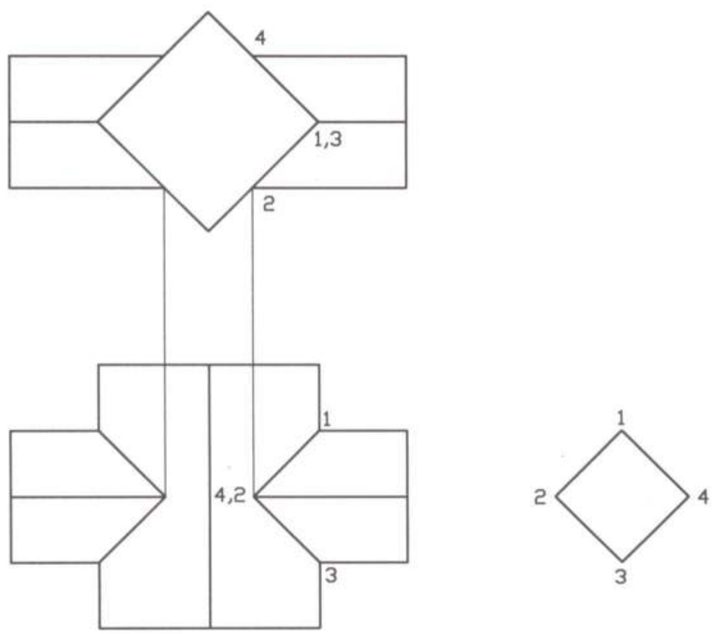


Figure 8.6: Intersection of Two Prisms at Right Angles

8.5 ***Intersection of Two Cylinders***

In Figure 8.7, the intersection of two circular cylinders at right angles has been shown. The intersection may be obtained in accordance with the following steps:

- In the right side view the circular part of the horizontal cylinder is divided into a reasonable number of equal spaces and they are marked with 1, 2, 3, 4 etc.
- Now in the top view on the circular part of the vertical cylinder, the projections are drawn corresponding to the points 1, 2, 3, 4 etc. in the right side view. They are respectively represented by 1, 2, 3, 4 etc.
- Next from the top and right side views, projections are drawn in the front view to find the required points of intersections, which are also marked with 1, 2, 3, 4 etc.
- Finally to complete the intersection, smooth curves are drawn through the points 1, 2, 3, 4 etc. in the front view with the help of irregular curve.

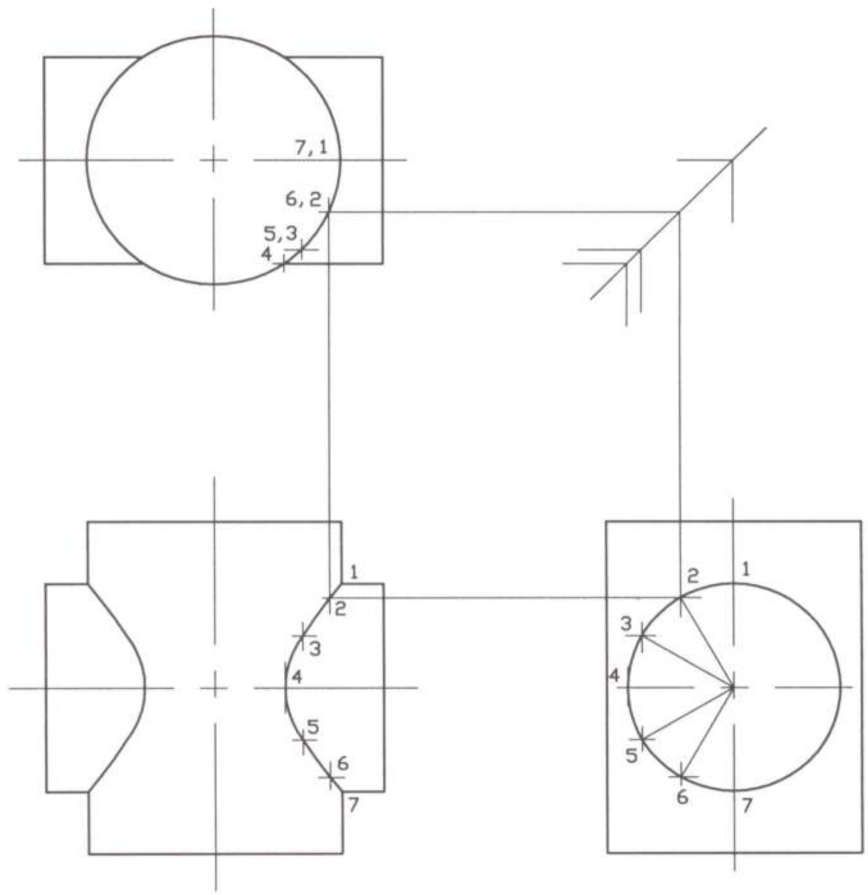


Figure 8.7: Intersection of Two Circular Cylinders at Right Angles

8.6 ***Intersection of a Cylinder and a Cone***

In Figure 8.8, the intersection of a circular cylinder and a right circular cone has been shown. The steps to draw the intersection are as follows:

- First in the top view equal divisions are made on the base circle of the cone. They are represented by the lines 01, 02, 03, 04 etc.
- Now in the front and right side views the corresponding lines 01, 02, 03, 04 etc. are drawn in respect of projections from the top view.
- In the right side view the lines 01, 02, 03, 04 etc. intersect with the circular part of the horizontal cylinder at the points a, b, c, d.
- Then the projections from the points a, b, c, d are drawn in both the top and the front views which intersect with the division lines 02, 03. These intersecting points are joined with the help of irregular curve.

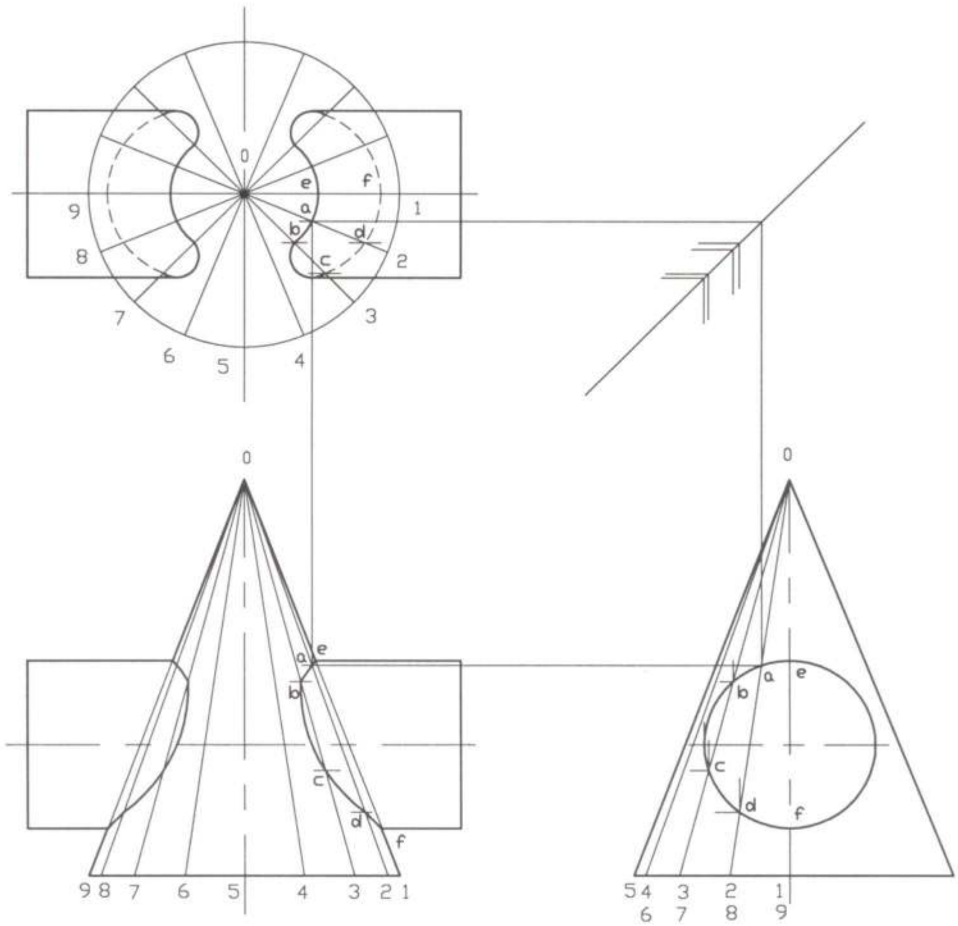


Figure 8.8: Intersection of a Circular Cylinder and a Right Circular Cone

8.7 ***Intersection of a Prism and a Cone***

Intersection of a prism and a right circular cone has been shown in Figure 8.9. The steps of drawing line of intersection are as follows:

- First the division lines are drawn in the right side view such that they pass through the edge points of the prism a, d, f which intersect the surface of the cone and the intermediate points b and e. The division lines are marked with 1, 2, 3 etc. at the base of the cone.
- Now the division lines are drawn in the top view taking projection from the division lines in the right side view and they are marked with 1, 2, 3 etc.
- Next the division lines in the front view are drawn with respect to the division lines in the top view.
- Then taking projection from the right side view the points of intersection in the top and front views are obtained and they are joined with the help of irregular curve.

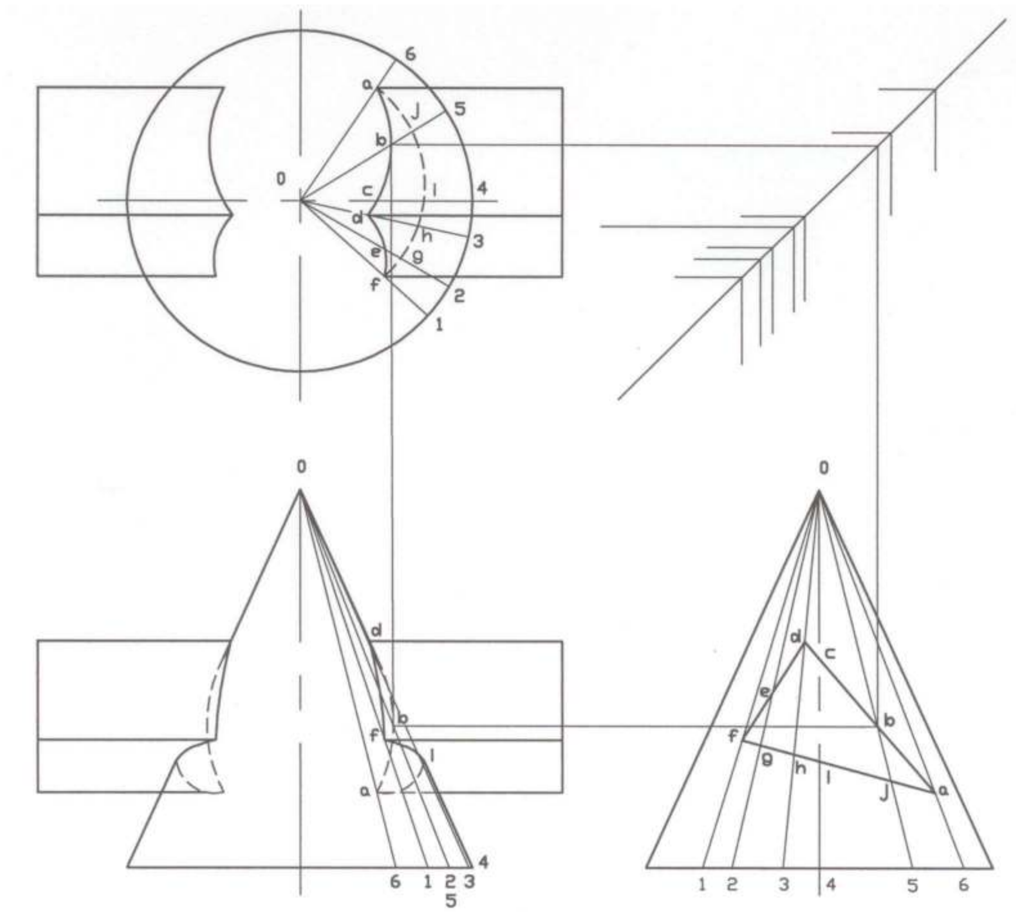


Figure 8.9: Intersection of a Prism and a Right Circular Cone

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.***

Prob. 8.1: Find the line of intersection of a right circular cone and a plane as shown in Fig. P8.1.

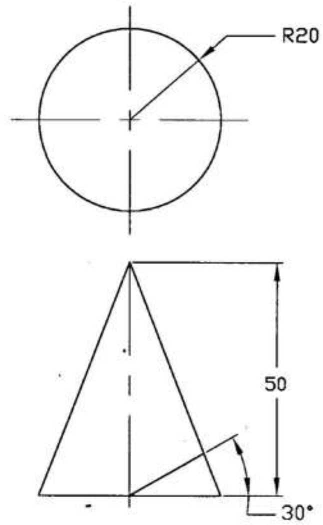


Fig. P 8.1

Prob. 8.2: Find the line of intersection of two circular cylinders at right angles to each other as shown in Fig. P8.2.

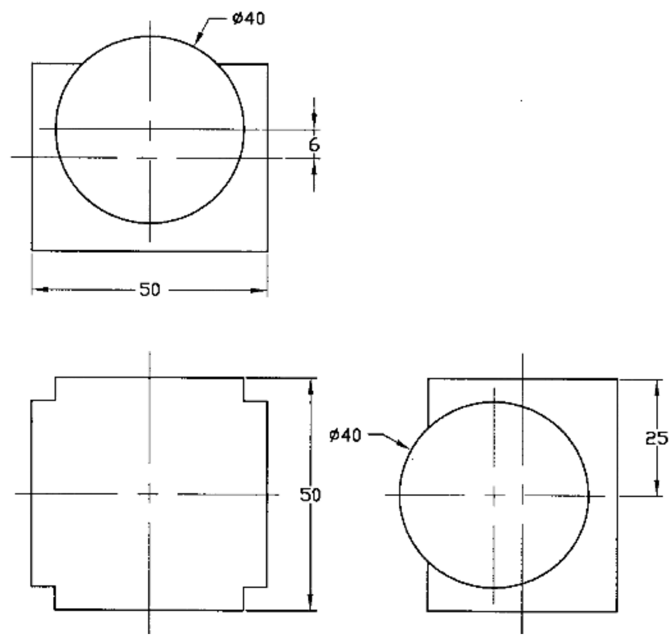


Fig. P8.2

Prob. 8.3: Find the line of intersection of the two cylinders as shown in Fig. P8.3.

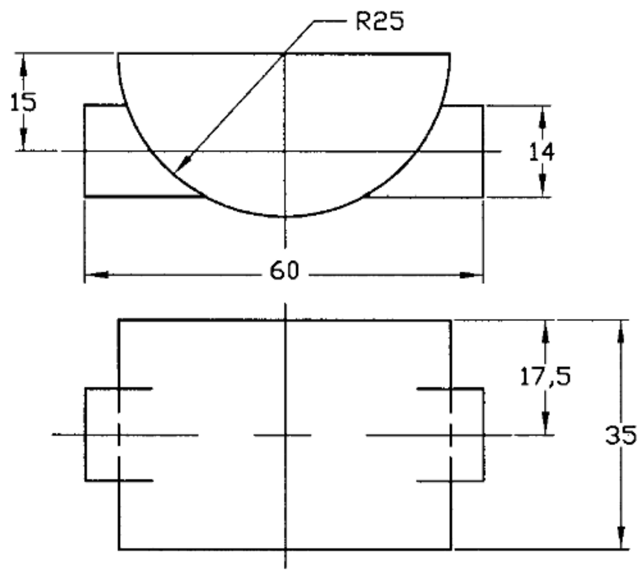


Fig. P8.3

Prob. 8.4: Find the line of intersection of a circular cylinder and a triangular prism as shown in Fig. P8.4.

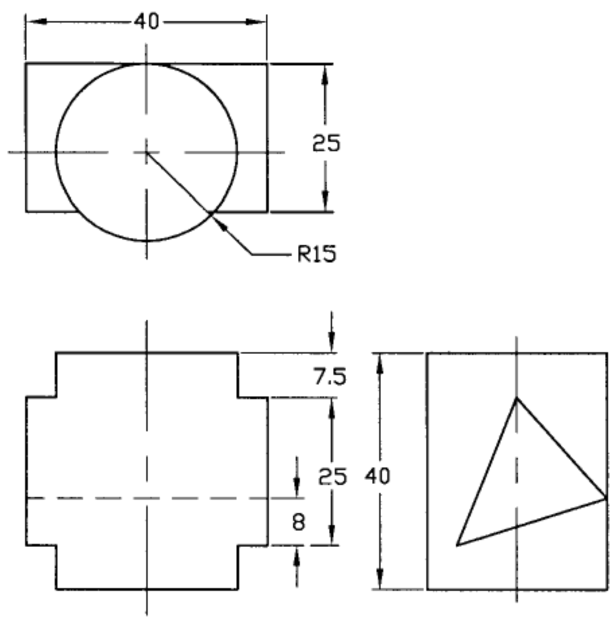


Fig. P8.4

Prob. 8.5: Find the line of intersection of a hexagonal prism and a square prism at right angle to each other as shown in Fig. P8.5.

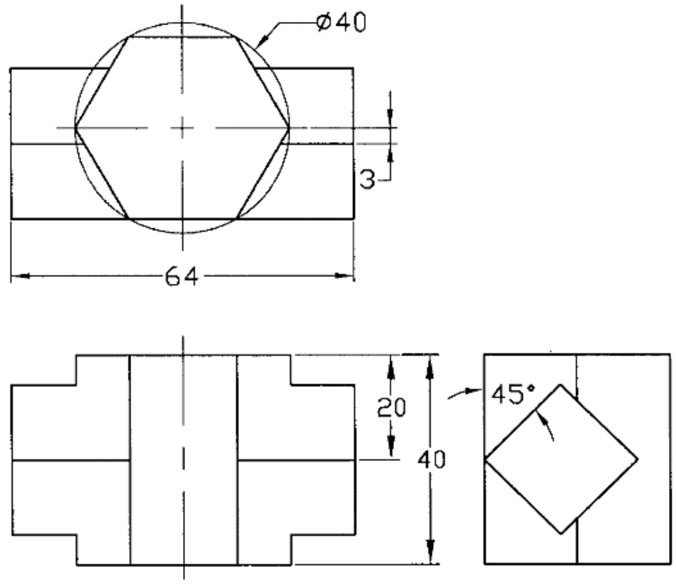


Fig. P8.5

Prob. 8.6: Find the line of intersection of a rectangular prism and a triangular prism at right angle to each other as shown in Fig. P8.6.

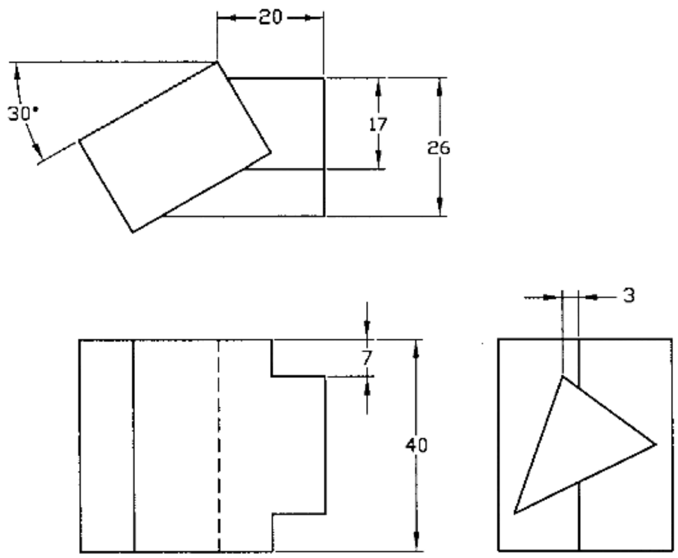


Fig. P8.6

Prob. 8.7: Find the line of intersection of two square prisms at right angle to each other as shown in Fig. P8.7.

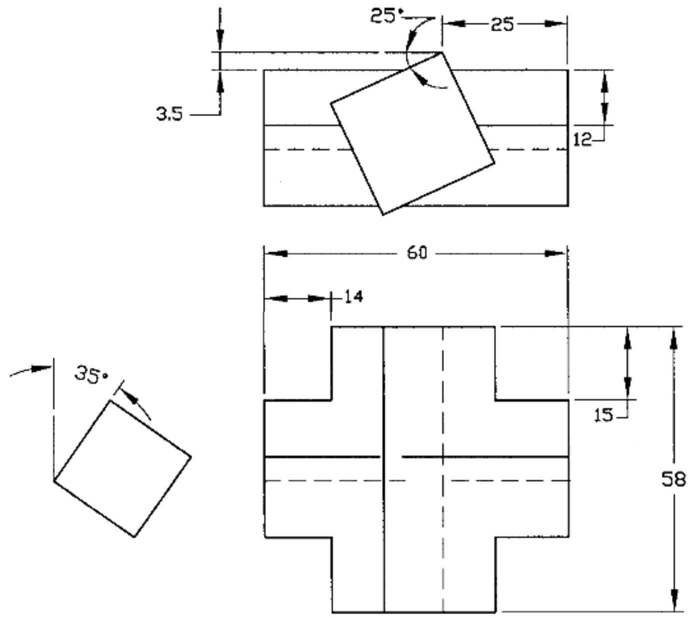


Fig. P8.7

Prob. 8.8: Find the line of intersection of a hexagonal prism and a triangular prism whose axes are at 45^0 angle to each other as shown in Fig. P8.8.

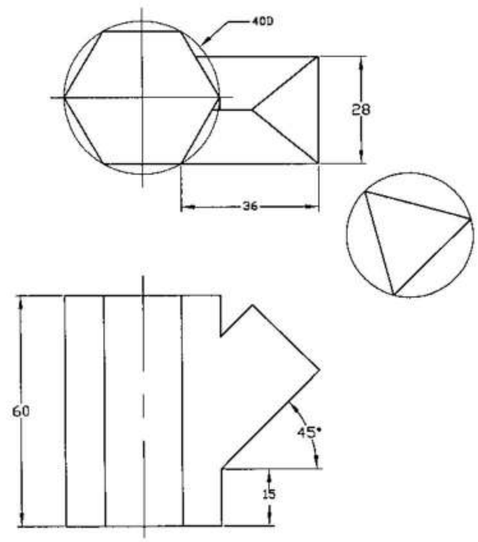
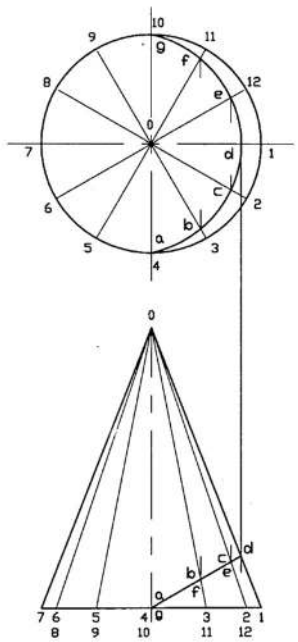
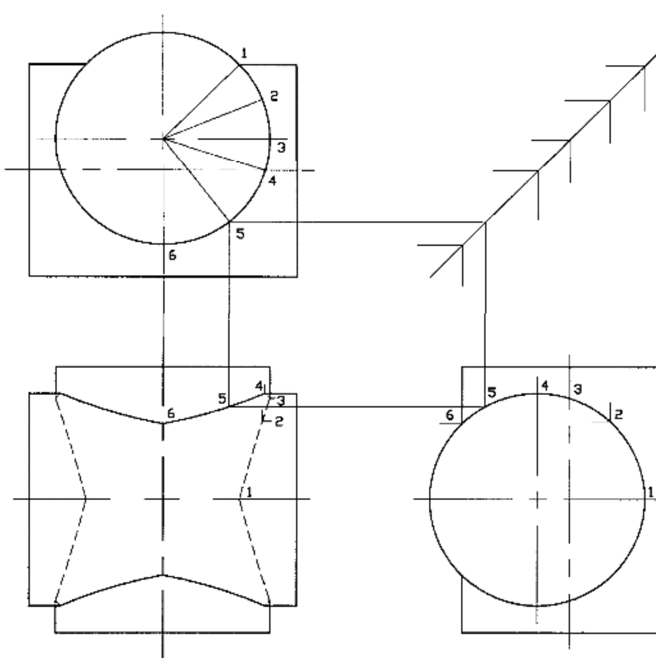


Fig. P8.8

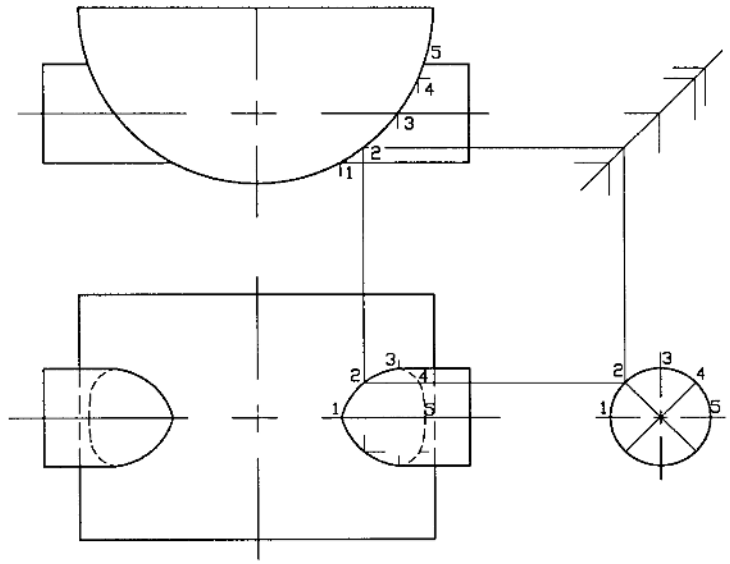
Solutions for Example Problems



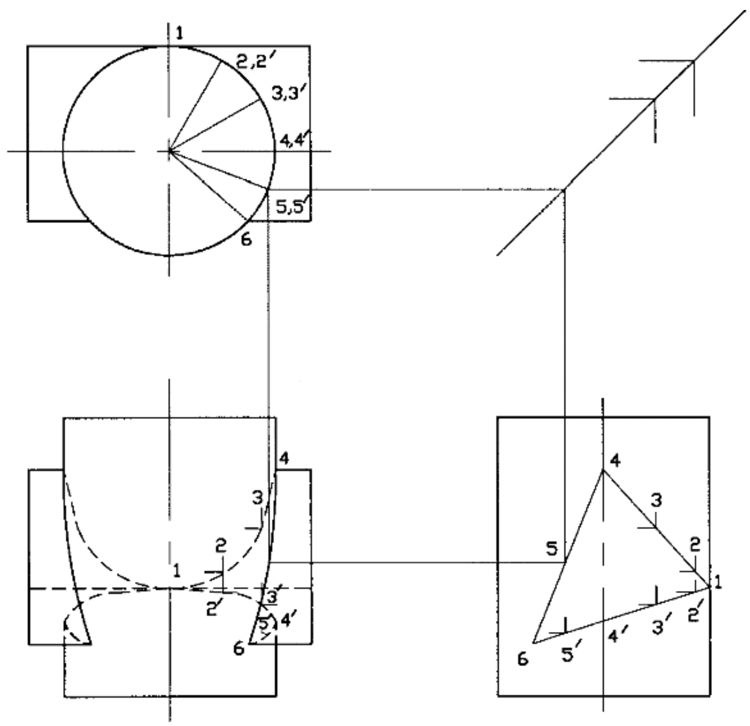
Solution of P8.1



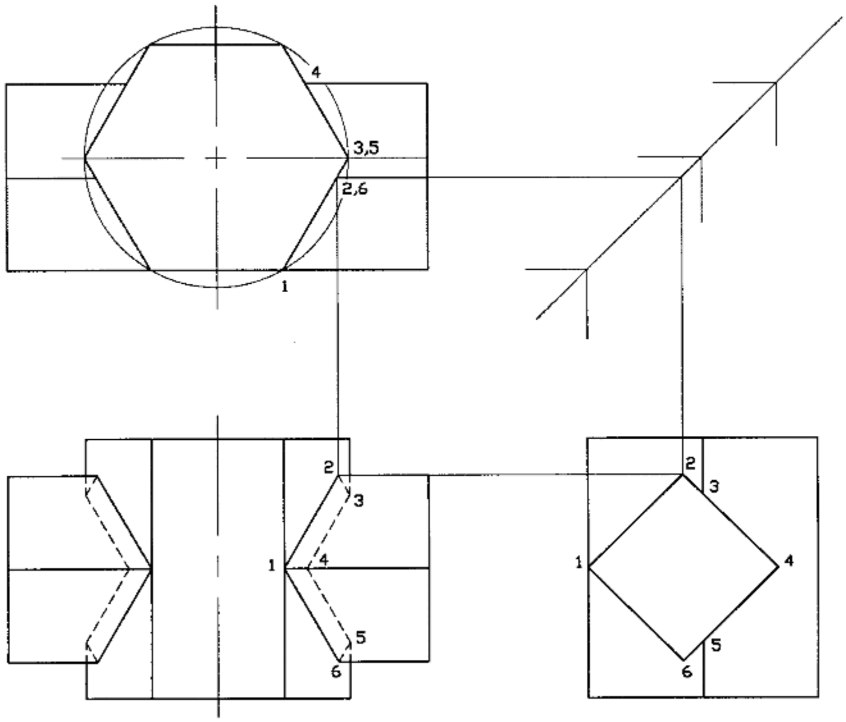
Solution of P8.2



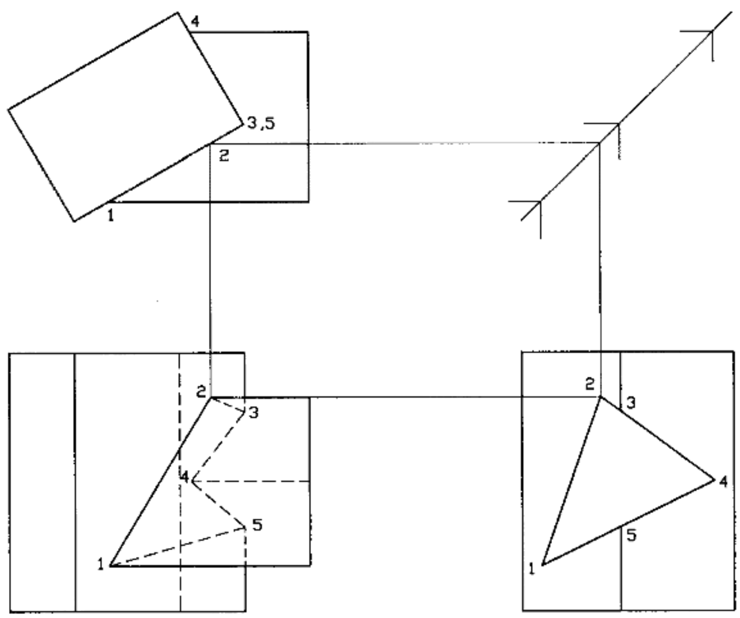
Solution of P8.3



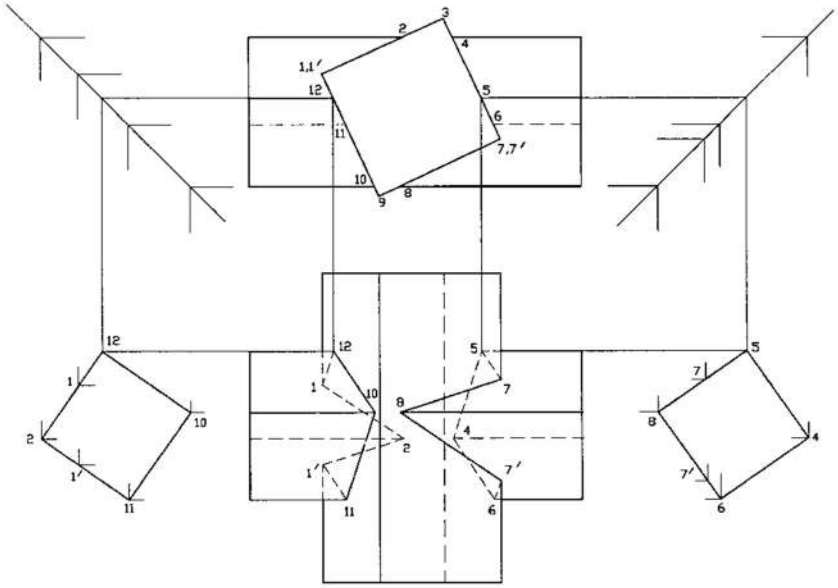
Solution of P8.4



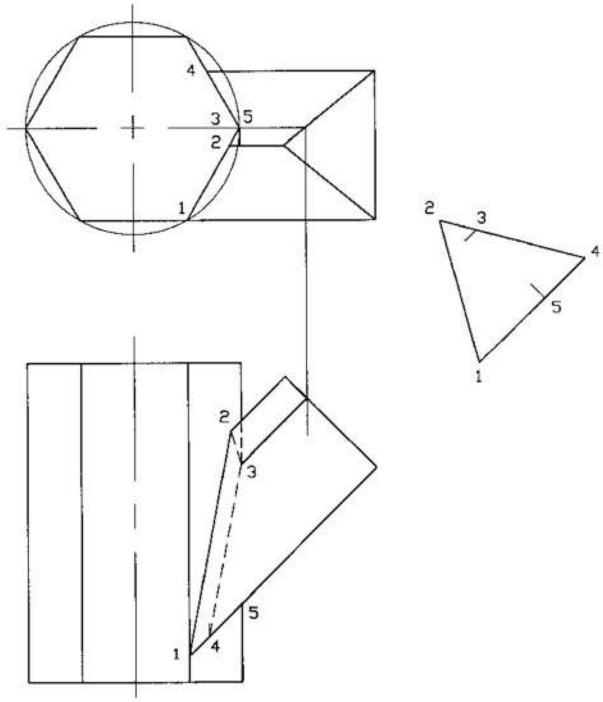
Solution of P8.5



Solution of P8.6



Solution of P8.7



Solution of P8.8

Problems

Prob. 8.9- 8.12: Determine the line of intersection between two prisms as shown in Fig. P8.9 to P8.12.

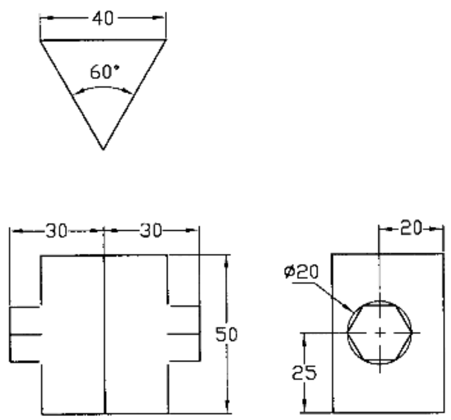


Fig. P8.9

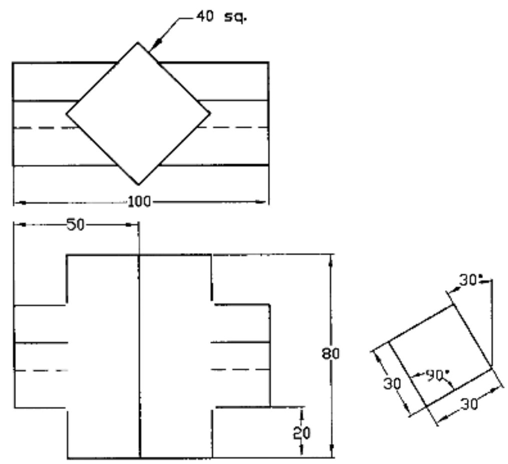


Fig. P8.10

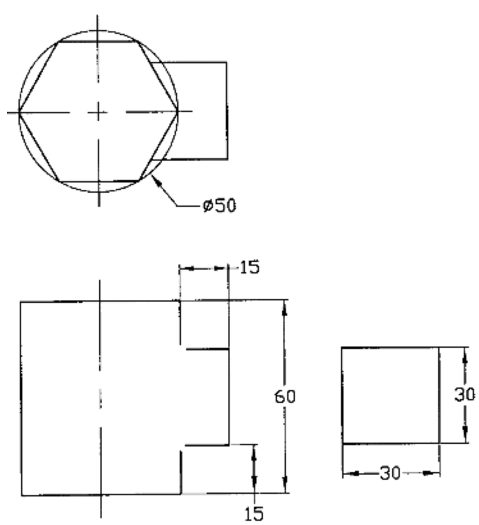


Fig. P8.11

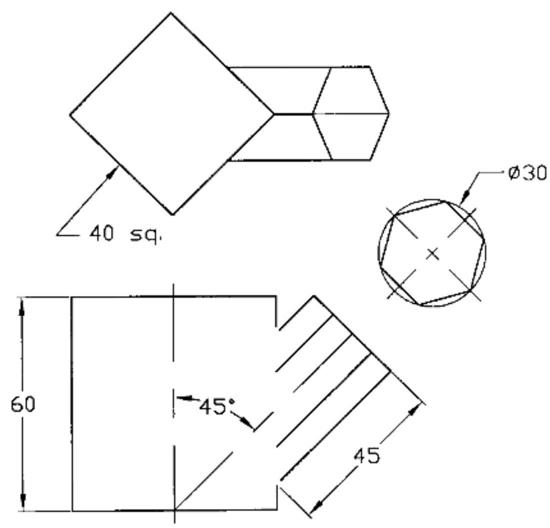


Fig. P8.12

Prob. 8.13 – 8.16: Determine the line of intersection between two cylinders as shown in Fig. P8.13 to P8.16.

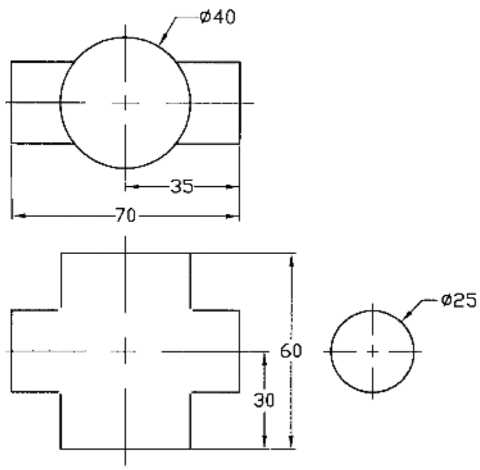


Fig. P8.13

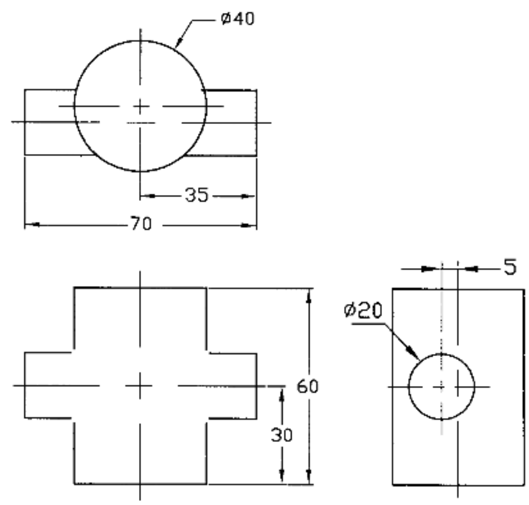


Fig. P8.14

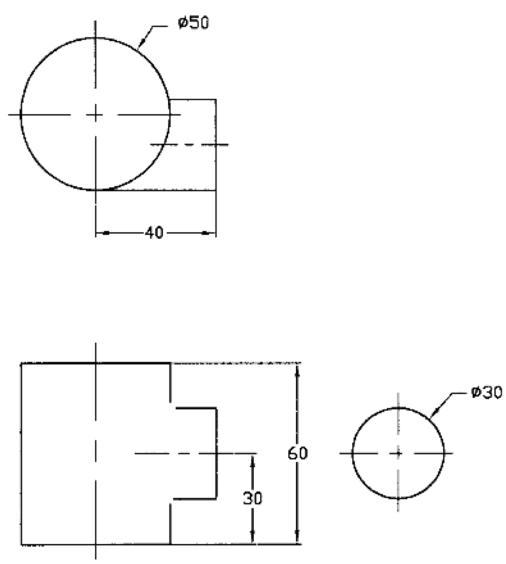


Fig. P8.15

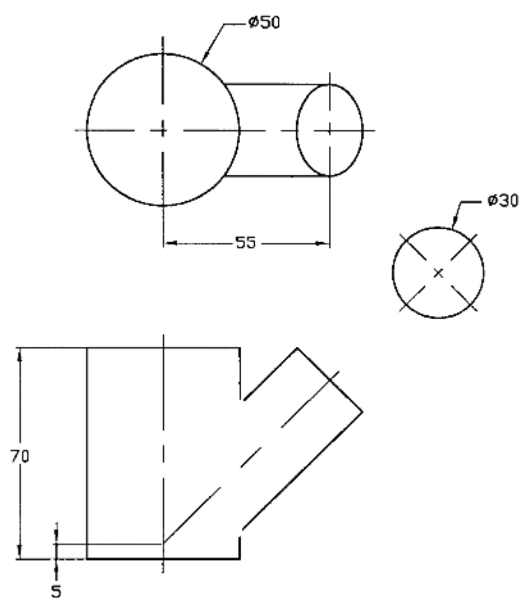


Fig. P8.16

Prob. 8.17 – 8.20: Determine line of intersection between a cylinder and a prism as shown in Fig. P8.17 to P8.20.

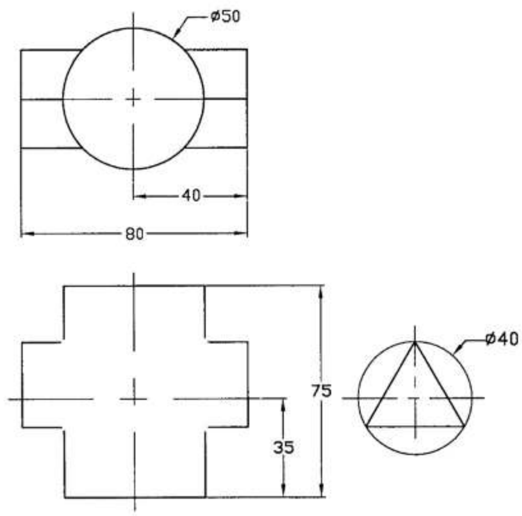


Fig. P8.17

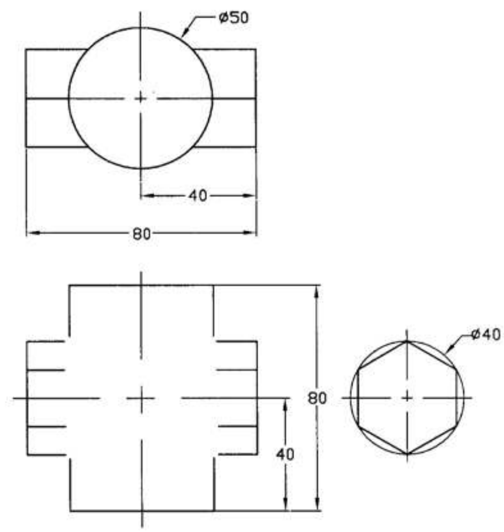


Fig. P8.18

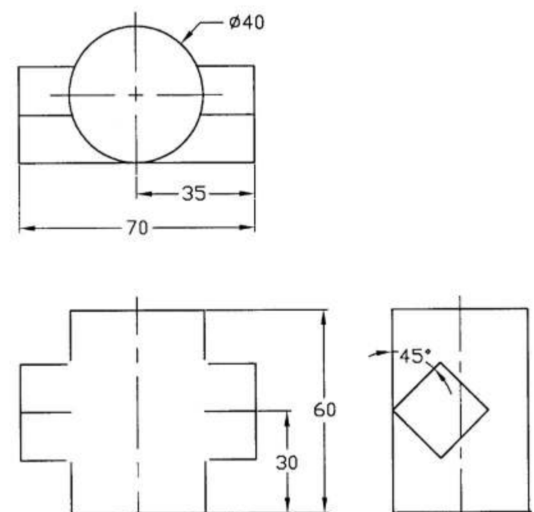


Fig. P8.19

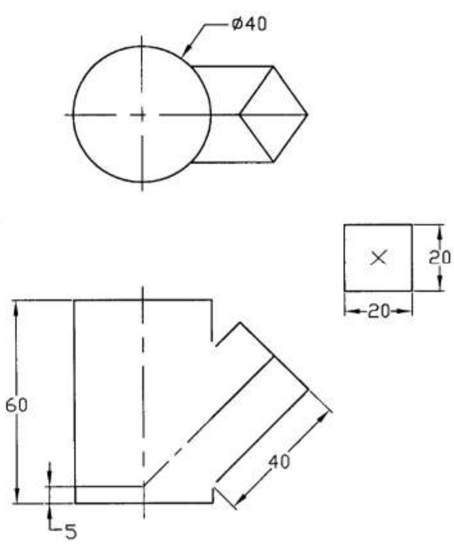


Fig. P8.20

Prob. 8.21 – 8.24: Determine line of intersection between a cone and a cylinder as shown in Fig. P8.21 to P8.24.

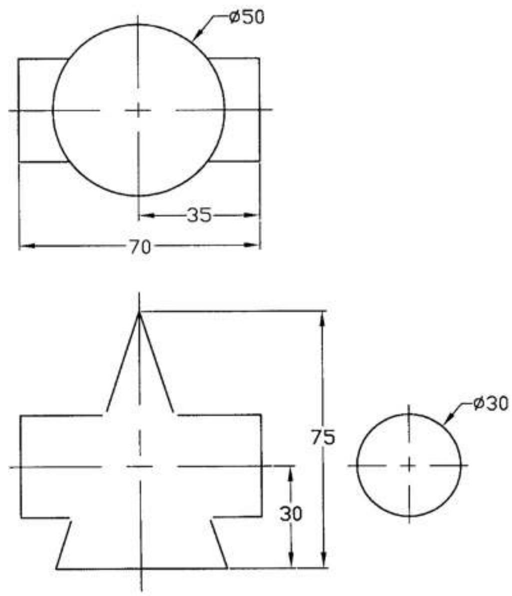


Fig. P8.21

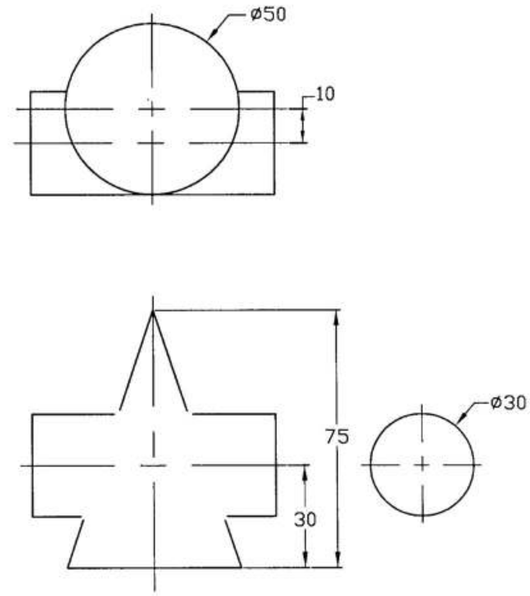


Fig. P8.22

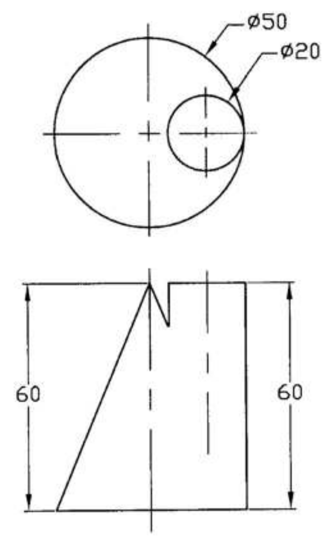


Fig. P8.23

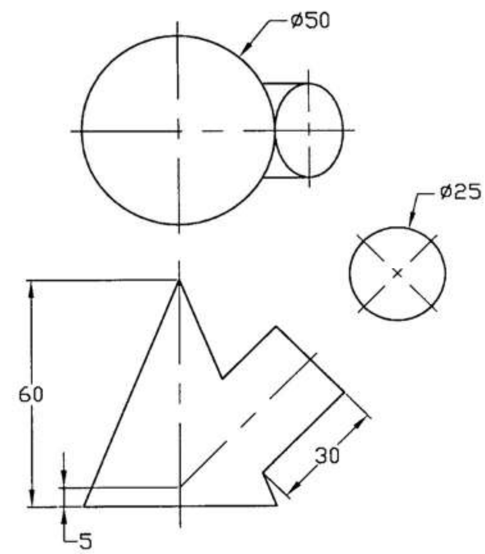


Fig. P8.24

Prob. 8.25 – 8.26: Determine the line of intersection between a cone and a prism as shown in Fig. P8.25 to P8.26.

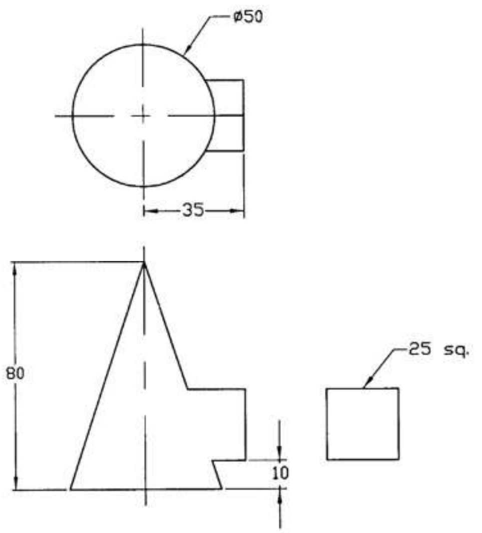


Fig. P8.25

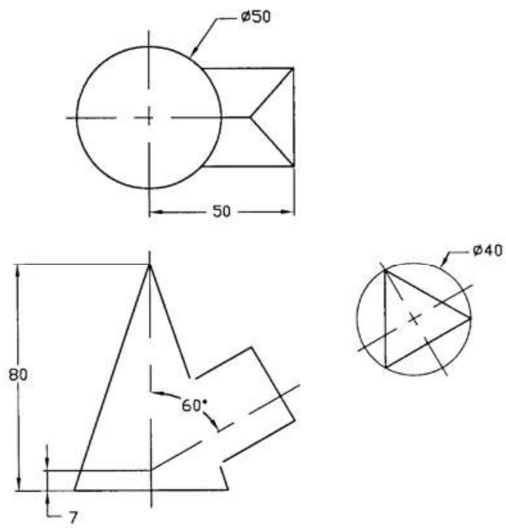


Fig. P8.26

Prob. 8.27 – 8.28: Determine the line of intersection between a cone and a prism as shown in Fig. P8.27 to P8.28.

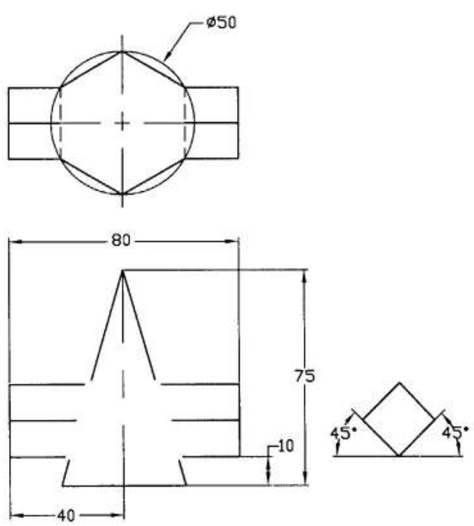


Fig. P8.27

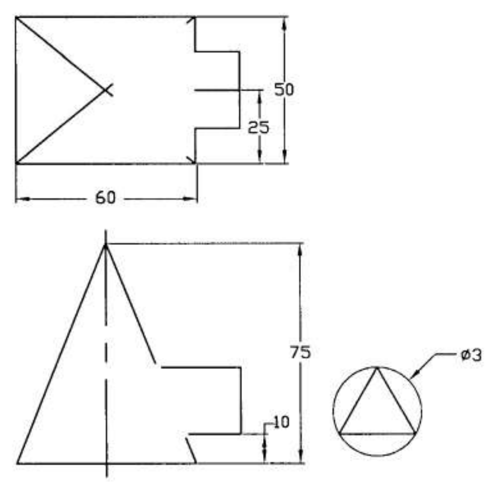


Fig. P8.28

CHAPTER 9

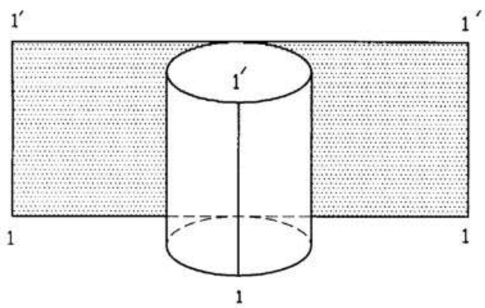
SURFACE DEVELOPMENT

9.1 *Introduction*

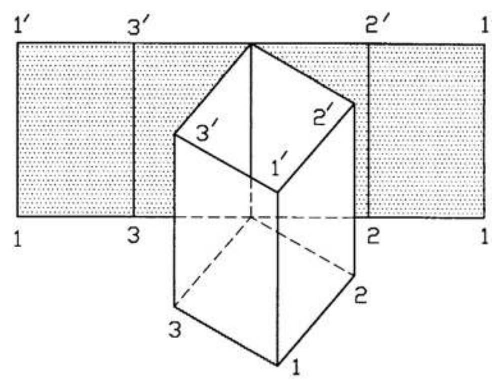
Surface developments are required for the manufacture of many engineering products. The complete surface of an object laid out in a plane is called the development of the surface. Surface development is essential for the sheet-metal work, stone cutting and pattern making etc. For the construction of duct, tunnel, chimneys etc. from the sheet, the surface development is necessary. When a sheet is properly cut and rolled, folded or formed, it makes the required object. There are some objects for which exact surface development cannot be made. For example, the surface of a ball cannot be developed exactly. However, it can be done approximately with many pieces of surfaces. In this chapter some simple examples in regard to the surface developments will be presented.

9.2 *Objects in Unfolding Condition*

In order to give the concept of the surface development, few examples will be discussed in this section systematically. The lateral surfaces of a cylinder and a square prism in unfolding condition are shown in Figures 9.1(a) and 9.1(b) respectively. The length and breadth of the lateral surface of the circular cylinder in unfolding condition are equal to the circumference and the height of the cylinder respectively as shown in Figure 9.1(a). While the length and breadth of the lateral surface of the square prism in unfolding condition, are equal to the perimeter and the height of the square prism respectively as shown in Figure 9.1(b).



(a)



(b)

Figure 9.1: Lateral Surface of a Cylinder and a Square Prism in Unfolding Condition

9.3 *Development of a Hexagonal Prism*

In Figure 9.2 the development of lateral surface of a truncated hexagonal prism has been shown. The steps of the development are as follows:

- First the base a-b-c-d-e-f-a is drawn in the bottom view.
- Next the elevation is drawn taking projection from the bottom view.

- Then the perimeter of the base is drawn on line a-a. This line is called stretch-out or girth line.
- Now the line is divided into six equal divisions.
- Then the perpendicular line at each point a, b, c etc. is drawn. This line is often called measuring or bend-line. Along this line folding is to be done in order to construct the prism.
- Now taking projection from the elevation, the length of each bend line is found and the lateral surface of the hexagonal prism is completed.

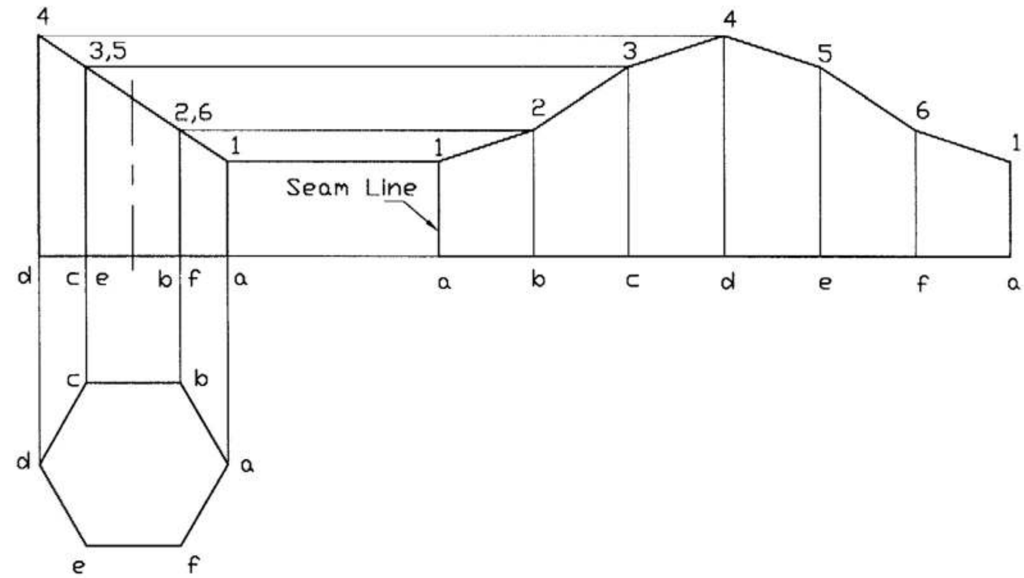


Figure 9.2: Development of Lateral Surface of a Hexagonal Prism

9.4 ***Development of a Right Cylinder***

In Figure 9.3 the development of the lateral surface of a truncated right cylinder has been shown. The following steps are to be considered to develop the surface.

- First the base of the cylinder is drawn in the bottom view and then the elevation is drawn taking projection from the bottom view.
- Now the circumference of the base is divided into 12 equal divisions. For higher number of divisions accuracy increases and for lower divisions it decreases.
- Then the base of the cylinder is developed into a straight-line a-a, which is equal to the circumference of the base. It is equal to π times the diameter of the base.
- Now the straight-line a-a has to be divided into 12 equal divisions.
- Then from each division perpendicular line has to be drawn.
- Next the length of each perpendicular line has to be taken corresponding to the projection of the elevation.
- Then the lateral surface is to be completed.

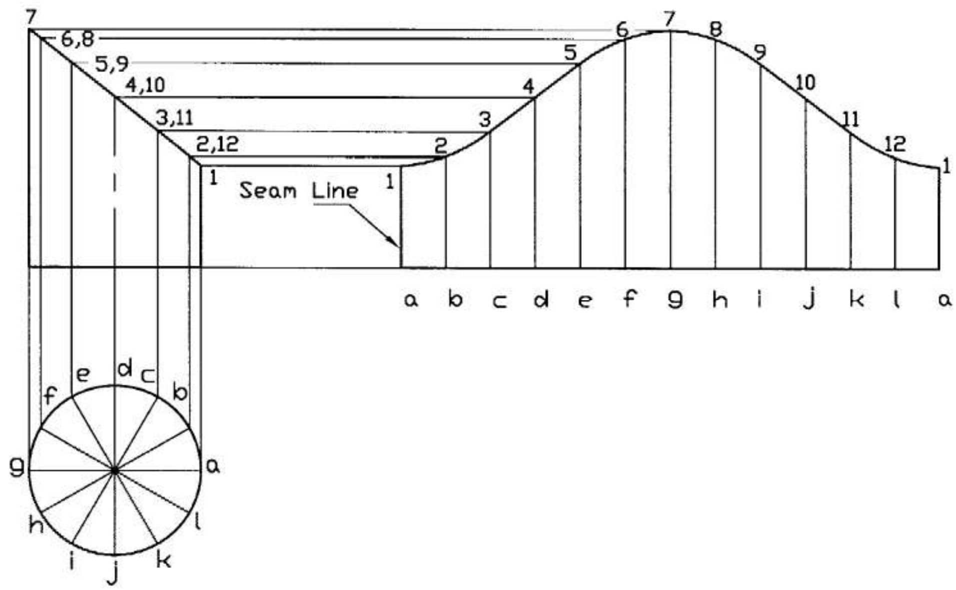


Figure 9.3: Development of Lateral Surface of a Truncated Right Cylinder

9.5 ***Development of a Right Pyramid***

The development of the lateral surface of a truncated right pyramid has been shown in Figure 9.4. The steps to develop the surface are as follows:

- First the hexagonal base abcdefa is drawn in the top view. Then the elevation is drawn taking projection from the top view.
- Now the truncated portion is drawn in the elevation and then in the top view taking projection from the elevation.
- Next an arc a-a is drawn with radius oa, which is the true length of the lateral edge of the pyramid. Since the pyramid is regular, all the lateral edges are equal.
- Now from the top view six equal sides of the hexagonal base ab, bc, cd etc. are drawn on the arc a-a.
- Next from each intersecting point of the sides and the arc, lines ao, bo, co etc. are drawn up to the vertex o. Thus the six lateral surface of the pyramid is developed.
- Then in the elevation to obtain the true lengths, projections $22^{/}$, $33^{/}$ etc. are drawn on the true length of the lateral edge ao.
- Now the lines ao, bo, co etc. passing through the intersecting points in the arc a-a are marked respectively with the points 1, $2^{/}$, $3^{/}$ etc. by the true lengths a1, $a2^{/}$, $a3^{/}$ etc. from the elevation.
- Then the consecutive points are added to obtain the lateral surface of the truncated pyramid.

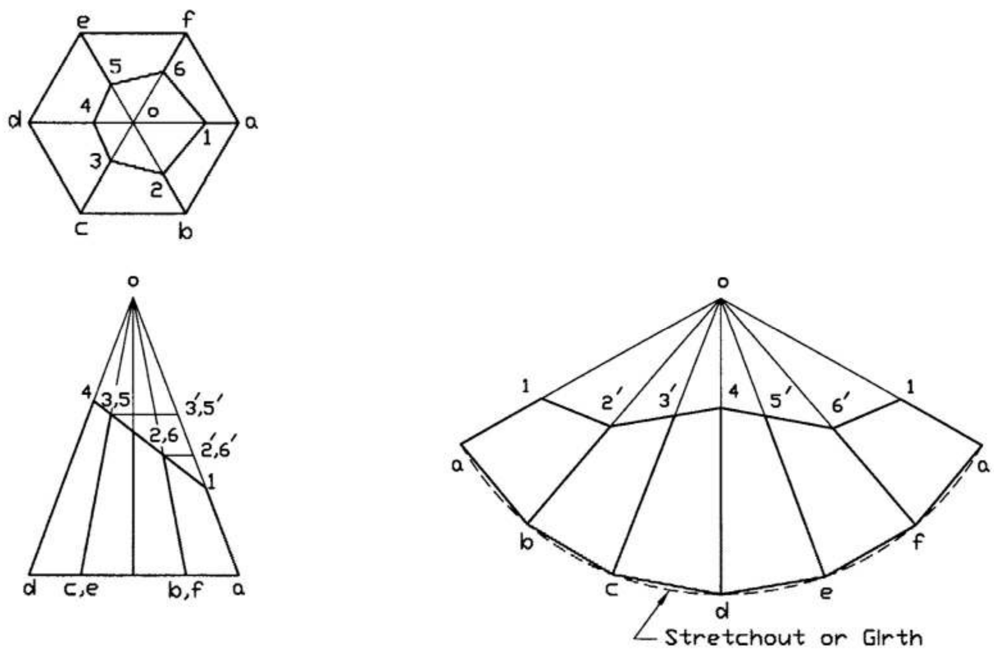


Figure 9.4: Development of Lateral Surface of a Truncated Right Pyramid

9.6 *Development of a Right Cone*

In Figure 9.5 the development of the lateral surface of a truncated right cone is shown. The development of the surface may be done according to the following steps.

- First the circular base is drawn in the top view, and then the elevation is drawn taking projection from the top view.
- Next the truncated portion is drawn in the elevation and then in the top view taking projection from the elevation.
- Now the circular base is divided into a sufficient number of equal divisions (say 12) so that the chordal distance of each division equals approximately the arc of the division.
- Now an arc a-a is drawn with radius oa, which is the true length of the lateral edge of the cone.
- Then 12 equal divisions are made in the arc a-a, each division being equal to the chordal distance of the division already made in the base circle in the top view. Then from each division point lines ao, bo etc. are drawn up to the point o.
- Next in the elevation true lengths are projected up to the points $b^{/}$, $c^{/}$, $d^{/}$ etc. on the true length of the lateral edge ao.
- Next the lines ao, bo, co passing through the division points in the arc a-a are marked respectively with the points $a^{/}$, $b^{/}$, $c^{/}$ etc. by the true lengths equal to $aa^{/}$, $bb^{/}$, $cc^{/}$ etc. from the elevation.
- Through the marked points a smooth curve is drawn and thus the lateral surface of the truncated right cone is developed.

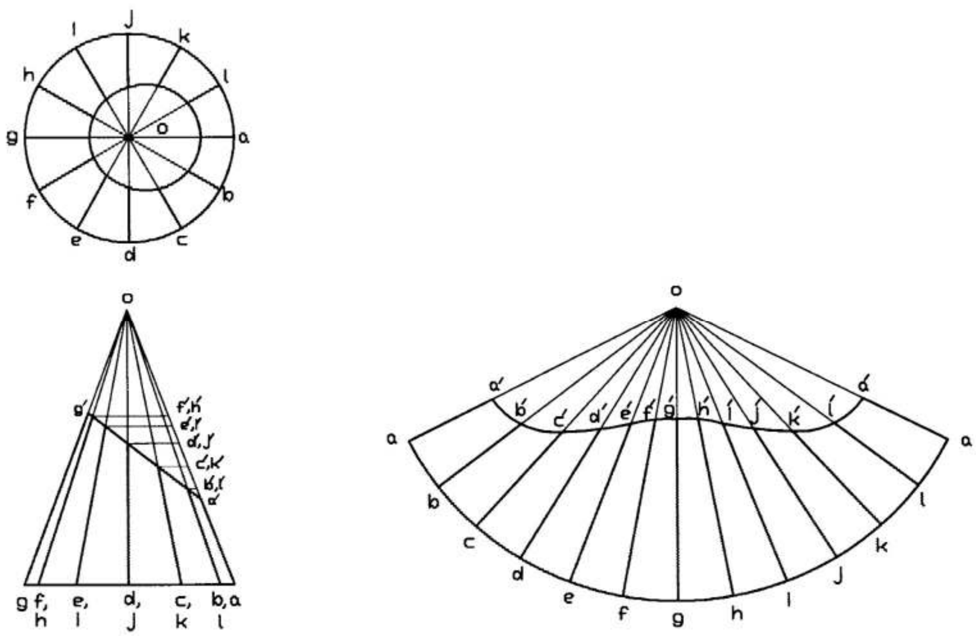


Figure 9.5: Development of Lateral Surface of a Truncated Right Cone

9.7 ***Development of an Oblique Pyramid***

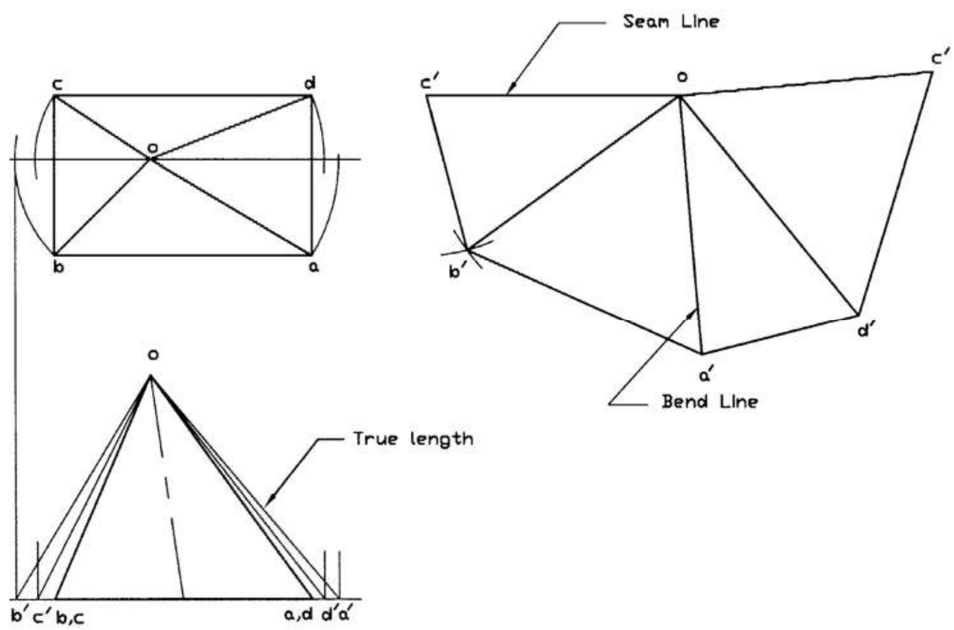


Figure 9.6: Development of Lateral Surface of an Oblique Pyramid

In Figure 9.6, the development of the lateral surface of a rectangular oblique pyramid is shown. The steps to develop the surface are as follows:

- First the base of the oblique pyramid is drawn in the top view. Then the elevation of the pyramid is drawn.
- Next the true lengths of the four unequal lateral edges are obtained rotating them parallel to the frontal plane as shown. The true lengths are $oa^{/}$, $ob^{/}$, $oc^{/}$ and $od^{/}$.
- Then a line equal to $oc^{/}$, the smallest true length of the four, is drawn.
- Now taking the point o as center with radius $ob^{/}$ and $c^{/}$ as center with radius cb two arcs are drawn so that they intersect each other. Then the points o and $b^{/}$ as well as $c^{/}$ and $b^{/}$ are connected to find the lateral surface of ocb. In the similar ways the three other lateral surfaces of oba, oad and odc are drawn in sequence. Thus the development of the lateral surface of the rectangular oblique pyramid is completed.

9.8 *Development of an Oblique Cone*

In Figure 9.7 the development of the lateral surface of an oblique cone has been presented. The surface development of the oblique cone is done approximately using the method of triangulation. That is, the surface is assumed consisting of a large number triangular strips. The triangles are so chosen that the bases of them are very small. The following steps are followed to develop the surface of the oblique cone.

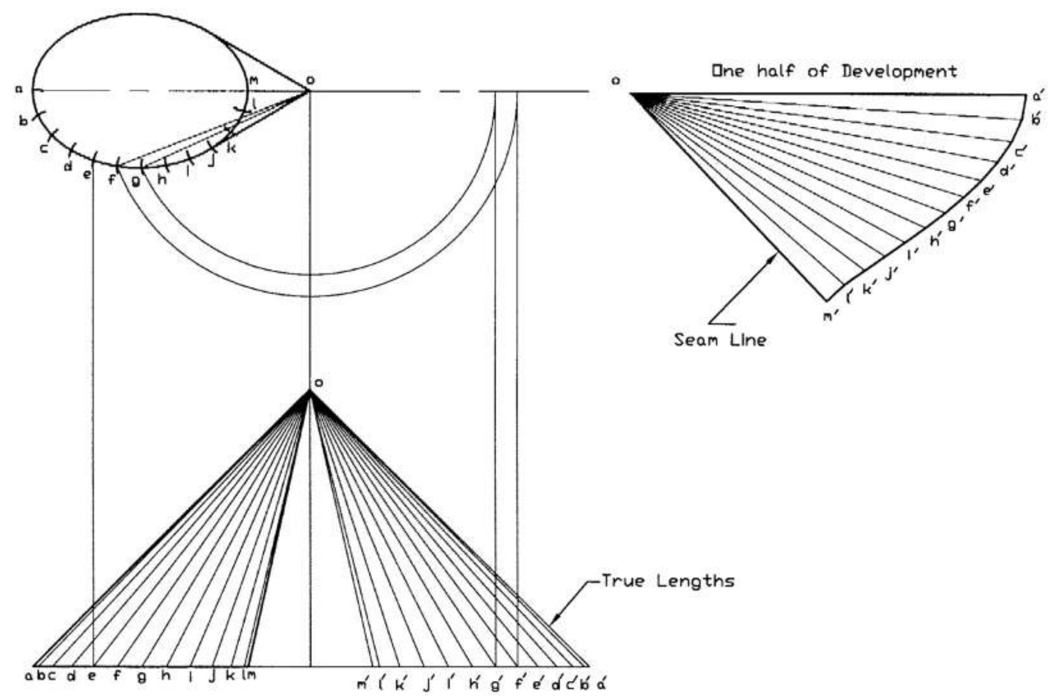


Figure 9.7: Development of Lateral Surface of an Oblique Cone

- First the base of the oblique cone is drawn in the top view and then the elevation of it, is drawn.

- Now the base of the oblique is divided into a sufficient number of equal divisions so that the sum of the chordal distances is approximately equal to the true length of the base curve. Then the true lengths are drawn as shown in the figure.
- Now the triangle $oa^/b^/$ is drawn with one side $a^/b^/$ (=ab on the top view) and the other sides equal to true lengths $oa^/$ and $ob^/$. Similarly all the other triangles $ob^/c^/$, $oc^/d^/$ etc. are drawn to develop one half lateral surface of the oblique cone.

Example Problems

Note: For solutions see the following section of ***Solutions for Example Problems.***

Prob. 9.1: Develop the lateral surfaces of the truncated right rectangular prism as shown in Fig. P9.1.

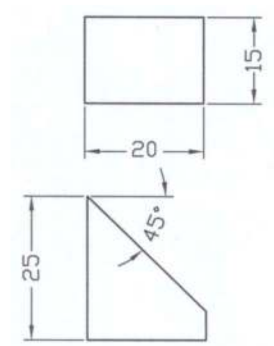


Fig. P9.1

Prob. 9.2: Develop the lateral surfaces of the right square prism as shown in Fig. P9.2.

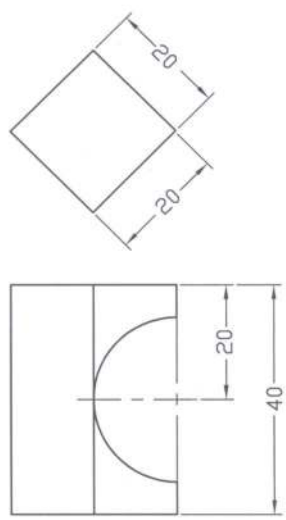


Fig. P9.2

Prob. 9.3: Develop the lateral surfaces of the right pentagonal prism as shown in Fig. P9.3.

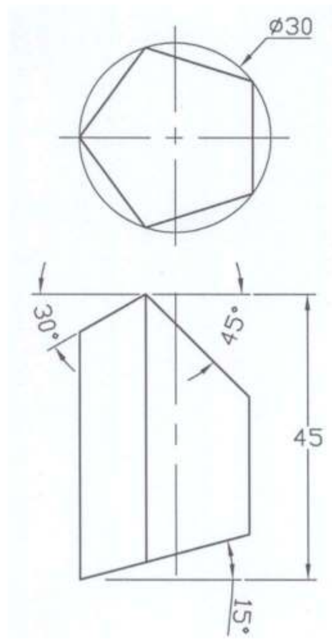


Fig. P9.3

Prob. 9.4: Develop the lateral surfaces of the right hexagonal prism as shown in Fig. P9.4.

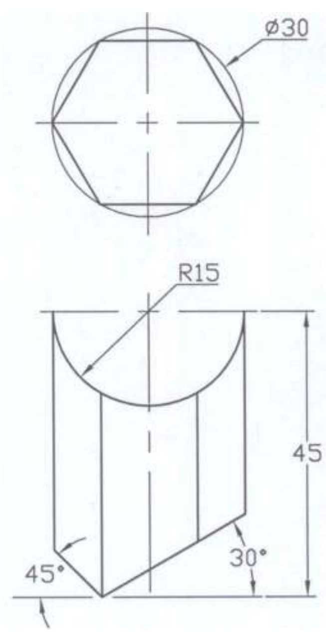


Fig. P9.4

Prob. 9.5: Develop the surfaces of the oblique pyramid with square base as shown in Fig. P9.5.

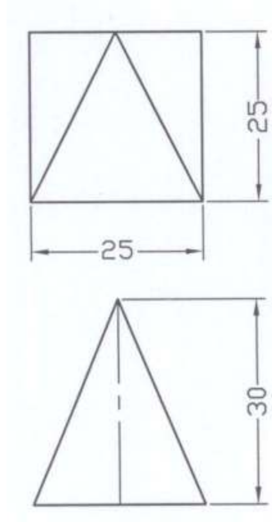


Fig. P9.5

Prob. 9.6: Develop the lateral surfaces of the right rectangular pyramid as shown in Fig. P9.6.

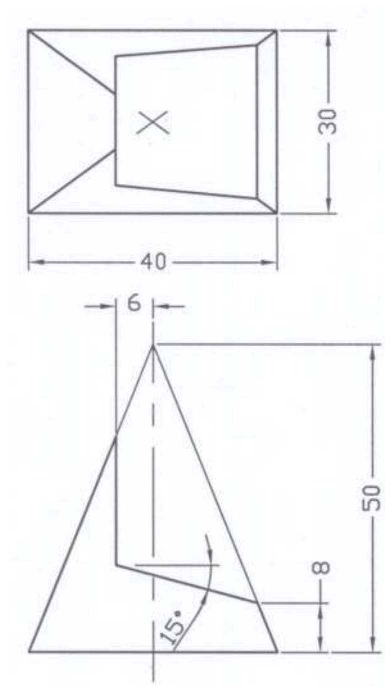


Fig. P9.6

Prob. 9.7: Develop the lateral surfaces of the truncated oblique pyramid given in Fig. P9.7.

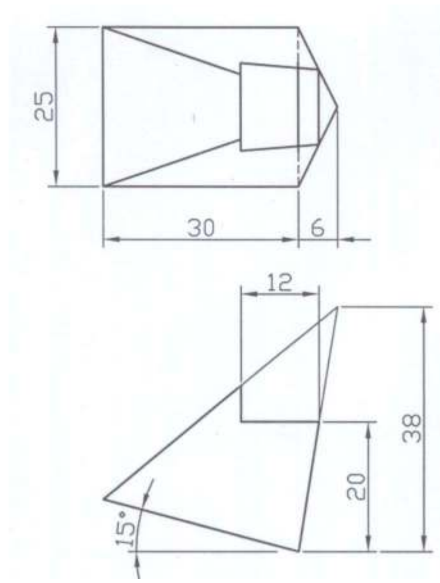


Fig. P9.7

Prob. 9.8: Develop the lateral surfaces of the truncated oblique hexagonal pyramid given in Fig. P9.8.

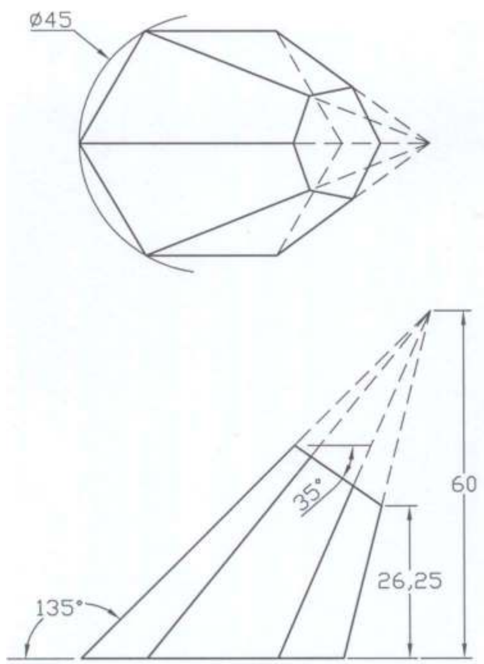


Fig. P9.8

Prob. 9.9: Develop the lateral surfaces of the right cylinder as shown in Fig. P9.9.

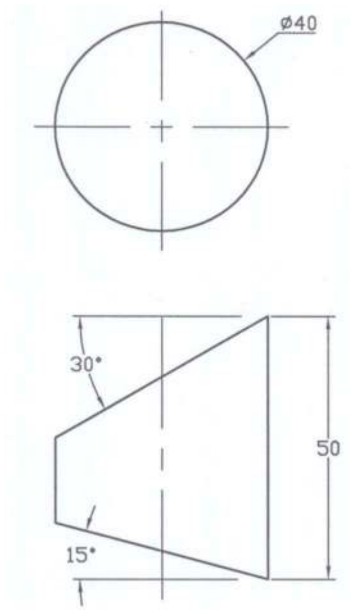


Fig. P9.9

Prob. 9.10: Develop the lateral surfaces of the right cylinder with a hole as shown in Fig. P9.10.

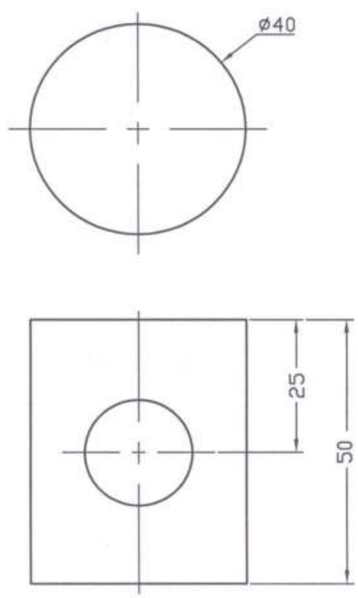


Fig. P9.10

Prob. 9.11: Develop the lateral surfaces of the right cylinder given in Fig. P9.11.

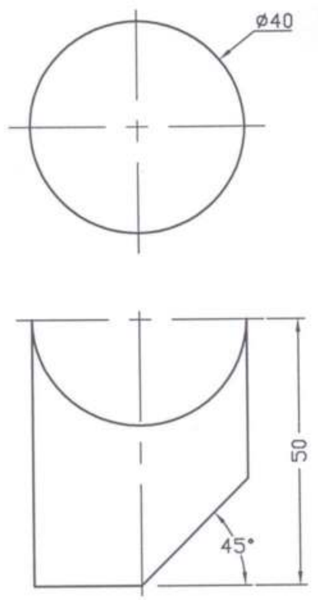


Fig. P9.11

Prob. 9.12: Develop the lateral surfaces of the right cylinder as shown in Fig. P9.12.

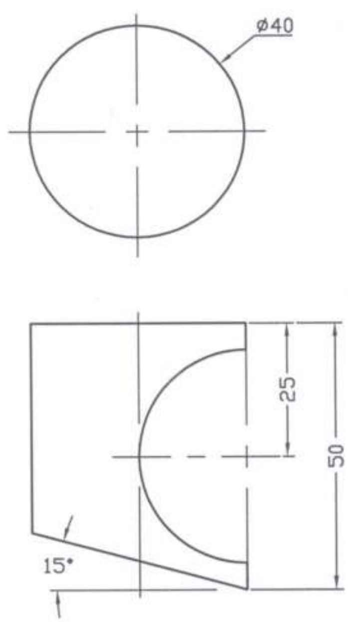


Fig. P9.12

Prob. 9.13: Develop the lateral surfaces of the truncated right circular cone given in Fig. P9.13.

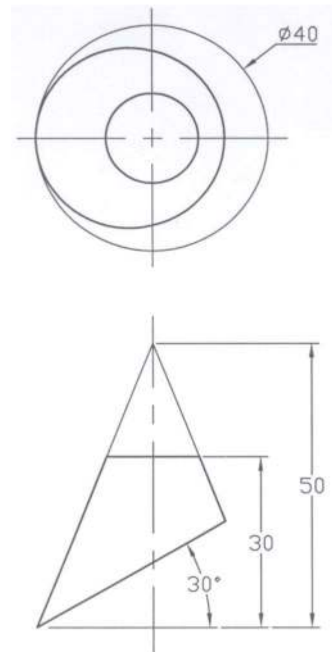


Fig. P9.13

Prob. 9.14: Develop the lateral surfaces of the right cone with a hole as shown in Fig. P9.14.

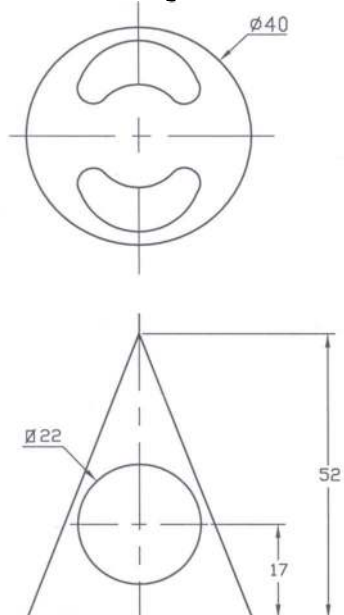


Fig. P9.14

Prob. 9.15: Develop the lateral surfaces of the right cone given in Fig. P9.15.

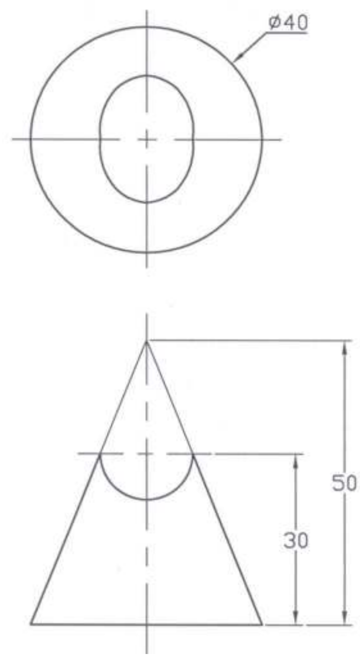


Fig. P9.15

Prob. 9.16: Develop the lateral surfaces of the oblique cone with circular base given in Fig. P9.16.

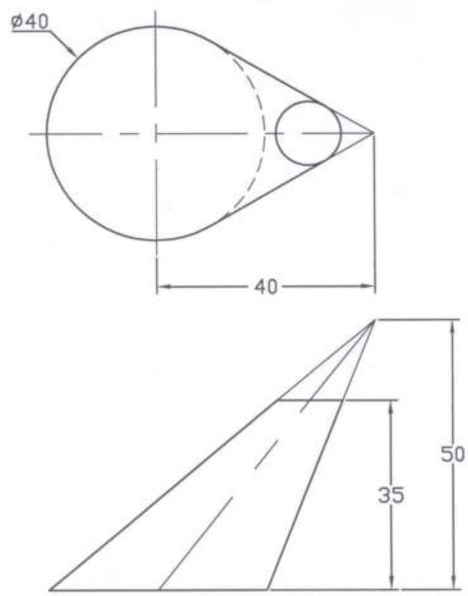
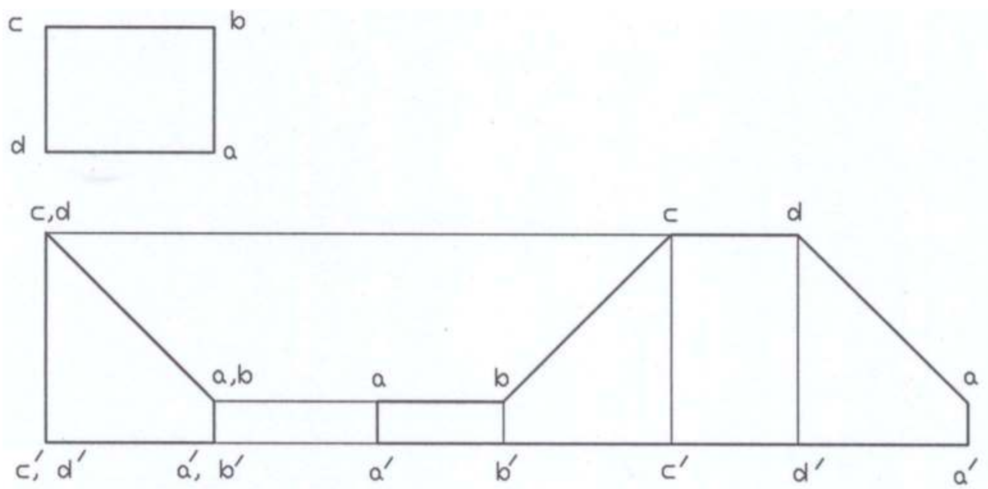
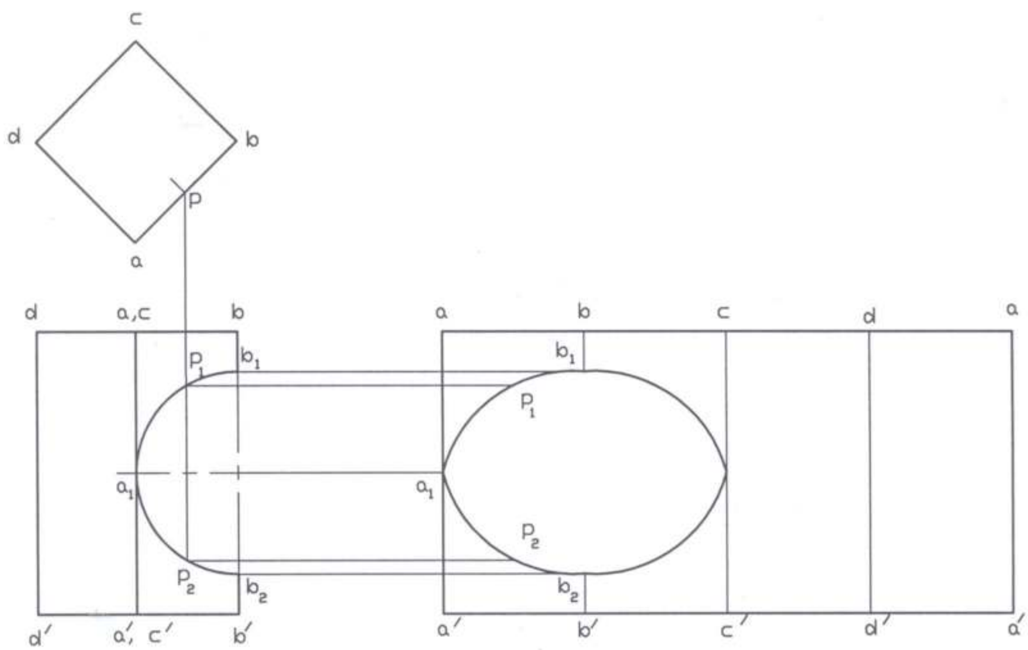


Fig. P9.16

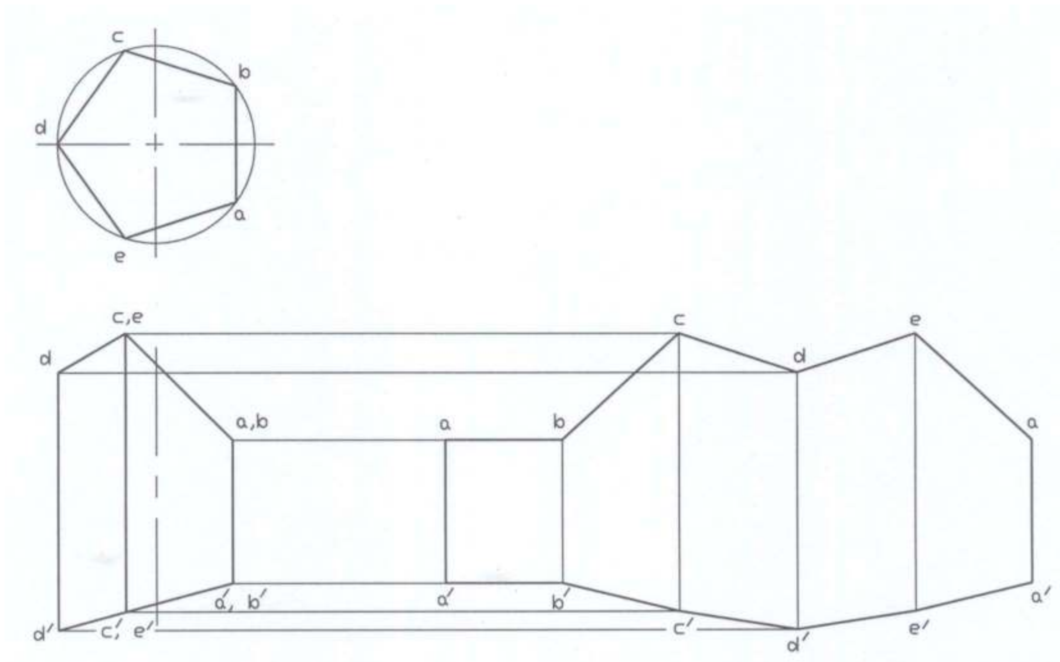
Solutions for Example Problems



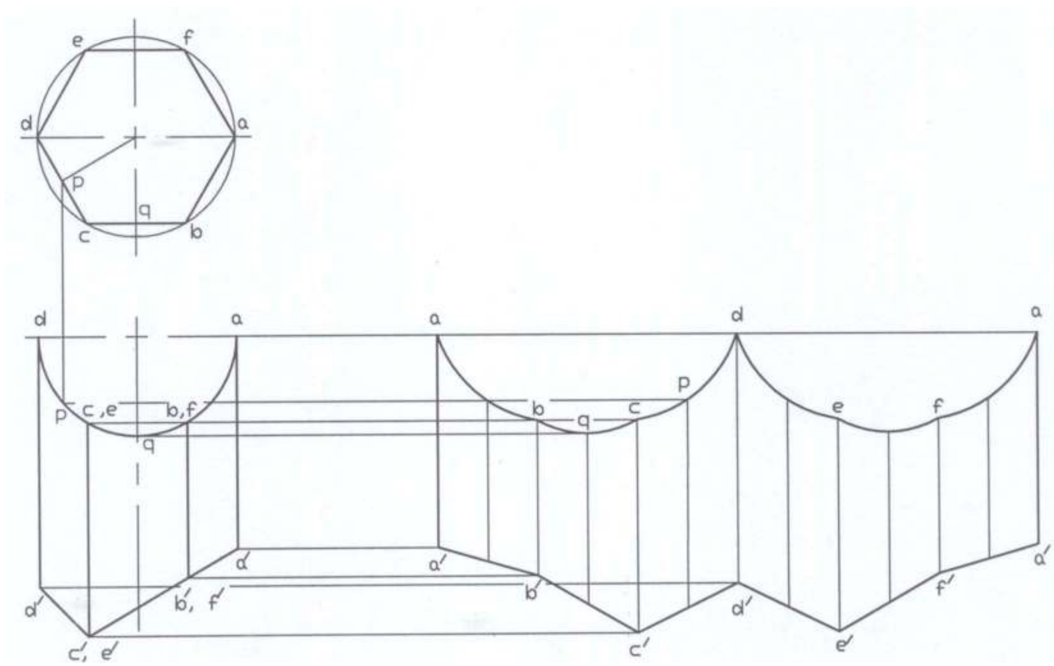
Solution of P9.1



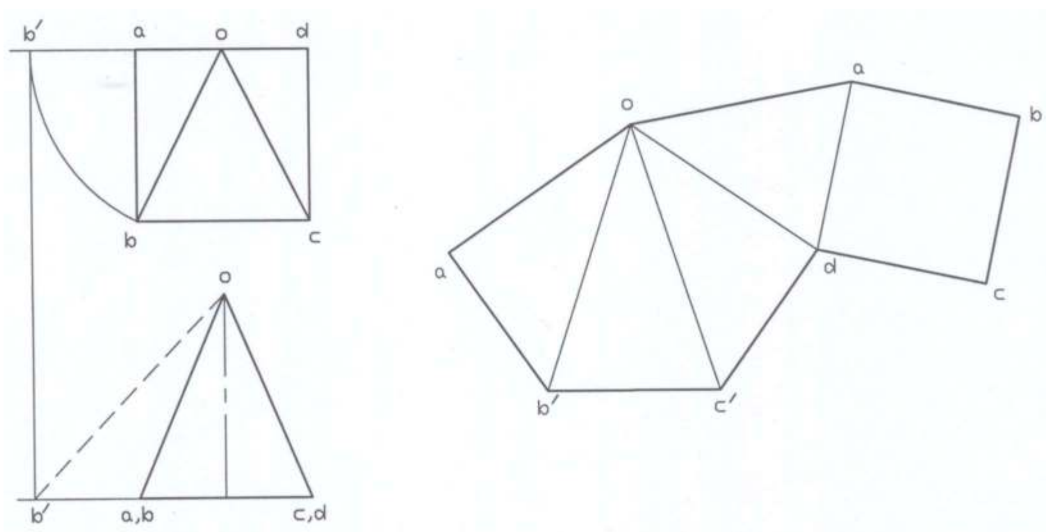
Solution of P9.2



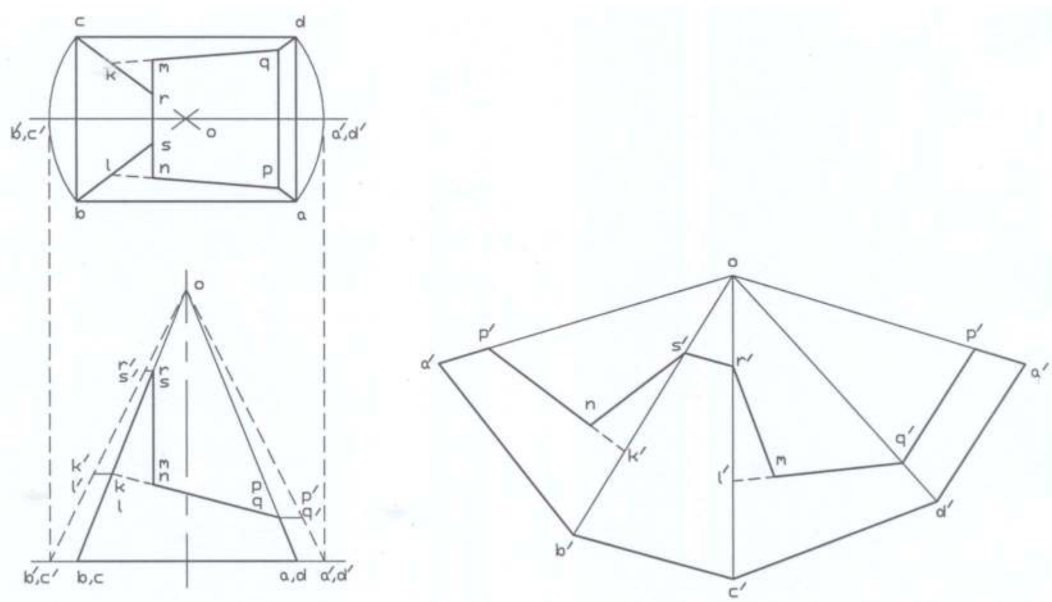
Solution of P9.3



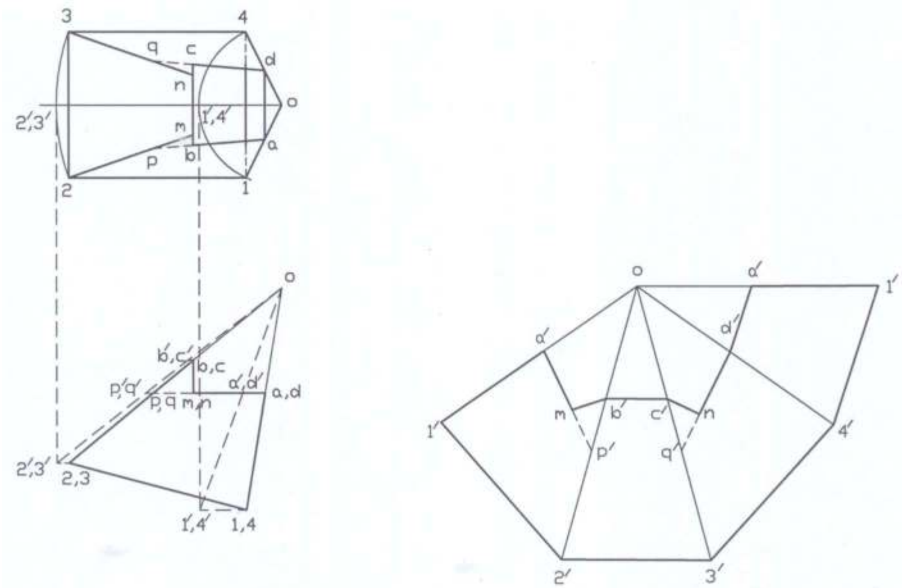
Solution of P9.4



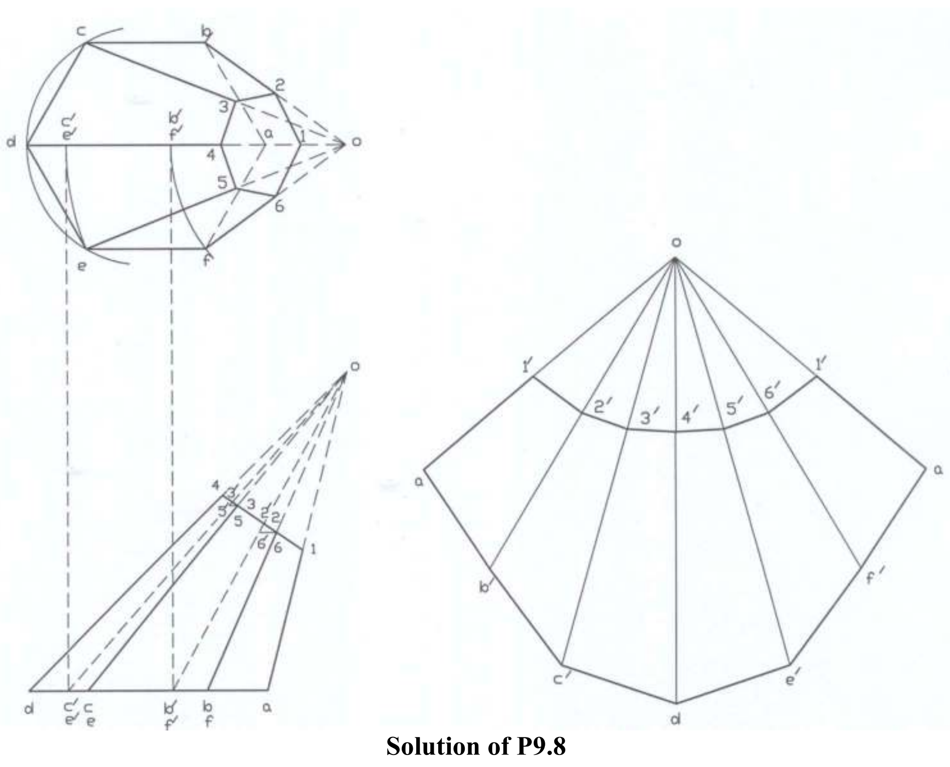
Solution of P9.5



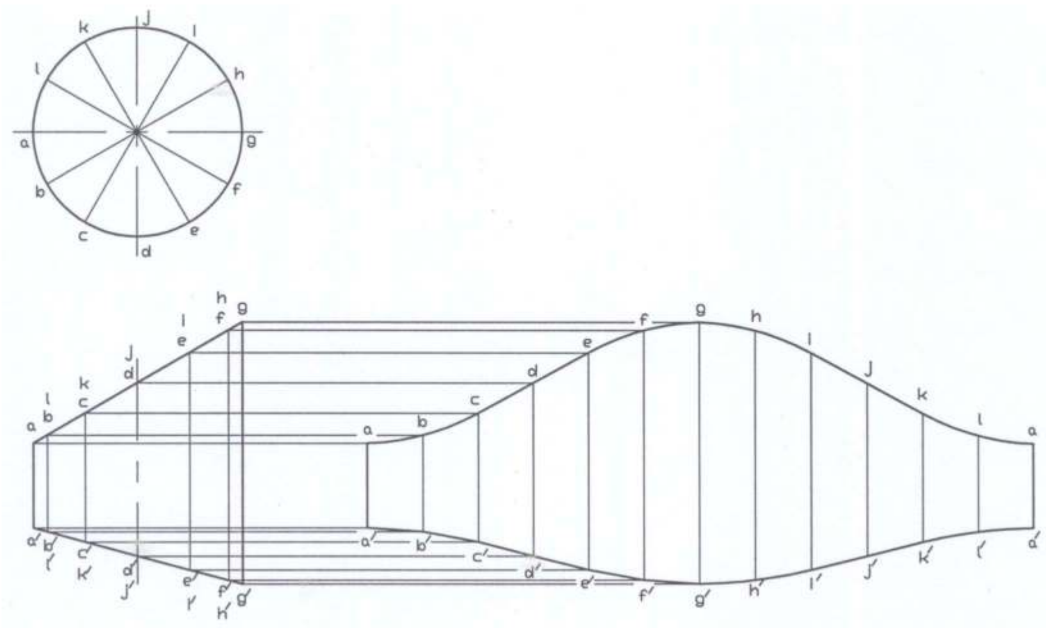
Solution of P9.6



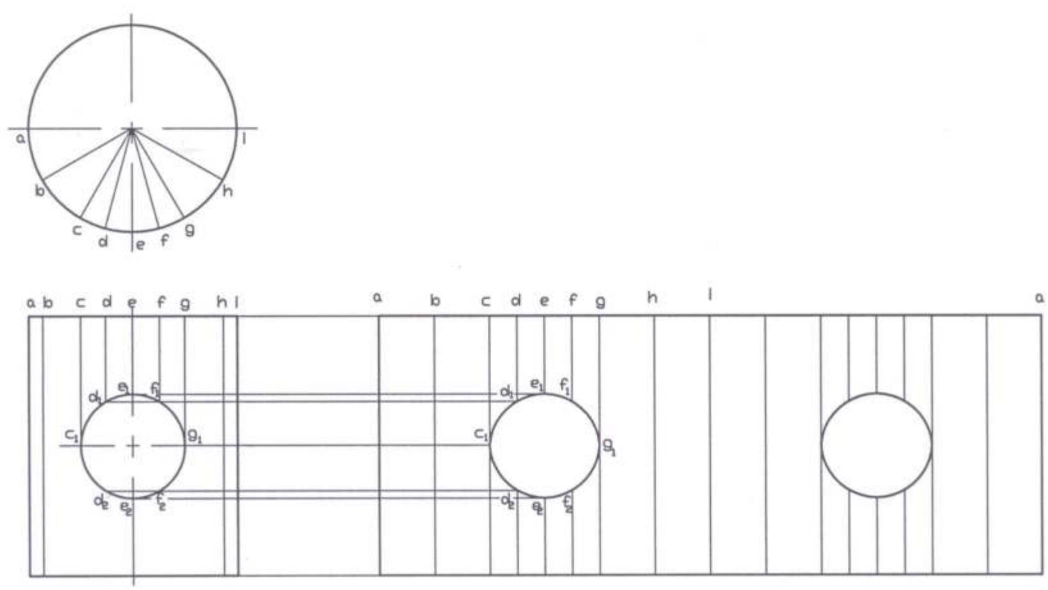
Solution of P9.7



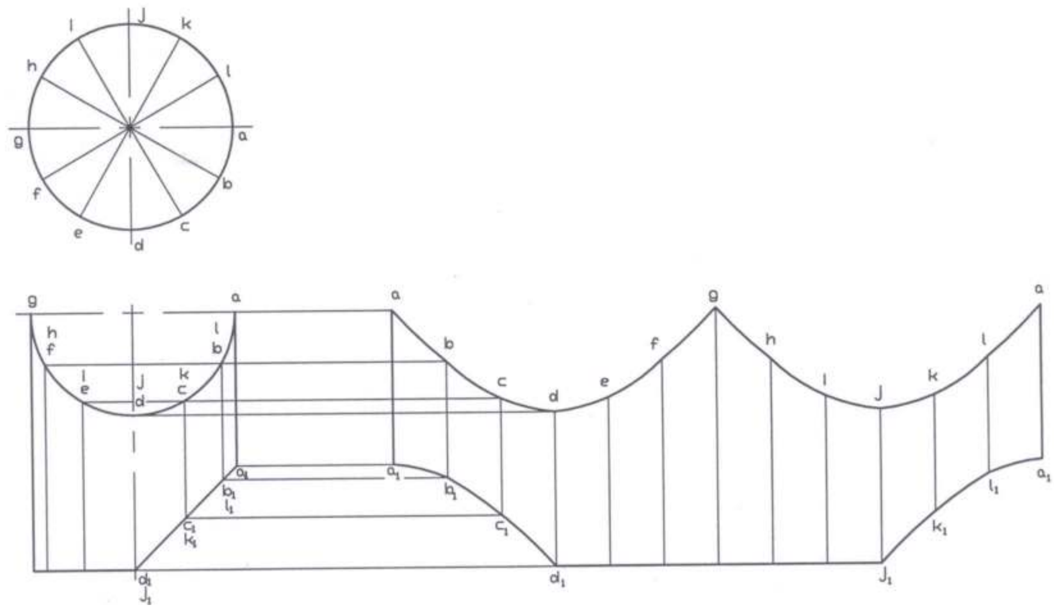
Solution of P9.8



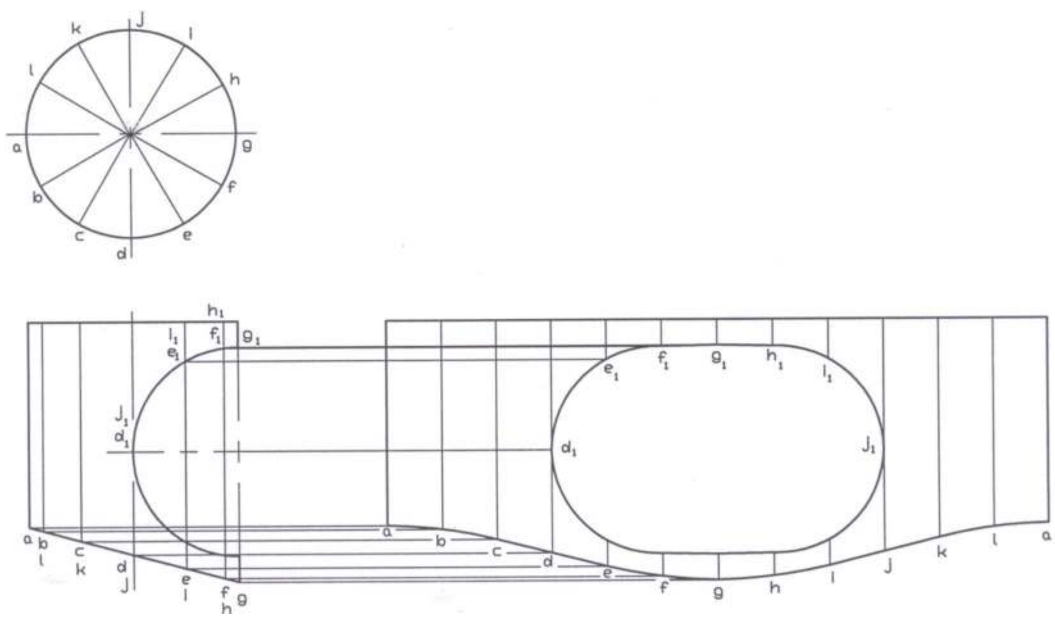
Solution of P9.9



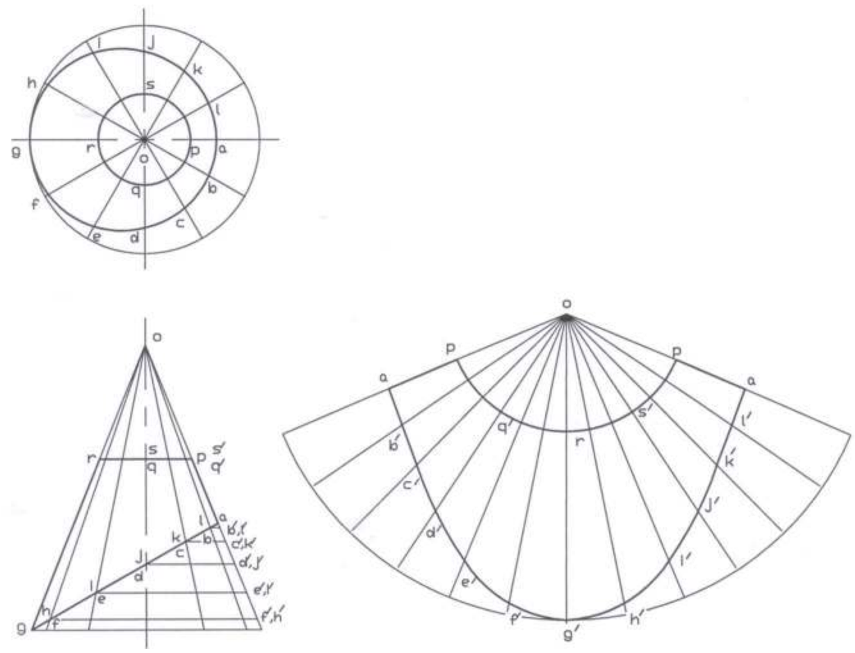
Solution of P9.10



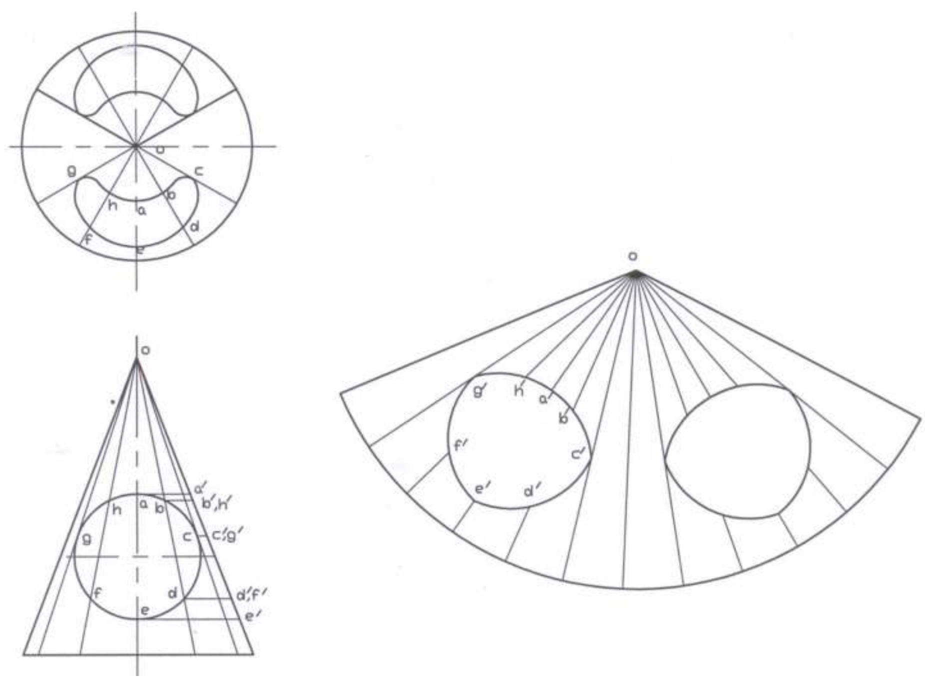
Solution of P9.11



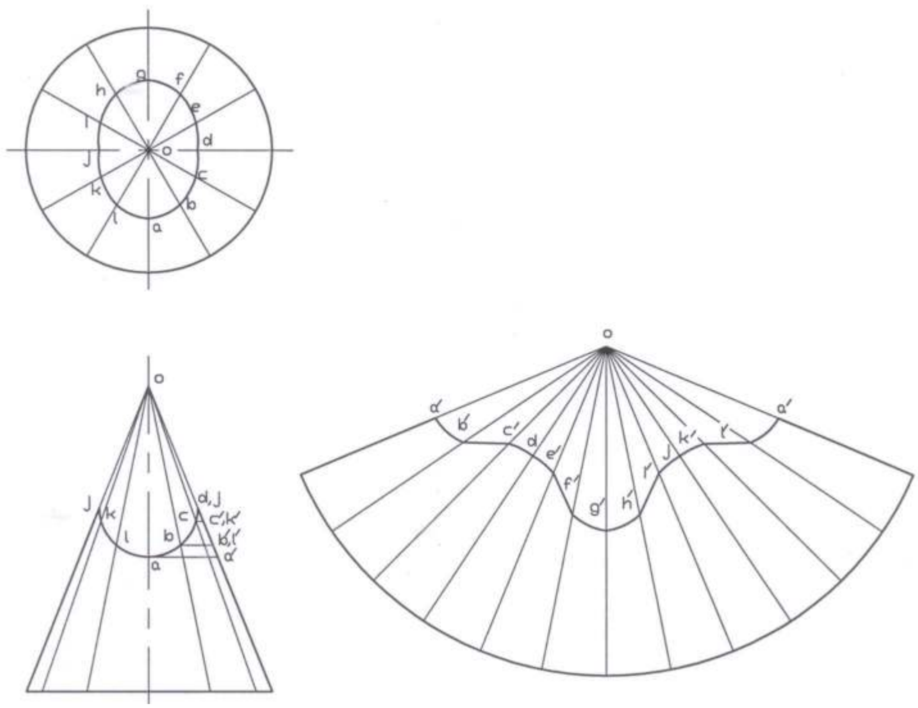
Solution of P9.12



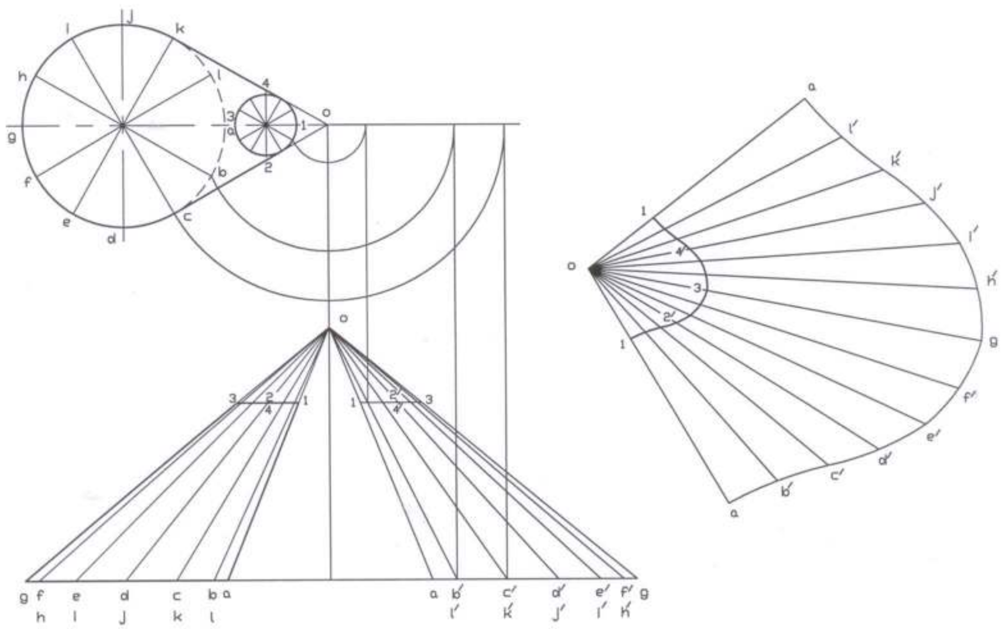
Solution of P9.13



Solution of P9.14



Solution of P9.15



Solution of P9.16

Problems

Prob. 9.17 – 9.22: Develop the lateral surfaces of the prisms as shown in Fig. P9.17 to P9.22.

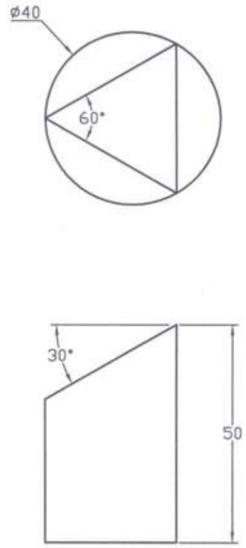


Fig. P9.17

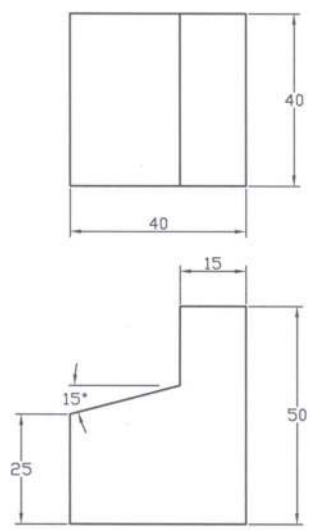


Fig. P9.18

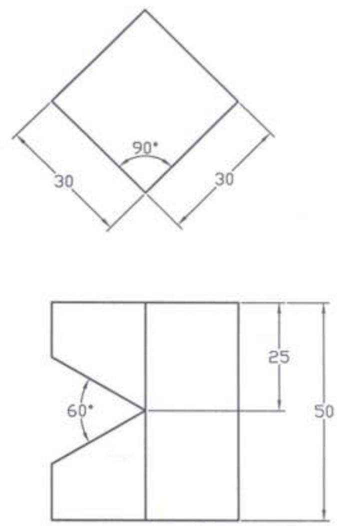


Fig. P9.19

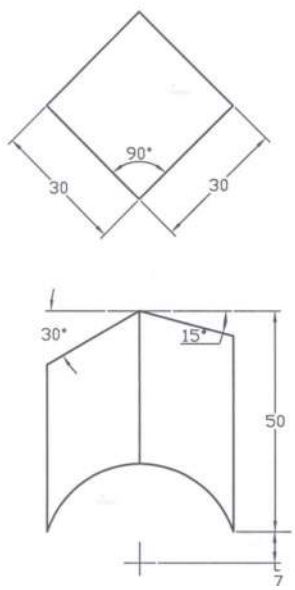


Fig. P9.20

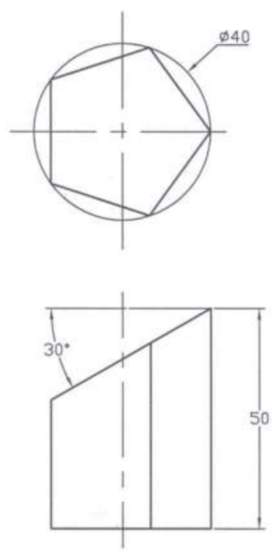


Fig. P9.21

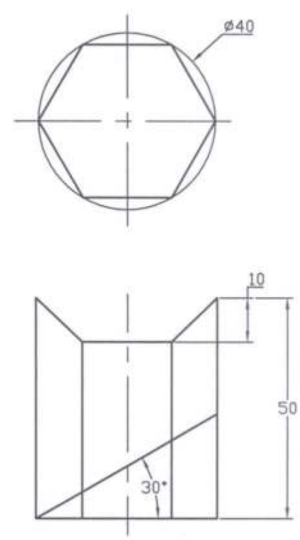


Fig. P9.22

Prob. 9.23 – 9.28: Develop the lateral surfaces of the pyramids as shown in Fig. P9.23 to P9.28.

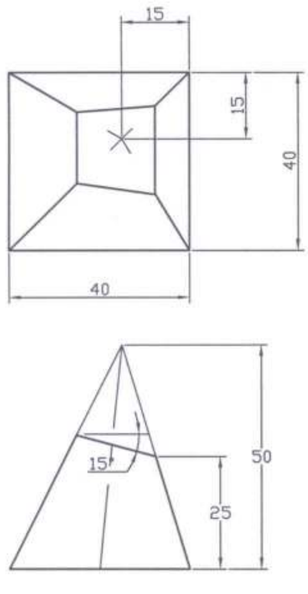


Fig. P9.23

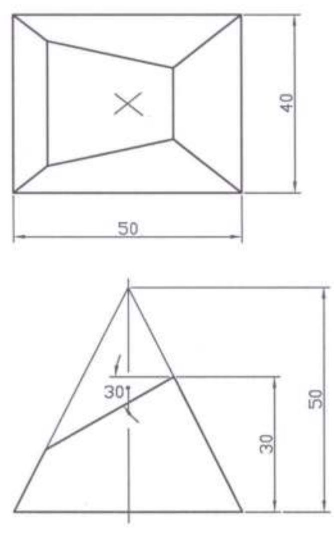


Fig. P9.24

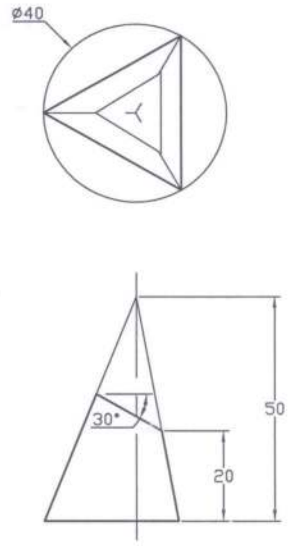


Fig. P9.25

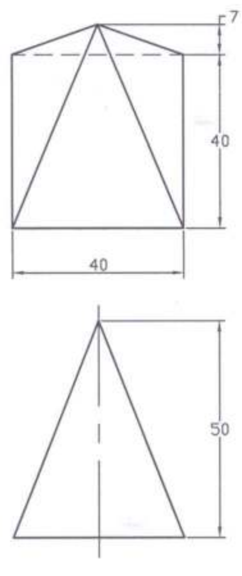


Fig. P9.26

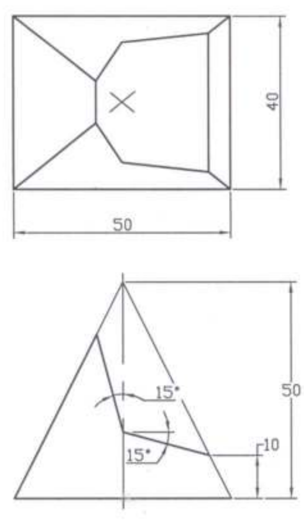


Fig. P9.27

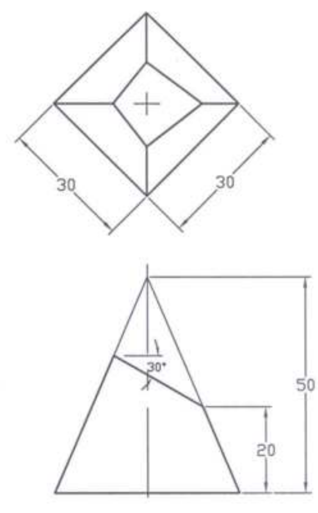


Fig. P9.28

Prob. 9.29 – 9.34: Develop the lateral surfaces of the cylinders as shown in Fig. P9.29 to P9.34.

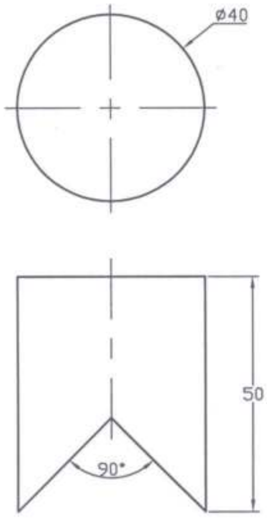


Fig. P9.29

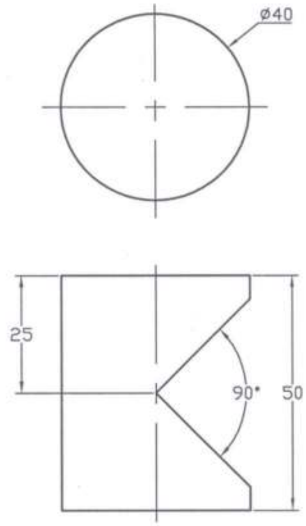


Fig. P9.30

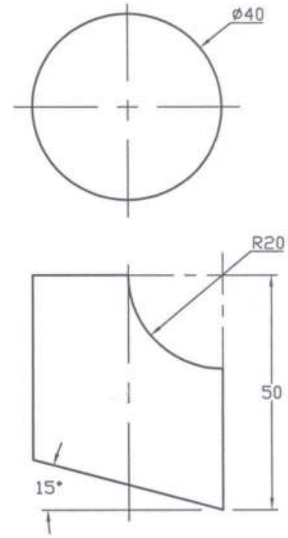


Fig. P9.31

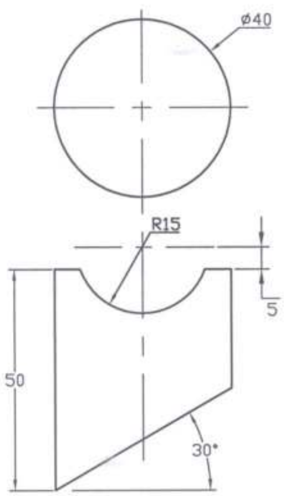


Fig. P9.32

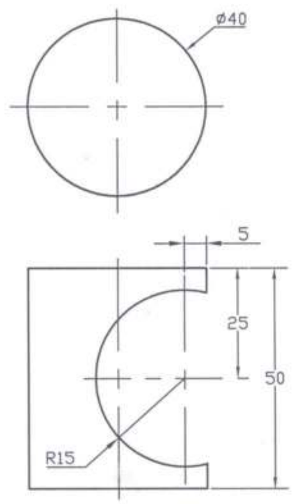


Fig. 9.33

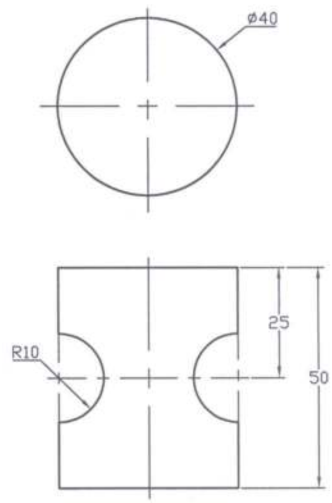


Fig.P9.34

Prob. 9.35 – 9.40: Develop the lateral surfaces of the cones as shown in Fig. P9.35 to Fig. P9.40.

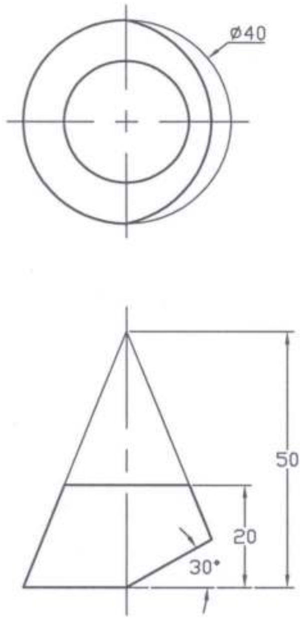


Fig. P9.35

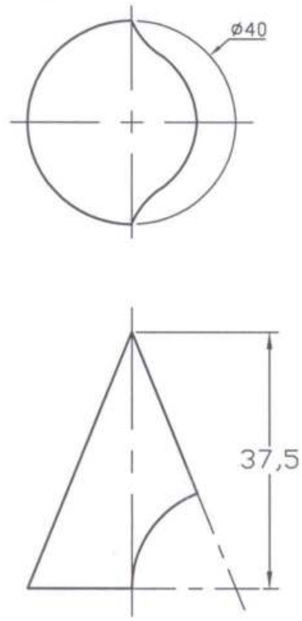


Fig. P9.36

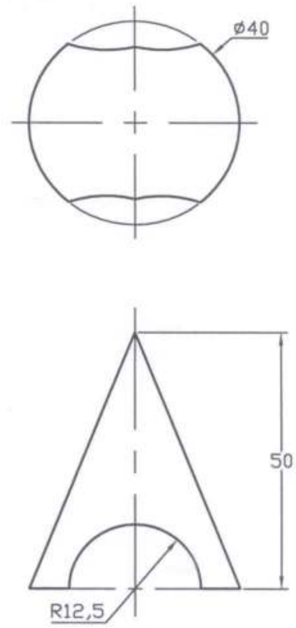


Fig. P9.37

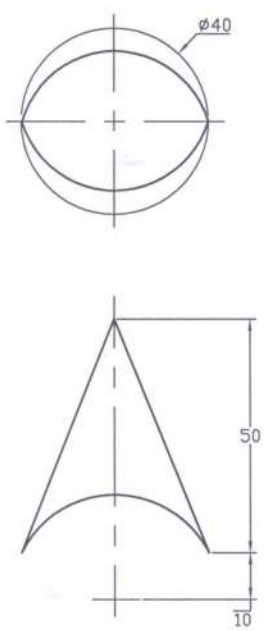


Fig. P9.38

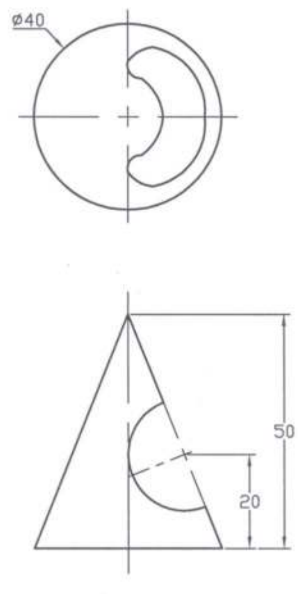


Fig. P9.39

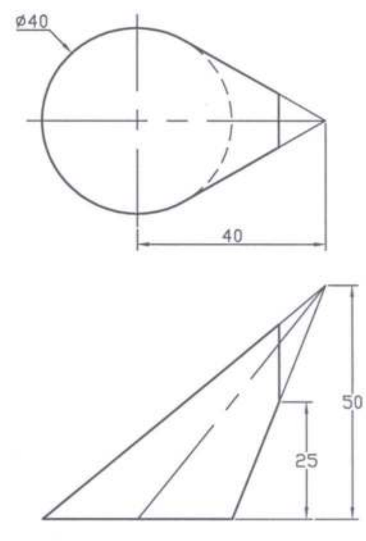


Fig. P9.40

CHAPTER 10

WORKING DRAWINGS

10.1 *Working Drawings*

The methodology to manufacture a complete machine will be discussed in a nutshell in this chapter. Before going to manufacture a machine, working drawings of the machine are necessary. A working drawing provides with the complete information for the manufacture of a machine. Working drawings may be categorized into two classes: detail drawings and assembly drawings. Detail drawings give full information for the manufacture of the individual part. On the other hand the assembly drawings supply the required information to assemble the various parts to make the complete machine.

10.2 *Detail Drawings*

Detail drawings furnish with the full information for the construction of the individual machine part. The following information is usually provided in the detail drawings.

- The necessary orthographic projections of the various machine parts to present them very distinctly. If necessary, sometimes the pictorial views may be provided. The sectional views, auxiliary views and enlarged views may also be supplied, if necessary.
- The required dimensions of the individual machine part are to be provided for their manufacture. Tolerances may be mentioned, where necessary.
- The detail specification of the machine part has to be given. It will include the material by which the machine part will be manufactured. The heat treatment and surface finish are to be mentioned in the detail drawing as well. Sometimes general notes may be necessary.
- The other information such as, the quantity of the machine part, scale of the drawing, method of projection, date, part number, the names of the drafter, checker, approver etc. are to be given.
- The name of the Company and drawing sheet size are to be provided. Besides these revisions or modifications are to be given, if there are any.

It is the usual practice to show the drawing of the individual machine part on a single drawing sheet. The title block is shown on the bottom right corner of the drawing sheet in general. During drawing it has to be taken care of about use of the standard components. The checker may follow a checklist in order to be sure that all the necessary information has been incorporated properly.

10.3 *Assembly Drawing*

Assembly drawing provides the necessary views of the complete machine showing the relative positions of all the machine parts so that they can be assembled easily according to the assembly drawing. Sometimes the machine may be of complicated shape and bigger size consisting of a large number of machine parts. In that case the assembly may be divided into several sub-assemblies for convenience. Then it becomes suitable to show the relative positions of the different parts in the sub-assembly. Later all the sub-assemblies are combined together to obtain the assembly drawing. This assembly drawing consisting of the several sub-assemblies, provides the relative location of the sub-assemblies but cannot give the distinct feature of the individual machine parts as in the sub-assembly drawing. Usually in the assembly drawing the sectional view is shown so that the hidden lines are avoided and the parts can be identified easily.

In the assembly drawing the individual machine part is identified by the number, which is encircled including a leader pointing towards the individual part. At the bottom right corner a list of the machine parts are provided showing the part number, part name, material and quantity usually. A revision table is provided to record modification of the individual machine part. In the assembly drawing usually no dimension is shown but sometimes for convenience overall dimensions may be given between the centers of the machine parts or from one part to another one.

10.4 *Standard parts*

To perform drawings it is necessary to use the standard items, where applicable. There are many standard components such as, angles, channels, sheets, threads, keys, bearings, etc. It is necessary to have the clear idea about the standard components for performing drawings. The tables providing the dimensions of the various essential standard items are given in the appendices.

Example Problems

Prob.10.1: Make a complete working drawing of a belt-drive as shown in the following Fig. P10.1. Give the necessary limits and fits consistent with the work for smooth operation in accordance with ANSI/ISO. Also choose the appropriate materials for each of the components. Fillets and rounds are 3 mm, where applicable.

Note: For solution see Figures S10.1a to S10.1h in the following section of ***Solutions for Example Problems.***

Prob.10.2: An assembly drawing of a cranking mechanism is shown in Figure P10.2 as follows. Make the detail drawings of the parts to be manufactured (Part nos. 1 to 8) in accordance with the list in the drawing shown. Choose suitable dimensions for the detail drawings to make them compatible with the assembly. Make any reasonable assumption, if necessary for your solution.

Note: For solution see Figures S10.2a to S10.2g in the following section of ***Solutions for Example Problems.***

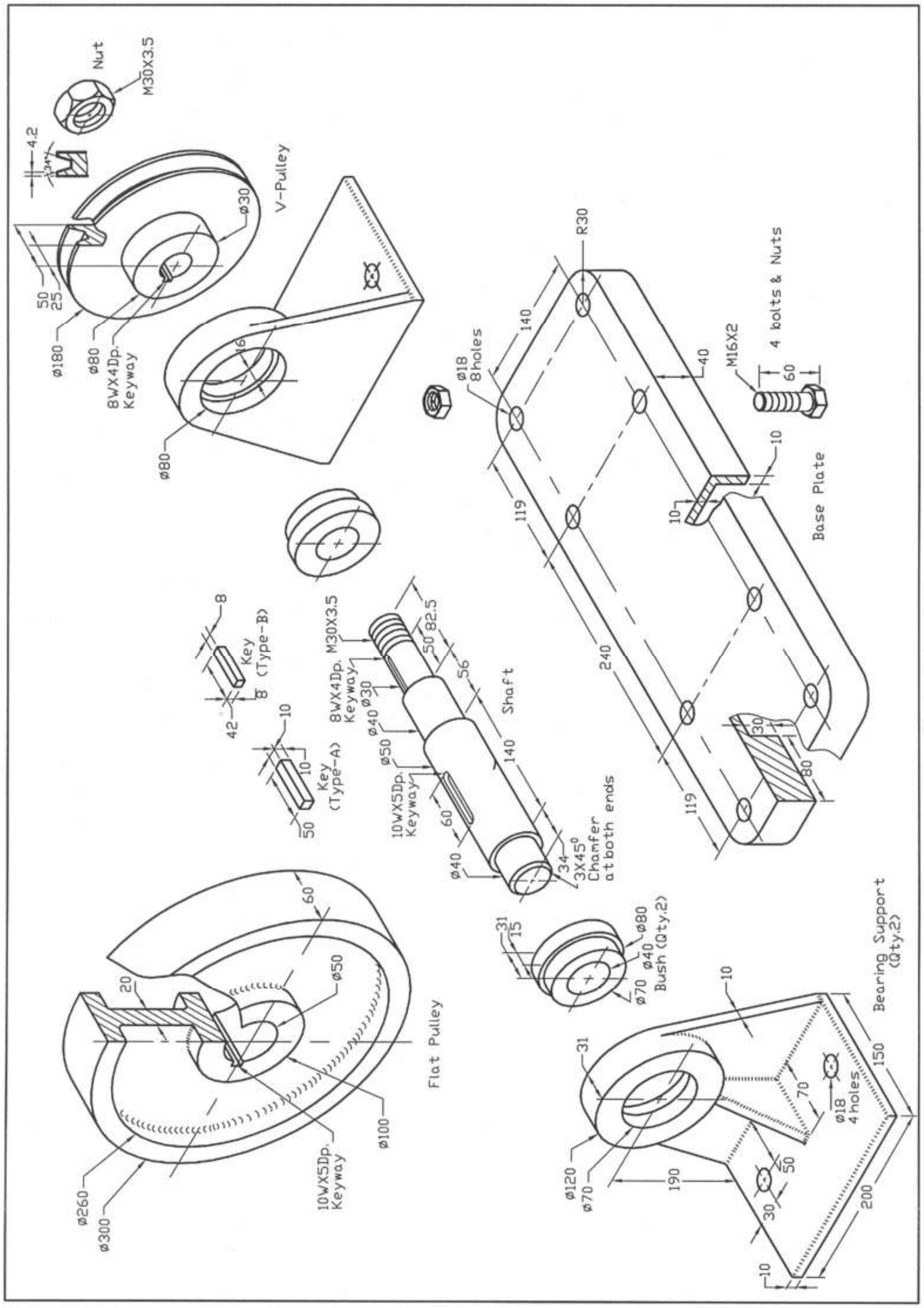
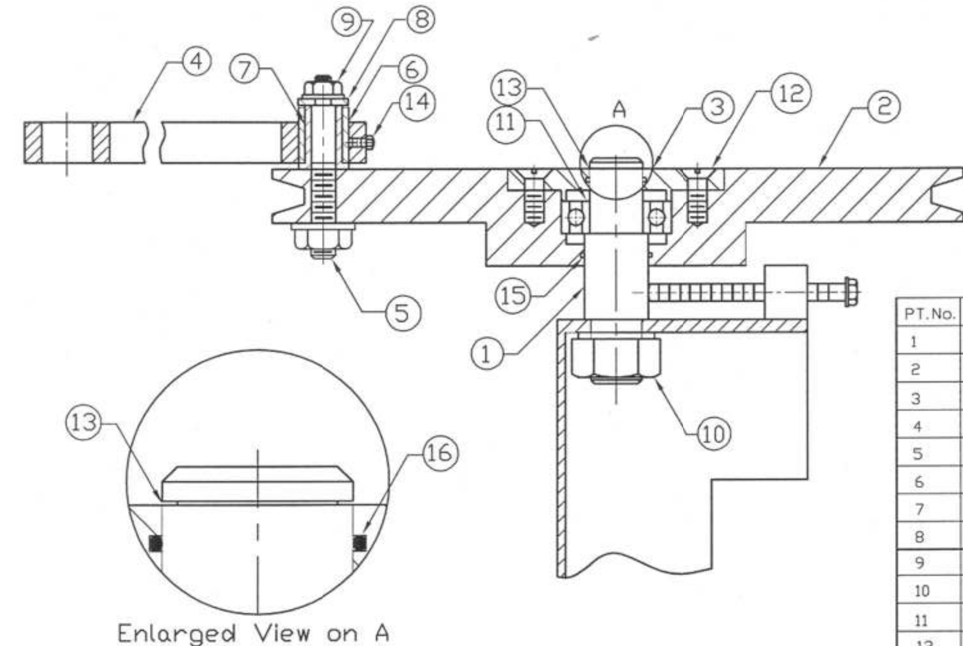


Fig. P10.1: Belt-Drive



PT.No.	Part Name	Mat.	Qty.	Notes
1	Pulley Shaft	SAE 304	1	
2	Pulley	SAE 304	1	
3	Bearing Cover	SAE 304	1	
4	Link	SAE 304	1	
5	Roller Shaft	SAE 304	1	
6	Roller	SAE 304	1	
7	Bushing	Gunmetal	1	
8	Washer	Gunmetal	1	
9	Nut & washer For Roller Shaft	SAE 304	1	M8x1.25
10	Nut & washer	SAE 304	1	
11	Bearing	SAE 304	1	Br.No.6205
12	Fixing Screw	SAE 304	4	M8x1.25
13	Retaining Ring	SAE 304	1	M5100-25
14	Shear Screw	SAE 304	1	M4x0.7,L=15mm
15	O-ring	Rubber	1	ID=28mm W=3mm
16	O-ring	Rubber	1	ID=24mm W=3mm

Name: Cranking Mechanism		Scale: 1:4
Drawn by: ACM	Checked by: MQI	Approved by: DKD
BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		

Fig. P10.2: Assembly Drawing of Cranking Mechanism

Solutions for Example Problems

Solution of P10.1 (Contd.)

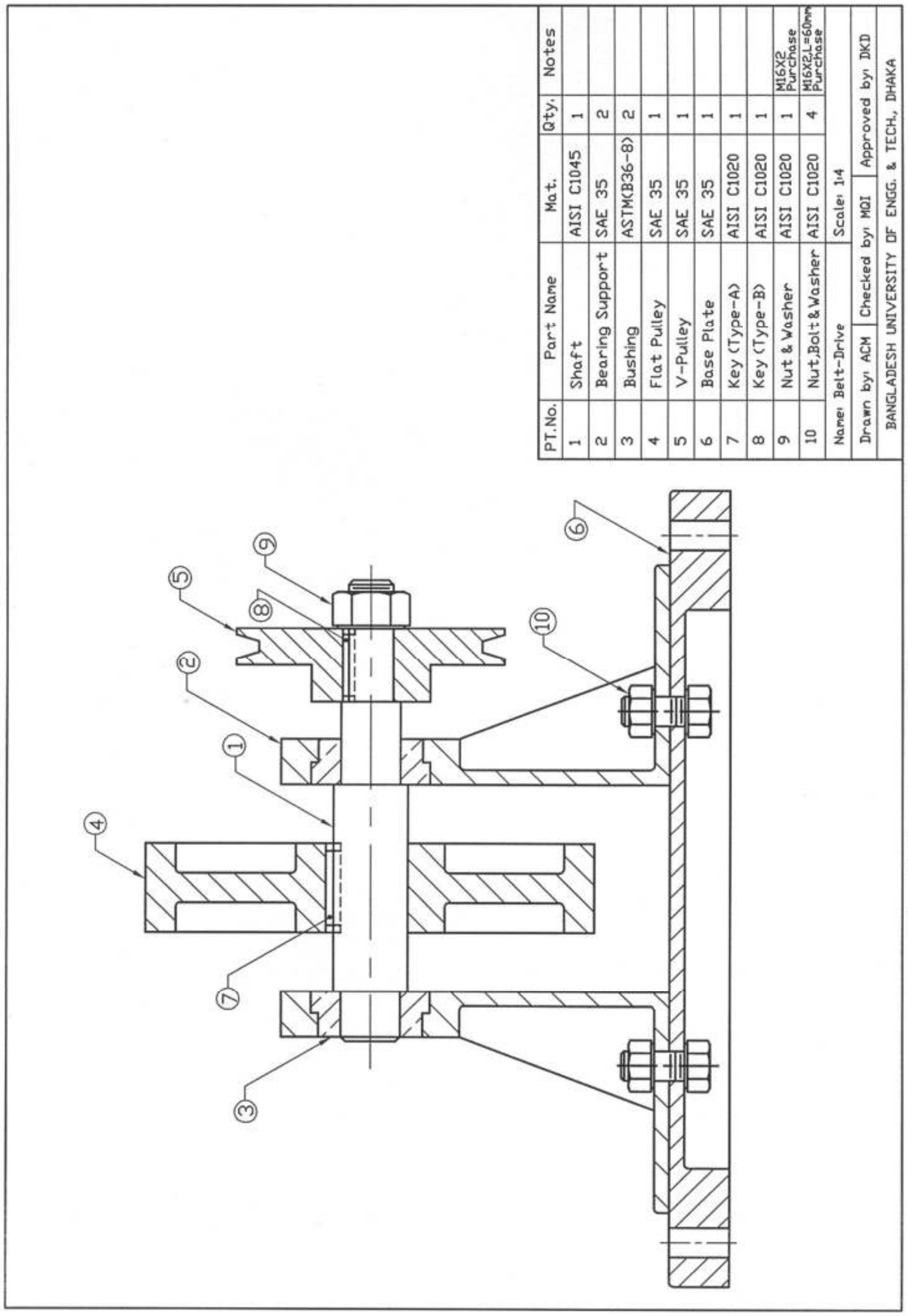


Fig. S10.1a: Assembly Drawing of Belt-Drive

Solution of P10.1 (Contd.)

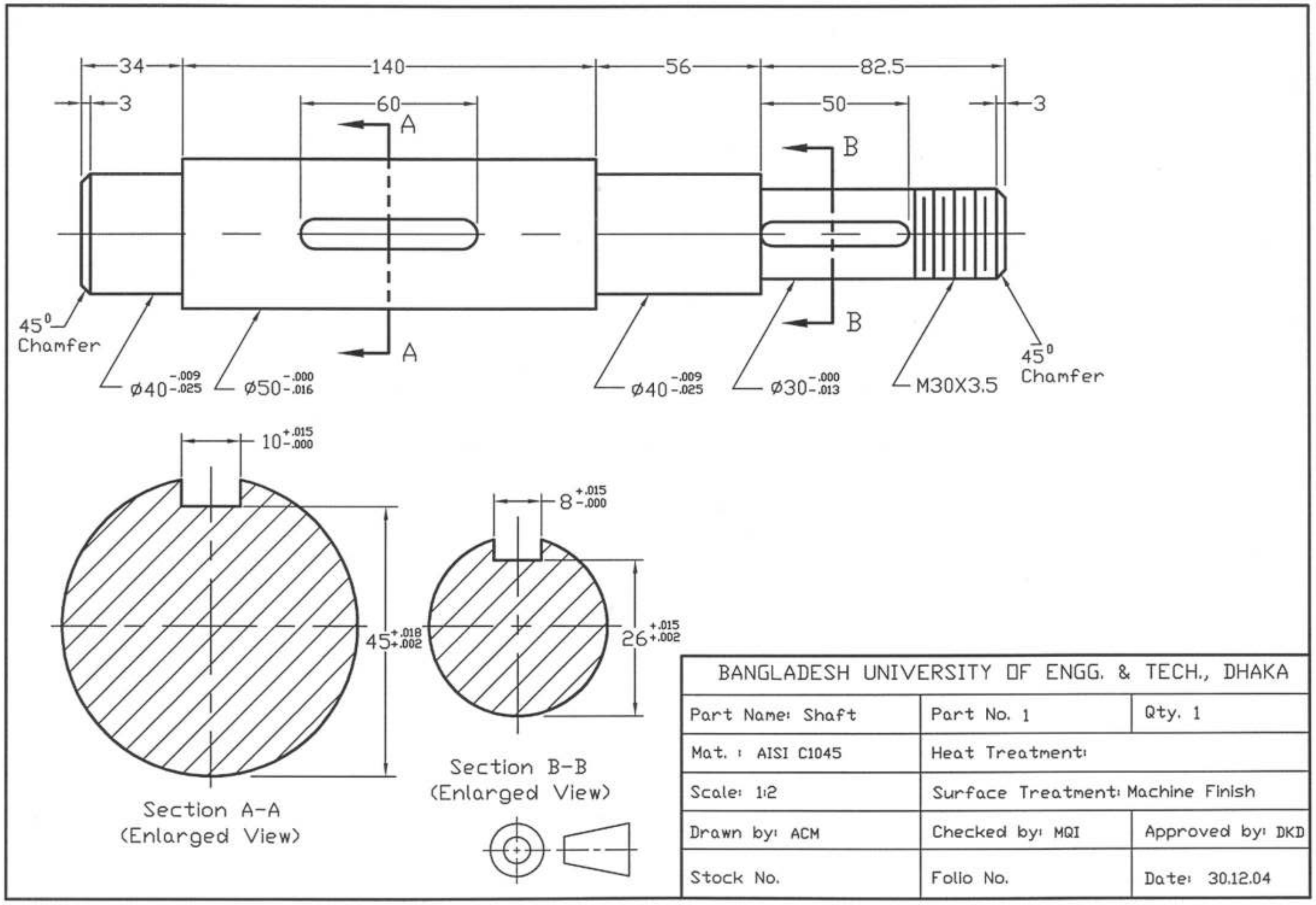
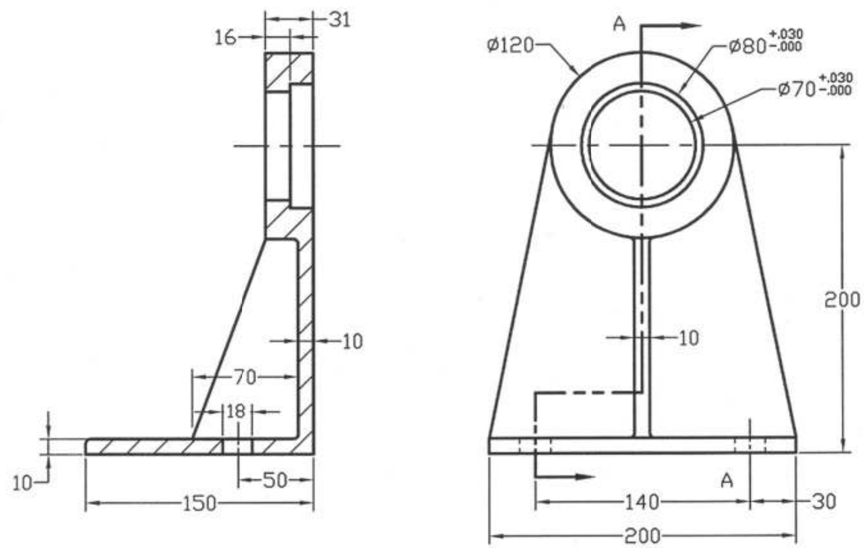
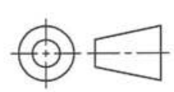


Fig. S10.1b: Detail Drawing of Shaft

Solution of P10.1 (Contd.)





BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: Bearing Support	Part No. 2	Qty. 2
Mat. : SAE 35	Heat Treatment:	
Scale: 1:2	Surface Treatment: Enamel Painted	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 30.12.04

Fig. S10.1c: Detail Drawing of Bearing Support

Solution of P10.1 (Contd.)

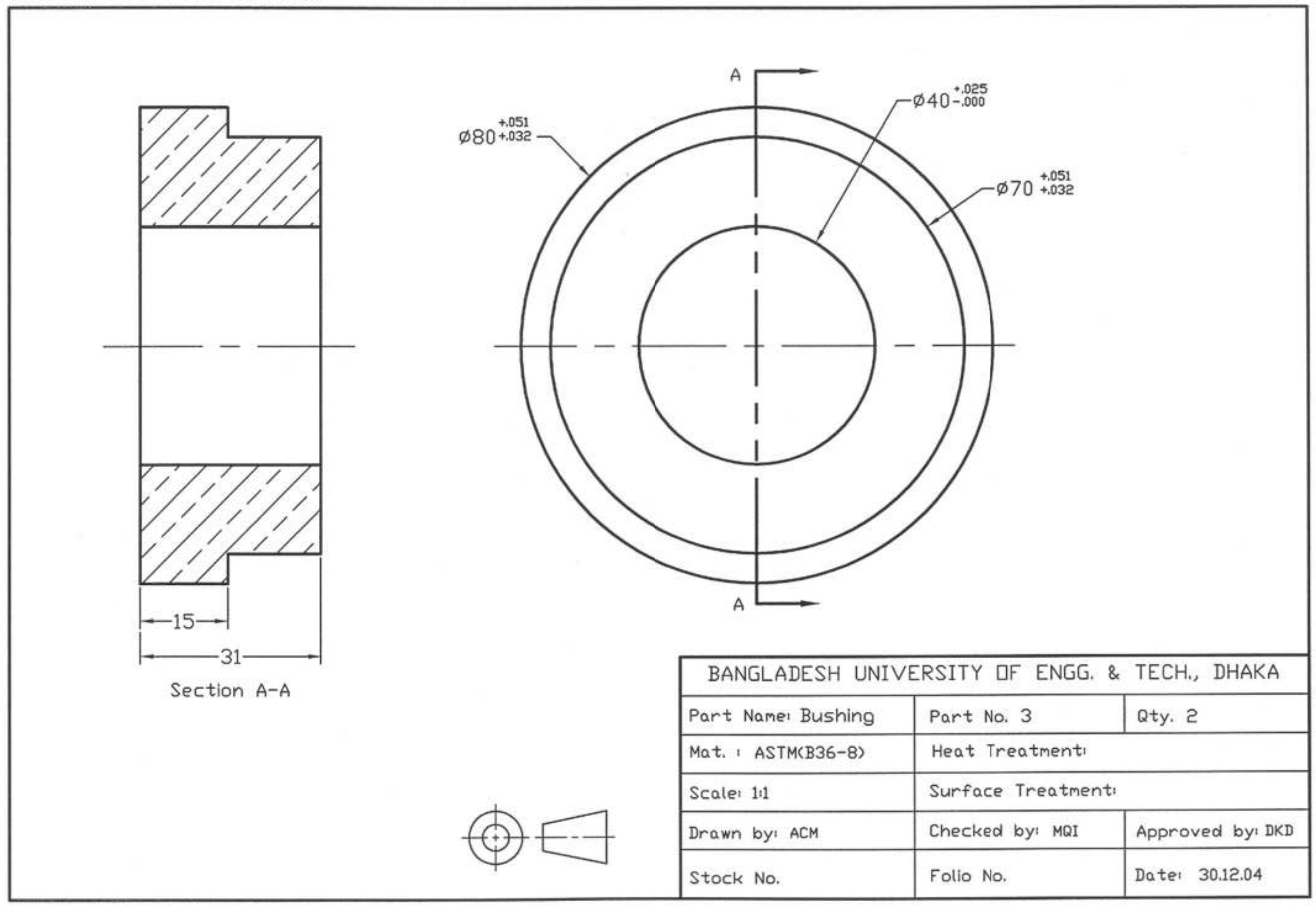
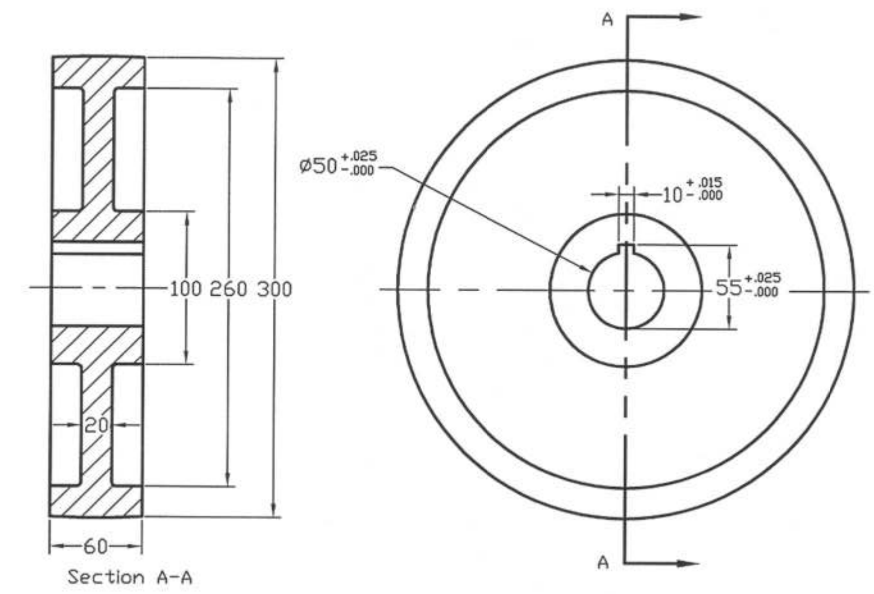
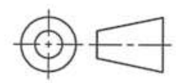


Fig. S10.1d: Detail Drawing of Bushing

Solution of P10.1 (Contd.)

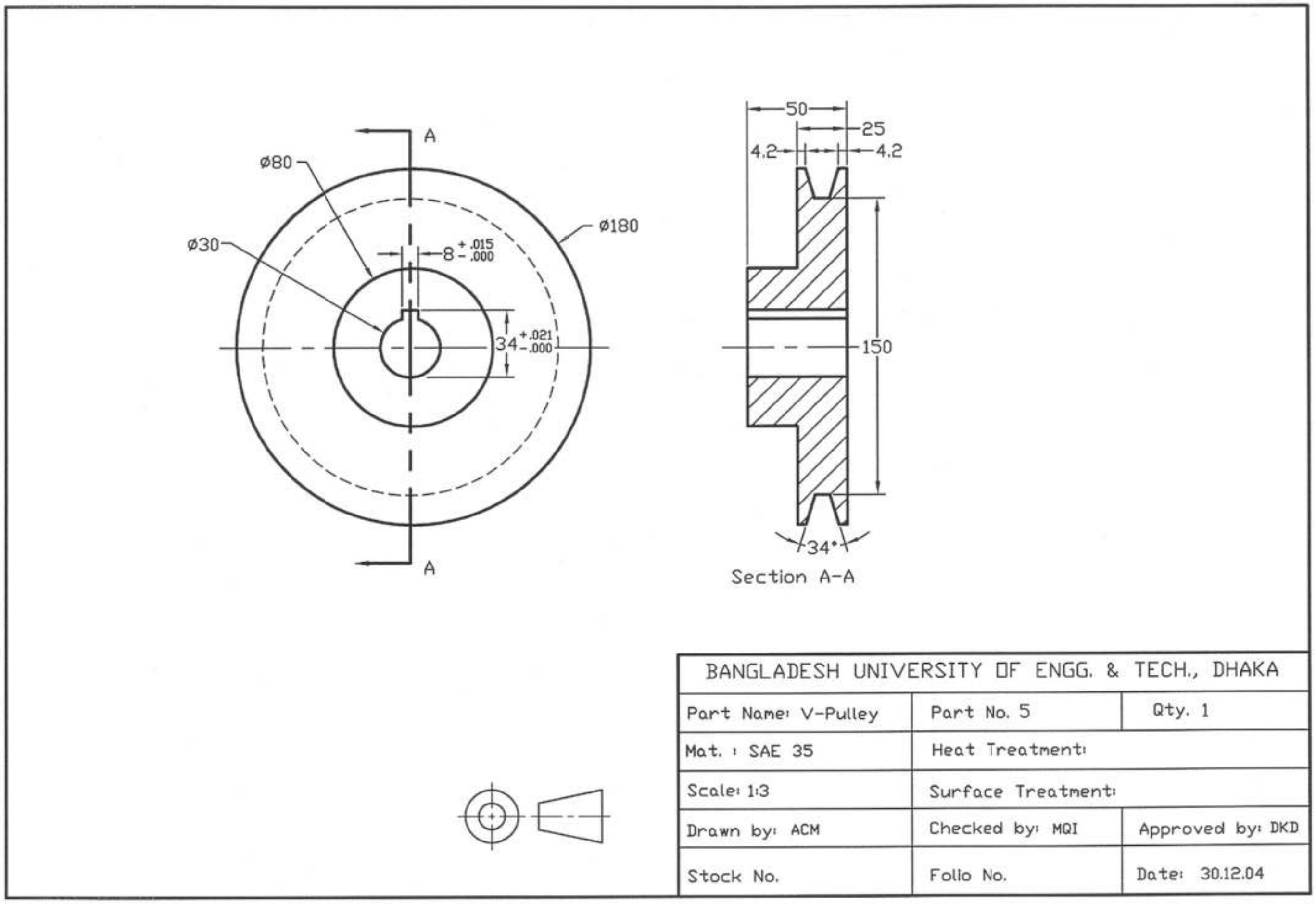




BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: Flat Pulley	Part No. 4	Qty. 1
Mat. : SAE 35	Heat Treatment:	
Scale: 1:4	Surface Treatment: Enamel Painted	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 30.12.04

Fig. S10.1e: Detail Drawing of Flat Pulley

Solution of P10.1 (Contd.)



BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: V-Pulley	Part No. 5	Qty. 1
Mat. : SAE 35	Heat Treatment:	
Scale: 1:3	Surface Treatment:	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 30.12.04

Fig. S10.1f: Detail Drawing of V-Pulley

Solution of P10.1 (Contd.)

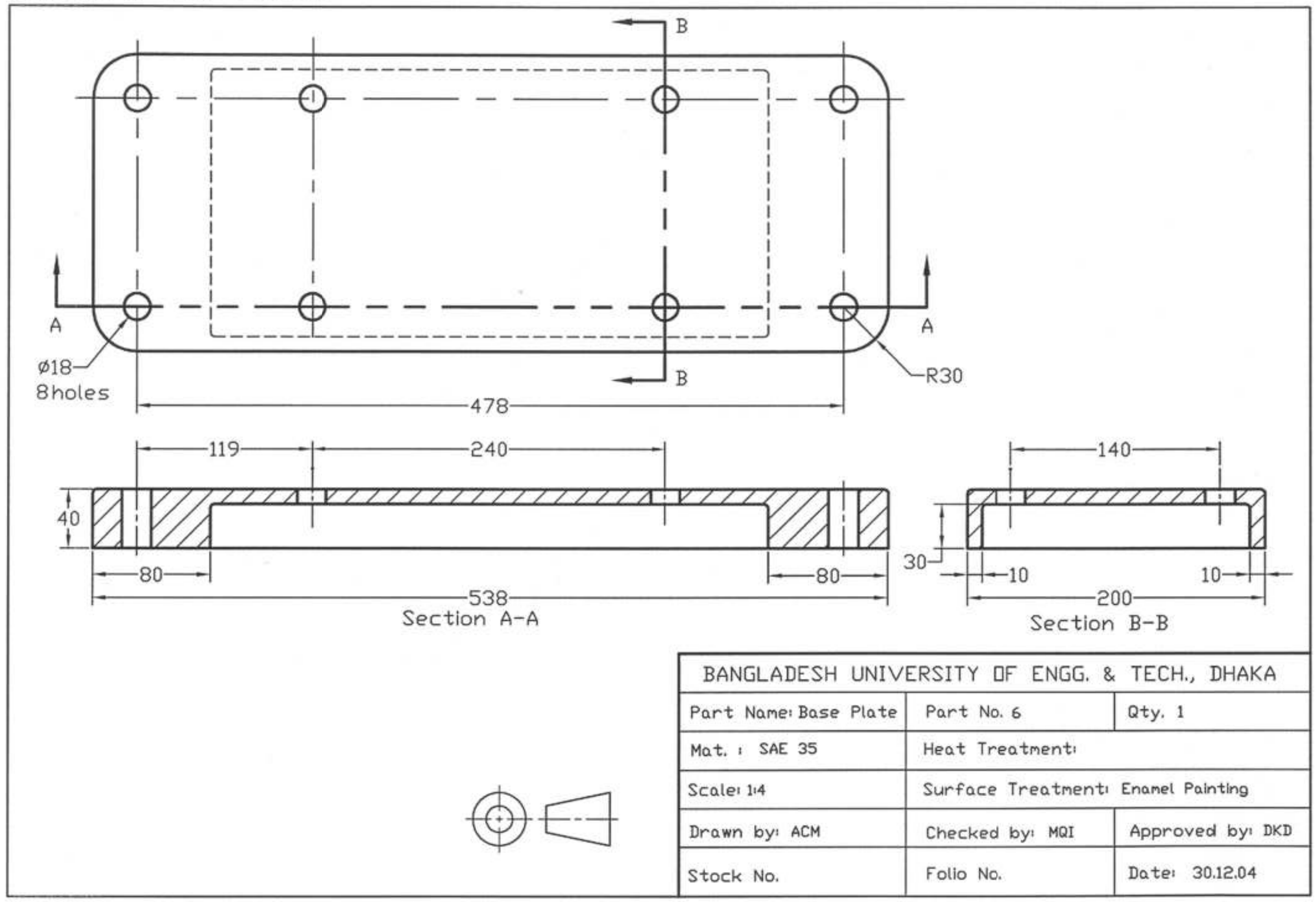
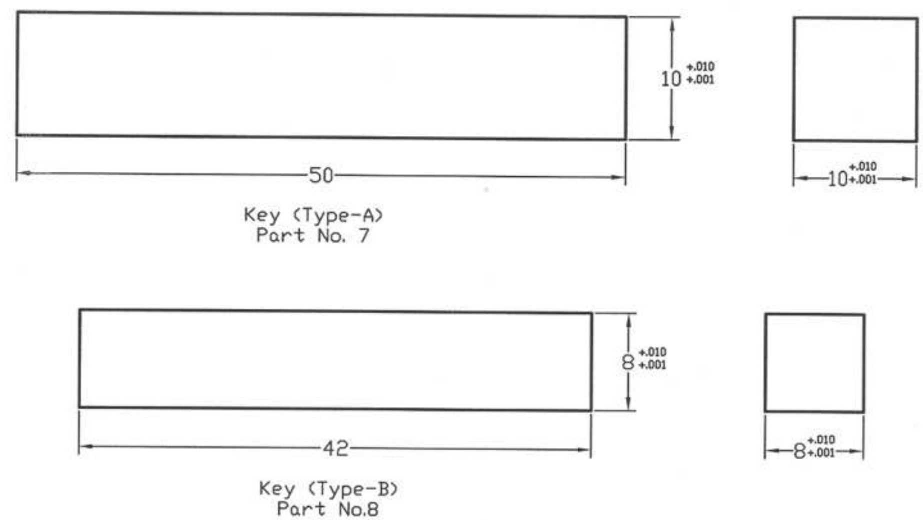
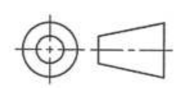


Fig. S10.1g: Detail Drawing of Base Plate

Solution of P10.1 (Contd.)

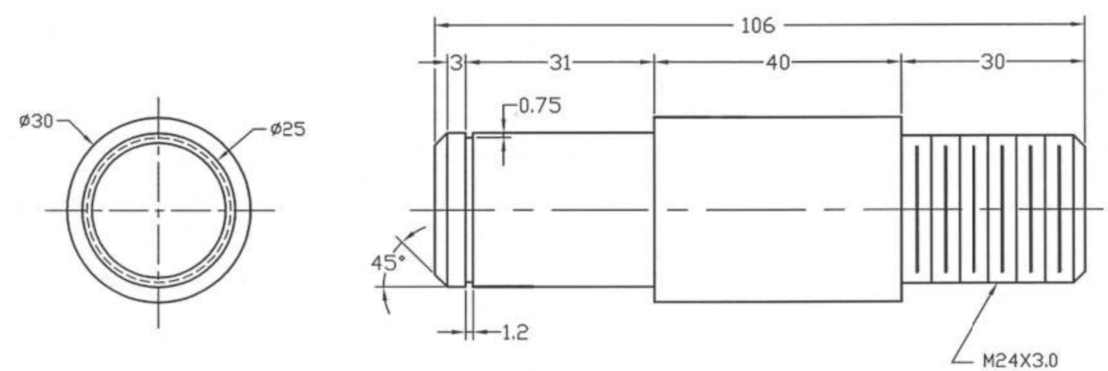


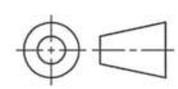


BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: Key	Part No. 7,8	Qty. 2
Mat. : AISI C1020	Heat Treatment:	
Scale: 2:1	Surface Treatment:	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 30.12.04

Fig. S10.1h: Detail Drawing of Key

Solution of P10.2 (Contd.)





BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: Pulley Shaft	Part No. 1	Qty. 1
Mat. : SAE 304	Heat Treatment:	
Scale: 1:1	Surface Treatment: Machine Finish	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 27.06.05

Fig. S10.2a: Detail Drawing of Pulley Shaft

Solution of P10.2 (Contd.)

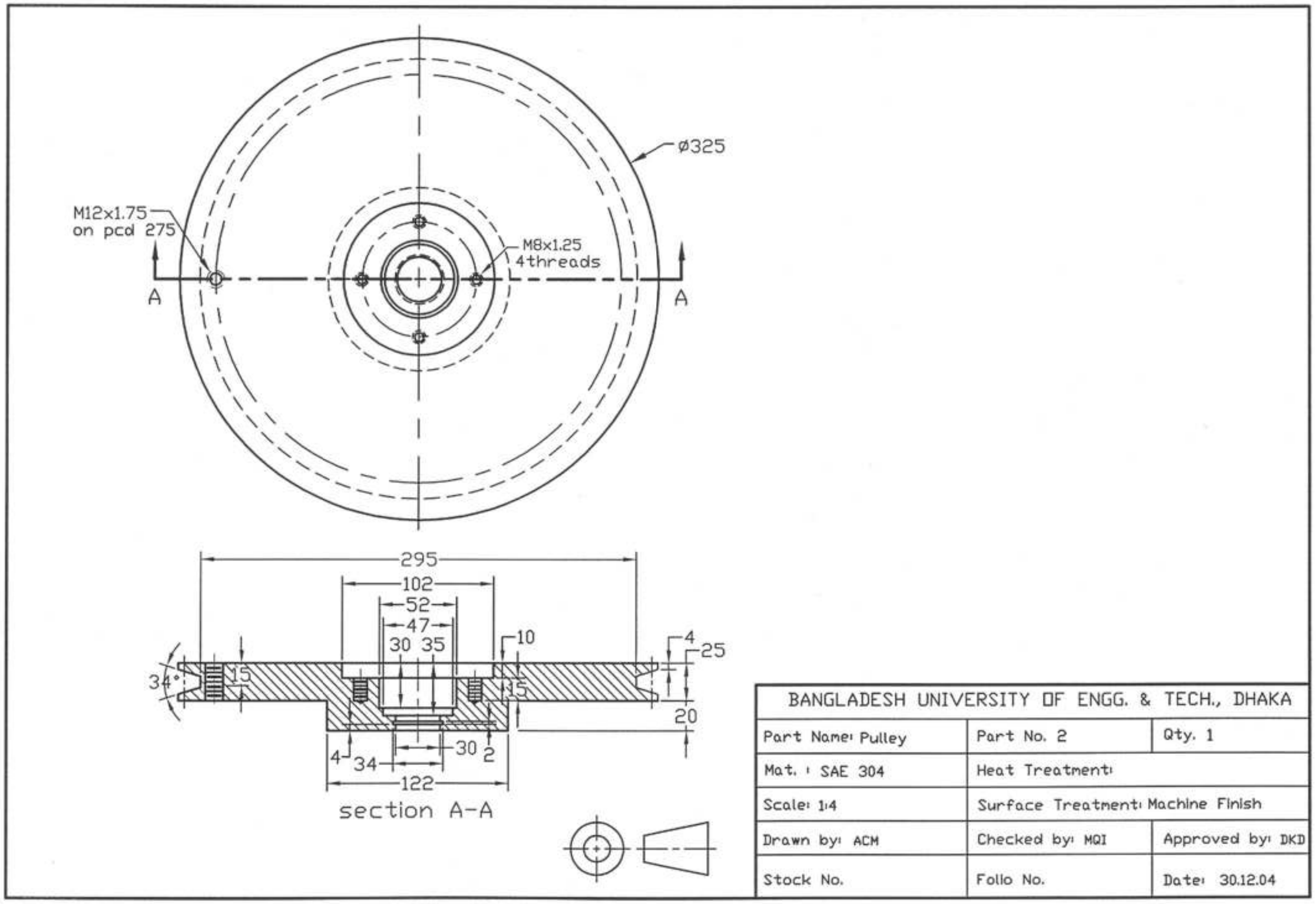
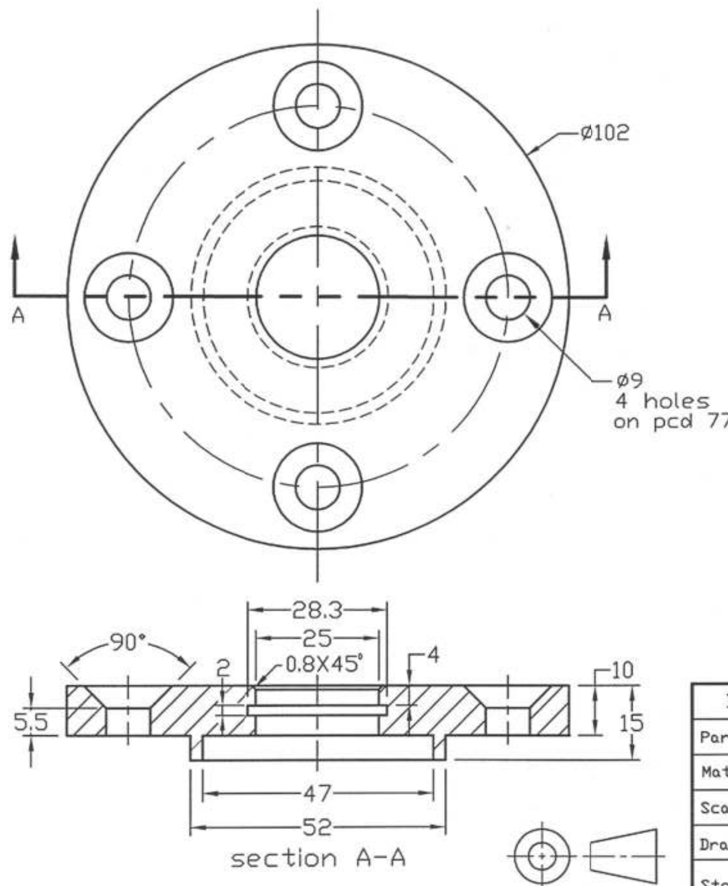


Fig. S10.2b: Detail Drawing of Pulley

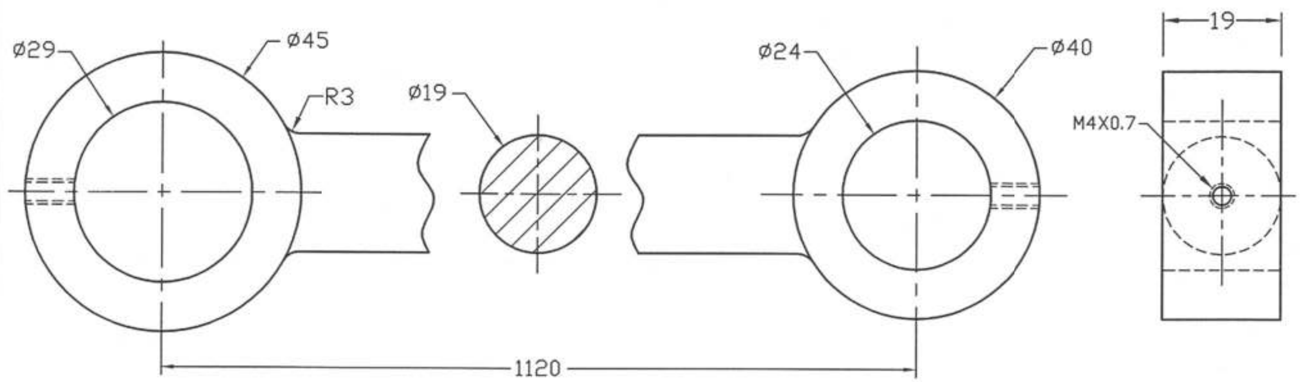
Solution of P10.2 (Contd.)

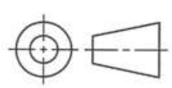


BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: Bearing Cover	Part No. 3	Qty. 1
Mat. : SAE 304	Heat Treatment:	
Scale: 1: 1.25	Surface Treatment: Machine Finish	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 30.12.04

Fig. S10.2c: Detail Drawing of Bearing Cover

Solution of P10.2 (Contd.)





BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: Link	Part No. 4	Qty. 1
Mat. : SAE 304	Heat Treatment:	
Scale: 1:1	Surface Treatment: Machine Finish	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 30.12.04

Fig. S10.2d: Detail Drawing of Link

Solution of P10.2 (Contd.)

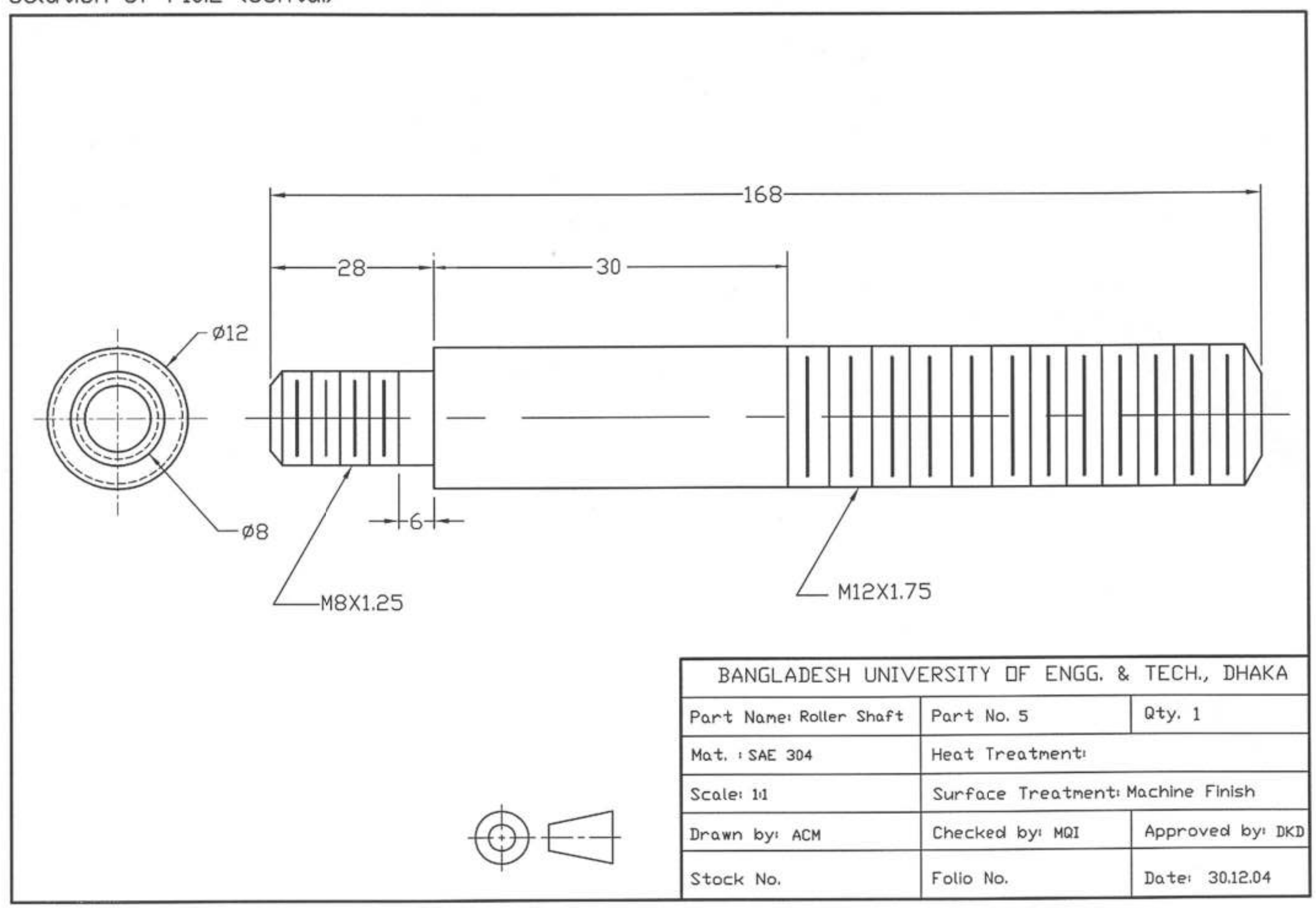
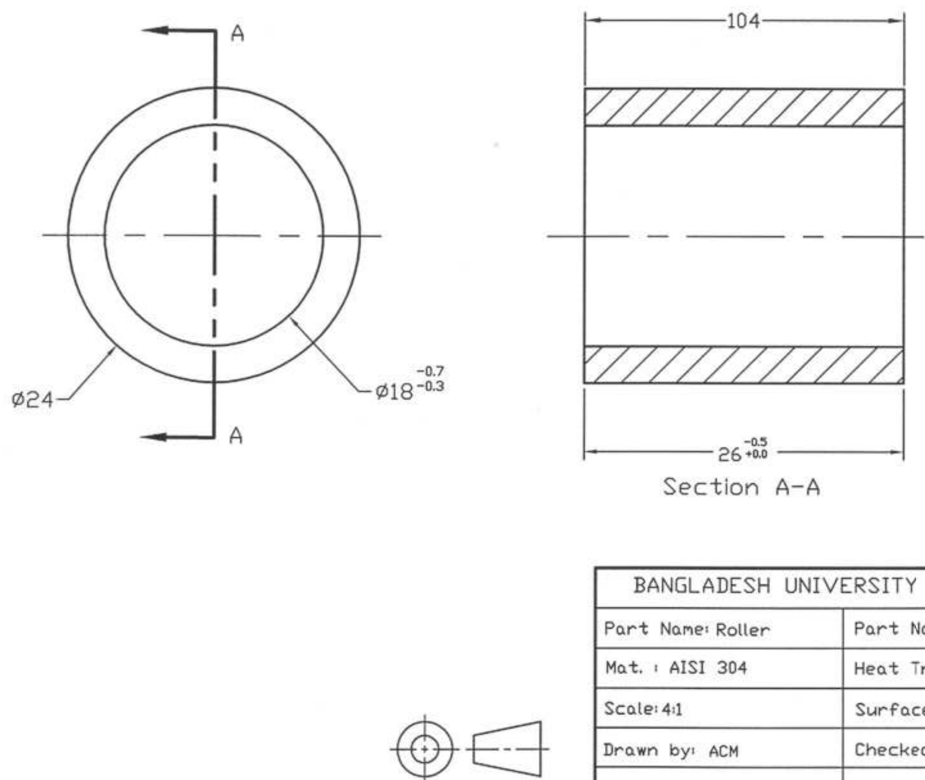


Fig. S10.2e: Detail Drawing of Roller Shaft

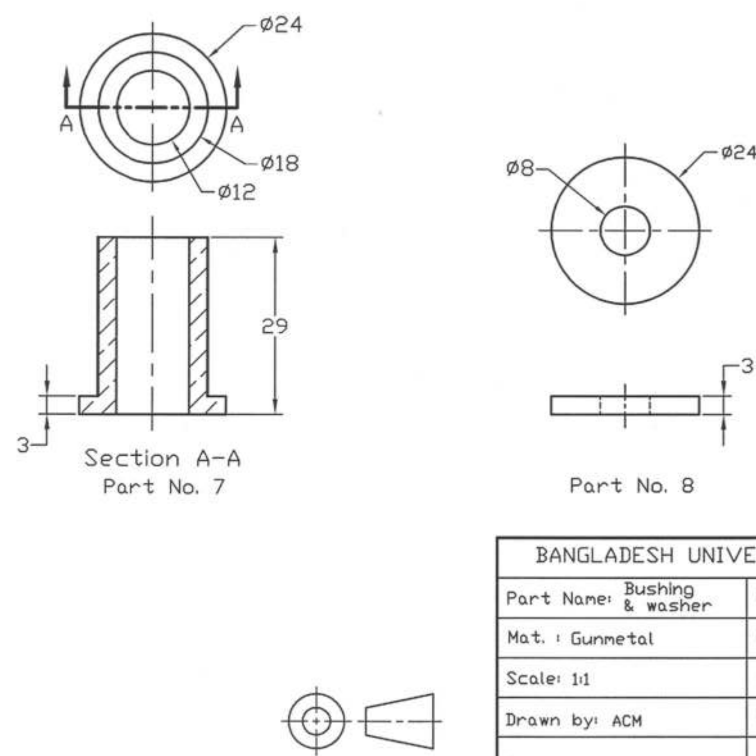
Solution of P10.2 (Contd.)



BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: Roller	Part No. 6	Qty. 1
Mat. : AISI 304	Heat Treatment:	
Scale: 4:1	Surface Treatment: Machine Finish	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 30.12.04

Fig. S10.2f: Detail Drawing of Roller

Solution of P10.2 (Contd.)



BANGLADESH UNIVERSITY OF ENGG. & TECH., DHAKA		
Part Name: Bushing & washer	Part No. 7, 8	Qty. 1
Mat. : Gunmetal	Heat Treatment:	
Scale: 1:1	Surface Treatment: Machine Finish	
Drawn by: ACM	Checked by: MQI	Approved by: DKD
Stock No.	Folio No.	Date: 30.12.04

Fig. S10.2g: Detail Drawing of Bushing and Washer

Problems

Prob.10.3: Give a complete working drawing of the relief valve shown in the following Fig.P10.3.

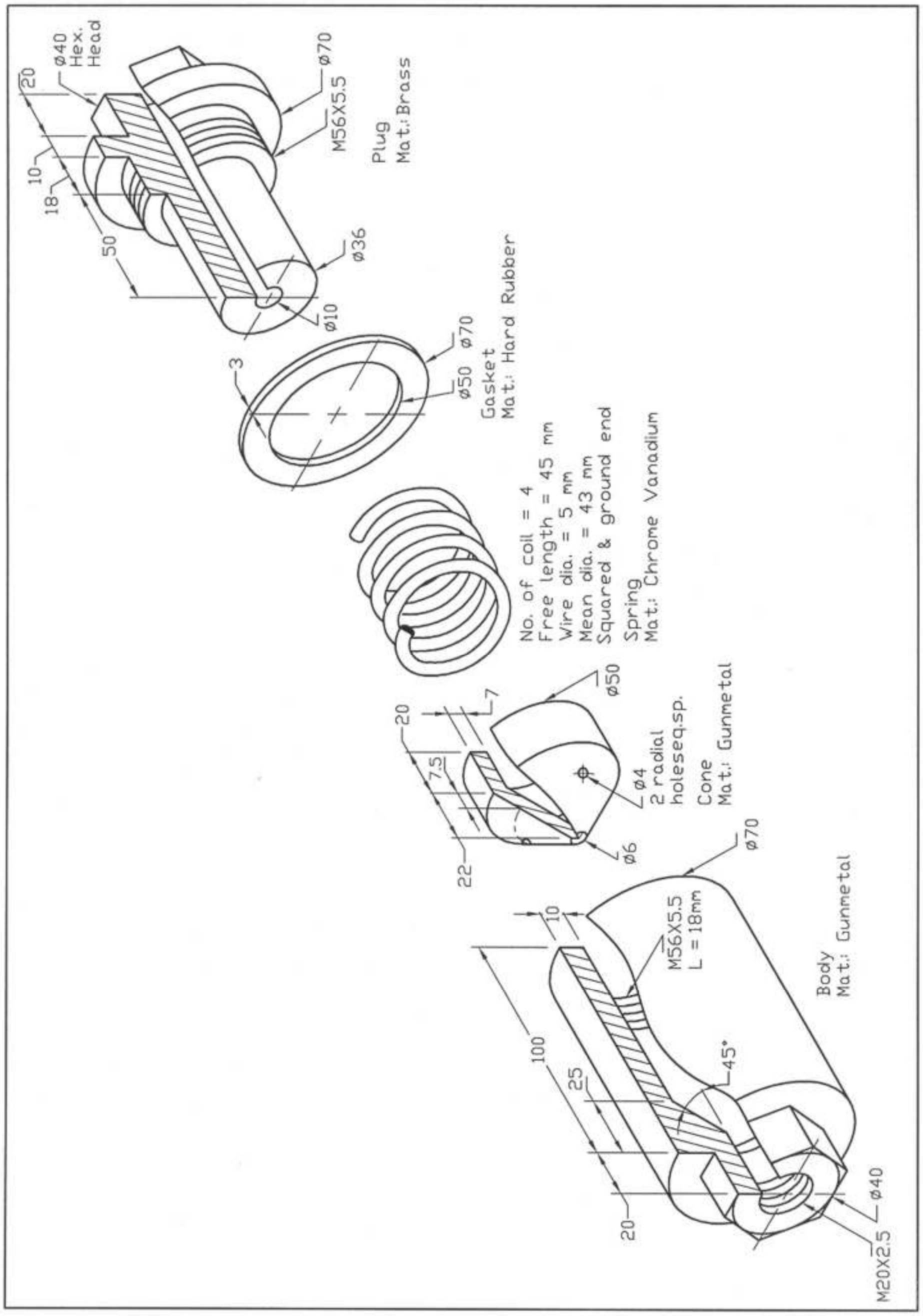


Fig. P10.3: Releif Valve

CHAPTER 11

BASICS OF AUTOCAD

11.1 *Introducing AutoCAD*

AutoCAD, created by Autodesk Corporation, is the most widely used technical drawing program. All over the world people are using this AutoCAD to generate different kinds of drawings. The major disciplines that use AutoCAD are:

- Mechanical
- Architecture
- GIS (Geographic Information Systems)
- Facilities Management
- Electrical/Electronic
- Multimedia etc.

AutoCAD creates and saves file in 'vector' format. Vector files store an image by mathematically defining an object that makes up the image. Opening AutoCAD and the different basic components of AutoCAD will be introduced in short in this section.

11.1.1 *Opening AutoCAD*

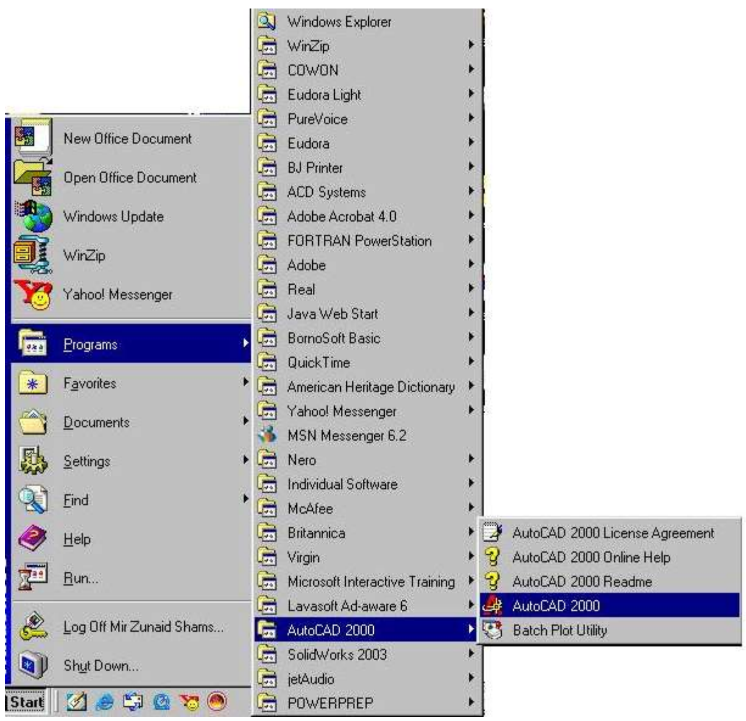


Figure 11.1: Steps to Open Auto CAD

Having AutoCAD installed in computer, one can easily open AutoCAD by clicking on the AutoCAD icon, which looks like for AutoCAD 2000 version. It can also be opened selecting the following steps as mentioned below (Figure 11.1).

Start → Programs → AutoCAD 2000 → AutoCAD 2000

Opening AutoCAD will display AutoCAD window showing Startup dialogue box (Figure 11.2). Now selecting the **Start from Scratch** button and the radio button for the desired unit (English or Metric) followed by the **OK** button in the Startup dialog box, the Startup dialog box disappears from the window and the window becomes ready for the desired drawing.

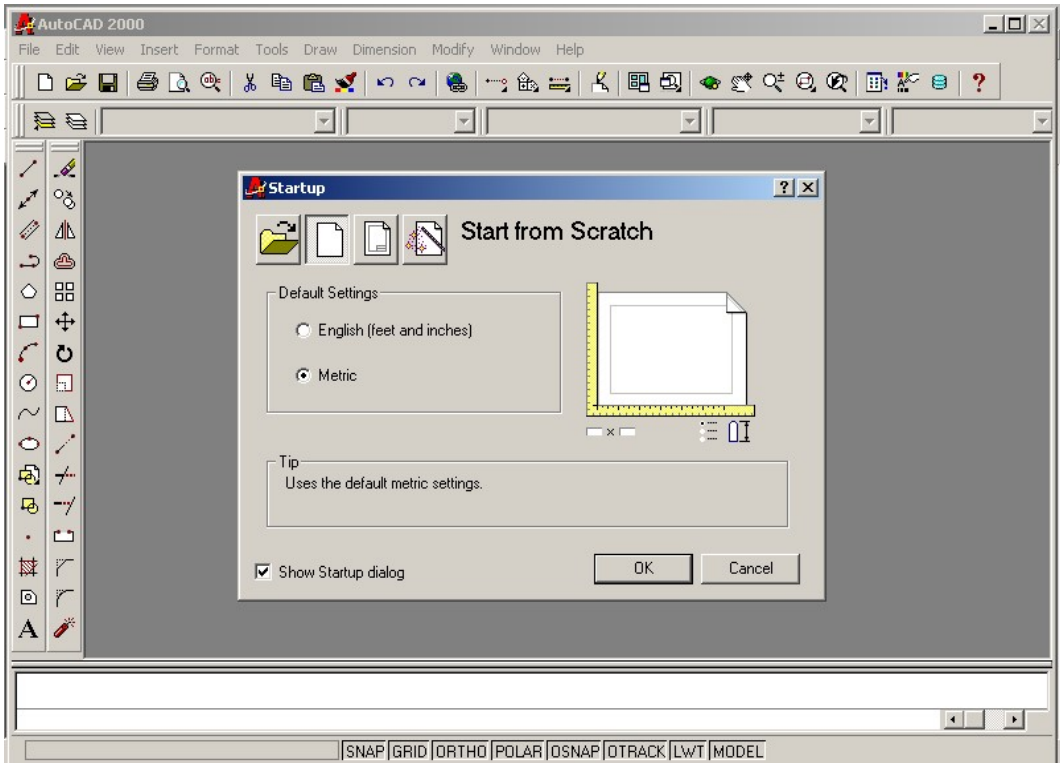


Figure 11.2: AutoCAD Window Showing Startup Dialogue Box

A portion of the Startup dialog box is shown in Figure 11.3. Of the four buttons located at the top left corner of the Startup dialog box, the first one from the left is for **Open a Drawing**, the second one is for **Start from Scratch**, the third one is for **Use a Template** and the fourth one is for **Use a Wizard**. By clearing check box of **Show Startup dialog** of the Startup dialog box (Figure 11.2), it will disappear from the window. However, it may be displayed again by selecting the following steps as mentioned below.

Tools → **Options** → **System** tab → **Show Startup dialog** check box

AutoCAD window showing **Open a Drawing** is presented in Figure 11.4. When the Open a Drawing button is selected as shown in this figure the four recent files will be displayed. Now one can select the desired one and click the **OK** button, the file will be ready for work. If the desired file is not available here one has to click on the **Browse** button, a Select File dialog box will appear where from the required file has to be selected followed by **Open** button.

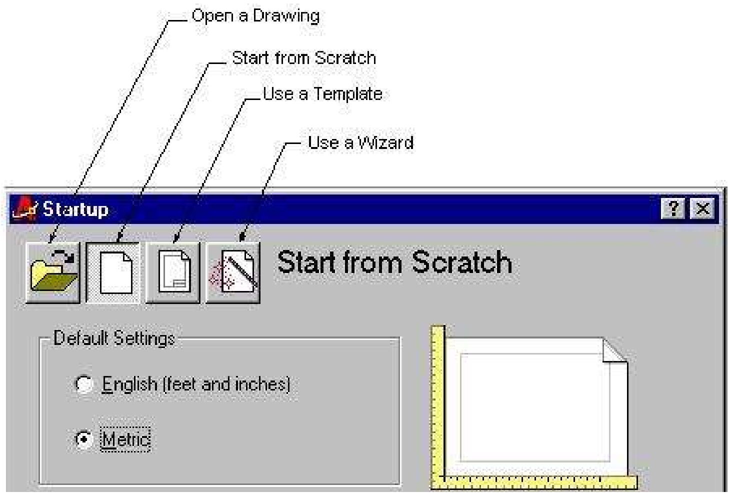


Figure 11.3: A Portion of Startup Dialog Box

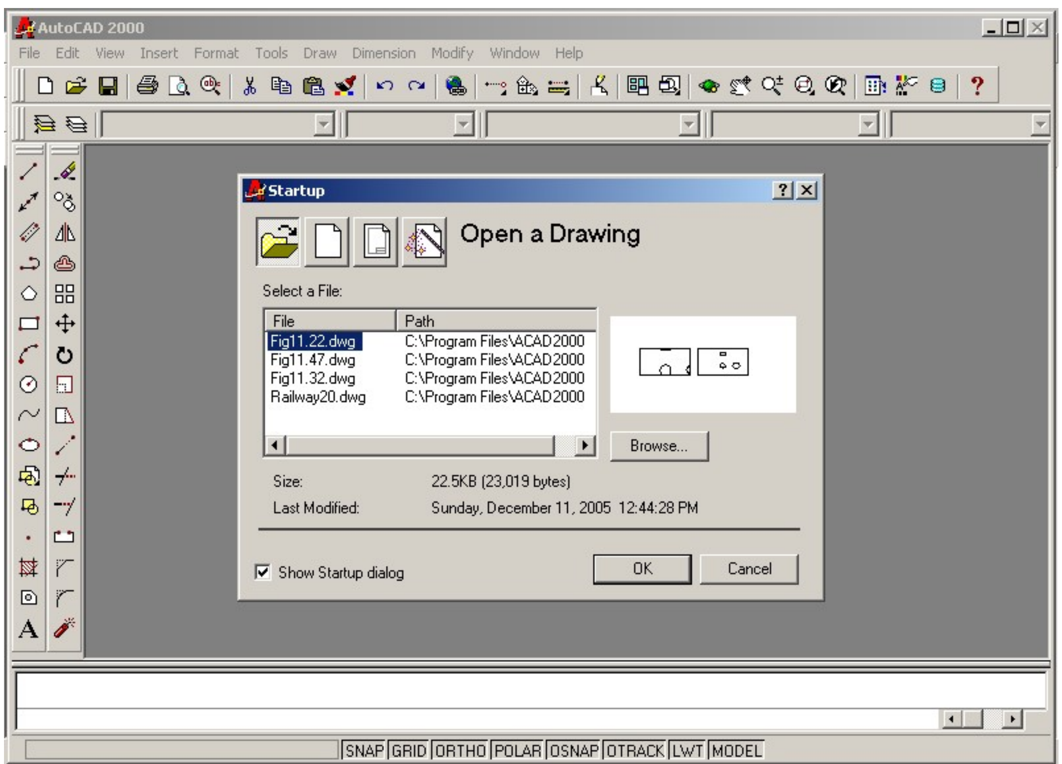


Figure 11.4: AutoCAD Window Showing Open a Drawing

11.1.2 ***Components of AutoCAD Window***

In Figure 11.5 the various components of the AutoCAD window have been introduced. It is the basic view of AutoCAD window. However, depending on the user's preference there may be slight difference in orientation and arrangement of the toolbars. It contains title bar, menu bar, standard toolbar, object properties toolbar, draw toolbar, modify toolbar, command area (command prompts area) and status bar etc. as shown. The area at the middle of the window is known as the drawing area.

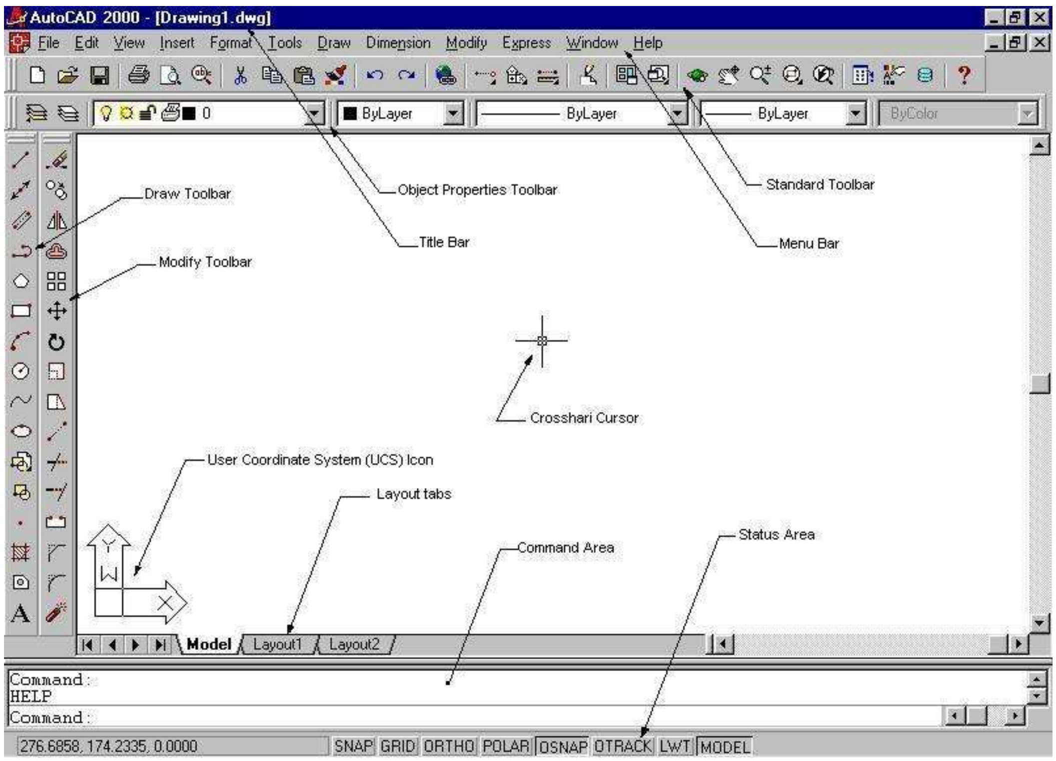


Figure 11.5: Components of AutoCAD Window

11.1.3 *Crosshair Cursor*

In AutoCAD drawing area, the crosshair looks like two intersecting lines with a small square box at their intersection. At the bottom of the AutoCAD screen, at the left end of the status bar, the X, Y coordinates are seen, the values of them change as the cross hair moves within the drawing area indicating its location as shown in Figure 11.6. As the cross hair cursor is moved to select any command, the cursor turns into the shape of an arrow.

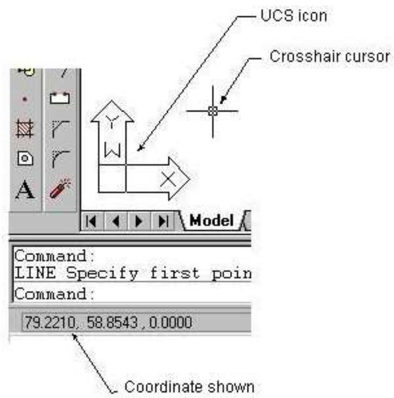


Figure 11.6: Crosshair Cursor

11.1.4 *Menu Bar*

The menu bar of AutoCAD window contains a number of titles such as, File, Edit, View, Insert, Format, Tools, Draw, Dimension, Modify, Window and Help. If any one of the menu bar titles is selected, it is highlighted and a menu is displayed just below the title consisting of a number of commands, which is called the **pulled-down menu**. In Figure 11.7 AutoCAD window displaying **Tools menu** is shown, which is obtained selecting Tools title from the menu bar.

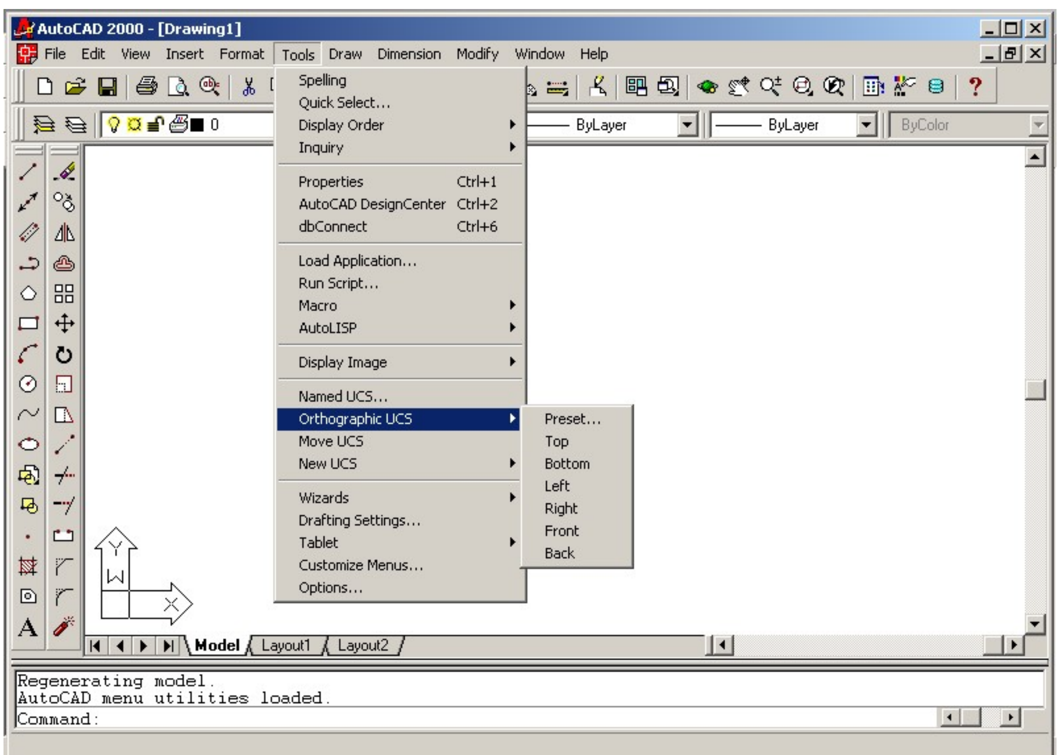


Figure11.7: AutoCAD window Displaying Tools Menu

If one selects Draw title, draw menu will be displayed just below the draw title and so on. At the right side of some of the commands in the pull-down menu there are right arrows. If one takes the cursor on the command having an arrow at its right side, a parallel menu is displayed (Figure 11.7). It is called the **cascading menu**, which provides some command options.

11.1.5 *Short Cut Menu*

When right mouse button is clicked in the drawing area a menu is displayed, which is called the **short cut menu** (Figure 11.8). One can invoke some useful commands from the short cut menu. Once any command is initiated from the toolbar or menu or inserted in the command area, next the same command becomes available in the short cut menu at the top of the list of commands. Thus the available command in the short cut menu is changed according to the requirement of the user. It saves time to use the command from the shortcut menu directly.

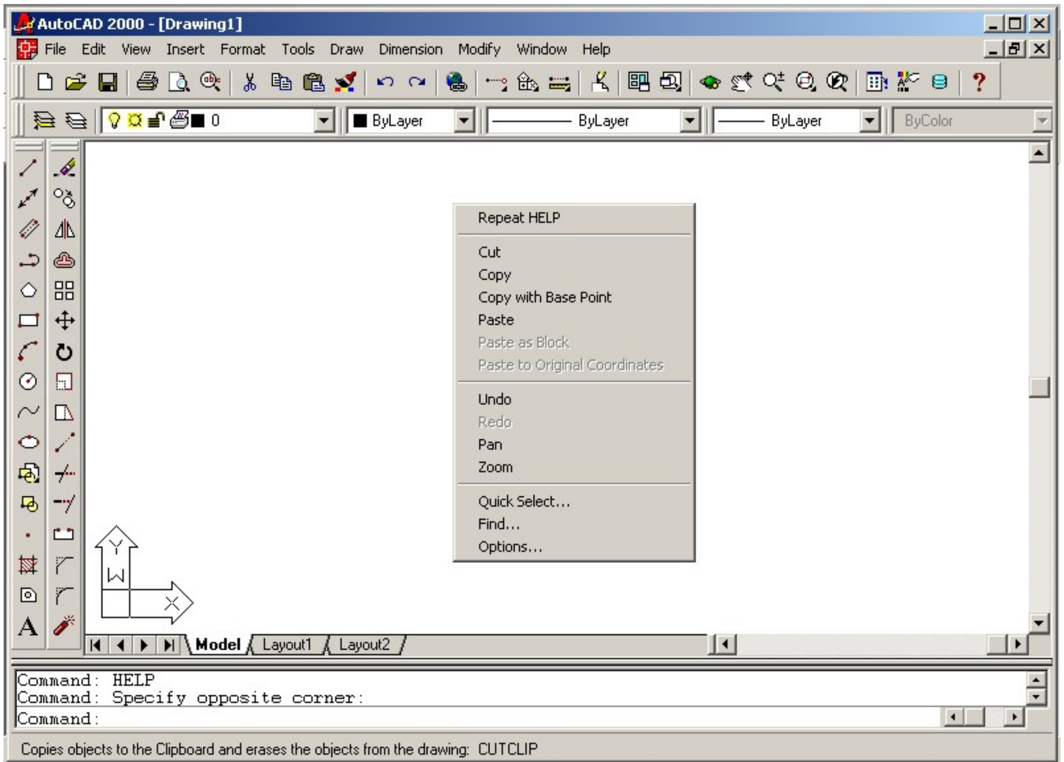


Figure 11.8: AutoCAD Window Showing Shortcut Menu

11.1.6 *Toolbar*

AutoCAD window consists of a number of toolbars. It becomes easier and convenient to invoke command from the toolbar directly. Each toolbar contains a number of icons for various commands. When the cursor is brought on any icon for selection, the command prompts are displayed in the command area. There is a standard toolbar at the top of the window. It contains a number of very useful commands, which are introduced in Figure 11.9.

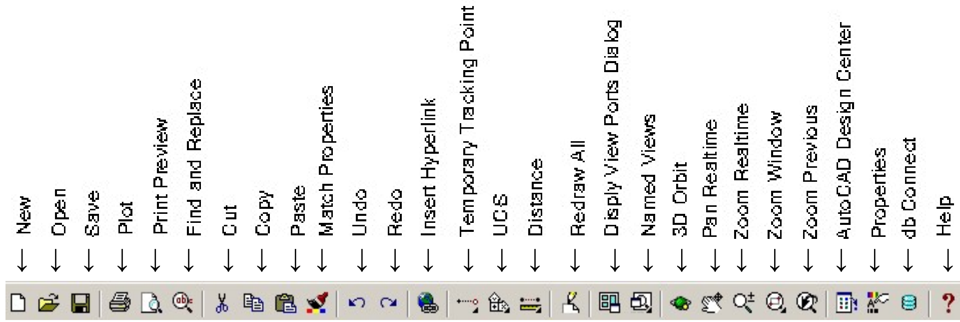


Figure 11.9: Standard Toolbar

In Figure 11.10 and Figure 11.11, draw and modify toolbars are presented respectively. The various icons of the draw and modify commands have been introduced in these figures. One can choose the required command say, line, construction line, erase, copy etc. from these toolbars directly. It becomes time saving and efficient to invoke the command from these toolbars.

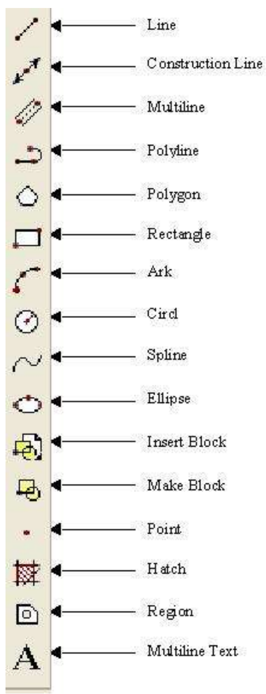


Figure 11.10: Draw Toolbar

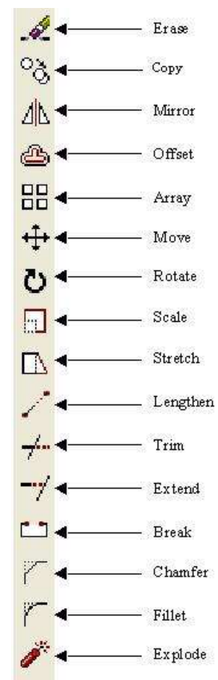


Figure 11.11: Modify Toolbar

In order to turn a Toolbars on and off one has to click on View title in the menu bar, a view menu will appear (Figure 11.12). Now clicking on Toolbars in view menu a Toolbars dialog box will be displayed (Figure 11.13). There are many toolbars in the list of Toolbars dialog box. When the crosscheck box beside the desired toolbar in the toolbars list is selected, that toolbar appears in the AutoCAD window. On the other hand if the crosscheck box is deselected, the toolbar disappears from the window.

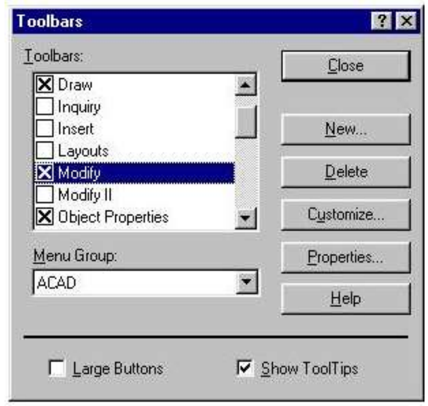


Figure 11.12: View Menu

Figure 11.13: Toolbars Dialog Box

The view of AutoCAD window for every user may not look the same. Some toolbars (Draw and Modify toolbars) as shown in Figure 11.5 may be present; on the other hand some other toolbars may be absent, which are not shown in the figure. One can select the dimension toolbar and insert toolbar etc. from the Toolbars dialog box, if necessary.

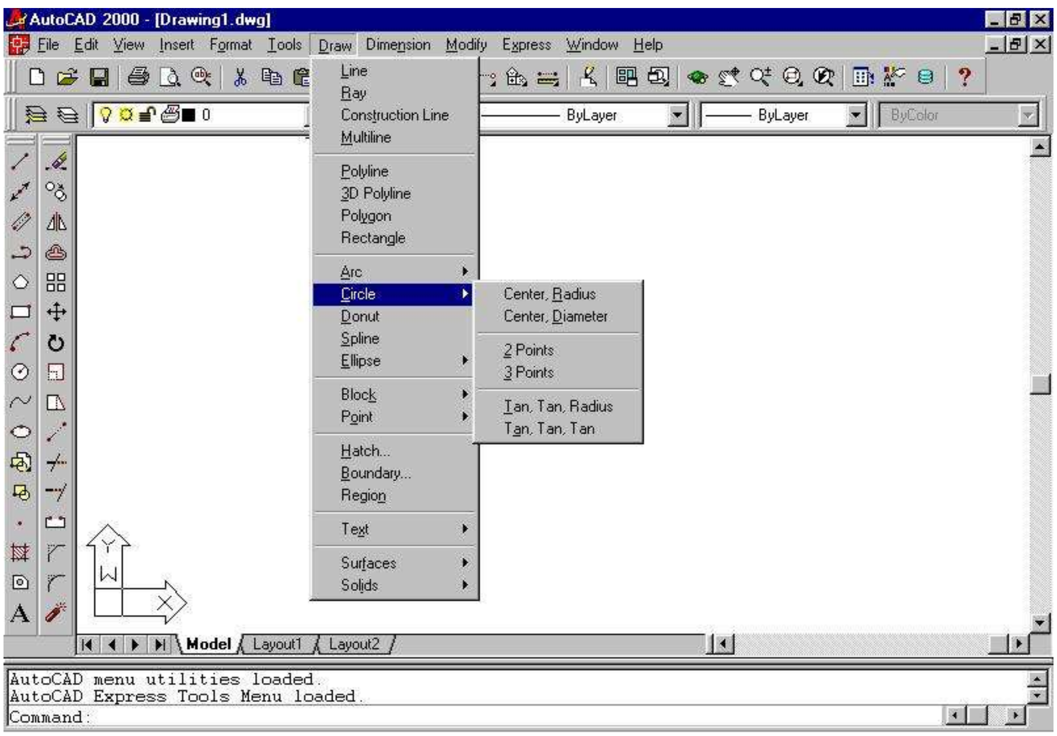


Figure 11.14: Draw Commands from Menu

Commands of Draw Toolbar and Modify Toolbar can also be picked up from the menu bar. In Figure 11.14 the Draw commands in the draw menu is shown. One can find the Modify commands from the modify menu by selecting Modify title in the menu bar. Pull down menus offers more variations and detail in some menus.

11.1.7 *Command Area*

For the beginners it is suggested always to keep an eye on the command area because once an operation is in process, AutoCAD prompts the operator and mentions in the command area what the next step would be. For example, when the user performs an operation to draw a line by clicking the Line command either from the toolbar or from the menu bar, AutoCAD prompts the command in the command area as shown in Figure 11.15. Any command can also be initiated by inserting it directly in the command area.

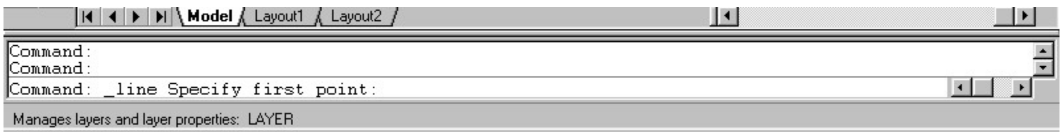


Figure 11.15: Command Area of Auto CAD Window

11.1.8 *Options Dialogue Box*

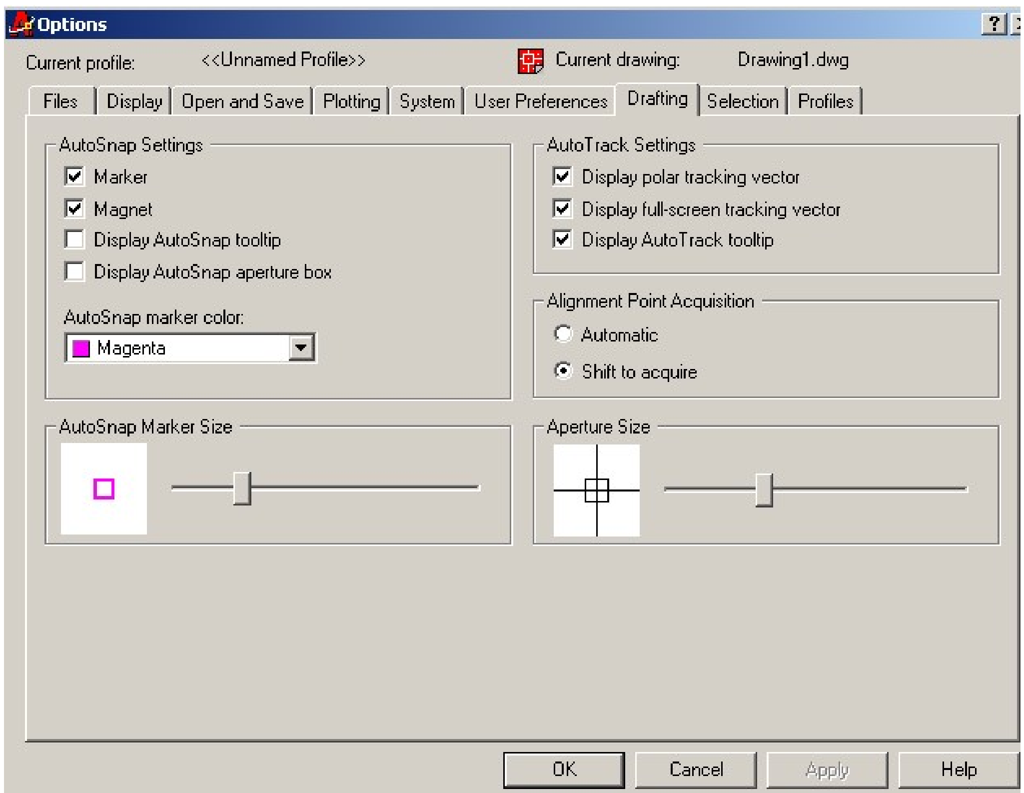
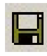


Figure 11.16: Options Dialog Box

Menu: **Tools→ Options**
Command: **OPTIONS**

Options dialog box can be invoked from the pull-down menu clicking **Options** command. By default Drafting tab is set as shown in Figure 11.16. One can change the AutoSnap Settings using this dialogue box. AutoSnap marker color can be changed from the drop down list displayed by clicking down arrow button. AutoSnap marker size can be adjusted by moving the slider bar horizontally in the AutoSnap marker size area, which is displayed in the image box at the left of the AutoSnap marker size area. The aperture size can also be adjusted by moving the slider bar horizontally in the aperture size area. The aperture size is displayed at the left image box in the aperture size area.

11.1.9 ***Saving File***

Standard Toolbar: 
Menu: **File** → **Save** or **Save As**
Command: **SAVE** or **SAVEAS**

One can invoke the **Save** command either from the toolbar directly or from the menu. It can also be initiated by inserting in the command area. To create a duplicate file one has to click on **Save As** command. To save in a new file or a duplicate file, a Save Drawing As dialogue box (Figure 11.17) will be displayed, where the new file name is to be inserted. Then click on the **Save** button, the file will be saved in the new file name.

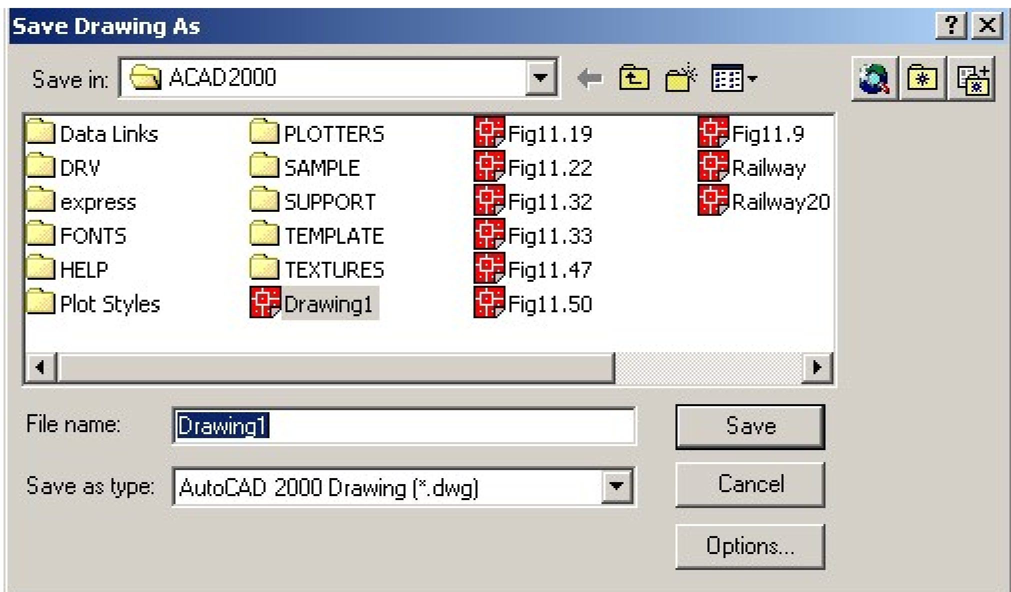


Figure 11.17: Save Drawing As Dialog Box

Save as type list box (Figure 11.18) is used to specify the drawing format in which it is needed to save the file. To save the file as AutoCAD 2000 drawing file, select AutoCAD 2000 drawing from the drop-down list.

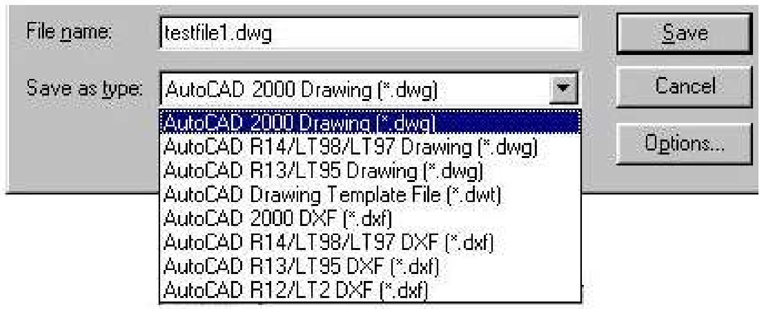


Figure 11.18: Save as Type List Box

11.1.10 *Opening Existing File*

Standard Toolbar: 
Menu: **File → open**
Command: **OPEN**

One can open the existing file either from the menu or the standard toolbar. It can also be initiated by inserting the **Open** command in the command area. A Select File dialogue box (Figure 11.19) will be displayed. Now the required file can be selected from the list of the files. When selected the preview of the drawing is displayed in the preview area as shown in the figure. The required files may be in the different locations, e.g. Desktop, and My Documents etc., which can be obtained from the drop down list by clicking the down arrow button of Look in area.

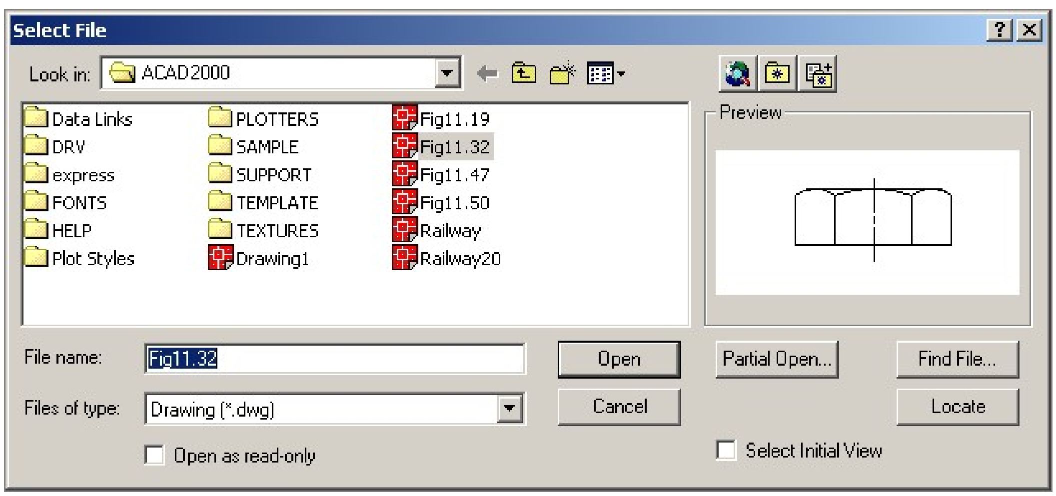


Figure 11.19: Select File Dialogue Box

11.1.11 *Starting New Drawing*

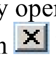
Standard Toolbar: 
Menu: **File → New**

Command: **NEW**

For starting a new drawing, one can either use the standard toolbar or the menu. It can also be initiated by entering the **New** command in the command prompt area. A Create New Drawing dialogue box will appear, which is identical to the Start Up dialogue box (Figure 11.2). Now to accept the Start from Scratch option after selection of the required units, when the **OK** button is clicked, the new drawing area is displayed, where the necessary drawing can be performed.

11.1.12 ***Closing Drawing***

Menu: **File → Close**
Menu: **File → Exit**
Command: **CLOSE** or **EXIT**

The current drawing file can be closed from the menu or one can close it by clicking the close window button  located at the top right corner of the drawing area. It can also be initiated by entering the **Close** command in the command area. If the drawing has not been saved in case of a new drawing or changes in the existing drawing, an AutoCAD dialogue box will appear showing warning "**Save changes to**". To save the drawing click on the **Yes** button, for no save click on the **No** button and for cancel click on the **Cancel** button. Now the user may open another drawing file to work. To exit from the AutoCAD, one can click the close upper button  located at the top right corner of the screen or it can be done from the menu or inserting **Exit** command in the command area.

11.2 ***Drawing Aids***

This section contains some basic drawing aids, which are necessary to perform drawing precisely and quickly. Setting units, setting limits and drafting settings are described in short in this section. Various snap modes are taken into consideration for discussion in a nutshell.

11.2.1 ***Setting Units***

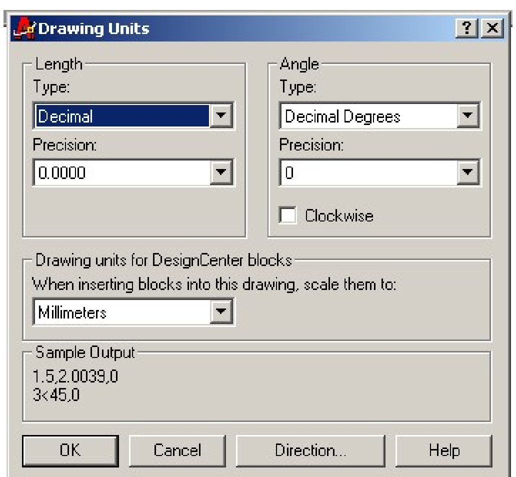


Figure 11.20: Drawing Units Dialog Box

Menu: **Format → Units**
Command: **UNITS**

When the **Units** command is invoked from the menu, a Drawing Units dialogue box (Figure 11.20) is displayed. The user can use this dialogue box to set the desired units. The desired units, length, angle can be selected from the drop-down list by clicking the down arrow button. The precision for length and angle can also be selected from the drop down list by clicking the down arrow button.

11.2.2 *Setting Limits*

Menu: **Format → Limits**
Command: **LIMITS**

To perform the drawing in AutoCAD it is necessary to set limits of the drawing area. Depending on the space required for the different views, dimensions, distance between views, border, title block and notes etc. the limits of the drawing area are calculated. Then the final limits are specified matching with the standard sheet size so that the calculated limits can be accommodated in the standard sheet, as a result during printing in the standard sheet no problem arises. The **Limits** command can be invoked from the menu or it can be initiated inserting in the command area. In the prompt sequence in the command area the values of the co-ordinates of the lower left corner and the upper right corner of the drawing area have to be inserted. For example, the limits of the A3 sheet are **0,0** for the lower left corner and **420,297** for the upper right corner.

11.2.3 *Status Bar*

To make the drawing easier and convenient, AutoCAD has provided some drawing aids. Various snap modes on status bar are shown in Figure 11.21. These snap modes such as, SNAP, GRID, ORTHO (orthogonal mode), POLAR (polar mode) and OSNAP (object snap mode) etc. are useful drawing aids to perform the drawing smoothly. One can easily toggle any desired mode on and off by clicking on it in the status bar.

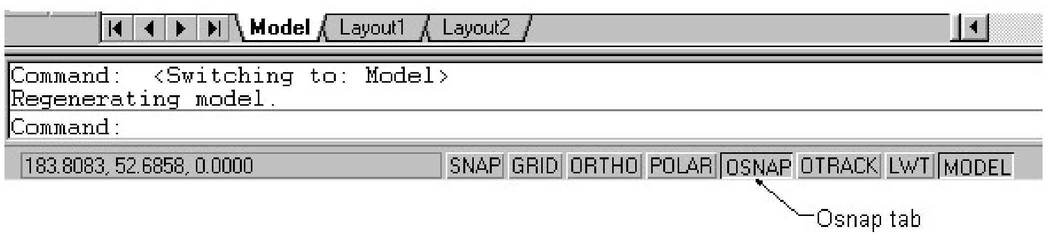


Figure 11.21: Various Snap Modes on Status Bar

11.2.4 *Drafting Settings*

Menu: **Tools → Drafting Settings**
Command: **DSETTINGS**

The Drafting Settings dialog box can be invoked from the menu. A Drafting Settings dialog box for snap and grid mode is shown in Figure 11.22. To display it **Snap and Grid** button of the Drafting Settings dialog box is selected. Clicking right mouse button on any tab except ORTHO and MODEL tabs in the status bar and then clicking settings, one can also display Drafting Settings dialog box to select the necessary option. The different snap modes in the status bar can be customized and controlled with the help of the Drafting Settings dialog box.

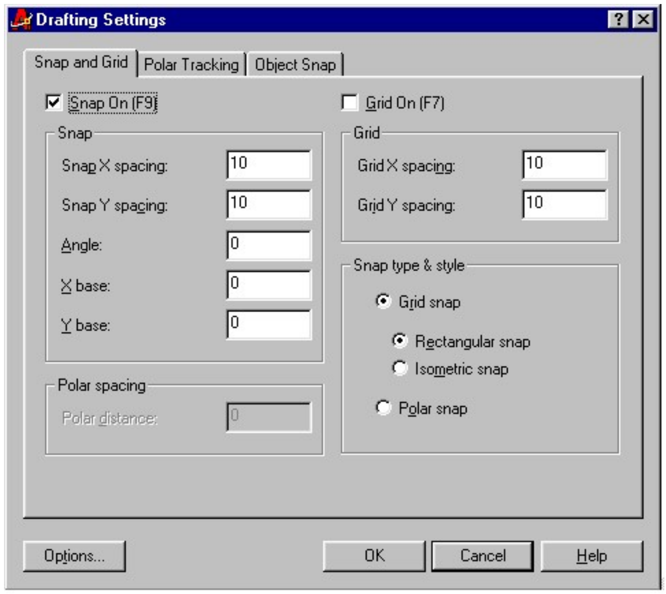


Figure 11.22: Drafting Settings Dialog Box (For Snap And Grid)

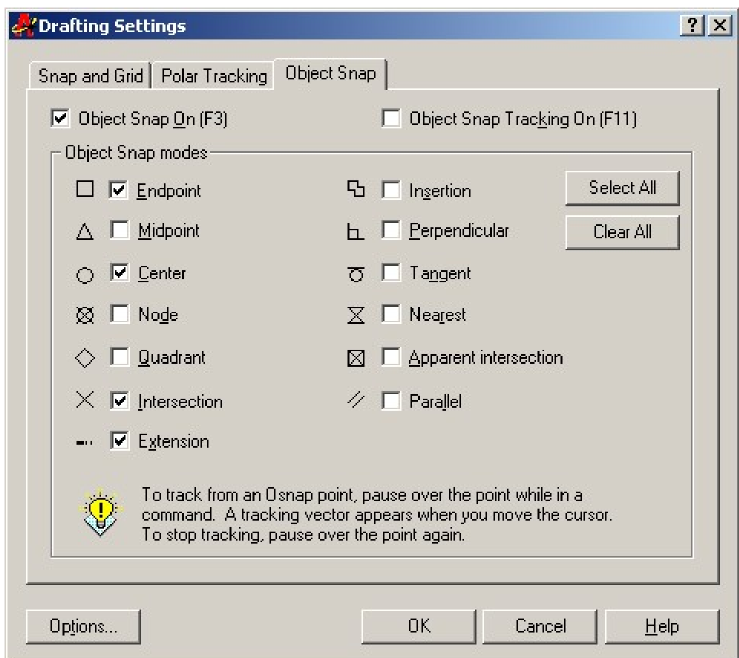


Figure 11.23: Drafting Settings Dialog Box (For Object Snap)

In Figure 11.23 a Drafting Settings dialog box for object snap mode is shown. It appears when the **Object Snap** button is selected in the dialog box. There are thirteen different snap modes in this dialog box. One can easily enable or disable any particular snap mode in this dialog box by selecting or deselecting the check box against each snap mode.

11.2.5 ***Object Snap Mode***

Status Bar: **OSNAP**
Menu: **Tools → Drafting Settings**
Command: **OSNAP**

The OSNAP (object snap) is one of the useful drawing aids, which is used in conjunction with other commands to perform the drawing accurately. One can pick up a specific point using the OSNAP mode. The usage of OSNAP for location of the center point of a circle (Figure 11.24a), the end point of a line (Figure 11.24b), the mid-point of a line (Figure 11.24c) and intersecting point of two lines (Figure 11.24d) are shown. There are many other options as well. It can be noticed that when the cursor is brought near to the point to be picked up, the OSNAP marker is highlighted and locates that desired point accurately. Each OSNAP has different shapes of marker e.g. circle, square, triangle, cross etc. (Figure 11.24).

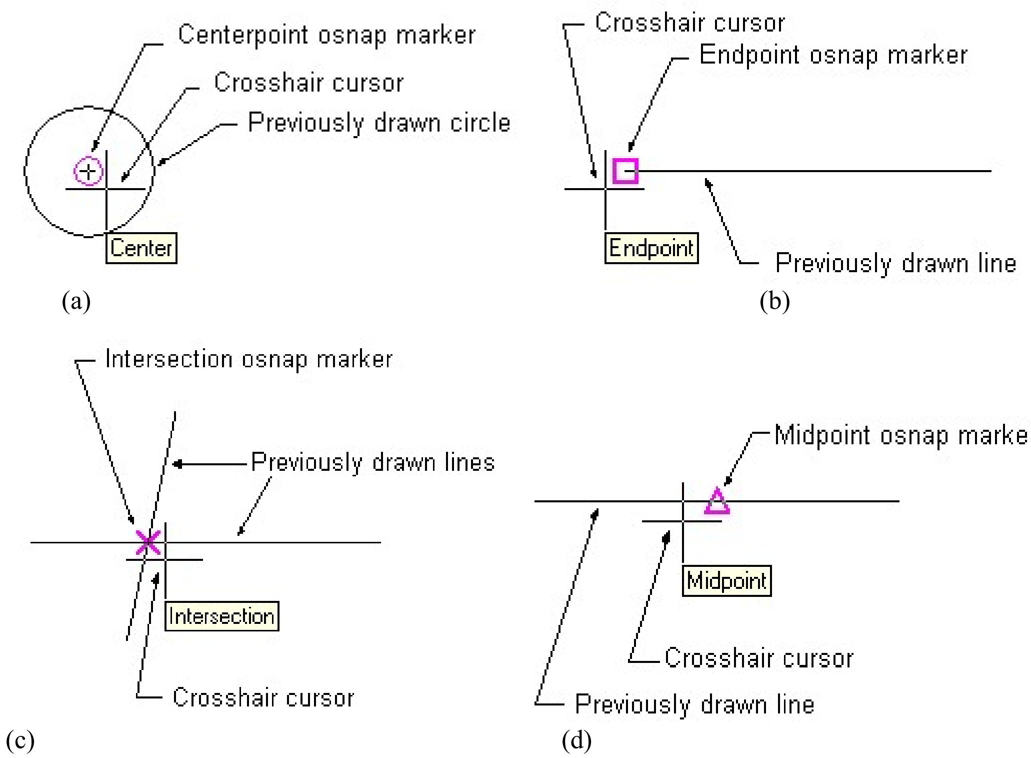


Figure 11.24: Use of OSNAP Mode

11.2.6 ***Orthogonal Mode***

Status Bar: **ORTHO**
Command: **ORTHO**

When the ORTHO (orthogonal mode) is made on, one can only draw orthogonal lines. Moreover, with orthogonal mode on, one can only move the objects vertically or horizontally. Click ORTHO on the status bar to toggle orthogonal mode on and off. It is not allowed to have polar tracking on, at the same time when orthogonal mode is on. Picking a point with object snap or by entering the values of coordinate will override ORTHO mode activity.

11.2.7 ***Drawing Grid***

Status Bar: **GRID**
Menu: **Tools → Drafting Settings**
Command: **GRID**

The drawing grid (Figure 11.25) is a regular pattern of dots displayed on the screen, which acts as a visual aid; it is tantamount to making a drawing on the graph paper placed on the drawing board. Drawing grid can be turned on or off by clicking on GRID tab in the status bar. It can also be selected in the Drafting Settings dialog box. Selecting the grid mode on, a drawing grid appears in the drawing area. The Grid X spacing and Grid Y spacing can be inserted in the dialog box (Figure 11.22). The grid spacing is 10 units in Figure 11.25a, while it is 5 units in Figure 11.25b. The grid is available in orthogonal mode (Figure 11.25c) and in isometric mode (Figure 11.25d). The grid gives the idea about the size of the drawing objects.

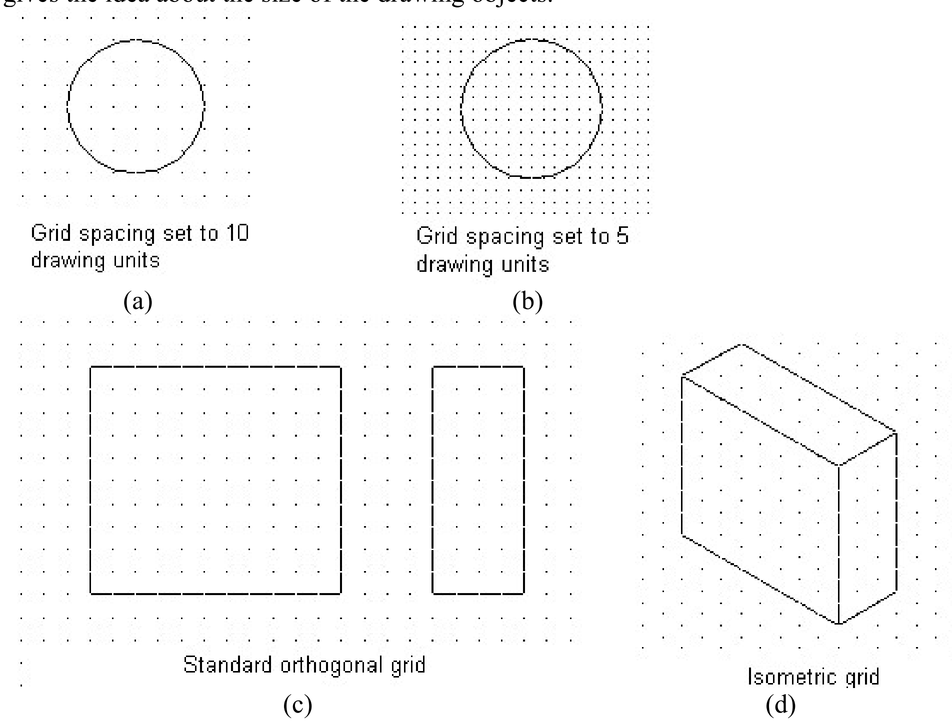


Figure 11.25: Drawing Grid

11.2.8 ***Snap Mode***

Status Bar: **SNAP**
Menu: **Tools → Drafting Settings**
Command: **SNAP**

Snap mode can be made on and off by clicking SNAP button in the status bar. It can be selected in the Drafting Settings dialog box also. The values of Snap X spacing and Snap Y spacing can be inserted in the dialog box (Figure 11.22). The snap mode is used to set increments for the cursor movement. The snap spacing is independent of the grid spacing; as a result they can have same or different values. Usually the snap and grid increments are set to the same value. With the snap mode turned on, AutoCAD only allows to pick points, which lie on a regular grid.

11.3 *Use of Draw Command*

In AutoCAD, the Draw commands can be invoked in a number of ways. They can be initiated entering the required command (e.g. line, circle etc.) in the command prompt area with the help of the keyboard. Often it is necessary to insert any value in the command area as well. After entering any command or inserting any value in the command area, **press enter** every time. Draw commands can be invoked from the Draw toolbar (Figure 11.10) directly or they can be obtained clicking on Draw title in the menu bar (Figure11.14).

11.3.1 *Line*

Draw Toolbar:  (Figure 11.10)
Menu: **Draw → Line** (Figure 11. 14)
Command: **LINE** or **L**

Using the Line command, a line can be drawn between any two points within the drawing area. Click on the **line** icon  at Draw toolbar **or** click **Draw→Line** in the menu. It can also be initiated by entering the line command in the command prompt area. Now click on any point at P_1 in the drawing area and move the cross hair to the next point at P_2 and click, P_1P_2 will turn into a line. Next move the cross hair to the point P_3 as in Figure 11.26, the dashed line along P_2P_3 is known as the rubber band line. After clicking at P_3, P_2P_3 will also turn into a line.

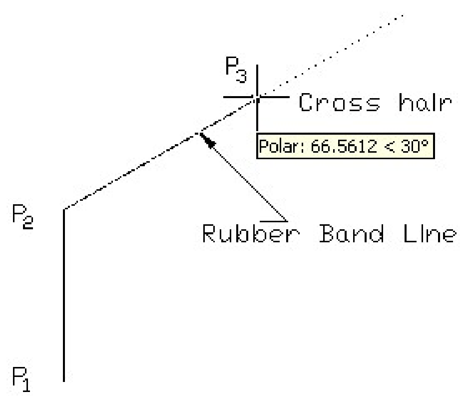


Figure 11.26: Rubber Band Lines

Drawing lines by Cartesian co-ordinates

In two dimensional Cartesian co-ordinate system, one can draw lines (Figure 11.27) specifying x- and y-coordinates in the command area following the sequence as mentioned below (example).

Insert first point **10,20** and press enter	Insert next point **80,100** and press enter	Insert next point **150,100** and press enter	To end press enter

Click Line → → → →

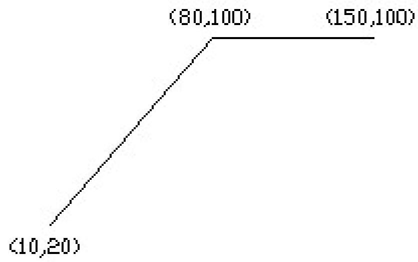


Figure 11.27: Lines by Cartesian Co-ordinates

Drawing Line by Direct Distance Entry

Using direct distance entry one can draw lines very efficiently. It is the most useful and timesaving approach for drawing lines. Following the sequence as mentioned below (example) lines are drawn (Figure 11.28) in ORTHO mode (located at the status bar). In the ORTHO mode lines are drawn vertically and horizontally. Line at an angle can also be drawn using direct distance entry. To draw lines at an angle the cross hair has to be moved at a desired angle and the similar sequence as shown below has to be followed.

Click Line → Click at any point P → Move cursor to the right, insert **80** and press enter → Move cursor to upward, insert **100** and press enter → To end press enter

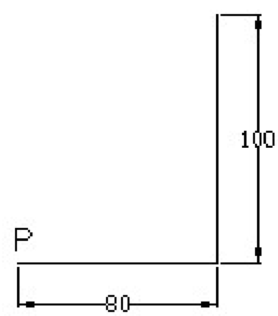


Figure 11.28: Lines by Direct Distance Entry

Drawing Lines by Relative Co-ordinates

In this system after specifying the first point, the next point is measured with respect to the previous point. In AutoCAD the relative co-ordinate system is identified by the symbol @. As an example, the steps as shown below are followed to draw lines by the relative co-ordinate system (Figure 11.29).

Insert first point **40,50** and press enter | Insert next point **@40,0** and press enter | Insert next point **@30,50** and press enter | To end press enter

Click Line → → → →

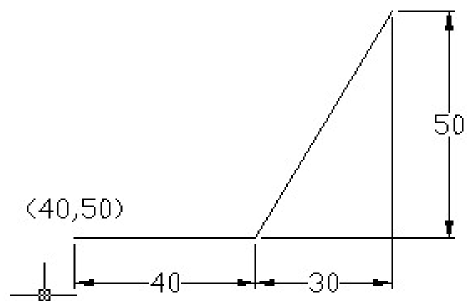


Figure 11.29: Lines by Relative Co-ordinates

Drawing Lines by Polar Co-ordinates

In the Polar co-ordinate system, one can draw a line at an angle. After specifying the first point, the next point is specified at a distance and an angle with the x-axis. The distance is measured in reference to the first point (Figure 11.30). The sequence of drawing a line in polar co-ordinate is shown below as an example.

Click Line → Insert first point **30,40** and press enter → Insert next point **@50<55** and press enter → Insert next point **@40<22** and press enter → To end press enter

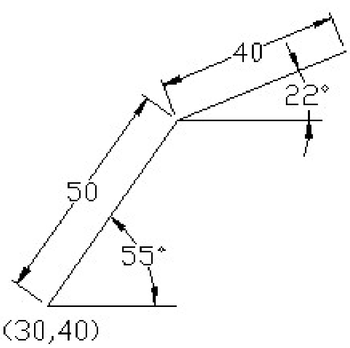


Figure 11.30: Lines by Polar Co-ordinates

There is very efficient and useful method of drawing lines at an angle using Polar Tracking. On menu bar click on **Tools** title, a Tools menu (Figure 11.7) will appear, then click on **Drafting Settings** on Tools menu, a Drafting Settings dialogue box for polar tracking (Figure 11.31) will be displayed after invoking **Polar Tracking** tab on it.

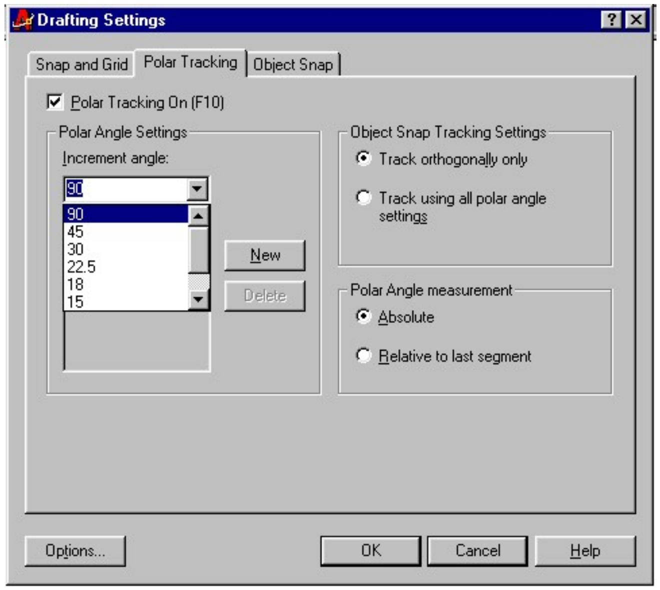


Figure 11.31: Drafting Settings Dialogue Box For Polar Tracking

On this Drafting Settings dialogue box there is a drop down list named Increment Angle showing several increment angles available by default. Any desired value of increment angle (say 15, 30 etc.) can be selected from the drop down list. Once the desired increment angle is set, the line at an angle (Figure 11.32) can be drawn easily from the following steps as shown below.

Click Line	→	Click at first point in	→	Move the cross hair, notice the angle and	→	To end press enter

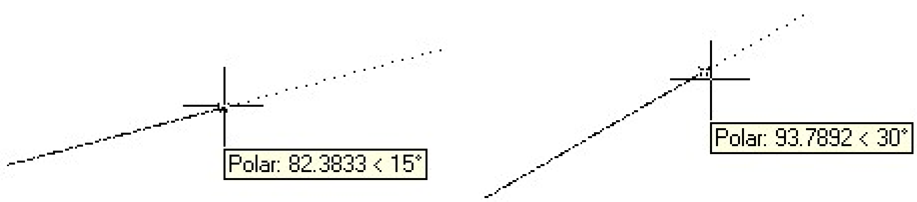


Figure 11.32: Drawing Lines at an Angle

Setting the increment angle at 15^0 a line at an angle of 15^0, 30^0, 45^0 etc. can be drawn with a step of 15^0. In the drawing area the angle will be indicated showing the value of the angle (Figure 11.32). A rubber band line with the dotted tracking line also will guide to draw the line at an angle. The length of the line can be inserted in the command prompt area.

11.3.2 ***Polyline***

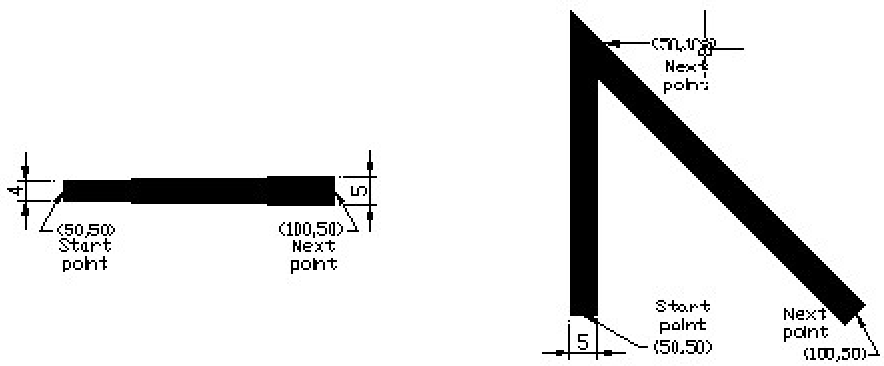
Draw Toolbar: 
Menu: **Draw → Polyline**
Command: **PLINE** or **PL**

Polyline command can be used to draw different types of features. Drawing can be done using **Polyline** command in several options. (Poly means many).

Width Option: In this option a polyline can be drawn with the desired thickness of the line. The sequence to draw a polyline (Figure 11.33a) is shown below as an example. Clicking at the points by the cross hair one can also specify the start point and end point. Another example of a polyline is shown in Figure 11.33b.

Click Polyline → Insert start point **50,50** and press enter → Enter W to change current width and press enter → Insert starting width **4.0** and press enter → Insert ending width **5.0** and press enter

→ Insert next point **75,50** and press enter



(a) (b)

Figure 11.33: Polylines Using Width Option

Half Width Option: This option is similar to the width option. In place of the full width, the half width is inserted and W for width option is replaced by H for half width option.

Arc Option: One can draw a polyarc using arc option. There are many ways to draw an arc. Here only one way is taken into consideration. The sequence to draw an arc (Figure 11.34) is given below as an example. The values of the co-ordinates can be inserted for the start and end points but here the start and end points have been selected by clicking with the cross hair.

Click Polyline → Click at start point → Enter W to change current width and press enter → Insert starting width **6.0** and press enter

→ Insert ending width **3.0** and press enter → Enter A for arc and press enter → Click at end point of arc and press enter

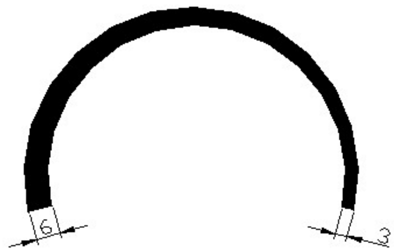


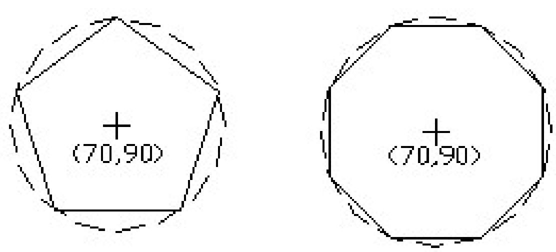
Figure 11.34: Polyarc Using Arc Option

11.3.3 ***Polygon***

Draw Toolbar: 
Menu: **Draw → Polygon**
Command: **POLYGON**

To draw a regular two-dimensional polygon, **Polygon** command is used. The regular polygon has the equal sides and equal angles. The center of the polygon may be specified either by clicking only at the center point by the cross hair or by inserting values of co-ordinates in the command area. The polygon may be drawn with the inscribed or circumscribed option. In the inscribed option (insert **I** for inscribed), the polygon remains within the imaginary circle and each corner of the polygon touches the circumference of the circle. While in the circumscribed option (insert **C** for circumscribed), the polygon remains outside the imaginary circle and each side of the polygon touches the circumference of the circle. The steps of drawing a polygon are given below. Examples of drawing polygons in inscribed and circumscribed options are given in Figure 11.35. The circles shown are for clarification only; they will not appear in the drawing.

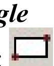
Click Polygon → Insert no. of sides and press → Insert center of polygon **70,90** and press enter → Enter option Inscribed/Circumscribed and press enter → Insert radius of circle **25** and



Inscribed Circumscribed

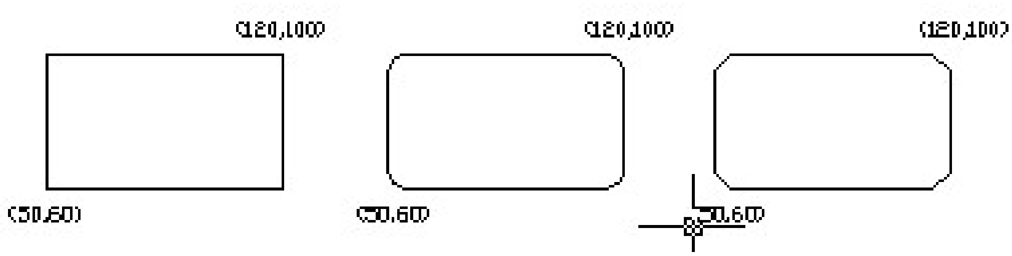
Figure 11.35: Regular Polygon

11.3.4 ***Rectangle***

Draw Toolbar: 
Menu: **Draw → Rectangle**
Command: **RECTANG**

Using **Rectangle** command one can draw a rectangle. The two corner points at the opposite ends of the diagonal of the rectangle are to be specified either by clicking only at the points by the cross hair or inserting the values of co-ordinates in the command area. The steps to draw a rectangle with fillet are given below. Examples of drawing rectangles are shown in Figure 11.36. The chamfer, fillet, width etc. can be set inserting the desired values in the command area. By default they have zero values. In case of sharp corner the second and third steps have to be omitted in the following sequence.

Click Rectangle → Insert F for fillet and press → Insert fillet radius **5.0** and press enter → Insert first corner point **50,60** and press enter → Insert other corner point **120,100** and press enter



With Sharp Corner | With Fillet 5mm | With Chamfer 5mm

Figure 11.36: Rectangle

11.3.5 ***Arc***

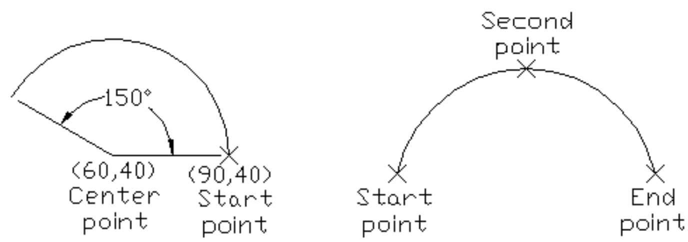
Draw Toolbar: 
Menu: **Draw → Arc**
Command: **ARC** or **A**

An arc is a part of a circle. It can be drawn using eleven different options. Of them the most useful one is the three points option. One can obtain the various options from the menu by clicking on **Draw** → **Arc**. To draw the arc by the three points option, one can specify the points either by clicking only with cross hair at the points or by inserting the values of co-ordinates in the command area. An arc by the **Three Points Option** (Figure 11.37) can be drawn following the steps as shown below.

Click Arc → Click at start point of arc → Click at second point of arc → Click at end point of arc

An arc by **Start, Center, and Angle Option** can be drawn (Figure 11.37) in accordance with the following steps.

Click Arc → Insert **CE** for center and press enter → Insert center point **60,40** and press enter → Insert start point **90,40** and press enter → Enter **A** for angle and press enter → Insert included angle **150** and press enter



Start Center Angle Option Three Points Option

Figure 11.37: Arcs

11.3.6 ***Circle***

Draw Toolbar: 
Menu: **Draw → Circle**
Command: **CIRCLE** or **C**

In AutoCAD a circle can be drawn by several options.

Center and Radius/Diameter Option: In this option the center of the circle is to be specified. The center may be specified either by clicking only with the cross hair at the center point or inserting the values of co-ordinates in the command area. By default the radius mode appears. For diameter the mode has to be changed inserting D. To draw the circle (Figure 11.38) the sequence as mentioned below is to be followed. For radius mode the third step in the following sequence is to be omitted.

Click Circle → Click at center of the circle → Insert D for Diameter mode and press enter → Insert value of radius/Diameter and press enter

Two/Three Points (2P/3P) Option: In this option circles are drawn (Figure 11.38) specifying two points for two points (2P) option and three points for three points (3P) option on the circumference of the circle. The points may be specified either by clicking only with the cross hair at the points or inserting the values of co-ordinates in the command area. The steps to be followed to draw the circle are provided below. For the two points option the last step has to be omitted.

Click Circle | Enter 2P/3P and press enter | Click at first point | Click at second point | Click at third point

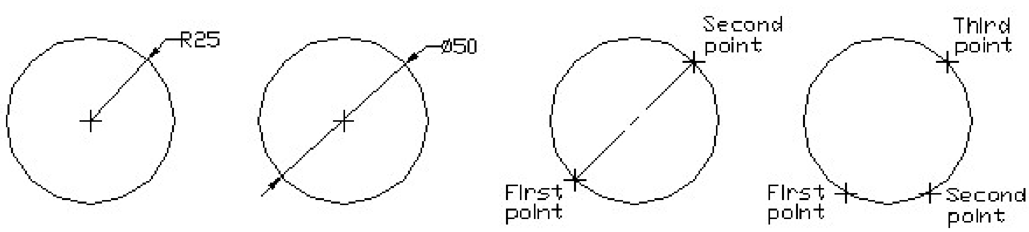
→ → → →

Tangent, Tangent Radius Option: In this option one can draw a circle (Figure 11.38) by clicking on two objects at any point on them which will be the tangents to the circle and inserting the value of the radius of the circle. The steps that are followed to draw the circle in this option are given below.

Click Circle → Enter TTR and press enter → Click on object for first tangent → Click on object for second tangent → Insert radius and press enter

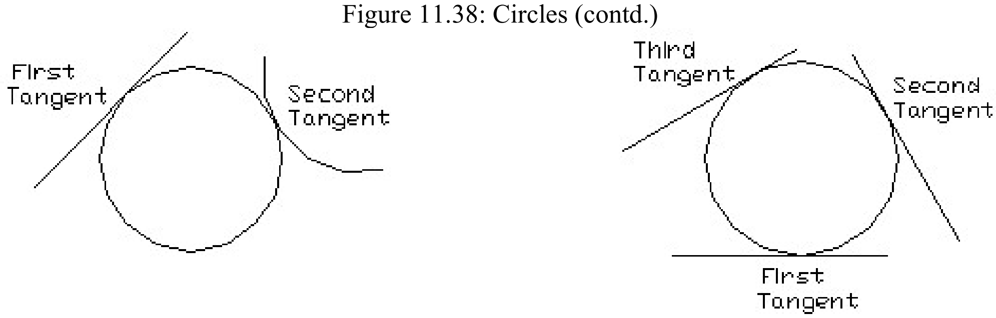
Tangent, Tangent, Tangent Option: In this option a circle (Figure 11.38) can be drawn by clicking on three objects at any point on them which will be the tangents to the circle. It has to be invoked from the menu clicking **Draw → Circle → Tan, Tan, Tan**. The steps as mentioned below are followed to draw the circle in this option.

Click Circle → Click on Tan Tan Tan → Click on object for first tangent → Click on object for second tangent → Click on object for third tangent



Center Radius Option | Center Diameter Option | Two Points Option | Three Points Option

Figure 11.38: Circles (contd.)



Tangent, Tangent Radius Option | Tangent, Tangent Tangent Option

Figure 11.38: Circles (contd.)

11.3.7 *Spline*

Draw Toolbar: 
Menu: **Draw → Spline**
Command: **SPLINE** or **SPL**

Spline command can be used to draw a curved line of any desired shape. One can specify the points for drawing the curve by inserting the values of co-ordinates of the points in the command area or by clicking only at the points by the cross hair. A curved line is drawn (Figure 11.39) using **Spline** command according to the following sequence as mentioned below. Inserting the values of the co-ordinates in the command prompt area, a curve of any regular shape e.g. sinusoidal one may be drawn easily.

Click Spline → Click at first point → Click at next point → Click at next point → Click at next point → To end press enter thrice

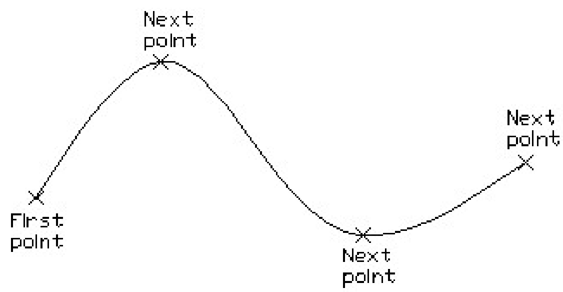


Figure 11.39: Curved Line

11.3.8 *Ellipse*

Draw Toolbar: 
Menu: **Draw → Ellipse**
Command: **ELLIPSE** or **EL**

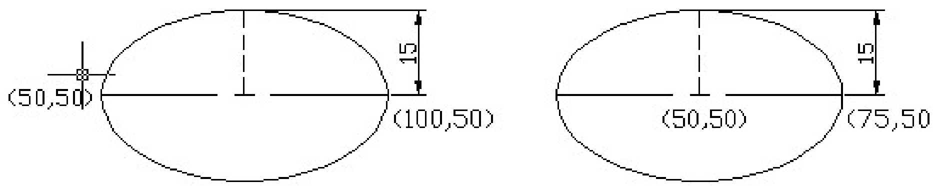
An ellipse is the most common geometric feature. It can be drawn using **Ellipse** command. Some options are there to draw ellipses. The most common two options will be presented here only.

Axis and Eccentricity Option: In this option an ellipse is drawn inserting the values of the co-ordinates at the two end points of an axis of the ellipse and the distance to the other axis (Figure 11.40). By clicking at the points with the cross hair only the ellipse can be drawn. The steps are shown below to draw the ellipse by inserting values in command area.

Click Ellipse → Insert axis end point **50,50** and press enter → Insert other end point **100,50** and press enter → Insert distance to other axis **15** and press enter

Center and Two Axes options: In this option to draw an ellipse (Figure 11.40) the values of the co-ordinates of the center point and end point of an axis and the distance to the other axis have to be inserted. By clicking at the points with the cross hair only the ellipse can be drawn. The sequence of the steps to draw the ellipse is shown below.

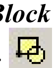
Click Ellipse	→	Enter C for center and	→	Insert center point **50,50** and press	→	Insert end point of axis **75,50** and press enter	→	Insert distance to other axis **15** and press enter



Axis and Eccentricity Option Center and Two Axes Option

Figure 11.40: Ellipse

11.3.9 ***Make Block***

Draw Toolbar: 

Menu: **Draw → Block**

Command: **BLOCK** or **B**

When a part of a drawing, the whole one or any symbol is necessary to be redrawn time and again in the different location in the same drawing or another one, **Make Block** command can be used to store that under a block name. All the objects within the block will be treated as the single object. The entire block can be selected by picking a point on it. The block can be placed in a drawing in the desired orientation and with the desired scale factor. It can be moved, erased or listed as the single object. Invoking **Make Block** command, a Block Definition dialogue box (Figure 11.41) will be displayed. The sequence of the steps for making a block (Figure 11.42) using **Make Block** command is given below.

In the Block Definition dialogue box in the base point area clicking on **Pick point** button insertion base point can be specified. This point is used as the reference point to insert the block. Usually the lower left corner or the center point of the objects to be converted to block is chosen as the insertion base point. When the block is inserted, it is observed that the block appears at the insertion point. Any suitable unit can also be chosen from the drop down list in the dialogue box.

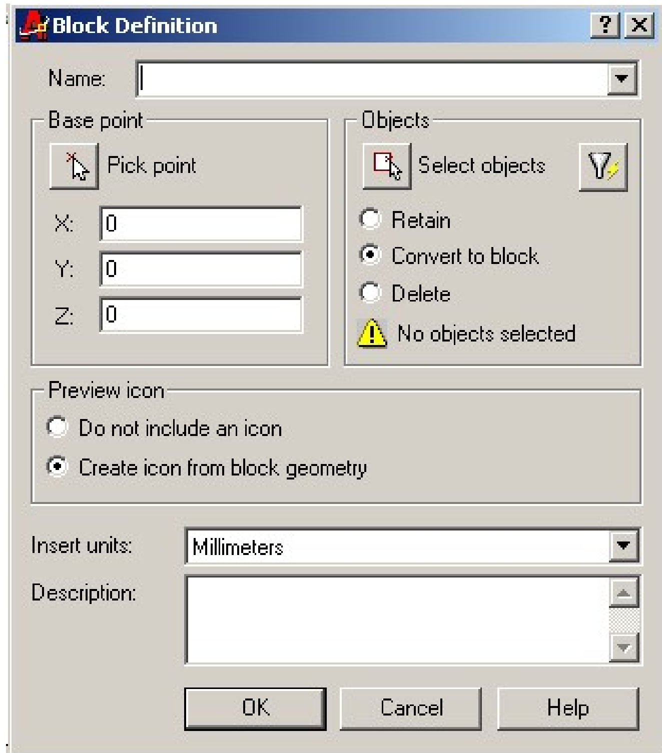


Figure 11.41: Block Definition Dialogue Box

If a block name is given, which already exists, an AutoCAD dialogue box will appear showing warning. It occurs during closing the Block Definition dialogue box clicking **OK** button. One can redefine block name clicking on the **Yes** button or exit clicking on the **No** button.

Click Make Block → Specify block name in dialogue box → Click on pick point button in base point area → Click on base point either at left lower corner or center point of the objects to make block

→ Click on select objects button in objects area → Select objects by pick box to make block → Click right mouse button or → Click OK in dialogue box

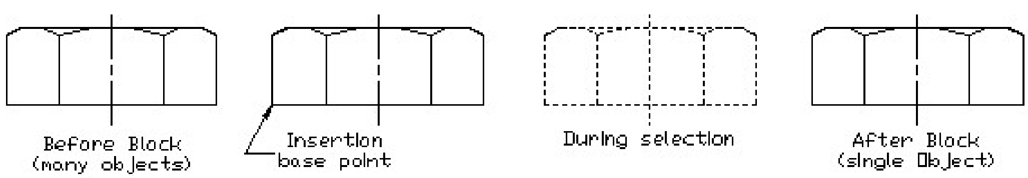


Figure 11.42: Making a Block

11.3.10 *Insert Block*

Draw Toolbar: 
Command: **INSERT**

With the help of **Insert Block** command one can insert the block in any desired location of the drawing. Invoking the **Insert Block** command, Insert dialogue box (Figure 11.43) will be displayed. Now in the dialogue box one can put the values of the co-ordinates of the insertion point, scale and rotation angle directly or they can be specified on the screen by selecting **Specify On-screen** with tick. For the insertion point by default **Specify On-screen** is selected. There is option in the dialogue box to insert different scales in the x-, y- and z- directions separately. Uniform scale may also be chosen. In Figure 11.44 the uses of scale and rotation are shown to insert block. The steps as mentioned below are to be followed to insert any block in the drawing.

Click Insert Block → Enter block name to be inserted in dialogue box → Specify insertion point → Insert scale if there is any → Insert rotation angle if there is any → Click on OK button in dialogue box → Click at insertion point on screen

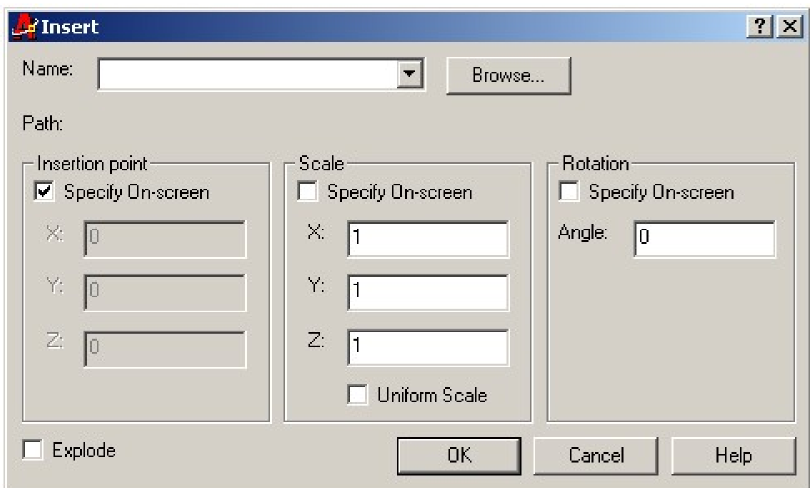
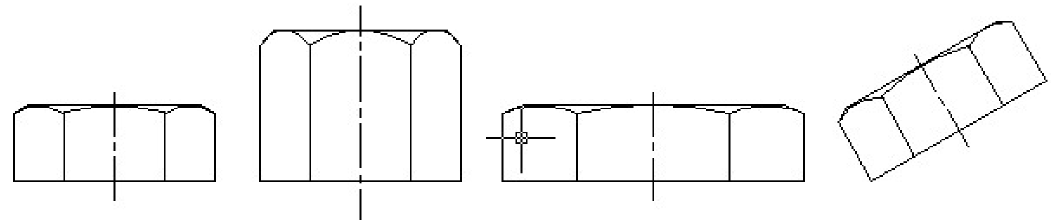


Figure 11.43: Insert Dialogue Box



Original Block | Block with Scale X = 1.0, Y =2.0 | Block with Scale X =1.5, Y = 1.0 | Block with Rotation Angle = 30^0

Figure 11.44: Block Inserted with Scale and Rotation

11.3.11 ***Point***

Draw Toolbar:
Menu: **Draw → Point**
Command: **POINT** or **PO**

The point is the basic drawing object. It can be drawn on the screen by using **Point** command. If the **Point** command is invoked from the menu by clicking Draw title, single or multiple point option can be selected. But when the **Point** command is invoked from the Draw toolbar only the multiple point option is selected. Using single point option one point can be drawn only while using multiple point option many points can be drawn. In case of the multiple point option one can exit by pressing **ESC** button while for the single point option after drawing the single point, it comes to exit automatically.

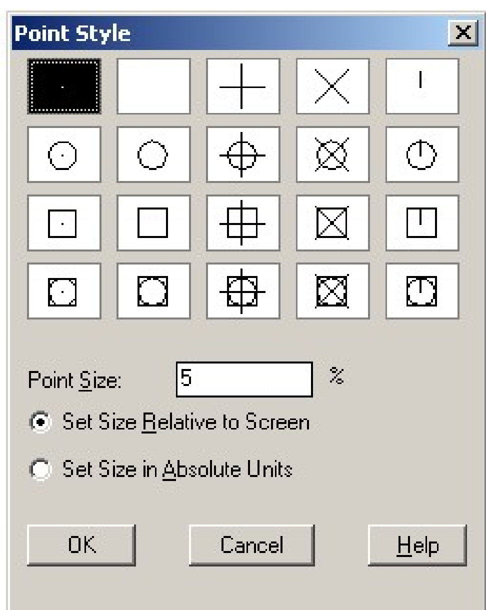


Figure 11.45: Point Style Dialogue Box

The type of point to be drawn can be changed from the Point Style dialogue box. Point Style dialogue box (Figure 11.45) can be displayed from the menu clicking **Format → Point Style**. By default the dot point is selected. Any type of desired point can be selected from the Point Style dialogue box and the size of it can also be changed according to the requirement. Then clicking **OK** button in the Point Style dialogue box the type of point is set for drawing.

11.3.12 ***Hatch***

Draw Toolbar: 
Menu: **Draw → Hatch**
Command: **HATCH** or **H**

Hatching is very important to represent section in mechanical drawing. This hatching can be done using the **Hatch** command. AutoCAD provides with a number of hatch patterns for some common materials. When the **Hatch** command is invoked, a Boundary Hatch dialogue box (Figure 11.46) appears.

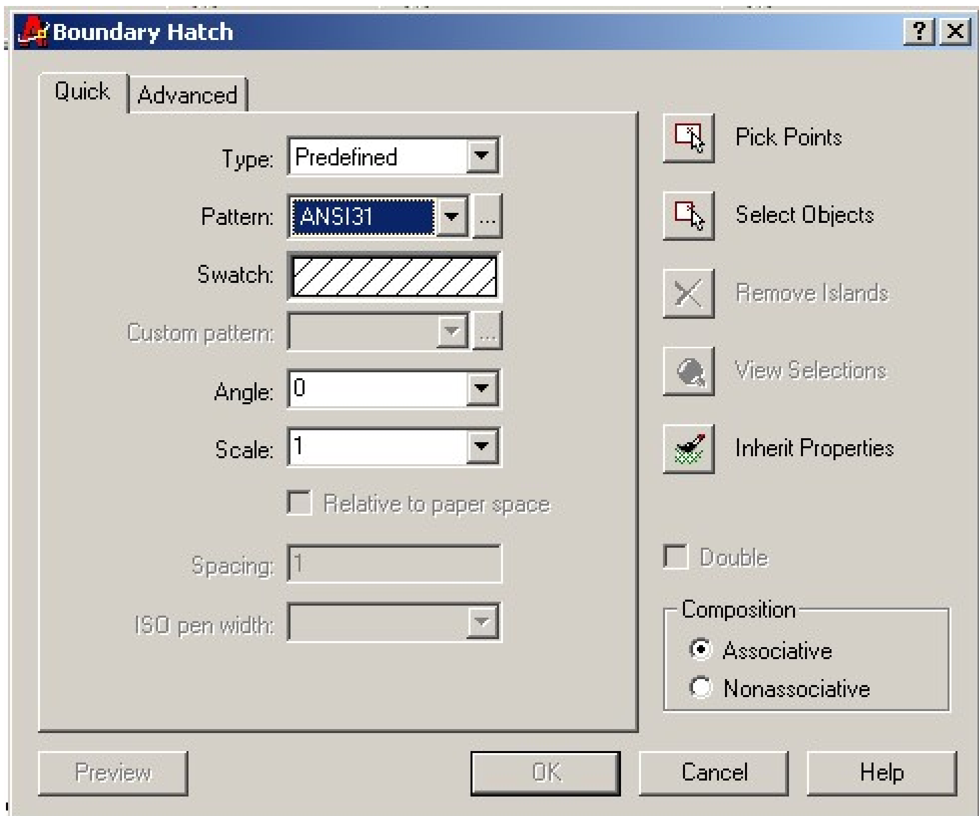


Figure 11.46: Boundary Hatch Dialogue Box

The steps as mentioned below are followed to give hatch line in an area (Figure 11.47). The objects to be selected for hatching may be done by clicking on the **Pick Points** tab or the **Select Objects** tab in the dialogue box. In case of the **Pick Points** tab, the area is selected for hatching by picking a point within the area while for the **Select Objects** tab; the area to be selected is kept within the window made by the pick box. Pattern, angle and scale may be chosen from the drop down list displayed by clicking on the down arrow button of the respective area in the Boundary Hatch dialogue box. Different patterns of the section can also be made visible for selection by clicking on the tab indicated by three dots, which is located at the right side of Pattern area in the dialogue box.

Click Hatch → Set hatch pattern, angle, scale in the dialogue box → Click on pick points tab in the dialogue → Click at point P within the area to select for → Press enter or click right mouse button and enter → Click on OK tab in dialogue box

Before Hatch During Selection After Hatch

Figure 11.47: Hatching an Area

11.4 *Use of Modify Command*

Sometimes it is necessary to modify the basic drawing object e.g. line, circle, rectangle, polygon etc. in order to obtain the final object. AutoCAD provides a number of modify commands such as, **Erase**, **Copy**, **Mirror**, **Offset**, **Array**, **Move** and **Rotate** etc., the uses of them are described below in a nutshell. One can invoke the modify command either from the modify toolbar (Figure 11.11) directly or from the menu (Figure 11.48). It can also be invoked entering the desired command in the command prompt area. Sometimes it may be necessary to insert any value in the command area. After entering any command or inserting any value in the command area, **press enter** every time.

When any modify command will be invoked, the cross-hair cursor will turn into the shape of a pick box (Figure 11.49a) with some exception. In case of the offset command, the shape of a cross hair will appear in place of a pick box. In order to select the object to be modified, it is brought within the window, which is done by picking at point P_1 and then at point P_2 by using the pick box. The whole objects will be selected (Figure 11.49c). An object is selected by picking on it directly with the pick box (Figure 11.49d).

The selection of the object can be done before invoking any modify command. One can do this bringing the whole objects within the window with the help of the cross-hair cursor picking at point P_1 and then at point P_2, the whole objects will be selected (Figure 11.50b). An object may also be selected by picking on it directly with the cross-hair cursor (Figure 11.50c). The selected lines turn into the dashed lines.

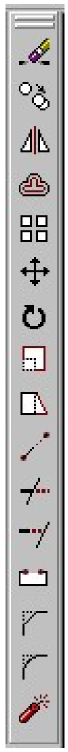


Figure 11.11: Modify Toolbar

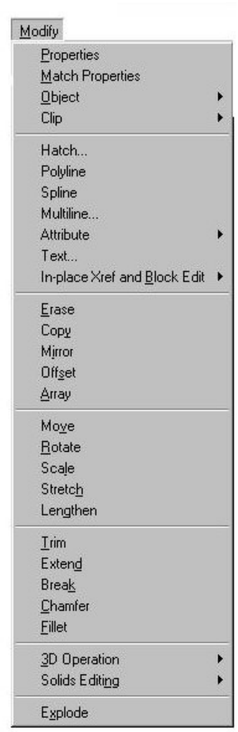
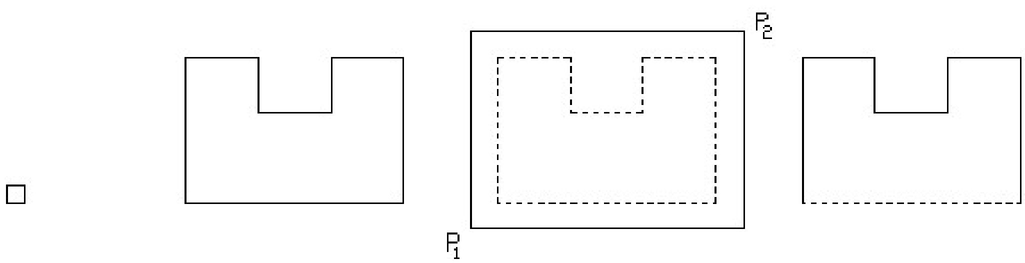


Figure 11.48: Modify Commands From Menu



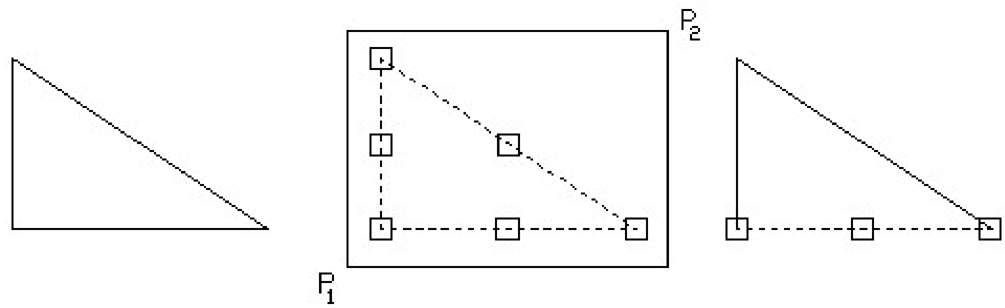
(a) Pick box

(b) Object Before Selection

(c) Whole Objects Selected

(d) An Object Selected

Figure 11.49: Selection of Object with Pick Box



Object Before Selection (a) | Whole object Selected (b) | An Object Selected (c)

Figure 11.50: Selection of Object with Cross hair Cursor

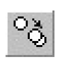
11.4.1 ***Erase***

Modify Toolbar:  (Figure 11.11)
Menu: **Modify → Erase** (Figure 11. 48)
Command: **ERASE** or **E**

The **Erase** command is one of the simplest and most useful AutoCAD commands. The command erases (deletes) any selected object(s) from the drawing. The delete key of keyboard can also be used to erase a selected object by cross-hair cursor in AutoCAD. The following sequence may be followed to erase any object(s).

Click Erase → Select object(s) → Click right mouse button or press enter

11.4.2 ***Copy***

Modify Toolbar: 
Menu: **Modify → Copy**
Command: **COPY** or **CP**

The **Copy** command can be used to create one or more duplicates of any object that was previously created. Copy is a very useful and timesaving command because if any complex drawing is needed to be redrawn, it can be done easily using copy command as many times as are required (Figure 11.51).

The steps as mentioned below are followed to make the copy of any object(s).

Click Copy | Select object | Click right mouse button or press enter | Click at any base point (P1) | Drag the cursor to a second point of displacement (P2) and click

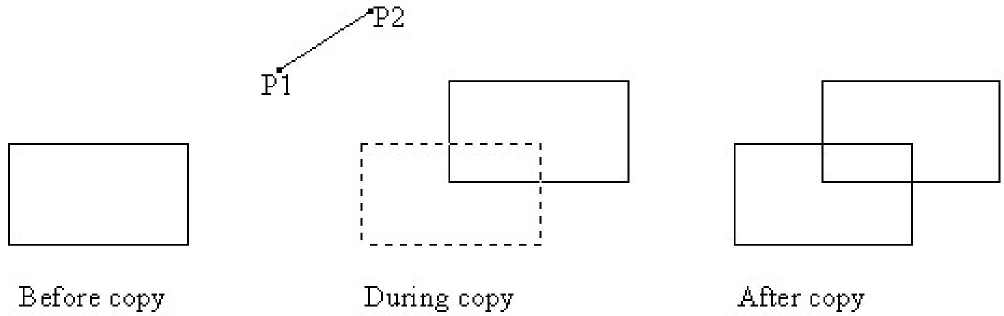
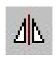


Figure 11.51: Use of Copy Command

11.4.3 ***Mirror***

Modify Toolbar: 
Menu: **Modify → Mirror**
Command: **MIRROR** or **MI**

When it is necessary to draw the image of an object the **Mirror** command is used. It makes the mirror copy of the selected object(s) (Figure 11.52). The mirror copy can also be made at any angle. The sequence as shown below is followed to make the mirror of any object(s).

Click Mirror → Select object → Click right mouse button or → Click at two ref. points (P1 & P2) of mirror line → Click right mouse button and enter or press

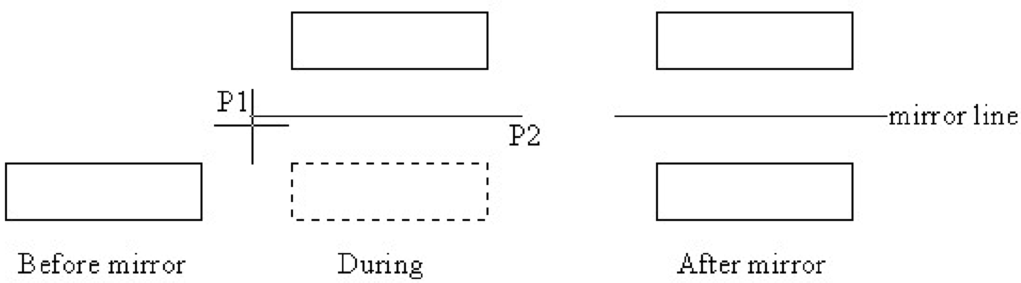


Figure 11.52: Use of Mirror Command

11.4.4 ***Offset***

Modify Toolbar: 
Menu: **Modify → Offset**
Command: **OFFSET** or **O**

To draw the parallel lines, polylines, concentric circles, ellipses, arcs, curves, etc., **Offset** command may be used. An object similar to the selected object is created using offset command (Figure 11.53).

To offset an object the offset distance is to be inserted in the command prompt area. The steps as mentioned below are followed to make the offset of any object(s).

Click Offset → Insert offset distance → Click right mouse button and enter or press enter → Select object → Click point on the side to offset → Click right mouse button or press enter

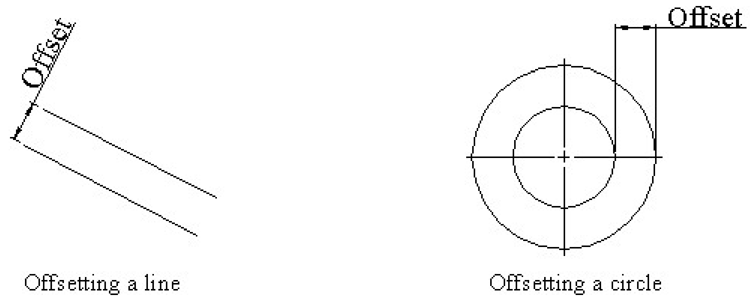


Figure 11.53: Use of Offset Command

11.4.5 ***Array***

Modify Toolbar:
Menu: **Modify → Array**
Command: **ARRAY** or **AR**

The **Array** command makes multiple copies of selected object(s) either in a rectangular matrix (columns and rows) (Figure 11.54) or in a polar (circular) pattern (Figure 11.55). The necessary data is to be inserted in the command prompt area. The steps of rectangular array are as follows. In AutoCAD 2004 the array command is activated in a different way.

Click Array → Select object and click right mouse button or press → Insert type of array (Rectangular/Polar) and press enter → Insert number of rows and press enter → Insert number of columns and press enter → Insert distance between rows and press enter → Insert distance between columns and press enter

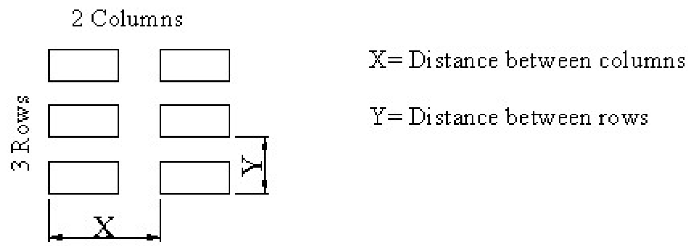


Figure 11.54: Use of Rectangular Array Command

The steps to make the polar array of any object(s) are shown below.

Click Array → Select object and click right mouse button or press enter → Insert type of array (Rectangular/Polar) and press enter → Click at center point of

→ Insert number of items in array and press → Specify the angle to fill and press → Rotate arrayed object (Yes/No) and press enter

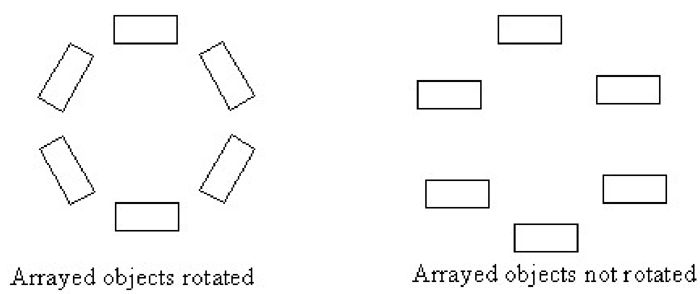


Figure 11.55: Use of Polar Array Command

11.4.6 *Move*

Modify Toolbar: 
Menu: **Modify → Move**
Command: **MOVE** or **M**

Sometimes it is necessary to move object from one place to another (Figure 11.56), which can be achieved by using **Move** command following the steps as shown below.

Click Move → Select object → Click right mouse button or press enter → Click at base point (P1) → Drag the cursor to second point of displacement (P2) and click

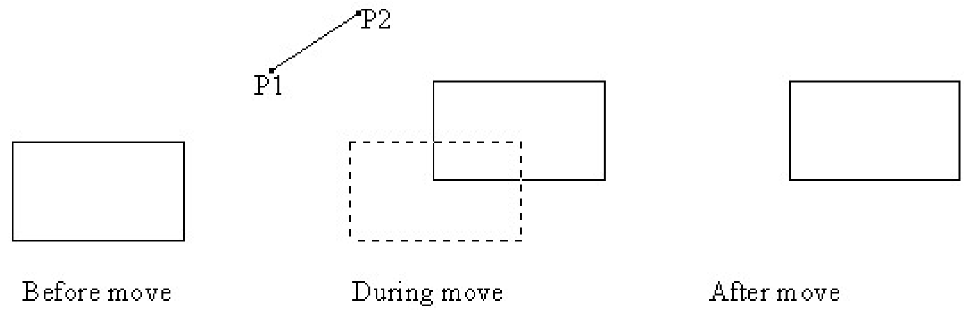


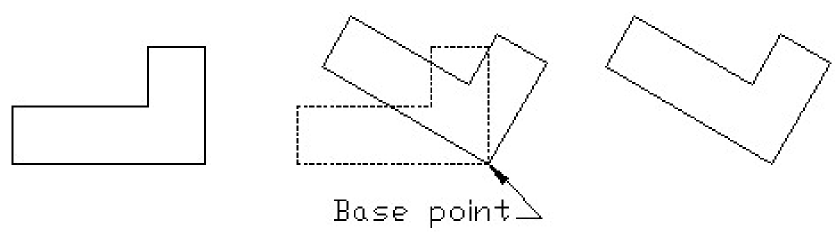
Figure 11.56: Use of Move Command

11.4.7 ***Rotate***

Modify Toolbar: 
Menu: **Modify → Rotate**
Command: **ROTATE** or **RO**

One can rotate an object at any angle either in anticlockwise or clockwise direction about any specified point (base point) by using the **Rotate** command (Figure 11.57). For anticlockwise rotation insert positive angle and for clockwise rotation insert negative angle in degree in the command prompt area. The steps to rotate any object(s) are as follows.

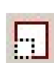
Click Rotate → Select object → Click right mouse button or → Click base point about which object is to be rotated → Insert rotation angle and



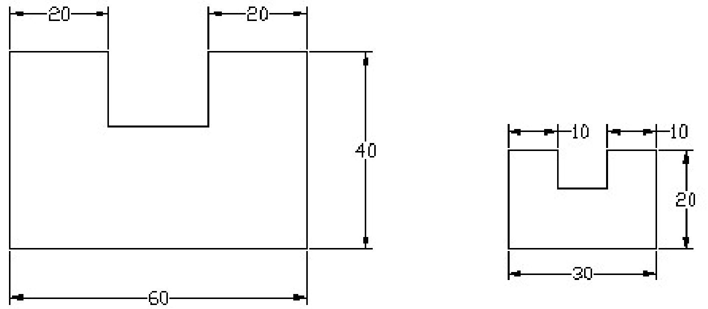
Before Rotate During Rotate After Rotate

Figure 11.57: Use of Rotate Command

11.4.8 ***Scale***

Modify Toolbar: 
Menu: **Modify → Scale**
Command: **SCALE** or **SC**

The **Scale** command is used to enlarge or shorten the size of an object in accordance with the desired scale. By doing this, the shape of the object is not changed at all. It is a timesaving command to edit. If the scale of the object is changed, the dimensions of the object are also changed according to the scale (Figure 11.58).



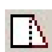
Before Scaling After Scaling (Scale factor = 0.5)

Figure 11.58: Use of Scale Command

The base point is that point in respect of which, the scaling is done. The base point may be at any point on the corner, inside or outside of the object. The value of the desired scale factor is to be inserted in the command prompt area. The sequence to scale any object(s) is as follows.

Click Scale → Select object → Click right mouse button or press enter → Click at Base point → Insert scale factor and press enter

11.4.9 ***Stretch***

Modify Toolbar: 
Menu: **Modify → Stretch**
Command: **STRETCH**

The **Stretch** command can be used to stretch the object(s) (Figure 11.59). One can either lengthen or shorten the objects using this command.

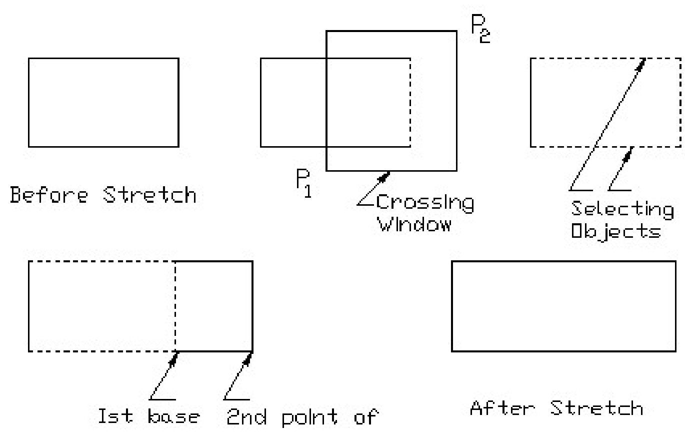


Figure 11.59: Use of Stretch Command

The steps to stretch any object(s) are as follows.

Click Stretch → Select objects by crossing window picking at P_1 and P_2 → Select objects to be stretched → Click right mouse button or press enter

→ Specify base point → Drag cursor to the second point of displacement and click

11.4.10 ***Lengthen***

Modify Toolbar: 
Menu: **Modify → lengthen**
Command: **LENGTHEN**
One can use the **Lengthen** command to extend or shorten lines, polylines, arcs and elliptical arcs. The Lengthen command has several options to change the length of the object. The side at which it is necessary to extend or shorten, picking has to be done on that side to select the object. In case of the closed object like circle lengthen command has no effect. While the lengthen command is invoked, the following prompt sequence is displayed.

_Lengthen
Select an object or [Delta/Percent/Total/Dynamic]:

Delta Option: In this option the length of the angle of an object is either increased or decreased by Delta length or Delta angle (Figure 11.60). The Delta value is inserted either numerically or specifying two points. The positive value will increase and the negative value will decrease the length of the selected object. If the value of the delta length is inserted in the third step in place of inserting A for delta angle, the fourth step is to be omitted in the following sequence.

Click Lengthen → Insert **DE** for delta and press enter → Insert value of delta length or A for delta angle → Insert value of delta angle

→ Select object picking on it → Click right mouse button and enter or press enter

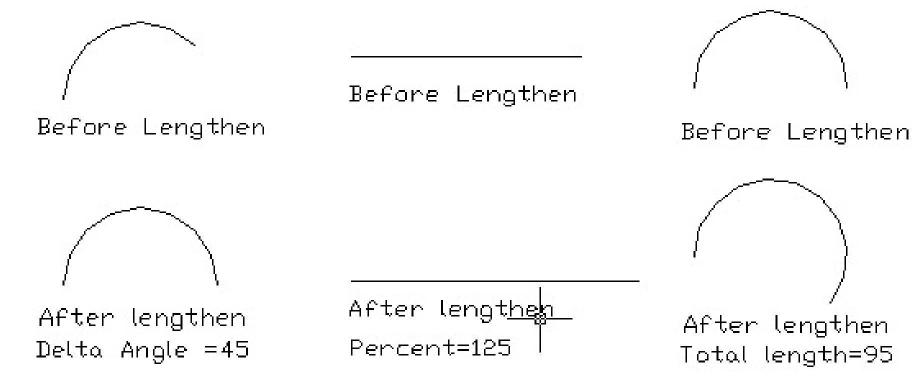
Percent Option: In this option the percentage of the original line is to be specified in respect of which the line will be extended or shortened (Figure 11.60). For percent of 125 the original length will be increased by 25 percent while for percent of 75 the original length will be decreased by 25 percent. The steps are as follows.

Click Lengthen | Insert P for Percent and press enter | Insert percent and press | Select object picking on it | Click right mouse button and enter or press

→ → → →

Total Option: In this option the value of the total length is specified, as a result the original length of the line is either increased or decreased to make it equal to the total length (Figure 11.60). If the inserted value of the total length is 95, the original length will either increase or decrease to make its length equal to 95. The value can be inserted either numerically or specifying two points. The steps are as follows.

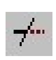
Click Lengthen → Insert T for Total and press → Insert value of total length and press enter → Select object picking on it → Click right mouse button and enter or press enter



(a) Delta (Angle) Option (b) Percent Option (c) Total Option

Figure 11.60: Use of Lengthen Command

11.4.11 ***Trim***

Modify Toolbar: 
Menu: **Modify → Trim**
Command: **TRIM** or **TR**

The **Trim** command is used in order to trim a part of an object. An object is trimmed with respect to another object known as the cutting edge (Figure 11.61). The sequence to trim any object is as follows.

Click Trim → Select cutting edge → Click right mouse button or press enter → Select the portion of the object to be trimmed → Click right mouse button and enter or press

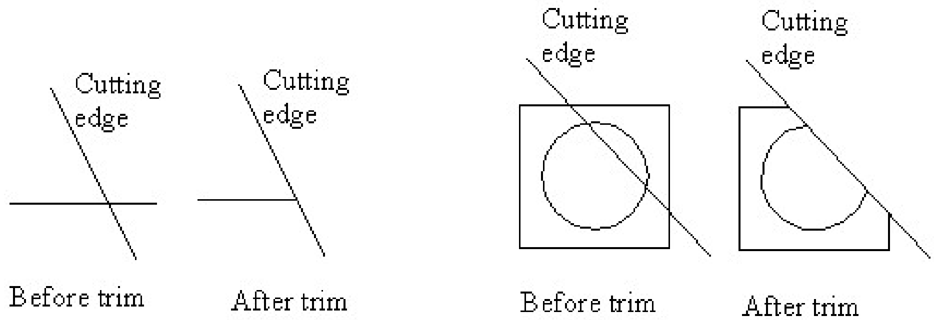


Figure 11.61: Use of Trim Command

11.4.12 ***Extend***

Modify Toolbar: 
Menu: **Modify → Extend**
Command: **EXTEND** or **EX**
Using the **Extend** command one can extend a line, polyline or arc up to another object known as the boundary edge (Figure 11.62). This command may be considered as opposite to the Trim command. The steps to extend any object are shown below.

Click Extend	→	Select boundar y edge	→	Click right mouse button or	→	Select the object to be	→	Click right mouse button and enter or press enter

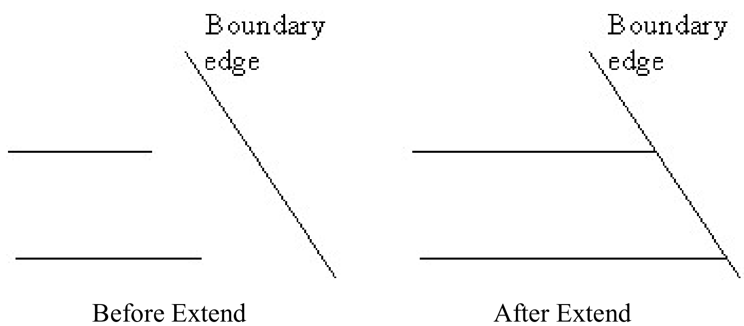


Figure 11.62: Use of Extend Command

11.4.13 B***reak***

Modify Toolbar: 
Menu: **Modify → Break**
Command: **BREAK** or **BR**

The **Break** command can be used to break or remove a part of the object, e.g. line, polyline, arc, circle, ellipse, etc. The object is thus divided into two parts (Figure 11.63). For the circle with position of the first and second break points as shown, the feature of Figure 11.50b will appear and interchanging the break points the feature of Figure 11.50c will appear. The steps to break any object are as follows.

Click Break → Pick on first break point → Click on second break point

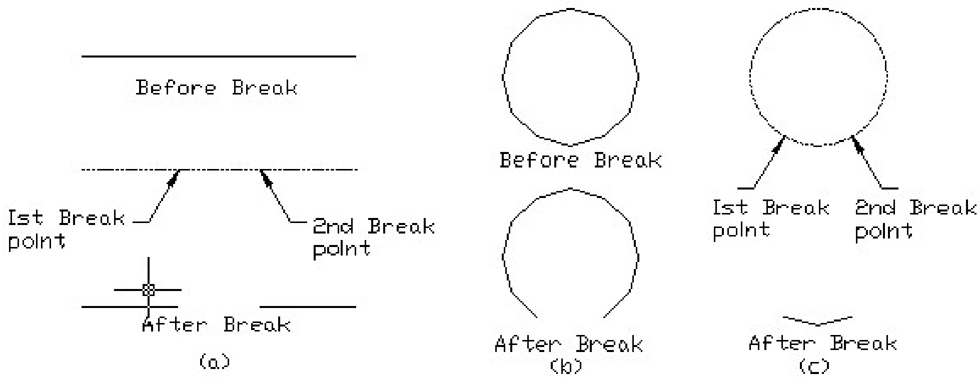


Figure 11.63: Use of Break Command

11.4.14 ***Chamfer***

Modify Toolbar: 
Menu: **Modify → Chamfer**
Command: **CHAMFER**

With the help of the **Chamfer** command the sharp edge of an object can be chamfered according to the requirement. By invoking chamfer command the following prompt sequence will be displayed.

(TRIM mode) Current chamfer distance Dist1 = 10.0000, Dist2 = 10.0000
Select first line or [Polyline/Distance/Angle/Trim/Method]:

Necessary option as required by the user can be set from the prompt sequence before chamfering the object.

Distance Option: In this option the values of the distances are inserted, if the current chamfer distances do not match with the requirement of the user. The steps to make chamfer by distance option are as follows.

Click Chamfer → Insert D for Distance and press enter → Insert first chamfer distance and press enter → Insert second chamfer distance and press enter

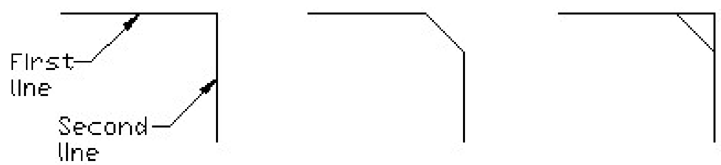
Angle Option: In this option the values of distance on one side and angle are inserted. The value of the angle has to be inserted in degree. The steps in angle option are as follows.

Insert A for Angle and press enter | Insert chamfer length on first line and press enter | Insert chamfer angle from first line and press enter

Click Chamfer → → →

Trim Option: One can set Trim or No Trim mode in this option by entering T for Trim mode and No for No Trim mode. With Trim mode the extended part beyond the chamfer will be removed while with No Trim mode, it will be retained (Figure 11.64). To make the desired chamfer, if the current Trim mode matches with the requirement of the user, the second and third steps should be avoided from the following sequence.

Click Chamfer → Insert T for Trim mode change and press enter → Enter Trim mode option Trim/No Trim and press enter → Select first line picking on it by pick box → Select second line picking on it by pick box



Before Chamfer | After Chamfer (With Trim) | After Chamfer (With No Trim)

Figure 11.64: Use of Chamfer Command

11.4.15 ***Fillet***

Modify Toolbar: 
Menu: **Modify → Fillet**
Command: **FILLET**

The **Fillet** command is used to make the fillet (round arc) or round connecting two objects. The fillet is the inside round corner and the round is the outside one. By invoking Fillet command the following prompt sequence will be displayed.

Current Settings: Mode = TRIM, Radius = 10.000
Select first object or [Polyline/Radius/Trim]:

Radius Option: If the required fillet radius of the user does not match with the current fillet radius, one has to change the current fillet radius in this option. To obtain the required fillet radius the steps as mentioned below are followed.

Click Fillet → Insert R for fillet radius and press enter → Insert fillet radius and press enter

Trim Option: One can set Trim or No Trim mode in this option by entering T for Trim mode and No for No Trim mode. With Trim mode the extended part beyond the fillet or round will be removed while with No Trim mode, it will be retained (Figure 11.65). To make the fillet, if the current Trim

mode matches with the requirement of the user, the second and third steps should be avoided from the following sequence. The fillet command can be used between two parallel lines, where the fillet will turn into the shape of a semicircle, the radius of which is equal to half the distance between the parallel lines.
The sequence to make fillet by trim option is as follows.

Click Fillet → Insert T for Trim mode change and press enter → Enter Trim/No Trim mode and press enter → Select first object → Select second object

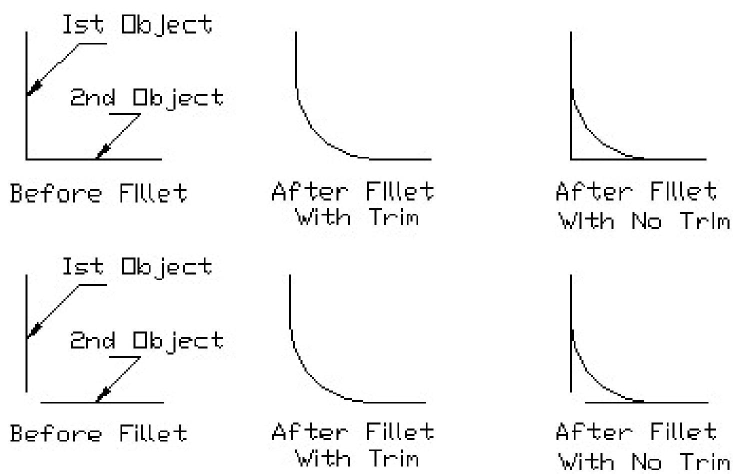


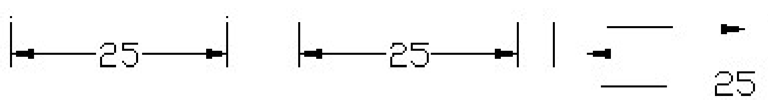
Figure 11.65: Use of Fillet Command

11.4.16 ***Explode***

Modify Toolbar: 
Menu: **Modify → Explode**
Command: **EXPLODE**

Using the **Explode** command, any dimension or block can be broken into separate objects (Figure 11.66). In case of a dimension line, it is separated into the text, arrowhead and line objects. Then each object can be selected and edited individually. When a block is exploded, it turns into a group of objects that can be edited separately. To perform the explode command; the steps as mentioned below are to be followed.

Click Explode → Select object(s) to be exploded → Click right mouse button or press



Before Explode During Selection After Explode

Figure 11.66: Use of Explode Command

11.5 ***Object Properties Toolbar***

In Figure 11.67 an object properties toolbar is shown. It gives the provision to change the color, line weight, and line types of the object(s). This toolbar also contains another important layers option.

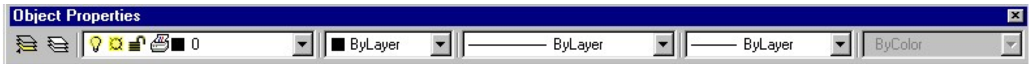


Figure 11.67: Object Properties Toolbar

11.5.1 ***Color***

A drop down list of various available colors is shown in Figure 11.68. Any one color may be chosen from this list according to the requirement. This drop down list is obtained by clicking the down arrow button just above the list. If one selects the option **Other** in the list, a Select Color dialog box will be displayed from where the desired color can also be chosen. The Select Color dialog box can also be displayed inserting COLOR in the command area.

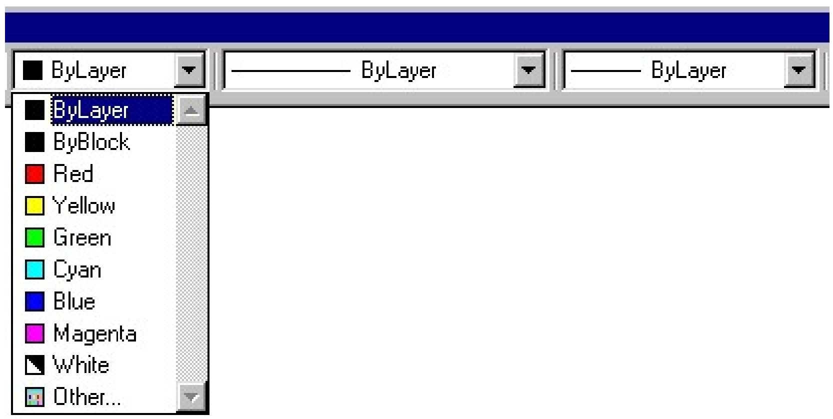


Figure 11.68: Drop Down List Showing Various Colors

11.5.2 ***Line Weights***

The thickness of the various lines (e.g. continuous line, center line and hidden line etc.) in the drawing can be changed by selecting the required thickness from the drop down list as shown in Figure 11.69. The drop down list is obtained by clicking the down arrow button just above the drop down list.

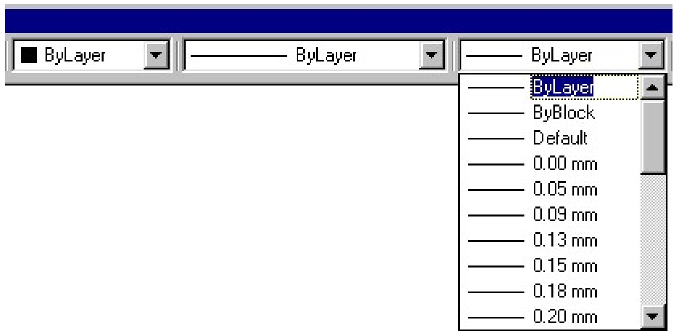


Figure 11.69: Drop Down List Showing Various Line Thickness

11.5.3 ***Line Types***

From the drop down list as shown in Figure 11.70, one can select any desired line type. When the option **Other** is selected from the list, a Linetype Manager dialogue box as shown in Figure 11.71 will be displayed. Now clicking on the **Load** tab located at top right side of this dialogue box, another Load or Reload Linetypes dialogue box will also appear as shown in Figure 11.72.

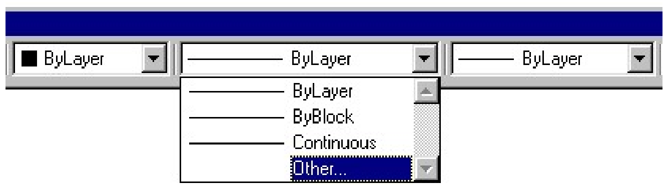


Figure 11.70: Drop Down List Showing Line Types

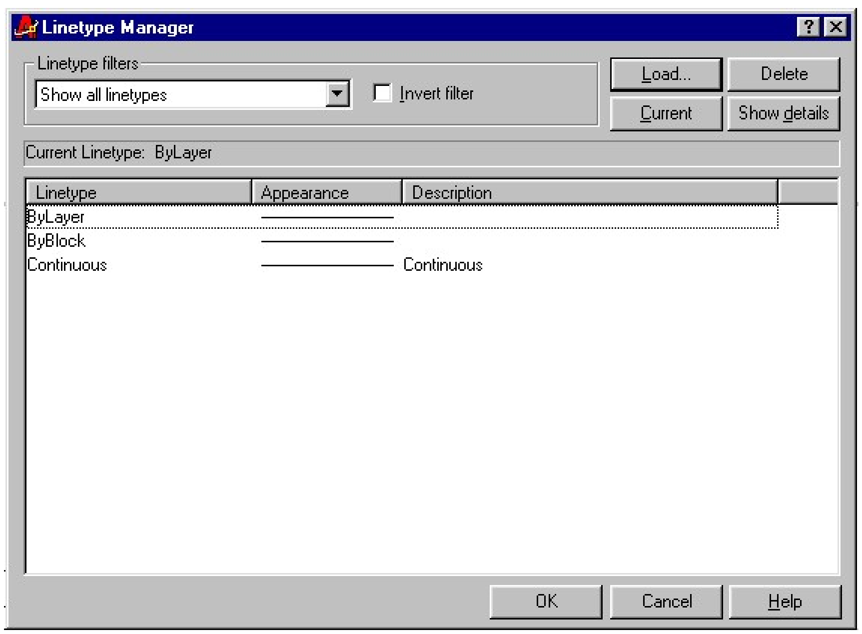


Figure11.71: Linetype Manager Dialog Box

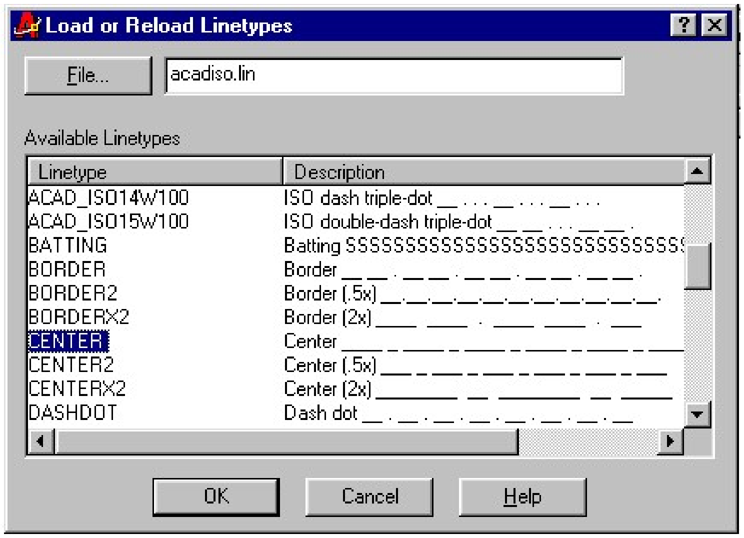


Figure 11.72: Load or Reload Linetypes Dialog Box

There are different types of lines available in the Load or Reload Linetypes dialog box from where one can select the required one. After selecting the required line, click on the **OK** button first in this dialog box, then on the **OK** button in Linetype Manager dialog box. This line will be loaded and visible in the drop down list for selection.

11.5.4 *Layers*

Toolbar: **Object Properties → Layers**
Menu: **Format → Layers**
Command: **LAYER**

Layers are like transparent sheets. When the drawing becomes too much complicated due to the presence of lots of drawing objects then it becomes easier to draw different set of objects in different layers shown in Figure 11.73.

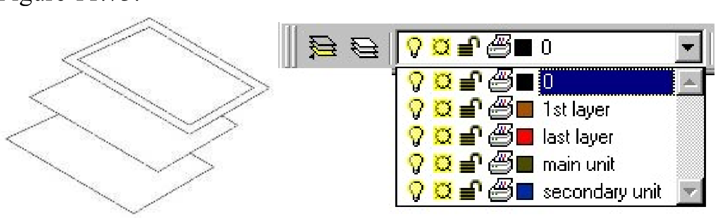


Figure 11.73: Different Layers

The Layer Properties Manager dialog box as shown in Figure 11.74 is displayed by clicking **Format → Layer** from the menu. To generate layer, click on the **New** tab located at the top right corner of the Layer Properties Manager dialogue box. After generation of the different layers, they become as displayed in the drop down list as shown in Figure 11.73. Now the user may work in the different layers after selecting them from this drop down list.

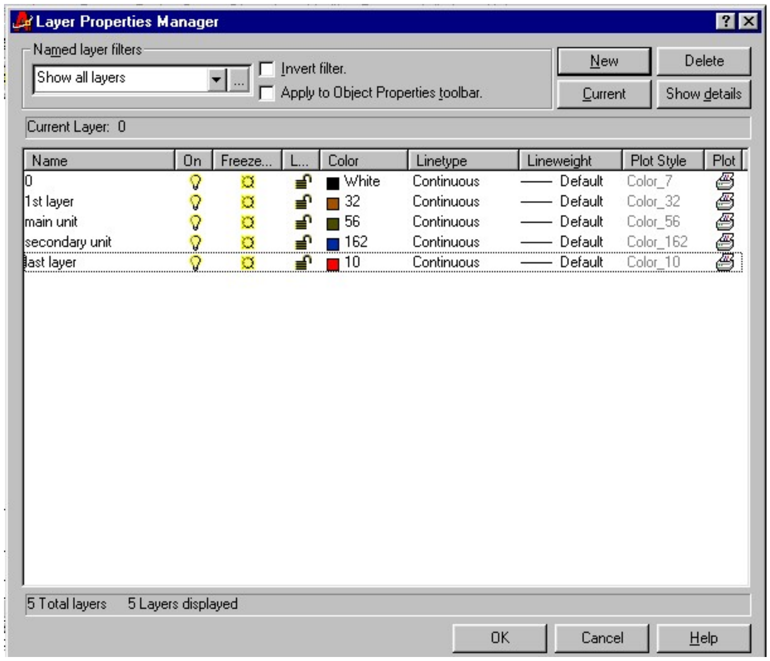


Figure 11.74: Layer Properties Manager Dialog Box

11.5.5 *Properties Dialog Box*

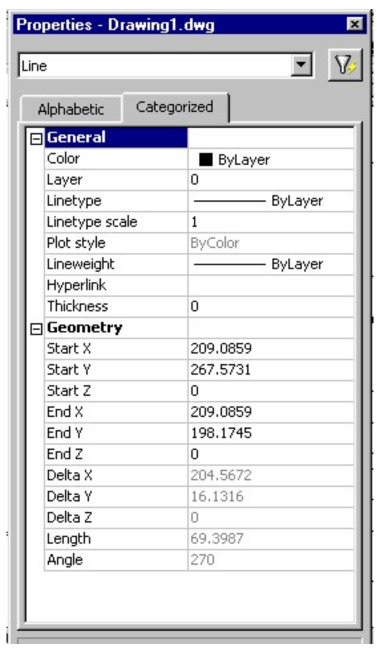


Figure 11.75: Properties Dialogue Box

The properties (thickness, scale, layer, color etc.) of the different objects such as, line, circle, arc etc. can be modified using the Properties dialogue box as shown in Figure 11.75. This dialog box can be displayed by clicking **Tools → Properties** from the menu. One can find this dialog box by clicking **Properties** in the short cut menu as well. When the object will be selected for changing its properties the name of it (e.g. Line, Circle, Arc etc.) will be displayed at the top of this dialog box.

11.6 *Drawing Display*

In AutoCAD some commands such as, Redraw, Zoom are necessary to perform the drawing precisely and smoothly. These commands are very useful; as a result they have been taken into consideration for discussion in short in this section.

11.6.1 *Redraw*

Standard Toolbar: 
Menu: **View → Redraw**
Command: **REDRAW** or **R**

During drawing in AutoCAD it is necessary to pick at different points and erase different objects. When picking is made at different points, small cross marks may appear. These small marks are called blip marks or **blips**. These blip marks are not the part of the drawing. After deleting an object from the drawing, some spots may remain there in the drawing as well. In order to clean them from the drawing **Redraw** command is used. This command may be either invoked from the standard toolbar or from the menu. When **Redraw** command is clicked the whole drawing is cleaned eliminating these undesired marks. While clicking on the command no prompt sequence is displayed in the command prompt area. In AutoCAD for some commands redraw action takes place automatically.

11.6.2 *Zoom*

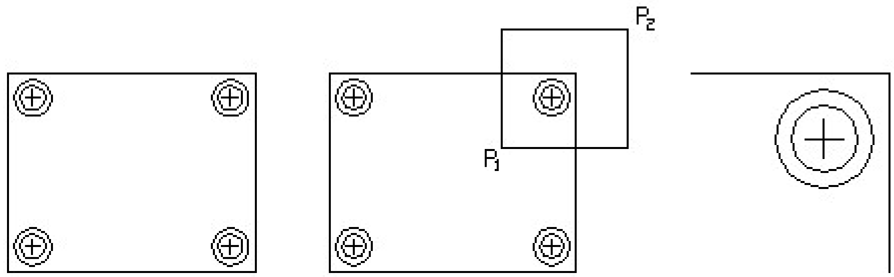
Sometimes it happens that it is necessary to draw a small object or modify it in a drawing. It is difficult to accomplish that manually but in AutoCAD it can be done easily by making enlarged view of that portion of the drawing. In AutoCAD there is **Zoom** command, which can be used to enlarge or reduce size of the drawing, displayed on the screen. **Zoom** command has several options. Of them the most useful options are only taken into consideration here.

Zoom Window Option

Standard Toolbar: 
Menu: **View → Zoom → Window**
Command: **ZOOM** or **Z**

Zoom window option is the most useful one. The specific area, which is needed to be magnified, is brought within the rectangular window under this option. The window is created by two specified points at the opposite corners of the rectangular window. One may specify the two opposite corners of the window either clicking at the points with the cross hair or inserting values of the co-ordinates of the points in the command area. The center of the window becomes the center of the new display screen. The window option of **Zoom** command can be invoked either from the standard toolbar or from the menu. It can also be initiated by entering the ZOOM in the command prompt area. The steps for zooming objects (Figure 11.76) are given below.

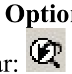
Click	→	Click at point P_1	→	Click at point P_2



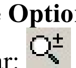
Before Zoom During Zoom After Zoom

Figure 11.76: Zooming Object with Window Option

Zoom Previous Option

Standard Toolbar: 
Menu: **View → Zoom → Previous**
Command: **ZOOM** or **Z**
To edit a portion of a drawing it may be required to magnify it by **Zoom** command. At the end of the edit it becomes necessary to go back to the previous view, which can be done by using the previous option of **Zoom** command. This command can be invoked either from the standard toolbar or from the menu. One can also initiate this **Zoom** command by entering ZOOM in the command prompt area. Ten previous views are stored in the view port; as such one may come back to the required view by using previous option of **Zoom** command successively until it appears on the screen.

Zoom Realtime Option

Standard Toolbar: 
Menu: **View → Zoom → In/Out**
Command: **ZOOM** or **Z**

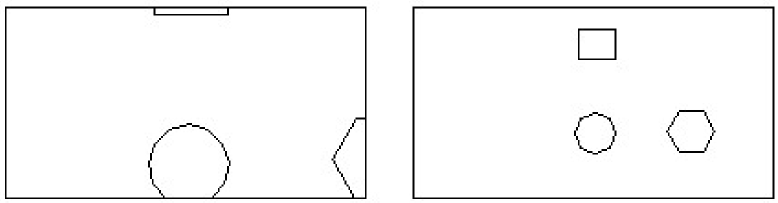
Realtime option of **Zoom** command is the most useful command for magnification or reduction of size of the drawing. During performing drawing, sometimes this command becomes necessary. This command can be invoked from the standard toolbar or from the menu or by entering ZOOM in the command area. When it is initiated from the standard tool bar, its positive sign indicates **Zoom in** while its negative sign indicates **Zoom out**. To activate the positive sign pressing the left mouse button drag the cursor to the upward direction, the drawing will be magnified. To further magnify bring back the cursor and drag it in the same way in the upward direction.

To reduce the size of the drawing the opposite action has to be done. That is, press the left mouse button and drag the cursor to the downward direction. To further reduce the size bring back the cursor and drag it in the same way in the downward direction. If the cursor is dragged vertically up from the mid point of the screen to the top of it, the drawing will be magnified by 100% (Zoom in 2x magnification). On the other hand if the cursor is dragged vertically downward from the mid point of the screen to the bottom of it, the drawing will be reduced by 100% (Zoom out 0.5x magnification). If the command is invoked from the menu, for **In** option the drawing is magnified by 100% and for **Out** option it is reduced by 100%. While invoking the command from the toolbar the sequence as mentioned below has to be followed.

Click Zoom	→	Press left mouse button and drag it up or down to magnify or reduce drawing size	→	Click right mouse button and exit or press enter

Zoom All Option
Menu: **View → Zoom → All**
Command: **ZOOM** or **Z**

All option of **Zoom** command can be invoked from the menu. It can also be initiated by entering ZOOM in the command area. It is the useful command for drawing. Sometimes some portion of a drawing in a file may remain outside the boundary limit of the screen, which is not displayed. Using this command all the objects of the file appear on the screen (Figure 11.77).



Before Zooming After Zooming

Figure 11.77: Application of Zoom Command

11.6.3 ***Realtime Pan***

Standard Toolbar: 
Menu: **View → Pan → Realtime**
Command: **PAN** or **P**

With the help of Realtime option of **Pan** command one can slide the drawing to the left, right, up or down to bring the portion of the drawing inside the window for work. This command can be invoked either from the standard toolbar or from the menu. It can also be initiated by entering PAN in the command area. Pressing the left mouse button drag the cursor to the desired direction until the desired portion of the drawing appears in the window. The steps as mentioned below are followed to slide the drawing.

Click Pan	→	Press left mouse button and drag it to slide drawing to desired direction	→	Click right mouse button and exit or press enter

11.7 ***Dimensioning***

Menu: **View → Toolbars → Dimension**
Menu: **Dimension**

Dimension indicates the measurement of the object, which is used for manufacturing. The Dimension commands can be invoked either from the Dimension toolbar or from the Dimension menu. Dimension toolbar (Figure 11.78) can be obtained selecting the crosscheck box beside Dimension in the Toolbars dialog box (Figure 11.13).

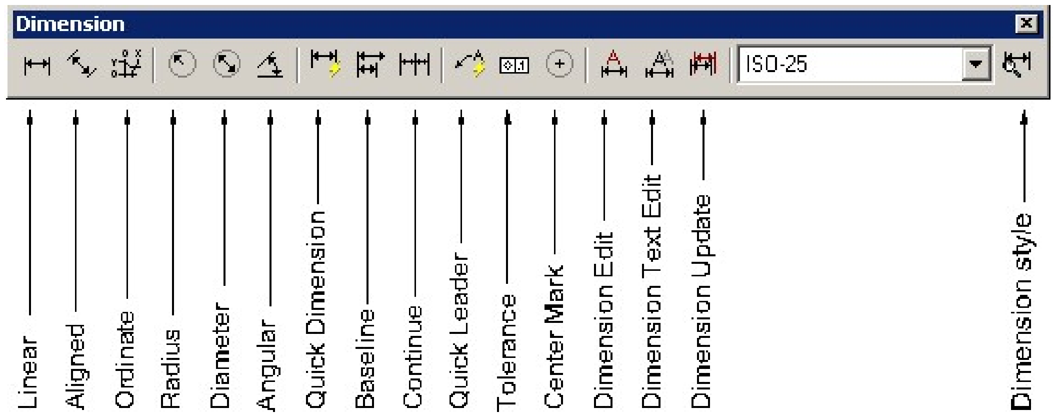


Figure 11.78: Dimension Toolbar

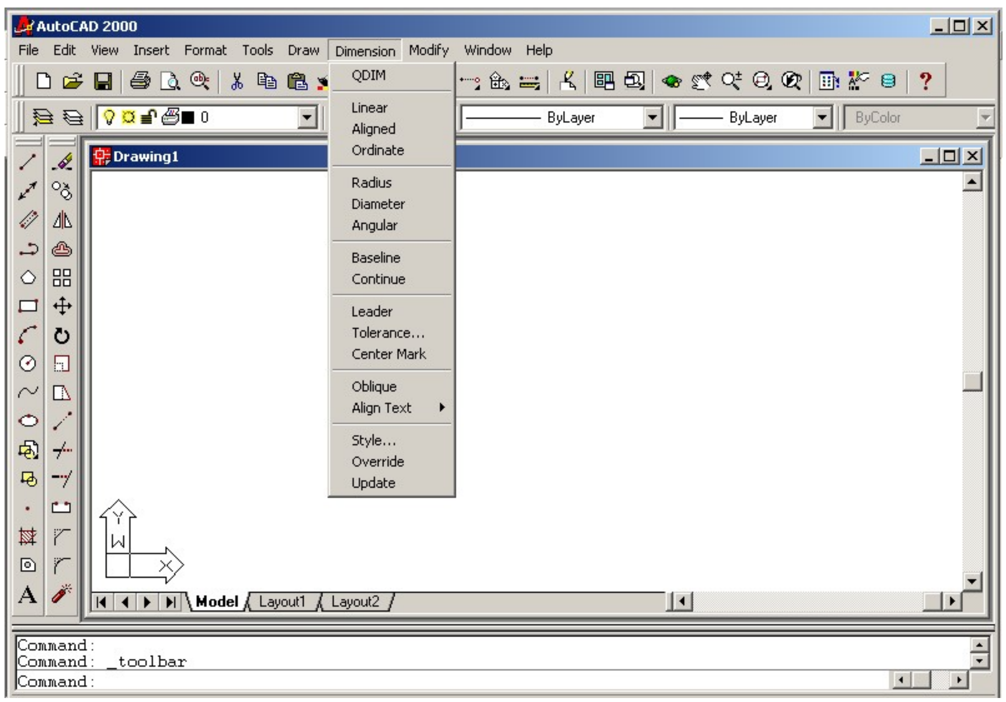


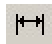
Figure 11.79: AutoCAD Window Showing Dimension Menu

On the other hand, Dimension menu (Figure 11.79) can be obtained selecting **Dimension** title in the menu bar. Dimension commands can also be initiated by entering command in the command prompt area.

11.7.1 *Types of Dimensioning*

As can be seen from the Dimension menu or Dimension toolbar that they consist of many dimension commands. But here only the most common types of dimension commands (e.g. Linear, Aligned, Baseline, Continue, Radius, Diameter, Angular, Center Mark and Leader) will be taken into consideration for discussion in short.

Linear

Dimension Toolbar: 
Menu: **Linear**
Command: **DIMLIN** or **DIMLINEAR**

To give the dimension between two points of a line either horizontally or vertically (Figure 11.80) **Linear** command is used. Object snap (OSNAP) mode is made on to locate the points accurately. The sequence to give linear dimension on an object, invoking command from the toolbar or menu is as follows.

Click Linear → Click on first extension line origin P_1 → Click on second extension line origin P_2 → Move cursor to the location of dimension line and click left button

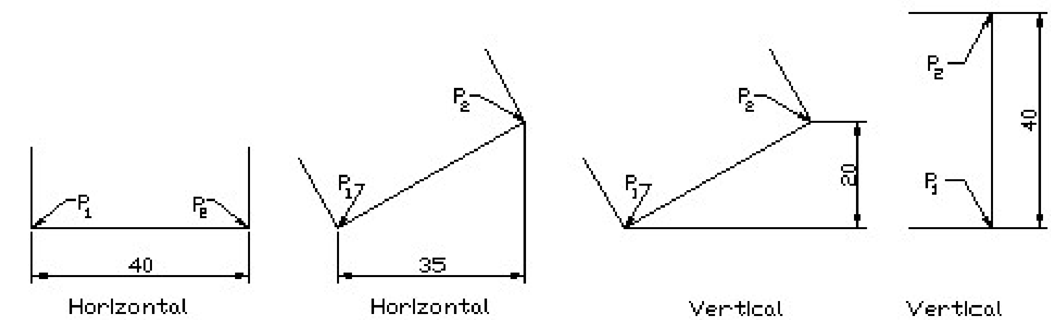


Figure 11.80: Linear Dimension

Aligned

Dimension Toolbar: 
Menu: **Aligned**
Command: **DIMALIGNED**

Using **Aligned** command, dimensioning between two end points of an inclined line can be given (Figure 11.81). With the help of this command linear dimensioning can also be given. The steps to give aligned dimension on an object, invoking command from the toolbar or menu are as follows.

Click Aligned → Click on first extension line origin P_1 → Click on second extension line origin P_2 → Move cursor to the location of dimension line and click left button

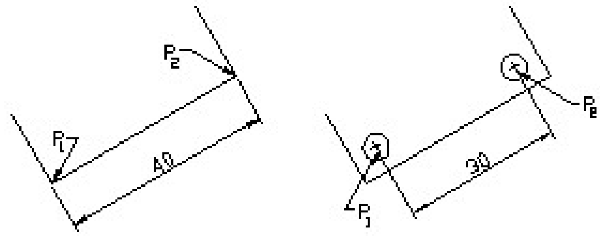


Figure 11.81: Aligned Dimension

Baseline

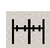
Dimension Toolbar: 
Menu: **Baseline**
Command: **DIMBASE** or **DIMBASELINE**

When dimensions (Figure 11.82a) are given in reference to a fixed point (base point or reference point), it is called baseline dimensioning. **Baseline** command is used for this dimension. The sequence for the baseline dimension invoking command from the toolbar or menu is mentioned below.

Click Linear → Click on first extension line origin P_1 → Click on second extension line origin P_2 → Move cursor to dimension line location and click left button

→ Click Baseline → Click on second extension line origin P_3 → Click on second extension line origin P_4 and so on → To end Press enter twice or click right mouse button and enter and press enter / right click

Continue

Dimension Toolbar: 
Menu: **Continue**
Command: **DIMCONT** or **DIMCONTINUE**

Using **Continue** command linear dimensioning can be continued from the second extension line of the previous dimension. This is a kind of chain dimensioning (Figure 11.82b). When the command is invoked from the toolbar or menu, the steps for the continue dimensioning become as follows.

Click Linear → Click on first extension line origin P_1 → Click on second extension line origin P_2 → Move cursor to dimension line location and click left button

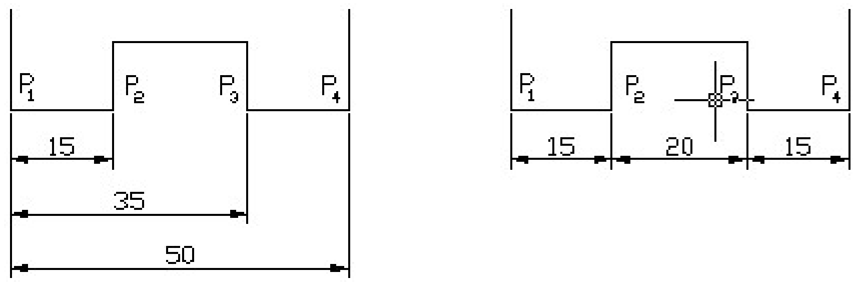
Click Continue

Click on second extension line origin P_3

Click on second extension line origin P_4 and so on

To end Press enter twice or click right mouse button and enter and press enter / right click

→ → → →



(a) (b)

Figure 11.82: Baseline and Continue Dimensions

Radius

Dimension Toolbar: 
Menu: **Radius**
Command: **DIMRAD** or **DIMRADIUS**

With the help of **Radius** command, the radius of an arc or the circle (Figure 11.83a) can be given. The steps to give the radius dimension invoking command from the toolbar or menu are as follows.

Click Radius → Select arc or circle by pick box → Move cursor to dimension line location and click

Diameter

Dimension Toolbar: 
Menu: **Diameter**
Command: **DIMDIA** or **DIMDIAMETER**

Using **Diameter** command, the diameter of an arc or the circle (Figure 11.83b) can be given. The steps to give the diameter dimension invoking command from the toolbar or menu are as follows

Click Diameter → Select arc or circle by pick box → Move cursor to dimension line location and click

Angular

Dimension Toolbar: 
Menu: **Angular**
Command: **DIMANG** or **DIMANGULAR**

The angle between two intersecting lines can be given using **Angular** Command (Figure 11.83c). The sequence to insert the angular dimension invoking command from the toolbar or menu is shown below.

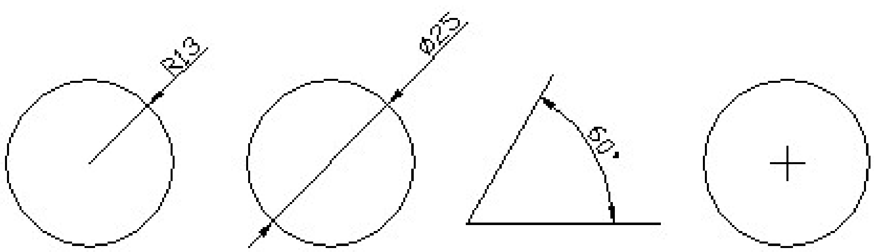
Click Angular → Select first line by pick box → Select second line by pick box → Move cursor to dimension line location and click

Center Mark

Dimension Toolbar: 
Menu: **Center Mark**
Command: **DIMCENTER**

Sometimes it is necessary to give the center mark of an arc or circle (Figure 11.83d), which can be done by using **Center Mark** Command. The sequence to provide center mark is shown below.

Click Center Mark → Select arc or circle



(a) (b) (c) (d)

Figure 11.83: Radius, Diameter, Angular Dimensions and Center Mark

Leader

Dimension Toolbar: 
Menu: **Leader**
Command: **LEADER** or **QLEADER**

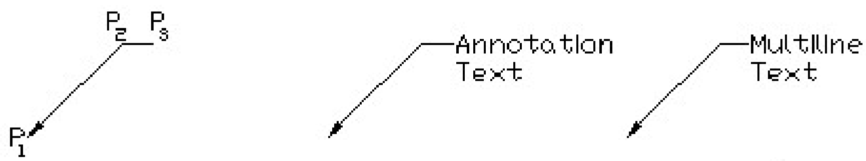
Often it becomes necessary to show dimensioning without using another dimension command or attach annotations to an object, which can be done using **Leader** command. The leader consists of an arrow and a leader line. The leader command offers several options as well. The sequence to draw a leader is shown below.

Click at first leader point P_1 | Click at next point P_2 | Click at next point P_3 | Insert text width if any and press enter

Click Leader → → → →

→ Insert first line of annotation and press enter → Insert next line of annotation if any and press enter and so on or press enter to exit → Press enter -Multiline Text Editor dialog box displayed → Enter Text and select **OK** button to exit

In option for no text (Figure 11.84a), omit 6th and 7th steps in the above sequence, and select **OK** button only in 9th step to exit. In option for annotation text (Figure 11.84b) omit the last two steps. In option for Multiline text (Figure 11.84c) omit 6th and 7th steps.



(a) (b) (c)

Figure 11.84: Leader in Different Options

11.7.2 ***Setting Dimension Properties***

Dimension Toolbar: 
Menu: **Format→ Dimension Style**
or **Dimension → Style**
Command: **DDIM** or **DIMSTYLE**

Invoking **Dimension Style** command from the dimension toolbar or menu a Dimension Style Manager dialog box as shown in Figure 11.85 will be displayed. It can also be initiated by entering DDIM/DIMSTYLE command in the command area.

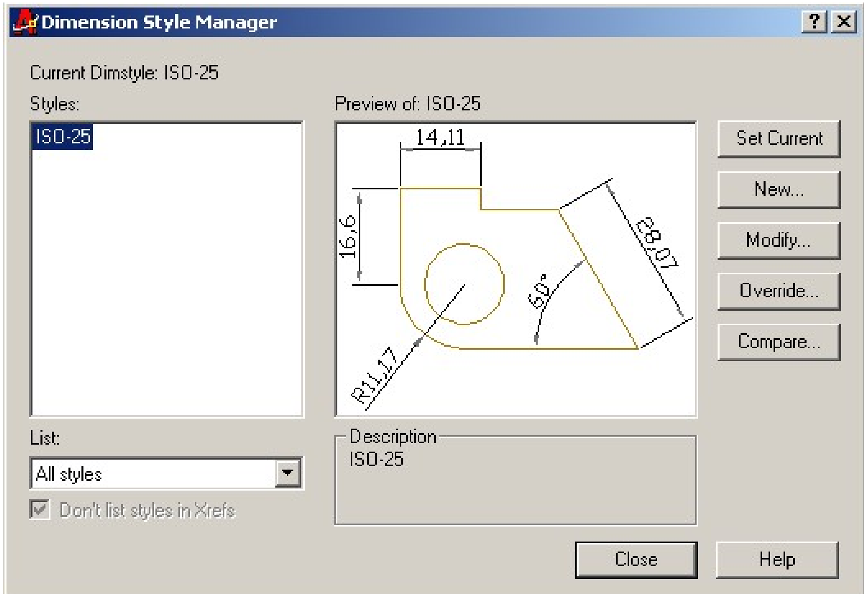


Figure 11.85: Dimension Style Manager Dialogue Box

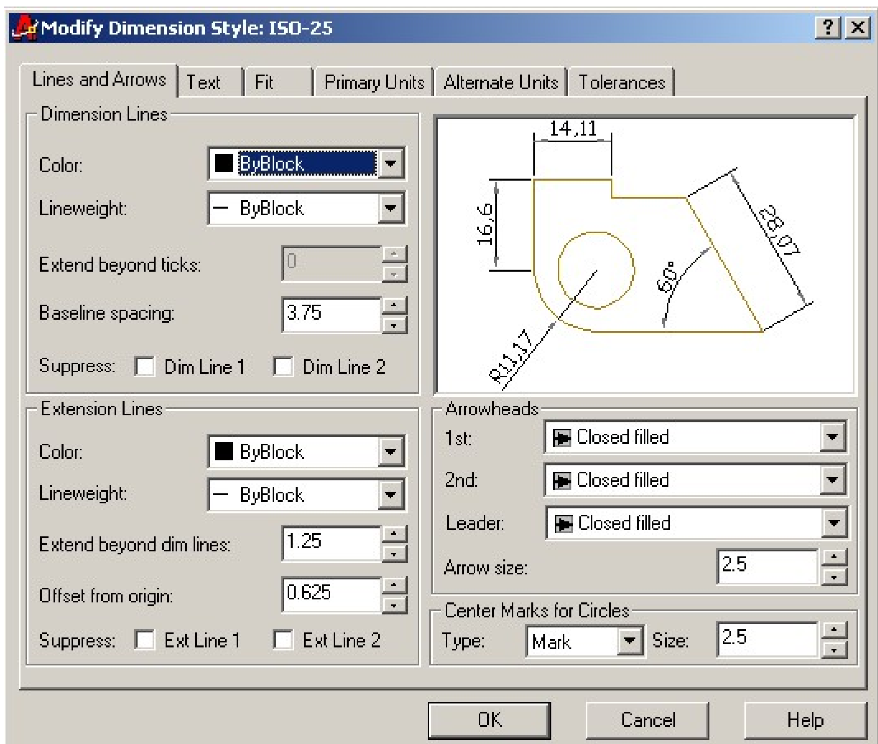


Figure 11.86: Modify Dimension Style Dialogue Box (For Lines and Arrows)

Now to set the different properties of the dimensions, click the **Modify** button in this dialog box. Then a Modify Dimension Style dialog box (For Lines and Arrows) will appear by default (Figure 11.86). This dialogue box consists of a number of tabs such as, Lines and Arrows, Text, Fit, Primary units, Alternate units, Tolerances from where any one can be selected according to the requirement. When the Text tab is selected a ModifyDimension Style dialog box (For Text) is displayed (Figure 11.87). The Modify Dimension Style dialog box (For Lines and Arrows) contains a number of areas such as, Dimension Lines, Extension Lines, Arrowheads and Center Marks for Circles. On the other hand the Modify Dimension Style dialog box (For Text) contains a number of areas such as, Text Appearance, Text Placement and Text Alignment.

According to the requirement of the user necessary settings can be made from here. When the Primary Units tab of the Modify Dimension Style dialog box will be selected that dialog box will provide different areas like Linear Dimensions, Measurement Scale, Angular Dimensions and Zero Suppressions from where one can set the scale of drawing, precision and decimal separator etc. After necessary settings select the **OK** button in the Modify Dimension Style dialog box followed by **Close** button in the Dimension Style Manager dialog box.

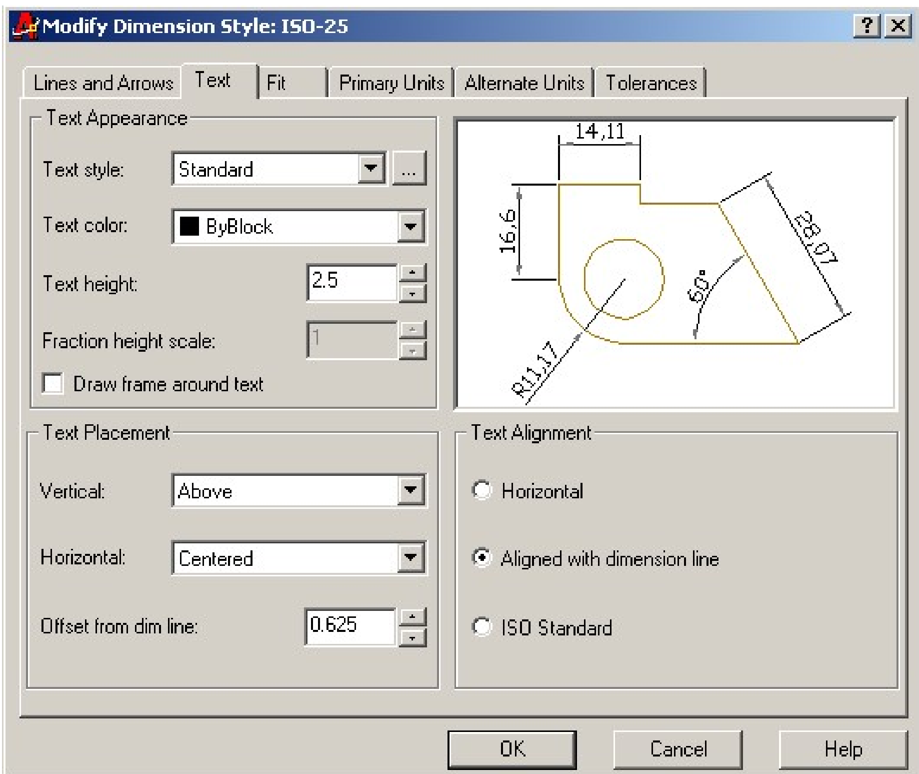
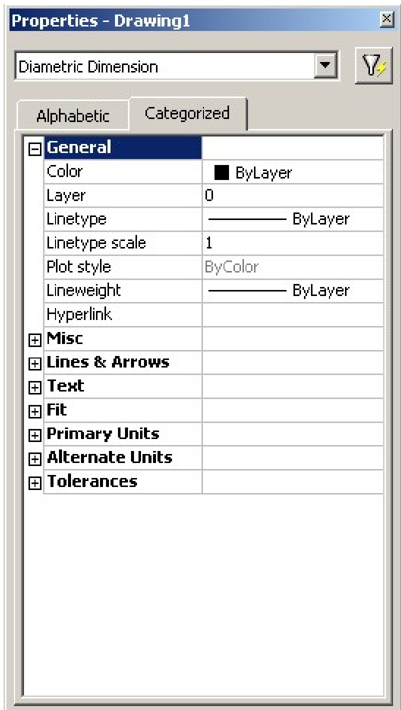
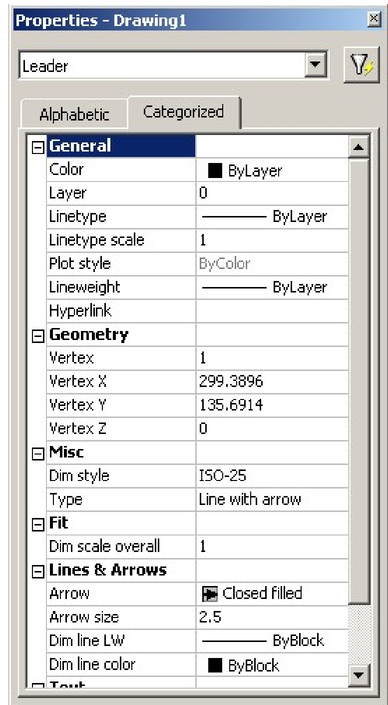


Figure 11.87: Modify Dimension Style dialogue Box (For Text)

11.7.3 ***Editing Dimensions***
Menu: **Tools → Properties** or **Modify → Properties**
Command: **PROPERTIES**

The Properties window can be invoked from the menu or by entering **Properties** command in the command area. To change the properties of dimension, dimension text style, geometry, format etc. Properties window can be used. The Properties window for dimension (Figure 11.88a) consists of a number of sections such as, General, Misc, Primary Units, Fit, Lines & Arrows, Text, Tolerances and Tolerances & Text. While the Properties window for leader (Figure 11.88b) contains various sections like General, Geometry, Misc, Fit, Lines & Arrows and Text.





(a) (b)

Figure 11.88: Properties Window

When any dimension is selected to edit, that dimension name appears at the top of the Properties window. In order to use any of them select it and perform the necessary change in the dimension.

11.8 *Isometric Drawing*

In this section a brief idea to make the isometric drawing will be provided. The dimensioning in the isometric drawing will also be taken into consideration for discussion.

11.8.1 *Generating Isometric Drawing*

Menu: **Tools → Drafting Settings**
Command: **DSETTINGS**

Drafting Settings dialogue box can be obtained from the menu or by entering command in the command prompt area. Drafting Settings dialog box can also be invoked from the status bar by right clicking on any of the button except the ORTHO and MODEL button, a short cut menu will appear from where settings can be selected. Now clicking on **Snap and Grid** button at the top left corner of the Drafting Settings dialogue box, one finds the dialog box (Figure 11.22) where click on **Isometric snap** radio button in the Snap type & style area. Next click on **Polar Tracking** button, the dialogue box will turn into the feature as shown in Figure 11.89. Now in the area of Polar Angle Settings of this dialogue box, select the increment angle of 30 from the drop down list, which is found clicking on the down arrow button. In this condition the cross hair cursor is set for the isometric drawing.

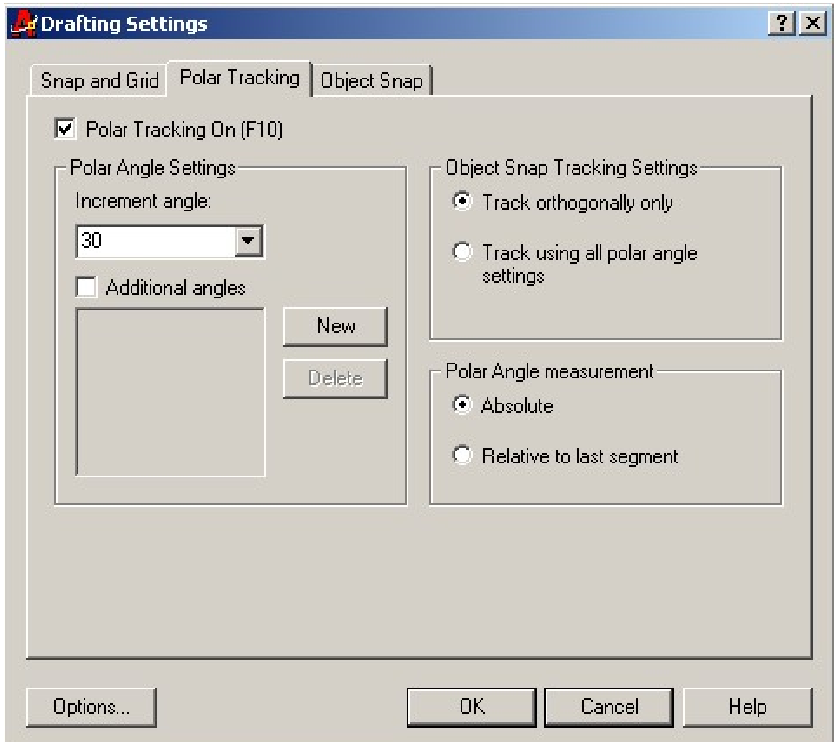


Figure 11.89: Drafting Settings Dialogue Box (For Polar Tracking)

The orientation of the crosshair for the isoplane left is shown in Figure11.90.

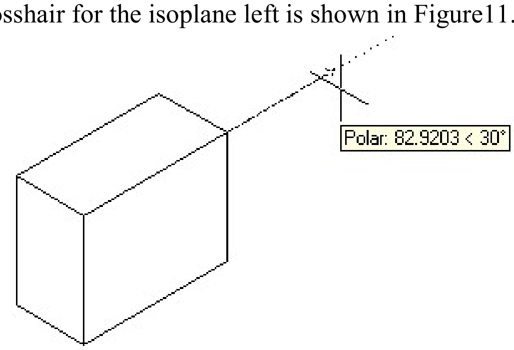


Figure 11.90: Crosshair Position For Isoplane Left

However, a user can easily change the orientation of the crosshair cursor for the isoplane right and isoplane top by pressing **Ctrl** and **E** keys simultaneously (**Ctrl** + **E**) or using the function key F5. The different isoplanes can also be obtained by entering **Isoplane** command in the command prompt area followed by L for left, R for right and T for top.

In the isometric view the circle in the different isoplanes turns into the shape of ellipse, which is called isocircle. To draw the isocircle the steps as mentioned below are followed. However, before doing that, the position of the crosshair has to be changed for the respective isoplane left, isoplane right and isoplane top as shown in Figure 11.91 from left to right.

Click Ellipse → Insert I for isocircle option → Click at center point of isocircle → Insert value of circle radius and press enter

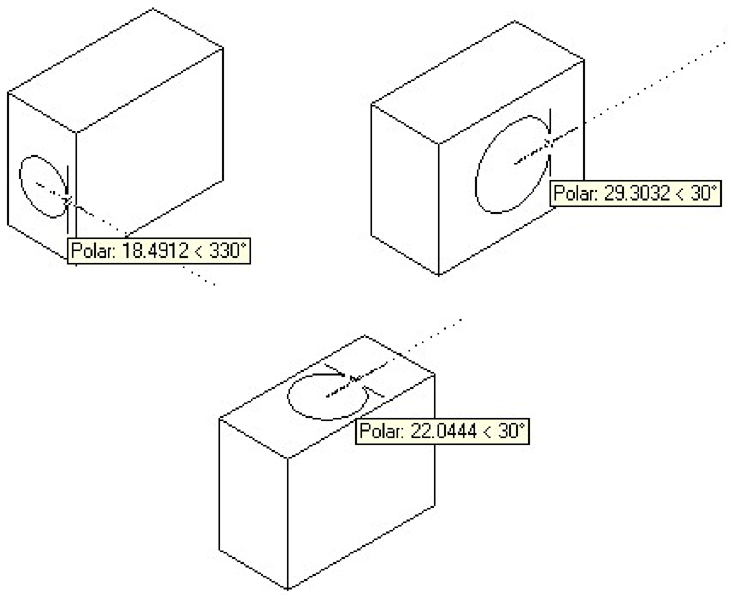
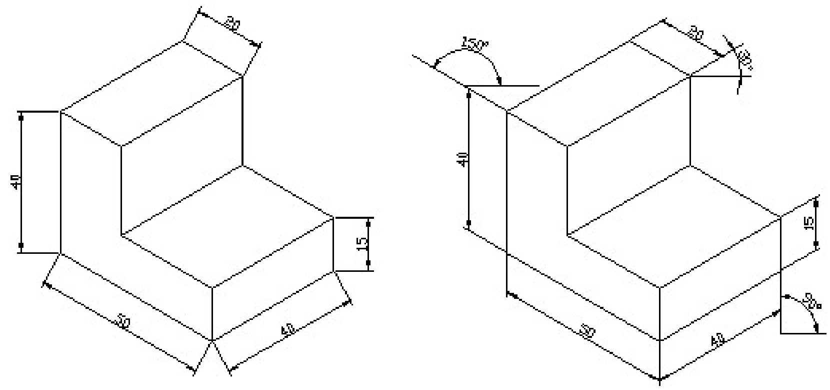


Figure 11.91: Isocircle in Different Isoplanes

11.8.2 *Dimensioning Isometric Objects*

In order to give the dimension in the isometric objects two steps are followed. In the first step using **Aligned** command (Section 11.7.1), the necessary dimensions are given (Figure 11.92a). Then in the second step these dimensions are edited with the help of **Oblique** command (Figure 11.92b). The sequence to edit the dimensions is as follows. In the third step of the sequence the value of oblique angle is to be inserted. The angle between the extension line of the isometric dimension and positive x-axis is called the oblique angle. The oblique angles 30^0, 90^0 and 150^0 are shown in Figure 11.92b.

Click Dimension → Click oblique → Select dimension by pick box → Press enter or right → Insert value of oblique angle and



(a) (b)

Figure 11.92: Dimensioning in Isometric Drawing

11.9 ***Solid Object Drawing***

Menu: **View →Toolbars → Solids/Solids Editing**
Menu: **Draw → Solids**
Menu: **Modify → Solids Editing**

AutoCAD provides the scope to draw a great variety of solid objects. Some standard geometrical shapes are available in **Solids** toolbar. Solid drawings are edited with some tools that are available in **Solids Editing** toolbar. These toolbars can be obtained by selecting the crosscheck box beside **Solids** and **Solids Editing** in the Toolbars dialog box (Figure 11.13). Then the Solids and Solids Editing toolbars will be displayed, which are shown in Figure 11.93 and Figure 11.94 respectively. One can also obtain **Solids** commands from Draw menu (Figure 11.95) and **Solids Editing** commands (Figure 11.96) from Modify menu.

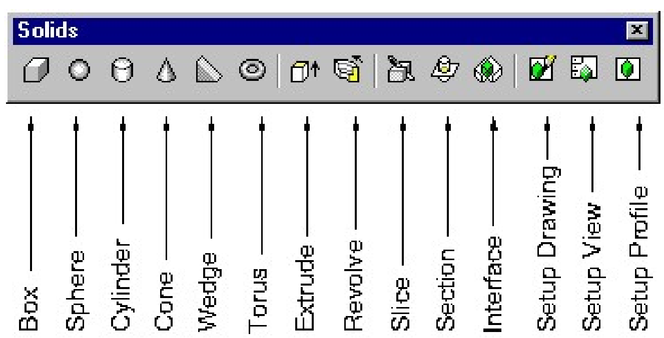
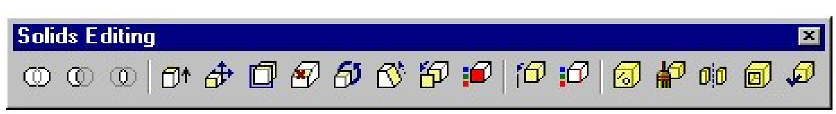


Figure 11.93: Solids Toolbars



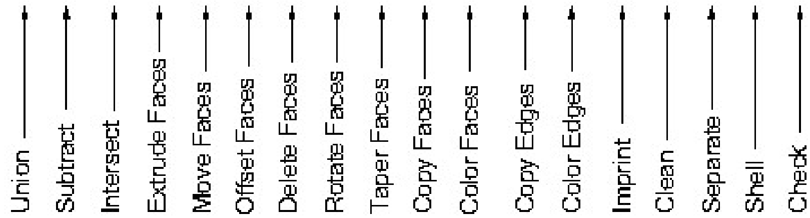


Figure 11.94: Solids Editing Toolbars

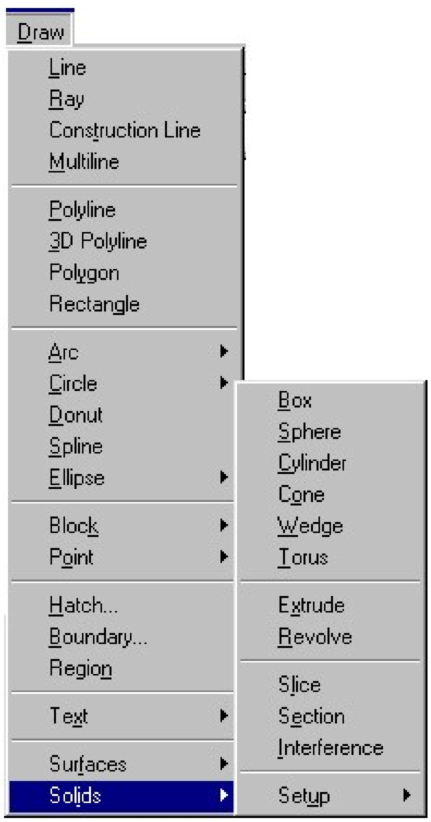


Figure 11.95: Solids in Draw Menu

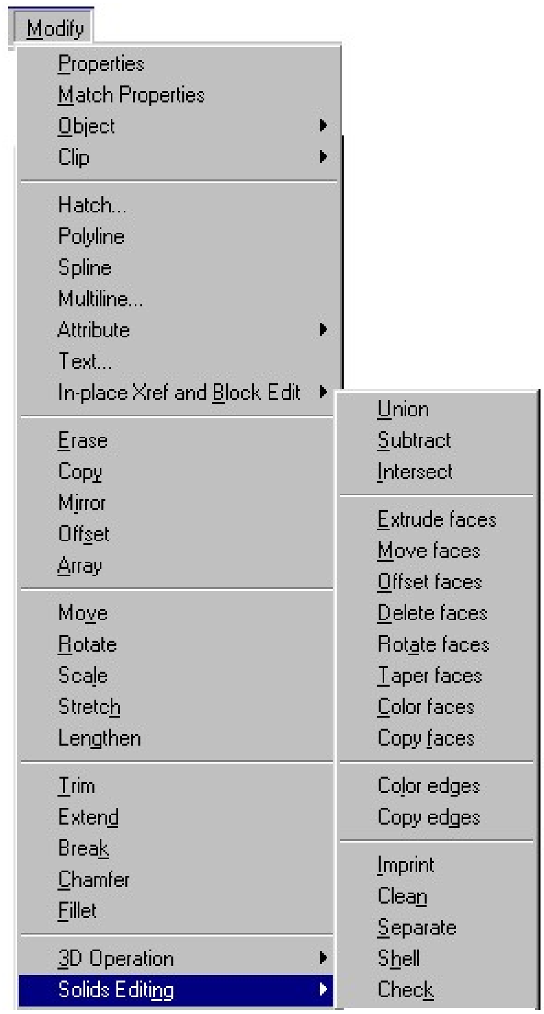


Figure 11.96: Solids Editing in Modify Menu

The most common solid objects available in the **Solids** toolbar are box, sphere, cylinder, cone, wedge and torus, which are shown from left to right in the **Solids** toolbar. Some additional solids drawing options are included in this toolbar amongst which extrude, revolve and slice may be mentioned. In the **Solids Editing** toolbar the first three tools are union, subtract and intersect. These three tools are of utmost importance while working with solid objects in AutoCAD.

To make the 3D drawing, the UCS orientation is changed beforehand. To do that steps as mentioned below are to be followed.

Click View → Click 3D Views → Click SW Isometric or SE Isometric or NE Isometric or NW Isometric as needed

11.9.1 ***Generating Solid Box***

Solids Toolbar: 
Menu: **Draw → Solids → Box**
Command: **BOX**

To generate a solid box (Figure 11.97a), **Box** command is used. This command has two options: **Corner** and **Center**. By default **Corner** option appears. The steps to generate a solid box by using corner option are given below as an example.

Click Box → Insert corner of box **50,50,0** or select corner by clicking at any desired point and → Enter L for length and press enter → Insert length **80** and press enter → Insert width **60** and press enter → Insert height **30** and press enter

11.9.2 ***Generating Solid Cone***

Solids Toolbar: 
Menu: **Draw → Solids → Cone**
Command: **CONE**

Using **Cone** command a solid cone is generated (Figure 11.97b). This command has two options: **Elliptical** and **Center point**. By default Center point option appears. The height of the cone occurs along the z-axis and the base in the xy-plane (current). However, UCS command option can be used to change the base plane. The sequence to generate a solid cone by using Center option is given below as an example.

Click Cone → Insert center point for base of cone **50,50,0** or select center by clicking at any desired point and press enter → Insert radius for base of cone **30** and press enter → Insert height of cone **60** and press enter

11.9.3 ***Generating Solid Cylinder***

Solids Toolbar: 
Menu: **Draw → Solids → Cylinder**
Command: **CYLINDER**

With the help of **Cylinder** command, a solid cylinder is generated (Figure 11.97c). This command has two options: **Elliptical** and **Center point**. By default **Center point** option appears. The base of the cylinder occurs in the xy-plane and the height occurs in the z-direction (current). However, UCS command option may be used to change the base point. The steps as mentioned below are followed to generate a solid cylinder by using Center point option as an example.

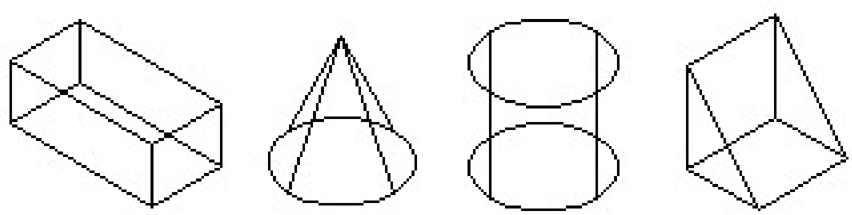
Click Cylinder	→	Insert center point for base of cylinder **50,50,0** or select center by clicking at any desired point and press enter	→	Insert radius for base of cylinder **30** and press enter	→	Insert height of cylinder **50** and press enter

11.9.4 ***Generating Solid Wedge***

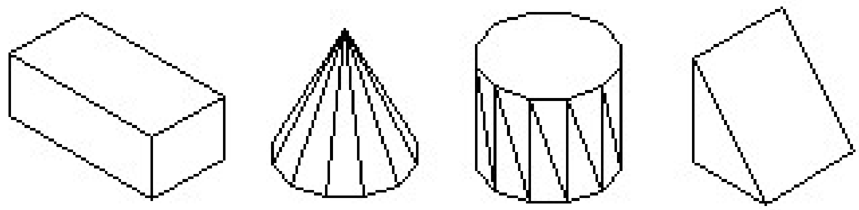
Solids Toolbar: 
Menu: **Draw → Solids → Wedge**
Command: **WEDGE**

Wedge command is used to generate a solid wedge (Figure 11.97d). It has two options: **Corner** and **Center**. By default **Corner** option appears. The sequence to generate a solid wedge by using the Corner option is given below as an example. After making the solid objects some hidden lines remain behind them, which are removed by using the **Hide** command. This command can be invoked from the menu **View → Hide** or one can initiate it by entering HIDE command in the command area followed by press enter. Then the hidden lines will be removed (Figure 11.97e to Figure 11.97h).

Click Wedge	→	Insert first corner wedge **50,50,0** and press enter	→	Insert corner **90,100,0** and press enter	→	Insert height **50** and press enter



(a) (b) (c) (d)



(e) (f) (g) (h)

Figure 11.97: Solids Objects

11.9.5 ***Revolve***

By using **Revolve** command a solid object is generated in accordance with the following steps provided as an example.

Step 1:
Menu: **Tools → New UCS**
Command: **UCS**

In this step the orientation of the UCS (SE Isometric), which was created before (Figure 11.98a) is changed into the new orientation (Figure 11.98b) by entering UCS command in the command area from the following sequence. It can also be invoked from the menu by clicking on **New UCS** command.

Enter UCS → Enter option **N** for New → Insert origin of new UCS **X** for x-axis and press → Insert rotation angle about x-axis **90** and press enter

Step 2:
In this step by using the **Polyline** command (section 11.3.2), the area is generated following the sequence as given below (Figure 11.98c).

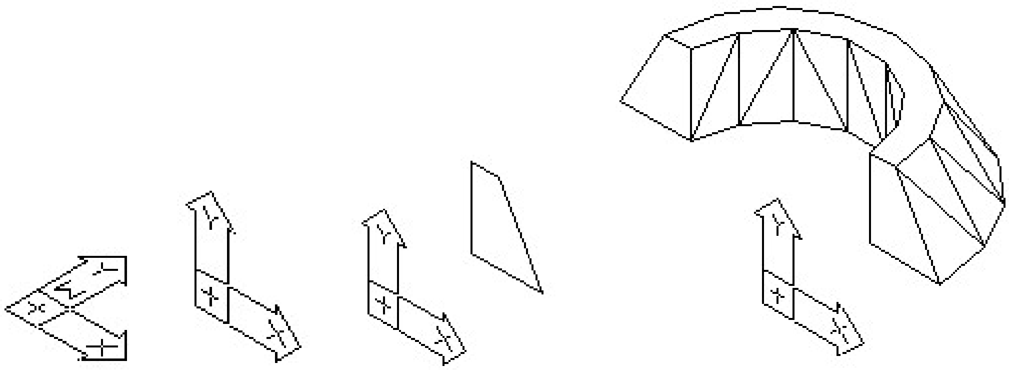
Click Polyline → Insert start point **50.50** → Insert next point **50.950** → Insert next point **65.95** → Insert next point **90.50** → Enter **C** for close

Step 3:
Solids Toolbar: 
Menu: **Draw → Solids → Revolve**
Command: **REVOLVE**

By using the **Revolve** command, one can generate the solid object revolving the area (Figure 11.98c) in accordance with the following sequence. Now after using HIDE command, one finds the solid object as shown in Figure 11.98d.

Select objects and press enter | Enter axis **Y** and press enter | Insert rotation angle **180** and press enter

Click Revolve → → →



(a) (b) (c) (d)

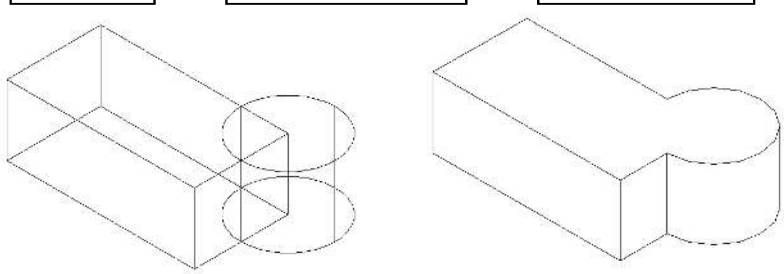
Figure 11.98: Use of Revolve Command

11.9.6 ***Union***

Solids Editing Toolbar: 
Menu: **Modify → Solids Editing → Union**
Command: **UNION**

The **Union** command in the Solids Editing toolbar can be used to unite two solids. A union of a box and a cylinder is shown in Figure 11.99. The sequence as mentioned below is followed to make the union. Then the command DISPSILH is entered in the command prompt area followed by press enter. Next insert **1** for new value of DISPSILH. Now enter the command HIDE as a result the final solid object is obtained as shown in Figure 11.99.

Click Union → Select objects to be combined → Press enter or right-click



Before Union After Union

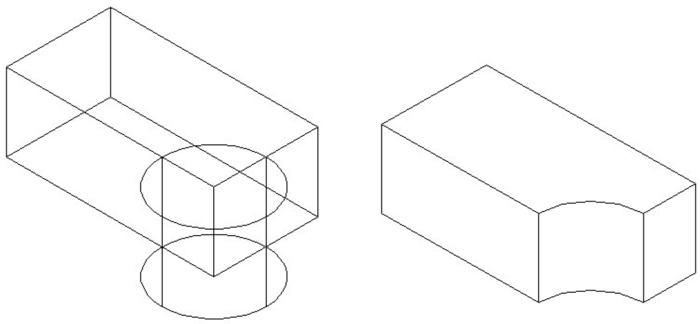
Figure 11.99: Use of Union Command

11.9.7 ***Subtract***

Solids Editing Toolbar: 
Menu: **Modify → Solids Editing → Subtract**
Command: **SUBTRACT**

One solid object or a portion of it can be subtracted from another solid object by using the **Subtract** command from the **Solids Editing** toolbar. A subtraction of a portion of cylinder from a box is shown in Figure 11.100. The steps for subtraction are as follows. The object which is selected later, the shared or intersected part of that object is subtracted from the first object. Now using DISPSILH and HIDE command the final object as shown in Figure 11.100 is obtained.

Click Subtract → Select one object → Press enter or right-click → Select other object(s) → Press enter or right-click



Before Subtract　　　　After Subtract

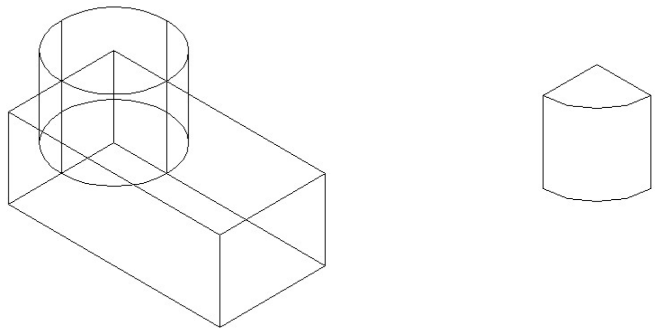
Figure 11.100: Use of Subtract Command

11.9.8 ***Intersect***

Solids Editing Toolbar: 
Menu: **Modify → Solids Editing → Intersect**
Command: **INTERSECT**

When two solids are intersected or overlapped, the **Intersect** command results in keeping only the portion of solid they share with each other. Intersection of a box and a cylinder is shown in Figure 11.101. The sequence for intersection is mentioned below. Now using DISPSILH and HIDE command the final object as shown in Figure 11.101 is obtained.

Click Intersect → Select objects to be intersected → Press enter or right-click



Before Intersect After Intersect

Figure 11.101: Use of Intersect Command

11.10 *Creating Text*

In AutoCAD a text can be written in two ways. One is the Single Line Text and the other is the Multiline Text. In this section both of them will be discussed in short. Some idea in regard to editing text and text style will also be taken into consideration in a nutshell.

11.10.1 *Single Line Text*

Menu: **Draw → Text → Single Line Text**
Command: **DTEXT or TEXT**

Single line text can be written by invoking **Text** command from the menu or it can be initiated by entering the TEXT command in the command area. There are many options in writing the text. Of them the most common ones will be considered here for discussion only. To write the single line text, the sequence as mentioned below has to be followed.

Click Single Line Text in menu or enter TEXT in command area → Click at start point of text → Insert height and press enter → Insert rotation angle and

→ Enter text (1st line) and press enter → Enter text (2nd line) if any and press enter or to end press enter

If th[...]e writt[...]he text, the first and second steps of the above sequence are to be replaced by the following three steps. When J is inserted for justify, there appears 14 different options to place the text, C is one of them. Other options are TL (Top Left), TC (Top Center), TR (Top Right), ML (Middle Left), MC (Middle Center), MR (Middle Right), BL (Bottom Left), BC (Bottom Center) and BR (Bottom Right) etc. According to the requirement any one can be inserted.

Insert J for Justify and press enter → Insert C for center of text and press enter → Click at center point of text

11.10.2 ***Multiline Text***

Draw Toolbar: 
Menu: **Draw → Text → Multiline Text**

Multiline text can be written by invoking command from the toolbar or menu. The steps as shown below are followed to write Multiline text. When the window for the Multiline text is created in the drawing area, a Multiline Text Editor dialog box is displayed (Figure 11.102), where the text is to be written. After clicking on OK button in the dialog box, the text will appear within the window created previously.

Click Multiline Text	→	Click at first corner of window	→	Click at opposite corner of window	→	Enter text in dialog box	→	Click on **OK** button

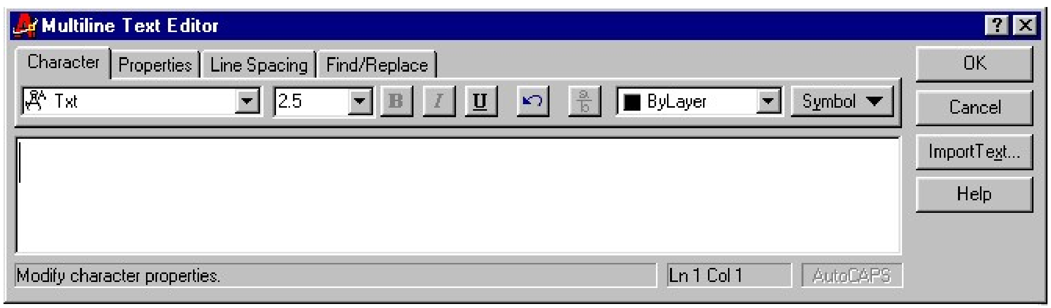


Figure 11.102: Multiline Text Editor Dialog Box

11.10.3 ***Editing Text***

Modify II Toolbar: 
Menu: **Modify → Text**
Command: **DDEDIT**

From the toolbars dialog box (Figure 11.13) the modify II Toolbar can be obtained by selecting crosscheck box beside Modify II. After selecting the text object to be edited, one can invoke the command from the toolbar or menu. It can also be initiated by entering DDEDIT command in the command area. Then an Edit Text dialog box (Figure 11.103) will appear containing the selected text to edit. After required edit, click on the **OK** button in the dialog box, the necessary edit will be done. The text object can also be edited by using editing commands **Move**, **Erase**, **Rotate**, **Copy**, and **Mirror** etc. One can edit the text object by using the short-cut menu. As the text object is selected and right-clicked in the drawing area a short-cut menu appears where clicking **Text Edit**, the Edit Text dialog box (Figure 11.103) is displayed to edit the text.

Figure 11.103: Edit Text Dialog Box

11.10.4 ***Text Style***

Menu: **Format → Text Style**
Command: **STYLE** or DDSTYLE

Invoking command in the menu one can change the current text style or it can be initiated by entering STYLE command in the command area. Then a Text Style dialog box (Figure 11.104) will appear.

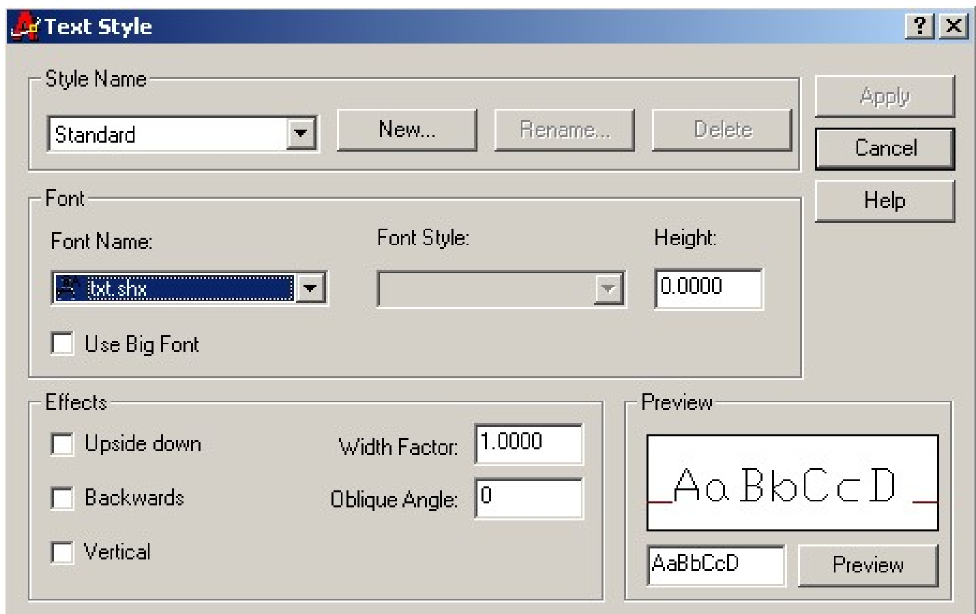


Figure 11.104: Text Style Dialog Box

In the Font area of this dialog box clicking the drop-down arrow, a drop down list consisting a number of Fonts will be displayed from where any desired Font can be selected. As the Font is selected, it is displayed in the Preview area of this dialog box. Once the selection is completed, click on the **Apply** button followed by **Close** button located at the top right corner of the dialog box. In this dialog box there are other options as well which can be selected according to the requirement.

11.11 ***Plotting Drawing***

Standard Toolbar: 
Menu: **File → Plot**

Command: **PLOT** or **PRINT**

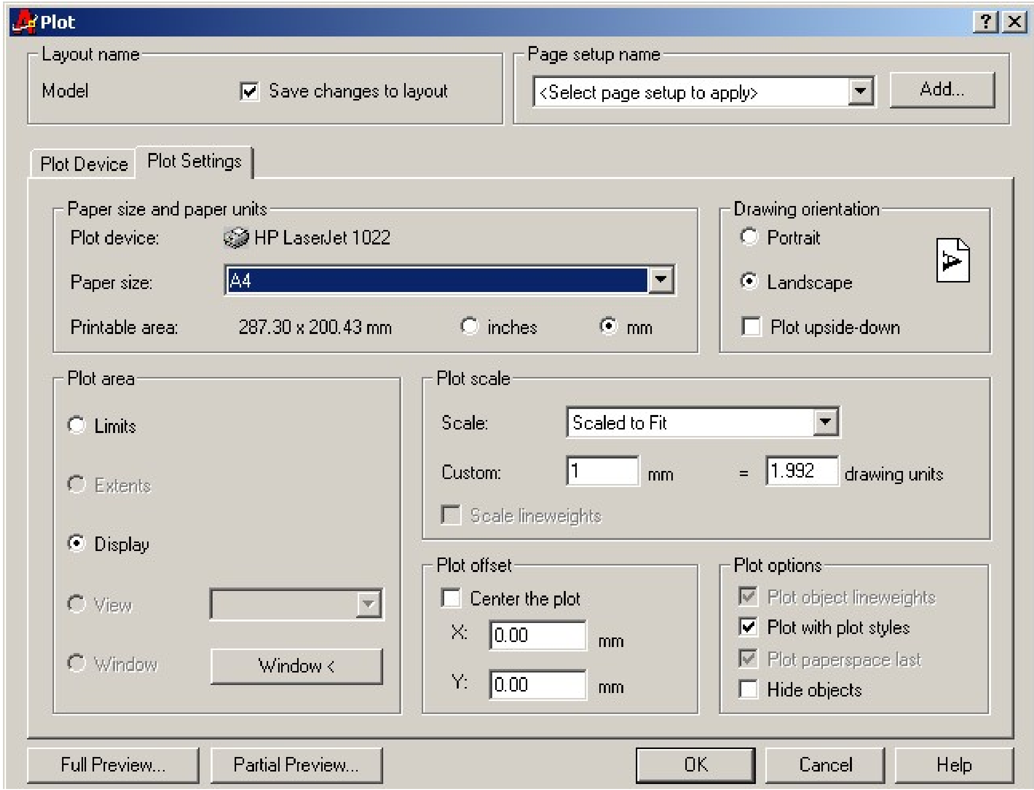


Figure 11.105: Plot Dialog Box (For Plot Settings)

Most drawing jobs remain incomplete until the final outcome is presented on paper. Many printers and plotters can handle a wide range of drawing sizes and paper types. When **Plot** command is invoked in the toolbar or menu, **Plot** dialogue box (Figure 11.105) is displayed containing many options for selection. It can also be initiated by entering PLOT command in the command area. In this dialog box the Plot Settings tab is selected. This dialog box contains different areas such as, Paper size and paper units area, Drawing orientation area, Plot area, Plot scale area, Plot offset and Plot options area. Selecting Plot Device tab in the dialog box, one can select the appropriate Plot Device in the Plotter configuration area.

11.11.1 ***Paper Size and Paper Units***

In Figure 11.106, a drop-down list consisting of various paper sizes is shown. It is obtained when the down arrow button in the Paper size and paper units area of the Plot dialog box is selected. Now the required paper size can be chosen from the drop down list. The measurements can be set either in inches or mm.

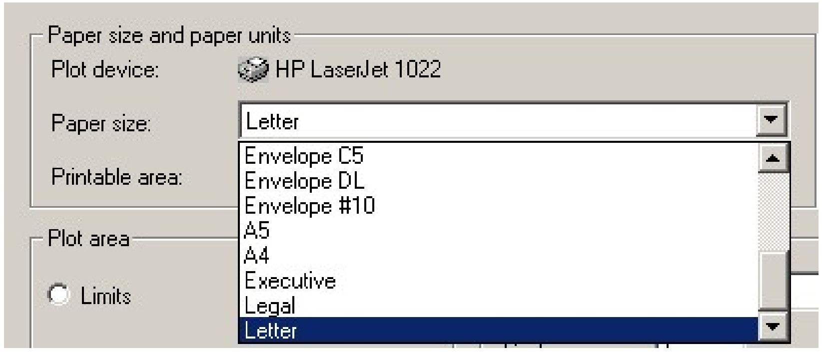


Figure 11.106: Paper Size in Drop Down List

11.11.2 ***Plot Scale***

When the down arrow button in the Plot scale area of the Plot dialog box is selected a drop-down list containing a number of standard scale as shown in Figure 11.107 is displayed. Now the required scale can be chosen from this drop-down list.

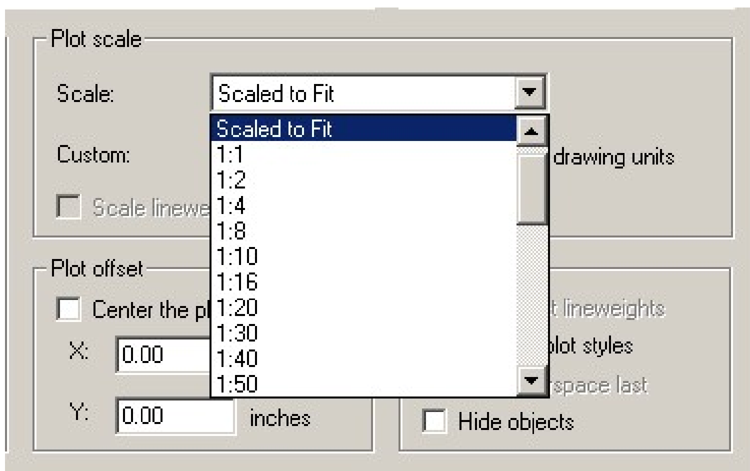


Figure 11.107: Drop Down List in Plot Scale

The scale is displayed in the custom edit box (Figure 11.105). The option **Scaled to Fit** is set by default, which will enable to plot the desired drawing accommodating in the paper selected previously. The scale factor can be changed manually in the custom edit box.

11.11.3 ***Drawing Orientation***

In the drawing orientation area of the Plot dialog box (Figure 11.105) one can choose either the Portrait or Landscape option for the paper orientation by selecting the radio button. A plot upside-down option can also be chosen here if needed.

11.11.4 *Plot Area*

By default the plot is set to the layout. However, it can be set to Limits, Extents and Display in the **Plot** dialogue box by selecting the radio button in the Plot area of the Plot dialog box (Figure 11.105). By clicking on the **Window** tab the drawing will be displayed in the window from where a portion of the drawing may be selected within the window for printing.

11.11.5 *Plot offset*

In the Plot offset area (Figure 11.105) if **Center the plot** option is selected the plotting will be done in reference to the center of the paper by adjusting the values of X and Y. When the printing is not desired at the lower-left corner of the printable area on the paper the plot can be made offset using the X and Y text boxes. For example, if both X and Y are set to 2, the plot will be moved up and right by two units.

11.11.6 *Plot options*

In the Plot options area of the Plot dialog box (Figure 11.105), if the **Plot object line weights** is cleared, in the plotting line weights will not be shown in spite of using line weights in the drawing. When **Plot with Plot Styles** is cleared the plot styles will not be shown in the plotting in spite of using plot styles in the drawing. Clearing the **Plot paper space Last** option the objects are plotted on the paper space layout first. The hidden lines of 3D objects will not be shown if the checkbox of **Hide objects** is selected.

11.11.7 *Plot Preview*

The view of the drawing, which will be plotted on the specified paper size, can be seen before plotting by selecting the **Full Preview** or **Partial Preview** button located at the left bottom corner of the Plot dialog box (Figure 11.105).

Appendix 1: Lettering [ISO 3098 – 2: 2000] (Contd.)

For application on technical Drawings and associated documents, the characteristics of lettering specified by ISO 3098 – 2: 2000 are given here. In order to facilitate lettering the line thickness for lower-case and capital letters has been made same as well as uniform. The distance between the two adjacent lines or the space between letters or numerals has been made at least twice the line thickness. When the thickness of the two adjacent lines is different, the spacing is made twice the thickness of the heavier line. For microfilming and other photographic reproductions it is necessary.

In Figure A1.1 the relative dimensions of the letters and numerals have been presented. The height h of the capital letter is considered as the base of dimensioning. The values of the standard heights are 2.5, 3.5, 5.0, 7.0, 10.0, 14.0, and 20.0 mm. The height h and c shall never be less than 2.5 mm. When both the capitals and lower-case letters will be combined, then if c is 2.5 mm, h will be 3.5 mm. In Tables A1.1 and A1.2 the different recommended ratios for the height of lower-case letters, the space between the characters, the minimum space of the base lines and the minimum space of the words for two standard ratios of d/h are given. Two types of lettering one-inclined 15^0 to the right and another vertical (upright) are shown in Figures A1.2 and A1.3 respectively for line thickness of (1/14)h known as lettering A and in Figures A1.4 and A1.5 respectively for the line thickness of (1/10)h known as lettering B.

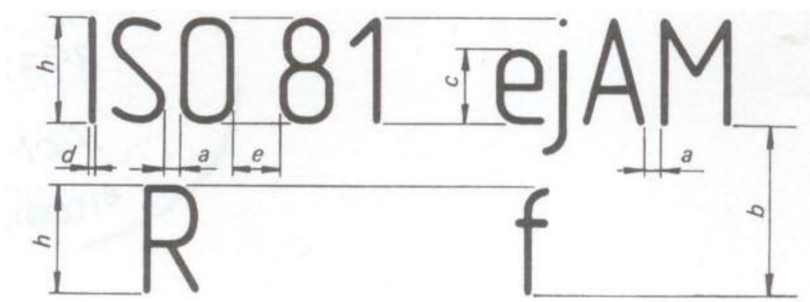


Figure A1.1: Relative Dimensions of Letters and Numerals

Table A1.1: Lettering A (d = h/14)

Characteristic	Ratio	Dimensions (mm)						
Height of capitals h	(14/14)h	2.5	3.5	5	7	10	14	20
Height of lower-case letters c (without stem or tail)	(10/14)h	-	2.5	3.5	5	7	10	14
Spacing between characters a	(2/14)h	0.35	0.5	0.7	1	1.4	2	2.8
Min. spacing of base lines b	(20/14)h	3.5	5	7	10	14	20	28
Min. spacing between words e	(6/14)h	1.05	1.5	2.1	3	4.2	6	8.4
Thickness of lines d	(1/14)h	0.18	0.25	0.35	0.5	0.7	1	1.4

Appendix 1: Lettering [ISO 3098 – 2: 2000] (Contd.)

Table A1.2: Lettering B (d = h/10)

Characteristic		Ratio	Dimensions (mm)						
Height of capitals	h	(10/10)h	2.5	3.5	5	7	10	14	20
Height of lower-case letters (without stem or tail)	c	(7/10)h	-	2.5	3.5	5	7	10	14
Spacing between characters	a	(2/10)h	0.5	0.7	1	1.4	2	2.8	4
Min. spacing of base lines	b	(14/10)h	3.5	5	7	10	14	20	28
Min. spacing between words	e	(6/10)h	1.5	2.1	3	4.2	6	8.4	12
Thickness of lines	d	(1/10)h	0.25	0.35	0.5	0.7	1	1.4	2

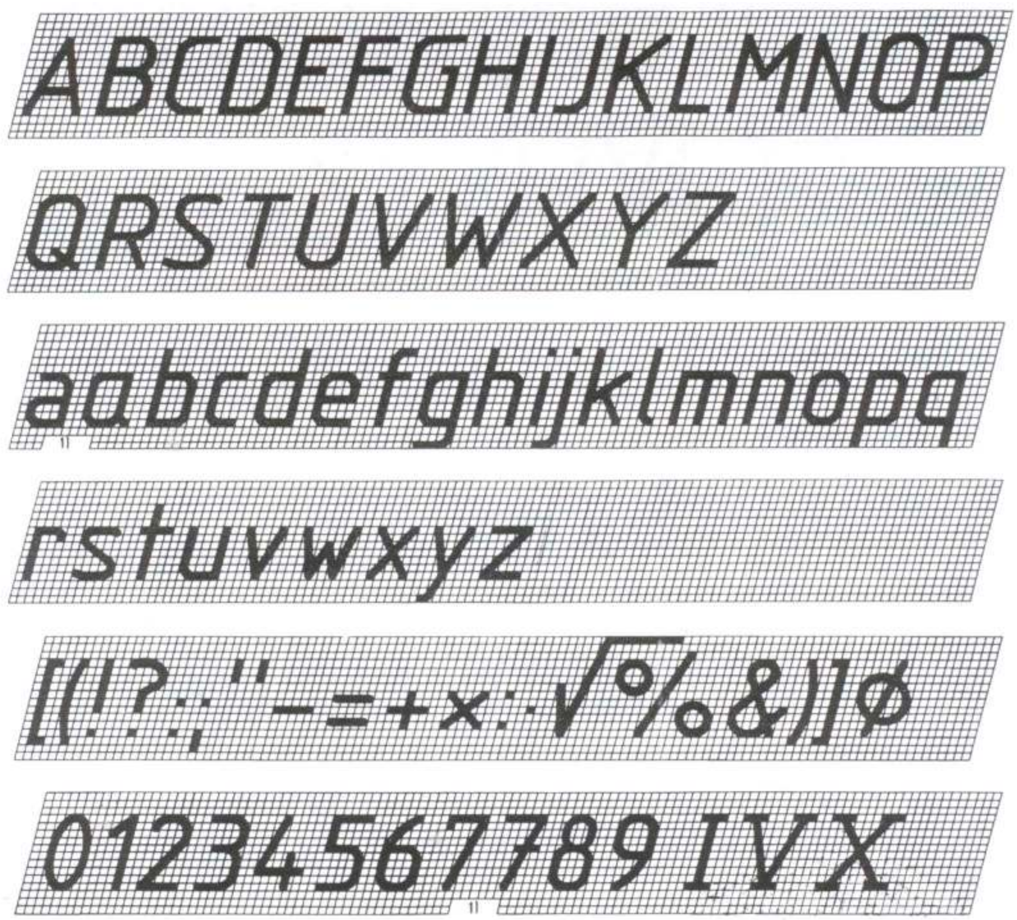


Figure A1.2: Lettering A Inclined

Appendix 1: Lettering [ISO 3098 – 2: 2000] (Contd.)

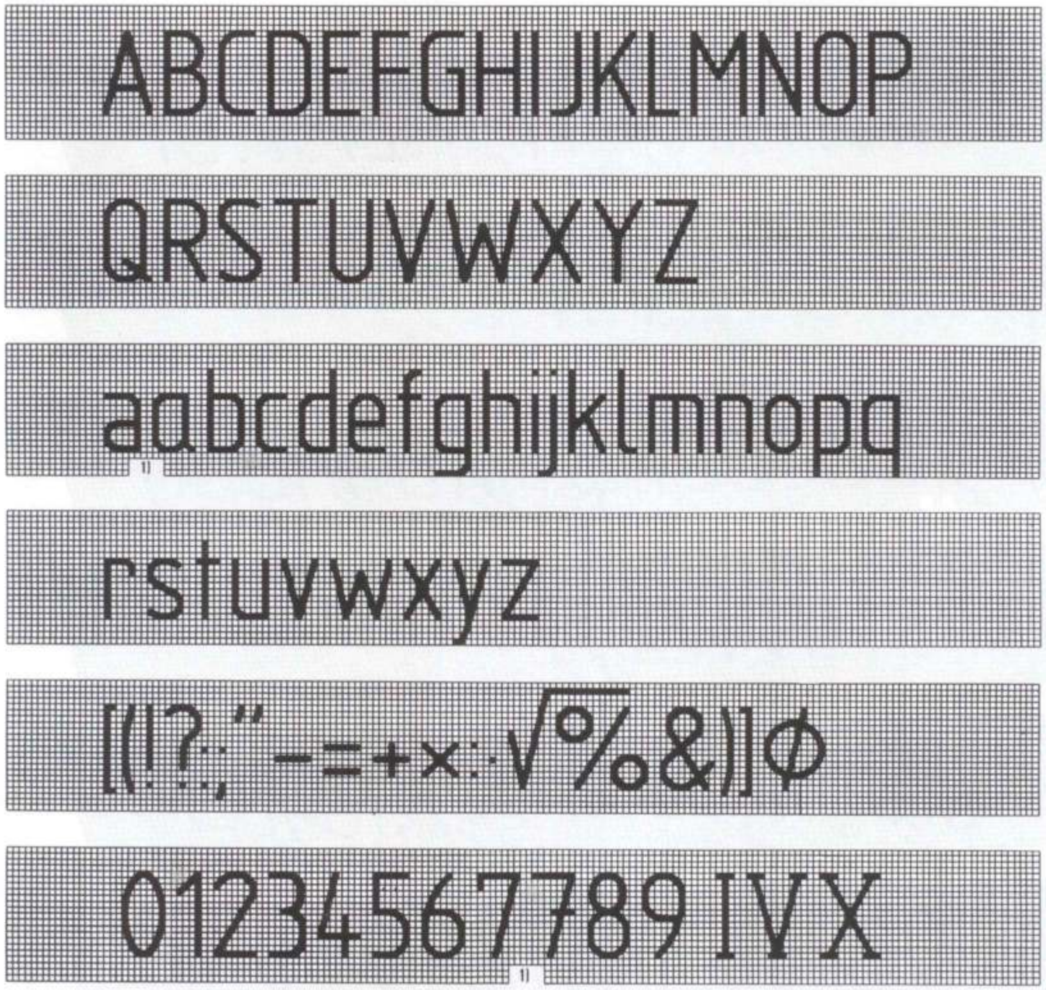


Figure A1.3: Lettering A Vertical

Appendix 1: Lettering [ISO 3098 – 2: 2000] (Contd.)

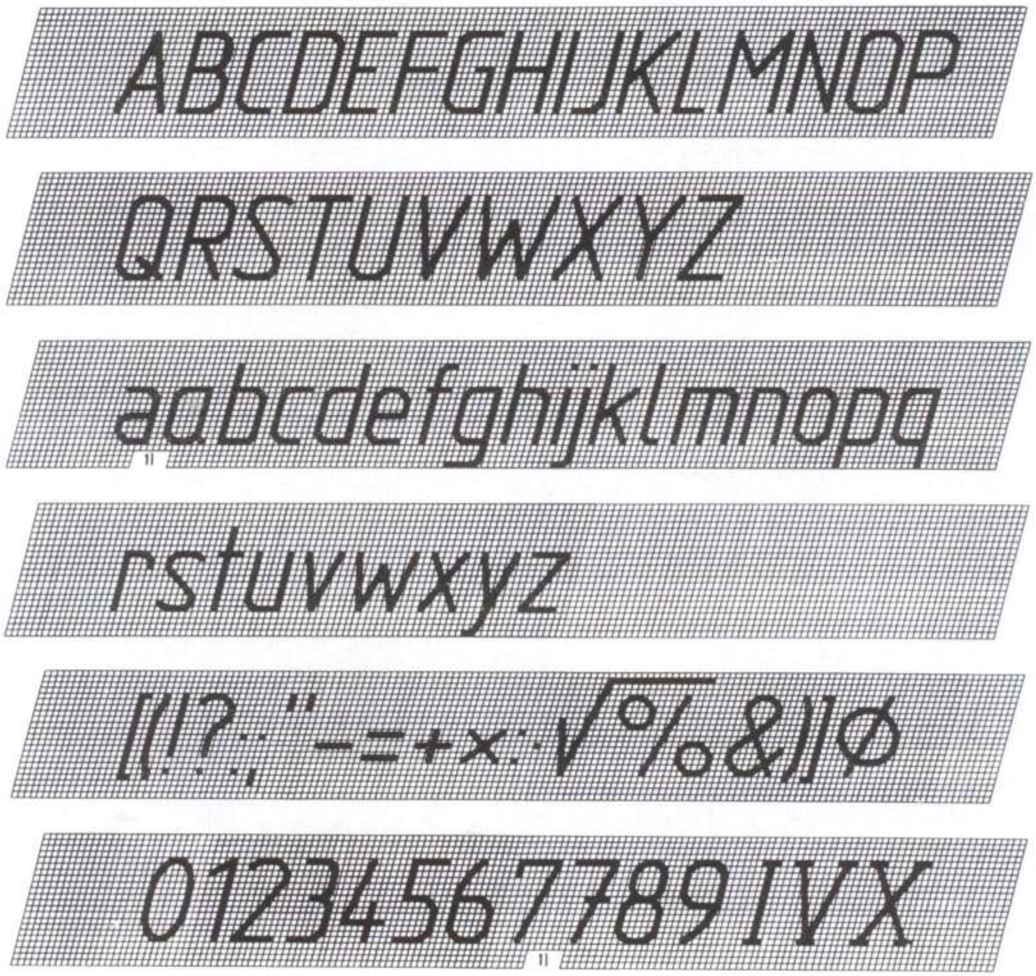


Figure A1.4: Lettering B Inclined

Appendix 1: Lettering [ISO 3098 – 2: 2000] (Contd.)

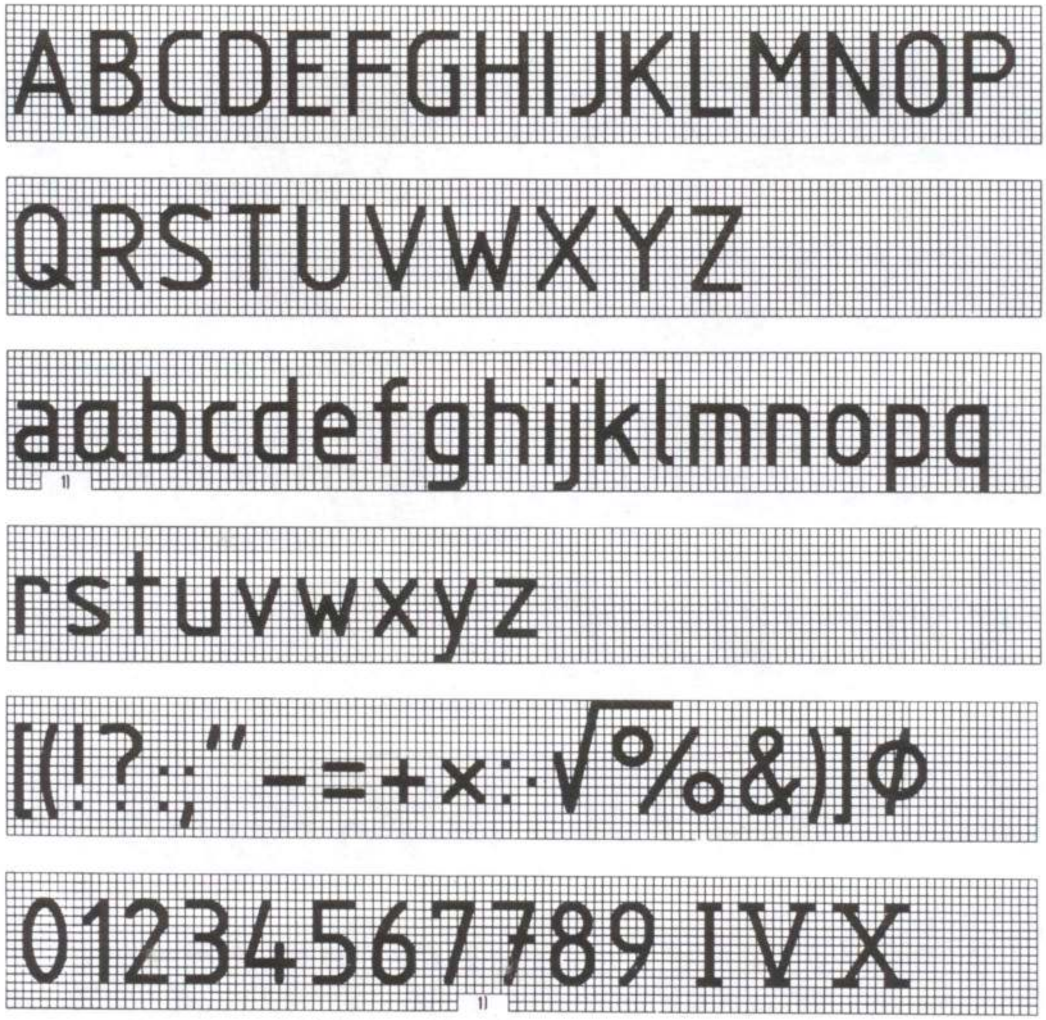


Figure A1.5: Lettering B Vertical

Appendix 2: SI Conversion Table (Contd.)

Length

millimeter (mm)	1 mm = 0.0394 in.	1 in. = 25.4 mm

centimeter (mm)	1 cm = 0.3937 in.	1 ft = 30.48 cm
meter (m)	1 m = 39.37 in. = 3.28 ft	1 yd = 0.914 m
kilometer (km)	1 km = 0.621 mile	1 mile = 1.61 km

Area

square millimeter (mm^2)	1 mm^2 = 0.001 55 sq. in.	1 sq. in. = 6452 mm^2
square centimeter (cm^2)	1 cm^2 = 0.155 sq. in.	1 sq. ft = 0.0929 m^2
square meter (m^2)	1 m^2 = 10.8 sq. ft	1 sq. yd = 0.8361 m^2
	= 1.2 sq. yd	
square kilometer (km^2)	1 km^2 = 0.386 sq. mile	1 sq. mile = 2.59 km^2

Volume

cubic millimeter (mm^3)	1 mm^3 = 0.000 061 cu. in.	1 cu. In = 16387 mm^3
cubic centimeter (cm^3)	1 cm^3 = 0.061 cu. in.	1 cu. In = 16.387 cm^3
cubic meter (m^3)	1 m^3 = 35.32 cu. ft	1 cu. Ft = 0.0283 m^3
	= 1.308 cu. yd	1 cu. Yd = 0.756 m^3
milliliter (ml)	1 ml = 0.035 fl. oz	1 fl. Oz = 28.57 ml

Mass

milligram (mg)	1 mg = 0.000 035 oz	1 oz = 28300 mg
gram (gm)	1 gm = 0.0353 oz	1 oz = 28.3 gm
kilogram (kg)	1 kg = 2.205 lb	1 lb = 0.454 kg
	= 0.0685 slug	1 slug = 14.594 kg
		1 ton(metric) = 1000 kg
		1 ton(short) = 907.18 kg
		1 ton(long) = 1017 kg
		= 2240 lb

Capacity

Liter (L)	1 L = 2.113 pt. (US)	1 pt. (US) = 0.473 L
	= 1.057 qt. (US)	1 qt. (US) = 0.946 L
	= 0.264 gal (US)	1 gal (US) = 3.785 L
	= 1.76 pt. (imp)	1 pt. (imp) = 0.568 L
	= 0.88 qt. (imp)	1 qt. (imp) = 1.137 L
	= 0.22 gal (imp)	1 gal (imp) = 4.546 L

Appendix 2: SI Conversion Table (contd.)

Temperature

Celsius degree (^0C)	$^0C = (5/9)(^0F - 32)$	$^0F = (9/5)^0C + 32$

Density

gram per cubic centimeter (gm/cm^3)	1 gm/cm^3 = $6.237x10^{-5}$ lb/ft^3	1lb/ft^3 = $16.03x10^3$ gm/cm^3
kilogram per cubic meter (kg/m^3)	1 kg/m^3 = $6.237x10^{-2}$ lb/ft^3	1 lb/ft^3 = 16.03 kg/m^3

Force

Newton (N)	1 N	= 0.2248 lb_f	1 lb_f = 4.448 N
		= 0.1019 kg_f	1 kg_f = 9.807 N
		= 10^5dynes	1 dyne = 10^{-5} N
		= 2.248 x 10^{-4} kip	1 kip = 4448 N

Energy/Work

Joule (J)	1 J	= 0.737 ft-lb	1 ft-lb = 1.355 J
		= 0.2387 calorie	1 calorie = 4.19 J
Kilo joule (kJ)	1 kJ	= 0.948 Btu	1 Btu = 1.055 kJ
Mega joule (MJ)	1MJ	= 0.278 kWh	1 kWh = 3.6 MJ

Power

Kilowatt (kW)	1 kW	= 1.34 hp	1 hp = 0.746 kW
		= 0.952 Btu/s	1 Btu/s = 1.05 kW
Watt (W)	1 W	= 0.0226 ft-lb/min	1 ft-lb/min = 44.2537 W

Pressure

Kilopascal (kPa)	1 kPa	= 0.145 psi	1 psi = 6.895 kPa
		= 20.885 psf	1 psf = 47.88 Pa
		= 0.01 ton_f/sq. ft	1 ton_f/sq. ft = 45.76 kPa

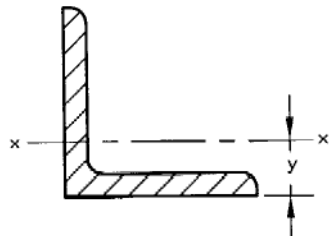
Torque

Newton meter (N.m)	1 N.m	=0.74 lb-ft	1 lb-ft = 1.36 N.m

Speed/Velocity

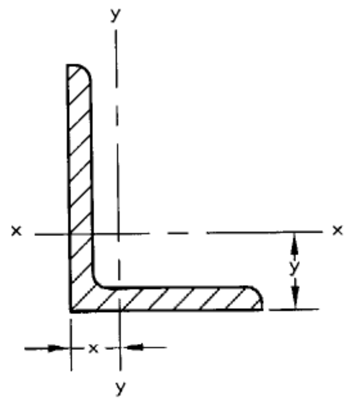
Meter per second (m/s)	1 m/s = 3.28 ft/sec	1 ft/sec = 0.305 m/s
Kilometer per hour (km/hr)	1 km/hr = 0.62 mph	1 mph = 1.61 km/hr

Appendix 3: Properties of Equal Angles.



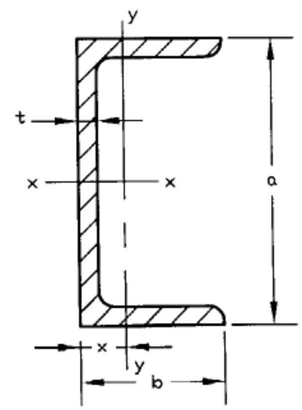
Section Size (mm)	Mass per Meter, m (kg)	Area, A (mm^2)	Moment of Inertia, I_{x-x} (mm^4)	Radius of Gyration, k (mm)	Section Modulus, z (mm^3)	Centroidal Distance, y (mm)
25 x 25 x 3	1.11	142	0.80×10^4	7.5	0.45×10^3	7.2
x 4	1.45	185	1.01×10^4	7.4	0.58×10^3	7.6
x 5	1.77	226	1.20×10^4	7.3	0.71×10^3	8.0
40 x 40 x 4	2.42	308	4.47×10^4	12.1	1.55×10^3	11.2
x 5	2.97	379	5.43×10^4	12.0	1.91×10^3	11.6
x 6	3.52	448	6.31×10^4	11.9	2.26×10^3	12.0
50 x 50 x 5	3.77	480	11.0×10^4	15.1	3.05×10^3	14.0
x 6	4.47	559	12.8×10^4	15.0	3.61×10^3	14.5
x 8	5.82	741	16.3×10^4	14.8	4.68×10^3	15.2
60 x 60 x 5	4.57	582	19.4×10^4	18.2	4.45×10^3	16.4
x 6	5.42	691	22.8×10^4	18.2	5.29×10^3	16.9
x 8	7.09	903	29.2×10^4	18.0	6.89×10^3	17.7
x 10	8.69	1110	34.9×10^4	17.8	8.41×10^3	18.5
80 x 80 x 6	7.34	935	55.8×10^4	24.4	9.57×10^3	21.7
x 8	9.63	1230	72.2×10^4	24.3	12.6×10^3	22.6
x 10	11.9	1510	87.5×10^4	24.1	15.4×10^3	23.4
100 x 100 x 8	12.2	1550	145×10^4	30.6	19.9×10^3	27.4
x 12	17.8	2270	207×10^4	30.2	29.1×10^3	29.0
x 15	21.9	2790	249×10^4	29.8	35.6×10^3	30.2
150 x 150 x 10	23.0	2930	624×10^4	46.2	56.9×10^3	40.3
x 12	27.3	3480	737×10^4	46.0	67.7×10^3	41.2
x 15	33.8	4300	898×10^4	45.7	83.5×10^3	42.5
x 18	40.1	4100	1050×10^4	45.4	98.7×10^3	43.7

Appendix 4: Properties of Unequal Angles.



Section Size (mm)	Mass per Meter, m (kg)	Area, A (mm^2)	Moment of Inertia, I_{x-x} (mm^4)	Moment of Inertia, I_{y-y} (mm^4)	Centroidal Distance, y (mm)	Centroidal Distance, x (mm)
65 x 50 x 5	4.35	554	23.2×10^4	11.9×10^4	19.9	12.5
x 6	5.16	658	27.2×10^4	14.0×10^4	20.4	12.9
x 8	6.75	860	34.8×10^4	17.7×10^4	21.1	13.7
75 x 50 x 6	5.65	719	40.5×10^4	14.4×10^4	24.4	12.1
x 8	7.39	941	52.0×10^4	18.4×10^4	25.2	12.9
80 x 60 x 6	6.37	811	51.4×10^4	24.8×10^4	24.7	14.8
x 7	7.36	938	59.0×10^4	28.4×10^4	25.1	15.2
x 8	8.34	1060	66.3×10^4	31.8×10^4	25.5	15.6
100 x 65 x 7	8.77	1120	113×10^4	37.6×10^4	32.3	15.1
x 8	9.94	1270	127×10^4	42.2×10^4	32.7	15.5
x 10	12.3	1560	154×10^4	51.0×10^4	33.6	16.3
100 x 75 x 8	10.6	1350	133×10^4	64.1×10^4	31.0	18.7
x 10	13.0	1660	162×10^4	77.6×10^4	31.9	19.5
x 12	15.4	1970	189×10^4	90.2×10^4	32.7	20.3
150 x 75 x 10	17.0	2160	501×10^4	85.8×10^4	53.2	16.1
x 12	20.2	2570	589×10^4	99.9×10^4	54.1	16.9
x 15	24.8	3160	713×10^4	120.0×10^4	55.3	18.1

Appendix 5: Properties of Channels



Section Size a x b (mm)	Mass per Meter, m (kg)	Thickness, t (mm)	Area, A (mm^2)	Moment of Inertia, I_{x-x} (mm^4)	Moment of Inertia, I_{y-y} (mm^4)	Centroidal Distance, x (mm)
76 x 38	6.70	5.1	85.3	74.14×10^4	10.66×10^4	11.9
102 x 51	10.42	6.1	132.8	207.7 x 104	29.10×10^4	15.1
127 x 64	14.90	6.4	189.8	482.5×10^4	67.23×10^4	19.4
152 x 76	17.88	6.4	227.7	851.5×10^4	113.8×10^4	22.1
152 x 89	23.84	7.1	303.6	1166×10^4	215.1×10^4	28.6
178 x 76	20.84	6.6	265.4	1337×10^4	134.0×10^4	22.0
178 x 89	26.81	7.6	341.5	1753×10^4	241.0×10^4	27.6
203 x 76	23.82	7.1	303.4	1950×10^4	151.3×10^4	21.3
203 x 89	29.78	8.1	379.4	2491×10^4	264.4×10^4	26.5
229 x 76	26.06	7.6	332.0	2610×10^4	158.7×10^4	20.0
229 x 89	32.76	8.6	417.3	3387×10^4	285.0×10^4	25.3
254 x 76	28.29	8.1	360.3	3367×10^4	162.6×10^4	18.6
254 x 89	35.74	9.1	454.2	4448×10^4	302.4×10^4	24.2
305 x 89	41.69	10.2	531.1	7061×10^4	325.4×10^4	21.8
305 x 102	46.18	10.2	588.3	8214×10^4	499.5×10^4	26.6

Appendix 6: Metric Twist Drill Sizes

Drill Diameters (mm)						
0.40	1.03	**2.20**	**5.00**	**10.00**	21.50	**48.00**
0.42	**1.05**	2.30	5.20	10.30	**22.00**	**50.00**
0.45	1.08	**2.40**	**5.30**	**10.50**	23.00	51.50
0.48	**1.10**	**2.50**	5.40	10.80	**24.00**	**53.00**
0.50	1.15	**2.60**	**5.60**	**11.00**	**25.00**	54.00
0.52	**1.20**	2.70	5.80	11.50	**26.00**	**56.00**
0.55	**1.25**	**2.80**	**6.00**	**12.00**	27.00	58.00
0.58	**1.30**	2.90	6.20	**12.50**	**28.00**	**60.00**
0.60	1.35	**3.00**	**6.30**	**13.00**	29.00	
0.62	**1.40**	3.10	6.50	13.50	**30.00**	
0.65	1.45	**3.20**	**6.70**	**14.00**	31.00	
0.68	**1.50**	3.30	6.80	14.50	**32.00**	
0.70	1.55	**3.40**	6.90	**15.00**	33.00	
0.72	**1.60**	3.50	**7.10**	15.50	**34.00**	
0.75	1.65	**3.60**	7.30	**16.00**	35.00	
0.78	**1.70**	3.70	**7.50**	16.50	**36.00**	
0.80	1.75	**3.80**	7.80	**17.00**	37.00	
0.82	**1.80**	3.90	**8.00**	17.50	**38.00**	
0.85	1.85	**4.00**	8.20	**18.00**	39.00	
0.88	**1.90**	4.10	**8.50**	18.50	**40.00**	
0.90	1.95	**4.20**	8.80	**19.00**	41.00	
0.92	**2.00**	4.40	**9.00**	19.50	**42.00**	
0.95	2.05	**4.50**	9.20	**20.00**	43.50	
0.98	**2.10**	4.60	**9.50**	20.50	**45.00**	
1.00	2.15	**4.80**	9.80	**21.00**	46.50	

Note: Preferred size drills are made **boldface**

Appendix 7: Number and Letter-Size Drills

Number	Size (mm)	Number	Size (mm)	Number	Size (mm)	Number	Size (mm)
80	0.343	50	1.778	20	4.089	A	5.944
79	0.368	49	1.854	19	4.216	B	6.045
78	0.406	48	1.930	18	4.305	C	6.147
77	0.457	47	1.994	17	4.394	D	6.248
76	0.508	46	2.057	16	4.496	E	6.350
75	0.533	45	2.083	15	4.572	F	6.528
74	0.572	44	2.184	14	4.623	G	6.629
73	0.610	43	2.261	13	4.700	H	6.756
72	0.635	42	2.375	12	4.800	I	6.909
71	0.660	41	2.438	11	4.851	J	7.036
70	0.711	40	2.489	10	4.915	K	7.137
69	0.742	39	2.527	9	4.978	L	7.366
68	0.787	38	2.578	8	5.080	M	7.493
67	0.813	37	2.642	7	5.105	N	7.671
66	0.838	36	2.705	6	5.182	O	8.026
65	0.889	35	2.794	5	5.220	P	8.204
64	0.914	34	2.819	4	5.309	Q	8.433
63	0.940	33	2.870	3	5.410	R	8.611
62	0.965	32	2.946	2	5.613	S	8.839
61	0.991	31	3.048	1	5.791	T	9.093
60	1.016	30	3.264			U	9.347
59	1.041	29	3.354			V	9.576
58	1.069	28	3.569			W	9.804
57	1.092	27	3.658			X	10.084
56	1.181	26	3.734			Y	10.262
55	1.321	25	3.797			Z	10.490
54	1.397	24	3.861				
53	1.511	23	3.912				
52	1.613	22	3.988				
51	1.702	21	4.039				

Appendix 8: Metric Screw Threads (ISO724: 1993) (Contd.)

Nominal Diameter !st choice	Nominal Diameter 2nd choice	Coarse Pitch	Fine Pitch 6	4	3	2	1.5	1.25	1	0.75	0.5	0.35	0.25	0.2
1.6		0.35												0.2
	1.8	0.35												0.2
2		0.4											0.25	
	2.2	0.45											0.25	
2.5		0.45										0.35		
3		0.5										0.35		
	3.5	0.6										0.35		
4		0.7									0.5			
	4.5	0.75									0.5			
5		0.8									0.5			
6		1								0.75				
8		1.25							1	0.75				
10		1.5						1.25	1	0.75				
12		1.75					1.5	1.25	1					
	14	2					1.5	1.25	1					
16		2					1.5		1					
	18	2.5				2	1.5		1					
20		2.5				2	1.5		1					
	22	2.5				2	1.5		1					
24		3				2	1.5		1					
	27	3				2	1.5		1					
30		3.5				2	1.5		1					
	33	3.5				2	1.5							
36		4			3	2	1.5							
	39	4			3	2	1.5							
42		4.5		4	3	2	1.5							
	45	4.5		4	3	2	1.5							
48		5		4	3	2	1.5							
	52	5		4	3	2	1.5							
56		5.5		4	3	2	1.5							
	60	5.5		4	3	2	1.5							
64		6		4	3	2	1.5							
	68	6		4	3	2	1.5							
72		6	6	4	3	2	1.5							
	76		6	4	3	2	1.5							
80		6	6	4	3	2	1.5							

Appendix 8: Metric Screw Threads (ISO724: 1993)(Contd.)

Nominal Diameter !st choice	Nominal Diameter 2nd choice	Coarse Pitch	Fine Pitch 6	4	3	2	1.5	1.25	1	0.75	0.5	0.35	0.25	0.2
	85		6	4	3	2								
90		6	6	4	3	2								
	95		6	4	3	2								
100		6	6	4	3	2								
	105		6	4	3	2								
110			6	4	3	2								
	115		6	4	3	2								
	120		6	4	3	2								
125			6	4	3	2								
	130		6	4	3	2								
140			6	4	3	2								
	150		6	4	3	2								
160			6	4	3									
	170		6	4	3									
180			6	4	3									
	190		6	4	3									
200			6	4	3									
	210		6	4	3									
220			6	4	3									
	240		6	4	3									
250			6	4	3									
	260		6	4										
280			6	4										
	300		6	4										

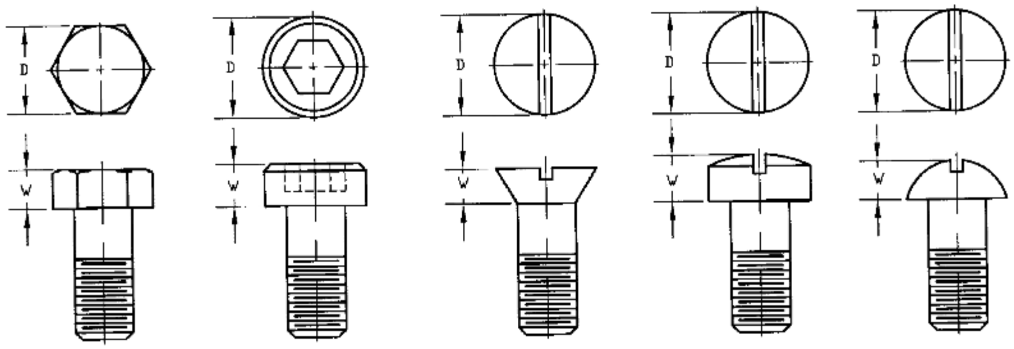
Note: all dimensions are in mm

Appendix 9: Tap Drill Sizes for Specific Metric Screw Threads

Nominal Diameter	Coarse		Fine							
	Pitch	Tap Drill	Pitch	Tap Drill	Pitch	Tap Drill	Pitch	Tap Drill	Pitch	Tap Drill
1.6	0.35	1.25								
1.8	0.35	1.45								
2	0.4	1.6								
2.2	0.45	1.75								
2.5	0.45	2.05	0.35	2.15						
3	0.5	2.5	0.35	2.65						
3.5	0.6	2.9	0.35	3.15						
4	0.7	3.3	0.5	3.5						
4.5	0.75	3.7	0.5	4						
5	0.8	4.2	0.5	4.5						
6	1	5	0.75	5.2						
8	1.25	6.8	1	7	0.75	7.2				
10	1.5	8.5	1.25	8.7	1	9.0	0.75	9.2		
12	1.75	10.2	1.5	10.5	1.25	10.7	1	11		
14	2	12	1.5	12.5	1.25	12.7	1	13		
16	2	14	1.5	14.5	1	15				
18	2.5	15.5	2	16	1.5	16.5	1	17		
20	2.5	17.5	2	18	1.5	18.5	1	19		
22	2.5	19.5	2	20	1.5	20.5	1	21		
24	3	21	2	22	1.5	22.5	1	23		
27	3	24	2	25	1.5	25.5	1	26		
30	3.5	26.5	2	28	1.5	28.5	1	29		
33	3.5	29.5	2	31	1.5	31.5				
36	4	32	2	34	1.5	34.5				
39	4	35	2	37	1.5	37.5				
42	4.5	37.5	4	38	3	39	2	40	1.5	40.5
45	4.5	39	4	41	3	42	2	43	1.5	43.5
48	5	43	4	44	3	45	2	46	1.5	46.5

Note: all dimensions are in mm

Appendix 10: Common Cap Screws

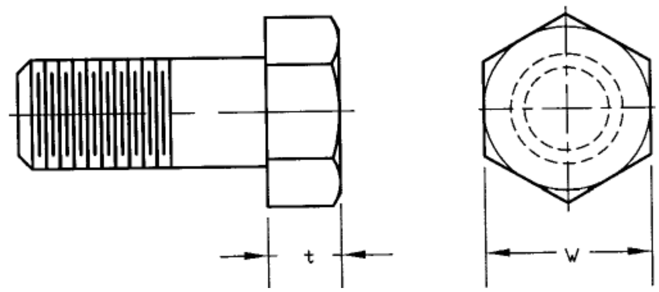


Hexagon Head | Socket Head | Flat Head | Fillister Head | Round or Oval Head

Nominal Size	Hexagon Head		Socket Head		Key Size	Flat Head		Fillister Head		Round or Oval Head	
	D	W	D	W		D	W	D	W	D	W
M3	5.5	2	5.5	3	2.5	5.6	1.6	6	2.4	5.6	
M4	7	2.8	7	4	3	7.5	2.2	8	3.1	7.5	
M5	8.5	3.5	9	5	4	9.2	2.5	10	3.8	9.2	
M6	10	4	10	6	5	11	3	12	4.6	11	
M8	13	5.5	13	8	6	14.5	4	16	6	14.5	
M10	17	7	16	10	8	18	5	20	7.6	18	
M12	19	8	18	12	10						
M14	22	9	22	14	12						
M16	24	10	24	16	14						

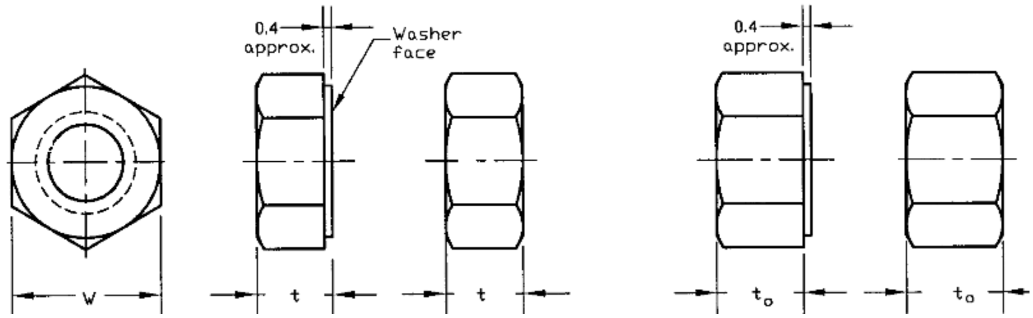
Note: all dimensions are in mm

Appendix 11: Hexagon-Head Bolts



Nominal Bolt Diameter & Thread Pitch	Width Across Flats W (mm)	Thickness t(mm)
M5 x 0.8	8	3.9
M6 x 1	10	4.7
M8 x 1.25	13	5.7
M10 x 1.5	15	6.8
M12 x 1.75	18	8
M14 x 2	21	9.3
M16 x 2	24	10.5
M20 x 2.5	30	13.1
M24 x 3	36	15.6
M30 x 3.5	46	19.5
M36 x 4	55	23.4

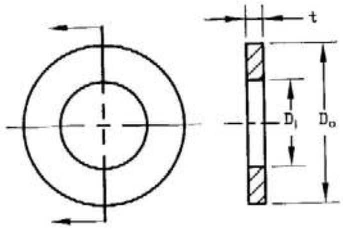
Appendix 12: Hexagon-Head Nuts



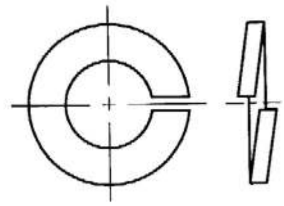
Regular Thick

Nominal Size (mm)	Width Across Flats, W (mm)	Thickness max.	
		Regular, t (mm)	Thick, t_o (mm)
M4 x 0.7	7.0	-	3.2
M5 x 0.8	8.0	4.5	5.3
M6 x 1.0	10.0	5.6	6.5
M8 x 1.25	13.0	6.6	7.8
M10 x 1.5	15.0	9.0	10.7
M12 x 1.75	18.0	10.7	12.8
M14 x 2.0	21.0	12.5	14.9
M16 x 2.0	24.0	14.5	17.4
M20 x 2.5	30.0	18.4	21.2
M24 x 3.0	36.0	22.0	25.4
M30 x 3.5	46.0	26.7	31.0
M36 x 4.0	55.0	32.0	37.6

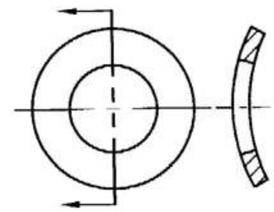
Appendix 13: Common Washer Sizes



Flat Washer



Lock washer

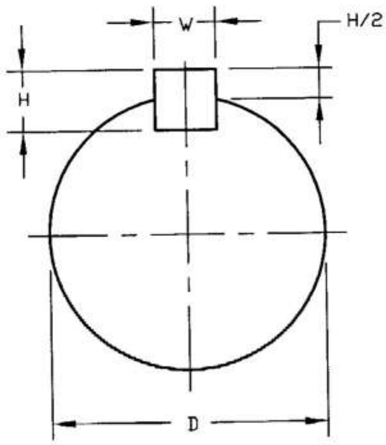


Spring Lock Washer

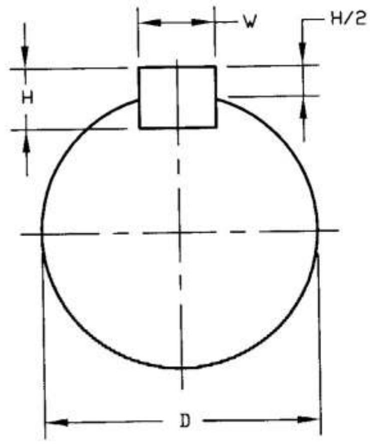
Bolt Size	Flat Washer			Lock Washer			Spring Lock Washer		
	D_i	D_o	t	D_i	D_o	t	D_i	D_o	t
2	2.2	5.5	0.5	2.1	3.3	0.5			
3	3.2	7.0	0.5	3.1	5.7	0.8			
4	4.3	9.0	0.8	4.1	7.1	0.9	4.2	8.0	0.3/0.4
5	5.3	11.0	1.0	5.1	8.7	1.2	5.2	10.0	0.4/0.5
6	6.4	12.0	1.5	6.1	11.1	1.6	6.2	12.5	0.5/0.7
7	7.4	14.0	1.5	7.1	12.1	1.6	7.2	14.0	0.5/0.8
8	8.4	17.0	2.0	8.2	14.2	2.0	8.2	16.0	0.6/0.9
10	10.5	21.0	2.5	10.2	17.2	2.2	10.2	20.0	0.8/1.1
12	13.0	24.0	2.5	12.3	20.2	2.5	12.2	25.0	0.9/1.5
14	15.0	28.0	2.5	14.2	23.2	3.0	14.2	28.0	1.0/1.5
16	17.0	30.0	3.0	16.2	26.2	3.5	16.3	31.5	1.2/1.7
18	19.0	34.0	3.0	18.2	28.2	3.5	18.3	35.5	1.2/2.0
20	21.0	36.0	3.0	20.2	32.2	4.0	20.4	40.0	1.5/2.25
22	23.0	39.0	4.0	22.5	34.5	4.0	22.4	45.0	1.75/2.5
24	25.0	44.0	4.0	24.5	38.5	5.0			
27	28.0	50.0	4.0	27.5	41.5	5.0			
30	31.0	56.0	4.0	30.5	46.5	6.0			

Note: all dimensions are in mm

Appendix 14: Square and Flat Stock Keys



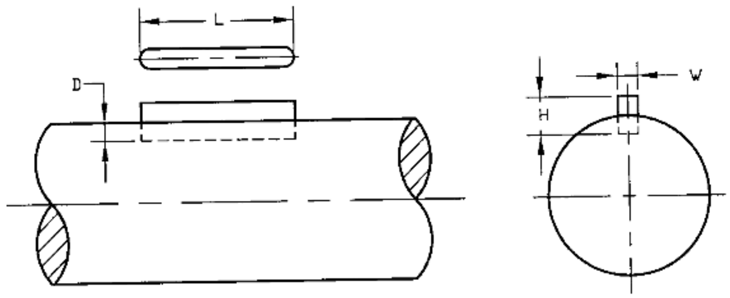
Square



Flat

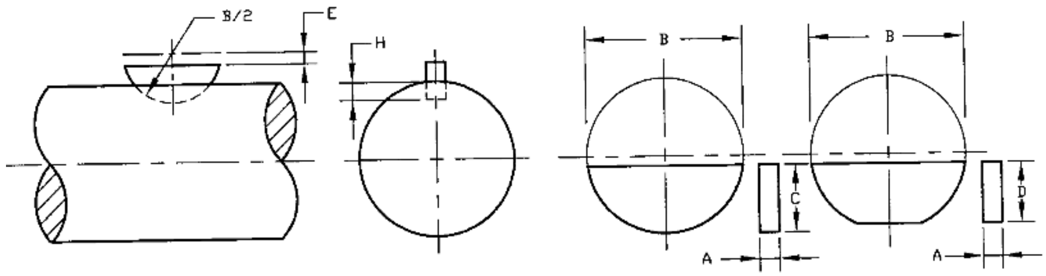
Diameter of Shaft D, (mm)		Square Key Nominal Size (mm)		Flat Key Nominal Size (mm)	
Over	Up To	W	H	W	H
6	8	2	2		
8	10	3	3		
10	12	4	4		
12	17	5	5		
17	22	6	6		
22	30	7	7	8	7
30	38	8	8	10	8
38	44	9	9	12	8
44	50	10	10	14	9
50	58	12	12	16	10

Appendix 15: Pratt and Whitney Keys



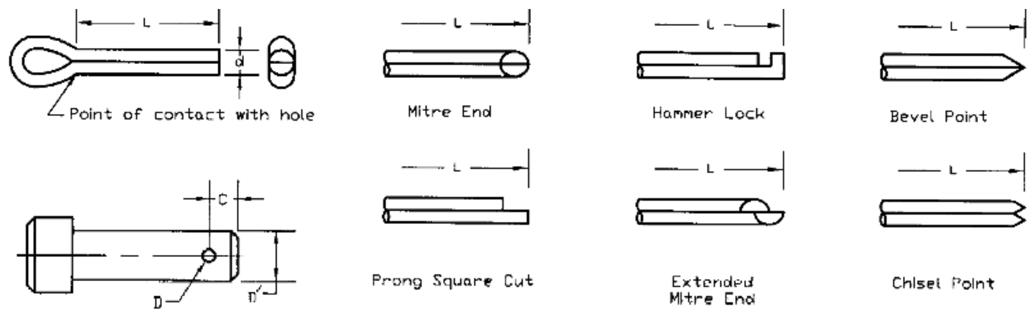
Key No.	Different parameters (mm)				Key No.	Different Parameters (mm)			
	L	W	H	D		L	W	H	D
1	12	1.6	2.4	1.6	22	35	6.5	10.0	6.0
2	12	2.4	3.6	2.4	23	35	10.0	12.0	8.0
3	12	3.2	5.0	3.2	F	35	12.0	14.0	10.0
4	16	2.4	3.6	2.4	24	38	6.5	10.0	6.0
5	16	3.2	5.0	3.2	25	38	10.0	12.0	8.0
6	16	4.0	6.0	4.0	G	38	12.0	14.0	10.0
7	20	3.2	5.0	3.2	51	45	6.5	10.0	6.0
8	20	4.0	6.0	4.0	52	45	10.0	12.0	8.0
9	20	5.0	7.0	5.0	53	45	12.0	14.0	10.0
10	22	4.0	6.0	4.0	26	50	5.0	7.0	5.0
11	22	5.0	7.0	5.0	27	50	6.0	10.0	6.5
12	22	6.0	8.4	7.0	28	50	8.0	12.0	8.0
A	22	6.5	10.0	6.5	29	50	10.0	14.0	10.0
13	25	5.0	7.0	5.0	54	55	6.0	10.0	6.5
14	25	6.0	8.4	6.0	55	55	8.0	12.0	8.0
15	25	6.5	10.0	6.5	56	55	10.0	14.0	10.0
B	25	8.0	12.0	8.0	57	55	11.0	16.0	12.0
16	28	5.0	7.0	5.0	58	65	8.0	12.0	8.0
17	28	7.0	8.0	5.0	59	65	10.0	14.0	10.0
18	28	6.5	10.0	6.0	60	65	11.0	16.0	12.0
C	28	10.0	12.0	8.0	61	65	12.0	20.0	13.0
19	32	5.0	7.0	5.0	30	75	10.0	14.0	10.0
20	32	7.0	8.0	5.0	31	75	11.0	16.0	12.0
21	32	6.5	10.0	6.0	32	75	12.0	20.0	13.0
D	32	10.0	12.0	8.0	33	75	14.0	22.0	14.0
E	32	12.0	14.0	10.0	34	75	16.0	24.0	16.0

Appendix 16: Woodruff Keyways



Key No.	Nominal Size (mm)	Key dimensions (mm)			Key Seat Dimension (mm)
	A x B	E	C	D	H
204	1.6 x 6.4	0.5	2.8	2.8	4.3
304	2.4 x 12.7	1.3	5.1	4.8	3.8
305	2.4 x 15.9	1.5	6.4	6.1	5.1
404	3.2 x 12.7	1.3	5.1	4.8	3.6
405	3.2 x 15.9	1.5	6.4	6.1	4.6
406	3.2 x 19.1	1.5	7.9	7.6	6.4
505	4.0 x 15.9	1.5	6.4	6.1	4.3
506	4.0 x 19.1	1.5	7.9	7.6	5.8
507	4.0 x 22.2	1.5	9.7	9.1	7.4
606	4.8 x 19.1	1.5	7.9	7.6	5.3
607	4.8 x 22.2	1.5	9.7	9.1	7.1
608	4.8 x 25.4	1.5	11.2	10.9	8.6
609	4.8 x 28.6	2.0	12.2	11.9	9.9
807	6.4 x 22.2	1.5	9.7	9.1	6.4
808	6.4 x 25.4	1.5	11.2	10.9	7.9

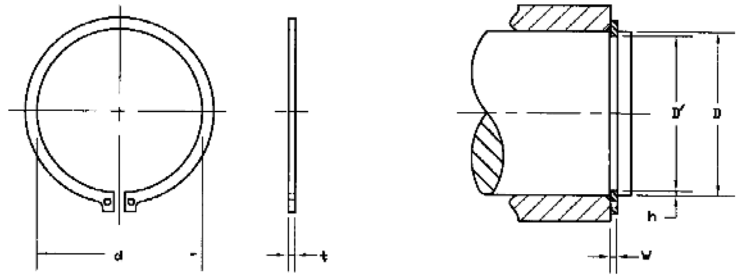
Appendix 17: Cotter Pins

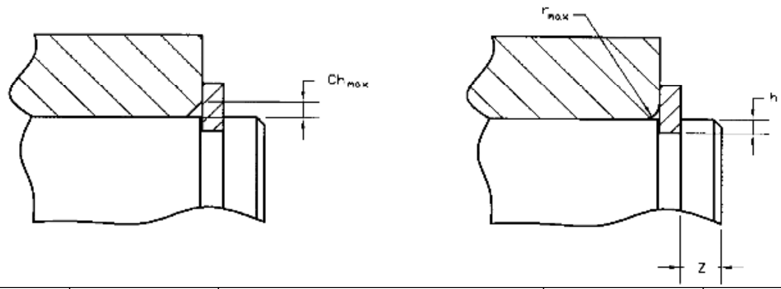


Nominal Bolt or Thread Size Range $D^{/}$	Nominal Cotter-Pin Size d	Cotter-Pin Hole D	Min. End Clearance* C
– 2.5	0.6	0.8	1.5
2.5 – 3.5	0.8	1	2
3.5 – 4.5	1	1.2	2
4.5 – 5.5	1.2	1.4	2.5
5.5 – 7.0	1.6	1.8	2.5
7.0 – 9.0	2	2.2	3
9.0 – 11	2.5	2.8	3.5
11 – 14	3.2	3.6	5
14 – 20	4	4.5	6
20 – 27	5	5.6	7
27 – 39	6.3	6.7	10
39 – 56	8.0	8.5	15
56 - 80	10	10.5	20

Note: all dimensions are in mm
* Distance between end of bolt to center of hole

Appendix18: Retaining Rings – External

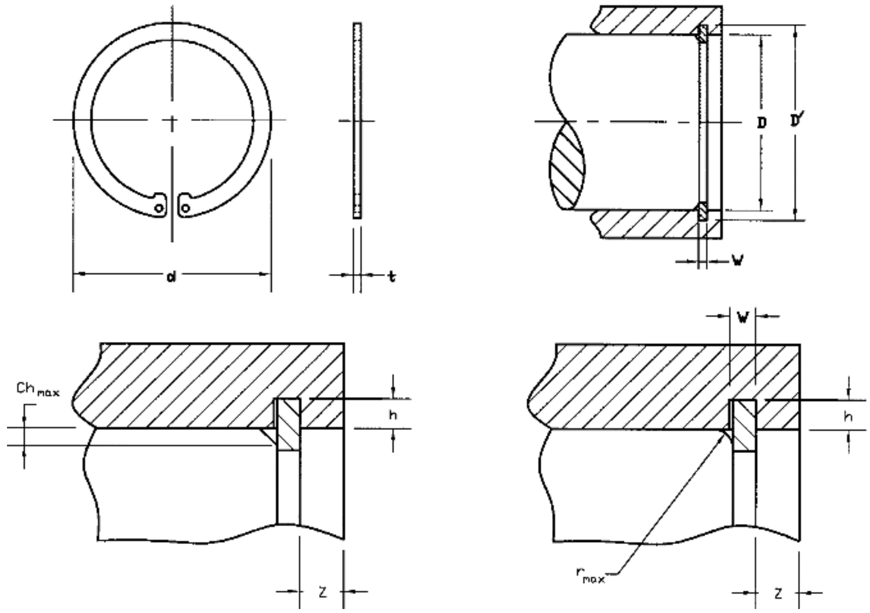




Shaft Dia.	External Series	Retaining Ring Dimensions		Groove Dimensions				Corner Dimensions		Edge Margin	Nomi-nal Groove Depth
D	Size – No.	d	t	$D^{/}$	Tol	W	Tol	r_{max}	Ch_{max}	Z	h
4	M5100-4	3.6	0.25	3.80	-0.08	0.32	+0.05	0.35	0.25	0.30	0.10
6	M5100-6	5.5	0.40	5.70	-0.08	0.50	+0.10	0.35	0.25	0.50	0.15
8	M5100-8	7.2	0.60	7.50	-0.1	0.70	+0.15	0.50	0.35	0.80	0.25
10	M5100-10	9.0	0.60	9.40	-0.1	0.70	+0.15	0.70	0.40	0.90	0.30
12	M5100-12	10.9	0.60	11.35	-0.12	0.70	+0.15	0.80	0.45	1.00	0.33
14	M5100-14	12.9	0.90	13.25	-0.12	1.00	+0.15	0.90	0.50	1.20	0.38
16	M5100-16	14.7	0.90	15.10	-0.15	1.00	+0.15	1.10	0.60	1.40	0.45
18	M5100-18	16.7	1.1	17.00	-0.15	1.20	+0.15	1.20	0.70	1.50	0.50
20	M5100-20	18.4	1.1	18.85	-0.15	1.20	+0.15	1.20	0.70	1.70	0.58
22	M5100-22	20.3	1.1	20.70	-0.15	1.20	+0.15	1.30	0.80	1.90	0.65
24	M5100-24	22.2	1.1	22.60	-0.15	1.20	+0.15	1.40	0.80	2.10	0.70
25	M5100-25	23.1	1.1	23.50	-0.15	1.20	+0.15	1.40	0.80	2.30	0.75
30	M5100-30	27.9	1.3	28.35	-0.20	1.40	+0.15	1.60	1.00	2.50	0.83
35	M5100-35	32.3	1.3	32.90	-0.20	1.40	+0.15	1.80	1.10	3.10	1.05
40	M5100-40	36.8	1.6	37.70	-0.30	1.75	+0.20	2.10	1.20	3.40	1.15
45	M5100-45	41.6	1.6	42.40	-0.30	1.75	+0.20	2.30	1.40	3.90	1.30
50	M5100-50	46.2	1.6	47.20	-0.30	1.75	+0.20	2.40	1.40	4.20	1.40

Note: all dimensions are in mm

Appendix 19: Retaining Rings – Internal



Housing Dia.	Internal Series	Retaining Ring Dimensions		Groove Dimensions				Corner Dimensions		Edge Margin	Nominal Groove Depth
D	Size – No.	d	t	D′	Tol	W	Tol	r_{max}	Ch_{max}	Z	h
8	MN5000-8	8.80	0.4	8.40	+0.06	0.5	+0.1	0.4	0.30	0.6	0.2
10	MN5000-10	11.10	0.6	10.50	+0.1	0.7	+0.15	0.5	0.35	0.8	0.25
12	MN5000-12	13.30	0.6	12.65	+0.1	0.7	+0.15	0.6	0.4	1.0	0.33
14	MN5000-14	15.45	0.9	14.80	+0.1	1.0	+0.15	0.7	0.5	1.2	0.40
16	MN5000-16	17.70	0.9	16.90	+0.1	1.0	+0.15	0.7	0.5	1.4	0.45
18	MN5000-18	20.05	0.9	19.05	+0.1	1.0	+0.15	0.75	0.6	1.6	0.53
20	MN5000-20	22.25	0.9	21.15	+0.15	1.0	+0.15	0.9	0.7	1.7	0.57
22	MN5000-22	24.40	1.1	23.30	+0.15	1.2	+0.15	0.9	0.7	1.9	0.65
24	MN5000-24	26.55	1.1	25.40	+0.15	1.2	+0.15	1.0	0.8	2.1	0.70
25	MN5000-25	27.75	1.1	26.60	+0.15	1.2	+0.15	1.0	0.8	2.4	0.80
30	MN5000-30	33.40	1.3	31.90	+0.20	1.4	+0.15	1.2	1.0	2.9	0.95
35	MN5000-35	38.75	1.3	37.20	+0.20	1.4	+0.15	1.2	1.0	3.3	1.10
40	MN5000-40	44.25	1.6	42.40	+0.20	1.75	+0.20	1.7	1.3	3.6	1.20
45	MN5000-45	49.95	1.6	47.60	+0.20	1.75	+0.20	1.7	1.3	3.9	1.30
50	MN5000-50	55.35	1.6	53.10	+0.20	1.75	+0.20	1.7	1.3	4.6	1.55

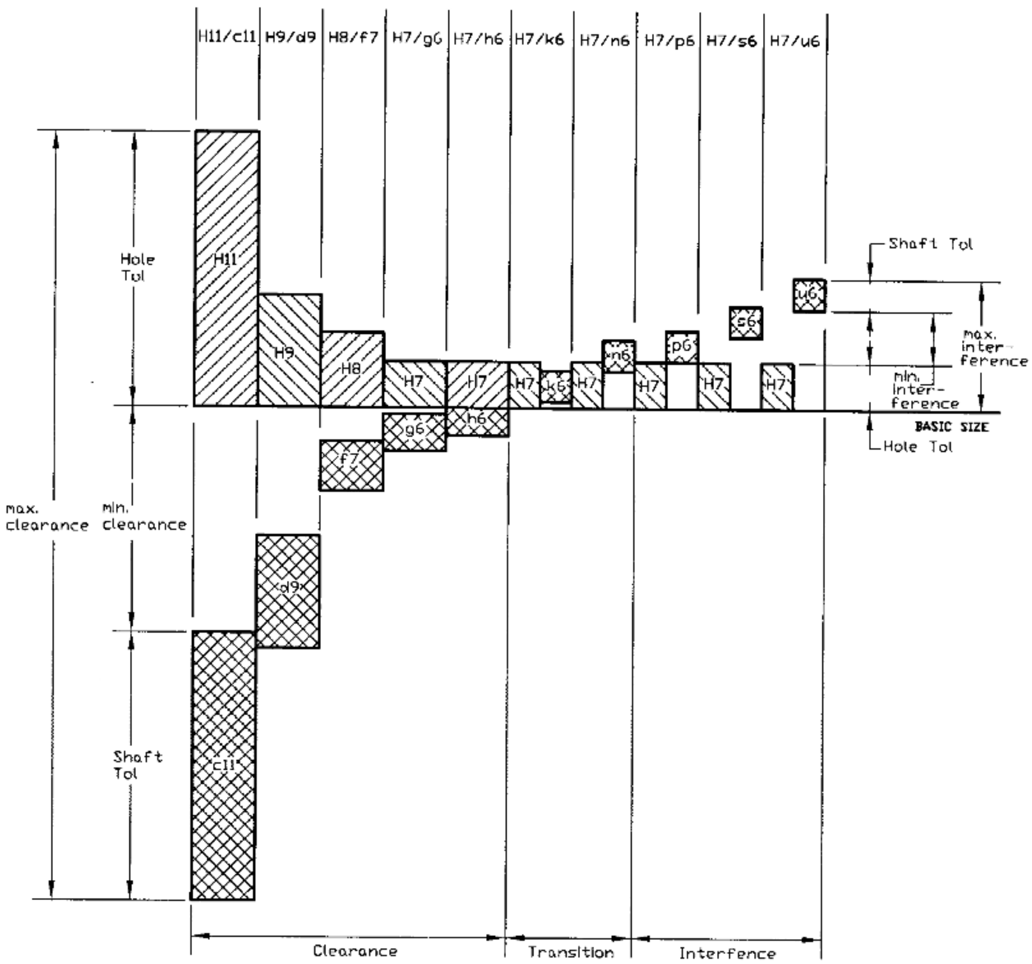
Note: all dimensions are in mm

Appendix 20: Description for Preferred Hole-Basis and Shaft-Basis Fits

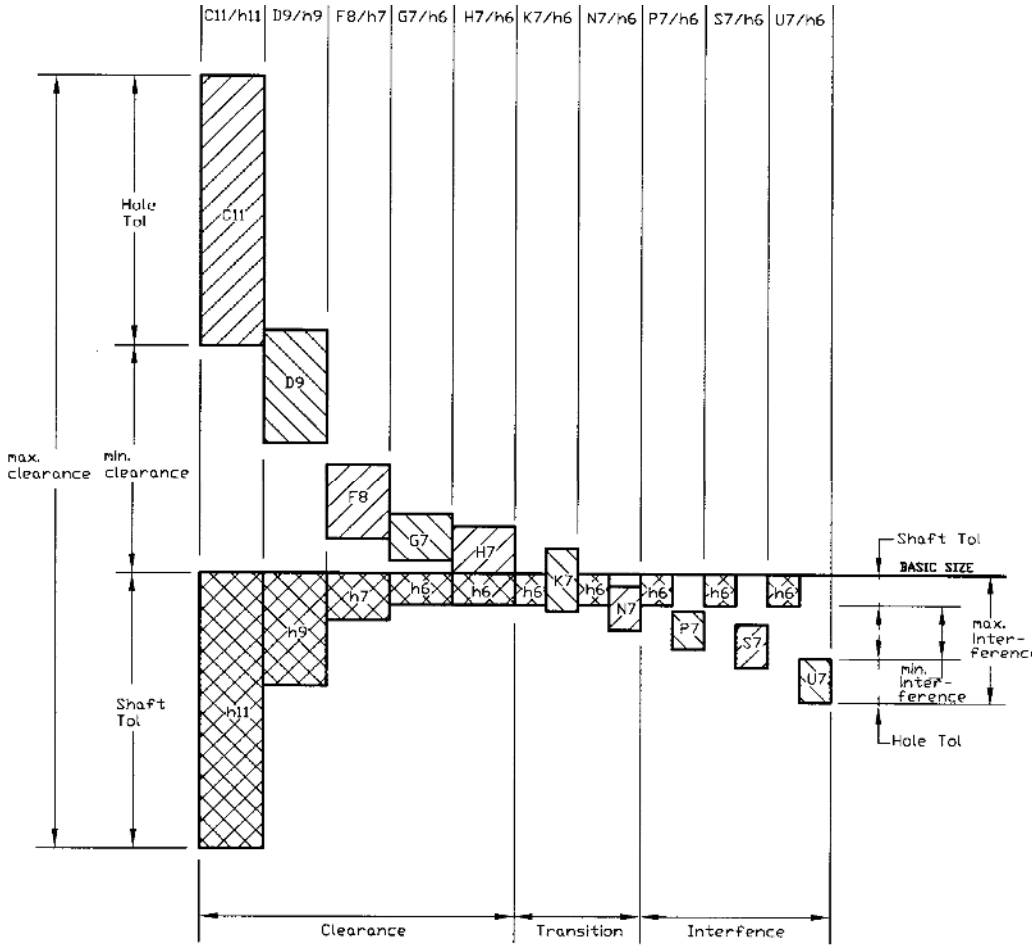
Type of Fit	Description	Symbol	
		Hole Basis	Shaft Basis
Clearance	**Loose running fit** for wide commercial tolerances or allowances on external members	H11/c11	C11/h11
	Free running fit not for use where accuracy is essential, but good for large temperature variations, high running speeds, or heavy journal pressures	H9/d9	D9/h9
	Close running fit for running on accurate machines and for accurate location at moderate speeds and journal pressures	H8/f7	F8/h7
	Sliding fit not intended to run freely, but to move and turn freely and locate accurately	H7/g6	G7/h6
	Locational clearance fit provide snug fit for location of stationary parts, but can be freely assembled and disassembled	H7/h6	H7/h6
Transition	**Locational transition fit** for accurate location, a compromise between clearance and interference	H7/k6	K7/h6
	Locational transition fit for more accurate location where greater interference is permissible	H7/n6	N7/h6
Interference	**Locational interference fit** for parts requiring rigidity and alignment with prime accuracy of location but without special bore pressure requirements	H7/p6	P7/h6
	Medium drive fit for ordinary steel parts or shrink fits on light sections, the tightest fit usable with cast iron	H7/s6	S7/h6
	Force fit suitable for parts which can be highly stressed or for shrink fits where the heavy pressing forces required are impractical	H7/u6	U7/h6

Source: Preferred Metric Limits and Fits, ANSI B4.2-1978. See also BS 4500

Appendix 21: Graphical Representation of Preferred Hole Basis Fits



Appendix 22: Graphical Representation of Preferred Shaft Basis Fits



Appendix 23: Preferred Hole Basis Fits [ANSI B4.2 – 1978 (R 1984)] (Contd.)

Clearance Fits (dimensions in mm)										
Basic Size	Loose Running		Free Running		Close Running		Sliding		Locational Clearance	
	Hole Fit H11	Shaft c11	Hole Fit H9	Shaft d9	Hole Fit H8	Shaft f7	Hole Fit H7	Shaft g6	Hole Fit H7	Shaft h6
1	1.060	0.940	1.025	0.980	1.014	0.994	1.010	0.998	1.010	1.000
Max	0.180		0.070		0.030		0.018		0.016	
	1.000	0.880	1.000	0.955	1.000	0.984	1.000	0.992	1.000	0.994
Min	0.060		0.020		0.006		0.002		0.000	
1.2	1.260	1.140	1.225	1.180	1.214	1.194	1.210	1.198	1.210	1.200
Max	0.180		0.070		0.030		0.018		0.016	
	1.200	1.080	1.200	1.155	1.200	1.184	1.200	1.192	1.200	1.194
Min	0.060		0.020		0.006		0.002		0.000	
1.6	1.660	1.540	1.625	1.580	1.614	1.594	1.610	1.598	1.610	1.600
Max	0.180		0.070		0.030		0.018		0.016	
	1.600	1.480	1.600	1.555	1.600	1.584	1.600	1.592	1.600	1.594
Min	0.060		0.020		0.006		0.002		0.000	
2	2.060	1.940	2.025	1.980	2.014	1.994	2.010	1.998	2.010	2.000
Max	0.180		0.070		0.030		0.018		0.016	
	2.000	1.880	2.000	1.955	2.000	1.984	2.000	1.992	2.000	1.994
Min	0.060		0.020		0.006		0.002		0.000	
2.5	2.560	2.440	2.525	2.480	2.514	2.494	2.510	2.498	2.510	2.500
Max	0.180		0.070		0.030		0.018		0.016	
	2.500	2.380	2.500	2.455	2.500	2.484	2.500	2.492	2.500	2.494
Min	0.060		0.020		0.006		0.002		0.000	
3	3.060	2.940	3.025	2.980	3.014	2.994	3.010	2.998	3.010	3.000
Max	0.180		0.070		0.030		0.018		0.016	
	3.000	2.880	3.000	2.955	3.000	2.984	3.000	2.992	3.000	2.994
Min	0.060		0.020		0.006		0.002		0.000	
4	4.075	3.930	4.030	3.970	4.018	3.990	4.012	3.996	4.012	4.000
Max	0.220		0.090		0.040		0.024		0.020	
	4.000	3.855	4.000	3.940	4.000	3.978	4.000	3.988	4.000	3.992
Min	0.070		0.030		0.010		0.004		0.000	
5	5.075	4.930	5.030	4.970	5.018	4.990	5.012	4.996	5.012	5.000
Max	0.220		0.090		0.040		0.024		0.020	
	5.000	4.855	5.000	4.940	5.000	4.978	5.000	4.988	5.000	4.992
Min	0.070		0.030		0.010		0.004		0.000	
6	6.075	5.930	6.030	5.970	6.018	5.990	6.012	5.996	6.012	6.000
Max	0.220		0.090		0.040		0.024		0.020	
	6.000	5.855	6.000	5.940	6.000	5.978	6.000	5.988	6.000	5.992
Min	0.070		0.030		0.010		0.004		0.000	

8	8.090	7.920	8.036	7.960	8.022	7.987	8.015	7.995	8.015	8.000
Max	0.260		0.112		0.050		0.129		0.024	
	8.000	7.830	8.000	7.924	8.000	7.972	8.000	7.986	8.000	7.991
Min	0.080		0.040		0.013		0.005		0.000	
10	10.090	9.920	10.036	9.960	10.022	9.987	10.015	9.995	10.015	10.000
Max	0.260		0.112		0.050		0.029		0.124	
	10.000	9.830	10.000	9.924	10.000	9.972	10.000	9.986	10.000	9.991
Min	0.080		0.040		0.013		0.005		0.000	

Appendix 23: Preferred Hole Basis Fits [ANSI B4.2 – 1978 (R 1984)] (contd.)

Clearance Fits (dimensions in mm)										
	Loose Running		Free Running		Close Running		Sliding		Locational Clearance	
Basic Size	Hole Fit H11	Shaft c11	Hole Fit H9	Shaft d9	Hole Fit H8	Shaft f7	Hole Fit H7	Shaft g6	Hole Fit H7	Shaft h6
12	12.110	11.905	12.043	11.950	12.027	11.984	12.018	11.994	12.018	12.000
Max	0.315		0.136		0.061		0.035		0.029	
	12.000	11.795	12.000	11.907	12.000	11.966	12.000	11.983	12.000	11.989
Min	0.095		0.050		0.016		0.006		0.000	
16	16.110	15.905	16.043	15.950	16.027	15.984	16.018	15.994	16.018	16.000
Max	0.315		0.136		0.061		0.035		0.029	
	16.000	15.795	16.000	15.907	16.000	15.966	16.000	15.983	16.000	15.989
Min	0.095		0.050		0.016		0.006		0.000	
20	20.130	19.890	20.052	19.935	20.033	19.980	20.021	19.993	20.021	20.000
Max	0.370		0.169		0.074		0.041		0.034	
	20.000	19.760	20.000	19.883	20.000	19.959	20.000	19.980	20.000	19.987
Min	0.110		0.065		0.020		0.007		0.000	
25	25.130	24.890	25.052	24.935	25.033	24.980	25.021	24.993	25.021	25.000
Max	0.370		0.169		0.074		0.041		0.034	
	25.000	24.760	25.000	24.883	25.000	24.959	25.000	24.980	25.000	24.987
Min	0.110		0.065		0.020		0.007		0.000	
30	30.130	29.890	30.052	29.935	30.033	29.980	30.021	29.993	30.021	30.000
Max	0.370		0.169		0.074		0.041		0.034	
	30.000	29.760	30.000	29.883	30.000	29.959	30.000	29.980	30.000	29.987
Min	0.110		0.065		0.020		0.007		0.000	
40	40.160	39.880	40.062	39.920	40.039	39.975	40.025	39.991	40.025	40.000
Max	0.440		0.204		0.089		0.050		0.041	
	40.000	39.720	40.000	39.858	40.000	39.950	40.000	39.975	40.000	39.984
Min	0.120		0.080		0.025		0.009		0.000	
50	50.160	49.870	50.062	49.920	50.039	49.975	50.025	49.991	50.025	50.000
Max	0.450		0.204		0.089		0.050		0.041	
	50.000	49.710	50.000	49.858	50.000	49.950	50.000	49.975	50.000	49.984
Min	0.130		0.080		0.025		0.009		0.000	

60	60.190	59.860	60.074	59.900	60.046	59.970	60.030	59.990	60.030	60.000
Max	0.520		0.248		0.106		0.059		0.049	
	60.000	59.670	60.000	59.826	60.000	59.940	60.000	59.971	60.000	59.981
Min	0.140		0.100		0.030		0.010		0.000	
80	80.190	79.850	80.074	79.900	80.046	79.970	80.030	79.990	80.030	80.000
Max	0.530		0.248		0.106		0.059		0.049	
	80.000	79.660	80.000	79.826	80.000	79.940	80.000	79.971	80.000	79.981
Min	0.150		0.100		0.030		0.010		0.000	
100	100.220	99.830	100.087	99.880	100.054	99.964	100.035	99.988	100.035	100.000
Max	0.610		0.294		0.125		0.069		0.057	
	100.000	99.610	100.000	99.793	100.000	99.929	100.000	99.966	100.000	99.978
Min	0.170		0.120		0.036		0.012		0.000	
120	120.220	119.820	120.087	119.880	120.054	119.964	120.035	119.988	120.035	120.000
Max	0.620		0.294		0.125		0.069		0.057	
	120.000	119.600	120.000	119.793	120.000	119.929	120.000	119.966	120.000	119.978
Min	0.180		0.120		0.036		0.012		0.000	

Appendix 23: Preferred Hole Basis Fits [ANSI B4.2 – 1978 (R 1984)] (contd.)

Transition and Interference Fits (dimensions in mm)										
Basic Size	Locational Transition		Locational transition		Locational Interference		Medium Drive		Force	
	Hole Fit H7	Shaft k6	Hole Fit H7	Shaft n6	Hole Fit H7	Shaft p6	Hole Fit H7	Shaft s6	Hole Fit H7	Shaft u6
1	1.010	1.006	1.010	1.010	1.010	1.012	1.010	1.020	1.010	1.024
Max	0.010		0.006		0.004		-0.004		-0.008	
	1.000	1.000	1.000	1.004	1.000	1.006	1.000	1.014	1.000	1.018
Min	-0.006		-0.010		-0.012		-0.020		-0.024	
1.2	1.210	1.206	1.210	1.210	1.210	1.212	1.210	1.220	1.210	1.224
Max	0.010		0.006		0.004		-0.004		-0.008	
	1.200	1.200	1.200	1.204	1.200	1.206	1.200	1.214	1.200	1.218
Min	-0.006		-0.010		-0.012		-0.020		-0.024	
1.6	1.610	1.606	1.610	1.610	1.610	1.612	1.610	1.620	1.610	1.624
Max	0.010		0.006		0.004		0.004		-0.008	
	1.600	1.600	1.600	1.604	1.600	1.606	1.600	1.614	1.600	1.618
Min	-0.006		-0.010		-0.012		-0.020		-0.024	
2	2.010	2.006	2.010	2.010	2.010	2.012	2.010	2.020	2.010	2.024
Max	0.010		0.006		0.004		-0.004		-0.008	
	2.000	2.000	2.000	2.004	2.000	2.006	2.000	2.014	2.000	2.018
Min	-0.006		-0.010		-0.012		-0.020		-0.024	
2.5	2.510	2.506	2.510	2.510	2.510	2.512	2.510	2.520	2.510	2.524
Max	0.010		0.006		0.004		-0.004		-0.008	
	2.500	2.500	2.500	2.504	2.500	2.506	2.500	2.514	2.500	2.518
Min	-0.006		-0.010		-0.012		-0.020		-0.024	

3	3.010	3.006	3.010	3.010	3.010	3.012	3.010	3.020	3.010	3.024
Max	0.010		0.006		0.004		-0.004		-0.008	
	3.000	3.000	3.000	3.004	3.000	3.006	3.000	3.014	3.000	3.018
Min	-0.006		-0.010		-0.012		-0.020		-0.024	
4	4.012	4.009	4.012	4.016	4.012	4.020	4.012	4.027	4.012	4.031
Max	0.011		0.004		0.000		-0.007		-0.011	
	4.000	4.001	4.000	4.008	4.000	4.012	4.000	4.019	4.000	4.023
Min	-0.009		-0.016		-0.020		-0.027		-0.031	
5	5.012	5.009	5.012	5.016	5.012	5.020	5.012	5.027	5.012	5.031
Max	0.011		0.004		0.000		-0.007		-0.011	
	5.000	5.001	5.000	5.008	5.000	5.012	5.000	5.019	5.000	5.023
Min	-0.009		-0.016		-0.020		-0.027		-0.031	
6	6.012	6.009	6.012	6.016	6.012	6.020	6.012	6.027	6.012	6.031
Max	0.011		0.004		0.000		-0.007		-0.011	
	6.000	6.001	6.000	6.008	6.000	6.012	6.000	6.019	6.000	6.023
Min	-0.009		-0.016		-0.020		-0.027		-0.031	
8	8.015	8.010	8.015	8.019	8.015	8.024	8.015	8.032	8.015	8.037
Max	0.014		0.005		0.000		-0.008		-0.013	
	8.000	8.001	8.000	8.010	8.000	8.015	8.000	8.023	8.000	8.028
Min	-0.010		-0.019		-0.024		-0.032		-0.037	
10	10.015	10.010	10.015	10.019	10.015	10.024	10.015	10.032	10.015	
Max	0.014		0.005		0.000		-0.008		10.037	-0.013
	10.000	10.001	10.000	10.010	10.000	10.015	10.000	10.023	10.000	
Min	- 0.010		-0.019		-0.024		-0.032		10.028	-0.037

Appendix 23: Preferred Hole Basis Fits [ANSI B4.2 – 1978 (R 1984)] (contd.)

Transition and Interference Fits (dimensions in mm)																
Basic Size		Locational Transition			Locational transition			Locational Interference			Medium Drive			Force		
		Hole H7	Shaft k6	Fit	Hole H7	Shaft n6	Fit	Hole H7	Shaft p6	Fit	Hole H7	Shaft s6	Fit	Hole H7	Shaft u6	Fit
12	Max	12.018	12.012	0.017	12.018	12.023	0.006	12.018	12.029	0.000	12.018	12.039	-0.010	12.018	12.044	-0.015
	Min	12.000	12.001	-0.012	12.000	12.012	-0.023	12.000	12.018	-0.029	12.000	12.028	-0.039	12.000	12.033	-0.044
16	Max	16.018	16.012	0.017	16.018	16.023	0.006	16.018	16.029	0.000	16.018	16.039	-0.010	16.018	16.044	-0.015
	Min	16.000	16.001	-0.012	16.000	16.012	-0.023	16.000	16.018	-0.029	16.000	16.028	-0.039	16.000	16.033	-0.044
20	Max	20.021	20.015	0.019	20.021	20.028	0.006	20.021	20.035	-0.001	20.021	20.048	-0.014	20.021	20.054	-0.020
	Min	20.000	20.002	-0.015	20.000	20.015	-0.028	20.000	20.022	-0.035	20.000	20.035	-0.048	20.000	20.041	-0.054
25	Max	25.021	25.015	0.019	25.021	25.028	0.006	25.021	25.035	-0.001	25.021	25.048	-0.014	25.021	25.061	-0.027
	Min	25.000	25.002	-0.015	25.000	25.015	-0.028	25.000	25.022	-0.035	25.000	25.035	-0.048	25.000	25.048	-0.061
30	Max	30.021	30.015	0.019	30.021	30.028	0.006	30.021	30.035	-0.001	30.021	30.048	-0.014	30.021	30.061	-0.027
	Min	30.000	30.002	-0.015	30.000	30.015	-0.028	30.000	30.022	-0.035	30.000	30.035	-0.048	30.000	30.048	-0.061
40	Max	40.025	40.018	0.023	40.025	40.033	0.008	40.025	40.042	-0.001	40.025	40.059	-0.018	40.025	40.076	-0.035
	Min	40.000	40.002	-0.018	40.000	40.017	-0.033	40.000	40.026	-0.042	40.000	40.043	-0.059	40.000	40.060	-0.076

50	Max	50.025	50.018	0.023	50.025	50.033	0.008	50.025	50.042	-0.001	50.025	50.059	-0.018	50.025	50.086	-0.045
	Min	50.000	50.002	-0.018	50.000	50.017	-0.033	50.000	50.026	-0.042	50.000	50.043	-0.059	50.000	50.070	-0.086
60	Max	60.030	60.021	0.028	60.030	60.039	0.010	60.030	60.051	-0.002	60.030	60.072	-0.023	60.030	60.106	-0.057
	Min	60.000	60.002	-0.021	60.000	60.020	-0.039	60.000	60.032	-0.051	60.000	60.053	-0.072	60.000	60.087	-0.106
80	Max	80.030	80.021	0.028	80.030	80.039	0.010	80.030	80.051	-0.002	80.030	80.078	-0.029	80.030	80.121	-0.072
	Min	80.000	80.002	-0.021	80.000	80.020	-0.0.39	80.000	80.032	-0.0.51	80.000	80.059	-0.078	80.000	80.102	-0.121
100	Max	100.035	100.025	0.032	100.035	100.045	0.012	100.035	100.059	-0.002	100.035	100.093	-0.036	100.035	100.146	-0.089
	Min	100.000	100.003	-0.025	100.000	100.023	-0.045	100.000	100.037	-0.059	100.000	100.071	-0.093	100.000	100.124	-0.146
120	Max	120.035	120.025	0.032	120.035	120.045	0.012	120.035	120.059	-0.002	120.035	120.101	-0.044	120.035	120.166	-0.109
	Min	120.000	120.003	-0.025	120.000	120.023	-0.045	120.000	120.037	-0.059	120.000	120.079	-0.101	120.000	120.144	-0.166

Appendix 24: Preferred Shaft Basis Fits [ANSI B4.2 – 1978 (R 1984)] (Contd.)

Clearance Fits (dimensions in mm)										
	Loose Running		Free Running		Close Running		Sliding		Locational Clearance	
Basic Size	Hole C11	Shaft h11	Hole D9	Shaft h9	Hole F8	Shaft h7	Hole G7	Shaft h6	Hole H7	Shaft h6
	Fit		Fit		Fit		Fit		Fit	
1	1.120	1.000	1.045	1.000	1.020	1.000	1.012	1.000	1.010	1.000
Max	0.180		0.070		0.030		0.018		0.016	
	1.060	0.940	1.020	0.975	1.006	0.990	1.002	0.994	1.000	0.994
Min	0.060		0.020		0.006		0.002		0.000	
1.2	1.320	1.200	1.245	1.200	1.220	1.200	1.212	1.200	1.210	1.200
Max	0.180		0.070		0.030		0.018		0.016	
	1.260	1.140	1.220	1.175	1.206	1.190	1.202	1.194	1.200	1.194
Min	0.060		0.020		0.006		0.002		0.000	
1.6	1.720	1.600	1.645	1.600	1.620	1.600	1.612	1.600	1.610	1.600
Max	0.180		0.070		0.030		0.018		0.016	
	1.660	1.540	1.620	1.575	1.606	1.590	1.602	1.594	1.600	1.594
Min	0.060		0.020		0.006		0.002		0.000	
2	2.120	2.000	2.045	2.000	2.020	2.000	2.012	2.000	2.010	2.000
Max	0.180		0.070		0.030		0.018		0.016	
	2.060	1.940	2.020	1.975	2.006	1.990	2.002	1.994	2.000	1.994
Min	0.060		0.020		0.006		0.002		0.000	
2.5	2.620	2.500	2.545	2.500	2.520	2.500	2.512	2.500	2.510	2.500
Max	0.180		0.070		0.030		0.018		0.016	
	2.560	2.440	2.520	2.475	2.506	2.490	2.502	2.494	2.500	2.494
Min	0.060		0.020		0.006		0.002		0.000	
3	3.120	3.000	3.045	3.000	3.020	3.000	3.012	3.000	3.010	3.000
Max	0.180		0.070		0.030		0.018		0.016	
	3.060	2.940	3.020	2.975	3.006	2.990	3.002	2.994	3.000	2.994
Min	0.060		0.020		0.006		0.002		0.000	

4	4.145	4.000	4.060	4.000	4.028	4.000	4.016	4.000	4.012	4.000
Max	0.220		0.090		0.040		0.024		0.020	
	4.070	3.925	4.030	3.970	4.010	3.988	4.004	3.992	4.000	3.992
Min	0.070		0.030		0.010		0.004		0.000	
5	5.145	5.000	5.060	5.000	5.028	5.000	5.016	5.000	5.012	5.000
Max	0.220		0.090		0.040		0.024		0.020	
	5.070	4.925	5.030	4.970	5.010	4.988	5.004	4.992	5.000	4.992
Min	0.070		0.030		0.010		0.004		0.000	
6	6.145	6.000	6.060	6.000	6.028	6.000	6.016	6.000	6.012	6.000
Max	0.220		0.090		0.040		0.024		0.020	
	6.070	5.925	6.030	5.970	6.010	5.988	6.004	5.992	6.000	5.992
Min	0.070		0.030		0.010		0.004		0.000	
8	8.170	8.000	8.076	8.000	8.035	8.000	8.020	8.000	8.015	8.000
Max	0.260		0.112		0.050		0.029		0.024	
	8.080	7.910	8.040	7.964	8.013	7.985	8.005	7.991	8.000	7.991
Min	0.080		0.040		0.013		0.005		0.000	
10	10.170	10.000	10.076	10.000	10.035	10.000	10.020	10.000	10.015	10.000
Max	0.260		0.112		0.050		0.029		0.024	
	10.080	9.910	10.040	9.964	10.013	9.985	10.005	9.991	10.000	9.991
Min	0.080		0.040		0.013		0.005		0.000	

Appendix 24: Preferred Shaft Basis Fits [ANSI B4.2 – 1978 (R 1984)] (contd.)

Clearance Fits (dimensions in mm)										
	Loose Running		Free Running		Close Running		Sliding		Locational Clearance	
Basic Size	Hole Fit C11	Shaft h11	Hole Fit D9	Shaft h9	Hole Fit F8	Shaft h7	Hole Fit G7	Shaft h6	Hole Fit H7	Shaft h6
12	12.205	12.000	12.093	12.000	12.043	12.000	12.024	12.000	12.018	12.000
Max	0.315		0.136		0.061		0.035		0.029	
	12.095	11.890	12.050	11.957	12.016	11.982	12.006	11.989	12.000	11.989
Min	0.095		0.050		0.016		0.006		0.000	
16	16.205	16.000	16.093	16.000	16.043	16.000	16.024	16.000	16.018	16.000
Max	0.315		0.136		0.061		0.035		0.029	
	16.095	15.890	16.050	15.957	16.016	15.982	16.006	15.989	16.000	15.989
Min	0.095		0.050		0.016		0.006		0.000	
20	20.240	20.000	20.117	20.000	20.053	20.000	20.028	20.000	20.021	20.000
Max	0.370		0.169		0.074		0.041		0.034	
	20.110	19.870	20.065	19.948	20.020	19.979	20.007	19.987	20.000	19.987
Min	0.110		0.065		0.020		0.007		0.000	
25	25.240	25.000	25.117	25.000	25.053	25.000	25.028	25.000	25.021	25.000
Max	0.370		0.169		0.074		0.041		0.034	
	25.110	24.870	25.065	24.948	25.020	24.979	25.007	24.987	25.000	24.987
Min	0.110		0.065		0.020		0.007		0.000	

30	30.240	30.000	30.117	30.000	30.053	30.000	30.028	30.000	30.021	30.000
Max	0.370		0.169		0.074		0.041		0.034	
	30.110	29.870	30.065	29.948	30.020	29.979	30.007	29.987	30.000	29.987
Min	0.110		0.065		0.020		0.007		0.000	
40	40.280	40.000	40.142	40.000	40.064	40.000	40.034	40.000	40.025	40.000
Max	0.440		0.204		0.089		0.050		0.041	
	40.120	39.840	40.080	39.938	40.025	39.975	40.009	39.984	40.000	39.984
Min	0.120		0.080		0.025		0.009		0.000	
50	50.290	50.000	50.142	50.000	50.064	50.000	50.034	50.000	50.025	50.000
Max	0.450		0.204		0.089		0.050		0.041	
	50.130	49.840	50.080	49.938	50.025	49.975	50.009	49.984	50.000	49.984
Min	0.130		0.080		0.025		0.009		0.000	
60	60.330	60.000	60.174	60.000	60.076	60.000	60.040	60.000	60.030	60.000
Max	0.520		0.248		0.106		0.059		0.049	
	60.140	59.810	60.100	59.926	60.030	59.970	60.010	59.981	60.000	59.981
Min	0.140		0.100		0.030		0.010		0.000	
80	80.340	80.000	80.174	80.000	80.076	80.000	80.040	80.000	80.030	80.000
Max	0.530		0.248		0.106		0.059		0.049	
	80.150	79.810	80.100	79.926	80.030	79.970	80.010	79.981	80.000	79.981
Min	0.150		0.100		0.030		0.010		0.000	
100	100.390	100.000	100.207	100.000	100.090	100.000	100.047	100.000	100.035	100.000
Max	0.610		0.294		0.125		0.069		0.057	
	100.170	99.780	100.120	99.913	100.036	99.965	100.012	99.978	100.000	99.978
Min	0.170		0.120		0.036		0.012		0.000	
120	120.400	120.000	120.207	120.000	120.090	120.000	120.047	120.000	120.035	120.000
Max	0.620		0.294		0.125		0.069		0.057	
	120.180	119.780	120.120	119.913	120.036	119.965	120.012	119.978	120.000	119.978
Min	0.180		0.120		0.036		0.012		0.000	

Appendix 24: Preferred Shaft Basis Fits [ANSI B4.2 – 1978 (R 1984)] (contd.)

Transition and Interference Fits (dimensions in mm)										
Basic Size	Locational Transition		Locational transition		Locational Interference		Medium Drive		Force	
	Hole Fit K7	Shaft h6	Hole Fit N7	Shaft h6	Hole Fit P7	Shaft h6	Hole Fit S7	Shaft h6	Hole Fit U7	Shaft h6
1	1.000	1.000	0.996	1.000	0.994	1.000	0.986	1.000	0.982	
Max	0.006		0.002		0.000		-0.008		1.000	-0.012
	0.990	0.994	0.986	0.994	0.984	0.994	0.976	0.994	0.972	
Min	-0.010		-0.014		-0.016		-0.024		0.994	-0.028
1.2	1.200	1.200	1.196	1.200	1.194	1.200	1.186	1.200	1.182	
Max	0.006		0.002		0.000		-0.008		1.200	-0.012
	1.190	1.194	1.186	1.194	1.184	1.194	1.176	1.194	1.172	
Min	-0.010		-0.014		-0.016		-0.024		1.194	-0.028

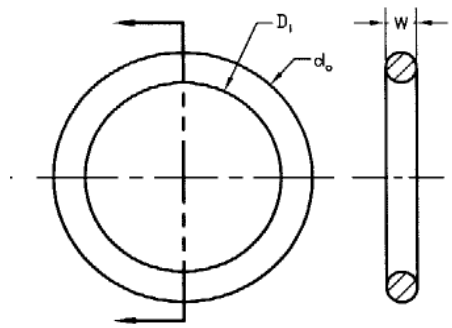
1.6	1.600	1.600	1.596	1.600	1.594	1.600	1.586	1.600	1.582	
Max	0.006		0.002		0.000		-0.008		1.600	-0.012
	1.590	1.594	1.586	1.594	1.584	1.594	1.576	1.594	1.572	
Min	-0.010		-0.014		-0.016		-0.024		1.594	-0.028
2	2.000	2.000	1.996	2.000	1.994	2.000	1.986	2.000	1.982	
Max	0.006		0.002		0.000		-0.008		2.000	-0.012
	1.990	1.994	1.986	1.994	1.984	1.994	1.976	1.994	1.972	
Min	-0.010		-0.014		-0.016		-0.024		1.994	-0.028
2.5	2.500	2.500	2.496	2.500	2.494	2.500	2.486	2.500	2.482	
Max	0.006		0.002		0.000		-0.008		2.500	-0.012
	2.490	2.494	2.486	2.494	2.484	2.494	2.476	2.494	2.472	
Min	-0.010		-0.014		-0.016		-0.024		2.494	-0.028
3	3.000	3.000	2.996	3.000	2.994	3.000	2.986	3.000	2.982	
Max	0.006		0.002		0.000		-0.008		3.000	-0.012
	2.990	2.994	2.986	2.994	2.984	2.994	2.976	2.994	2.972	
Min	-0.010		-0.014		-0.016		-0.024		2.994	-0.028
4	4.003	4.000	3.996	4.000	3.992	4.000	3.985	4.000	3.981	
Max	0.011		0.004		0.000		-0.007		4.000	-0.011
	3.991	3.992	3.984	3.992	3.980	3.992	3.973	3.992	3.969	
Min	-0.009		-0.016		-0.020		-0.027		3.992	-0.031
5	5.003	5.000	4.996	5.000	4.992	5.000	4.985	5.000	4.981	
Max	0.011		0.004		0.000		-0.007		5.000	-0.011
	4.991	4.992	4.984	4.992	4.980	4.992	4.973	4.992	4.969	
Min	-0.009		-0.016		-0.020		-0.027		4.992	-0.031
6	6.003	6.000	5.996	6.000	5.992	6.000	5.985	6.000	5.981	6.000
Max	0.011		0.004		0.000		-0.007		-0.011	
	5.991	5.992	5.984	5.992	5.980	5.992	5.973	5.992	5.969	5.992
Min	-0.009		-0.016		-0.020		-0.027		-0.031	
8	8.005	8.000	7.996	8.000	7.991	8.000	7.983	8.000	7.978	8.000
Max	0.014		0.005		0.000		-0.008		-0.013	
	7.990	7.991	7.981	7.991	7.976	7.991	7.968	7.991	7.963	7.991
Min	-0.010		-0.019		-0.024		-0.032		-0.037	
10	10.005	10.000	9.996	10.000	9.991	10.000	9.983	10.000	9.978	
Max	0.014		0.005		0.000		- 0.008		10.000	- 0.013
	9.990	9.991	9.981	9.991	9.976	9.991	9.968	9.991	9.963	9.991
Min	- 0.010		- 0.019		- 0.024		- 0.032		- 0.037	

Appendix 24: Preferred Shaft Basis Fits [ANSI B4.2 – 1978 (R 1984)] (contd.)

Transition and Interference Fits (dimensions in mm)																
Basic Size		Locational Transition			Locational transition			Locational Interference			Medium Drive			Force		
		Hole K7	Shaft h6	Fit	Hole N7	Shaft h6	Fit	Hole P7	Shaft h6	Fit	Hole S7	Shaft h6	Fit	Hole U7	Shaft h6	Fit
12	Max	12.006	12.000	0.017	11.995	12.000	0.006	11.989	12.000	0.000	11.979	12.000	-0.010	11.974	12.000	-0.015
	Min	11.988	11.989	-0.012	11.977	11.989	-0.023	11.971	11.989	-0.029	11.961	11.989	-0.039	11.956	11.989	-0.044
16	Max	16.006	16.000	0.017	15.995	16.000	0.006	15.989	16.000	0.000	15.979	16.000	-0.010	15.974	16.000	-0.015

	Min	15.988	15.989	-0.012	15.977	15.989	-0.023	15.971	15.989	-0.029	15.961	15.989	-0.039	15.956	15.989	-0.044
20	Max	20.006	20.000	0.019	19.993	20.000	0.006	19.986	20.000	-0.001	19.973	20.000	-0.014	19.967	20.000	-0.020
	Min	19.985	19.987	-0.015	19.972	19.987	-0.028	19.965	19.987	-0.035	19.952	19.987	-0.048	19.946	19.987	-0.054
25	Max	25.006	25.000	0.019	24.993	25.000	0.006	24.986	25.000	-0.001	24.973	25.000	-0.014	24.960	25.000	-0.027
	Min	24.985	24.987	-0.015	24.972	24.987	-0.028	24.965	24.987	-0.035	24.952	24.987	-0.048	24.939	24.987	-0.061
30	Max	30.006	30.000	0.019	29.993	30.000	0.006	29.986	30.000	-0.001	29.973	30.000	-0.014	29.960	30.000	-0.027
	Min	29.985	29.987	-0.015	29.972	29.987	-0.028	29.965	29.987	-0.035	29.952	29.987	-0.048	29.939	29.987	-0.061
40	Max	40.007	40.000	0.023	39.992	40.000	0.008	39.983	40.000	-0.001	39.966	40.000	-0.018	39.949	40.000	-0.035
	Min	39.982	39.984	-0.018	39.967	39.984	-0.033	39.958	39.984	-0.042	39.941	39.984	-0.059	39.924	39.984	-0.076
50	Max	50.007	50.000	0.023	49.992	50.000	0.008	49.983	50.000	-0.001	49.966	50.000	-0.018	49.939	50.000	-0.045
	Min	49.982	49.984	-0.018	49.967	49.984	-0.033	49.958	49.984	-0.042	49.941	49.984	-0.059	49.914	49.984	-0.086
60	Max	60.009	60.000	0.028	59.991	60.000	0.010	59.979	60.000	-0.002	59.958	60.000	-0.023	59.924	60.000	-0.057
	Min	59.979	59.981	-0.021	59.961	59.981	-0.039	59.949	59.981	-0.051	59.928	59.981	-0.072	59.894	59.981	-0.106
80	Max	80.009	80.000	0.028	79.991	80.000	0.010	79.979	80.000	-0.002	79.952	80.000	-0.029	79.909	80.000	-0.072
	Min	79.979	79.981	-0.021	79.961	79.981	-0.039	79.949	79.981	-0.051	79.922	79.981	-0.078	79.879	79.981	-0.121
100	Max	100.010	100.000	0.032	99.990	100.000	0.012	99.976	100.000	-0.002	99.942	100.000	-0.036	99.889	100.000	-0.089
	Min	99.975	99.978	-0.025	99.955	99.978	-0.045	99.941	99.978	-0.059	99.907	99.978	-0.093	99.854	99.978	-0.146
120	Max	120.010	120.000	0.032	119.990	120.000	0.012	119.976	120.000	-0.002	119.934	120.000	-0.044	119.869	120.000	-0.109
	Min	119.975	119.978	-0.025	119.955	119.978	-0.045	119.941	119.978	-0.059	119.899	119.978	-0.101	119.834	119.978	-0.166

Appendix 25: O Rings (Contd.)

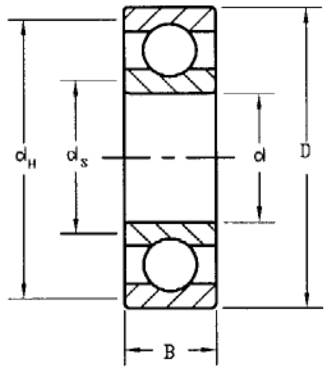


Inner Dia. D_i (mm)	Outer Diameter D_o (mm) for					
	W=1.5mm	W=2.0mm	W=3.0mm	W=4.0mm	W=5.0mm	W=6.0mm
3	6					
4	7					
5	8					
6	9					
7	10					
8	11	12	14			
9	12	13	15			
10	13	14	16	18	20	
12	15	16	18	20	22	24
14	17	18	20	22	24	26
15	18	19	21	23	25	27
16	19	20	22	24	26	28
18	21	22	24	26	28	30
20	23	24	26	28	30	32
22	25	26	28	30	32	34
24	27	28	30	32	34	36
25	28	29	31	33	35	37
26	29	30	32	34	36	38
28	31	32	34	36	38	40
30	33	34	36	38	40	42
32	35	36	38	40	42	44
34	37	38	40	42	44	46
35	38	39	41	43	45	47
36	39	40	42	44	46	48
38	41	42	44	46	48	50

Appendix 25: O Rings (Contd.)

Inner Dia. D_i (mm)	Outer Diameter D_o (mm) for					
	W=1.5mm	W=2.0mm	W=3.0mm	W=4.0mm	W=5.0mm	W=6.0mm
40	43	44	46	48	50	52
42	45	46	48	50	52	54
44	47	48	50	52	54	56
45	48	49	51	53	55	57
46	49	50	52	54	56	58
48	51	52	54	56	58	60
50	53	54	56	58	60	62
52	55	56	58	60	62	64
54	57	58	60	62	64	66
55	58	59	61	63	65	67
56	59	60	62	64	66	68
58	61	62	64	66	68	70
60	63	64	66	68	70	72
65	68	69	71	73	75	77
70	73	74	76	78	80	82
75	78	79	81	83	85	87
80	83	84	86	88	90	92

Appendix 26: Single Row Deep Groove Ball Bearings (Contd.)



Bearing No.	Fillet Radius (mm)	Nominal Bearing Dimensions (mm)			Basic Load Rating (Radial) (kN)		Shoulder Diameter (mm)	
		d	D	B	C	C_o	d_S	d_H
6200	0.6	10	30	9	5.4	2.36	12.5	27
6201	0.6	12	32	10	7.28	3.1	14.5	28
6202	0.6	15	35	11	8.06	3.75	17.5	31
6203	0.6	17	40	12	9.95	4.75	19.5	34
6204	1.0	20	47	14	13.5	6.55	25	41
6205	1.0	25	52	15	14.8	7.8	30	47
6206	1.0	30	62	16	20.3	11.2	35	55
6207	1.0	35	72	17	27	15.3	41	65
6208	1.1	40	80	18	32.5	19	46	72
6209	1.1	45	85	19	35.1	21.6	52	77
6210	1.1	50	90	20	37.1	23.2	56	82
6211	1.5	55	100	21	46.2	29	63	90
6212	1.5	60	110	22	55.3	36	70	99
6213	1.5	65	120	23	58.5	40.5	74	109
6214	1.5	70	125	24	63.7	45	79	114
6215	1.5	75	130	25	68.9	49	86	119
6216	2.0	80	140	26	72.8	55	93	127
6217	2.0	85	150	28	87.1	64	99	136
6218	2.0	90	160	30	101	73.5	104	146
6219	2.1	95	170	32	114	81.5	110	156

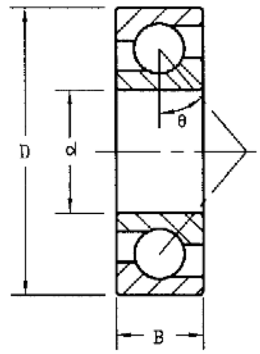
Appendix 26: Single Row Deep Groove Ball Bearings (contd.)

Bearing No.	Fillet Radius (mm)	Nominal Bearing Dimensions (mm)			Basic Load Ratings (radial) (kN)		Shoulder Diameter (mm)	
		d	D	B	C	C_o	d_S	d_H
6300	0.6	10	35	11	8.52	3.4	12.5	31
6301	1.0	12	37	12	10.1	4.15	16	32
6302	1.0	15	42	13	11.9	5.4	19	37
6303	1.1	17	47	14	14.3	6.55	21	41
6304	1.1	20	52	15	16.8	7.8	25	45
6305	1.1	25	62	17	23.4	11.6	31	55
6306	1.1	30	72	19	29.6	16	37	65
6307	1.5	35	80	21	35.1	19	43	70
6308	1.5	40	90	23	42.3	24	49	80
6309	1.5	45	100	25	55.3	31.5	54	89
6310	2.0	50	110	27	65	38	62	97
6311	2.0	55	120	29	74.1	45	70	106
6312	2.1	60	130	31	85.2	52	75	116
6313	2.1	65	140	33	97.5	60	81	125
6314	2.1	70	150	35	111	68	87	134
6315	2.1	75	160	37	119	76.5	93	144
6316	2.1	80	170	39	130	86.5	99	153
6317	3.0	85	180	41	140	96.5	106	161
6318	3.0	90	190	43	151	108	111	170
6319	3.0	95	200	45	159	118	117	179

Note: Values of basic load ratings in the Table are for SKF explorer bearings.

The bearing numbers as mentioned in the Table are for the bearings without shield and seal. However, the bearings are also available with shield on one side designated as (e.g. 6200-Z), shield on both sides designated as (e.g. 6200-2Z), seal on one side designated as (e.g. 6200-RS), seals on both sides designated as (e.g. 6200-2RS) provided that the dimensions and loads for each of them will remain identical as mentioned in the Table.

Appendix 27: Angular Contact Bearings



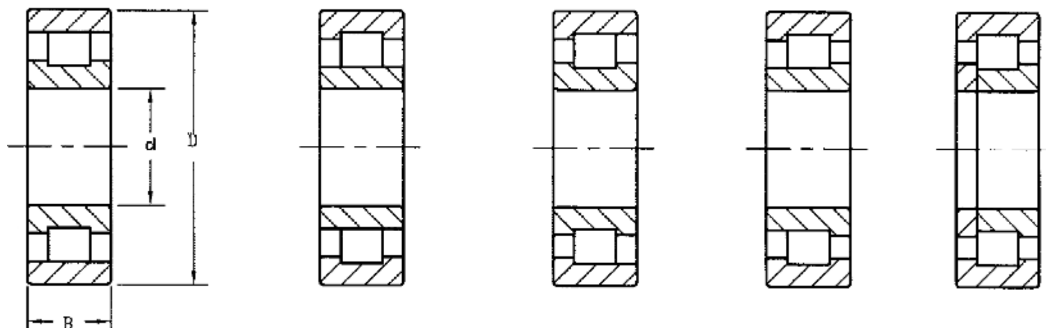
θ = Contact angle

Bearing Number	Bore d (mm)	OD D (mm)	Width B (mm)	Fillet Radius r (mm)	Basic Load ratings (kN)		Shoulder Dia. (mm)	
					C	C_o	d_S	d_H
7200B	10	30	9	0.6	4.68	2.25	12.5	27
7201B	12	32	10	0.6	6.90	3.39	14.5	28
7202B	15	35	11	0.6	7.48	3.93	17.5	31
7203A	17	40	12	0.6	9.3	5	19.5	34
7204A	20	47	14	1.0	12.5	7	25	41
7205A	25	52	15	1.0	14	8.69	30	47
7206A	30	62	16	1.0	19.5	12.5	35	55
7207A	35	72	17	1.0	25.7	17	41	65
7208A	40	80	18	1.0	30.4	21.4	46	72
7209A	45	85	19	1.0	34.4	24.4	52	77
7210A	50	90	20	1.0	35.9	26.7	56	82
7211A	55	100	21	1.5	44.2	33.9	63	90
7303A	17	47	14	0.6	13.9	7.2	19.5	41
7304A	20	52	15	1.0	16.0	8.7	25	47
7305A	25	62	17	1.0	23.0	13.4	30	55
7306A	30	72	19	1.0	29.2	17.6	35	65
7307A	35	80	21	1.0	34.9	22.2	41	72
7308A	40	90	23	1.0	42.2	27.8	46	82
7309A	45	100	25	1.0	54.8	36.8	52	90
7310A	50	110	27	1.0	64.1	44.2	56	99
7311A	55	120	29	1.5	74.8	52.1	63	109

Note: The designation and basic load ratings are for the Japanese bearings. Symbols B and A indicate 40^0 and 30^0 contact angles respectively. Pioneer also designates in the same way.

Appendix 28: Cylindrical Roller Bearings (Contd.)

The bearing number listed in the Table is general. Features of various series of bearings are shown below. According to the requirement the designation of the bearing has to be given as (e.g. N205, NU205 etc.).



N205	NU205	NF205	NJ205	NUP205
Separable outer ring, non-locating	Separable inner ring, non-locating	Separable outer ring, one direction location	Separable inner ring, one-direction location	Separable inner ring two direction location

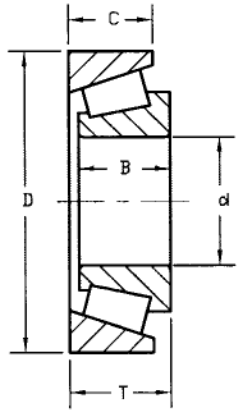
High capacity bearings are also available for each series, which are designated as (e.g. N205 E etc.). The dimensions of the high capacity bearings are identical as in the Table but the basic load rating is higher.

Bearing Number	Bore d (mm)	OD D (mm)	Width B (mm)	Basic Load Rating (kN)	Bearing Number	Bore d (mm)	OD D (mm)	Width B (mm)	Basic Load Rating (kN)
205	25	52	15	16.8	305	25	62	17	28.6
206	30	62	16	22.4	306	30	72	19	36.9
207	35	72	17	31.9	307	35	80	21	44.6
208	40	80	18	41.8	308	40	90	23	56.1
209	45	85	19	44.0	309	45	100	25	72.1
210	50	90	20	45.7	310	50	110	27	88
211	55	100	21	56.1	311	55	120	29	102
212	60	110	22	64.4	312	60	130	31	123
213	65	120	23	76.4	313	65	140	33	138
214	70	125	24	79.2	314	70	150	35	151

Appendix 28: Cylindrical Roller Bearings (Contd.)

Bearing Number	Bore d (mm)	OD D (mm)	Width B (mm)	Basic Load Rating (kN)	Bearing Number	Bore d (mm)	OD D (mm)	Width B (mm)	Basic Load Rating (kN)
215	75	130	25	91.3	315	75	160	37	183
216	80	140	26	106	316	80	170	39	190
217	85	150	28	119	317	85	180	41	212
218	90	160	30	142	318	90	190	43	242
219	95	170	32	165	319	95	200	45	264
220	100	180	34	183	320	100	215	47	303
221	110	200	38	229	321	110	240	50	391
222	120	215	40	260	322	120	260	55	457
223	130	230	40	270	323	130	280	58	539
224	140	250	42	391	324	140	300	62	682
225	150	270	45	446	425	150	320	65	781

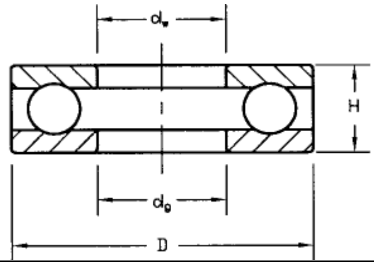
Appendix 29: Tapered Roller Bearings



Bearing Number	Nominal Bearing dimensions (mm)					Basic Load Ratings (Thrust) kN	
	d	D	B	C	T	C	C_o
30203	17	40	12	11	13.25	19.8	13.2
30204	20	47	14	12	15.25	26.8	18.2
30205	25	52	15	13	16.25	32.2	23
30206	30	62	16	14	17.25	41.2	29.5
30207	35	72	17	15	18.25	51.2	37.2
30208	40	80	18	16	19.75	59.8	42.8
30209	45	85	19	16	20.75	64.2	47.8
30210	50	90	20	17	21.75	72.2	55.2
30211	55	100	21	18	22.75	86.5	65.5
30212	60	110	22	19	23.75	97.8	74.5
30213	65	120	23	20	24.75	112	86.2
30214	70	125	24	21	26.25	125	97.5
30215	75	130	25	22	27.25	130	105
30216	80	140	26	22	28.25	150.8	120
30217	85	150	28	24	30.5	168	135
30218	90	160	30	26	31.5	188	152

Note: The values of basic load ratings are given by HENGDA. SKF or FAG designates the bearing number as (e.g. 30203A).

Appendix 30: Thrust Ball Bearings (Single Direction)



Bearing Number	Bearing dimensions (mm)					Basic Load Ratings (kN)	
	D	d_w	d_g	H	r	C	C_o
51100	24	10	10.2	9	0.3	10	14
51101	26	12	12.2	9	0.3	10.3	15.4
51102	28	15	15.2	9	0.3	10.5	16.8
51103	30	17	17.2	9	0.3	10.5	18.2
51104	35	20	20.2	10	0.3	14.2	24.7
51105	42	25	25.2	11	0.6	19.5	37.2
51106	47	30	30.2	11	0.6	20.4	42.2
51107	52	35	35.2	12	0.6	20.4	44.7
51108	60	40	40.2	13	0.6	26.9	62.8
51109	65	45	45.2	14	0.6	27.8	69.1
51110	70	50	50.2	14	0.6	28.8	75.4
51111	78	55	55.2	16	0.6	34.8	93.1
51112	85	60	60.2	17	1	41.4	113
51113	90	65	65.2	18	1	41.7	117
51114	95	70	70.2	18	1	42.3	127
51115	100	75	75.2	19	1	43	127
51116	105	80	80.2	19	1	44.6	141
51117	110	85	85.2	19	1	45.9	150
51118	120	90	90.2	22	1	59.7	190
51120	135	100	100.2	25	1	85	268
51122	145	110	110.2	25	1	87.1	288
51124	155	120	120.2	25	1	89	308
51126	170	130	130.2	30	1	104	352
51128	180	140	140.2	31	1	107	377
51130	190	150	150.2	31	1	109	402
51132	200	160	160.2	31	1	112	427

Note: The values of basic load ratings are provided by HENGDA. NTN designates bearing number in the same way.

Appendix 31: Wire and Sheet Metal Gage and Thickness (Contd.)

Name of Gage	American Standard Or Brown & Sharpe (B &S)	Birming-ham or Stubs Iron Wire (BWG)	US Standard (USS)	US Standard (revised) Manufac-tures Standard	Steel Wire or Washburn & Moen (W&M)	Music Wire	Imperial Wire Gage Imperial Standard (SWG)
Principal Use	Nonferrous Sheet, wire and rod	Tubing, Ferrous Strip, Flat wire	Ferrous sheet and plate	Ferrous sheet and plate	Ferrous Wire Except music wire	Music Wire	Nonferrous
7/0			12.700		12.446		
6/0	14.732		11.906		11.722	0.102	
5/0	13.119		11.113		10.935	0.127	
4/0	11.684	11.532	10.319		10.003	0.152	
3/0	10.404	10.795	9.525		9.208	0.178	
2/0	9.266	9.652	8.731		8.407	0.203	
0	8.253	8.636	7.938		7.785	0.229	
1	7.348	7.620	7.144		7.188	0.254	
2	6.543	7.214	6.747		6.668	0.279	
3	5.827	6.579	6.350	6.073	6.190	0.305	
4	5.189	6.045	5.953	5.695	5.723	0.330	5.89
5	4.620	5.588	5.556	5.314	5.258	0.356	5.39
6	4.115	5.156	5.159	4.935	4.877	0.406	4.88
7	3.665	4.572	4.763	4.554	4.496	0.457	4.47
8	3.264	4.191	4.366	4.176	4.115	0.508	4.06
9	2.906	3.759	3.969	3.797	3.767	0.559	3.66
10	2.588	3.404	3.572	3.416	3.429	0.610	3.25
11	2.305	3.048	3.175	3.038	3.061	0.660	2.95
12	2.053	2.769	2.778	2.657	2.680	0.737	2.64
13	1.828	2.413	2.381	2.278	2.324	0.787	2.34
14	1.628	2.108	1.984	1.897	2.032	0.838	2.03
15	1.450	1.829	1.786	1.709	1.829	0.889	1.83
16	1.290	1.651	1.588	1.519	1.588	0.940	1.63
17	1.150	1.473	1.429	1.367	1.372	0.991	1.42
18	1.024	1.245	1.270	1.214	1.207	1.041	1.22
19	0.912	1.067	1.111	1.062	1.041	1.092	1.02
20	0.812	0.889	0.953	0.912	0.884	1.143	0.91

Note: all sizes are in mm

Appendix 31: Wire and Sheet Metal Gage and Thickness (Contd.)

Name of Gage	American Standard Or Brown & Sharpe (B &S)	Birming-ham or Stubs Iron Wire (BWG)	US Standard (USS)	US Standard (revised) Manufac-tures Standard	Steel Wire or Washburn & Moen (W&M)	Music Wire	Imperial Wire Gage Imperial Standard (SWG)
Principal Use	Nonferrous Sheet, wire and rod	Tubing, Ferrous Strip, Flat wire	Ferrous sheet and plate	Ferrous sheet and plate	Ferrous Wire Except music wire	Music Wire	Nonferrous
21	0.723	0.813	0.873	0.836	0.805	1.194	0.81
22	0.644	0.711	0.794	0.760	0.726	1.245	0.71
23	0.573	0.635	0.714	0.683	0.655	1.295	0.61
24	0.511	0.559	0.635	0.607	0.584	1.397	0.56
25	0.455	0.508	0.556	0.531	0.518	1.499	0.51
26	0.405	0.457	0.476	0.455	0.460	1.600	0.46
27	0.361	0.406	0.437	0.417	0.440	1.702	0.42
28	0.321	0.356	0.397	0.379	0.412	1.803	0.38
29	0.286	0.330	0.357	0.343	0.381	1.905	0.35
30	0.255	0.305	0.318	0.305	0.356	2.032	0.32
31	0.227	0.254	0.278	0.267	0.335	2.159	….
32	0.202	0.229	0.258	0.246	0.325	2.286	0.27
33	0.180	0.203	0.238	0.229	0.230	2.413	0.25
34	0.160	0.178	0.218	0.208	0.264		0.23
35	0.143	0.127	0.198	0.191	0.241		0.21
36	0.127	0.102	0.179	0.170	0.229		…..
37	0.113		0.169	0.163	0.216		0.17
38	0.101		0.159	0.152	0.203		0.15
39	0.090				0.191		…..
40	0.080				0.178		o.12

Note: all sizes are in mm

Appendix 32: Dimensions of Steel Tubes (BS 1387:1967)

Series	Nominal Bore (mm)	Outside diameter (mm)		Thickness (mm)	Weight of Black Tube (kg/m)	
		Max.	Min.		Plain End	Screwed & Socketed
Light	6	10.1	9.7	1.8	0.361	0.384
	8	13.6	13.2	1.8	0.517	0.521
	10	17.1	15.7	1.8	0.674	0.680
	15	21.4	21.0	2.0	0.952	0.961
	20	26.9	26.4	2.35	1.41	1.42
	25	33.8	33.2	2.65	2.01	2.03
	32	42.5	41.9	2.65	2.58	2.61
	40	48.4	47.8	2.9	3.25	3.29
	50	60.2	59.6	2.9	4.11	4.18
	65	76.0	75.2	3.35	5.80	5.92
	80	88.7	87.9	3.35	6.81	6.98
	100	113.9	113.0	3.65	9.89	10.2
Medium	6	10.4	9.8	2.0	0.407	0.410
	8	13.9	13.3	2.35	0.650	0.654
	10	17.4	16.8	2.35	0.852	0.858
	15	21.7	21.1	2.65	1.22	1.23
	20	27.2	26.6	2.65	1.58	1.59
	25	34.2	33.4	3.25	2.44	2.46
	32	42.9	42.1	3.25	3.14	3.17
	40	48.8	48.0	3.25	3.61	3.65
	50	60.8	59.8	3.65	5.10	5.17
	65	76.6	75.4	3.65	6.51	6.63
	80	89.5	88.1	4.05	8.47	8.64
	100	114.9	113.3	4.5	12.1	12.4
	125	140.6	138.7	4.85	16.2	16.7
	150	166.1	164.3	4.85	19.2	19.8
Heavy	6	10.4	9.3	2.65	0.493	0.496
	8	13.9	13.3	2.9	0.769	0.773
	10	17.4	16.8	2.9	1.02	1.03
	15	21.7	21.1	3.25	1.45	1.46
	20	27.2	26.6	3.25	1.90	1.91
	25	34.2	33.4	4.05	2.97	2.99
	32	42.9	42.1	4.05	3.84	3.87
	40	48.8	48.0	4.05	4.43	4.47
	50	60.8	59.8	4.5	6.17	6.24
	65	76.6	75.4	4.5	7.90	8.02
	80	89.5	88.1	4.85	10.1	10.3
	100	114.9	113.3	5.4	14.4	14.7
	125	140.6	138.7	5.4	17.8	18.3
	150	166.1	164.1	5.4	21.2	21.8

Note: BS stands for British Standard

Appendix 33: Preferred Sizes and Renard Numbers

Millimeters
0.05, 0.06, 0.08, 0.10, 0.12, 0.16, 0.20, 0.25, 0.30, 0.40, 0.50, 0.60 0.70, 0.80, 0.90, 1.0, 1.1, 1.2, 1.4, 1.5, 1.6, 1.8, 2.0, 2.2, 2.5, 2.8, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 8.0, 9.0, 10, 11, 12, 14, 16, 18, 20, 22, 25, 28, 30, 32, 35, 40, 45, 50, 60, 80, 100, 120, 140, 160, 180, 200, 250, 300
Renard Numbers
1st Choice, R5 : 1, 1.6, 2.5, 4, 6.3, 10 2nd Choice, R10: 1.25, 2, 3.15, 5, 8 3rd Choice, R20: 1.12, 1.4, 1.8, 2.24, 2.8, 3.55, 4.5, 5.6, 7.1, 9 4th Choice, R40: 1.06, 1.18, 1.32, 1.5, 1.7, 1.9, 2.12, 2.36, 2.65, 3, 3.35, 3.75, 4.25, 4.75, 5.3, 6, 6.7, 7.5, 8.5, 9.5
Fraction Inches
1/64, 1/32, 1/16, 3/32, 1/8, 5/32, 3/16, ¼, 5/16, 3/8, 7/16, ½, 9/16, 5/8, 11/16, 3/4, 7/8, 1, 1¼, 1½, 1 ¾, 2, 2¼, 2½, 2 ¾, 3, 3¼, 3½, 3 ¾, 4, 4¼, 4½, 4 ¾, 5, 5¼, 5½, 5 ¾, 6, 6½, 7, 7½, 8, 8½, 9, 9½, 10, 10½, 11, 11½, 12, 12½, 13, 13½, 14, 14½, 15, 15½, 16, 16½, 17, 17½, 18, 18½, 19, 19½, 20
Decimal Inches
0.010, 0.012, 0.016, 0.020, 0.025, 0.032, 0.040, 0.05, 0.06, 0.08, 0.10, 0.12, 0.16, 0.20, 0.24, 0.30, 0.40, 0.50, 0.60, 0.80, 1.00, 1.20, 1.40, 1.60, 1.80, 2.0, 2.4, 2.6, 2.8, 3.0, 3.2, 3.4, 3.6, 3.8, 4.0, 4.2, 4.4, 4.6, 4.8, 5.0, 5.2, 5.4, 5.6, 5.8, 6.0, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0, 10.5, 11.0, 11.5, 12.0, 12.5, 13.0, 13.5, 14.0, 14.5, 15.0, 15.5, 16.0, 16.5, 17.0, 17.5, 18.0, 18.5, 19.0, 19.5, 20.0

Note: All parts or items are not available in all the sizes shown in the Table.